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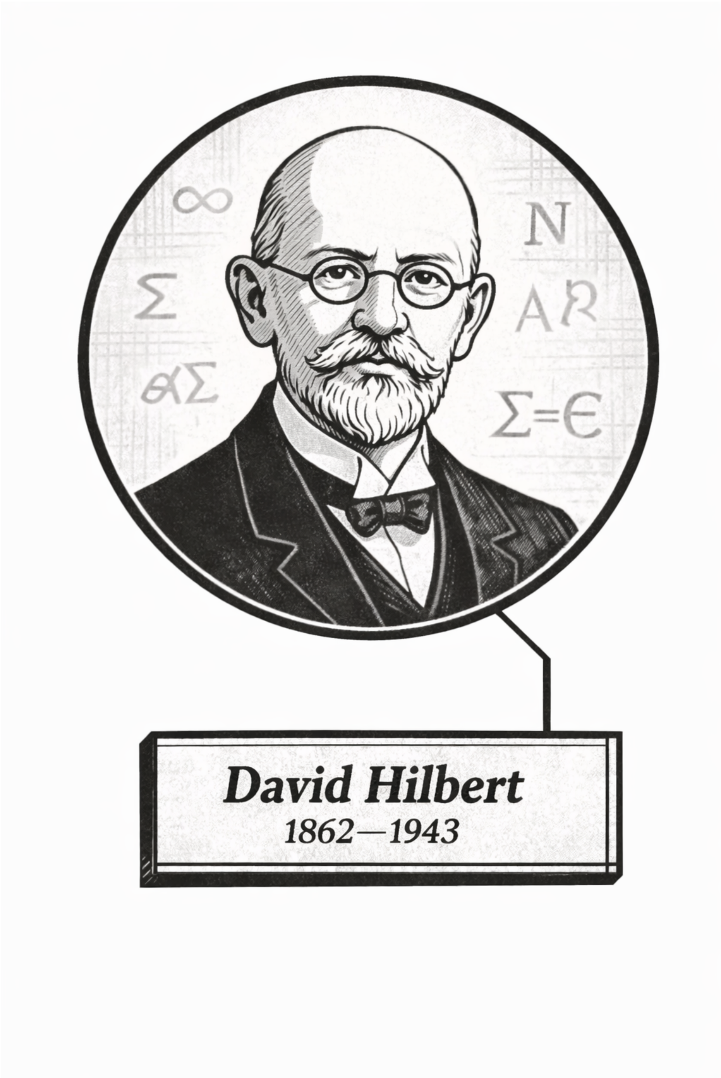


You get a room, you get a room, everybody gets a room...

David Hilbert (probably)

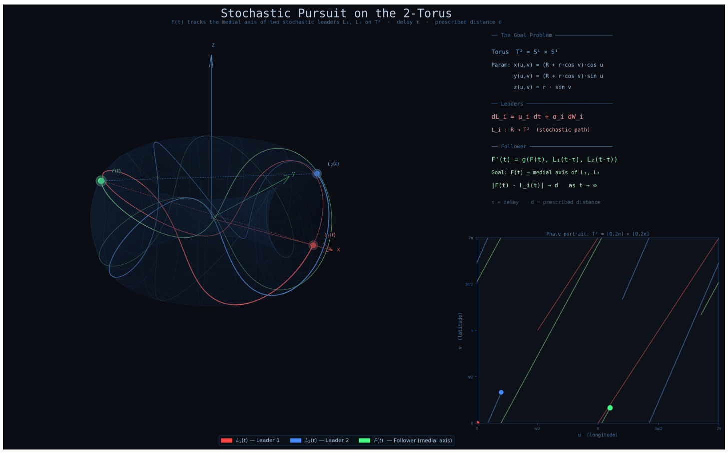
Volume III

Analysis

Self-Study Notes

William Sollers

June 18, 2026



First one to find a geodesic wins!

Torus Pursuit

Part I

Analysis

Chapter 1

Real Analysis

real-analysis

Structure of the Real Line $\rightarrow$ **Real Analysis** $\rightarrow$ Sequences

1.1 Foundational Analysis Properties

Summary of Foundational Analysis Properties

Concept	Goal	Proof mechanism / logical form
Well-definedness	Consistency	$x \sim y \Rightarrow F(x) = F(y)$
Existence	Feasibility	$\exists x\ P(x)$
Uniqueness	Singularity	$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2$
Linearity	Decomposition	$L(f+g) = L(f) + L(g)$ and $L(cf) = cL(f)$
Uniformity	One choice works globally	$\forall \varepsilon > 0, \exists \delta > 0, \forall x, y,\ Q(\varepsilon, \delta, x, y)$
Equivalence	Interchangeable descriptions	$A \Rightarrow B$ and $B \Rightarrow A$
Characterisation	Usable test	$P(x) \iff C(x)$
Estimate	Control	$A \leq B$, $\|A\| < \varepsilon$, or $d(A, B) < \varepsilon$
Boundedness	Two-sided control	$\exists l, u \in S\ \forall x \in A\ (l \leq x \leq u)$
Upper/lower control	One-sided domination	$\forall x \in A\ (x \leq u)$ or $\forall x \in A\ (l \leq x)$
Extremality	Realized optimality	$m \in A \wedge \mathrm{Bound}(m, A)$
Least/greatest character	Optimal bound	$\mathrm{Bound}(b, A) \wedge \forall c\, (\mathrm{Bound}(c, A) \Rightarrow b \preceq c)$
Monotonicity	Order preservation	Start with $x \leq y$ and prove $f(x) \leq f(y)$, or the reverse
Closure	Staying inside a class	$P(x) \wedge P(y) \Rightarrow P(x \star y)$
Approximation	Arbitrary precision	$\forall \varepsilon > 0\ \exists x\ d(x, a) < \varepsilon$

Many proofs in analysis are not isolated inventions. They prove recurring structural properties: a definition is independent of choices, an object exists, there is only one possible object, an operation respects addition and scalars, a single parameter works everywhere, or two definitions describe the same phenomenon. Naming the property first tells us what the proof must do.

Well-Definedness

Well-definedness appears whenever an object is defined using choices: representatives of equivalence classes, selected sequences, selected partitions, or selected approximations. The goal is to prove that different allowed choices give the same output.

$$x \sim x' \Rightarrow F(x) = F(x').$$

Remark 1.1.1 (Proof sketch).

1. Identify the choice built into the definition.
2. Take two allowed choices, usually x and x' with $x \sim x'$.
3. Compute or compare the two proposed outputs.
4. Show the outputs agree in the codomain.

Remark 1.1.2 (Crucial logic). The output must be independent of the chosen representative. If changing the representative changes the output, the proposed definition is not a function on the quotient or chosen domain.

Existence

Existence proves that at least one object satisfying a property is available:

$$\exists x\ P(x).$$

In analysis the object might be a limit, a supremum, a point in an interval, a subsequence, a bound, an index N, a radius δ, or a partition.

Remark 1.1.3 (Proof sketch).

1. Determine every requirement in the property P.
2. Produce a candidate explicitly, or invoke a theorem that guarantees one.
3. Verify the candidate belongs to the required domain.
4. Verify the candidate satisfies every part of P.

Remark 1.1.4 (Constructive and nonconstructive existence). A constructive proof names the object directly. A nonconstructive proof uses a theorem such as completeness, the intermediate value theorem, or Bolzano-Weierstrass. In both cases, the final burden is the same: the object must satisfy the stated property.

Uniqueness

Uniqueness proves that at most one object can satisfy a property:

$$P(x_1) \wedge P(x_2) \Rightarrow x_1 = x_2.$$

It does not by itself prove existence. It only proves that two successful candidates cannot be distinct.

Remark 1.1.5 (Proof sketch). 1. Assume x_1 and x_2 both satisfy P.

2. Use the defining property of x_1 and the defining property of x_2.

3. Derive two-sided comparison, zero distance, or identical defining data.

4. Conclude $x_1 = x_2$.

Remark 1.1.6 (Common analysis forms). For suprema, show $s \le t$ and $t \le s$. For limits, show $|L - M| < \varepsilon$ for every $\varepsilon > 0$, hence $L = M$. In metric spaces, show $d(x_1, x_2) = 0$.

Linearity

Linearity proves that an operation decomposes over addition and scalar multiplication:

$$L(f+g) = L(f) + L(g), \qquad L(cf) = cL(f).$$

The goal is not merely algebraic neatness. Linearity says that a complicated input can be analysed by splitting it into simpler pieces.

Remark 1.1.7 (Proof sketch). 1. Take arbitrary admissible inputs f, g and scalar c.

2. Expand $L(f + g)$ using the definition of L and the definition of addition.

3. Rearrange the expression until $L(f) + L(g)$ appears.

4. Repeat with $L(cf)$ and show it equals $cL(f)$.

Remark 1.1.8 (Typical analysis examples). Limits, derivatives, integrals, and many summation operators have linearity theorems. Each proof follows the same shape: expand the left side, split the expression, and identify the two pieces.

Uniformity

Uniformity means that one choice works for all relevant points at once. The prototype is not just uniform continuity, but any statement where a witness is chosen before the point is allowed to vary:

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y,\ Q(\varepsilon, \delta, x, y).$$

The key phrase is "one choice." The same N, δ, bound, or partition must work globally over the specified domain.

Remark 1.1.9 (Proof sketch). 1. Fix the tolerance or target requirement.

2. Choose a single witness that depends only on the allowed global data.

3. Let the point, pair of points, function, or index vary arbitrarily.

4. Verify that the same witness works in every permitted case.

Remark 1.1.10 (Contrast with pointwise statements). Pointwise statements allow the witness to depend on the point. Uniform statements do not. The order of quantifiers is the whole issue:

$$\forall x\,\exists\delta \quad \text{is pointwise, while} \quad \exists\delta\,\forall x \quad \text{is uniform.}$$

Equivalence

Equivalence proves that two statements are logically interchangeable:

$$A \iff B.$$

Such theorems are common because analysis often has several descriptions of the same phenomenon: epsilon-delta, sequential, order-theoretic, topological, or integral.

Remark 1.1.11 (Proof sketch). 1. Prove $A \Rightarrow B$.

2. Prove $B \Rightarrow A$.

3. Keep the assumptions for the two directions separate.

4. After both directions are proved, either description may be used.

Remark 1.1.12 (Typical analysis examples). Sequential and epsilon-delta definitions of continuity are equivalent. Riemann and Darboux integrability are equivalent under the appropriate hypotheses. A set is closed exactly when it contains the limits of all convergent sequences from the set.

Characterisation

A characterisation is an equivalence whose purpose is to turn a definition into a usable test:

$$P(x) \iff C(x).$$

The theorem does not merely say two statements agree. It supplies a criterion that can be used in later proofs.

Remark 1.1.13 (Proof sketch). 1. Identify the original definition.

2. Identify the proposed criterion.

3. Prove definition $\Rightarrow$ criterion.

4. Prove criterion $\Rightarrow$ definition.

Remark 1.1.14 (Typical analysis examples). The epsilon characterisation of the supremum turns least-upper-bound language into an approximation rule. Cauchy criteria turn convergence questions into internal distance estimates. Sequential criteria turn topological conditions into statements about sequences.

Estimates and Bounds

An estimate proves control:

$$A \leq B, \qquad |A| < \varepsilon, \qquad d(A, B) < \varepsilon.$$

The aim is not to compute the exact value. The aim is to put the target quantity inside a range that is strong enough for the theorem.

Remark 1.1.15 (Proof sketch).

1. Identify the quantity that must be controlled.
2. Replace it by a larger or simpler quantity using valid inequalities.
3. Use hypotheses to replace variable terms by constants or small errors.
4. End with the required bound.

Remark 1.1.16 (Typical analysis examples). Showing a sequence is bounded, proving an error term tends to zero, bounding a Riemann-sum error, or proving $|f(x) - L| < \varepsilon$ are all estimate problems. The proof is a chain of control.

Boundedness

Boundedness proves two-sided control. In an ordered setting, the usual form is that a set or sequence is bounded above and bounded below:

$$\exists l, u \in S \; \forall x \in A \; (l \leq x \leq u).$$

Remark 1.1.17 (Proof sketch).

1. Prove bounded above by finding a candidate upper bound.
2. Prove bounded below by finding a candidate lower bound.
3. Verify both candidates work for every element under consideration.
4. Conclude that the object is bounded.

Upper and Lower Control

Upper and lower control prove one-sided domination. The proof focuses on a candidate bound and checks that every element lies on the required side of it:

$$\forall x \in A \; (x \leq u), \qquad \forall x \in A \; (l \leq x).$$

Remark 1.1.18 (Proof sketch).

1. Name the candidate upper or lower bound.
2. Let an arbitrary element of the set be given.
3. Use the hypotheses or definition of the set to compare that element with the candidate.
4. Since the element was arbitrary, conclude the one-sided bound.

Extremality

Extremality proves that an optimal value is actually realized by the object being studied. A maximum is not merely an upper bound; it is an upper bound that belongs to the set. A minimum is the corresponding realized lower bound.

$$\mathrm{Maximum}(m, A) \iff (m \in A) \wedge \mathrm{UpperBound}(m, A),$$

$$\mathrm{Minimum}(m, A) \iff (m \in A) \wedge \mathrm{LowerBound}(m, A).$$

Remark 1.1.19 (Proof sketch).

1. Prove the candidate belongs to the set.
2. Prove the candidate is the appropriate one-sided bound.
3. Combine membership with the bound condition.
4. Conclude maximum or minimum.

Least and Greatest Character

Least and greatest character proves optimality among competitors. For a supremum, the candidate must first be an upper bound, and then it must be no larger than every other upper bound. Infimum proofs reverse the direction.

$$\mathrm{Supremum}(s, A) \iff \mathrm{UpperBound}(s, A) \wedge \forall u \in S \left(\mathrm{UpperBound}(u, A) \Rightarrow s \leq u\right),$$

$$\mathrm{Infimum}(i, A) \iff \mathrm{LowerBound}(i, A) \wedge \forall l \in S \left(\mathrm{LowerBound}(l, A) \Rightarrow l \leq i\right).$$

Remark 1.1.20 (Proof sketch).

1. Prove the candidate is a bound of the required kind.
2. Let an arbitrary competing bound be given.
3. Compare the candidate with that competing bound.
4. Conclude the candidate is the least upper bound or greatest lower bound.

Monotonicity

Monotonicity proves order preservation or order reversal:

$$x \leq y \Rightarrow f(x) \leq f(y), \qquad x \leq y \Rightarrow f(y) \leq f(x).$$

The proof begins with an arbitrary ordered pair and carries that order through the definitions.

Remark 1.1.21 (Proof sketch).

1. Take arbitrary points x and y with $x \leq y$.
2. Expand the definitions of the function or operation.
3. Use order rules and hypotheses to compare the outputs.
4. Conclude that the order is preserved or reversed.

Closure

Closure proves that a class is stable under an operation. One starts with objects already known to belong to the class, combines them, and then verifies that the result still satisfies the defining property.

$$P(x) \wedge P(y) \Rightarrow P(x \star y).$$

For closure under scalar operations, the same pattern becomes

$$P(x) \Rightarrow P(c \star x)$$

for every admissible scalar or parameter c.

Remark 1.1.22 (Proof sketch).

1. Take arbitrary objects from the class.
2. Form the object produced by the operation.
3. Check every condition in the definition of the class.
4. Conclude the result remains inside the class.

Approximation

Approximation proves arbitrary precision. The proof usually begins with a tolerance and constructs an object close enough to a target:

$$\forall \varepsilon > 0\ \exists x\ d(x, a) < \varepsilon.$$

Remark 1.1.23 (Proof sketch).

1. Fix an arbitrary tolerance $\varepsilon > 0$.
2. Choose or construct an approximating object.
3. Estimate the error between the approximation and the target.
4. Show the error is below ε.

A Classification Pass

Before proving a theorem, classify its job:

- Does the statement depend on choices? Check well-definedness.
- Does it ask for an object? Prove existence.
- Does it say only one object can work? Prove uniqueness.
- Does it say an operation splits across sums and scalars? Prove linearity.
- Does one witness have to work for every point? Prove uniformity.
- Does it say two descriptions agree? Prove equivalence.
- Does it give a test for a definition? Prove a characterisation.
- Does it ask for smallness or control? Prove an estimate.

- Does it ask for two-sided order control? Prove boundedness.
- Does it ask for one-sided order control? Prove an upper or lower bound.
- Does it ask for a realized best element? Prove extremality.
- Does it ask for the best bound among all bounds? Prove least or greatest character.
- Does it preserve or reverse order? Prove monotonicity.
- Does it stay inside a class after an operation? Prove closure.
- Does it ask for arbitrary precision? Prove approximation.

Remark 1.1.24 (The point). The classification does not solve the proof, but it tells us what a completed proof must contain. For example, an existence proof must exhibit or obtain an object, while a uniqueness proof must compare two candidates. An equivalence proof must have two directions, while a uniformity proof must choose one witness before the arbitrary point is fixed.

1.2 Manipulating Inequalities and the Logic of Bounding

Inequality Toolkit Quick Reference

Tool	Statement	Typical use
Triangle inequality	$\|a+b\| \le \|a\| + \|b\|$	Split a distance at an intermediate p
Distance form	$\|a-b\| \le \|a-c\| + \|c-b\|$	Introduce pivot c to separate two est
Reverse triangle ineq.	$\big\|\|a\| - \|b\|\big\| \le \|a-b\|$	Bound magnitudes using closeness
Absolute value interval	$\|x\| \le y \iff -y \le x \le y$	Convert $\|\cdot\|$ bound to two-sided ineq
Boundedness replacement	$\|b_n\| \le M$ (constant): replace $\|b_n\|$ by M	Eliminate variable factors in product
ε-budget split	$\|A\| < \frac{\varepsilon}{2}$ and $\|B\| < \frac{\varepsilon}{2} \Rightarrow \|A+B\| < \varepsilon$	Combine two independent estimates
$+1$ trick	Replace $\|L\|$ by $\|L\|+1$ in denominators	Avoid division by zero when L may

The Fundamental Asymmetry Between Equality and Inequality

Equality is bilateral: $a = b$ licenses substitution in either direction and carries no information about magnitude. An inequality is directional: $a \le b$ locates a relative to b on the line, says nothing about the gap, and does not support backwards substitution.

This asymmetry is the source of a strategic shift. In an equality argument the goal is to show that two expressions are the same object. In a bounding argument the goal is to show that one expression is *controlled* by another. Control is inherently one-sided: a ceiling above $|f(x)|$ is useful regardless of how far below it $|f(x)|$ actually falls.

Remark 1.2.1 (Consequence for proof strategy). The question in analysis is rarely "what is $|a_n - L|$?" It is "is $|a_n - L|$ smaller than ε?" A valid crude bound that answers the second question is more useful than an exact answer to the first. Deliberate over-estimation absent from algebra is a required habit.

Remark 1.2.2 (Common error). Treating an inequality as if it were an equality by reversing direction. Given $a \leq b$, negation gives $-a \geq -b$, not $-a \leq -b$. Multiplication by a negative constant reverses direction. Failing to track sign changes when multiplying or negating is the most frequent algebraic error in ε-δ computations.

The Basic Rules of Inequality Arithmetic

Proposition 1.2.3 (Order arithmetic). *Let* $a, b, c, d \in \mathbb{R}$.

(i) If $a \leq b$ *and* $c \leq d$*, then* $a + c \leq b + d$.

(ii) If $a \leq b$ *and* $c > 0$*, then* $ac \leq bc$.

(iii) If $a \leq b$ *and* $c < 0$*, then* $ac \geq bc$.

(iv) $|x| \leq y$ *(with* $y \geq 0$*) if and only if* $-y \leq x \leq y$.

Go to proof.

Remark.

Remark 1.2.4 (Fully quantified form). (i) $\forall a, b, c, d \in \mathbb{R},\ (a \leq b) \wedge (c \leq d) \Rightarrow a + c \leq b + d$. (iv) $\forall x \in \mathbb{R},\ \forall y \geq 0,\ |x| \leq y \Leftrightarrow -y \leq x \leq y$.

Remark 1.2.5 (Inequalities add but do not subtract). Rule (i) licenses adding inequalities in the same direction. There is no subtraction rule: $a \leq b$ and $c \leq d$ yields no conclusion about $a - c$ versus $b - d$ in general. The direction of $a - c$ versus $b - d$ depends on the relative sizes of $(b - a)$ and $(d - c)$, which are not controlled.

Remark 1.2.6 (Consequence for absolute values). Rule (iv) is the bridge between $|\cdot|$ bounds and two-sided linear inequalities. It is invoked silently every time an ε-N argument converts $|a_n - L| < \varepsilon$ into positional information about a_n.

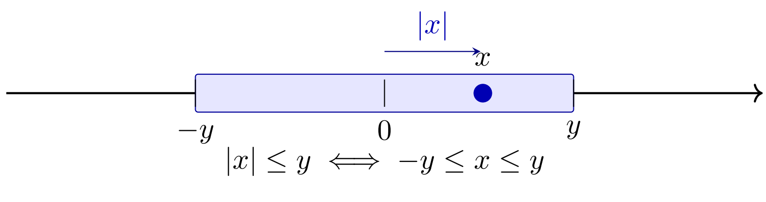


Figure 1.1: The absolute value characterisation as a symmetric interval. The shaded region is $[-y, y]$; the point x lies inside with $|x| \leq y$. The equivalence $|x| \leq y \Leftrightarrow -y \leq x \leq y$ (Proposition 1.2.3(iv)) converts an absolute value bound into a two-sided linear constraint.

The Triangle Inequality and the Pivot Move

Theorem (Triangle Inequality)

For all $a, b \in \mathbb{R}$, $|a+b| \leq |a| + |b|$. Equivalently, for all $a, b, c \in \mathbb{R}$, $|a-b| \leq |a-c| + |c-b|$.

Remark 1.2.7 (Fully quantified form). $\forall a, b \in \mathbb{R},\ |a+b| \leq |a| + |b|$.

Remark 1.2.8 (The pivot move). The distance form $|a-b| \leq |a-c| + |c-b|$ is the operative version in analysis. The element c is a ***pivot***: an intermediate point that decomposes a hard distance into two pieces, each controllable by hypothesis. In the proof that $a_n + b_n \to L + M$, the pivot $c = L + b_n$ gives

$$|(a_n + b_n) - (L + M)| = |(a_n - L) + (b_n - M)| \leq |a_n - L| + |b_n - M|,$$

and the two pieces are controlled by the individual convergences.

Remark 1.2.9 (Reverse triangle inequality). $\big||a| - |b|\big| \leq |a-b|$. Proof: apply the standard inequality to $a = (a-b) + b$ to get $|a| - |b| \leq |a-b|$; symmetry gives the other side. Consequence: $|\cdot| : \mathbb{R} \to \mathbb{R}$ is Lipschitz with constant 1, so closeness of inputs implies closeness of magnitudes.

Remark 1.2.10 (Common error). Writing $|a| + |b| \leq |a+b|$. The triangle inequality always produces an ***upper*** bound on a sum, never a lower bound. The reverse ($|1| + |-1| = 2 > |1 + (-1)| = 0$) is a one-line counterexample.

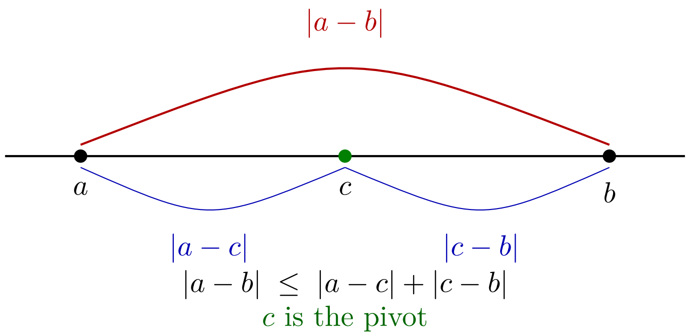


Figure 1.2: The triangle inequality in distance form. The direct distance $|a-b|$ (red) is bounded above by the sum of indirect distances $|a-c|$ and $|c-b|$ (blue) through pivot c. In analysis, c is chosen to split $|a-b|$ into two pieces each controlled by a separate hypothesis.

Bounding Is Not Solving

In algebra, a computation terminates when the exact value is found. In analysis, a bounding argument terminates when the expression is shown to fall within an acceptable range. These are structurally different goals.

Remark 1.2.11 (Deliberate crudeness is valid). To bound $|x^2 - 9x + 1|$ on $[-1, 5]$:

$$|x^2 - 9x + 1| \leq |x|^2 + 9|x| + 1 \leq 25 + 45 + 1 = 71.$$

The actual supremum is far smaller than 71. That is irrelevant. The bound is valid, explicit, and was obtained cheaply. The test for a bound is not "is this sharp?" but "is this true and sufficient for the argument's purpose?"

Remark 1.2.12 (Consequence for proof reading). A reader who checks that each inequality is valid can trust the conclusion without caring whether the bounds are optimal. Optimality is irrelevant to logical correctness.

The ε-Budget Split

The prototype of all two-piece bounding arguments:

$$|A+B| \le |A| + |B| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

The total tolerance ε is a budget allocated among the summands. With k summands, allocate ε/k each; with summands of unequal difficulty, split unevenly provided the shares sum to at most ε.

Remark 1.2.13 (The four-step procedure). Standard structure of an ε-N or ε-δ proof:

1. Apply the triangle inequality: express $|f-L|$ as a sum $|A_1|+\cdots+|A_k|$.
2. Assign budgets $\varepsilon_i > 0$ with $\sum_i \varepsilon_i = \varepsilon$.
3. For each piece, find N_i (or δ_i) so that $|A_i| < \varepsilon_i$ for all $n \ge N_i$ (or $|x-a| < \delta_i$).
4. Set $N = \max_i N_i$ (or $\delta = \min_i \delta_i$) to enforce all constraints simultaneously.

Step 4 reflects the logical structure of the conclusion: all conditions must hold at once, so the threshold is the most restrictive one.

Remark 1.2.14 (Common error). Taking $N = N_1$ and ignoring N_2. For $n \ge N_1$ the first piece is controlled; the second is not. Both pieces must be within budget simultaneously, which requires $n \ge \max\{N_1, N_2\}$.

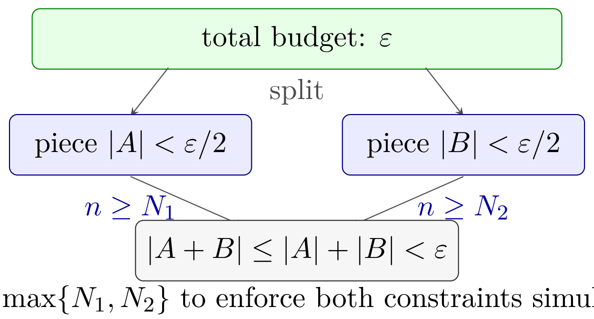


Figure 1.3: The ε-budget split. The total tolerance ε is distributed as shares $\varepsilon/2$ between two pieces. Each piece is controlled independently for $n \ge N_i$. Taking $N = \max\{N_1, N_2\}$ enforces both bounds simultaneously, giving $|A+B| < \varepsilon$ for all $n \ge N$. With k pieces, assign ε/k each and take $N = \max_i N_i$.

The Boundedness Replacement

Proposition 1.2.15 (Convergent sequences are bounded). *If $(b_n) \to b$, then there exists $M > 0$ such that $|b_n| \leq M$ for all $n \in \mathbb{N}$. Go to proof.*

Remark.

Remark 1.2.16 (Fully quantified form). $\forall (b_n) \subset \mathbb{R},\ \lim_{n\to\infty} b_n = b \ \Rightarrow\ \exists M > 0,\ \forall n \in \mathbb{N},\ |b_n| \leq M$.

Remark 1.2.17 (The replacement move). Given $|b_n| \leq M$:

$$|b_n| \cdot |a_n - L| \leq M \cdot |a_n - L|.$$

The variable factor is gone. Choose N so that $|a_n - L| < \varepsilon/M$ for $n \geq N$; the whole product is then less than ε. The constant M need not be tight.

Remark 1.2.18 (Common error). Choosing $|a_n - L| < \varepsilon/|b_n|$. This is circular: N would depend on n, since $|b_n|$ changes with n. The threshold N must be a fixed number, independent of n. Replacing $|b_n|$ by the constant M breaks the circularity.

The Work-Backwards Principle

ε-N proofs are *stated* forwards but *discovered* backwards. The proof asserts: "Let $\varepsilon > 0$. Choose $N = \lceil C/\varepsilon \rceil$." The value N was found by scratchwork: "the expression simplifies to C/n; this is less than ε when $n > C/\varepsilon$; so set $N = \lceil C/\varepsilon \rceil$."
The scratchwork is a private computation that produces the *witness*. The proof states the witness and verifies it, working forwards.

Scratchwork (private):	Work backwards from $|a_n - L| < \varepsilon$ to find what N must satisfy.
Proof (public):	State N explicitly. Verify $n \geq N \Rightarrow |a_n - L| < \varepsilon$.

Remark 1.2.19 (Common error). Allowing scratchwork to appear in the proof: "we need $n > C/\varepsilon$, so choose n this large." The proof must assert a fixed N and verify the conclusion holds. A threshold stated as "n large enough" is an incomplete scratchwork note, not a proof.

Remark 1.2.20 (Connection to the geometric phase). The work-backwards process is the analytic counterpart of the geometric phase in limit proofs. The geometric phase identifies structure and chooses a pivot; the backwards computation determines the explicit threshold. The proof states both and verifies the chain forwards.

Chaining Inequalities

A standard proof move builds a chain $|f(x)| \leq A(x) \leq B \leq \varepsilon$, where each step uses a different tool: the triangle inequality, a domain bound, and the choice of δ.

Remark 1.2.21 (Replacing by a larger quantity). A quantity on the right-hand side of $\leq$ may be freely replaced by anything larger, since this only weakens the inequality. If $|x| \leq |x-1| + 1$ and $|x-1| < \delta$, then $|x| < \delta + 1$. The replacement of $|x-1|$ by δ is valid because $|x-1|$ appears on the right. The same replacement on the *left* would be invalid.

Remark 1.2.22 (Common error). Including a step where the inequality points the wrong direction, or where a step holds only generically rather than pointwise. A chain is only as strong as its weakest link; every $\leq$ must be unconditionally valid.

The $+1$ Trick for Denominators

When $\varepsilon/|L|$ appears in an estimate and L may be zero, replace $|L|$ by $|L|+1 \geq 1$:

$$\frac{\varepsilon}{|L|+1} \leq \varepsilon.$$

Remark 1.2.23 (Fully quantified form). $\forall L \in \mathbb{R},\ |L|+1 \geq 1 > 0$. Hence $\varepsilon/(|L|+1)$ is well-defined and positive for all $\varepsilon > 0$ and all $L \in \mathbb{R}$.

Remark 1.2.24 (Generalisation). The $+1$ is an instance of a wider principle: any constant intended for a denominator must be guaranteed strictly positive. If the constant is a given quantity C that might be zero, replace it by $C + 1$. The choice of 1 is conventional; any fixed positive number serves the same purpose.

Remark 1.2.25 (Common error). Using $|L|$ as a denominator without first verifying $L \neq 0$. The product limit proof $a_n b_n \to LM$ is the canonical location for this error: the bound $\varepsilon/(2|L|)$ fails when $L = 0$, and the $L = 0$ case requires separate treatment or the use of $|L|+1$.

Summary: The Bounding Mindset

Algebra	**Analysis**
Find the exact value of x	Show $\lvert x - L\rvert < \varepsilon$
Equality: both sides are the same object	Inequality: one side controls the other
Work backwards and forwards symmetrically	Discover witness backwards; verify forwards
Exact answer is the goal	Any valid bound is sufficient
Fractions must be simplified	Crude over-estimates are acceptable
Multiply both sides freely	Track sign changes; direction is load-bearing

Remark 1.2.26 (The structure of a bounding proof). A bounding proof is a *chain of control*: inequalities connecting the target expression to ε, where each link uses one of the toolkit tools above. The proof is complete when the chain reaches ε. Every link must be unconditionally valid; every variable factor must be replaced by a constant before the chain can close.

1.3 Completeness as a Constructive Engine

Completeness Construction Toolkit Quick Reference

Tool	What it supplies	Detail
ε-characterisation of sup	$\forall\varepsilon > 0,\ \exists x \in S,\ x > \sup S - \varepsilon$	↓
Inductive selection	Monotone $(x_n) \subset S$ with $x_n \to \sup S$	↓
Monotone Approximation	Such a sequence exists for any nonempty bounded S	↓
LUB $\Leftrightarrow$ MCT	The two completeness statements are equivalent	↓

The ε-Characterisation of Suprema and Infima

Proposition 1.3.1 (ε-characterisation of sup). *Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then for every $\varepsilon > 0$ there exists $x \in S$ with $x > s - \varepsilon$.* *Go to proof.*

Remark.

Remark 1.3.2 (Fully quantified form). $\forall\varepsilon > 0,\ \exists x \in S,\ s - \varepsilon < x \leq s$. Equivalently: any $M < s$ is not an upper bound for S.

Remark 1.3.3 (Proof strategy). If no such x existed for some $\varepsilon_0 > 0$, then $s - \varepsilon_0$ would be an upper bound for S smaller than s, contradicting $s = \sup S$.

Remark 1.3.4 (Operative role). The ε-characterisation is the *existential engine* of constructive completeness arguments. At each inductive step it is invoked with a specific ε value, discharging the universal quantifier and yielding a concrete existence claim. Existential instantiation then names a witness call it x_{n+1} and the rest of the argument uses only the single property $x_{n+1} > s - \varepsilon$. The witness is underspecified; the argument makes no commitment about which element of S is chosen, and no further properties of x_{n+1} are used.

Remark 1.3.5 (Symmetric form for inf). For $t = \inf S$: for every $\varepsilon > 0$ there exists $y \in S$ with $y < t + \varepsilon$. The proof is symmetric: any $M > t$ is not a lower bound.

Inductive Selection at Bounds

Theorem (Inductive Selection at sup)

Let $S \subset \mathbb{R}$ be nonempty and bounded above, and let $s = \sup S$. Then there exists a strictly increasing sequence $(x_n)_{n=1}^{\infty} \subset S$ with $x_n \to s$.

Remark 1.3.6 (Fully quantified form). $\exists (x_n) \subset S$: $\forall n,\ x_n < x_{n+1}$; $\forall n,\ s - \frac{1}{n} < x_n \leq s$; $\lim_{n\to\infty} x_n = s$.

Remark 1.3.7 (Why no algebraic formula is possible in general). The set S may have large gaps; any fixed formula might land outside S entirely. The construction must be inductive, selecting from S at each stage using the ε-characterisation rather than evaluating a closed expression.

Remark 1.3.8 (Proof strategy the max device). The inductive step must simultaneously satisfy two lower bounds: $x_{n+1} > x_n$ (monotonicity) and $x_{n+1} > s - \frac{1}{n+1}$ (proximity). A single invocation of the existential can impose only one threshold. Setting

$$M_n := \max\Big(x_n,\ s - \tfrac{1}{n+1}\Big)$$

collapses both requirements into one: $x_{n+1} > M_n$ delivers both $x_{n+1} > x_n$ and $x_{n+1} > s - \frac{1}{n+1}$ simultaneously. Since $M_n < s$ (both entries are strictly below s whenever $s \notin S$, or the constant sequence $x_n = s$ handles the case $s \in S$), the ε-characterisation with $\varepsilon = s - M_n > 0$ supplies the required witness.

Base case. Set $\varepsilon = 1$. The ε-characterisation gives $x_1 \in S$ with $x_1 > s - 1$; since $x_1 \in S$, also $x_1 \leq s$.
Inductive step. Given x_n with $s - \frac{1}{n} < x_n \leq s$, form M_n as above and instantiate the ε-characterisation with $\varepsilon = s - M_n > 0$ to obtain $x_{n+1} \in S$ with $x_{n+1} > M_n$. Then:

$$\begin{aligned} x_{n+1} > M_n \geq x_n \quad &\Rightarrow \quad x_{n+1} > x_n, \\ x_{n+1} > M_n \geq s - \tfrac{1}{n+1} \quad &\Rightarrow \quad s - \tfrac{1}{n+1} < x_{n+1} \leq s. \end{aligned}$$

Convergence. $0 \leq s - x_n < \frac{1}{n}$; the squeeze theorem and the Archimedean Property give $x_n \to s$.

Remark 1.3.9 (What the construction requires). Three ingredients: (i) completeness of $\mathbb{R}$ ($\sup S$ exists); (ii) the ε-characterisation (Proposition 1.3.1), supplying the witness at each step; (iii) the Archimedean Property (??), forcing $|s - x_n| \leq \frac{1}{n} \to 0$. The internal structure of S its cardinality, openness, density plays no role.

Remark 1.3.10 (Common error). Instantiating the existential with only the proximity threshold $s - \frac{1}{n+1}$ and ignoring x_n. The resulting witness satisfies proximity but not monotonicity: one might obtain $x_1 = 0.9$, $x_2 = 0.4$, $x_3 = 0.85$, each close to s but not increasing. The max is not a convenience; it is the minimal device that encodes both requirements into one threshold.

Remark 1.3.11 (Infimum case). Replace max by min, $s - \frac{1}{n}$ by $t + \frac{1}{n}$, and reverse all strict inequalities. The resulting sequence $(y_n) \subset S$ is strictly decreasing with $y_n \to t = \inf S$.

Monotone Approximation of Bounds

Proposition 1.3.12 (Monotone Approximation of Bounds). *Let $S \subset \mathbb{R}$ be nonempty and bounded. Then there exist monotone sequences $(x_n), (y_n) \subset S$ with $x_n \to \sup S$ and $y_n \to \inf S$. Go to proof.*

Remark.

Remark 1.3.13 (Fully quantified form). $\forall S \neq \emptyset$ bounded: $\exists (x_n) \subset S$, $x_n \uparrow \sup S$; $\exists (y_n) \subset S$, $y_n \downarrow \inf S$.

Remark 1.3.14 (Proof strategy). The strictly increasing sequence from Section 1.3 is already monotone and converges to $\sup S$. Taking $X_n = \max\{x_1, \dots, x_n\}$ makes it explicitly non-decreasing (useful when the inductive step does not guarantee strict increase). The infimum case is symmetric.

Remark 1.3.15 (Consequence). Every supremum is the limit of a sequence from the set. The sequential characterisation of sup and inf that they are realised as limits of internal sequences follows from this construction, making Proposition 1.3.12 the canonical proof of that characterisation.

Remark 1.3.16 (Forward connections). The inductive selection pattern reappears in:

- **Monotone Convergence Theorem** every increasing bounded sequence converges to the supremum of its range; this construction exhibits that the supremum is always realised as such a limit.
- **Nested Interval Property** the left and right endpoints of nested intervals form monotone sequences converging to a common point by exactly this mechanism.
- **Existence of square roots** the standard proof sets $s = \sup\{x \in \mathbb{R} : x^2 < 2\}$ and extracts an approximating sequence from that set by inductive selection.

LUB $\iff$ MCT: The Completeness Equivalence

Definition (Least Upper Bound Property)

$\mathbb{R}$ has the **least upper bound property** if every nonempty set $A \subset \mathbb{R}$ that is bounded above has a supremum in $\mathbb{R}$.

Theorem (LUB $\iff$ MCT)

The following are equivalent over the Archimedean ordered field axioms:

(i) $\mathbb{R}$ has the least upper bound property (LUB).

(ii) Every bounded monotone sequence of real numbers converges (MCT).

Remark 1.3.17 (Fully quantified form). (i) $\forall A \subset \mathbb{R},\ (A \neq \emptyset \wedge \exists M,\ \forall a \in A,\ a \leq M) \Rightarrow \exists \sup A \in \mathbb{R}$.
(ii) $\forall (x_n) \subset \mathbb{R},\ (\forall n,\ x_n \leq x_{n+1}) \wedge (\exists M,\ \forall n,\ x_n \leq M) \Rightarrow \exists L \in \mathbb{R},\ \lim_{n\to\infty} x_n = L$.

Remark 1.3.18 (Proof strategy LUB $\Rightarrow$ MCT). Let (x_n) be increasing and bounded. The image set $A = \{x_n : n \in \mathbb{N}\}$ is bounded above, so LUB gives $\alpha = \sup A$. The ε-characterisation produces N with $x_N > \alpha - \varepsilon$. Monotonicity promotes this to a tail statement: $\alpha - \varepsilon < x_n \leq \alpha$ for all $n \geq N$. Hence $x_n \to \alpha$.
The quantifier structure is:

$$\alpha = \sup A \ \Rightarrow\ (\forall \varepsilon > 0)\ \exists N\ (x_N > \alpha - \varepsilon) \xrightarrow{\text{monotone}} (\forall \varepsilon > 0)\ \exists N\ (\forall n \geq N)\ (|x_n - \alpha| < \varepsilon).$$

Remark 1.3.19 (Proof strategy MCT $\Rightarrow$ LUB). Given $A \neq \emptyset$ and bounded above by u_1, pick $l_1 \in A$. Build a bisection bracket: at each stage, let $m_n = (l_n + u_n)/2$; if m_n is an upper bound for A set $u_{n+1} = m_n$, otherwise set $l_{n+1} = m_n$ (and pick a witnessing $a \in A$ with $a > l_{n+1}$). The sequences (l_n) and (u_n) are monotone and bounded, so MCT gives limits $l_n \to s$ and $u_n \to s$ (since $u_n - l_n = (u_1 - l_1)/2^n \to 0$). The invariants "$u_n$ is an upper bound for A" and "l_n is not" force $s = \sup A$.

The construction maintains:

- a ***universal*** invariant: $(\forall a \in A)(a \leq u_n)$,
- a ***failure*** invariant: $(\exists a \in A)(a > l_n)$.

These two invariants, together with $u_n - l_n \to 0$, pin s as the least upper bound.

Remark 1.3.20 (Why this matters structurally). The chain LUB $\Leftrightarrow$ MCT $\Leftrightarrow$ NIP $\Leftrightarrow$ BW $\Leftrightarrow$ CC (Cauchy Criterion) is the backbone of the equivalence cluster at the heart of real analysis. All five statements assert that $\mathbb{R}$ has no gaps, each in its own language. The inductive selection technique building sequences from sets using the ε-characterisation is the constructive face of all five equivalences.

Remark 1.3.21 (Downstream reuse). The quantifier pattern

$$(\forall \varepsilon > 0)(\exists \text{witness}) \Rightarrow (\exists \text{sequence with controlled tail})$$

reappears in Bolzano-Weierstrass (extracting convergent subsequences), $\limsup/\liminf$ (tail suprema as a decreasing sequence), compactness (finite subcovers as finite witnesses), and approximation theorems throughout analysis.

1.4 Walking a Sequence with a Predicate

Predicate-Walking Toolkit Quick Reference

Theorem	Predicate $P(n)$	Tool	Detail
Monotone Subseq.	n is a peak of (x_n)	Dichotomy on peak count	↓
Bolzano–Weierstrass	interval has ∞ many terms	Bisection + pigeonhole	↓
Arzelà–Ascoli	converges at $d_1, \ldots, d_m$	Diagonal extraction	↓

The Infinitely-Often / Eventually Dichotomy

Definition (Infinitely Often / Eventually)

Let (x_n) be a sequence and P a predicate on $\mathbb{N}$. P holds **infinitely often** if $\{n \in \mathbb{N} : P(n)\}$ is infinite; P holds **eventually** if there exists $N \in \mathbb{N}$ such that $P(n)$ holds for all $n \geq N$.

Remark 1.4.1 (Fully quantified form). P infinitely often: $\forall N \in \mathbb{N},\ \exists n \geq N,\ P(n)$.
P eventually: $\exists N \in \mathbb{N}, \forall n \geq N,\ P(n)$.

Proposition 1.4.2 (The dichotomy). *For any predicate P on $\mathbb{N}$, exactly one of the following holds:*

(i) P holds infinitely often.

(ii) $\neg P$ holds eventually.

Go to proof.

Remark.

Remark 1.4.3 (Fully quantified form). $\big(\forall N,\ \exists n \geq N,\ P(n)\big) \ \veebar\ \big(\exists N,\ \forall n \geq N,\quad \neg P(n)\big)$. The two cases are exhaustive because $\mathbb{N}$ is infinite: if P fails for all but finitely many n, those indices have a maximum N, and $\neg P(n)$ holds for all $n > N$.

Remark 1.4.4 (Relation to Infinite Pigeonhole). The index set $\mathbb{N}$ is partitioned into $\{n : P(n)\}$ and $\{n : \neg P(n)\}$; an infinite set cannot have both parts finite. The "infinitely often" case provides an inexhaustible reservoir of indices from which to select subsequence terms inductively.

The Abstract Extraction Template

Remark 1.4.5 (Four-step pattern). Every predicate-walking argument follows the same skeleton:

1. *Define the predicate.* Identify the property $P(n)$ to propagate: peak, infinite-occupancy, pointwise convergence, etc.

2. *Apply the dichotomy.* Either P holds infinitely often, or $\neg P$ holds eventually. Handle each case.

3. *Extract the subsequence.* In the "infinitely often" case, enumerate indices satisfying P in increasing order: $n_1 < n_2 < \cdots$. In the "eventually" case, work in the $\neg P$-tail and build inductively, using $\neg P$ as the fuel.

4. *Verify the property.* Show (x_{n_k}) has the desired property, using the predicate and any additional hypotheses (boundedness, MCT, NIP, equicontinuity).

The critical structural requirement: at each inductive stage, one must find $n_{k+1} > n_k$ satisfying the relevant condition. The "infinitely often" guarantee ensures the reservoir is never exhausted.

The Peak Argument: Monotone Subsequence Theorem

Definition (Peak Index)

An index $m \in \mathbb{N}$ is a **peak** of (x_n) if $x_m \geq x_n$ for all $n \geq m$.

Remark 1.4.6 (Fully quantified form). m is a peak: $\forall n \in \mathbb{N},\ n \geq m \Rightarrow x_m \geq x_n$.

Theorem (Monotone Subsequence)

Every sequence (x_n) in $\mathbb{R}$ has a monotone subsequence.

Remark 1.4.7 (Fully quantified form). $\forall (x_n) \subset \mathbb{R},\ \exists (n_k)$ strictly increasing, (x_{n_k}) is mon

Remark 1.4.8 (Proof strategy -- peak dichotomy). Apply the dichotomy to $P(m) =$ "m is a peak."

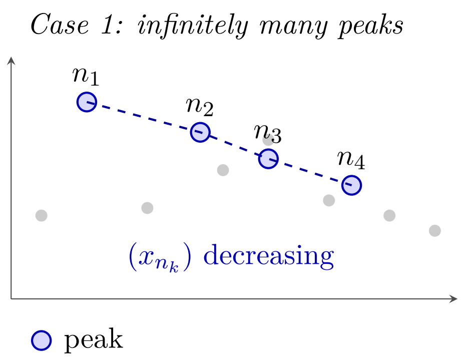
Case 1: infinitely many peaks. Let $n_1 < n_2 < \cdots$ enumerate the peaks. Since n_k is a peak and $n_{k+1} > n_k$: $x_{n_k} \geq x_{n_{k+1}}$. The subsequence is decreasing.

Case 2: finitely many peaks. Let N be an index beyond all peaks. No $n \geq N$ is a peak, so for each such n there exists $m > n$ with $x_m > x_n$. Set $n_1 = N$. Since n_1 is not a peak, $\exists n_2 > n_1$ with $x_{n_2} > x_{n_1}$. Inductively, this produces a strictly increasing subsequence.

The no-peak condition $\neg P(n_k)$ is the fuel for each inductive step: it is precisely the statement "there exists a later term exceeding x_{n_k}," which supplies n_{k+1}.

Remark 1.4.9 (Consequence: Bolzano-Weierstrass). Every bounded sequence has a bounded monotone subsequence. By MCT, that subsequence converges. Bolzano-Weierstrass follows immediately.

Remark 1.4.10 (Common error). Assuming Case 2 produces an increasing subsequence "by default." It does, but only because $\neg P(n)$ supplies a strictly larger term. If the predicate were defined differently, $\neg P$ might provide no useful structure. The conclusion depends on the specific content of $\neg P$, not the abstract case structure.



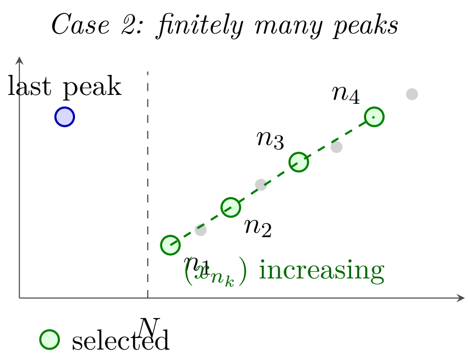


Figure 1.4: The two cases of the Monotone Subsequence Theorem (Section 1.4). *Left:* Infinitely many peaks (blue) yield a decreasing subsequence (dashed blue). *Right:* After the last peak, every $n \geq N$ is a non-peak, so a later term always exceeds the current one; this fuels the inductive construction of an increasing subsequence (dashed green).

The Infinite-Occupancy Predicate: Bolzano-Weierstrass via Bisection

Proposition 1.4.11 (Bolzano-Weierstrass via bisection). *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence. Go to proof.*

Remark.

Remark 1.4.12 (Proof strategy -- infinite-occupancy predicate). The predicate at each stage: $P(I)$ ="interval I contains infinitely many terms of (x_n)." Since $(x_n) \subseteq [a_0, b_0]$ for some bounded interval, $P([a_0, b_0])$ holds.

Inductive step. Given $I_k = [a_k, b_k]$ with $P(I_k)$, bisect at $m_k = (a_k + b_k)/2$. The two halves cover I_k; at least one contains infinitely many terms. Set I_{k+1} to be that half. Pick $x_{n_{k+1}} \in I_{k+1}$ with $n_{k+1} > n_k$. The predicate $P(I_{k+1})$ holds by construction.

Convergence. The intervals have $|I_k| = (b_0 - a_0)/2^k \to 0$. By NIP there exists a unique $L \in \bigcap I_k$. Since $a_k \leq x_{n_k} \leq b_k$ and both endpoints converge to L, the squeeze theorem gives $x_{n_k} \to L$.

Remark 1.4.13 (Comparison with the peak proof). The peak proof is combinatorial: it operates on the values of the sequence directly and produces a monotone subsequence without requiring boundedness. Convergence follows from MCT.
The bisection proof is geometric: it operates on intervals containing the sequence's range and produces a subsequence squeezed to a limit directly. Boundedness is required from the outset to produce the initial interval.
Both proofs use the same dichotomy, but in the peak proof it is applied to an index predicate; in the bisection proof, to an interval predicate.

Diagonal Extraction: Arzelà-Ascoli

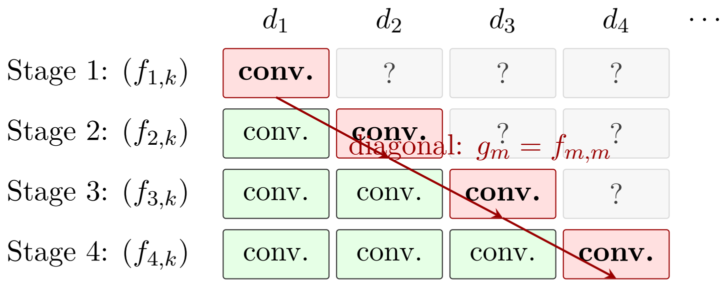
Remark 1.4.14 (Diagonal extraction pattern). The diagonal argument generalises predicate-walking to sequences of functions. Let (f_n) be bounded and equicontinuous in $C^0([a,b], \mathbb{R})$, and let $D = \{d_1, d_2, \ldots\} = \mathbb{Q} \cap [a,b]$.

Stage 1. $(f_n(d_1))$ is bounded; extract by BW a subsequence $(f_{1,k})$ converging at d_1.
Stage 2. $(f_{1,k}(d_2))$ is bounded; extract a sub-subsequence $(f_{2,k})$ converging at d_2. As a subsequence of $(f_{1,k})$, it still converges at d_1.
Stage m. Extract $(f_{m,k})$ as a subsequence of $(f_{m-1,k})$ converging at d_m. By induction, $(f_{m,k})$ converges at $d_1, \ldots, d_m$.

Diagonal step. Define $g_m = f_{m,m}$. For any fixed j and $m \geq j$, the term $g_m = f_{m,m}$ is a term of $(f_{j,k})$ with index $k = m \geq j$. Hence $g_m(d_j) \to y_j$. The sequence (g_m) converges pointwise on all of D.

Equicontinuity then propagates pointwise convergence on D to uniform convergence on $[a, b]$ via an $\varepsilon/3$ argument.

Remark 1.4.15 (Why the diagonal works). At stage m, the predicate "converges at $d_1, \ldots, d_m$" holds for all terms of $(f_{m,k})$. The diagonal term $g_m = f_{m,m}$ is a tail term of every earlier row $f_{j,k}$ for $j \leq m$, so it satisfies the predicate for all $j \leq m$ simultaneously. Sending $m \to \infty$ gives convergence at every d_j.



(g_m) converges pointwise on all of $D = \{d_1, d_2, \ldots\}$

Figure 1.5: Diagonal extraction in the Arzelà-Ascoli argument. Each row $(f_{m,k})$ converges at $d_1, \ldots, d_m$ (green) and is a subsequence of the row above. The diagonal $g_m = f_{m,m}$ (red) is a tail of every row, so it converges at every d_j. Equicontinuity then extends pointwise convergence on the dense set D to uniform convergence on $[a, b]$.

Limitations of the Technique

Remark 1.4.16 (The predicate must be decidable at each stage). At each inductive step, one must determine which branch of the dichotomy applies. If the predicate requires knowledge of the limit a quantity not yet known the construction cannot proceed. Predicate-walking requires a finitary, combinatorial, or topological condition on the terms already seen.

Remark 1.4.17 (Multiple simultaneous predicates may exhaust the reservoir). The conjunction $P \wedge Q$ can hold only finitely often even when P and Q each hold infinitely often. The diagonal argument avoids this by handling one predicate at a time, composing via the diagonal.

Remark 1.4.18 (Subsequences, not sequences). Predicate-walking yields existence of a subsequence with the desired property, not a conclusion about the full sequence. The Monotone Subsequence Theorem does not assert (x_n) is monotone, or that the extracted subsequence is unique.

Remark 1.4.19 (Convergence requires a separate argument). The extracted subsequence has structural properties but not yet a limit. Converting structure into convergence requires a separate theorem: monotone subsequence $\to$ MCT (requires boundedness); bisection subsequence $\to$ NIP and squeeze; pointwise convergence on D $\to$ equicontinuity and $\varepsilon/3$.

Remark 1.4.20 (Quantitative control is absent). Predicate-walking proofs are pure existence arguments. They specify neither the rate of convergence nor which indices are selected. Explicit rates or constructive witnesses require a different approach.

Remark 1.4.21 (Countability is essential for the diagonal). The Arzelà-Ascoli diagonal requires a countable dense D so that points can be enumerated and handled one at a time. In non-separable metric spaces (those without a countable dense subset) the argument breaks down.

The Unifying Principle

Remark 1.4.22 (Infinite Pigeonhole as the common core). Every predicate-walking argument rests on the fact that an infinite set cannot be partitioned into two finite parts. Applied to $\mathbb{N}$: if P holds only finitely often, the P-indices have a maximum, and $\neg P$ holds from that point on. The method reduces a complex existence question to a sequence of binary choices, each justified by the Infinite Pigeonhole Principle. The construction is inductive, automatic, and never runs dry, provided the predicate is well-chosen and the reservoir is genuinely infinite at each stage.

Remark 1.4.23 (Dependency summary).
1. Monotone Subsequence Theorem peak dichotomy alone; no completeness.
2. Bolzano-Weierstrass via MCT Monotone Subseq. + MCT (requires completeness via AoC).
3. Bolzano-Weierstrass via bisection NIP + squeeze (requires completeness via NIP $\equiv$ AoC).
4. Arzelà-Ascoli BW at each diagonal stage + equicontinuity.

The Monotone Subsequence Theorem is the only result in the cluster requiring no completeness. All others depend on completeness via MCT or NIP.

1.5 Interval Bisection and Nested Intervals

Bisection and NIP Toolkit Quick Reference

Label	Statement	Proof method	Detail
NIP	$\bigcap_{n=1}^{\infty} I_n \neq \emptyset$ for nested closed bounded intervals	Supremum of left endpoints	↓
IVT	f continuous, $f(a) < L < f(b) \Rightarrow \exists\, c \in (a,b),\ f(c) = L$	Bisection + NIP	↓
UNC	$\mathbb{R}$ is uncountable	Nested interval diagonalisation	↓

The Nested Interval Property

Theorem (Nested Interval Property)

For each $n \in \mathbb{N}$, let $I_n = [a_n, b_n]$ be a closed bounded interval with $I_1 \supseteq I_2 \supseteq I_3 \supseteq \cdots$. Then $\bigcap_{n=1}^{\infty} I_n \neq \emptyset$.

Remark 1.5.1 (Fully quantified form). $\forall n \in \mathbb{N},\ I_n = [a_n, b_n] \neq \emptyset,\ [a_{n+1}, b_{n+1}] \subseteq [a_n, b_n] \implies \exists x \in \mathbb{R},\ \forall n,\ x \in I_n$.

Remark 1.5.2 (Proof strategy). Let $A = \{a_n : n \in \mathbb{N}\}$. Nestedness implies every b_n is an upper bound for A. By AoC, $x = \sup A$ exists. Then $a_n \leq x \leq b_n$ for all n, so $x \in I_n$ for all n.

Remark 1.5.3 (Closedness is essential). The open intervals $(0, 1/n)$ are nested but $\bigcap(0, 1/n) = \emptyset$. Closedness ensures the endpoints can serve as witnesses in the AoC argument.

Remark 1.5.4 (Boundedness is essential). The unbounded closed intervals $[n, \infty)$ are nested but their intersection is empty. Both hypotheses are necessary.

Remark 1.5.5 (Singleton intersection). If additionally $b_n - a_n \to 0$, the intersection is a singleton $\{x\}$, since $a_n \leq x \leq b_n$ and the squeeze theorem forces $a_n \to x$ and $b_n \to x$. This strengthened form is used in most applications.

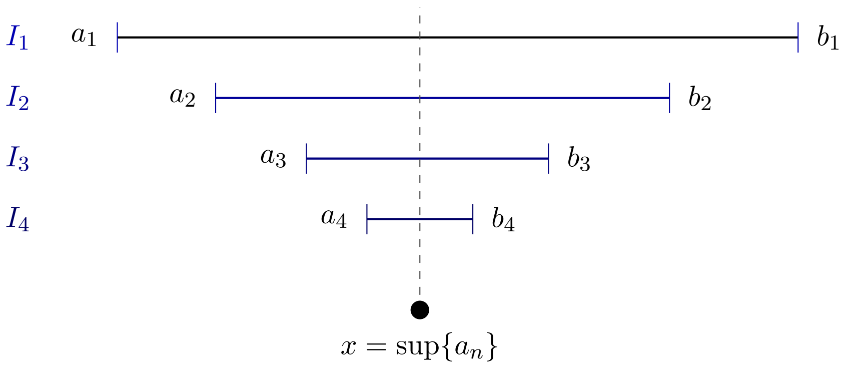


Figure 1.6: The Nested Interval Property (Section 1.5). Left endpoints a_n increase; right endpoints b_n decrease; each b_n is an upper bound for $\{a_n\}$. By the Axiom of Completeness, $x = \sup\{a_n\}$ exists and satisfies $a_n \leq x \leq b_n$ for all n, so $x \in \bigcap I_n$.

Remark 1.5.6 (Equivalence with completeness). NIP is equivalent to AoC over the Archimedean ordered field axioms. In particular, NIP fails over $\mathbb{Q}$: the intervals $[1, \sqrt{2}\,] \cap \mathbb{Q}$ shrink to a gap that $\mathbb{Q}$ cannot fill.

The Bisection Technique

Definition (Bisection Sequence)

Let $I_0 = [a_0, b_0]$ carry a property P. The **bisection sequence** (I_n) is defined inductively: given $I_n = [a_n, b_n]$, let $m_n = (a_n + b_n)/2$ and set

$$I_{n+1} = \begin{cases} [a_n, m_n] & \text{if } [a_n, m_n] \text{ inherits } P, \\ [m_n, b_n] & \text{otherwise.} \end{cases}$$

Remark 1.5.7 (Fully quantified form). $I_0 = [a_0, b_0]$; $\forall n,\ I_{n+1} \subseteq I_n$; $|I_n| = (b_0 - a_0)/2^n$.

Remark 1.5.8 (Length control). Bisection gives automatic length control: $|I_n| = (b_0 - a_0)/2^n \to 0$. No separate argument for the length condition of NIP is required.

Remark 1.5.9 (Abstract template). Every bisection proof has the same five-step shape:

1. *Initialize.* Identify P and starting interval I_0.
2. *Bisect.* Check which half inherits P.
3. *Extract.* Apply NIP to obtain $x \in \bigcap I_n$; note $a_n \to x$ and $b_n \to x$.
4. *Conclude.* Pass P through the limit, using continuity, MCT, or squeeze as appropriate.
5. *Verify hypotheses.* Confirm intervals are closed and nested; length control is free from bisection.

Remark 1.5.10 (Connection to predicate-walking). The bisection technique is a special case of predicate-walking: the predicate $P(I)$ = "interval I inherits the relevant property" is propagated at each stage by bisecting and keeping the half that satisfies P. The infinite-pigeonhole dichotomy appears in the Bolzano-Weierstrass proof (Proposition 1.4.11) but not in the IVT proof, where neither half is discarded based on cardinality instead the sign-test determines the choice deterministically.

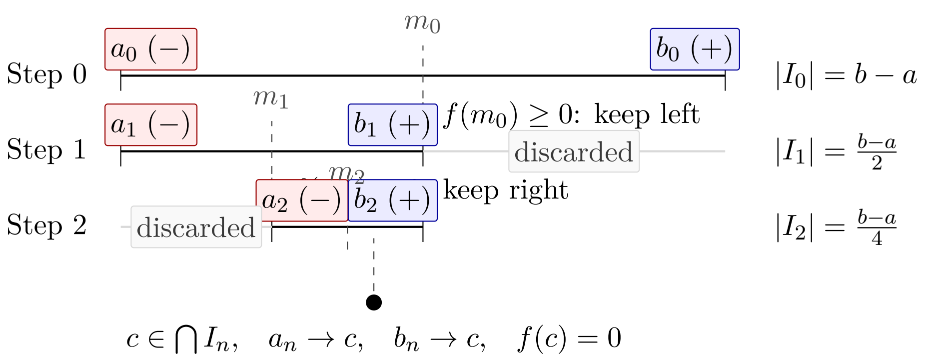


Figure 1.7: Three steps of the bisection proof of the Intermediate Value Theorem (Proposition 1.5.11) for $f(a) < 0 < f(b)$. At each step the half preserving a sign change is retained and the other discarded; lengths halve as $(b - a)/2^n \to 0$. By NIP, $c \in \bigcap I_n$ exists uniquely; continuity forces $f(c) = 0$.

The Intermediate Value Theorem via Bisection

Proposition 1.5.11 (Intermediate Value Theorem). *Let $f : [a,b] \to \mathbb{R}$ be continuous. If $L \in \mathbb{R}$ satisfies $f(a) < L < f(b)$ or $f(a) > L > f(b)$, then there exists $c \in (a,b)$ with $f(c) = L$. Go to proof.*

Remark.

Remark 1.5.12 (Proof strategy -- bisection approach). Reduce to $L = 0$, $f(a) < 0 < f(b)$ (replace f by $f - L$).

Property to propagate. $f(a_n) < 0$ and $f(b_n) \geq 0$ a sign change across the interval.
Branching rule. Let $m_n = (a_n + b_n)/2$: if $f(m_n) \geq 0$ set $[a_{n+1}, b_{n+1}] = [a_n, m_n]$; otherwise set $[a_{n+1}, b_{n+1}] = [m_n, b_n]$. The sign-change property is preserved in both cases.
Closing argument. NIP yields $c \in \bigcap I_n$ with $a_n \to c$ and $b_n \to c$. Since $f(a_n) < 0$ for all n: continuity gives $f(c) \leq 0$. Since $f(b_n) \geq 0$ for all n: continuity gives $f(c) \geq 0$. Therefore $f(c) = 0$.

Remark 1.5.13 (Two completeness paths). IVT admits two proofs from completeness:

1. *AoC path.* Set $K = \{x \in [a,b] : f(x) \leq 0\}$, $c = \sup K$; rule out $f(c) > 0$ and $f(c) < 0$ using the LUB property.

2. *NIP path.* Bisection as above.

The two paths are equivalent over the Archimedean ordered field axioms.

Remark 1.5.14 (Applied interpretation). The bisection construction is directly algorithmic: it locates a root to any desired precision in finitely many steps. After n bisections the root is known to within $(b - a)/2^n$. This is the bisection method of numerical analysis.

Uncountability of $\mathbb{R}$ via Nested Intervals

Proposition 1.5.15. *$\mathbb{R}$ is uncountable. Go to proof.*

Remark.

Remark 1.5.16 (Proof strategy -- nested interval diagonalisation).
Suppose for contradiction $\mathbb{R} = \{x_1, x_2, \ldots\}$. Build nested intervals (I_n) inductively with $I_{n+1} \subseteq I_n$ and $x_n \notin I_{n+1}$: at each step I_n contains two disjoint closed subintervals; x_{n+1} lies in at most one, so choose the other.
By NIP, $\bigcap I_n \neq \emptyset$: pick $y \in \bigcap I_n$. Then $y \neq x_n$ for every n (since $y \in I_n$ but $x_n \notin I_n$), contradicting the assumption that $\{x_n\}$ enumerates $\mathbb{R}$.

Remark 1.5.17 (Relation to Cantor's diagonal argument). This is the geometric form of Cantor's diagonalisation: instead of altering a decimal digit, the nth real is excluded from the nth nested interval. The witness y is the interval analogue of the "diagonal" real.

Summary: When to Use Each Technique

Remark 1.5.18 (Bisection vs. general nested intervals).

Feature	**Bisection**	**General nested intervals**
How built	Halving at midpoints	Any shrinking closed intervals
Length control	Automatic: $(b_0 - a_0)/2^n$	Must ensure $\lvert I_n \rvert \to 0$ separately
Typical use	Property propagation via sign test	Existence, uncountability, co
Canonical examples	IVT, BW	Uncountability, Baire's theor
Guarantor	NIP $\equiv$ AoC	NIP $\equiv$ AoC

Remark 1.5.19 (Completeness is indispensable). Neither technique works over $\mathbb{Q}$. The bisection procedure for $f(x) = x^2 - 2$ on $[1,2]$ runs mechanically over $\mathbb{Q}$, but the limiting point $\sqrt{2}$ is not in $\mathbb{Q}$. Every bisection proof is, at bottom, an appeal to the completeness of $\mathbb{R}$.

Remark 1.5.20 (Cross-section of the chapter). The four sections of this chapter are not independent: each builds on the last. Inequality bounding (§1.2) requires only the ordered field axioms. Completeness as a constructive engine (§1.3) requires AoC and the Archimedean Property. Predicate-walking (§1.4) requires completeness via MCT or NIP, depending on the variant. Bisection and NIP (§1.5) is the geometric synthesis: it encapsulates completeness in a form that drives the IVT, BW, and uncountability proofs simultaneously, connecting back to both the inductive-selection constructions of §1.3 and the predicate-walking patterns of §1.4.

1.6 Subsequence Partition via Residue Classes

Residue Partition Toolkit — Quick Reference

Tool	**Statement**	**Use**
k-Periodicity Criterion	$(a_n) \to L \iff \forall r,\ (a_{kn+r}) \to L$	Prove convergence by analysing
Forward direction	Subseqs. of convergent seqs. converge to L	Reduce to subsequence analysis
Divergence corollary	Two classes $\to$ different limits $\Rightarrow$ (a_n) diverges	$\downarrow$
Index translation	$m = kn + r \Rightarrow n \geq N_r$ once $m \geq k \max_r N_r$	Core of the backward direction

The Residue-Class Partition

For any sequence (a_n) and any integer $k \geq 2$, the division algorithm partitions $\mathbb{N}$ into k disjoint residue classes:

$$\mathbb{N} = \bigsqcup_{r=0}^{k-1} \{kn + r : n \geq 0\}.$$

Each class defines a subsequence: $(a_{kn+r})_{n\geq 0}$ selects every kth term of (a_n) beginning at position r. These k subsequences partition (a_n) exactly every term appears in precisely one class, order within each class is preserved, and their union recovers (a_n).

Remark 1.6.1 (Fully quantified form of the partition). $\forall n \in \mathbb{N}$, $\exists!\, r \in \{0, \ldots, k-1\}$, $\exists!\, q \in \mathbb{N}$, $n = kq + r$. Hence $\{(a_{kn+r})_{n\geq 0} : r = 0, \ldots, k-1\}$ is a partition of (a_n) into k subsequences.

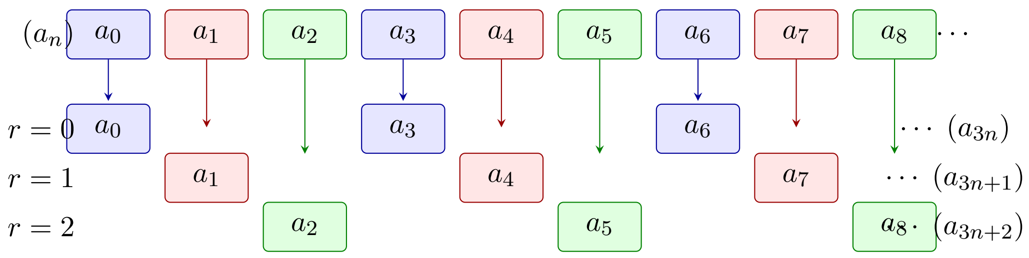


Figure 1.8: The residue-class partition of (a_n) with modulus $k = 3$. Each term is coloured by residue class: blue ($r = 0$), red ($r = 1$), green ($r = 2$). The partition is exact; the k-Periodicity Criterion (Figure 1.8) states that $(a_n) \to L$ if and only if each coloured subsequence converges to the same L.

The k-Periodicity Convergence Criterion

Theorem (k-Periodicity Convergence Criterion)

Let (a_n) be a sequence, $k \geq 2$ an integer, and $L \in \mathbb{R}$. The following are equivalent:

(i) $(a_n) \to L$.

(ii) For every $r \in \{0, 1, \ldots, k-1\}$, the subsequence $(a_{kn+r}) \to L$.

Remark 1.6.2 (Fully quantified form). $(i) \Leftrightarrow (ii)$, where:
(i): $\forall \varepsilon > 0$, $\exists N \in \mathbb{N}$, $\forall m \geq N$, $|a_m - L| < \varepsilon$.
(ii): $\forall r \in \{0, \ldots, k-1\}$, $\forall \varepsilon > 0$, $\exists N_r \in \mathbb{N}$, $\forall n \geq N_r$, $|a_{kn+r} - L| < \varepsilon$.

Remark 1.6.3 (Proof strategy). **Forward direction** $(i) \Rightarrow (ii)$. Each (a_{kn+r}) is a subsequence of (a_n). Subsequences of a convergent sequence converge to the same limit (Proposition 1.2.15).

Backward direction $(ii) \Rightarrow (i)$. The operative direction. Given $\varepsilon > 0$, for each residue r let N_r satisfy $|a_{kn+r} - L| < \varepsilon$ for all $n \geq N_r$. Set $N^* = k \cdot \max_r N_r$. For any $m \geq N^*$, write $m = kn + r$ via the division algorithm. Then:

$$n = \frac{m - r}{k} \geq \frac{N^* - (k-1)}{k} \geq \frac{k \max_r N_r - (k-1)}{k} \geq N_r,$$

so $|a_m - L| = |a_{kn+r} - L| < \varepsilon$.

Remark 1.6.4 (The index translation). The key step converts a global index m to a local index n within its residue class via $m = kn + r$. The factor k in the definition of N^* absorbs the offset $r \le k-1$ from every class simultaneously, ensuring that $n \ge N_r$ holds for whichever class m lands in.

The Divergence Corollary

Proposition 1.6.5 (Residue divergence criterion). *If there exist residues $r, s \in \{0, \ldots, k-1\}$ such that $(a_{kn+r}) \to L$ and $(a_{kn+s}) \to M$ with $L \ne M$, then (a_n) diverges. Go to proof.*

Remark.

Remark 1.6.6 (Fully quantified form). $(a_{kn+r}) \to L \ \wedge\ (a_{kn+s}) \to M \ \wedge\ L \ne M \ \Rightarrow\ (a_n)$ diverges.

Remark 1.6.7 (Proof strategy). Contrapositively: if $(a_n) \to L$, then every subsequence converges to L (forward direction of Figure 1.8). If two residue-class subsequences converge to different limits, (a_n) cannot converge.

Remark 1.6.8 (Canonical example). The sequence $((-1)^n)$ has $a_{2n} = 1 \to 1$ and $a_{2n+1} = -1 \to -1$. Since $1 \ne -1$, the sequence diverges established immediately without any ε-N computation.

The Alternating Series Test via Even/Odd Partition

The $k = 2$ case of Figure 1.8 drives the cleanest proof of the Alternating Series Test.

Theorem (Alternating Series Test)

Let (a_n) be a positive decreasing sequence with $a_n \to 0$. Then the alternating series $\sum_{n=1}^{\infty} (-1)^{n+1} a_n$ converges.

Remark 1.6.9 (Fully quantified form). $(\forall n,\ a_n > 0) \wedge (\forall n,\ a_{n+1} \le a_n) \wedge (a_n \to 0) \Rightarrow \exists L,\ \sum_{n=1}^{\infty} (-1)^{n+1} a_n = L$.

Remark 1.6.10 (Proof strategy -- even/odd partition). Let s_n denote the nth partial sum. Apply Figure 1.8 with $k = 2$: it suffices to show $(s_{2k}) \to L$ and $(s_{2k+1}) \to L$ for the same L.

Even subsequence (s_{2k}): decreasing and bounded. Group consecutive pairs:

$$s_{2k} = \sum_{j=1}^{k} (-a_{2j-1} + a_{2j}).$$

Since (a_n) is decreasing, each summand $-a_{2j-1} + a_{2j} \le 0$, so (s_{2k}) is decreasing. Also:

$$s_{2k} = -a_1 + \sum_{j=1}^{k-1} (a_{2j} - a_{2j+1}) + a_{2k} \ge -a_1,$$

since each term in the sum is non-negative. Thus (s_{2k}) is decreasing and bounded below by $-a_1$. By MCT, $s_{2k} \to L$ for some $L \geq -a_1$.

Odd subsequence (s_{2k+1}): linked to the even. Observe $s_{2k+1} = s_{2k} + a_{2k+1}$. Since $s_{2k} \to L$ and $a_{2k+1} \to 0$, the Algebraic Limit Theorem gives $s_{2k+1} \to L$.

Applying the criterion. Both residue-class subsequences converge to the same L. By Figure 1.8 with $k = 2$, $(s_n) \to L$.

Remark 1.6.11 (Why the partition is the right tool). The even and odd partial sums have structurally different characters: the even sums are monotone (enabling MCT), the odd sums are then controlled via the algebraic relationship $s_{2k+1} = s_{2k} + a_{2k+1}$. No single uniform argument handles both. The partition separates the two cases into independent analyses, then the criterion reassembles the conclusion.

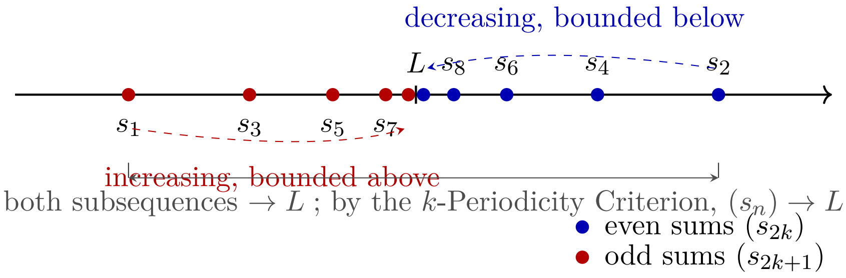


Figure 1.9: Even and odd partial sums of an alternating series $\sum(-1)^{n+1}a_n$ with (a_n) positive, decreasing, and tending to zero. Even sums (s_{2k}) (blue) decrease and are bounded below; by MCT they converge to L. Odd sums (s_{2k+1}) (red) satisfy $s_{2k+1} = s_{2k} + a_{2k+1} \to L$. Since both residue classes (mod $k = 2$) converge to L, the k-Periodicity Criterion (Figure 1.8) gives $(s_n) \to L$.

Remark 1.6.12 (Alternative proofs). The Alternating Series Test admits three proofs from different completeness avatars:

1. *Cauchy criterion* -- show (s_n) is eventually ε-steady directly.
2. *NIP* -- the nested intervals $[s_{2k}, s_{2k-1}]$ shrink to a point by the Nested Interval Property.
3. *MCT + residue partition* -- as above.

The partition proof is the most transparent structurally: the monotone behaviour and the limit identity are isolated into the two residue classes rather than handled simultaneously.

Higher k and Choosing the Right Period

For sequences with period $k > 2$, the criterion applies directly. The sequence $a_n = \cos(2\pi n/3)$ takes values $1, -\frac{1}{2}, -\frac{1}{2}, 1, -\frac{1}{2}, -\frac{1}{2}, \ldots$, cycling with period 3. Partitioning modulo 3:

$$\begin{aligned} a_{3n} &= 1 \to 1, \\ a_{3n+1} &= -\tfrac{1}{2} \to -\tfrac{1}{2}, \\ a_{3n+2} &= -\tfrac{1}{2} \to -\tfrac{1}{2}. \end{aligned}$$

The three limits differ ($1 \neq -\frac{1}{2}$), so the sequence diverges by Proposition 1.6.5.

Remark 1.6.13 (Choosing k to match the period). The right choice of k is the natural period of the sequence's structure. Using $k = 2$ for a period-3 sequence produces residue classes that are themselves period-3 subsequences; convergence of each class then requires a further partition with $k = 3$. Choosing k to match the period compresses the argument to a single application of the criterion.

Remark 1.6.14 (Applying the criterion to prove convergence). To prove $(a_n) \to L$ via the criterion, one must:

1. Select k matching the period of the sequence's behaviour.
2. Prove $(a_{kn+r}) \to L$ for each $r \in \{0, \ldots, k-1\}$, using whatever tool fits that residue class.
3. Invoke Figure 1.8 to conclude $(a_n) \to L$.

The method does not produce L -- it confirms a candidate. Identifying L requires separate work (direct computation, MCT applied to a class, etc.) before the criterion is invoked.

Contrast with Predicate-Walking

Remark 1.6.15 (Dual logical directions). The residue partition technique and predicate-walking (§1.4) operate in opposite logical directions.

Feature	Predicate-walking	Residue partition
Flow	Whole $\to$ one part	All parts $\to$ whole
Goal	Property of extracted subseq.	Convergence of full (a_n)
Coverage	One "good" subsequence	All k residue classes
Divergence use	Two classes, different limits	Same: two classes, different limit
Completeness role	Via MCT or NIP	Via MCT or algebraic limit theorem

Predicate-walking extracts what is needed; the residue partition reassembles from all pieces. The divergence criterion is available in both frameworks: two subsequences with different limits immediately imply divergence.

Limitations

Remark 1.6.16 (The same-limit condition is non-negotiable). The backward direction of Figure 1.8 requires all k residue-class subsequences to converge to the *same* L. There is no weakening: a convergent sequence has all its subsequences converging to the identical limit, so any two classes converging to different limits already certify divergence. The technique cannot produce convergence from mixed limits.

Remark 1.6.17 (The limit must be known in advance). The criterion confirms a candidate limit L; it does not produce L from scratch. In practice, L is identified by direct computation on one residue class (e.g., the even partial sums converge to L by MCT), then the criterion assembles global convergence.

Remark 1.6.18 (Exact periodicity is required). The technique applies when the sequence's qualitative behaviour repeats with exact period k in the index. For sequences whose oscillation decays without exact periodicity, the residue classes are not individually tractable, and a direct ε-N argument or a squeeze is more appropriate.

Remark 1.6.19 (Common error). Applying the criterion with k smaller than the true period, then concluding convergence after verifying only the even residue class. If the odd class is not verified, the conclusion does not follow. Every residue class must be checked.

Remark 1.6.20 (Forward connections). The residue partition technique reappears in:

- **Cesàro means** -- the Cesàro average of a sequence (a_n) can be analysed by partitioning the partial sums of the average into residue classes.
- **Fourier series** -- the convergence of partial sums of a Fourier series is often established by analysing even and odd partial sums (or partial sums along specific subsequences of the approximation order).
- **Recursively defined sequences** -- when a recurrence has period k, the residue-class subsequences each satisfy a simpler recurrence, enabling separate analysis.

1.7 Asymptotic Notation

Toolkit: Asymptotic Notation

Asymptotic notation records relative size near a limiting process. The atomic notions formalised below are:

- **little-o at a point** — one function is negligible compared with another as the input approaches a point;
- **little-o at infinity** — the same negligible comparison along an unbounded input range;
- **increment form** — the notation $r(h) = o(h)$ as $h \to 0$;
- **algebra rules** — sums and bounded factors preserve negligible errors;

Little-o Notation

Asymptotic notation separates the main term of an expression from the part that becomes negligible in a limiting process. The notation $f = o(g)$ means that f is smaller than every fixed multiple of g near the limiting point. It does not say that f is zero. It says that f is negligible relative to the chosen scale g.

Definition 1.7.1 (Little-o at a Point). Let $a \in \mathbb{R}$, let f and g be real-valued functions defined on a punctured neighbourhood of a, and suppose $g(x) \neq 0$ for all x sufficiently close to a with $x \neq a$. We write

$$f(x) = o(g(x)) \qquad (x \to a)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement).

$$f(x) = o(g(x)) \;\; (x \to a) \Longleftrightarrow \forall \varepsilon > 0 \; \exists \delta > 0 \; \forall x \\ \big(0 < |x - a| < \delta \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation). The scale g is the unit of comparison. The assertion $f = o(g)$ says that no matter how small a tolerance multiple εg is chosen, f eventually fits inside that multiple.

Definition 1.7.2 (Little-o at Infinity). Let f and g be real-valued functions defined on some interval (M, ∞), and suppose $g(x) \neq 0$ for all sufficiently large x. We write

$$f(x) = o(g(x)) \qquad (x \to \infty)$$

if and only if for every $\varepsilon > 0$ there exists $R > 0$ such that

$$x > R \quad \Longrightarrow \quad |f(x)| \leq \varepsilon |g(x)|.$$

Remark (Standard quantified statement).

$$f(x) = o(g(x)) \;\; (x \to \infty) \Longleftrightarrow \forall \varepsilon > 0 \; \exists R > 0 \; \forall x \\ \big(x > R \Rightarrow |f(x)| \leq \varepsilon |g(x)|\big).$$

Remark (Interpretation). The point version and the infinity version have the same logical shape. Only the neighbourhood changes: small punctured intervals around a are replaced by tails (R, ∞).

Definition 1.7.3 (Increment Little-o). Let r be a real-valued function defined for all sufficiently small nonzero h. We write

$$r(h) = o(h) \qquad (h \to 0)$$

if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$0 < |h| < \delta \quad \Longrightarrow \quad |r(h)| \leq \varepsilon |h|.$$

Remark (Standard quantified statement).

$$r(h) = o(h) \ \ (h \to 0) \iff \forall \varepsilon > 0 \, \exists \delta > 0 \, \forall h \, \big(0 < |h| < \delta \Rightarrow |r(h)| \le \varepsilon |h|\big).$$

Remark (Interpretation). This is the form used in first-order approximation. A remainder $r(h)$ is smaller than the increment h itself, so the quotient $r(h)/h$ tends to zero.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.7.4 (Little-o Quotient Characterisation). *Let $a \in \mathbb{R}$, and let f and g be real-valued functions defined on a punctured neighbourhood of a, with $g(x) \neq 0$ for x sufficiently close to a and $x \neq a$. Then*

$$f(x) = o(g(x)) \ \ (x \to a) \quad \iff \quad \frac{f(x)}{g(x)} \to 0 \ \ (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$[\forall \varepsilon > 0 \, \exists \delta > 0 \; 0 < |x - a| < \delta \Rightarrow |f(x)| \le \varepsilon |g(x)|] \iff \frac{f(x)}{g(x)} \to 0.$$

Remark (Interpretation). The quotient form is often the fastest test for little-o. The epsilon form is often the safest proof form because it avoids manipulating a quotient until the nonvanishing of the scale has been checked.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.7.5 (Sum Rule for Little-o). *If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$ as $x \to a$, then*

$$f_1(x) + f_2(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f_1 = o(g) \wedge f_2 = o(g) \implies f_1 + f_2 = o(g).$$

Remark (Interpretation). Negligible errors can be added. The proof is the usual epsilon-budget split: make each error smaller than half the final tolerance.

Remark (Dependencies).

- Little-o at a Point

Proposition 1.7.6 (Bounded Factor Rule for Little-o). *If $f(x) = o(g(x))$ as $x \to a$ and m is bounded near a, then*

$$m(x)f(x) = o(g(x)) \qquad (x \to a).$$

Go to proof.

Remark (Standard quantified statement).

$$f = o(g) \wedge \exists M > 0\ \exists \eta > 0\ \forall x \left(0 < |x - a| < \eta \Rightarrow |m(x)| \leq M\right) \Longrightarrow mf = o(g).$$

Remark (Interpretation). A bounded multiplier cannot turn a negligible error into a main-order term. It only changes the constant in the epsilon estimate.

Remark (Dependencies).

- Little-o at a Point

1.8 Inequality

Toolkit: Inequality

The algebra of inequalities governs how strict and nonstrict order relations behave under addition, multiplication, reciprocation, and composition. The atomic notions formalised below are:

- **addition on both sides** — adding the same term preserves strict and nonstrict inequalities;
- **addition of inequalities** — strict, nonstrict, and mixed pairwise addition;
- **multiplication by a positive scalar** — preserves direction;
- **multiplication by a negative scalar** — reverses direction;
- **nonstrict multiplication by a nonneg scalar** — the zero edge case;
- **squeeze theorem** — equality forced by double bounding;
- **transitivity** — strict and mixed chaining;
- **reciprocal inequalities** — order reversal under reciprocation of positive quantities;
- **square monotonicity** — ordering of squares for nonneg inputs;
- **square-root monotonicity** — ordering of square roots (requires `def:square-root`);
- **AM-GM for two terms** — arithmetic–geometric mean inequality;
- **Bernoulli's inequality** — $(1 + x)^n \geq 1 + nx$.

Inequality

Theorem 1.8.1 (Addition Preserves Strict Inequality). *For all* $a, b, c \in \mathbb{R}$,

$$a < b \Rightarrow a + c < b + c.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \Rightarrow a + c < b + c.$$

Remark (Dependencies).
Ordered field, ordered field axioms, additive cancellation.

Theorem 1.8.2 (Addition Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$,

$$a \leq b \Rightarrow a + c \leq b + c.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \Rightarrow a + c \leq b + c.$$

Remark (Dependencies).
Ordered field, addition preserves strict inequality.

Theorem 1.8.3 (Addition of Strict Inequalities). *For all* $a, b, c, d \in \mathbb{R}$,

$$a < b \wedge c < d \Rightarrow a + c < b + d.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a < b \wedge c < d \Rightarrow a + c < b + d.$$

Remark (Dependencies).
addition preserves strict inequality, transitivity of strict inequality.

Theorem 1.8.4 (Addition of Nonstrict Inequalities). *For all* $a, b, c, d \in \mathbb{R}$,

$$a \leq b \wedge c \leq d \Rightarrow a + c \leq b + d.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \wedge c \leq d \Rightarrow a + c \leq b + d.$$

Remark (Dependencies).
addition preserves nonstrict inequality.

Theorem 1.8.5 (Mixed Addition of Inequalities). *For all* $a, b, c, d \in \mathbb{R}$,

$$a \leq b \wedge c < d \Rightarrow a + c < b + d.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c, d \in \mathbb{R}, \quad a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Remark (Dependencies).
addition preserves nonstrict inequality, addition preserves strict inequality, mixed transitivity.

Theorem 1.8.6 (Multiplication by a Positive Scalar Preserves Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ 0 < c \ \Rightarrow \ ac < bc.$$

Remark (Dependencies).
Ordered field, multiply positive, positive cone.

Theorem 1.8.7 (Multiplication by a Negative Scalar Reverses Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \ \wedge \ c < 0 \ \Rightarrow \ ac > bc.$$

Remark (Dependencies).
Ordered field, multiply negative, positive cone.

Theorem 1.8.8 (Multiplication by a Positive Scalar Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \ \wedge \ 0 < c \ \Rightarrow \ ac \leq bc.$$

Remark (Dependencies).
multiplication by positive preserves strict inequality.

Theorem 1.8.9 (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ 0 \leq c \ \Rightarrow \ ac \leq bc.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge 0 \leq c \Rightarrow ac \leq bc.$$

Remark (Interpretation). This extends the positive-scalar result to include $c = 0$, where both sides become 0 and the inequality holds trivially.

Remark (Dependencies).
multiplication by positive preserves nonstrict inequality, ordered field.

Theorem (Squeeze)

Theorem 1.8.10 (Squeeze). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \leq c \wedge a = c \Rightarrow b = a.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \leq c \wedge a = c \Rightarrow b = a.$$

Remark (Interpretation). The squeeze principle forces equality when a value is simultaneously bounded above and below by equal quantities. The continuous analogue (if $f(x) \leq g(x) \leq h(x)$ and $f(x) \to L$ and $h(x) \to L$ then $g(x) \to L$) is the central tool for computing limits. The algebraic version stated here is its foundation.

Remark (Dependencies).
Ordered field, additive cancellation.

Theorem 1.8.11 (Transitivity of Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \wedge b < c \Rightarrow a < c.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a < b \wedge b < c \Rightarrow a < c.$$

Remark (Dependencies).
Ordered field, positive cone.

Theorem 1.8.12 (Mixed Transitivity). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \wedge b < c \Rightarrow a < c.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b, c \in \mathbb{R}, \quad a \leq b \wedge b < c \Rightarrow a < c.$$

Remark (Dependencies).
Ordered field, transitivity of strict inequality.

Theorem 1.8.13 (Reciprocal Reverses Strict Inequality for Positives). *For all* $a, b \in \mathbb{R}$*,*

$$0 < a < b \;\Rightarrow\; 0 < \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 < a < b \;\Rightarrow\; 0 < \tfrac{1}{b} < \tfrac{1}{a}.$$

Remark (Dependencies).
Ordered field, inverse positive, reciprocal order for positive elements, multiplication by positive preserves strict inequality.

Theorem 1.8.14 (Reciprocal Equivalence for Positives). *For all* $a, b \in \mathbb{R}$ *with* $a > 0$ *and* $b > 0$*,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R},\; a > 0 \;\wedge\; b > 0 \;\Rightarrow\; \big(a < b \iff \tfrac{1}{b} < \tfrac{1}{a}\big).$$

Remark (Dependencies).
reciprocal reverses strict inequality for positives, reciprocal order for positive elements.

Theorem 1.8.15 (Square Is Strictly Monotone on Nonneg Reals). *For all* $a, b \in \mathbb{R}$*,*

$$0 \leq a < b \;\Rightarrow\; a^2 < b^2.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a < b \;\Rightarrow\; a^2 < b^2.$$

Remark (Dependencies).
Ordered field, addition of strict inequalities, multiplication by positive preserves strict inequality, squares are nonnegative.

Theorem 1.8.16 (Square Root Is Monotone on Nonneg Reals). *For all* $a, b \in \mathbb{R}$*,*

$$0 \leq a \leq b \;\Rightarrow\; \sqrt{a} \leq \sqrt{b}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad 0 \leq a \leq b \;\Rightarrow\; \sqrt{a} \leq \sqrt{b}.$$

Remark (Audit note). This theorem depends on def:square-root, which does not yet appear in the repository label inventory. That definition must be created and registered before this result can be fully formalised.

Remark (Dependencies).
def:square-root (not yet registered — see audit note above), square is strictly monotone on nonneg reals.

Theorem 1.8.17 (AM-GM Inequality for Two Terms). *For all* $a, b \geq 0$*,*

$$\sqrt{ab} \leq \frac{a+b}{2}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R},\ a \geq 0 \ \wedge\ b \geq 0 \ \Rightarrow\ \sqrt{ab} \leq \frac{a+b}{2}.$$

Remark (Interpretation). The arithmetic-geometric mean inequality for two terms is the simplest case of the AM-GM family. The standard proof proceeds by squaring both sides (using square monotonicity) and reducing to $(\sqrt{a} - \sqrt{b})^2 \geq 0$, which follows from squares being nonnegative.

Remark (Dependencies).
squares are nonnegative, square is strictly monotone on nonneg reals, def:square-root (not yet registered — see audit note in square root monotonicity).

Theorem (Bernoulli's Inequality)

Theorem 1.8.18 (Bernoulli's Inequality). *For all* $x \in \mathbb{R}$ *with* $x \geq -1$ *and all* $n \in \mathbb{N}$*,*

$$(1+x)^n \geq 1 + nx.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R},\ \forall n \in \mathbb{N},\ x \geq -1 \ \Rightarrow\ (1+x)^n \geq 1 + nx.$$

Remark (Interpretation). Bernoulli's inequality is proved by induction and is a key estimate in analysis: it provides a linear lower bound for exponential growth and is used to establish the divergence of the harmonic series, bounds on compound interest, and properties of the exponential function.

Remark (Dependencies).
Ordered field, multiplication by nonneg preserves nonstrict inequality, addition preserves nonstrict inequality.

1.9 Modulus

Toolkit: Modulus

The absolute value (modulus) of a real number is the fundamental measure of its magnitude, stripped of sign. The atomic notions formalised below are:

- **absolute value** — the piecewise definition $|a| := \max(a, -a)$;
- **nonnegativity** — $|a| \geq 0$ for all $a \in \mathbb{R}$;
- **zero characterisation** — $|a| = 0$ if and only if $a = 0$;
- **self-or-negation** — $|a|$ equals either a or $-a$;
- **symmetry** — $|-a| = |a|$;
- **multiplicativity** — $|ab| = |a||b|$;
- **quotient rule** — $|a/b| = |a|/|b|$ for $b \neq 0$;
- **bound by modulus** — $-|a| \leq a \leq |a|$;
- **nonstrict modulus inequality** — $|a| \leq r \iff -r \leq a \leq r$;
- **strict modulus inequality** — $|a| < r \iff -r < a < r$;
- **triangle inequality** — $|a + b| \leq |a| + |b|$;
- **reverse triangle inequality** — $||a| - |b|| \leq |a - b|$;
- **generalised triangle inequality** — $|\sum_{k=1}^{n} a_k| \leq \sum_{k=1}^{n} |a_k|$.

Modulus

Definition (Absolute Value)

Definition 1.9.1 (Absolute Value). Let $a \in \mathbb{R}$. The ***absolute value*** (or ***modulus***) of a is

$$|a| := \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Equivalently, $|a| := \max(a, -a)$.

Remark (Standard quantified statement).

$$|a| = \max(a, -a) := \begin{cases} a & \text{if } a \geq 0, \\ -a & \text{if } a < 0. \end{cases}$$

Remark (Interpretation). The absolute value measures magnitude independent of sign. On the real line it equals the distance from a to 0: $|a| = d(a, 0)$ under the standard metric. The two-branch definition partitions $\mathbb{R}$ at 0 and reflects negative values to positive ones.

Remark (Dependencies).

Ordered field, $\mathbb{R}$ is an ordered field.

Theorem 1.9.2 (Absolute Value Is Nonnegative). *For all $a \in \mathbb{R}$,*

$$|a| \geq 0.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| \geq 0.$$

Remark (Dependencies).
Absolute value, squares are nonnegative, ordered field.

Theorem 1.9.3 (Absolute Value Zero Iff Zero). *For all $a \in \mathbb{R}$,*

$$|a| = 0 \iff a = 0.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| = 0 \iff a = 0.$$

Remark (Dependencies).
Absolute value, absolute value is nonnegative, ordered field.

Theorem 1.9.4 (Absolute Value Equals Self or Negation). *For all $a \in \mathbb{R}$, either $|a| = a$ or $|a| = -a$.*

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |a| = a \lor |a| = -a.$$

Remark (Dependencies).
Absolute value.

Theorem 1.9.5 (Absolute Value Is Symmetric). *For all $a \in \mathbb{R}$,*

$$|-a| = |a|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad |-a| = |a|.$$

Remark (Dependencies).
Absolute value, double negation.

Theorem 1.9.6 (Absolute Value of a Product). *For all $a, b \in \mathbb{R}$,*

$$|ab| = |a|\,|b|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |ab| = |a|\,|b|.$$

Remark (Dependencies).
Absolute value, negation of product, product of negatives, squares are nonnegative.

Theorem 1.9.7 (Absolute Value of a Quotient). *For all $a \in \mathbb{R}$ and $b \in \mathbb{R}$ with $b \neq 0$,*

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall b \in \mathbb{R} \setminus \{0\}, \quad \left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Remark (Dependencies).
Absolute value, absolute value of a product, absolute value zero iff zero, uniqueness of multiplicative inverse.

Theorem 1.9.8 (Bound by Modulus). *For all $a \in \mathbb{R}$,*

$$-|a| \leq a \leq |a|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R}, \quad -|a| \leq a \leq |a|.$$

Remark (Dependencies).
Absolute value, absolute value is nonnegative, ordered field.

Theorem (Nonstrict Modulus Inequality)

Theorem 1.9.9 (Nonstrict Modulus Inequality). *For all $a \in \mathbb{R}$ and $r \geq 0$,*

$$|a| \leq r \iff -r \leq a \leq r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r \geq 0 \Rightarrow \big(|a| \leq r \iff -r \leq a \leq r\big).$$

Remark (Interpretation). This biconditional is the primary tool for replacing a modulus inequality with a pair of simultaneous linear inequalities. It underlies the ε-δ manipulations throughout analysis: the condition $|x - a| \leq \varepsilon$ is equivalent to $a - \varepsilon \leq x \leq a + \varepsilon$.

Remark (Dependencies).
Absolute value, bound by modulus, ordered field.

Theorem 1.9.10 (Strict Modulus Inequality). *For all $a \in \mathbb{R}$ and $r > 0$,*

$$|a| < r \iff -r < a < r.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a \in \mathbb{R},\ \forall r \in \mathbb{R},\ r > 0 \ \Rightarrow\ \big(|a| < r \iff -r < a < r\big).$$

Remark (Dependencies).
Absolute value, nonstrict modulus inequality, ordered field.

Theorem (Triangle Inequality)

Theorem (Triangle Inequality). *For all $a, b \in \mathbb{R}$,*

$$|a + b| \leq |a| + |b|.$$

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad |a + b| \leq |a| + |b|.$$

Remark (Interpretation). The triangle inequality is the foundational estimate of analysis. Every convergence argument, continuity proof, and metric-space result that requires bounding a sum reduces, at some step, to this inequality. The name reflects its geometric meaning: the length of one side of a triangle does not exceed the sum of the other two.

Theorem 1.9.11 (Reverse Triangle Inequality). *For all $a, b \in \mathbb{R}$,*

$$\big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}, \quad \big|\,|a| - |b|\,\big| \leq |a - b|.$$

Remark (Interpretation). The reverse triangle inequality provides a lower bound on $|a - b|$ in terms of the difference of moduli. It is used to bound the modulus of a difference from below, particularly when proving that a function is not uniformly continuous or that a sequence does not converge.

Remark (Dependencies).
Absolute value, triangle inequality, absolute value is symmetric.

Theorem 1.9.12 (Generalised Triangle Inequality). *For all $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in \mathbb{R}$,*

$$\left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Proof). Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N},\ \forall a_1, \dots, a_n \in \mathbb{R}, \quad \left| \sum_{k=1}^{n} a_k \right| \leq \sum_{k=1}^{n} |a_k|.$$

Remark (Dependencies).
Absolute value, triangle inequality.

Proofs

Remark (Return). Return to Theorem

Proposition (Little-o Quotient Characterisation). *Let $a \in \mathbb{R}$, and let f and g be real-valued functions defined on a punctured neighbourhood of a, with $g(x) \neq 0$ for x sufficiently close to a and $x \neq a$. Then*

$$f(x) = o(g(x)) \ (x \to a) \quad \iff \quad \frac{f(x)}{g(x)} \to 0 \ (x \to a).$$

Professional Standard Proof.
By definition, $f = o(g)$ as $x \to a$ means that for every $\varepsilon > 0$, eventually $|f(x)| \leq \varepsilon |g(x)|$. Since $g(x) \neq 0$ eventually, this is equivalent to

$$\left| \frac{f(x)}{g(x)} \right| \leq \varepsilon$$

eventually. That is precisely the assertion that $f(x)/g(x) \to 0$ as $x \to a$. ☐

Detailed Learning Proof.
Assume first that $f = o(g)$. Given $\varepsilon > 0$, choose $\delta > 0$ so that $0 < |x - a| < \delta$ implies $|f(x)| \leq \varepsilon |g(x)|$. On a smaller punctured neighbourhood, if necessary, $g(x) \neq 0$. Dividing by $|g(x)|$ gives $|f(x)/g(x)| \leq \varepsilon$, so $f(x)/g(x) \to 0$.
Conversely, assume $f(x)/g(x) \to 0$. Given $\varepsilon > 0$, choose $\delta > 0$ so that $0 < |x - a| < \delta$ implies $|f(x)/g(x)| \leq \varepsilon$, again within a punctured neighbourhood where g is nonzero. Multiplying by $|g(x)|$ gives $|f(x)| \leq \varepsilon |g(x)|$, hence $f = o(g)$. ☐

Remark (Dependencies).

- Little-o at a Point

Remark (Return). Return to Theorem

Proposition (Sum Rule for Little-o). *If $f_1(x) = o(g(x))$ and $f_2(x) = o(g(x))$ as $x \to a$, then*

$$f_1(x) + f_2(x) = o(g(x)) \qquad (x \to a).$$

Professional Standard Proof.
Let $\varepsilon > 0$. Since $f_1 = o(g)$ and $f_2 = o(g)$, each term is eventually bounded by $(\varepsilon/2)|g(x)|$ in absolute value. On the intersection of the two punctured neighbourhoods,

$$|f_1(x) + f_2(x)| \leq |f_1(x)| + |f_2(x)| \leq \varepsilon|g(x)|.$$

Therefore $f_1 + f_2 = o(g)$. □

Detailed Learning Proof.
Choose a target tolerance $\varepsilon > 0$. The first little-o hypothesis gives a punctured neighbourhood on which $|f_1(x)| \leq (\varepsilon/2)|g(x)|$. The second gives a punctured neighbourhood on which $|f_2(x)| \leq (\varepsilon/2)|g(x)|$. Taking the smaller radius makes both estimates hold simultaneously. The triangle inequality then gives the desired bound for the sum. □

Remark (Dependencies).

- Little-o at a Point

Remark (Return). Return to Theorem

Proposition (Bounded Factor Rule for Little-o). *If $f(x) = o(g(x))$ as $x \to a$ and m is bounded near a, then*

$$m(x)f(x) = o(g(x)) \qquad (x \to a).$$

Professional Standard Proof.
Let $\varepsilon > 0$. Since m is bounded near a, choose $M > 0$ and a punctured neighbourhood on which $|m(x)| \leq M$. Since $f = o(g)$, choose a smaller punctured neighbourhood on which $|f(x)| \leq (\varepsilon/M)|g(x)|$. Then

$$|m(x)f(x)| \leq M|f(x)| \leq \varepsilon|g(x)|.$$

Thus $mf = o(g)$. □

Detailed Learning Proof.
The bounded factor can cost only a fixed multiplicative constant. If $|m(x)| \leq M$ near a, ask the little-o estimate for f to be ε/M instead of ε. Multiplying the two estimates gives the final tolerance ε. □

Remark (Dependencies).

- Little-o at a Point

Remark (Return). Return to Theorem

Theorem (Addition Preserves Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \;\Rightarrow\; a + c < b + c.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-add-both-sides. ☐

Detailed Learning Proof.
TODO: detailed learning proof for ineq-add-both-sides. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Addition Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \Rightarrow \ a + c \leq b + c.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-nonstrict-add-both-sides. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-nonstrict-add-both-sides. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Addition of Strict Inequalities). *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a < b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-add-inequalities. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-add-inequalities. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Addition of Nonstrict Inequalities). *For all* $a, b, c, d \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ c \leq d \ \Rightarrow \ a + c \leq b + d.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-nonstrict-add-inequalities. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-nonstrict-add-inequalities. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Mixed Addition of Inequalities). *For all* $a, b, c, d \in \mathbb{R}$,

$$a \leq b \ \wedge \ c < d \ \Rightarrow \ a + c < b + d.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-mixed-add. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-mixed-add. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Multiplication by a Positive Scalar Preserves Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge\ 0 < c \ \Rightarrow\ ac < bc.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-multiply-positive. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-multiply-positive. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Multiplication by a Negative Scalar Reverses Strict Inequality).
For all $a, b, c \in \mathbb{R}$,

$$a < b \wedge c < 0 \Rightarrow ac > bc.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-multiply-negative. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-multiply-negative. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Multiplication by a Positive Scalar Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \;\wedge\; 0 < c \;\Rightarrow\; ac \leq bc.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-nonstrict-multiply-positive. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-nonstrict-multiply-positive. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Multiplication by a Nonneg Scalar Preserves Nonstrict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ 0 \leq c \ \Rightarrow \ ac \leq bc.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-nonstrict-multiply-nonneg. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-nonstrict-multiply-nonneg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Squeeze). *For all* $a, b, c \in \mathbb{R}$,

$$a \leq b \leq c \ \land \ a = c \ \Rightarrow \ b = a.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-squeeze. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-squeeze. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Transitivity of Strict Inequality). *For all* $a, b, c \in \mathbb{R}$*,*

$$a < b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-transitivity-strict. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-transitivity-strict. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Mixed Transitivity). *For all* $a, b, c \in \mathbb{R}$*,*

$$a \leq b \ \wedge \ b < c \ \Rightarrow \ a < c.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-transitivity-mixed. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-transitivity-mixed. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Reciprocal Reverses Strict Inequality for Positives). *For all $a, b \in \mathbb{R}$,*

$$0 < a < b \Rightarrow 0 < \frac{1}{b} < \frac{1}{a}.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-reciprocal-positive. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-reciprocal-positive. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Reciprocal Equivalence for Positives). *For all $a, b \in \mathbb{R}$ with $a > 0$ and $b > 0$,*

$$a < b \iff \frac{1}{b} < \frac{1}{a}.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-reciprocal-flip. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-reciprocal-flip. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Square Is Strictly Monotone on Nonneg Reals). *For all* $a, b \in \mathbb{R}$,

$$0 \leq a < b \;\Rightarrow\; a^2 < b^2.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-square-monotone. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-square-monotone. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Square Root Is Monotone on Nonneg Reals). *For all* $a, b \in \mathbb{R}$,

$$0 \leq a \leq b \Rightarrow \sqrt{a} \leq \sqrt{b}.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-square-root-monotone. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-square-root-monotone. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (AM-GM Inequality for Two Terms). *For all* $a, b \geq 0$*,*

$$\sqrt{ab} \leq \frac{a+b}{2}.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-am-gm-two. ☐

Detailed Learning Proof.
TODO: detailed learning proof for ineq-am-gm-two. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bernoulli's Inequality). *For all $x \in \mathbb{R}$ with $x \geq -1$ and all $n \in \mathbb{N}$,*

$$(1+x)^n \geq 1+nx.$$

Professional Standard Proof.
TODO: professional standard proof for ineq-bernoulli. □

Detailed Learning Proof.
TODO: detailed learning proof for ineq-bernoulli. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value Is Nonnegative). *For all* $a \in \mathbb{R}$*,*

$$|a| \geq 0.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-nonneg. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-nonneg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value Zero Iff Zero). *For all* $a \in \mathbb{R}$*,*

$$|a| = 0 \iff a = 0.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-zero-iff-zero. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-zero-iff-zero. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value Equals Self or Negation). *For all $a \in \mathbb{R}$, either $|a| = a$ or $|a| = -a$.*

Professional Standard Proof.
TODO: professional standard proof for absolute-value-self-or-neg. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-self-or-neg. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value Is Symmetric). *For all* $a \in \mathbb{R}$*,*

$$|-a| = |a|.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-symmetric. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-symmetric. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value of a Product). *For all* $a, b \in \mathbb{R}$*,*

$$|ab| = |a|\,|b|.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-product. ☐

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-product. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Value of a Quotient). *For all $a \in \mathbb{R}$ and $b \in \mathbb{R}$ with $b \neq 0$,*

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-quotient. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-quotient. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bound by Modulus). *For all* $a \in \mathbb{R}$,

$$-|a| \leq a \leq |a|.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-bounds. $\square$

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-bounds. $\square$

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Nonstrict Modulus Inequality). *For all $a \in \mathbb{R}$ and $r \geq 0$,*

$$|a| \leq r \iff -r \leq a \leq r.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-le-iff. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-le-iff. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Strict Modulus Inequality). *For all* $a \in \mathbb{R}$ *and* $r > 0$,

$$|a| < r \iff -r < a < r.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-lt-iff. ☐

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-lt-iff. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Reverse Triangle Inequality). *For all* $a, b \in \mathbb{R}$*,*

$$\big|\,|a| - |b|\,\big| \le |a - b|.$$

Professional Standard Proof.
TODO: professional standard proof for reverse-triangle-inequality. $\square$

Detailed Learning Proof.
TODO: detailed learning proof for reverse-triangle-inequality. $\square$

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Generalised Triangle Inequality). *For all $n \in \mathbb{N}$ and $a_1, \ldots, a_n \in \mathbb{R}$,*

$$\left|\sum_{k=1}^{n} a_k\right| \leq \sum_{k=1}^{n} |a_k|.$$

Professional Standard Proof.
TODO: professional standard proof for absolute-value-sum-bound. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-sum-bound. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Real Analysis, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 2

Bounds

bounding

Sequences $\rightarrow$ **Bounds** $\rightarrow$ Continuity

Completeness Axiom

Axiom of Completeness

Remark (Why completeness is needed). The ordered field axioms alone do not prevent "holes" in the number line. The rationals $\mathbb{Q}$ satisfy all field and order axioms, yet fail completeness. Consider the set

$$S = \{x \in \mathbb{Q} : x^2 < 2\}.$$

This set is nonempty (e.g. $1 \in S$) and bounded above in $\mathbb{Q}$ (e.g. 2 is an upper bound). Yet $\sup S$ does not exist as a rational number: the candidate $\sqrt{2}$ is irrational. The set S has no least upper bound in $\mathbb{Q}$ -- a hole sits exactly where $\sqrt{2}$ should be.
The Completeness Axiom asserts that $\mathbb{R}$ has no such holes: every nonempty bounded-above set has a supremum *inside* $\mathbb{R}$. This single axiom is what distinguishes $\mathbb{R}$ from $\mathbb{Q}$, and it underlies every major theorem in analysis.

Definition (Least Upper Bound Property)

Definition 2.0.1 (Least Upper Bound Property). Let $(S, \leq)$ be an ordered set. The set S has the *Least Upper Bound Property* if and only if every nonempty subset of S that is bounded above in S has a supremum in S.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set.
S has the Least Upper Bound Property

$$\iff \forall A \subseteq S \, \Big[\big(A \neq \varnothing \land \exists u \in S \; \forall x \in A \; (x \leq u) \big) \rightarrow \exists s \in S \; \mathrm{Supremum}(s, A) \Big].$$

Remark (Definition predicate reading).

$$\mathrm{LeastUpperBoundProperty}(S) := \forall A \subseteq S \, \Big[\big(A \neq \varnothing \land \exists u \in S \; \forall x \in A \; (x \leq u) \big) \rightarrow \exists s \in S \; \mathrm{Supremum}(s,$$

Remark (Negated quantified statement).

Let $(S,\leq)$ be an ordered set.
S does not have the Least Upper Bound Property

$$\iff \exists A \subseteq S\ \Big[A \neq \varnothing \wedge \exists u \in S\ \forall x \in A\ (x \leq u) \wedge \forall s \in S\ \neg\,\mathrm{Supremum}(s,A)\Big].$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{LeastUpperBoundProperty}(S) \iff \exists A \subseteq S\ \Big[A \neq \varnothing \wedge \exists u \in S\ \forall x \in A\ (x \leq u) \wedge \forall s \in S\ \neg\,\mathrm{Supremum}(s$$

Remark (Failure modes). The property fails when there is a nonempty subset of the ambient ordered set that has at least one upper bound but has no least upper bound inside that same ambient set.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{LeastUpperBoundProperty}(S) = \underbrace{\exists A \subseteq S\ (A \neq \varnothing)}_{\texttt{nonempty witness set}} \wedge \underbrace{\exists u \in S\ \forall x \in A\ (x \leq u)}_{\texttt{bounded above in } S} \wedge \underbrace{\forall s \in S\ \neg\,\mathrm{Supremum}}_{\texttt{no supremum in } S}$$

Remark (Interpretation). The Least Upper Bound Property says that upper-bounded nonempty subsets do not point toward missing ceiling elements. Whenever the order supplies at least one upper bound, it also supplies a smallest upper bound in the same ambient set. Failure therefore indicates a hole in the ordered structure rather than a defect in the subset alone.

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound
- Supremum

Axiom (Completeness of $\mathbb{R}$)

Axiom 2.0.2 (Completeness of $\mathbb{R}$). The ordered field $\mathbb{R}$ has the Least Upper Bound Property.

Remark (Standard quantified statement).

$$\mathrm{LeastUpperBoundProperty}(\mathbb{R}).$$

Equivalently,

$$\forall A \subseteq \mathbb{R}\ \Big[\big(A \neq \varnothing \wedge \exists u \in \mathbb{R}\ \forall x \in A\ (x \leq u)\big) \to \exists s \in \mathbb{R}\ \mathrm{Supremum}(s,A)\Big].$$

Remark (Axiom predicate reading).

$$\mathrm{LeastUpperBoundProperty}(\mathbb{R}).$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R})$$
$$\iff \exists A \subseteq \mathbb{R}\ \Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\ \forall x \in A\ (x \leq u) \wedge \forall s \in \mathbb{R}\ \neg\,\mathrm{Supremum}(s, A)\Big].$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R}) \iff \exists A \subseteq \mathbb{R}\ \Big[A \neq \varnothing \wedge \exists u \in \mathbb{R}\ \forall x \in A\ (x \leq u) \wedge \forall s \in \mathbb{R}\ \neg\,\mathrm{Supremum}(s$$

Remark (Failure modes). Completeness would fail if some nonempty subset of $\mathbb{R}$ were bounded above in $\mathbb{R}$ but had no supremum in $\mathbb{R}$.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{LeastUpperBoundProperty}(\mathbb{R}) = \underbrace{\exists A \subseteq \mathbb{R}\ (A \neq \varnothing)}_{\text{nonempty witness set}} \wedge \underbrace{\exists u \in \mathbb{R}\ \forall x \in A\ (x \leq u)}_{\text{bounded above in } \mathbb{R}} \wedge \underbrace{\forall s \in \mathbb{R}\ \neg\,\mathrm{Supremun}}_{\text{missing real supre}}$$

Remark (Interpretation). The completeness axiom asserts that the real line has no order gaps of the least-upper-bound kind. Nonempty sets of real numbers that are bounded above always determine a real number sitting exactly at the least possible ceiling. In Volume II, $\mathbb{R}$ is constructed as the complete ordered field arising from Dedekind cuts; in this volume, completeness is used as the standing least-upper-bound principle for real analysis. This is the structural property that distinguishes $\mathbb{R}$ from ordered fields such as $\mathbb{Q}$.

Remark (Dependencies).

- The Real Numbers
- Ordered Field
- Least Upper Bound Property

Nested Intervals

Nested Interval Property

Theorem (Nested Interval Property)

Theorem 2.0.3 (Nested Interval Property). *Let (I_n) be a sequence of nonempty closed intervals in $\mathbb{R}$. If $I_{n+1} \subseteq I_n$ for every $n \in \mathbb{N}$, then*

$$\bigcap_{n=1}^{\infty} I_n \neq \varnothing.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (a_n) \subseteq \mathbb{R}\ \forall (b_n) \subseteq \mathbb{R}$$
$$\Big[\big(\forall n \in \mathbb{N}\ (a_n \leq b_n)\big) \wedge \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big)$$
$$\to \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x \in [a_n, b_n])\Big].$$

Remark (Predicate reading).

$$\mathrm{NestedClosedIntervals}(I_n) \implies \mathrm{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\exists (a_n) \subseteq \mathbb{R}\ \exists (b_n) \subseteq \mathbb{R}\\ &\Big[\big(\forall n \in \mathbb{N}\ (a_n \le b_n)\big) \wedge \big(\forall n \in \mathbb{N}\ ([a_{n+1}, b_{n+1}] \subseteq [a_n, b_n])\big)\\ &\qquad \wedge \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin [a_n, b_n])\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{NestedClosedIntervals}(I_n) \wedge \neg\,\mathrm{NonemptyIntersection}\left(\bigcap_{n=1}^{\infty} I_n\right).$$

Remark (Failure modes). The theorem fails when a nested sequence of nonempty closed real intervals has no common real point.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{NestedClosedIntervals}(I_n)}_{\texttt{closed intervals remain nested}} \wedge \underbrace{\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x \notin I_n)}_{\texttt{no real point lies in every interval}}.$$

Remark (Interpretation). Nested closed intervals cannot keep shrinking or relocating in a way that loses every real point. Nesting keeps the intervals aligned, nonemptiness prevents trivial collapse at each stage, and completeness supplies a real point that survives all stages at once.

Remark (Dependencies).
Axiom 2.0.2.

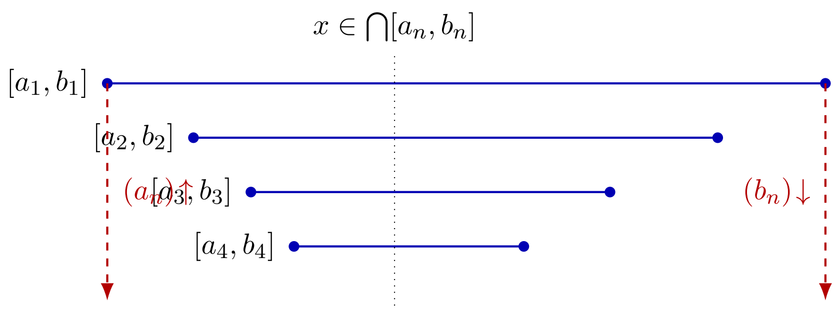


Figure 2.1: Nested intervals $[a_1, b_1] \supseteq [a_2, b_2] \supseteq \cdots$ pinch toward a common point. The sequence of left endpoints is increasing and bounded above; its supremum x lies in every interval.

Archimedean Property

Archimedean Property

Remark (Why this requires proof). The Archimedean Property states that the natural numbers are unbounded in $\mathbb{R}$. That conclusion is not a formal consequence of the ordered-field axioms alone. There exist ordered fields satisfying all ordered-field axioms in which the natural numbers are bounded above, so the statement must be justified from stronger structure.

The needed structure is completeness. If $\mathbb{N}$ were bounded above in $\mathbb{R}$, then the Least Upper Bound Property would produce a supremum $\alpha \in \mathbb{R}$ for $\mathbb{N}$. But then $\alpha - 1$ cannot be an upper bound, so there exists $n \in \mathbb{N}$ with $n > \alpha - 1$. Hence $n + 1 > \alpha$, contradicting the fact that α is an upper bound of $\mathbb{N}$.

Go to proof.

Theorem (Archimedean Property)

Theorem 2.0.4 (Archimedean Property). *For every real number x, there exists a natural number n such that $n > x$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Predicate reading).

$$\text{ArchimedeanProperty}(\mathbb{R}) \iff \forall x \in \mathbb{R}\ \exists n \in \mathbb{N}\ (n > x).$$

Remark (Negated quantified statement).

$$\exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \leq x).$$

Remark (Negation predicate reading).

$$\neg\,\text{ArchimedeanProperty}(\mathbb{R}) \iff \exists x \in \mathbb{R}\ \forall n \in \mathbb{N}\ (n \leq x).$$

Remark (Failure modes). The theorem fails exactly when the natural numbers are bounded above by some real number.

Remark (Failure mode decomposition).

$$\neg\,\text{ArchimedeanProperty}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}}_{\text{real upper bound}} \quad \underbrace{\forall n \in \mathbb{N}\ (n \leq x)}_{\text{all natural numbers remain below it}} \quad .$$

Remark (Interpretation). The Archimedean Property says that the natural numbers do not stop below any real ceiling. No real number is so large that every natural number remains below it. In this chapter, the result is a consequence of completeness: a bounded copy of $\mathbb{N}$ would have a supremum, and that supremum is contradicted by shifting one natural number upward.

Remark (Dependencies).

Axiom 2.0.2, Definition 2.1.9.

Corollary (Archimedean Reciprocal Corollary)

Corollary 2.0.5 (Archimedean Reciprocal Corollary). *For every $x > 0$ and every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $nx > y$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}\ \big(x > 0 \to \exists n \in \mathbb{N}\ (nx > y)\big).$$

Remark (Predicate reading).

$$\mathrm{ArchimedeanReciprocal}(\mathbb{R}) \iff \forall x, y \in \mathbb{R}\ \big(x > 0 \to \exists n \in \mathbb{N}\ (nx > y)\big).$$

Remark (Negated quantified statement).

$$\exists x, y \in \mathbb{R}\ \big(x > 0 \land \forall n \in \mathbb{N}\ (nx \le y)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ArchimedeanReciprocal}(\mathbb{R}) \iff \exists x, y \in \mathbb{R}\ \big(x > 0 \land \forall n \in \mathbb{N}\ (nx \le y)\big).$$

Remark (Failure modes). The corollary fails when a positive real step size has all of its natural multiples bounded above by some real threshold.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ArchimedeanReciprocal}(\mathbb{R}) = \underbrace{\exists x \in \mathbb{R}\ (x > 0)}_{\text{positive step size}} \land \underbrace{\exists y \in \mathbb{R}\ \forall n \in \mathbb{N}\ (nx \le y)}_{\text{all natural multiples stay bounded}}.$$

Remark (Interpretation). Any fixed positive real step eventually passes any real threshold when repeated enough times. The statement is the Archimedean Property applied after rescaling by the positive number x: growth by units becomes growth by steps of size x.

Remark (Dependencies).
Theorem 2.0.4.

Density

Density

Definition (Dense Subset)

Definition 2.0.6 (Dense Subset). Let $D \subseteq S \subseteq \mathbb{R}$. The set D is *dense in* S if and only if for every $x \in S$ and every $\varepsilon > 0$ there exists $d \in D$ such that $|x - d| < \varepsilon$.

Remark (Standard quantified statement).

$$\text{Let } D \subseteq S \subseteq \mathbb{R}.$$
$$D \text{ is dense in } S \iff \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{DenseSubset}(D, S) := \forall x \in S\ \forall \varepsilon > 0\ \exists d \in D\ \big(|x - d| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } D \subseteq S \subseteq \mathbb{R}.\\ &D \text{ is not dense in } S \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x-d| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DenseSubset}(D,S) \iff \exists x \in S\ \exists \varepsilon_0 > 0\ \forall d \in D\ \big(|x-d| \geq \varepsilon_0\big).$$

Remark (Failure modes). Density fails when some point of S has a positive-radius neighborhood that contains no point of D.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DenseSubset}(D,S) = \underbrace{\exists x \in S}_{\text{point not approximated}}\ \underbrace{\exists \varepsilon_0 > 0}_{\text{positive exclusion radius}}\ \underbrace{\forall d \in D\ \big(|x-d| \geq \varepsilon_0\big)}_{\text{no point of } D \text{ enters the neighborhood}}.$$

Remark (Interpretation). Density is an approximation property inside S. It does not say that every point of S already belongs to D; it says that points of D occur arbitrarily close to every point of S. Thus D can be a proper subset of S while still leaving no positive-width gap inside S.

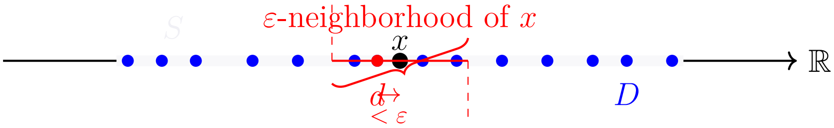


Figure ??. Illustration of a Dense Subset $D \subseteq S$. Given any $x \in S$ (black dot) and any $\varepsilon > 0$ (red interval width), there exists $d \in D$ (red-filled blue dot) such that $|x-d| < \varepsilon$.

Definition (Order Dense Subset)

Definition 2.0.7 (Order Dense Subset). Let X be a linearly ordered set and let $A \subseteq X$. The subset A is *order dense in* X if and only if for every $a, b \in X$ with $a < b$ there exists $c \in A$ such that $a < c < b$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is order dense in } X \iff \forall a, b \in X\ \big(a < b \to \exists c \in A\ (a < c < b)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{OrderDenseSubset}(A,X) :\Longleftrightarrow \forall a, b \in X\ \big(a < b \to \exists c \in A\ (a < c < b)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } X \text{ be a linearly ordered set and } A \subseteq X.\\ &A \text{ is not order dense in } X \iff \exists a, b \in X\ \big(a < b \wedge \forall c \in A\ (c \leq a \vee b \leq c)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OrderDenseSubset}(A, X) \iff \exists a, b \in X\ \big(a < b \wedge \forall c \in A\ (c \leq a \vee b \leq c)\big).$$

Remark (Failure modes). Order density fails when there is a nonempty open interval of X that contains no point of A.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(A, X) = \underbrace{\exists a, b \in X\ (a < b)}_{\text{nonempty interval}} \wedge \underbrace{\forall c \in A\ (c \leq a \vee b \leq c)}_{\text{no point of } A \text{ lies between } a \text{ and } b}.$$

Remark (Interpretation). Order density is an interval-penetration property. It ignores metric distance and asks only whether every ordered gap contains a point of the subset. A subset can therefore be order dense in X precisely when no nonempty open interval of X is missed entirely by that subset.

Remark (Dependencies).

- Total Order
- Subset

Theorem (Density of the Rational Numbers in $\mathbb{R}$)

Theorem 2.0.8 (Density of the Rational Numbers in $\mathbb{R}$). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a < b \to \exists q \in \mathbb{Q}\ (a < q < b)\big).$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R}\ \big(a < b \to \exists q \in \mathbb{Q}\ (a < q < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R}\ \big(a < b \wedge \forall q \in \mathbb{Q}\ (q \leq a \vee b \leq q)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R}\ \big(a < b \wedge \forall q \in \mathbb{Q}\ (q \leq a \vee b \leq q)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no rational number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R}\ (a < b)}_{\text{nonempty real interval}} \wedge \underbrace{\forall q \in \mathbb{Q}\ (q \leq a \vee b \leq q)}_{\text{no rational lies inside}}.$$

Remark (Interpretation). Rational numbers penetrate every real interval. The theorem does not say that every real number is rational; it says that rational numbers are never separated from a real interval by a positive order gap. This is the interval form of rational density in the real line.

Remark (Dependencies).
Definition 2.0.7, Theorem 2.0.4.

Theorem (Density of the Irrational Numbers in $\mathbb{R}$)

Theorem 2.0.9 (Density of the Irrational Numbers in $\mathbb{R}$). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q}\ (a < s < b)\big).$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \forall a, b \in \mathbb{R}\ \big(a < b \to \exists s \in \mathbb{R} \setminus \mathbb{Q}\ (a < s < b)\big).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R}\ \big(a < b \wedge \forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \vee b \leq s)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) \iff \exists a, b \in \mathbb{R}\ \big(a < b \wedge \forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \vee b \leq s)\big).$$

Remark (Failure modes). The theorem would fail precisely if some nonempty real interval contained no irrational number.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}) = \underbrace{\exists a, b \in \mathbb{R}\ (a < b)}_{\texttt{nonempty real interval}} \wedge \underbrace{\forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \vee b \leq s)}_{\texttt{no irrational lies inside}}.$$

Remark (Interpretation). Irrational numbers also penetrate every real interval. One way to see the result is to insert a rational point into a translated and rescaled interval, then shift it by an irrational amount. The outcome is that the irrational numbers, like the rational numbers, leave no interval gap in $\mathbb{R}$.

Remark (Dependencies).
Definition 2.0.7, Theorem 2.0.8.

Corollary (Rational and Irrational Points in Every Open Interval)

Corollary 2.0.10 (Rational and Irrational Points in Every Open Interval). *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R} \left[a < b \to \big(\exists q \in \mathbb{Q}\ (a < q < b) \wedge \exists s \in \mathbb{R} \setminus \mathbb{Q}\ (a < s < b)\big)\right].$$

Remark (Predicate reading).

$$\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \wedge \mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R}).$$

Remark (Negated quantified statement).

$$\exists a, b \in \mathbb{R} \left[a < b \wedge \big(\forall q \in \mathbb{Q}\ (q \leq a \vee b \leq q) \vee \forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \vee b \leq s)\big)\right].$$

Remark (Negation predicate reading).

$$\neg\left[\mathrm{OrderDenseSubset}(\mathbb{Q}, \mathbb{R}) \wedge \mathrm{OrderDenseSubset}(\mathbb{R} \setminus \mathbb{Q}, \mathbb{R})\right].$$

Remark (Failure modes). The corollary fails if a nonempty real interval misses all rational points or misses all irrational points.

Remark (Failure mode decomposition).

$$\exists a, b \in \mathbb{R}\ (a < b)\ \wedge\ \Big[\underbrace{\forall q \in \mathbb{Q}\ (q \leq a \vee b \leq q)}_{\texttt{no rational point}}\ \vee\ \underbrace{\forall s \in \mathbb{R} \setminus \mathbb{Q}\ (s \leq a \vee b \leq s)}_{\texttt{no irrational point}}\Big].$$

Remark (Interpretation). The corollary combines the two density theorems. Every real interval, no matter how small, contains representatives of both arithmetic types. Thus neither the rational numbers nor the irrational numbers occupy isolated regions of the real line.

Remark (Dependencies).
Theorem 2.0.8, Theorem 2.0.9.

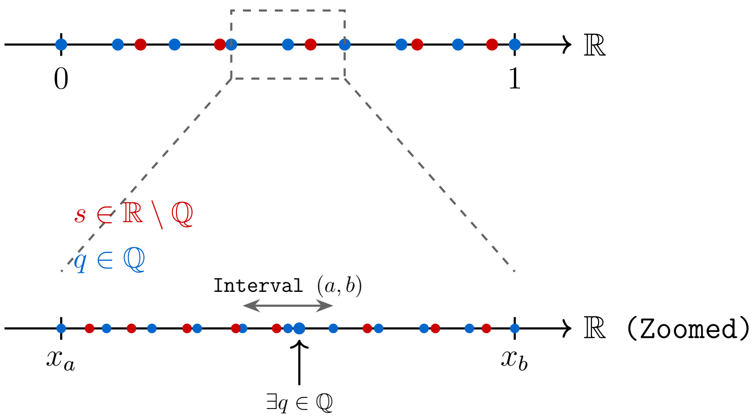


Figure 2.2: Visualization of the Density of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ in $\mathbb{R}$. Zooming into any interval reveals that both sets are always present.

Remark (Zoom invariance and structural consequences).
Density in $\mathbb{R}$ has a scale-invariant consequence: no matter how small a nonempty open interval becomes, it still contains both rational and

irrational points. The interleaving of $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ is therefore not a large-scale phenomenon. It persists at every interval scale.

This has three important consequences. First, the continuum does not become locally discrete under magnification. For every nonempty open interval (a, b), both

$$\mathbb{Q} \cap (a, b) \qquad \text{and} \qquad (\mathbb{R} \setminus \mathbb{Q}) \cap (a, b)$$

remain nonempty, and in fact infinite. Density is therefore stable under repeated restriction to smaller intervals.

Second, this behavior sharply distinguishes the real line from discrete numerical models. Floating-point systems form a discrete set of representable values, so repeated magnification eventually reaches a scale at which adjacent representable numbers leave no further point strictly between them. At that stage, interval penetration fails in the discrete model even though it continues to hold in $\mathbb{R}$.

Third, the Archimedean Property supplies the scale mechanism behind this behavior. Given any positive scale ε, there exists $n \in \mathbb{N}$ with

$$\frac{1}{n} < \varepsilon.$$

This makes it possible to construct rational approximations at arbitrarily small interval scales, which is one of the basic engines behind density arguments.

The resulting contrast is structural. Smooth mathematical models are built in a setting where interval penetration persists at every scale, whereas finite computational models eventually encounter a smallest representable separation. The transition from continuum behavior to discrete behavior occurs exactly where nonempty intervals in the numerical model cease to contain further representable points.

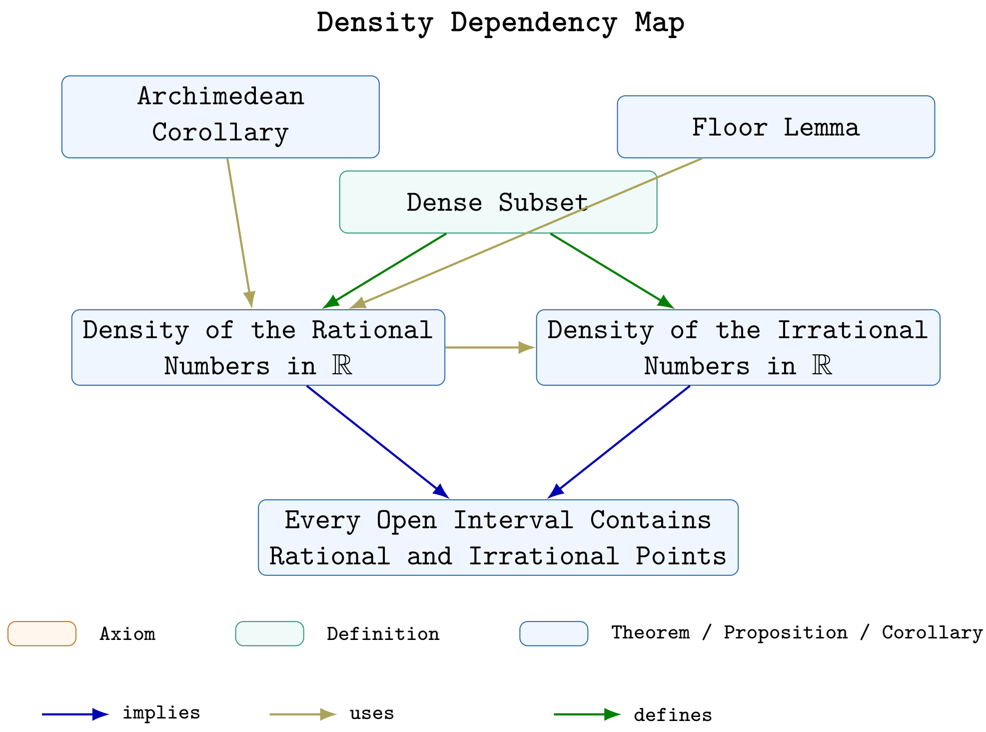


Figure 2.3: Dependency map for the density subchapter. The Floor Lemma and the Archimedean Corollary support the proof of the density of the rational numbers in $\mathbb{R}$. The irrational density theorem is then derived from the rational density result, and the final corollary combines both density theorems into a single interval statement. The definition records the local vocabulary of the subchapter.

Completeness Summary

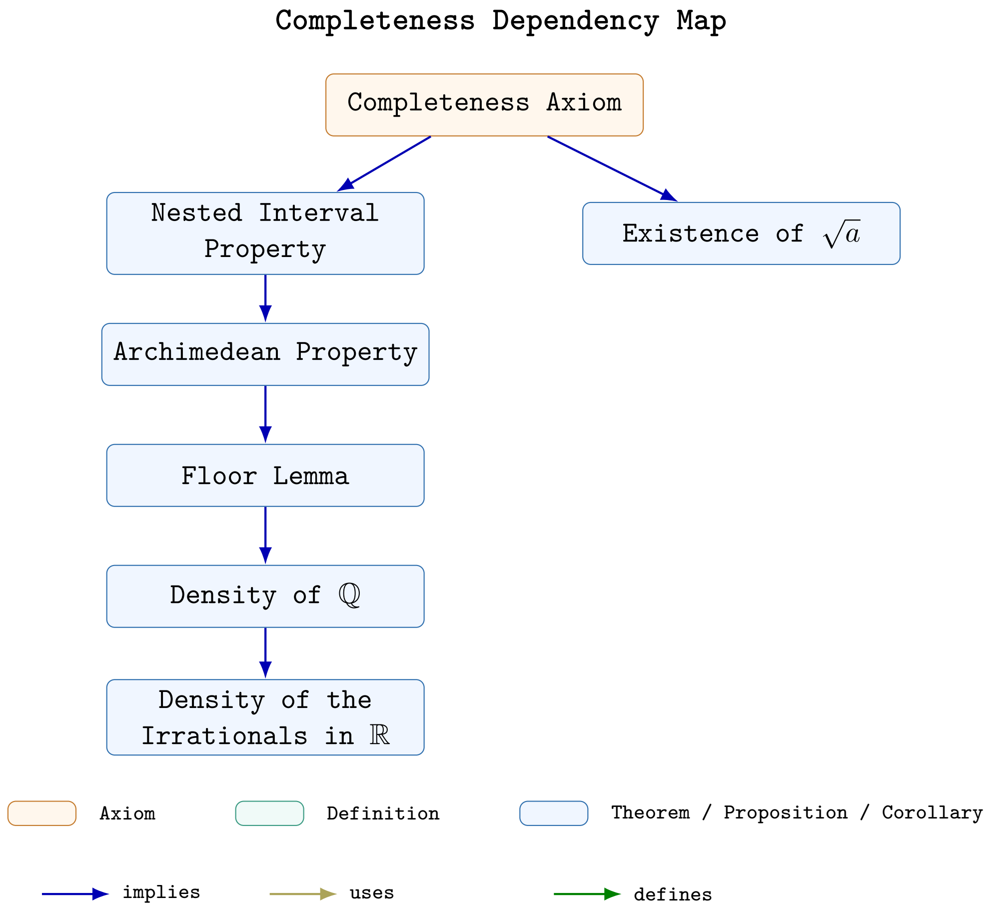


Figure 2.4: Dependency map for the completeness subchapter. The Completeness Axiom yields the Nested Interval Property and the existence of $\sqrt{a}$; the Nested Interval Property then supports the Archimedean Property, the Floor Lemma, and the density results for $\mathbb{Q}$ and for the irrational numbers.

Remark (Connections forward). Completeness appears in every major limit theorem in the notes. The Monotone Convergence Theorem (convergence chapter) constructs the limit of a bounded monotone sequence as $\sup\{a_n : n \in \mathbb{N}\}$; existence of this supremum requires completeness. The Bolzano-Weierstrass Theorem applies the Nested Interval Property directly to extract a convergent subsequence from any bounded sequence. The Cauchy Criterion identifies convergent sequences purely in terms of their own spacing, without naming the limit in advance; its proof constructs the limit as a supremum and again requires completeness. In metric spaces (Distance chapter), completeness in the sense of Cauchy sequences is the natural generalization of the same idea to abstract settings. Every ε-δ proof that locates a limit or extremum is drawing on this chapter's infrastructure.

2.1 Bounds and Extremal Values

Remark (Predicate vs. Existence: Structural Template). Each bound-type concept (maximum, minimum, upper bound, supremum, etc.) is defined by a predicate relative to an ambient ordered set S. Given $A \subseteq S$, let $P_A(x)$ denote such a predicate.

Predicate form. The statement "x is a P-object of A" means that $P_A(x)$ holds for a specific identified element $x \in S$. To prove this, one exhibits the element and verifies each condition in $P_A(x)$.

Existence form. The statement "A has a P-object" means that

$$\exists x \in S\ P_A(x).$$

Once established, one may fix such an x and reason with the fact that $P_A(x)$ holds.

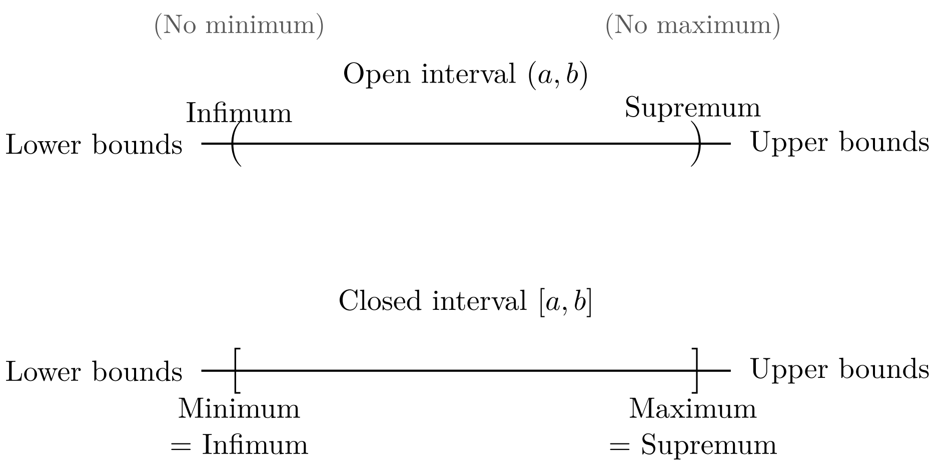
Notational glosses.

- "x is a P-object of A" abbreviates $P_A(x)$.
- "A has a P-object" abbreviates $\exists x \in S\ P_A(x)$.

Proof strategy summary.

Goal	**Form**	**Strategy**
$P_A(x)$	Predicate	Exhibit x and verify $P_A(x)$ directly.
$\exists x \in S\ P_A(x)$	Existence	Construct or identify a candidate and verify $P_A(x)$.
Hypothesis: $\exists x \in S\ P_A(x)$	--	Fix such an x and use the fact that $P_A(x)$ holds.

Note. The hypothesis row is not a proof goal. It records the standard move of fixing an existential witness once existence has been established.



Concept	**Predicate $P_A(x)$**	**Predicate Form**	**Existence Form**
Bound	$[\forall a \in A\ (a \leq x)] \vee [\forall a \in A\ (x \leq a)]$	x is a bound for A	A has a bound
Upper bound	$\forall a \in A\ (a \leq x)$	x is an upper bound of A	A is bounded above
Lower bound	$\forall a \in A\ (x \leq a)$	x is a lower bound of A	A is bounded below
Bounded	$\mathrm{BoundedAbove}(A) \wedge \mathrm{BoundedBelow}(A)$	–	A is bounded
Maximum	$(x \in A) \wedge \mathrm{UpperBound}(x, A)$	x is a maximum of A	A has a maximum
Minimum	$(x \in A) \wedge \mathrm{LowerBound}(x, A)$	x is a minimum of A	A has a minimum
Supremum	$\mathrm{UpperBound}(x, A) \wedge \forall u \in S\ (\mathrm{UpperBound}(u, A) \to x \leq u)$	x is the supremum of A	A has a supremum
Infimum	$\mathrm{LowerBound}(x, A) \wedge \forall \ell \in S\ (\mathrm{LowerBound}(\ell, A) \to \ell \leq x)$	x is the infimum of A	A has an infimum

Remark (Reading the summary table). The predicate column gives the condition that must hold for a specific identified element $x \in S$. The existence column records the corresponding existential statement. The rows are grouped by logical complexity: bounds use a single universal condition, extrema add membership, and suprema and infima add the least or greatest comparison against all competing bounds.

Bounds and Extremal Elements

Definition (Bound)

Definition 2.1.1 (Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $b \in S$. The element b is a ***bound*** for A if and only if b is a uniform one-sided comparison point for all elements of A:

$$\mathrm{Bound}(b, A) \iff [\forall x \in A\ (x \leq b)] \vee [\forall x \in A\ (b \leq x)].$$

Remark (Orientation). A bound always has a direction. The first alternative bounds A from above, and the second alternative bounds A from below. Thus "bound" is the common template, while upper and lower bounds are the two oriented cases.

Remark (Dependencies).

- Ordered Set
- Subset

Definition (Upper Bound)

Definition 2.1.2 (Upper Bound). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $u \in S$. The element u is an *upper bound* of A if and only if u is a bound for A from above, meaning $x \leq u$ for every $x \in A$.

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } u \in S.\\ u \text{ is an upper bound of } A \iff \forall x \in A\ (x \leq u).\end{gathered}$$

Remark (Definition predicate reading).

$$\operatorname{UpperBound}(u, A) := \forall x \in A\ (x \leq u).$$

Remark (Negated quantified statement).

$$\begin{gathered}\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } u \in S.\\ u \text{ is not an upper bound of } A \iff \exists x \in A\ (u < x).\end{gathered}$$

Remark (Negation predicate reading).

$$\neg \operatorname{UpperBound}(u, A) \iff \exists x \in A\ (u < x).$$

Remark (Failure modes). The attempt to declare u an upper bound collapses precisely when some element of A lies strictly above u, providing a concrete counterexample to the domination condition.

Remark (Failure mode decomposition).

$$\neg \operatorname{UpperBound}(u, A) = \underbrace{\exists x \in A\ (u < x)}_{\text{witness in } A \text{ exceeding } u}.$$

Remark (Interpretation). An upper bound caps a subset of an ordered structure from above, so every element of the subset fits beneath the chosen guard u. The negation highlights the standard obstruction: locating even one element of the set beyond u demolishes the claim. Geometrically, u places a horizontal ceiling over A on the number line, and the only failure mode consists of punching through that ceiling with a higher point of A.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound

Definition (Bounded Above)

Definition 2.1.3 (Bounded Above). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded above* in S if and only if there exists $u \in S$ such that u is an upper bound of A.

Remark (Existence reading).

$$A \text{ is bounded above in } S \iff \exists u \in S \ \mathrm{UpperBound}(u, A).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound

Definition (Lower Bound)

Definition 2.1.4 (Lower Bound). Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $l \in S$. The element l is a *lower bound* of A if and only if l is a bound for A from below, meaning every $x \in A$ satisfies $l \leq x$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } l \in S.\\ &l \text{ is a lower bound of } A \iff \forall x \in A \ (l \leq x).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{LowerBound}(l, A) := \forall x \in A \ (l \leq x).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } l \in S.\\ &\neg\big[l \text{ is a lower bound of } A\big] \iff \exists x \in A \ (x < l).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{LowerBound}(l, A) \iff \exists x \in A \ (x < l).$$

Remark (Failure modes). Failure occurs precisely when some element of A lies strictly below l, so l cannot bound A from below.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{LowerBound}(l, A) = \underbrace{\exists x \in A \text{ with } x < l}_{\text{witness below } l}.$$

Remark (Interpretation). A lower bound clamps the set A from below: starting at l and moving upward along the order one encounters all of A, but no point of A sits beneath l. Failure means that at least one element of A pokes under l, revealing that l was chosen too high to constrain the set from below.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound

Definition (Bounded Below)

Definition 2.1.5 (Bounded Below). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded below* in S if and only if there exists $l \in S$ such that l is a lower bound of A.

Remark (Existence reading).

$$A \text{ is bounded below in } S \iff \exists l \in S \ \mathrm{LowerBound}(l, A).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Lower Bound

Definition (Bounded)

Definition 2.1.6 (Bounded). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. The set A is *bounded* in S if and only if A is bounded above in S and bounded below in S.

Remark (Two-sided existence reading).

$$A \text{ is bounded in } S \iff \big(\exists u \in S \ \mathrm{UpperBound}(u, A)\big) \wedge \big(\exists l \in S \ \mathrm{LowerBound}(l, A)\big).$$

Remark (Dependencies).

- Ordered Set
- Subset
- Bounded Above
- Bounded Below

Definition (Minimum)

Definition 2.1.7 (Minimum). Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $m \in S$. The element m is the *minimum* of A if and only if $m \in A$ and m is a lower bound of A.

Remark (Standard quantified statement).

$$\text{Let } (S, \leq) \text{ be an ordered set, } A \subseteq S, \text{ and } m \in S.$$
$$m \text{ is the minimum of } A \iff \big(m \in A \wedge \forall x \in A \ (m \leq x)\big).$$

Remark (Definition predicate reading).

$$\text{Minimum}(m, A) :\equiv (m \in A) \wedge \forall x \in A\ (m \leq x).$$

$$\text{Minimum}(m, A) \iff (m \in A) \wedge \text{LowerBound}(m, A).$$

Remark (Negated quantified statement).

Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $m \in S$.
$\neg\big(m$ is the minimum of $A\big) \iff \big(m \notin A \vee \exists x \in A\ (m \not\leq x)\big)$.

Remark (Negation predicate reading).

$$\neg\,\text{Minimum}(m, A) \iff \big(m \notin A \vee \exists x \in A\ (m \not\leq x)\big).$$

Remark (Failure modes). Failure occurs either because m is not a member of A, or because some element of A lies strictly below m, violating the global lower-bound condition.

Remark (Failure mode decomposition).

$$\neg\,\text{Minimum}(m, A) = \underbrace{m \notin A}_{\text{not even a candidate}} \vee \underbrace{\exists x \in A\ (m \not\leq x)}_{\text{candidate but not below all elements}}.$$

Remark (Interpretation). A minimum point both belongs to the set and sits at its absolute bottom: every other element rests above it in the ambient order. The definition splits into the membership requirement, ensuring the bound is realized, and the universal inequality that promotes the element from a mere lower bound to a genuine extremal. Failure arises if the candidate lies outside the set or if another element dips below it; geometrically, the set either has no realized floor at that point or its supposed floor is undercut elsewhere.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound
- Lower Bound

Definition (Maximum)

Definition 2.1.8 (Maximum). Let $(S, \leq)$ be an ordered set and let $A \subseteq S$. An element $M \in S$ is the maximum of A if and only if $M \in A$ and M is an upper bound of A.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set and $A \subseteq S$.
M is the maximum of $A \iff (M \in A) \wedge \forall x \in A\ (x \leq M)$.

Remark (Definition predicate reading).

$$\operatorname{Maximum}(M, A) :\equiv (M \in A) \wedge \forall x \in A\ (x \leq M).$$

$$\operatorname{Maximum}(M, A) \iff (M \in A) \wedge \operatorname{UpperBound}(M, A).$$

Remark (Negated quantified statement).

Let $(S, \leq)$ be an ordered set and $A \subseteq S$.
$\neg\big(M$ is the maximum of $A\big) \iff (M \notin A) \vee \exists x \in A\ (M < x)$.

Remark (Negation predicate reading).

$$\neg \operatorname{Maximum}(M, A) \iff (M \notin A) \vee \exists x \in A\ (M < x).$$

Remark (Failure modes). The property fails either because the candidate value is missing from A or because some element of A sits strictly above it, violating the domination requirement.

Remark (Failure mode decomposition).

$$\neg \operatorname{Maximum}(M, A) = \underbrace{(M \notin A)}_{\text{membership failure}} \vee \underbrace{\big(\exists x \in A\ (M < x)\big)}_{\text{domination failure}}.$$

Remark (Interpretation). A maximum is an element of the set that simultaneously belongs to the set and dominates it under the ambient order. The domination clause guarantees that no point of A lies above the maximum, so the element captures the extreme right edge of the subset. Failure occurs either when the supposed extremal is not actually part of the set or when another member protrudes farther to the right, showing the candidate was not truly maximal. In geometric terms, the maximum is the endpoint where the subset comes to rest when viewed along the ordering axis.

Remark (Dependencies).

- Ordered Set
- Subset
- Bound
- Upper Bound

Definition (Supremum)

Definition 2.1.9 (Supremum). Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $s \in S$. The element s is the supremum of A if and only if s is an upper bound of A and every upper bound u of A satisfies $s \leq u$.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $s \in S$.
s is the supremum of $A \iff \Big[(\forall x \in A)(x \leq s) \wedge \forall u \in S\ \big((\forall x \in A)(x \leq u) \rightarrow s \leq u\big)\Big]$.

Remark (Definition predicate reading).

$$\mathrm{Supremum}(s, A) :\equiv \mathrm{UpperBound}(s, A) \land \forall u \in S \left(\mathrm{UpperBound}(u, A) \to s \le u\right).$$

Remark (Negated quantified statement).

Let $(S, \le)$ be an ordered set, $A \subseteq S$, and $s \in S$.
$\neg(s$ is the supremum of $A) \iff \left[\exists x \in A\,(s < x) \lor \exists u \in S\left((\forall x \in A)(x \le u) \land u < s\right)\right].$

Remark (Negation predicate reading).

$$\neg\,\mathrm{Supremum}(s, A) \iff \left(\exists x \in A\,(s < x)\right) \lor \left(\exists u \in S\,(\mathrm{UpperBound}(u, A) \land u < s)\right).$$

Remark (Failure modes). The definition fails in exactly two ways: either s is not an upper bound of A, or there exists a strictly smaller upper bound u of A, showing that s is not the least among all upper bounds.

Remark (Failure mode decomposition).

Let $(S, \le)$ be an ordered set, $A \subseteq S$, and $s \in S$.

$$\neg\,\mathrm{Supremum}(s, A) \iff \underbrace{\exists x \in A\,(s < x)}_{\text{fails to be an upper bound}} \lor \underbrace{\exists u \in S\left(\mathrm{UpperBound}(u, A) \land u < s\right)}_{\text{upper bound lies below } s}.$$

Remark (Interpretation). The supremum of A is the smallest ambient element that dominates the whole set. It exists precisely when s bounds A from above and no stricter bound sits below it. Failure occurs either because A pokes above s or because another upper bound nestles strictly beneath s, contradicting minimality. Geometrically, the supremum is the leftmost point in the order that caps the set, so every other cap must lie to its right.

Remark (Dependencies).

- Ordered Set
- Subset
- Upper Bound

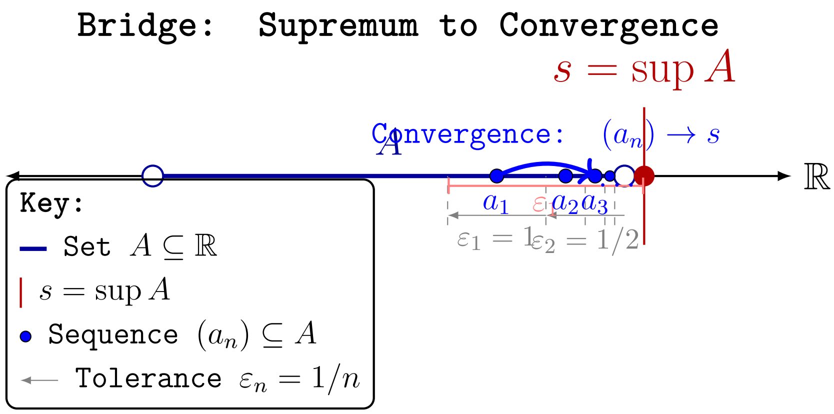


Figure 2.5: Illustrating the constructive connection between the static existence of a supremum (s) and the dynamic process of convergence. The ε-characterization acts as a sequence generator. By selecting decreasing tolerances $\varepsilon_n = 1/n$, we are guaranteed to find corresponding elements $a_n \in A$ that satisfy $s - \frac{1}{n} < a_n \leq s$. The sequence (a_n), contained strictly within A, is forced by the 'squeeze' of the tolerances to converge to s as $n \to \infty$. This construction provides the required sequence $(a_n) \subseteq A$ such that $\lim_{n\to\infty} a_n = \sup A$.

Definition (Infimum)

Definition 2.1.10 (Infimum)**.** Let $(S, \leq)$ be an ordered set, let $A \subseteq S$, and let $i \in S$. The element i is the infimum of A if and only if i is a lower bound of A and every lower bound l of A satisfies $l \leq i$.

Remark (Standard quantified statement).

Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $i \in S$.
i is the infimum of A

$$\iff \Big[\forall x \in A\ (i \leq x)\Big] \wedge \forall l \in S\ \Big([\forall x \in A\ (l \leq x)] \Rightarrow l \leq i\Big).$$

Remark (Definition predicate reading).

$$\text{Infimum}(i, A) :\!= \text{LowerBound}(i, A) \wedge \forall l \in S\ \big(\text{LowerBound}(l, A) \Rightarrow l \leq i\big).$$

Remark (Negated quantified statement).

Let $(S, \leq)$ be an ordered set, $A \subseteq S$, and $i \in S$.
i is not the infimum of A

$$\iff \Big[\exists x \in A\ (i \nleq x)\Big] \vee \exists l \in S\ \Big([\forall x \in A\ (l \leq x)] \wedge l \nleq i\Big).$$

Remark (Negation predicate reading).

$$\neg\,\text{Infimum}(i, A) \iff \neg\,\text{LowerBound}(i, A) \vee \exists l \in S\ (\text{LowerBound}(l, A) \wedge l \nleq i).$$

Remark (Failure modes). The definition fails either because i is not a lower bound of A, or because some other lower bound l is not beneath i, preventing i from being the greatest among all lower bounds.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{Infimum}(i, A) = \underbrace{\neg\,\mathrm{LowerBound}(i, A)}_{\texttt{does not bound } A \texttt{ below}} \vee \underbrace{\exists l \in S\ (\mathrm{LowerBound}(l, A) \wedge l \not\leq i)}_{\texttt{competing lower bound not below } i}.$$

Remark (Interpretation). The infimum of A is the greatest lower bound: it still lies below every element of A, yet no other lower bound stands above it. If i fails to be an infimum, either it already misses some element of A or there exists a better lower bound lying strictly higher in the order. Geometrically, in a number line picture, the infimum is the highest point lying to the left of A that is still a universal left barrier for the set.

Remark (Dependencies).

- Lower Bound
- Least Upper Bound Property

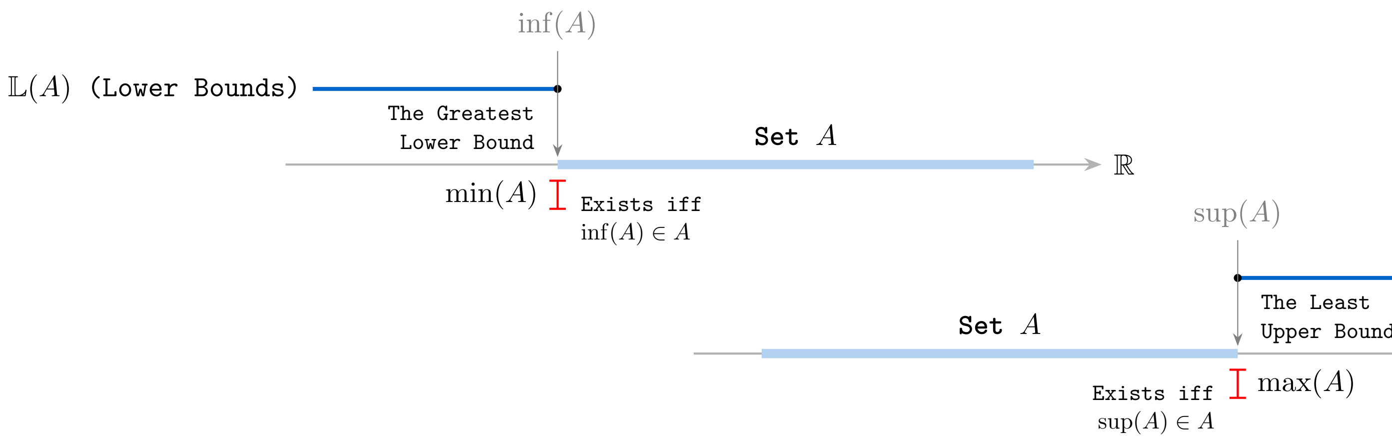


Figure 2.6: Separated geometric relationships for lower and upper bounds.

Contrast: Maximum vs. Supremum.

Property	**Maximum** $m = \max A$	**Supremum** $s = \sup A$
Must belong to A?	Yes. $m \in A$ is required.	No. s may or may not belong to A.
Must be an upper bound?	Yes. $\forall x \in A\ (x \leq m)$.	Yes. $\forall x \in A\ (x \leq s)$.
Must be the least upper bound?	Not stated separately, but membership together with the upper-bound condition forces it.	Yes. This is required explicitly.
Existence guaranteed?	Only when the set attains its supremum.	For nonempty sets bounded above in $\mathbb{R}$, yes, by the Completeness Axiom.

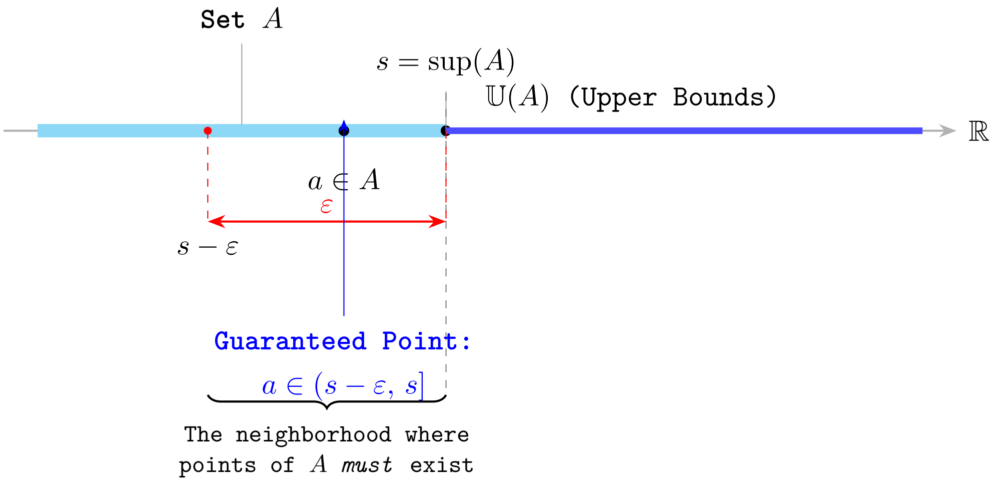


Figure 2.7: Refined visualization of the ε-characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

Remark (Maximum versus supremum). A maximum is an element of the set, whereas a supremum is an extremal upper bound that may lie outside the set. The two coincide exactly when the supremum belongs to the set.

Theorem (Least Upper Bound Property Implies Existence of Suprema)

Theorem 2.1.11 (Least Upper Bound Property Implies Existence of Suprema). *Let S be an ordered set with the least upper bound property. If $A \subseteq S$ is nonempty and bounded above in S, then A has a supremum in S.*

Go to proof.

Remark (Standard quantified statement).

Let S be an ordered set with the least upper bound property.
$\forall A \subseteq S\ \big[A \neq \varnothing \wedge \exists u \in S\ \forall x \in A\ (x \leq u) \implies \exists s \in S\ (s = \sup_S A)\big].$

Remark (Predicate reading).

$$\mathrm{LeastUpperBoundProperty}(S) \implies \forall A \subseteq S\ \big[A \neq \varnothing \wedge \mathrm{BoundedAbove}_S(A) \implies \exists s \in S\ \mathrm{Supremum}_S(s, A$$

Remark (Negated quantified statement).

$\mathrm{LeastUpperBoundProperty}(S)$ and $\exists A \subseteq S : A \neq \varnothing,$ A bounded above in S, and $\sup_S A$ does

is impossible.

Remark (Negation predicate reading).

$$\mathrm{LeastUpperBoundProperty}(S) \wedge \exists A \subseteq S\ \big[A \neq \varnothing \wedge \mathrm{BoundedAbove}_S(A) \wedge \neg \exists s \in S\ \mathrm{Supremum}_S(s, A)\big]$$

is a contradictory configuration.

Remark (Failure modes). The theorem could fail only if an admissible subset were nonempty and bounded above while still lacking a least upper bound in S. That is precisely the situation ruled out by the least upper bound property.

Remark (Failure mode decomposition).

$$\mathrm{LeastUpperBoundProperty}(S) = \underbrace{\forall A \subseteq S \left[A \neq \varnothing \wedge \mathrm{BoundedAbove}_S(A) \implies \exists s \in S \ \mathrm{Supremum}_S(s, A)\right]}_{\text{every admissible subset has a least upper bound in } S}$$

Remark (Interpretation). This theorem is the operational use of the least upper bound property. Once the ambient ordered set has that property, every nonempty bounded-above subset receives the least ceiling that the definition promises.

Remark (Dependencies).
Least Upper Bound Property; Upper Bound; Supremum.

Proposition (Upper Bound Property of Supremum)

Proposition 2.1.12 (Upper Bound Property of Supremum). *Let $A \subseteq \mathbb{R}$ and let $s = \sup A$. Then $x \leq s$ for every $x \in A$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s = \sup A.$$
$$\forall x \in A\ (x \leq s).$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \implies \mathrm{UpperBound}(s, A).$$

Remark (Negated quantified statement).

$$s = \sup A \quad \text{and} \quad \exists x \in A\ (s < x)$$

is impossible.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \neg\, \mathrm{UpperBound}(s, A)$$

is a contradictory configuration.

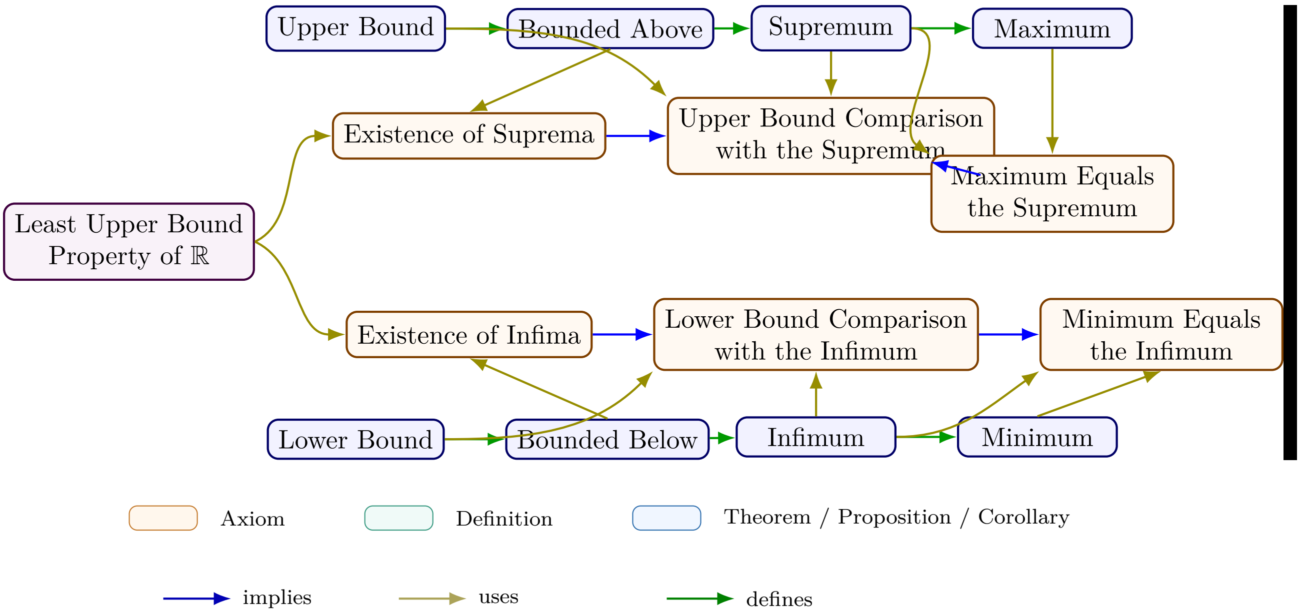
Remark (Failure modes). The proposition could fail only if a point of A sat strictly above its claimed supremum. Such a point would show that the claimed supremum is not even an upper bound.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{UpperBound}(s, A) \iff \underbrace{\exists x \in A\ (s < x)}_{\text{element above the claimed supremum}}.$$

Remark (Interpretation). The supremum has two jobs: it must bound the set from above, and it must be least among all such bounds. This proposition isolates the first job.

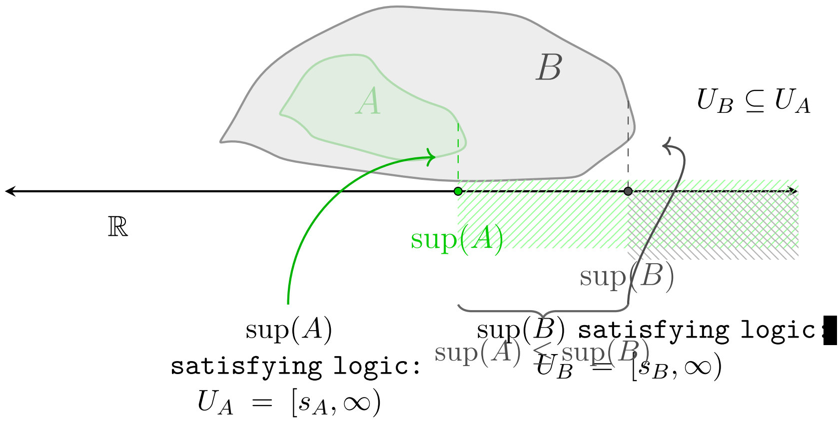
Remark (Dependencies).
Upper Bound; Supremum.



Consequences

Remark (Logical structure). Bounds, extrema, and least/greatest bounds form a hierarchy of increasing logical complexity. Upper and lower bounds are defined by a universal quantifier alone. Maximum and minimum add a membership condition: the extremum must belong to the set. Supremum and infimum relax membership but add a minimality (or maximality) condition: the bound must be the *least* (or *greatest*) among all bounds. Maximum implies supremum -- when $\max A$ exists, it equals $\sup A$ (Proposition 2.1.17 and the proof at maximum-implies-supremum) -- but the converse fails: $(0,1)$ has $\sup = 1$ but no maximum.

Remark (Connections forward). The ε-characterization (Proposition ??) is the workhorse tool of the bounding chapter. Every proof in the Bound Arithmetic section (§??) invokes it to verify that a candidate value is the least upper bound, not merely an upper bound. The Monotone Convergence Theorem (convergence chapter) uses the same mechanism: the limit of a bounded monotone sequence is identified as $\sup\{a_n\}$, and the ε-characterization produces the required N.



Caption (R7.5, R13.8): Figure ??: The Monotonicity of Supremum Under Set Inclusion. This visualization illustrates the arithmetic consequence of set inclusion, $A \subseteq B \subset \mathbb{R}$. The topology of the sets is represented by nested blobs: A (green) is contained within B (gray). Vertically dashed projections link the supremal boundary of each set down to the shared real axis, defining the unique points $s_A = \sup(A)$ and $s_B = \sup(B)$. The figure demonstrates that because the territory of A is restricted by B, its maximum guard point s_A cannot exceed s_B. Finally, the shaded regions to the right illustrate the territories of upper bounds (U_A and U_B), clearly showing that U_B is a proper subset of U_A. Any value guarding the larger set B necessarily satisfies the guard condition for the smaller set A.

Epsilon Characterization

ε-Characterization of Supremum

Definition (Epsilon Characterization of Supremum)

Definition 2.1.13 (Epsilon Characterization of Supremum). Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. The element s satisfies the *epsilon characterization of the supremum* of A if and only if s is an upper bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $s - \varepsilon < a$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s \text{ satisfies the epsilon characterization of } \sup A\\ &\iff \Big[\forall x \in A\ (x \leq s)\Big] \wedge \Big[\forall \varepsilon > 0\ \exists a \in A\ (s - \varepsilon < a)\Big].\end{aligned}$$

Remark (Definition predicate reading).

$$\text{EpsilonSupremum}(s, A) :\equiv \text{UpperBound}(s, A) \wedge \forall \varepsilon > 0\ \exists a \in A\ (s - \varepsilon < a).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.\\ &s \text{ does not satisfy the epsilon characterization of } \sup A\\ &\iff \Big[\exists x \in A\ (s < x)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (a \leq s - \varepsilon_0)\Big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{EpsilonSupremum}(s, A) \iff \Big[\exists x \in A\ (s < x)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (a \leq s - \varepsilon_0)\Big].$$

Remark (Failure modes). The characterization fails either because s is not an upper bound of A, or because A remains at least some positive distance below s.

Remark (Failure mode decomposition).

$$\neg\,\text{EpsilonSupremum}(s, A) = \underbrace{\exists x \in A\ (s < x)}_{\texttt{upper-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall a \in A\ (a \leq s - \varepsilon_0)}_{\texttt{positive gap below } s}.$$

Remark (Interpretation). This condition says that s caps the set from above while points of A approach s from below at every positive scale. The upper-bound clause prevents A from crossing above s, and the epsilon clause prevents s from floating strictly above the set. Geometrically, s is a ceiling that the set presses against arbitrarily closely.

Remark (Dependencies).
Definition 2.1.2.

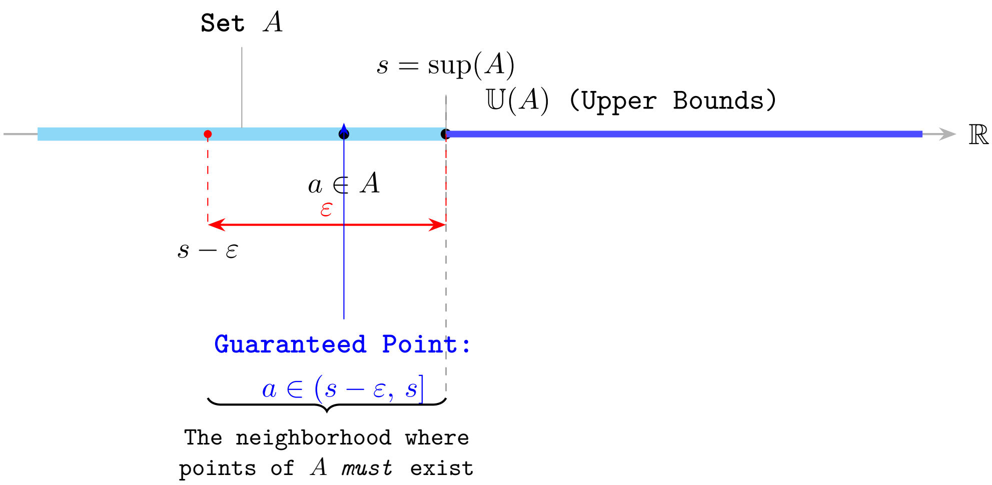


Figure 2.8: Refined visualization of the ε-characterization. Labels occupy staggered vertical lanes with leader lines to eliminate all horizontal collision, while the dashed spine preserves the geometric correspondence between the real line and the probe below it.

Definition (Epsilon Characterization of Infimum)

Definition 2.1.14 (Epsilon Characterization of Infimum). Let $A \subseteq \mathbb{R}$ and let $t \in \mathbb{R}$. The element t satisfies the *epsilon characterization of the infimum* of A if and only if t is a lower bound of A and for every $\varepsilon > 0$ there exists $a \in A$ such that $a < t + \varepsilon$.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$.

t satisfies the epsilon characterization of $\inf A$

$$\iff \Big[\forall x \in A\ (t \leq x)\Big] \wedge \Big[\forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon)\Big].$$

Remark (Definition predicate reading).

$$\mathrm{EpsilonInfimum}(t, A) := \mathrm{LowerBound}(t, A) \wedge \forall \varepsilon > 0\ \exists a \in A\ (a < t + \varepsilon).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R}$ and $t \in \mathbb{R}$.
t does not satisfy the epsilon characterization of $\inf A$
$$\iff \Big[\exists x \in A\ (x < t)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{EpsilonInfimum}(t, A) \iff \Big[\exists x \in A\ (x < t)\Big] \vee \Big[\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)\Big].$$

Remark (Failure modes). The characterization fails either because t is not a lower bound of A, or because A remains at least some positive distance above t.

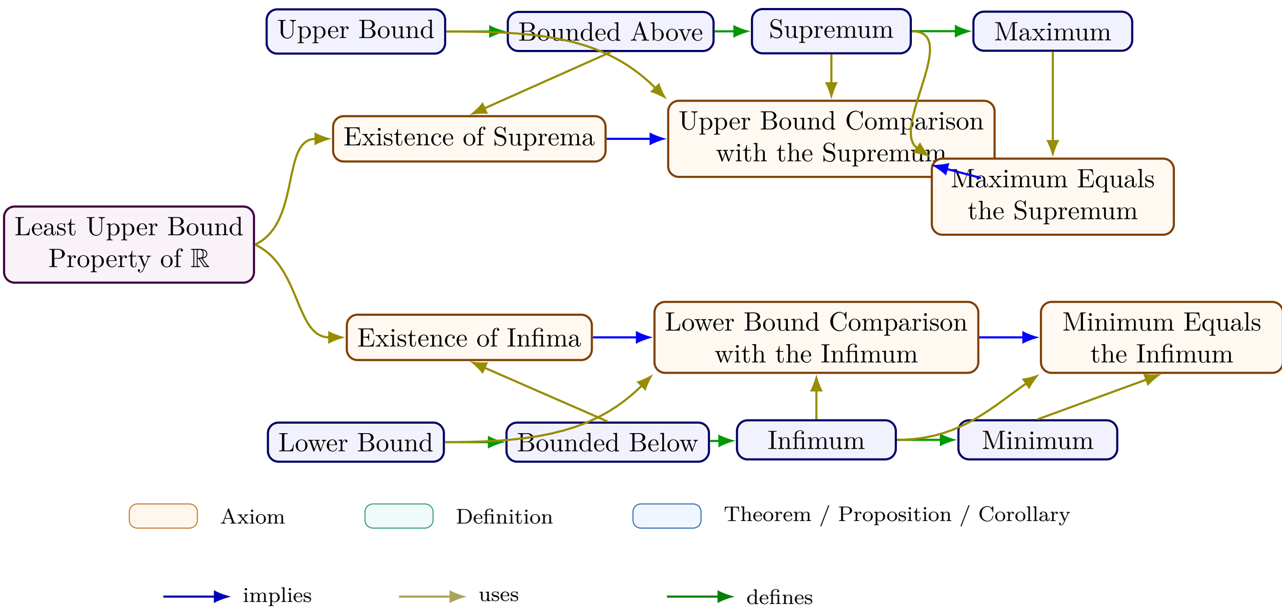
Remark (Failure mode decomposition).

$$\neg\,\mathrm{EpsilonInfimum}(t, A) = \underbrace{\exists x \in A\ (x < t)}_{\texttt{lower-bound failure}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall a \in A\ (t + \varepsilon_0 \le a)}_{\texttt{positive gap above } t}.$$

Remark (Interpretation). This condition says that t sits below the set while points of A approach t from above at every positive scale. The lower-bound clause prevents A from crossing below t, and the epsilon clause prevents t from sitting strictly below the set with a positive gap. Geometrically, t is a floor that the set presses against arbitrarily closely from above.

Remark (Dependencies).
Definition 2.1.4.

Extremal Bounds Summary



Remark (Equivalent formulations).
Over the ordered-field axioms, completeness is equivalent to each of the following:

- Every nonempty set bounded below has an infimum.
- Every Cauchy sequence in $\mathbb{R}$ converges.
- *(Nested Interval Property)* Every nested sequence of nonempty closed bounded intervals has nonempty intersection.

Algebra of Bounds

Algebraic Properties of Supremum and Infimum

Proposition (Uniqueness of the Maximum)

Proposition 2.1.15 (Uniqueness of the Maximum). *Let $A \subseteq \mathbb{R}$. If m_1 and m_2 are both maximum elements of A, then $m_1 = m_2$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } m_1, m_2 \in A.$$
$$\big[m_1 \text{ is a maximum of } A \wedge m_2 \text{ is a maximum of } A\big] \implies m_1 = m_2.$$

Remark (Predicate reading).

$$\text{Maximum}(m_1, A) \wedge \text{Maximum}(m_2, A) \implies m_1 = m_2.$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } m_1, m_2 \in A.$$
$$m_1 \text{ and } m_2 \text{ are both maximum elements of } A \text{ and } m_1 \neq m_2$$

is impossible.

Remark (Negation predicate reading).

$$\text{Maximum}(m_1, A) \wedge \text{Maximum}(m_2, A) \wedge m_1 \neq m_2$$

is a contradictory configuration.

Remark (Failure modes). The uniqueness claim could fail only if two distinct elements of A each dominated all elements of A. In the real order this cannot occur because each maximum must be less than or equal to the other.

Remark (Failure mode decomposition).

$$\begin{aligned}&\text{Maximum}(m_1, A) \wedge \text{Maximum}(m_2, A)\\ &\implies \underbrace{m_1 \leq m_2}_{m_2 \text{ dominates } A} \wedge \underbrace{m_2 \leq m_1}_{m_1 \text{ dominates } A} \implies m_1 = m_2.\end{aligned}$$

Remark (Interpretation). A maximum is not merely a high point of the set; it is the greatest point of the set. If two elements both claim that role, each must sit above the other, and antisymmetry of the real order forces them to be the same number.

Remark (Dependencies).
Maximum.

Proposition (Uniqueness of the Supremum)

Proposition 2.1.16 (Uniqueness of the Supremum). *Let $A \subseteq \mathbb{R}$. If s_1 and s_2 are both suprema of A, then $s_1 = s_2$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s_1, s_2 \in \mathbb{R}.$$
$$\big[s_1 = \sup A \ \wedge\ s_2 = \sup A\big] \implies s_1 = s_2.$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s_1, A) \wedge \mathrm{Supremum}(s_2, A) \implies s_1 = s_2.$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s_1, s_2 \in \mathbb{R}.$$
$$s_1 = \sup A, \quad s_2 = \sup A, \quad \text{and} \quad s_1 \neq s_2$$

is impossible.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s_1, A) \wedge \mathrm{Supremum}(s_2, A) \wedge s_1 \neq s_2$$

is a contradictory configuration.

Remark (Failure modes). The uniqueness claim could fail only if two different real numbers were both least upper bounds of the same set. Since each supremum is less than or equal to every upper bound, each must be less than or equal to the other.

Remark (Failure mode decomposition).

$$\mathrm{Supremum}(s_1, A) \wedge \mathrm{Supremum}(s_2, A) \implies \underbrace{s_1 \leq s_2}_{s_2 \text{ is an upper bound and } s_1 \text{ is least}} \wedge \underbrace{s_2 \leq s_1}_{s_1 \text{ is an upper bound and } s_2 \text{ is least}} \implies s_1 = s_2.$$

Remark (Interpretation). The supremum is the least possible ceiling over a set. Two ceilings can both be least only if neither one is actually above the other, so the order on $\mathbb{R}$ identifies them as the same number.

Remark (Dependencies).
Supremum.

Proposition (Maximum Implies Supremum)

Proposition 2.1.17 (Maximum Implies Supremum). *Let $A \subseteq \mathbb{R}$ and let $m \in A$. If m is a maximum of A, then $m = \sup A$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } m \in A.$$
$$m \text{ is a maximum of } A \implies m = \sup A.$$

Remark (Predicate reading).

$$\text{Maximum}(m, A) \implies \text{Supremum}(m, A).$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } m \in A.$$
$$m \text{ is a maximum of } A \text{ and } m \neq \sup A$$

is impossible.

Remark (Negation predicate reading).

$$\text{Maximum}(m, A) \wedge \neg\,\text{Supremum}(m, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if the greatest element of A were not the least upper bound of A. But a maximum already bounds the set from above, and every upper bound must lie at least as high as that maximum.

Remark (Failure mode decomposition).

$$\begin{aligned}\text{Maximum}(m, A) \implies &\underbrace{\forall x \in A\ (x \le m)}_{m \text{ is an upper bound}} \\ &\wedge \underbrace{\forall u \in \mathbb{R}\left[(\forall x \in A\ (x \le u)) \implies m \le u\right]}_{m \text{ is least among upper bounds}}.\end{aligned}$$

Remark (Interpretation). A maximum belongs to the set and sits above every element of the set. That makes it an upper bound, and no smaller upper bound can exist because it would have to sit below the maximum while still bounding the maximum itself.

Remark (Dependencies).
Maximum; Supremum.

Proposition (Supremum in the Set Is the Maximum)

Proposition 2.1.18 (Supremum in the Set Is the Maximum). *Let $A \subseteq \mathbb{R}$ and let $s = \sup A$. If $s \in A$, then s is a maximum of A.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.$$
$$\big[s = \sup A \;\wedge\; s \in A\big] \Longrightarrow s \text{ is a maximum of } A.$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \wedge s \in A \Longrightarrow \mathrm{Maximum}(s, A).$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } s \in \mathbb{R}.$$
$$s = \sup A, \quad s \in A, \quad \text{and } s \text{ is not a maximum of } A$$

is impossible.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge s \in A \wedge \neg\mathrm{Maximum}(s, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if the least upper bound belonged to the set but did not dominate every element of the set. That cannot happen because every supremum is already an upper bound.

Remark (Failure mode decomposition).

$$\mathrm{Supremum}(s, A) \wedge s \in A \Longrightarrow \underbrace{\forall x \in A\ (x \le s)}_{s \text{ is an upper bound}} \wedge \underbrace{s \in A}_{s \text{ belongs to the set}} \Longrightarrow \mathrm{Maximum}(s, A).$$

Remark (Interpretation). A supremum is usually only a least ceiling. When that ceiling is actually a member of the set, it becomes the greatest element of the set, so the extremal value is attained rather than merely approached.

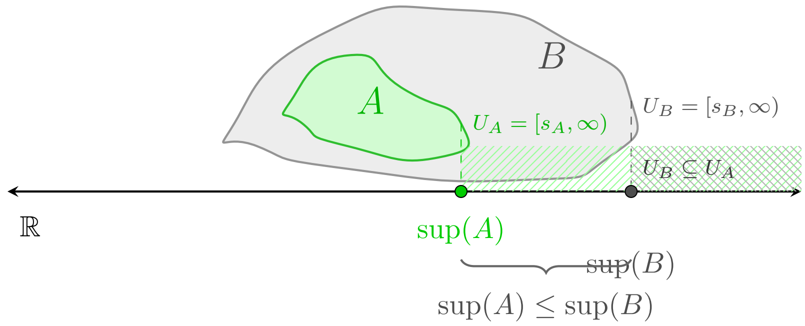
Remark (Dependencies).
Supremum; Maximum.

Lemma (Subset Inclusion Preserves Upper Bounds)

Lemma 2.1.19 (Subset Inclusion Preserves Upper Bounds). *Let $A \subseteq B \subseteq \mathbb{R}$. If u is an upper bound of B, then u is an upper bound of A.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq B \subseteq \mathbb{R} \text{ and } u \in \mathbb{R}.$$
$$\big[\forall y \in B\ (y \le u)\big] \Longrightarrow \big[\forall x \in A\ (x \le u)\big].$$



H]

Caption (R7.5, R13.8): Figure ??: The Monotonicity of Supremum Under Set Inclusion. This visualization illustrates the arithmetic consequence of set inclusion, $A \subseteq B \subset \mathbb{R}$. The topology of the sets is represented by nested blobs where A (green) is contained within B (gray). Vertically dashed projections link the supremal boundary of each set down to the shared real axis, defining the unique points $s_A = \sup(A)$ and $s_B = \sup(B)$. The figure demonstrates that because the territory of A is restricted by B, its maximum guard point s_A cannot exceed s_B. Finally, the shaded regions illustrate the territories of upper bounds (U_A and U_B), clearly showing that U_B is a proper subset of U_A. Any value guarding the larger set B necessarily satisfies the guard condition for the smaller set A.

Remark (Predicate reading).

$$A \subseteq B \wedge \mathrm{UpperBound}(u, B) \implies \mathrm{UpperBound}(u, A).$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad u \text{ is an upper bound of } B,$$
$$\text{and } u \text{ is not an upper bound of } A$$

is impossible.

Remark (Negation predicate reading).

$$A \subseteq B \wedge \mathrm{UpperBound}(u, B) \wedge \neg\,\mathrm{UpperBound}(u, A)$$

is a contradictory configuration.

Remark (Failure modes). The implication could fail only if some $x \in A$ exceeded u. Since $A \subseteq B$, that same x would belong to B, contradicting that u bounds B from above.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{UpperBound}(u, A) \iff \underbrace{\exists x \in A\ (u < x)}_{\texttt{witness in } A} \implies \underbrace{\exists x \in B\ (u < x)}_{A \subseteq B} \iff \neg\,\mathrm{UpperBound}(u, B).$$

Remark (Interpretation). A bound for a larger set automatically bounds every smaller set inside it. Inclusion can only remove possible counterexamples, so it cannot destroy an existing upper bound.

Remark (Dependencies).
Upper Bound.

Theorem (Supremum Is Monotone Under Inclusion)

Theorem 2.1.20 (Supremum Is Monotone Under Inclusion). *Let $A, B \subseteq \mathbb{R}$ be nonempty sets that are bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}$$
$$A \subseteq B \implies \sup A \leq \sup B.$$

Remark (Predicate reading).

$$A \subseteq B \wedge \text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \implies s_A \leq s_B.$$

Remark (Negated quantified statement).

$$A \subseteq B, \quad \sup B < \sup A$$

is impossible when A and B are nonempty and bounded above.

Remark (Negation predicate reading).

$$A \subseteq B \wedge \text{Supremum}(s_A, A) \wedge \text{Supremum}(s_B, B) \wedge s_B < s_A$$

is a contradictory configuration.

Remark (Failure modes). The monotonicity statement could fail only if the larger set had a strictly smaller least upper bound. But every upper bound for B is also an upper bound for A, so the least upper bound of A cannot sit above $\sup B$.

Remark (Failure mode decomposition).

$$\begin{aligned} A \subseteq B &\implies \underbrace{\text{UpperBound}(\sup B, A)}_{\sup B \text{ bounds the smaller set}} \\ &\implies \underbrace{\sup A \leq \sup B}_{\sup A \text{ is least among upper bounds of } A}. \end{aligned}$$

Remark (Interpretation). Making a set larger can only push its least ceiling upward or leave it unchanged. It cannot lower the supremum, because the larger set contains all of the old points that already had to be bounded.

Remark (Dependencies).
Upper Bound; Supremum; Subset Inclusion Preserves Upper Bounds.

Proposition (Upper Bound Comparison with the Supremum)

Proposition 2.1.21 (Upper Bound Comparison with the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. For $u \in \mathbb{R}$, the number u is an upper bound of A if and only if $s \leq u$.*
Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$.
$\forall u \in \mathbb{R}\ \big[u$ is an upper bound of $A \iff s \le u\big]$.

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \implies \forall u \in \mathbb{R}\ \big[\mathrm{UpperBound}(u, A) \iff s \le u\big].$$

Remark (Negated quantified statement).

$\exists u \in \mathbb{R}\ \big[(u$ is an upper bound of A and $u < s) \vee (s \le u$ and u is not an upper bound of $A)\big]$

is impossible when $s = \sup A$.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \exists u \in \mathbb{R}\ \big[(\mathrm{UpperBound}(u, A) \wedge u < s) \vee (s \le u \wedge \neg\, \mathrm{UpperBound}(u, A))\big]$$

is a contradictory configuration.

Remark (Failure modes). There are two apparent ways the equivalence could fail. Either an upper bound lies below the supremum, contradicting leastness, or a number above the supremum fails to bound the set, contradicting that the supremum already bounds the set.

Remark (Failure mode decomposition).

$$\begin{aligned} &\mathrm{UpperBound}(u, A) \implies s \le u, \\ &\qquad s \le u \implies \underbrace{\forall x \in A\ (x \le s)}_{s \text{ bounds } A} \implies \underbrace{\forall x \in A\ (x \le u)}_{u \text{ also bounds } A}. \end{aligned}$$

Remark (Interpretation). Once the supremum is known, all upper bounds are exactly the real numbers at or above it. The supremum is therefore the threshold where upper bounds begin.

Remark (Dependencies).
Upper Bound; Supremum.

Proposition (Every Element Lies Below the Supremum)

Proposition 2.1.22 (Every Element Lies Below the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$. Then $x \le s$ for every $x \in A$.*

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $s = \sup A$.
$\forall x \in A\ (x \le s)$.

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \implies \forall x \in A\ (x \le s).$$

Remark (Negated quantified statement).

$$s = \sup A \quad \text{and} \quad \exists x \in A\ (s < x)$$

is impossible.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \wedge \exists x \in A\ (s < x)$$

is a contradictory configuration.

Remark (Failure modes). The statement could fail only if some element of A sat strictly above $\sup A$. That would mean $\sup A$ is not even an upper bound of A.

Remark (Failure mode decomposition).

$$\neg[\forall x \in A\ (x \leq s)] \iff \underbrace{\exists x \in A\ (s < x)}_{\text{element above the alleged supremum}} \iff \neg\,\mathrm{UpperBound}(s, A).$$

Remark (Interpretation). The supremum is a ceiling for the set. It may or may not be a point of the set, but every actual element of the set must lie at or below it.

Remark (Dependencies).
Upper Bound; Supremum.

Theorem (Infimum Is Less Than or Equal to the Supremum)

Theorem 2.1.23 (Infimum Is Less Than or Equal to the Supremum).
Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above. Then $\inf A \leq \sup A$.

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty, bounded below, and bounded above.
$\inf A \leq \sup A$.

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \mathrm{Supremum}(s, A) \implies t \leq s.$$

Remark (Negated quantified statement).

$$\inf A > \sup A$$

is impossible for a nonempty set that has both bounds.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \mathrm{Supremum}(s, A) \wedge s < t$$

is a contradictory configuration.

Remark (Failure modes). The inequality could fail only if the greatest lower bound were strictly above the least upper bound. Any element of the nonempty set must lie between those two bounding values, which rules out this order.

Remark (Failure mode decomposition).

$$\text{Choose } a \in A. \quad \underbrace{\inf A \le a}_{\inf A \text{ is a lower bound threshold}} \quad \text{and} \quad \underbrace{a \le \sup A}_{\sup A \text{ is an upper bound threshold}} \implies \inf A \le \sup A.$$

Remark (Interpretation). A nonempty bounded set cannot have its floor above its ceiling. Any point of the set witnesses the bridge from the infimum up to the supremum.

Remark (Dependencies).
Infimum; Supremum.

Proposition (Order Comparison of Suprema)

Proposition 2.1.24 (Order Comparison of Suprema). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. If for every $x \in A$ there exists $y \in B$ such that $x \le y$, then $\sup A \le \sup B$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded above.}$$
$$\big[\forall x \in A\ \exists y \in B\ (x \le y)\big] \implies \sup A \le \sup B.$$

Remark (Predicate reading).

$$\big[\forall x \in A\ \exists y \in B\ (x \le y)\big] \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \implies s_A \le s_B.$$

Remark (Negated quantified statement).

$$\big[\forall x \in A\ \exists y \in B\ (x \le y)\big] \quad \text{and} \quad \sup B < \sup A$$

is impossible.

Remark (Negation predicate reading).

$$\big[\forall x \in A\ \exists y \in B\ (x \le y)\big] \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge s_B < s_A$$

is a contradictory configuration.

Remark (Failure modes). The comparison could fail only if the supremum of A rose above the supremum of B. But every element of A is dominated by some element of B, and every element of B lies below $\sup B$, so $\sup B$ is an upper bound for A.

Remark (Failure mode decomposition).

$$\forall x \in A\ \exists y \in B\ (x \le y) \implies \underbrace{\forall x \in A\ (x \le \sup B)}_{\sup B \text{ bounds } A}$$
$$\implies \underbrace{\sup A \le \sup B}_{\sup A \text{ is least among upper bounds of } A}.$$

Remark (Interpretation). This criterion compares two sets without requiring inclusion. If every point of A can be matched below some point of B, then B reaches at least as high as A in the supremum sense.

Remark (Dependencies).
Supremum; Every Element Lies Below the Supremum; Upper Bound Comparison with the Supremum.

Theorem (Translation Invariance of the Supremum)

Theorem 2.1.25 (Translation Invariance of the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

> Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$.
> $\sup\{a + c : a \in A\} = \sup A + c.$

Remark (Predicate reading).

$$\mathrm{Supremum}(s, A) \implies \mathrm{Supremum}(s + c, A + c).$$

Remark (Negated quantified statement).

$$\sup(A + c) \neq \sup A + c$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s, A) \land \neg\,\mathrm{Supremum}(s + c, A + c)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if translation either failed to preserve upper bounds or failed to preserve leastness. Adding the same real number to every comparison preserves the order, so neither failure mode can occur.

Remark (Failure mode decomposition).

$$\forall a \in A\ (a \leq s) \iff \forall a \in A\ (a + c \leq s + c),$$
$$\forall u\,[\mathrm{UpperBound}(u, A) \implies s \leq u] \iff \forall v\,[\mathrm{UpperBound}(v, A + c) \implies s + c \leq v].$$

Remark (Interpretation). Translation shifts the entire set and all of its ceilings by the same amount. The least ceiling therefore shifts by exactly that amount.

Remark (Dependencies).
Upper Bound; Supremum.

Theorem (Scalar Multiplication by a Positive Scalar)

Theorem 2.1.26 (Scalar Multiplication by a Positive Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$.
$\sup\{ka : a \in A\} = k \sup A.$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \Longrightarrow \text{Supremum}(ks, kA).$$

Remark (Negated quantified statement).

$$k > 0, \quad \sup(kA) \neq k \sup A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge k > 0 \wedge \neg\,\text{Supremum}(ks, kA)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if multiplying by a positive scalar failed to preserve upper bounds or failed to preserve leastness. Positive multiplication preserves the order, so both properties scale exactly.

Remark (Failure mode decomposition).

$$\forall a \in A\ (a \leq s) \Longleftrightarrow \forall a \in A\ (ka \leq ks),$$
$$\text{UpperBound}(u, A) \Longleftrightarrow \text{UpperBound}(ku, kA).$$

Remark (Interpretation). Positive scaling stretches the set without reversing order. The least ceiling stretches by the same positive factor.

Remark (Dependencies).
Upper Bound; Supremum.

Theorem (Scalar Multiplication by a Negative Scalar)

Theorem 2.1.27 (Scalar Multiplication by a Negative Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$.
$\sup\{ka : a \in A\} = k \inf A$.

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \wedge k < 0 \Longrightarrow \mathrm{Supremum}(kt, kA).$$

Remark (Negated quantified statement).

$$k < 0, \quad \sup(kA) \neq k \inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t, A) \wedge k < 0 \wedge \neg\,\mathrm{Supremum}(kt, kA)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if negative multiplication did not reverse lower bounds into upper bounds, or if the extremal value lost its leastness after reversal. Multiplication by a negative scalar reverses order exactly, so the infimum of A becomes the supremum of kA.

Remark (Failure mode decomposition).

$$\begin{aligned} \forall a \in A\ (t \leq a) &\Longleftrightarrow \forall a \in A\ (ka \leq kt), \\ \mathrm{LowerBound}(v, A) &\Longleftrightarrow \mathrm{UpperBound}(kv, kA) \qquad (k < 0). \end{aligned}$$

Remark (Interpretation). Negative scaling reflects the set and changes floors into ceilings. The lowest point threshold of A becomes the highest point threshold of the reflected and scaled set.

Remark (Dependencies).
Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Negation Exchanges Supremum and Infimum)

Theorem 2.1.28 (Negation Exchanges Supremum and Infimum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below. Then $-A = \{-a : a \in A\}$ is bounded above and*

$$\sup(-A) = -\inf A.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded below.
$\sup\{-a : a \in A\} = -\inf A$.

Remark (Predicate reading).

$$\mathrm{Infimum}(t, A) \Longrightarrow \mathrm{Supremum}(-t, -A).$$

Remark (Negated quantified statement).

$$\sup(-A) \neq -\inf A$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Infimum}(t, A) \wedge \neg\,\mathrm{Supremum}(-t, -A)$$

is a contradictory configuration.

Remark (Failure modes). The theorem could fail only if negation did not convert lower bounds into upper bounds or did not preserve the extremal threshold after reversing order. The map $x \mapsto -x$ reverses all inequalities, so the exchange is exact.

Remark (Failure mode decomposition).

$$\forall a \in A\ (t \leq a) \iff \forall a \in A\ (-a \leq -t),$$
$$\mathrm{LowerBound}(v, A) \iff \mathrm{UpperBound}(-v, -A).$$

Remark (Interpretation). Negation reflects the real line through the origin. Floors become ceilings, so the infimum of the original set becomes the negative of the supremum of the reflected set.

Remark (Dependencies).
Lower Bound; Upper Bound; Infimum; Supremum.

Theorem (Supremum of a Sum Set)

Theorem 2.1.29 (Supremum of a Sum Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above. Then*

$$\sup(A+B) = \sup A + \sup B, \qquad A+B = \{a+b : a \in A,\ \ b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded above.
$$\sup\{a+b : a \in A,\ \ b \in B\} = \sup A + \sup B.$$

Remark (Predicate reading).

$$\mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \implies \mathrm{Supremum}(s_A + s_B, A+B).$$

Remark (Negated quantified statement).

$$\sup(A+B) \neq \sup A + \sup B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge \neg\,\mathrm{Supremum}(s_A + s_B, A+B)$$

is a contradictory configuration.

Remark (Failure modes). The equality could fail only if the sum of the two suprema did not bound the sum set, or if it were not the least such bound. The first part follows from adding the two upper-bound inequalities; the second follows by approximating each supremum from within its set.

Remark (Failure mode decomposition).

$$a \le \sup A, \quad b \le \sup B \Longrightarrow a + b \le \sup A + \sup B,$$
$$\varepsilon > 0 \Longrightarrow \exists a \in A, \;\; b \in B : \sup A + \sup B - \varepsilon < a + b.$$

Remark (Interpretation). The highest limiting value of all pairwise sums is obtained by combining values that approach the separate ceilings of A and B.

Remark (Dependencies).
Supremum; Epsilon Characterization of Supremum.

Theorem (Supremum of a Difference Set)

Theorem 2.1.30 (Supremum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A, \;\; b \in B\}.$$

Go to proof.

Remark (Standard quantified statement).

> Let $A, B \subseteq \mathbb{R}$ be nonempty, A bounded above, and B bounded below. $\sup\{a - b : a \in A, \;\; b \in B\} = \sup A - \inf B.$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \Longrightarrow \text{Supremum}(s - t, A - B).$$

Remark (Negated quantified statement).

$$\sup(A - B) \neq \sup A - \inf B$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, B) \wedge \neg\, \text{Supremum}(s - t, A - B)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if subtracting elements of B did not convert the lower extremal behavior of B into an upper contribution. Since $a - b = a + (-b)$, this is the sum theorem combined with the negation exchange.

Remark (Failure mode decomposition).

$$A - B = A + (-B), \qquad \sup(-B) = -\inf B, \qquad \sup(A + (-B)) = \sup A + \sup(-B).$$

Remark (Interpretation). The largest possible difference comes from making the first term as large as possible and the subtracted term as small as possible.

Remark (Dependencies).
Supremum; Infimum; Negation Exchanges Supremum and Infimum; Supremum of a Sum Set.

Theorem (Supremum of a Dilation)

Theorem 2.1.31 (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$.
$\sup\{\lambda a : a \in A\} = \max\{\lambda \sup A, \lambda \inf A\}$.

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \implies \text{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A).$$

Remark (Negated quantified statement).

$$\sup(\lambda A) \neq \max\{\lambda \sup A, \lambda \inf A\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \wedge \neg\,\text{Supremum}(\max\{\lambda s, \lambda t\}, \lambda A)$$

is a contradictory configuration.

Remark (Failure modes). The formula separates into the cases $\lambda > 0$, $\lambda = 0$, and $\lambda < 0$. Positive dilation scales the supremum, zero dilation collapses the set to $\{0\}$, and negative dilation turns the infimum into the supremum.

Remark (Failure mode decomposition).

$$\sup(\lambda A) = \begin{cases} \lambda \sup A, & \lambda > 0, \\ 0, & \lambda = 0, \\ \lambda \inf A, & \lambda < 0. \end{cases}$$

Remark (Interpretation). A dilation either preserves the order, collapses all points to zero, or reverses the order. The displayed maximum packages those three cases into one formula.

Remark (Dependencies).
Supremum; Infimum; Maximum; Scalar Multiplication by a Positive Scalar; Scalar Multiplication by a Negative Scalar.

Theorem (Supremum of the Absolute-Value Image)

Theorem 2.1.32 (Supremum of the Absolute-Value Image). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be nonempty and bounded.
$$\sup\{|a| : a \in A\} = \max\{|\sup A|, |\inf A|\}.$$

Remark (Predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \implies \text{Supremum}(\max\{|s|, |t|\}, |A|).$$

Remark (Negated quantified statement).

$$\sup |A| \neq \max\{|\sup A|, |\inf A|\}$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\text{Supremum}(s, A) \wedge \text{Infimum}(t, A) \wedge \neg\,\text{Supremum}(\max\{|s|, |t|\}, |A|)$$

is a contradictory configuration.

Remark (Failure modes). The formula could fail only if the farthest absolute-value size were not achieved at one of the two extremal thresholds. Since every $a \in A$ lies between $\inf A$ and $\sup A$, its distance from zero is bounded by the larger endpoint magnitude.

Remark (Failure mode decomposition).

$$\inf A \leq a \leq \sup A \implies |a| \leq \max\{|\sup A|, |\inf A|\}.$$

Remark (Interpretation). The absolute-value image measures distance from the origin. For a bounded subset of the real line, the largest possible distance is controlled by whichever extremal endpoint lies farther from zero.

Remark (Dependencies).
Supremum; Infimum; Maximum; Every Element Lies Below the Supremum.

Proposition (Bounds for Suprema and Infima Under Set Addition)

Proposition 2.1.33 (Bounds for Suprema and Infima Under Set Addition). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \qquad \textit{and} \qquad \sup(A + B) \leq \sup A + \sup B.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } A, B \subseteq \mathbb{R} \text{ be nonempty and bounded.}$$
$$\inf A + \inf B \le \inf(A+B) \quad \wedge \quad \sup(A+B) \le \sup A + \sup B.$$

Remark (Predicate reading).

$$\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \wedge \mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \implies t_A + t_B \le \inf(A+B) \le \sup($$

Remark (Negated quantified statement).

$$\inf(A+B) < \inf A + \inf B \quad \text{or} \quad \sup A + \sup B < \sup(A+B)$$

is impossible under the stated hypotheses.

Remark (Negation predicate reading).

$$\big[\mathrm{Infimum}(t_A, A) \wedge \mathrm{Infimum}(t_B, B) \wedge \inf(A+B) < t_A + t_B\big] \vee \big[\mathrm{Supremum}(s_A, A) \wedge \mathrm{Supremum}(s_B, B) \wedge s_A + s_B$$

is a contradictory configuration.

Remark (Failure modes). The lower inequality could fail only if a sum fell below the sum of the two lower thresholds. The upper inequality could fail only if a sum rose above the sum of the two upper thresholds.

Remark (Failure mode decomposition).

$$\inf A \le a \le \sup A, \quad \inf B \le b \le \sup B \Longrightarrow \inf A + \inf B \le a + b \le \sup A + \sup B.$$

Remark (Interpretation). Every element of the sum set is built from one element of A and one element of B. Adding the individual lower and upper controls gives immediate lower and upper controls for every such sum.

Remark (Dependencies).
Infimum; Supremum; Supremum of a Sum Set.

Proposition (Upper Bounds Depend on the Ambient Order)

Proposition 2.1.34 (Upper Bounds Depend on the Ambient Order). *Let $(S, \le_S)$ and $(T, \le_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order.* *Go to proof.*

Remark (Standard quantified statement).

$$u \text{ is an upper bound of } A \text{ in } S \iff u \in S \text{ and } \forall x \in A\ (x \le_S u),$$
$$u \text{ is an upper bound of } A \text{ in } T \iff u \in T \text{ and } \forall x \in A\ (x \le_T u).$$

Remark (Predicate reading).

$$\mathrm{UpperBound}_S(u, A) \iff u \in S \wedge \forall x \in A\ (x \le_S u).$$

Remark (Negated quantified statement).

The phrase "u is an upper bound of A"

is not well-determined until the ambient ordered set and its order relation are fixed.

Remark (Negation predicate reading).

$$\neg\,\mathrm{WellDetermined}(\mathrm{UpperBound}(u, A))$$

when the ambient order is omitted.

Remark (Failure modes). Ambiguity occurs when the same set A is considered inside different ordered ambient sets or under different order relations. The candidate bound may belong to one ambient set and not another, or the relevant comparison relation may change.

Remark (Failure mode decomposition).

$$\mathrm{UpperBound}_S(u, A) \text{ and } \mathrm{UpperBound}_T(u, A)$$

are separate assertions unless $S = T$ and $\leq_S = \leq_T$ on the elements being compared.

Remark (Interpretation). An upper bound is not only about the subset; it is also about where the subset lives. Changing the ambient order changes the universe of possible bounds and the comparison used to test them.

Remark (Dependencies).
Upper Bound.

Proposition (Suprema Depend on the Ambient Set)

Proposition 2.1.35 (Suprema Depend on the Ambient Set). *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'.* *Go to proof.*

Remark (Standard quantified statement).

$$\sup_S A \text{ and } \sup_{S'} A$$

are ambient-relative objects and may differ in existence or value.

Remark (Predicate reading).

$$\mathrm{Supremum}_S(s, A) \quad \text{and} \quad \mathrm{Supremum}_{S'}(s, A)$$

are different assertions unless the ambient upper-bound sets agree.

Remark (Negated quantified statement).

$$\sup_S A = \sup_{S'} A$$

is not guaranteed by $A \subseteq S \subseteq S'$ alone.

Remark (Negation predicate reading).

$$\mathrm{Supremum}_{S'}(s, A) \wedge \neg\, \mathrm{Supremum}_{S}(s, A)$$

is possible when the larger ambient set contains the least upper bound but the smaller one does not.

Remark (Failure modes). The ambient-relative supremum can fail to transfer because the candidate least upper bound may not belong to the smaller ambient set, or because the smaller ambient set may have a different collection of upper bounds.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S \implies s \text{ cannot be the supremum of } A \text{ relative to } S.$$

Remark (Interpretation). Suprema are least upper bounds inside a chosen universe. Enlarging or shrinking that universe can create or remove the point that should serve as the least ceiling.

Remark (Dependencies).
Upper Bound; Supremum.

Theorem (Ambient Existence of Supremum)

Theorem 2.1.36 (Ambient Existence of Supremum). *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S.* *Go to proof.*

Remark (Standard quantified statement).

$$\exists S \subseteq S'\ \exists A \subseteq S : \quad \sup_{S'} A \text{ exists} \quad \text{and} \quad \sup_{S} A \text{ does not exist.}$$

Remark (Predicate reading).

$$\exists S \subseteq S'\ \exists A \subseteq S\ \exists s \in S' : \mathrm{Supremum}_{S'}(s, A) \wedge \neg \exists t \in S\ \ \mathrm{Supremum}_{S}(t, A).$$

Remark (Negated quantified statement).

Every ambient enlargement preserves and reflects existence of suprema.

This assertion is false.

Remark (Negation predicate reading).

$$\forall S \subseteq S'\ \forall A \subseteq S : \left[\exists s \in S'\ \mathrm{Supremum}_{S'}(s, A) \implies \exists t \in S\ \mathrm{Supremum}_{S}(t, A)\right]$$

is false.

Remark (Failure modes). The transfer from S' to S fails when the least upper bound exists in the larger ambient set but is missing from the smaller one. The smaller ambient set may contain upper bounds without containing a least upper bound.

Remark (Failure mode decomposition).

$$s = \sup_{S'} A, \quad s \notin S, \quad \text{and no } t \in S \text{ is least among the upper bounds of } A \text{ in } S.$$

Remark (Interpretation). Completeness is an ambient property. A larger ordered universe can fill a gap that remains absent in a smaller ordered universe.

Remark (Dependencies).
Upper Bound; Supremum; Suprema Depend on the Ambient Set.

Lemma (Rational Gap Example for Suprema)

Lemma 2.1.37 (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$. *Go to proof.*

Remark (Standard quantified statement).

$$A = \{q \in \mathbb{Q} : q^2 < 2\} \quad \Longrightarrow \quad \sup_{\mathbb{Q}} A \text{ does not exist.}$$

Remark (Predicate reading).

$$\neg \exists s \in \mathbb{Q} \ \mathrm{Supremum}_{\mathbb{Q}}(s, A), \qquad A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Remark (Negated quantified statement).

$$\exists s \in \mathbb{Q} \ (s = \sup_{\mathbb{Q}} A)$$

is false for this set.

Remark (Negation predicate reading).

$$\exists s \in \mathbb{Q} \ \mathrm{Supremum}_{\mathbb{Q}}(s, A)$$

is false.

Remark (Failure modes). A rational candidate below $\sqrt{2}$ fails to be an upper bound. A rational candidate above $\sqrt{2}$ is an upper bound but is not least, because another rational upper bound lies between it and $\sqrt{2}$.

Remark (Failure mode decomposition).

$$\begin{cases} s^2 < 2 & \Longrightarrow s \text{ is not an upper bound of } A, \\ s^2 > 2 & \Longrightarrow s \text{ is not the least upper bound of } A. \end{cases} \qquad (s \in \mathbb{Q})$$

Remark (Interpretation). The set wants $\sqrt{2}$ as its least ceiling, but that number is not rational. The rational ambient set therefore has a visible gap where the supremum should be.

Remark (Dependencies).
Upper Bound; Supremum; Ambient Existence of Supremum.

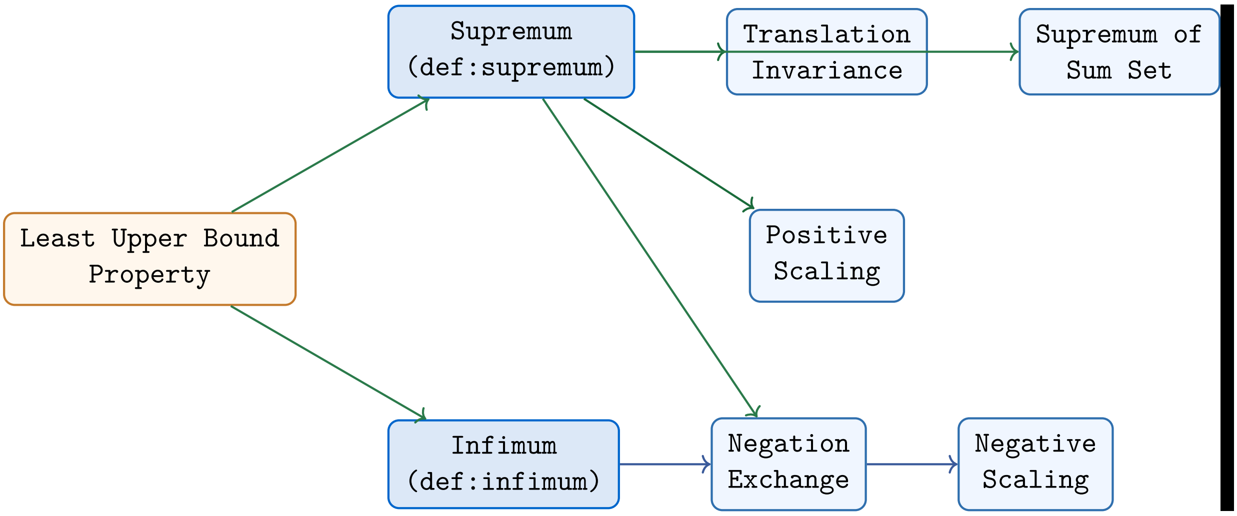
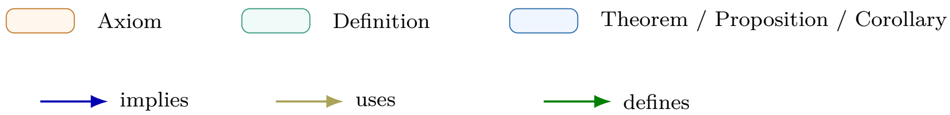


Figure 2.9: Bound Algebra Cluster Dependency Map. The map originates from the ‘Least Upper Bound Property‘ (Axiom) and branches into the dual properties of the continuum. The Negation Exchange serves as the structural bridge between supremum and infimum algebra.



Bound Algebra

Proofs

Remark (Return). Return to Theorem

Theorem (Archimedean Property). *For every $x \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $n > x$.*

Professional Standard Proof.
Assume for sake of contradiction that the property is false. Then there exists $x \in \mathbb{R}$ such that $n \leq x$ for all $n \in \mathbb{N}$. This implies $\mathbb{N}$ is bounded above. By the Least Upper Bound Property of $\mathbb{R}$, $s = \sup \mathbb{N}$ exists. Since s is the least upper bound, $s - 1$ is not an upper bound. Thus, there exists $m \in \mathbb{N}$ such that $m > s - 1$, which implies $m + 1 > s$. Since $m + 1 \in \mathbb{N}$, this contradicts that s is an upper bound for $\mathbb{N}$. Hence, $\mathbb{N}$ is unbounded. □

Detailed Learning Proof.
Step 1. Assume the negation for contradiction. We assume that the set of natural numbers $\mathbb{N}$ is bounded above. Formally:

$$\exists x \in \mathbb{R}\ \forall n \in \mathbb{N} : n \leq x$$

Step 2. Invoke the Least Upper Bound Property. Since $\mathbb{N}$ is non-empty ($1 \in \mathbb{N}$) and assumed to be bounded above, the completeness of $\mathbb{R}$ ensures that $s = \sup \mathbb{N}$ exists in $\mathbb{R}$.
Step 3. Apply the ε-characterization of the supremum. Let $\varepsilon = 1$. By the characterization of the supremum, there must be some element of the set strictly greater than $s - \varepsilon$. Thus, there exists $m \in \mathbb{N}$ such that:

$$m > s - 1$$

Step 4. Construct a contradiction within the set. Adding 1 to both sides of the inequality from Step 3:

$$m + 1 > s$$

By the inductive property of natural numbers, if $m \in \mathbb{N}$, then $m + 1 \in \mathbb{N}$. However, we have found a natural number $m + 1$ that is strictly greater than s, the supposed upper bound of $\mathbb{N}$.
Step 5. Conclude the proof. The existence of $m + 1 > s$ contradicts the definition of s as an upper bound. Therefore, our initial assumption that $\mathbb{N}$ is bounded must be false. Thus, for every $x \in \mathbb{R}$, there exists some $n \in \mathbb{N}$ such that $n > x$. □

Remark (Proof structure). The proof utilizes a contradiction strategy based on the completeness of the real numbers. It shows that if $\mathbb{N}$ were bounded, its "ceiling" would have to be surpassable by simply adding 1 to a value near that ceiling.

Remark (Dependencies).

- Least Upper Bound Property
- Definition of Supremum
- Definition of Upper Bound

Remark (Return). Return to Corollary

Corollary (Archimedean Corollary). *For every $\varepsilon \in \mathbb{R}$ such that $\varepsilon > 0$, there exists $n \in \mathbb{N}$ such that $\frac{1}{n} < \varepsilon$.*

Professional Standard Proof.
Let $\varepsilon > 0$. Since $\varepsilon \in \mathbb{R}$, its reciprocal $\frac{1}{\varepsilon}$ is also a real number. By the Archimedean Property, there exists $n \in \mathbb{N}$ such that $n > \frac{1}{\varepsilon}$. Since n and ε are both positive, we can invert the inequality to obtain $\frac{1}{n} < \varepsilon$. ☐

Detailed Learning Proof.
Step 1. Identify the target value for the Archimedean Property. We are given a fixed $\varepsilon > 0$. We want to find a natural number n that makes $\frac{1}{n}$ very small.
Step 2. Transform the inequality. The goal is $\frac{1}{n} < \varepsilon$. Since $n > 0$ and $\varepsilon > 0$, this is algebraically equivalent to:

$$n > \frac{1}{\varepsilon}$$

Let $x = \frac{1}{\varepsilon}$. Since ε is a real number, x is also a real number.
Step 3. Apply the Archimedean Property. The Archimedean Property states that for any real number x, there exists a natural number n such that $n > x$. Applying this to our $x = \frac{1}{\varepsilon}$:

$$\exists n \in \mathbb{N} : n > \frac{1}{\varepsilon}$$

Step 4. Revert to the reciprocal form. Starting with $n > \frac{1}{\varepsilon}$, we multiply both sides by $\frac{\varepsilon}{n}$ (which is positive, so the inequality direction is preserved):

$$\varepsilon > \frac{1}{n} \quad \text{or} \quad \frac{1}{n} < \varepsilon$$

Thus, for any positive tolerance ε, we have found a unit fraction $1/n$ that falls within that tolerance. ☐

Remark (Strategy: The Choice of Leastness Method). Because this corollary is a direct algebraic consequence of the Archimedean Property, we prove it through simple variable transformation ($x = 1/\varepsilon$) rather than using the supremum's ε-characterization directly. In the hierarchy of your repository, the Archimedean Property (which relies on "leastness") does the heavy lifting, allowing the corollary to be a "Side A" algebraic derivation.

Remark (Proof structure). This is a direct proof utilizing the Archimedean Property of $\mathbb{R}$.

Remark (Dependencies).

- Archimedean Property
- Ordered Field Multiplication Law

Remark (Return). Return to Theorem

Theorem (Density of the Rational Numbers in R). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $q \in \mathbb{Q}$ such that $a < q < b$.*

Professional Standard Proof.
TODO: professional standard proof for density-of-rationals-in-reals. □

Detailed Learning Proof.
TODO: detailed learning proof for density-of-rationals-in-reals. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Density of the Irrational Numbers in R). *For every $a, b \in \mathbb{R}$ with $a < b$, there exists $s \in \mathbb{R} \setminus \mathbb{Q}$ such that $a < s < b$.*

Professional Standard Proof.
TODO: professional standard proof for density-of-irrationals-in-reals. □

Detailed Learning Proof.
TODO: detailed learning proof for density-of-irrationals-in-reals. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Rational and Irrational Points in Every Open Interval). *Every open interval in $\mathbb{R}$ contains both a rational number and an irrational number.*

Professional Standard Proof.
TODO: professional standard proof for every-open-interval-contains-rational-and-irrational. ☐

Detailed Learning Proof.
TODO: detailed learning proof for every-open-interval-contains-rational-and-irrational. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Nested Interval Theorem). *Let $I_n = [a_n, b_n]$ be a sequence of non-empty closed bounded intervals such that $I_{n+1} \subseteq I_n$. Then $\bigcap_{n=1}^{\infty} I_n \neq \varnothing$.*

Professional Standard Proof.
Since $I_{n+1} \subseteq I_n$, the sequence of endpoints satisfies $a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$. Thus, (a_n) is monotone increasing and bounded above by b_1. By the Monotone Convergence Theorem, $a_n \to x$ where $x = \sup\{a_n\}$. For any fixed k, $a_n \leq b_k$ for all n (since if $n > k$, $a_n \leq b_n \leq b_k$, and if $n \leq k$, $a_n \leq a_k \leq b_k$). Thus, $x = \sup\{a_n\} \leq b_k$ for all k. Since $a_k \leq x$ for all k, we have $a_k \leq x \leq b_k$, implying $x \in I_k$ for all $k \in \mathbb{N}$. Thus $x \in \bigcap I_n$. □

Detailed Learning Proof.
Step 1. Analyze the endpoints of the nested intervals. By the definition of nesting, $[a_{n+1}, b_{n+1}] \subseteq [a_n, b_n]$. This implies the following ordering:

$$a_n \leq a_{n+1} \leq b_{n+1} \leq b_n$$

This tells us that (a_n) is a MonotoneIncreasingSequence and (b_n) is a MonotoneDecreasingSequence.
Step 2. Establish a bound for the sequence of left endpoints. Every b_k serves as an upper bound for the entire set $\{a_n : n \in \mathbb{N}\}$. To see this, if $n \leq k$, then $a_n \leq a_k \leq b_k$. If $n > k$, then $a_n \leq b_n \leq b_k$. Thus, $a_n \leq b_k$ for all n, k.
Step 3. Invoke the Least Upper Bound property. Let $A = \{a_n : n \in \mathbb{N}\}$. Since A is non-empty and bounded above (by b_1), $x = \sup A$ exists in $\mathbb{R}$.
Step 4. Show the supremum is contained in every interval. By definition of the supremum as an upper bound, $a_n \leq x$ for all n. From Step 2, we know that every b_n is an upper bound for A. By the "leastness" property of the supremum, $x \leq b_n$ for all n. Combining these, we have:

$$a_n \leq x \leq b_n \implies x \in [a_n, b_n] \text{ for all } n \in \mathbb{N}$$

Step 5. Conclude non-empty intersection. Since x is in every I_n, it belongs to their intersection. Thus, $\bigcap I_n$ is not empty. □

Remark (Proof structure). The proof utilizes the Monotone Convergence/Least Upper Bound property to identify the supremum of the left endpoints as a point that is caught between the "rising floor" and the "falling ceiling" of the intervals.

Remark (Dependencies).

- Least Upper Bound Property
- Monotone Sequence
- Infimum-Supremum Comparison

Remark (Return). Return to Theorem

Theorem (Least Upper Bound Property Implies Existence of Suprema). *Let S be a partially ordered set. If S has the Least Upper Bound Property, then every nonempty subset of S that is bounded above in S has a supremum in S.*

Professional Standard Proof.
Assume that S has the Least Upper Bound Property. Let $A \subseteq S$ be nonempty and bounded above in S. By the Least Upper Bound Property, there exists an element $s \in S$ that is a least upper bound of A in S. By definition, this means that s is a supremum of A in S. Hence

$$\exists s \in S \ \text{Supremum}(s, A).$$

Therefore every nonempty subset of S that is bounded above in S has a supremum in S. □

Detailed Learning Proof.
Step 1. Fix a subset to which the property applies.
Assume that S has the Least Upper Bound Property. Let $A \subseteq S$ be nonempty and bounded above in S.
Step 2. Match the hypotheses on A with the definition of the Least Upper Bound Property.
The Least Upper Bound Property says that every nonempty subset of S that is bounded above in S has a least upper bound in S. The subset A satisfies these hypotheses.
Step 3. Apply the property.
By the Least Upper Bound Property, there exists $s \in S$ such that s is a least upper bound of A.
Step 4. Translate to supremum.
By definition, this means $\text{Supremum}(s, A)$.
Step 5. Conclude.

$$\exists s \in S \ \text{Supremum}(s, A).$$

□

Remark (Proof structure). Direct application of the Least Upper Bound Property followed by unpacking the definition of supremum.

Remark (Dependencies).

- Definition of Least Upper Bound Property
- Definition of Supremum

Remark (Return). Return to Theorem

Proposition (Upper Bound Property of Supremum). *Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. If $s = \sup A$, then*

$$x \leq s \qquad \textit{for every } x \in A.$$

Professional Standard Proof.
Assume that $s = \sup A$. By the definition of supremum, s is an upper bound of A. Therefore

$$\forall x \in A\ (x \leq s).$$

Hence $x \leq s$ for every $x \in A$. □

Detailed Learning Proof.
Step 1. Fix the set and the candidate supremum.
Let $A \subseteq \mathbb{R}$ and let $s \in \mathbb{R}$. Assume that

$$s = \sup A.$$

The goal is to prove that every element of A lies below or at s.
Step 2. Recall the first part of the definition of supremum.
By definition, a supremum of a set is in particular an upper bound of that set. Thus from

$$s = \sup A$$

it follows that s is an upper bound of A.
Step 3. Unpack what upper bound means.
To say that s is an upper bound of A means exactly that

$$\forall x \in A\ (x \leq s).$$

This is already the desired conclusion.
Step 4. State the result in ordinary mathematical language.
Since

$$\forall x \in A\ (x \leq s),$$

it follows that

$$x \leq s \qquad \text{for every } x \in A.$$

Therefore every element of A lies below the supremum. □

Remark (Proof structure). This is a direct unpacking proof. The proposition isolates the upper-bound part of the definition of supremum, so the argument consists of invoking the definition and then rewriting the conclusion in ordinary quantified form.

Remark (Dependencies).

- Definition of Supremum
- Definition of Upper Bound

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Maximum). *If a set $A \subseteq \mathbb{R}$ has a maximum, then that maximum is unique.*

Professional Standard Proof.
Let $m_1, m_2 \in A$ be maxima of A. By the definition of maximum, m_1 is an upper bound of A; since $m_2 \in A$, it follows that $m_2 \leq m_1$. Similarly, m_2 is an upper bound of A; since $m_1 \in A$, it follows that $m_1 \leq m_2$. By the antisymmetry of the $\leq$ relation on $\mathbb{R}$, $m_1 \leq m_2$ and $m_2 \leq m_1$ imply $m_1 = m_2$. ☐

Detailed Learning Proof.
Step 1. Assume the existence of two maxima. Let $A \subseteq \mathbb{R}$ be a non-empty set. Suppose that m_1 and m_2 are both maxima of A. By the definition of maximum, this implies:

$$m_1 \in A \quad \text{and} \quad m_2 \in A.$$

Step 2. Use the upper bound property of the first maximum. Since m_1 is a maximum, it is an upper bound for A. Therefore, every element $a \in A$ satisfies $a \leq m_1$. Because m_2 is an element of A, it follows that:

$$m_2 \leq m_1.$$

Step 3. Use the upper bound property of the second maximum. Conversely, since m_2 is a maximum, every element $a \in A$ satisfies $a \leq m_2$. Because m_1 is an element of A, it follows that:

$$m_1 \leq m_2.$$

Step 4. Apply antisymmetry to conclude uniqueness. The results of Step 2 and Step 3 provide the pair of inequalities:

$$m_1 \leq m_2 \quad \text{and} \quad m_2 \leq m_1.$$

By the antisymmetry of the real ordering, this forces $m_1 = m_2$. Thus the maximum is unique. ☐

Remark (Proof structure). This is a uniqueness proof by antisymmetry. It assumes two candidates for the maximum and uses the double-inclusion logic of upper bounds to force their equality.

Remark (Dependencies).

- Definition of Maximum
- Ordered Field Trichotomy

Remark (Return). Return to Theorem

Proposition (Uniqueness of the Supremum). *If a set $A \subseteq \mathbb{R}$ has a supremum, then that supremum is unique.*

Professional Standard Proof.
Let $s_1 = \sup A$ and $s_2 = \sup A$. Since s_1 is a supremum, it is an upper bound of A, and because s_2 is the least upper bound, $s_2 \leq s_1$. Conversely, s_2 is an upper bound of A, and because s_1 is the least upper bound, $s_1 \leq s_2$. By the antisymmetry of the standard order on $\mathbb{R}$, $s_1 \leq s_2$ and $s_2 \leq s_1$ implies $s_1 = s_2$. □

Detailed Learning Proof.
Step 1. Define two potential candidates for the supremum. Let $A \subseteq \mathbb{R}$ be a non-empty subset with $s_1 = \sup A$ and suppose there exists a second supremum $s_2 = \sup A$.
Step 2. Apply the "Least" property to the first candidate. By the definition of supremum, s_2 is an upper bound of A. Since s_1 is the *least* upper bound, it must satisfy $s_1 \leq s_2$.
Step 3. Apply the "Least" property to the second candidate. Similarly, s_1 is an upper bound of A. Since s_2 is also a *least* upper bound, it must satisfy $s_2 \leq s_1$.
Step 4. Conclude using antisymmetry. In the real number system, the relation $\leq$ is antisymmetric. Because $s_1 \leq s_2$ and $s_2 \leq s_1$, the two values must be identical. Thus, $s_1 = s_2$. □

Remark (Proof structure). Direct proof utilizing the minimality of the supremum against itself to force equality via antisymmetry.

Remark (Dependencies).

- Definition of Supremum
- Definition of Upper Bound

Remark (Return). Return to Theorem

Proposition (Maximum Implies Supremum). *If a set $A \subseteq \mathbb{R}$ has a maximum m, then m is also the supremum of A.*

Professional Standard Proof.
Let $m = \max A$. By definition, $m \in A$ and m is an upper bound of A. Let u be any upper bound of A. Since $m \in A$ and u is an upper bound, it follows by the definition of an upper bound that $m \leq u$. Thus, m is the least upper bound of A, so $m = \sup A$. □

Detailed Learning Proof.
Step 1. Establish the upper bound property. Assume $\text{Maximum}(m, A)$. By the definition of a maximum:

$$m \in A \quad \text{and} \quad \forall x \in A, x \leq m$$

The second part of this statement satisfies the first requirement of being a supremum: m is an upper bound of A.
Step 2. Set up the "leastness" requirement. To show m is the *least* upper bound, we must show that any other upper bound u must be greater than or equal to m. Let u be an arbitrary upper bound of A.
Step 3. Use set membership as the pivot. By the definition of an upper bound:

$$\forall x \in A, x \leq u$$

Because $m \in A$ (from Step 1), we can substitute m for x:

$$m \leq u$$

Step 4. Conclude by definition. We have shown that m is an upper bound and that $m \leq u$ for any upper bound u. By the definition of the supremum, $m = \sup A$.
Step 5. Alternative via ε-characterization. Alternatively, let $\varepsilon > 0$ be arbitrary. We need to find $a \in A$ such that $m - \varepsilon < a$. Since $m \in A$ and $m - \varepsilon < m$, we can simply choose $a = m$. Thus, m satisfies the ε-characterization of the supremum. □

Remark (Proof structure). This is a direct proof utilizing the definition of an upper bound. It leverages the fact that a maximum is an element of the set to force any other upper bound to be at least as large as that maximum.

Remark (Dependencies).

- Definition of Maximum
- Definition of Upper Bound
- Definition of Supremum

Remark (Return). Return to Proposition

Proposition (Supremum in the Set Is the Maximum). *Let $s = \sup A$. If $s \in A$, then s is the maximum of A.*

Professional Standard Proof.
Let $A \subseteq \mathbb{R}$ be a non-empty set bounded above, and let $s = \sup A$. By the definition of the supremum, s is an upper bound of A, which implies $s \geq a$ for all $a \in A$.
The definition of a maximum m of a set A requires two conditions: 1. $m \in A$ 2. $m \geq a$ for all $a \in A$
We are given the hypothesis that $s \in A$, satisfying the first condition. From the definition of supremum, we have $s \geq a$ for all $a \in A$, satisfying the second condition. Therefore, s satisfies the definition of the maximum of A. Since the maximum is unique, we conclude that $s = \max A$. □

Remark (Detailed Learning Proof).
Step 1. Identify the "Guard" property of the supremum. By definition, $s = \sup A$ is an upper bound, meaning it stands as a ceiling for all elements in A, so $s \geq a$ for all $a \in A$.
Step 2. State the dual requirements for a maximum. To be a maximum, a value must be both an element of the set ($m \in A$) and an upper bound of the set ($\forall a \in A, m \geq a$).
Step 3. Synthesize the available information. The hypothesis provides $s \in A$, and the property of the supremum provides the upper bound condition $s \geq a$.
Step 4. Apply the definition of maximum. Because s meets both required criteria, it is the maximum. The uniqueness of the maximum ensures this identity is absolute.

Remark (Interpretation). This proof demonstrates the relationship between the "frontier" of a set and its members. While a supremum is the tightest possible guard (the Control), it does not usually have to be part of the set (e.g., $(0, 1)$ has supremum $1 \notin (0, 1)$). However, if that guard actually sits on a member of the set ($s \in A$), it becomes the "tallest" member, thereby satisfying the definition of a maximum.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Lemma

Lemma (Subset Inclusion Preserves Upper Bounds). *Let $A \subseteq B \subseteq \mathbb{R}$. If u is an upper bound of B, then u is an upper bound of A.*

Professional Standard Proof.
Let A and B be subsets of $\mathbb{R}$ such that $A \subseteq B$. Let $u \in \mathbb{R}$ be an upper bound of B. By the definition of an upper bound, $b \leq u$ for all $b \in B$. Since $A \subseteq B$, every element $a \in A$ is also an element of B. It follows that $a \leq u$ for all $a \in A$. Thus, u is an upper bound of A. □

Detailed Learning Proof.
Step 1. Identify the hypotheses and definitions. We are given the set inclusion $A \subseteq B$ and the fact that u is an upper bound for B. This means that for any b chosen from B, $b \leq u$.
Step 2. Relate elements of A to the elements of B. By the definition of a subset, the condition $A \subseteq B$ implies that every element a belonging to A must also be a member of B.
Step 3. Apply the upper bound condition to the subset. Because every $a \in A$ is also an element of B, and we know u is greater than or equal to every element in B, it must be that $a \leq u$ for every $a \in A$.
Step 4. Conclude using the definition of an upper bound. Since u satisfies the upper bound condition for all elements in A, u is, by definition, an upper bound for A. □

Remark (Interpretation). This result demonstrates the monotonicity of the upper bound property relative to set inclusion. Geometrically, if a point u lies to the right of every point in a set B, and A is entirely contained within B, then u must also lie to the right of every point in A. Consequently, the set of upper bounds for B is a subset of the set of upper bounds for A.

Remark (Proof structure). TODO: Record the proof strategy and major logical moves.

Remark (Dependencies).

- TODO: Add mathematical dependency links.

Remark (Return). Return to Theorem

Theorem (Supremum Is Monotone Under Inclusion). *Let A and B be non-empty subsets of $\mathbb{R}$ that are bounded above. If $A \subseteq B$, then $\sup A \leq \sup B$.*

Professional Standard Proof.
Let $s_B = \sup B$. Then s_B is an upper bound of B, so

$$\forall b \in B, \quad b \leq s_B.$$

Since $A \subseteq B$, it follows that

$$\forall a \in A, \quad a \leq s_B,$$

so s_B is an upper bound of A. Let $s_A = \sup A$. Since s_A is the least upper bound of A, and s_B is an upper bound of A, it follows that

$$s_A \leq s_B.$$

□

Detailed Learning Proof.
Step 1. Let $s_B = \sup B$. By definition of supremum, s_B is an upper bound of B, so $b \leq s_B$ for all $b \in B$.
Step 2. Since $A \subseteq B$, every element of A is an element of B. Therefore $a \leq s_B$ for all $a \in A$, so s_B is an upper bound of A.
Step 3. Let $s_A = \sup A$. By definition, s_A is the least upper bound of A.
Step 4. Since s_B is an upper bound of A, the defining property of the least upper bound gives $s_A \leq s_B$. □

Remark (Proof structure). Direct argument using the definition of upper bound and the minimality property of the supremum.

Remark (Dependencies).

- Definition of supremum
- Definition of upper bound
- Definition of subset

Remark (Return). Return to Proposition

Proposition (Upper Bound Comparison with the Supremum). *Let $A \subseteq \mathbb{R}$ be non-empty and bounded above, and let $s = \sup A$. For any $u \in \mathbb{R}$, u is an upper bound of A if and only if $s \leq u$.*

Professional Standard Proof.
($\Rightarrow$) Suppose u is an upper bound of A. Since $s = \sup A$, s is the least upper bound of A, so $s \leq u$.
($\Leftarrow$) Suppose $s \leq u$. Since s is an upper bound of A, we have $a \leq s$ for all $a \in A$. By transitivity of $\leq$ on $\mathbb{R}$, $a \leq u$ for all $a \in A$. Hence u is an upper bound of A. $\square$

Detailed Learning Proof.
Step 1. Let $s = \sup A$. Then s is an upper bound of A, and for every upper bound v of A, one has $s \leq v$.
Step 2. ($\Rightarrow$) Assume u is an upper bound of A. Then u belongs to the set of upper bounds of A. By the defining property of s as the least upper bound, $s \leq u$.
Step 3. ($\Leftarrow$) Assume $s \leq u$. Since s is an upper bound of A, $a \leq s$ for all $a \in A$.
Step 4. By transitivity of the order on $\mathbb{R}$, from $a \leq s$ and $s \leq u$, it follows that $a \leq u$ for all $a \in A$. Therefore u is an upper bound of A. $\square$

Remark (Proof structure). Equivalence proof: each direction follows directly from the definition of supremum and the transitivity of the order on $\mathbb{R}$.

Remark (Dependencies).

- Definition of supremum
- Definition of upper bound
- Order properties of $\mathbb{R}$

Remark (Return). Return to Proposition

Proposition (Every Element Lies Below the Supremum). *If $s = \sup A$, then for every $x \in A$, $x \leq s$.*

Professional Standard Proof.
Let $A \subseteq \mathbb{R}$ be a non-empty set that is bounded above, and let $s = \sup A$. By the definition of the supremum, s is the least upper bound of A. A necessary condition for s to be the supremum is that it satisfies $\mathrm{UpperBound}(s, A)$. By the definition of an upper bound, it follows that $x \leq s$ for every $x \in A$. □

Detailed Learning Proof.
Step 1. Identify the dual nature of the supremum. For a value s to be $\sup A$, it must simultaneously be an upper bound of A and the smallest of all such upper bounds.
Step 2. The condition $\mathrm{UpperBound}(s, A)$ means that s acts as a ceiling for the entire set.
Step 3. Translate the predicate into an inequality. By the formal definition of an upper bound, every element x belonging to A must be less than or equal to that bound. Therefore, $x \leq s$ for all $x \in A$. □

Remark (Proof structure). This is a direct proof utilizing the "Upper Bound" component of the supremum's definition.

Remark (Dependencies).

- Definition of Upper Bound
- Definition of Supremum

Remark (Return). Return to Theorem

Theorem (Infimum-Supremum Comparison). *For any non-empty bounded set $A \subseteq \mathbb{R}$, the infimum of A is less than or equal to the supremum of A.*

Professional Standard Proof.
Let A be a non-empty bounded set in $\mathbb{R}$. By the completeness of $\mathbb{R}$, $i = \inf A$ and $s = \sup A$ exist. Since $A \neq \varnothing$, choose $a \in A$. By the definition of infimum, $i \leq a$. By the definition of supremum, $a \leq s$. By the transitivity of $\leq$, $i \leq a \leq s$, which implies $i \leq s$. □

Detailed Learning Proof.
Step 1. Invoke Completeness. Because A is bounded and non-empty, the Least Upper Bound Property (and its counterpart for infima) ensures that $s = \sup A$ and $i = \inf A$ are well-defined real numbers.
Step 2. Utilize Non-emptiness. The assumption $A \neq \varnothing$ is necessary. It allows us to assert the existence of at least one element $a \in A$.
Step 3. Apply the Lower Bound property. By definition, the infimum is a lower bound for A. Therefore, for the specific element a chosen in Step 2:

$$i \leq a$$

Step 4. Apply the Upper Bound property. By definition, the supremum is an upper bound for A. Therefore, for the same element a:

$$a \leq s$$

Step 5. Chain by Transitivity. Combining the results of Step 3 and Step 4:

$$i \leq a \leq s$$

By the transitivity of the standard order on $\mathbb{R}$, we conclude $i \leq s$. □

Remark (Proof structure). This is a direct proof utilizing a single element of the set as a logical pivot between the lower bound property of the infimum and the upper bound property of the supremum.

Remark (Dependencies).

- Definition of Infimum
- Definition of Supremum
- Least Upper Bound Property of $\mathbb{R}$

Remark (Return). Return to Theorem

Proposition (Order Comparison of Supremum). *Let $A, B \subseteq \mathbb{R}$ be non-empty sets that are bounded above. If for every $x \in A$, there exists $y \in B$ such that $x \leq y$, then $\sup A \leq \sup B$.*

Professional Standard Proof.
Let $s_B = \sup B$. By definition, $y \leq s_B$ for all $y \in B$. By hypothesis, for any $x \in A$, there exists $y \in B$ such that $x \leq y$. By transitivity, $x \leq s_B$ for all $x \in A$, making s_B an upper bound for A. Since $\sup A$ is the least upper bound of A, it follows that $\sup A \leq s_B = \sup B$. □

Detailed Learning Proof.
Step 1. Identify the upper bound of the target set. Let $s_B = \sup B$. By the definition of the supremum as an upper bound, we have:

$$\forall y \in B, y \leq s_B$$

Step 2. Relate elements of A to the upper bound of B. Let $x \in A$ be arbitrary. The hypothesis states that there exists some $y \in B$ such that $x \leq y$. Combining this with Step 1, we have:

$$x \leq y \quad \text{and} \quad y \leq s_B$$

By the transitivity of the order relation on $\mathbb{R}$:

$$x \leq s_B$$

Step 3. Establish that s_B bounds set A. Since x was chosen arbitrarily from A, the inequality $x \leq s_B$ holds for all $x \in A$. Therefore, s_B is an upper bound for the set A.
Step 4. Apply the minimality of the supremum. By the definition of $\sup A$ as the *least* upper bound, it must be less than or equal to any other upper bound of A. Since s_B is an upper bound for A, we conclude:

$$\sup A \leq s_B$$

Substituting back $s_B = \sup B$, we reach the desired conclusion:

$$\sup A \leq \sup B$$

□

Remark (Proof structure). The proof employs a direct argument by showing that the supremum of B serves as an upper bound for A, then invoking the "leastness" property of the supremum of A.

Remark (Dependencies).

- Definition of Supremum
- Definition of Upper Bound
- Ordered Field Transitivity

Remark (Return). Return to Theorem

Theorem (Translation Invariance of the Supremum). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $c \in \mathbb{R}$. Then*

$$\sup(A + c) = \sup A + c, \qquad A + c = \{a + c : a \in A\}.$$

Professional Standard Proof.
TODO: professional standard proof for translation-invariance-supremum. □

Detailed Learning Proof.
TODO: detailed learning proof for translation-invariance-supremum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Scalar Multiplication by a Positive Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded above, and let $k > 0$. Then*

$$\sup(kA) = k \sup A, \qquad kA = \{ka : a \in A\}.$$

Professional Standard Proof.
TODO: professional standard proof for scalar-mult-positive. ☐

Detailed Learning Proof.
TODO: detailed learning proof for scalar-mult-positive. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Scalar Multiplication by a Negative Scalar). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded below, and let $k < 0$. Then*

$$\sup(kA) = k \inf A, \qquad kA = \{ka : a \in A\}.$$

Professional Standard Proof.
TODO: professional standard proof for scalar-mult-negative. □

Detailed Learning Proof.
TODO: detailed learning proof for scalar-mult-negative. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Negation Exchanges Supremum and Infimum). *Let $A \subseteq \mathbb{R}$ be a non-empty set. If A is bounded below, then* $\sup(-A) = -\inf A$*.*

Professional Standard Proof.
Let $i = \inf A$. By definition, $i \leq a$ for all $a \in A$. Multiplying by -1 reverses the inequality, giving $-i \geq -a$ for all $-a \in -A$. Thus, $-i$ is an upper bound of $-A$. To show it is the least upper bound, let u be any upper bound of $-A$. Then $u \geq -a$ for all $a \in A$, which implies $-u \leq a$. Thus, $-u$ is a lower bound of A. Since i is the greatest lower bound, $i \geq -u$, which implies $-i \leq u$. Hence $\sup(-A) = -i = -\inf A$. □

Detailed Learning Proof.
Step 1. Show $-i$ is an upper bound for the reflected set. Let $i = \inf A$. For any element $x \in -A$, there exists an element $a \in A$ such that $x = -a$. By the definition of the infimum, $i \leq a$. Multiplying both sides by -1 reverses the inequality:

$$-i \geq -a$$

Since $x = -a$, we have $-i \geq x$ for all $x \in -A$, meaning $-i$ is an upper bound of $-A$.
Step 2. Establish "Leastness" using the definition of infimum. To prove $-i$ is the *least* upper bound, we assume there is some other upper bound u for $-A$. By definition, $u \geq x$ for all $x \in -A$. This means $u \geq -a$ for all $a \in A$. Multiplying by -1 again:

$$-u \leq a \quad \forall a \in A$$

This identifies $-u$ as a lower bound for the original set A.
Step 3. Conclude by comparing the bounds. We know i is the **Greatest Lower Bound** of A. Since $-u$ is just ***a*** lower bound, it cannot exceed the greatest one:

$$i \geq -u$$

Multiplying by -1 a final time flips the inequality back:

$$-i \leq u$$

Because $-i$ is less than or equal to any arbitrary upper bound u, it must be the least upper bound of $-A$. □

Remark (Strategy: The Choice of Leastness Method). In this proof, we establish the "leastness" property by comparing $-i$ against an arbitrary upper bound u (the "Side A" method) rather than using the ε-characterization ("Side B"). While the ε-characterization is mathematically equivalent, it often requires "double negation" logic hereshowing that $a < i + \varepsilon$ implies $-a > -i - \varepsilon$. When a proof already involves a set transformation like negation, it is often procedurally cleaner to use the direct definition of the supremum to avoid the mental overhead of tracking flipped inequalities across a tolerance variable.

Remark (Proof structure). The proof follows a direct argument by showing $-\inf A$ satisfies both conditions of the supremum for set $-A$: it is an upper bound, and it is less than or equal to any other upper bound.

Remark (Dependencies).

- Definition of Infimum
- Definition of Supremum
- Order Reversal under Negation

Remark (Return). Return to Theorem

Theorem (Supremum of a Sum Set). *Let* $A, B \subseteq \mathbb{R}$ *be non-empty sets bounded above. Define* $A + B = \{a + b : a \in A, b \in B\}$. *Then* $\sup(A + B) = \sup A + \sup B$.

Professional Standard Proof.
Let $s_a = \sup A$ and $s_b = \sup B$. For any $x \in A + B$, $x = a + b$ for some $a \in A, b \in B$. Since $a \leq s_a$ and $b \leq s_b$, we have $a + b \leq s_a + s_b$, so $s_a + s_b$ is an upper bound of $A + B$. Thus $\sup(A + B) \leq s_a + s_b$. Conversely, for any $\varepsilon > 0$, there exist $a \in A$ and $b \in B$ such that $a > s_a - \varepsilon/2$ and $b > s_b - \varepsilon/2$. Then $a + b > s_a + s_b - \varepsilon$. Since $a + b \in A + B$, this shows $\sup(A + B) \geq s_a + s_b$. Therefore, $\sup(A + B) = s_a + s_b$. □

Detailed Learning Proof.
Step 1. Show $s_a + s_b$ is an upper bound for $A + B$. Let $s_a = \sup A$ and $s_b = \sup B$. By definition of the supremum:

$$\forall a \in A, a \leq s_a \quad \text{and} \quad \forall b \in B, b \leq s_b$$

Adding these inequalities, for any $a \in A, b \in B$:

$$a + b \leq s_a + s_b$$

Since every element of $A + B$ is of the form $a + b$, $s_a + s_b$ is an upper bound for $A + B$. Consequently:

$$\sup(A + B) \leq s_a + s_b$$

Step 2. Use ε-characterization to show $s_a + s_b$ is the least upper bound. Let $\varepsilon > 0$ be arbitrary. By the ε-characterization of $\sup A$ and $\sup B$, there exist elements $a \in A$ and $b \in B$ such that:

$$a > s_a - \frac{\varepsilon}{2} \quad \text{and} \quad b > s_b - \frac{\varepsilon}{2}$$

Summing these gives:

$$a + b > (s_a - \frac{\varepsilon}{2}) + (s_b - \frac{\varepsilon}{2}) = s_a + s_b - \varepsilon$$

Since $a + b \in A + B$, we have found an element in the sum set strictly greater than $s_a + s_b - \varepsilon$.
Step 3. Conclude the equality. Step 2 shows that no value smaller than $s_a + s_b$ can be an upper bound for $A + B$. Thus, the least upper bound must be $s_a + s_b$. □

Remark (Proof structure). The proof utilizes a double-inequality strategy: first proving $\sup(A + B) \leq \sup A + \sup B$ via the upper bound property, and then $\sup(A + B) \geq \sup A + \sup B$ via the ε-characterization of the supremum.

Remark (Dependencies).

- Definition of Supremum
- Definition of Upper Bound
- ε-characterization of Supremum

Remark (Return). Return to Theorem

Theorem (Supremum of a Difference Set). *Let $A, B \subseteq \mathbb{R}$ be nonempty, with A bounded above and B bounded below. Then*

$$\sup(A - B) = \sup A - \inf B, \qquad A - B = \{a - b : a \in A,\ b \in B\}.$$

Professional Standard Proof.
TODO: professional standard proof for supremum-difference-set. □

Detailed Learning Proof.
TODO: detailed learning proof for supremum-difference-set. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Supremum of a Dilation). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded, and let $\lambda \in \mathbb{R}$. Then*

$$\sup(\lambda A) = \max\{\lambda \sup A, \lambda \inf A\}, \qquad \lambda A = \{\lambda a : a \in A\}.$$

Professional Standard Proof.
TODO: professional standard proof for supremum-dilation. $\square$

Detailed Learning Proof.
TODO: detailed learning proof for supremum-dilation. $\square$

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Supremum of the Absolute-Value Image). *Let $A \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\sup |A| = \max\{|\sup A|, |\inf A|\}, \qquad |A| = \{|a| : a \in A\}.$$

Professional Standard Proof.
TODO: professional standard proof for supremum-absolute-value-image. ☐

Detailed Learning Proof.
TODO: detailed learning proof for supremum-absolute-value-image. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Bounds for Suprema and Infima Under Set Addition). *Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded. Then*

$$\inf A + \inf B \leq \inf(A + B) \qquad \textit{and} \qquad \sup(A + B) \leq \sup A + \sup B.$$

Professional Standard Proof.
TODO: professional standard proof for bounds-sum-set. ☐

Detailed Learning Proof.
TODO: detailed learning proof for bounds-sum-set. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Upper Bounds Depend on the Ambient Order). *Let $(S, \leq_S)$ and $(T, \leq_T)$ be ordered sets, and let $A \subseteq S \cap T$. Whether u is an upper bound of A must be interpreted relative to the chosen ambient order.*

Professional Standard Proof.
TODO: professional standard proof for upper-bounds-ambient-order. □

Detailed Learning Proof.
TODO: detailed learning proof for upper-bounds-ambient-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Suprema Depend on the Ambient Set). *Let $A \subseteq S \subseteq S'$ be subsets of an ordered set. The supremum of A relative to S, if it exists, need not agree with the supremum of A relative to S'.*

Professional Standard Proof.
TODO: professional standard proof for suprema-ambient-set. □

Detailed Learning Proof.
TODO: detailed learning proof for suprema-ambient-set. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Ambient Existence of Supremum). *There are ordered sets $S \subseteq S'$ and a subset $A \subseteq S$ such that A has a supremum relative to S' but has no supremum relative to S.*

Professional Standard Proof.
TODO: professional standard proof for ambient-existence-supremum. ☐

Detailed Learning Proof.
TODO: detailed learning proof for ambient-existence-supremum. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Lemma (Rational Gap Example for Suprema). *Let*

$$A = \{q \in \mathbb{Q} : q^2 < 2\}.$$

Then A is nonempty and bounded above in $\mathbb{Q}$, but A has no supremum relative to $\mathbb{Q}$.

Professional Standard Proof.
TODO: professional standard proof for rational-gap-suprema. ☐

Detailed Learning Proof.
TODO: detailed learning proof for rational-gap-suprema. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Bounding, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 3

Sequences

sequences

Functions → **Sequences** → Bounds

Sequence Definitions

3.1 Definitions

Definition (Sequence)

Definition 3.1.1 (Sequence). Let X be a nonempty set. A **sequence** in X is a function

$$x : \mathbb{N} \to X.$$

The value $x(n)$ is written x_n and called the **n-th term** of the sequence. The sequence itself is denoted $(x_n)_{n\in\mathbb{N}}$, or simply (x_n) when the index set is clear.

Remark (Standard quantified statement).

$$\forall X \neq \varnothing\ \forall x : \mathbb{N} \to X\ \big(\mathrm{Sequence}(x, X) \iff x : \mathbb{N} \to X\big).$$

Remark (Definition predicate reading).

$$\mathrm{Sequence}(x_n, X)\ :=\ x : \mathbb{N} \to X.$$

Remark (Interpretation). A sequence is not a set of terms but a function: the index carries independent weight. Two sequences (x_n) and (y_n) in X are equal precisely when $x_n = y_n$ for every $n \in \mathbb{N}$, so order and repetition are intrinsic to the object. The notational collapse from $x(n)$ to x_n is conventional suppression of function-application syntax; the underlying object remains a function $\mathbb{N} \to X$. When $X = \mathbb{R}$, the sequence is called a **real sequence**, which is the default ambient context for this chapter.

Remark (Dependencies).

- Function
- Natural Numbers

Example (Constant Sequence). Let $c \in \mathbb{R}$. The **constant sequence** is defined by

$$x_n := c \quad \forall n \in \mathbb{N}.$$

The constant sequence converges to c: given any $\varepsilon > 0$, take $N = 1$, and observe $|x_n - c| = 0 < \varepsilon$ for all $n \geq 1$. It is the degenerate case of convergence in which the tail is not merely eventually close to L but identically equal to it from the first term.

Example (The Sequence $(1/n)$). The sequence (x_n) defined by

$$x_n := \frac{1}{n} \quad \forall n \in \mathbb{N}$$

converges to 0. Given $\varepsilon > 0$, the Archimedean property supplies $N \in \mathbb{N}$ with $N > 1/\varepsilon$, and then for all $n \geq N$,

$$\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon.$$

This is the canonical example of a sequence that approaches its limit monotonically from above, never reaching it. It also serves as the standard witness in squeeze arguments and in the proof that $\inf\{1/n : n \in \mathbb{N}\} = 0$.

Example (The Sequence (n)). The sequence (x_n) defined by

$$x_n := n \quad \forall n \in \mathbb{N}$$

is divergent. For any proposed limit $L \in \mathbb{R}$, take $\varepsilon = 1$; the Archimedean property supplies $n \in \mathbb{N}$ with $n > L + 1$, so $|x_n - L| = |n - L| > 1 = \varepsilon$ for infinitely many n. The sequence is unbounded, and every convergent sequence is bounded, so divergence also follows from that theorem once it is established. This is the canonical example of divergence to $+\infty$: the terms are not oscillating outside a neighborhood of L but growing without bound in a single direction.

Null and Constant Sequences

Null and Constant Sequences

Definition (Constant Sequence)

Definition 3.1.2 (Constant Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **constant** exactly when there exists $c \in \mathbb{R}$ such that $x_n = c$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

Let (x_n) be a real sequence.
(x_n) is constant $\iff \exists c \in \mathbb{R}\, \forall n \in \mathbb{N}\, (x_n = c)$.

Remark (Definition predicate reading).

$$\mathrm{ConstantSequence}(x_n, c) := \forall n \in \mathbb{N}\, (x_n = c).$$

Remark (Negated quantified statement).

Let (x_n) be a real sequence.
(x_n) is not constant $\iff \forall c \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n \neq c)$.

Remark (Negation predicate reading).

$$\neg \exists c \in \mathbb{R}\ \mathrm{ConstantSequence}(x_n, c).$$

Remark (Failure modes). Every proposed constant value fails at some index.

Remark (Failure mode decomposition).

$$\forall c \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (x_n \neq c)}_{\text{a term differs from the proposed constant value}} \qquad .$$

Remark (Interpretation). A constant sequence has no genuine dependence on the index. The indexing is still part of the object, but every index is assigned the same real value.

Remark (Dependencies).

- Sequence

Definition (Null Sequence)

Definition 3.1.3 (Null Sequence). Let (x_n) be a real sequence. The sequence (x_n) is a **null sequence** exactly when for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for every $n \geq N$.

Remark (Standard quantified statement).

Let (x_n) be a real sequence.
(x_n) is a null sequence $\iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)$.

Remark (Definition predicate reading).

$$\mathrm{NullSequence}(x_n) :\equiv \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).$$

Remark (Negated quantified statement).

Let (x_n) be a real sequence.
(x_n) is not a null sequence $\iff \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ (|x_n| \geq \varepsilon)$.

Remark (Negation predicate reading).

$$\neg\,\mathrm{NullSequence}(x_n).$$

Remark (Failure modes). Some positive tolerance is missed infinitely far out in the sequence.

Remark (Failure mode decomposition).

$$\exists \varepsilon > 0\ \forall N \in \mathbb{N} \qquad \underbrace{\exists n \geq N\ (|x_n| \geq \varepsilon)}_{\text{the tail is not trapped inside the } \varepsilon\text{-band}} \qquad .$$

Remark (Interpretation). A null sequence is a sequence whose tail is eventually contained in every symmetric neighborhood of zero. The first finitely many terms are irrelevant; only the eventual behavior matters.

Remark (Dependencies).

- Sequence

Theorem (Constant Sequence Convergence)

Theorem 3.1.4 (Constant Sequence Convergence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and } c \in \mathbb{R}.$$
$$\big(\forall n \in \mathbb{N}\ (x_n = c)\big) \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - c| < \varepsilon).$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence is already equal to its proposed limit at every index. Every positive tolerance is therefore satisfied by any tail of the sequence.

Remark (Dependencies).

- Constant Sequence
- Convergent Sequence

Theorem (Zero Sequence is Null)

Theorem 3.1.5 (Zero Sequence is Null). *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\big(\forall n \in \mathbb{N}\ (x_n = 0)\big) \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

Remark (Interpretation). The zero sequence satisfies every null-sequence tolerance immediately because the distance from each term to zero is exactly zero.

Remark (Dependencies).

- Null Sequence
- Constant Sequence

Theorem (Constant Null Sequence)

Theorem 3.1.6 (Constant Null Sequence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

Let (x_n) be a real sequence and $c \in \mathbb{R}$.

$$\big(\forall n \in \mathbb{N}\ (x_n = c)\big) \implies \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \iff c = 0\Big).$$

Remark (Predicate reading).

$$\mathrm{ConstantSequence}(x_n, c) \implies \big(\mathrm{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). A sequence that never changes can approach zero only when its fixed value is already zero. Conversely, the fixed value zero gives the zero sequence.

Remark (Dependencies).

- Constant Sequence
- Null Sequence

Definition (Ultimately Constant Sequence)

Definition 3.1.7 (Ultimately Constant Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **ultimately constant** exactly when there exist $N \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N$.

Remark (Standard quantified statement).

Let (x_n) be a real sequence.

(x_n) is ultimately constant $\iff \exists N \in \mathbb{N}\ \exists c \in \mathbb{R}\ \forall n \geq N\ (x_n = c)$.

Remark (Definition predicate reading).

$$\mathrm{UltimatelyConstantSequence}(x_n, c) :\!= \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n = c).$$

Remark (Negated quantified statement).

Let (x_n) be a real sequence.

(x_n) is not ultimately constant $\iff \forall N \in \mathbb{N}\ \forall c \in \mathbb{R}\ \exists n \geq N\ (x_n \neq c)$.

Remark (Negation predicate reading).

$$\neg\exists c \in \mathbb{R}\ \mathrm{UltimatelyConstantSequence}(x_n, c).$$

Remark (Failure modes). No tail of the sequence becomes permanently equal to a single real value.

Remark (Failure mode decomposition).

$$\forall N \in \mathbb{N}\, \forall c \in \mathbb{R} \quad \underbrace{\exists n \geq N\ (x_n \neq c)}_{\text{the proposed tail value fails later in the sequence}}.$$

Remark (Interpretation). An ultimately constant sequence may vary on an initial finite segment, but its tail becomes fixed. The concept isolates eventual behavior from finite initial behavior.

Remark (Dependencies).

- Sequence

Theorem (Difference from the Limit is Null)

Theorem 3.1.8 (Difference from the Limit is Null). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence and } L \in \mathbb{R}.\\ &\big[\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\ (|x_n - L| < \varepsilon)\big]\\ &\qquad \Longleftrightarrow \qquad \big[\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\ (|x_n - L| < \varepsilon)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \Longleftrightarrow \mathrm{NullSequence}(x_n - L).$$

Remark (Interpretation). Subtracting the proposed limit translates the convergence problem to the special case of convergence to zero. Null sequences are therefore the local normal form for sequence convergence.

Remark (Dependencies).

- Convergent Sequence
- Null Sequence
- Pointwise Difference of Sequences

Theorem (Ultimately Constant Sequence Convergence)

Theorem 3.1.9 (Ultimately Constant Sequence Convergence). *Let (x_n) be a real sequence. If there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$, then (x_n) converges to c.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall c \in \mathbb{R} \Big([\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)] \implies [\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - c| < \varepsilon)] \Big).$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies \text{ConvergentSequence}(x_n, c).$$

Remark (Interpretation). Once the tail is equal to a fixed value, all sufficiently late terms are already exactly at the proposed limit.

Remark (Dependencies).

- Ultimately Constant Sequence
- Convergent Sequence

Theorem (Constant Implies Ultimately Constant)

Theorem 3.1.10 (Constant Implies Ultimately Constant). *Every constant real sequence is ultimately constant.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$[\exists c \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n = c)] \implies [\exists N \in \mathbb{N}\ \exists c \in \mathbb{R}\ \forall n \geq N\ (x_n = c)].$$

Remark (Predicate reading).

$$\text{ConstantSequence}(x_n, c) \implies \text{UltimatelyConstantSequence}(x_n, c).$$

Remark (Interpretation). A constant sequence has already stabilized from the first index, so it is automatically stabilized on some tail.

Remark (Dependencies).

- Constant Sequence
- Ultimately Constant Sequence

Theorem (Ultimately Zero Sequence is Null)

Theorem 3.1.11 (Ultimately Zero Sequence is Null). *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = 0)\big] \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon).$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, 0) \implies \text{NullSequence}(x_n).$$

Remark (Interpretation). If the tail is identically zero, then every sufficiently late term lies inside every positive tolerance around zero.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Ultimately Constant Null Sequence)

Theorem 3.1.12 (Ultimately Constant Null Sequence). *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall c \in \mathbb{R}\ \Big(\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = c)\big] \implies \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n| < \varepsilon)\big] \iff c = 0\Big)\Big).$$

Remark (Predicate reading).

$$\text{UltimatelyConstantSequence}(x_n, c) \implies \big(\text{NullSequence}(x_n) \iff c = 0\big).$$

Remark (Interpretation). Only the eventual value controls whether an ultimately constant sequence is null. A finite initial segment cannot prevent or create null behavior.

Remark (Dependencies).

- Ultimately Constant Sequence
- Null Sequence

Theorem (Tail Equality Preserves Convergence)

Theorem 3.1.13 (Tail Equality Preserves Convergence). *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

Let (x_n) and (y_n) be real sequences.

$$\begin{aligned}&\big[\exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (x_n = y_n)\big]\\ &\qquad \implies \quad \forall L \in \mathbb{R}\ \Big(\big[\forall \varepsilon > 0\ \exists N_x \in \mathbb{N}\ \forall n \geq N_x\ (|x_n - L| < \varepsilon)\big]\\ &\qquad\qquad \iff \quad \big[\forall \varepsilon > 0\ \exists N_y \in \mathbb{N}\ \forall n \geq N_y\ (|y_n - L| < \varepsilon)\big]\Big).\end{aligned}$$

Remark (Predicate reading).

$$\text{EventuallyEqual}(x_n, y_n) \implies \forall L \in \mathbb{R}\ \big(\text{ConvergentSequence}(x_n, L) \iff \text{ConvergentSequence}(y_n, L)\big).$$

Remark (Interpretation). Convergence is a tail property. Changing finitely many initial terms cannot change whether a sequence converges to a specified real number.

Remark (Dependencies).

- Convergent Sequence

Definition (Sequence Bounded Above)

Definition 3.1.14 (Sequence Bounded Above). Let (x_n) be a real sequence. The sequence (x_n) is **bounded above** exactly when there exists $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

Let (x_n) be a real sequence.
(x_n) is bounded above $\iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)$.

Remark (Definition predicate reading).

$$\text{BoundedAboveSequence}(x_n) :\equiv \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M).$$

Remark (Negated quantified statement).

Let (x_n) be a real sequence.
(x_n) is not bounded above $\iff \forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n)$.

Remark (Negation predicate reading).

$$\neg\,\text{BoundedAboveSequence}(x_n).$$

Remark (Failure modes). Every proposed upper bound is exceeded by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R} \qquad \underbrace{\exists n \in \mathbb{N}\ (M < x_n)}_{\texttt{a term rises above the proposed ceiling}} \qquad .$$

Remark (Interpretation). A bounded-above sequence has a finite ceiling. Failure means no real number serves as a ceiling for all terms.

Remark (Dependencies).

- Sequence

Definition (Sequence Bounded Below)

Definition 3.1.15 (Sequence Bounded Below). Let (x_n) be a real sequence. The sequence (x_n) is **bounded below** exactly when there exists $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$(x_n) \text{ is bounded below} \iff \exists m \in \mathbb{R}\, \forall n \in \mathbb{N}\, (m \leq x_n).$$

Remark (Definition predicate reading).

$$\text{BoundedBelowSequence}(x_n) \coloneqq \exists m \in \mathbb{R}\, \forall n \in \mathbb{N}\, (m \leq x_n).$$

Remark (Negated quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$(x_n) \text{ is not bounded below} \iff \forall m \in \mathbb{R}\, \exists n \in \mathbb{N}\, (x_n < m).$$

Remark (Negation predicate reading).

$$\neg\, \text{BoundedBelowSequence}(x_n).$$

Remark (Failure modes). Every proposed lower bound is undercut by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall m \in \mathbb{R} \quad \underbrace{\exists n \in \mathbb{N}\, (x_n < m)}_{\text{a term falls below the proposed floor}} .$$

Remark (Interpretation). A bounded-below sequence has a finite floor. Failure means no real number serves as a floor for all terms.

Remark (Dependencies).

- Sequence

Definition (Bounded Sequence)

Definition 3.1.16 (Bounded Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **bounded** exactly when there exists $M > 0$ such that $|x_n| \leq M$ for every $n \in \mathbb{N}$.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$(x_n) \text{ is bounded} \iff \exists M > 0\, \forall n \in \mathbb{N}\, (|x_n| \leq M).$$

Remark (Definition predicate reading).

$$\mathrm{BoundedSequence}(x_n) :\equiv \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&(x_n) \text{ is not bounded} \iff \forall M > 0\ \exists n \in \mathbb{N}\ (|x_n| > M).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedSequence}(x_n).$$

Remark (Failure modes). Every symmetric bound around zero is escaped by some term of the sequence.

Remark (Failure mode decomposition).

$$\forall M > 0 \qquad \underbrace{\exists n \in \mathbb{N}\ (|x_n| > M)}_{\text{a term escapes the symmetric band of radius } M} \qquad .$$

Remark (Interpretation). A bounded sequence is trapped in a finite symmetric band around zero. This is equivalent to having both a finite ceiling and a finite floor.

Remark (Dependencies).

- Sequence

Theorem (Eventually Bounded Above Tail Formulation)

Theorem 3.1.17 (Eventually Bounded Above Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \le M$ for every $n \ge N_0$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big]\\&\qquad \iff \qquad \big[\exists N_0 \in \mathbb{N}\ \exists M \in \mathbb{R}\ \forall n \ge N_0\ (x_n \le M)\big].\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BoundedAboveSequence}(x_n) \iff \exists M \in \mathbb{R}\ \mathrm{Eventually}\big(x_n,\ x_n \le M\big).$$

Remark (Interpretation). One-sided boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a larger upper bound.

Remark (Dependencies).

- Sequence Bounded Above

Theorem (Eventually Bounded Below Tail Formulation)

Theorem 3.1.18 (Eventually Bounded Below Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \le x_n$ for every $n \ge N_0$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\big[\exists m \in \mathbb{R}\, \forall n \in \mathbb{N}\, (m \le x_n)\big]$$
$$\iff \quad \big[\exists N_0 \in \mathbb{N}\, \exists m \in \mathbb{R}\, \forall n \ge N_0\, (m \le x_n)\big].$$

Remark (Predicate reading).

$$\text{BoundedBelowSequence}(x_n) \iff \exists m \in \mathbb{R}\ \text{Eventually}\big(x_n,\ m \le x_n\big).$$

Remark (Interpretation). One-sided lower boundedness is unchanged by ignoring finitely many initial terms, because finitely many exceptional values can be absorbed into a lower bound.

Remark (Dependencies).

- Sequence Bounded Below

Theorem (Eventually Bounded Tail Formulation)

Theorem 3.1.19 (Eventually Bounded Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \le M$ for every $n \ge N_0$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\big[\exists M > 0\, \forall n \in \mathbb{N}\, (|x_n| \le M)\big]$$
$$\iff \quad \big[\exists N_0 \in \mathbb{N}\, \exists M > 0\, \forall n \ge N_0\, (|x_n| \le M)\big].$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff \exists M > 0\ \text{Eventually}\big(x_n,\ |x_n| \le M\big).$$

Remark (Interpretation). Boundedness is a tail property up to finitely many exceptions. A finite initial segment can be enclosed by increasing the bound.

Remark (Dependencies).

- Bounded Sequence

Theorem (Bounded Sequence If and Only If Bounded Above and Below)

Theorem 3.1.20 (Bounded Sequence If and Only If Bounded Above and Below). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\big[\exists K > 0\, \forall n \in \mathbb{N}\, (|x_n| \le K)\big] \iff \big[\exists m \in \mathbb{R}\, \exists M \in \mathbb{R}\, \forall n \in \mathbb{N}\, (m \le x_n \wedge x_n \le M)\big].$$

Remark (Predicate reading).

$$\text{BoundedSequence}(x_n) \iff \big(\text{BoundedBelowSequence}(x_n) \wedge \text{BoundedAboveSequence}(x_n)\big).$$

Remark (Interpretation). An absolute-value bound gives a symmetric interval around zero. Having both a floor and a ceiling gives an interval that may not be symmetric, but it can be enlarged to a symmetric one.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Absolute Bound Gives Upper and Lower Bounds)

Theorem 3.1.21 (Absolute Bound Gives Upper and Lower Bounds). *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \le K$ for every $n \in \mathbb{N}$, then $-K \le x_n \le K$ for every $n \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall K > 0\, \big[\forall n \in \mathbb{N}\, (|x_n| \le K)\big] \implies \forall n \in \mathbb{N}\, (-K \le x_n \wedge x_n \le K).$$

Remark (Interpretation). An absolute-value inequality simultaneously controls the positive and negative directions. The upper barrier is K, and the lower barrier is $-K$.

Remark (Dependencies).

- Bounded Sequence
- Sequence Bounded Above

- Sequence Bounded Below

Theorem (Upper and Lower Bounds Give an Absolute Bound)

Theorem 3.1.22 (Upper and Lower Bounds Give an Absolute Bound).
Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } (x_n) \text{ be a real sequence.}\\&\big[\exists m \in \mathbb{R}\, \exists M \in \mathbb{R}\, \forall n \in \mathbb{N}\, (m \leq x_n \land x_n \leq M)\big]\\&\qquad\implies \qquad \big[\exists K > 0\, \forall n \in \mathbb{N}\, (|x_n| \leq K)\big].\end{aligned}$$

Remark (Interpretation). Two one-sided barriers can always be replaced by one symmetric barrier around zero. Any positive K at least as large as both $|m|$ and $|M|$ works.

Remark (Dependencies).

- Sequence Bounded Above
- Sequence Bounded Below
- Bounded Sequence

3.2 Examples and Counterexamples

Sequence Examples Toolkit

Concept family: Canonical model sequences and counterexamples.

Atomic items in this section:

- Example: Constant sequence
- Example: Reciprocal sequence
- Example: Alternating null sequence
- Example: Oscillating sequence
- Example: Geometric sequence
- Counterexample: Bounded does not imply convergent
- Counterexample: Successive differences tending to zero do not imply Cauchy

Examples and Counterexamples

Examples provide test objects for the definitions. A good sequence example usually isolates one behavior: settling to a limit, oscillating between several values, escaping to infinity, or satisfying a tempting condition that is still too weak to force convergence.

Example (Constant Sequence). Let $c \in \mathbb{R}$ and define $x_n = c$ for every $n \in \mathbb{N}$. Then (x_n) converges to c.

Remark (Interpretation). A constant sequence is the baseline example of convergence: every term is already equal to the proposed limit.

Example (Reciprocal Sequence). Define $x_n = 1/n$ for every $n \in \mathbb{N}$. Then (x_n) is decreasing, bounded below by 0, and converges to 0.

Remark (Interpretation). The reciprocal sequence is the standard positive null sequence. It is also the model for estimates of the form "eventually smaller than any fixed positive tolerance."

Example (Alternating Null Sequence). Define

$$x_n = \frac{(-1)^n}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is not monotone, but it converges to 0.

Remark (Interpretation). Oscillation in sign does not by itself prevent convergence. The amplitudes must be watched: here the oscillation collapses to zero.

Example (Oscillating Sequence). Define

$$x_n = (-1)^n \qquad \text{for every } n \in \mathbb{N}.$$

Then (x_n) is bounded and divergent. Its even-indexed subsequence converges to 1, and its odd-indexed subsequence converges to -1.

Remark (Interpretation). This is the canonical oscillator. The sequence never settles because two subsequences settle to different limits.

Example (Geometric Sequence). Let $r \in \mathbb{R}$ and define $x_n = r^n$ for every $n \in \mathbb{N}$. Then:

- if $|r| < 1$, then $x_n \to 0$;
- if $r = 1$, then $x_n \to 1$;
- if $r = -1$, then (x_n) oscillates and diverges;
- if $|r| > 1$, then (x_n) is unbounded.

Remark (Interpretation). The geometric sequence is the standard laboratory for rate, sign, and magnitude. The threshold $|r| = 1$ separates decay from growth.

Example (Bounded Does Not Imply Convergent). The sequence $x_n = (-1)^n$ is bounded, since $|x_n| \le 1$ for every $n \in \mathbb{N}$, but it does not converge.

Remark (Interpretation). Boundedness prevents escape to infinity, but it does not prevent persistent oscillation.

Example (Vanishing Successive Differences Do Not Imply Cauchy). Let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} \qquad \text{for every } n \in \mathbb{N}.$$

Then

$$x_{n+1} - x_n = \frac{1}{n+1} \to 0,$$

but (x_n) is not Cauchy.

Remark (Interpretation). Cauchy control requires all sufficiently late terms to be mutually close. Controlling only adjacent jumps is weaker.

Convergence

Definition (Convergence of a Sequence)

Definition 3.2.1 (Convergence of a Sequence). Let $(x_n) \subseteq \mathbb{R}$. The sequence (x_n) **converges** to $L \in \mathbb{R}$ if

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(|x_n - L| < \varepsilon\big).$$

The value L is called the **limit** of (x_n), written $x_n \to L$ or $\lim_{n\to\infty} x_n = L$. A sequence that converges to some $L \in \mathbb{R}$ is called **convergent**; otherwise it is called **divergent**.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\, \forall L \in \mathbb{R}$$
$$\Big(x_n \to L \iff \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(|x_n - L| < \varepsilon\big)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \;:=\; \forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\;\; \mathrm{OutputTolerance}(x_n, L, \varepsilon).$$

Remark (Negated quantified statement).

$$x_n \not\to L \iff \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N$$
$$\big(|x_n - L| \geq \varepsilon\big).$$

Remark (Negation predicate reading).

$$\mathrm{NotConvergentTo}(x_n, L) \;:=\; \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Failure modes). Failure of convergence to a fixed L means that some positive tolerance cannot be controlled on any tail. The escaped terms may oscillate, drift monotonically away, or grow without bound, but all such behavior is captured by the same persistent-tolerance obstruction.

Remark (Failure mode decomposition). The negation decomposes into a single structural obstruction:

$$\underbrace{\exists \varepsilon > 0}_{\texttt{a fixed tolerance}} \ \underbrace{\forall N \in \mathbb{N}}_{\texttt{no tail suffices}} \ \underbrace{\exists n \geq N}_{\texttt{terms escape infinitely often}} \ \underbrace{|x_n - L| \geq \varepsilon}_{\texttt{outside the target interval}} .$$

The sequence fails to converge to L not by drifting away once, but by returning outside the ε-interval infinitely often regardless of how far along the tail one looks.

Remark (Interpretation). The quantifier order $\forall \varepsilon \ \exists N \ \forall n \geq N$ encodes a strict dependency: N is permitted to depend on ε, but not on n. This asymmetry is load-bearing. Reversing the outer quantifiers to $\exists N \ \forall \varepsilon$ would assert that a single index N controls all tolerances simultaneously, which is a strictly stronger and generally false condition. The condition $|x_n - L| < \varepsilon$ is the membership assertion $x_n \in (L - \varepsilon, L + \varepsilon)$; convergence is therefore the statement that the tail of the sequence is eventually contained in every ε-neighborhood of L.

Remark (Dependencies).

- Definition of sequence

Definition (Convergence — Neighborhood Formulation)

Definition 3.2.2 (Convergence -- Neighborhood Formulation). Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The sequence (x_n) **converges** to L if for every $\varepsilon > 0$ the ε-neighborhood

$$V_\varepsilon(L) := (L - \varepsilon, L + \varepsilon)$$

contains all but finitely many terms of (x_n); that is,

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \, \forall L \in \mathbb{R} \\ \Big(x_n \to L \iff \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big)\Big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(x_n, L) := \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, \big(x_n \in V_\varepsilon(L)\big).$$

Remark (Negated quantified statement).

$$x_n \not\to L \iff \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \\ \big(x_n \notin V_\varepsilon(L)\big).$$

Remark (Negation predicate reading).

$$\text{NotConvergentTo}(x_n, L) := \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists n \geq N \, \big(x_n \notin V_\varepsilon(L)\big).$$

Remark (Failure modes). Failure of the neighborhood formulation is exactly failure to trap a tail in every neighborhood of L. Different examples may escape by oscillation, by monotone drift, or by unbounded growth, but the formal failure is always the existence of one neighborhood that every tail misses at least once.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon > 0}_{\text{a fixed neighborhood}} \quad \underbrace{\forall N \in \mathbb{N}}_{\text{no tail suffices}} \quad \underbrace{\exists n \geq N}_{\text{terms escape infinitely often}} \quad \underbrace{x_n \notin V_\varepsilon(L)}_{\text{outside every candidate neighborhood}}.$$

The sequence fails to converge to L precisely when some neighborhood of L contains only finitely many terms: infinitely many terms lie permanently outside it.

Remark (Interpretation). The neighborhood formulation reframes convergence as a set-containment condition rather than an inequality condition. The assertion $x_n \in V_\varepsilon(L)$ is identical to $|x_n - L| < \varepsilon$; the two definitions are logically equivalent and the choice between them is one of language, not content. The neighborhood language is the natural bridge to the topological formulation: in a general topological space, open sets replace ε-neighborhoods and the definition reads identically with $V_\varepsilon(L)$ replaced by any open set U containing L. The phrase *all but finitely many terms* is the canonical prose compression of the quantifier block $\exists N \in \mathbb{N}\, \forall n \geq N$, and is standard in the literature.

Remark (Dependencies).

- Convergence of a Sequence
- ε-neighbourhood

Theorem (Equivalence of Convergence Formulations)

Theorem 3.2.3 (Equivalence of Convergence Formulations). *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(x_n \in V_\varepsilon(L)\big)$.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\, \forall L \in \mathbb{R}$$
$$\Big((a) \iff (b) \iff (c) \iff (d)\Big).$$

Remark (Theorem predicate reading).

$$\mathrm{EquivalentConvergenceForms}(x_n, L) := \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{NeighborhoodConvergentSe}$$

Remark (Interpretation). The four conditions are the same geometric fact written in four notational registers. Condition (a) is the informal assertion. Condition (b) is the canonical ε-K definition: the absolute value formulation. Condition (c) makes the interval structure explicit by expanding $|x_n - L| < \varepsilon$ into its equivalent double inequality $L - \varepsilon < x_n < L + \varepsilon$; this is useful in proofs where upper and lower bounds are handled separately. Condition (d) is the neighborhood formulation: $x_n \in V_\varepsilon(L)$ is identical to conditions (b) and (c) but frames convergence as set membership, which is the natural language for the topological generalization. The index K plays the same role as N in the base definition Definition 3.2.1; the letter changes here to signal that K is the tail offset in the sense of Definition 3.2.36.

Remark (Dependencies).

- Convergence of a Sequence
- Convergence — Neighborhood Formulation

Limits

Theorem (Uniqueness of Limits)

Theorem 3.2.4 (Uniqueness of Limits). *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall K \in \mathbb{R}\ \Big((x_n \to L \ \wedge\ x_n \to K) \implies L = K \Big).$$

Remark (Theorem predicate reading).

$$\text{UniqueSequentialLimit}(x_n, L, K) := \big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K)\big) \implies L$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq \mathbb{R}\ \exists L \in \mathbb{R}\ \exists K \in \mathbb{R}\ \Big((x_n \to L \ \wedge\ x_n \to K) \ \wedge\ L \neq K \Big).$$

Remark (Negation predicate reading).

$$\neg\, \text{UniqueSequentialLimit}(x_n, L, K) := \text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(x_n, K) \wedge L \neq$$

Remark (Failure modes). The only failure mode is the existence of one real sequence with two distinct real limits. In $\mathbb{R}$ this is impossible because distinct real numbers can be separated by disjoint neighborhoods.

Remark (Failure mode decomposition).

$$\underbrace{\exists (x_n) \subseteq \mathbb{R}}_{\texttt{some sequence}} \; \underbrace{x_n \to L \;\wedge\; x_n \to K}_{\texttt{converges to two values}} \; \underbrace{L \neq K}_{\texttt{distinct limits}} \quad .$$

The theorem asserts this configuration is impossible in $\mathbb{R}$. The failure mode is not merely exotic: it occurs in non-Hausdorff topological spaces where distinct points cannot be separated by disjoint open sets. The proof that it cannot occur in $\mathbb{R}$ is precisely an argument that $\mathbb{R}$ has enough room to separate any two distinct points by disjoint ε-neighborhoods.

Remark (Interpretation). The theorem licenses the notation $\lim_{n\to\infty} x_n = L$: the definite article is justified precisely because the limit, when it exists, is unique. Without uniqueness the limit would be a relation rather than a function, and the notation would be ambiguous. The proof strategy is a separation argument: if $L \neq K$ then $|L - K| > 0$, and choosing $\varepsilon = |L - K|/2$ produces two disjoint neighborhoods that the tail of (x_n) cannot simultaneously occupy. The Hausdorff remark in the failure mode block is the forward pointer: when sequences are studied in general topological spaces, uniqueness of limits is equivalent to the Hausdorff separation axiom.

Remark (Dependencies).

- Convergence of a Sequence
- Triangle inequality

Theorem (Limit Preserves Eventual Order)

Theorem 3.2.5 (Limit Preserves Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \le y_n$ for every $n \ge N_0$, then $L \le M$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n),(y_n) \subseteq \mathbb{R}\; \forall L, M \in \mathbb{R}\\ &\quad \Big(x_n \to L \;\wedge\; y_n \to M \;\wedge\; \exists N_0 \in \mathbb{N}\; \forall n \ge N_0\; (x_n \le y_n)\Big)\\ &\quad \Longrightarrow \quad L \le M.\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(y_n, M) \wedge \mathrm{Eventually}(x_n,\ x_n \le y_n)\big) \Longrightarrow L \le M.$$

Remark (Interpretation). An inequality that holds on a tail survives passage to the limit, but only as a weak inequality. Strict inequalities on the terms may collapse at the limit, as in $0 < x_n$ with $x_n \to 0$.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Strict Limit Separation Gives Eventual Order)

Theorem 3.2.6 (Strict Limit Separation Gives Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R}\\ &\Big(x_n \to A\ \wedge\ y_n \to B\ \wedge\ A < B\Big)\\ &\implies\quad \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n).\end{aligned}$$

Remark (Theorem predicate reading).

$$\Big(\mathrm{ConvergentSequence}(x_n, A) \wedge \mathrm{ConvergentSequence}(y_n, B) \wedge A < B\Big) \implies \mathrm{Eventually}(x_n,\ x_n < y_n). \blacksquare$$

Remark (Interpretation). If the limiting values have a positive gap, then sufficiently late terms inherit the same order. The proof uses the open interval between A and B: eventually (x_n) lies near A and (y_n) lies near B, with room left between the two tails.

Remark (Dependencies).

- Convergence of a Sequence
- Uniqueness of Limits

Theorem (Eventual Strict Comparison Preserves Weak Limit Order)

Theorem 3.2.7 (Eventual Strict Comparison Preserves Weak Limit Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n), (y_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R}\\ &\quad\Big(x_n \to A\ \wedge\ y_n \to B\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < y_n)\Big) \implies A \leq B,\\ &\quad\Big(x_n \to A\ \wedge\ y_n \to B\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n > y_n)\Big) \implies A \geq B.\end{aligned}$$

Remark (Theorem predicate reading).

$$\big(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n,\ x_n < y_n)\big) \implies A \leq B,$$
$$\big(\text{ConvergentSequence}(x_n, A) \wedge \text{ConvergentSequence}(y_n, B) \wedge \text{Eventually}(x_n,\ x_n > y_n)\big) \implies A \geq B.$$

Remark (Interpretation). Strict comparison of terms does not force strict comparison of limits. The strict inequality can collapse in the limit, so the conclusion is weak: for example $1/n > 0$ for every $n \in \mathbb{N}$, but both sides have limit 0.

Remark (Dependencies).

- Limit Preserves Eventual Order

Theorem (Constant Comparison for Sequence Limits)

Theorem 3.2.8 (Constant Comparison for Sequence Limits). *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \leq B$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \geq N$, then $A \leq B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \geq B$ for every $n \geq N$, then $A \geq B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \geq N$, then $A \geq B$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall A, B \in \mathbb{R} \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq B)\Big) \implies A \leq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < B)\Big) \implies A \leq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \geq B)\Big) \implies A \geq B, \\
&\quad \Big(x_n \to A\ \wedge\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n > B)\Big) \implies A \geq B.
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n \leq B) \implies A \leq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n < B) \implies A \leq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n \geq B) \implies A \geq B, \\
&\text{ConvergentSequence}(x_n, A) \wedge \text{Eventually}(x_n,\ x_n > B) \implies A \geq B.
\end{aligned}$$

Remark (Interpretation). This is the constant version of the comparison theorem. A fixed real number B may be treated as the constant sequence with value B, so each conclusion is a direct application of eventual order preservation. Again, strict term inequalities produce only weak inequalities between limits.

Remark (Dependencies).

- Constant Sequence
- Limit Preserves Eventual Order
- Eventual Strict Comparison Preserves Weak Limit Order

Theorem (Constant Squeeze Theorem)

Theorem 3.2.9 (Constant Squeeze Theorem). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \le x_n \le L$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\\ &\quad \Big(\exists N_0 \in \mathbb{N}\ \forall n \ge N_0\ (L \le x_n \le L)\Big)\\ &\quad \Longrightarrow \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \ge N\ \big(|x_n - L| < \varepsilon\big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n,\ L \le x_n \le L) \Longrightarrow \text{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). This is the degenerate squeeze theorem: the lower and upper barriers are the same constant sequence. It says that eventual equality to a real number is enough for convergence to that real number.

Remark (Dependencies).

- Convergence of a Sequence
- Constant Sequence

Theorem (Sequence Squeeze Theorem)

Theorem 3.2.10 (Sequence Squeeze Theorem). *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \le x_n \le b_n$ for every $n \ge N_0$, then $x_n \to L$.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (a_n),(x_n),(b_n)\subseteq \mathbb{R}\ \forall L\in\mathbb{R}\\ &\quad\Big(a_n\to L\ \wedge\ b_n\to L\ \wedge\ \exists N_0\in\mathbb{N}\ \forall n\ge N_0\ (a_n\le x_n\le b_n)\Big)\\ &\quad\Longrightarrow\quad \forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall n\ge N\ \big(|x_n-L|<\varepsilon\big).\end{aligned}$$

Remark (Theorem predicate reading).

$\big(\mathrm{ConvergentSequence}(a_n,L)\wedge\mathrm{ConvergentSequence}(b_n,L)\wedge\mathrm{Eventually}(x_n,\ a_n\le x_n\le b_n)\big)\Longrightarrow \mathrm{Converg}$

Remark (Interpretation). The middle sequence is trapped between two tails that approach the same number. Once both barriers lie inside an ε-neighborhood of L, the trapped sequence must also lie inside that neighborhood.

Remark (Dependencies).

- Convergence of a Sequence
- Limit Preserves Eventual Order

Theorem (Absolute-Value Squeeze Theorem)

Theorem 3.2.11 (Absolute-Value Squeeze Theorem). *Let (x_n) and (u_n) be real sequences, and let $L\in\mathbb{R}$. Suppose that $u_n\to 0$. If there exists $N_0\in\mathbb{N}$ such that $|x_n-L|\le u_n$ for every $n\ge N_0$, then $x_n\to L$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n),(u_n)\subseteq \mathbb{R}\ \forall L\in\mathbb{R}\\ &\quad\Big(u_n\to 0\ \wedge\ \exists N_0\in\mathbb{N}\ \forall n\ge N_0\ (|x_n-L|\le u_n)\Big)\\ &\quad\Longrightarrow\quad \forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall n\ge N\ \big(|x_n-L|<\varepsilon\big).\end{aligned}$$

Remark (Theorem predicate reading).

$\big(\mathrm{ConvergentSequence}(u_n,0)\wedge\mathrm{Eventually}(x_n,\ |x_n-L|\le u_n)\big)\Longrightarrow \mathrm{ConvergentSequence}(x_n,L).$ ∎

Remark (Interpretation). This is the estimate form of the squeeze theorem. Instead of constructing two explicit bounding sequences, it bounds the error $|x_n-L|$ by a null sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- Sequence Squeeze Theorem

Algebra of Limits

Algebra of Convergent Sequences

The algebra of convergent sequences records that limits are compatible with the ordinary algebraic operations on real numbers. The pointwise operations below turn real sequences into new real sequences, and the theorem blocks that follow state how convergence passes through those operations. The reciprocal and quotient laws require nonzero denominator hypotheses, while the square-root law requires nonnegative terms.

Definition (Pointwise Sum of Sequences)

Definition 3.2.12 (Pointwise Sum of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise sum** of (x_n) and (y_n) is the real sequence $(x_n + y_n)$ defined by

$$(x+y)_n := x_n + y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

Let (x_n) and (y_n) be real sequences.
$$\forall n \in \mathbb{N}\ \big((x+y)_n = x_n + y_n\big).$$

Remark (Interpretation). The sum sequence is formed term by term. No limiting process is involved in the definition; convergence enters only when the behavior of the resulting sequence is studied.

Remark (Dependencies).

- Sequence

Definition (Pointwise Difference of Sequences)

Definition 3.2.13 (Pointwise Difference of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise difference** of (x_n) and (y_n) is the real sequence $(x_n - y_n)$ defined by

$$(x-y)_n := x_n - y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

Let (x_n) and (y_n) be real sequences.
$$\forall n \in \mathbb{N}\ \big((x-y)_n = x_n - y_n\big).$$

Remark (Interpretation). The difference sequence subtracts corresponding terms. It is equivalently the sum of (x_n) with the pointwise negation of (y_n).

Remark (Dependencies).

- Sequence
- Pointwise Sum of Sequences

Definition (Scalar Multiple of a Sequence)

Definition 3.2.14 (Scalar Multiple of a Sequence). Let (x_n) be a real sequence, and let $\alpha \in \mathbb{R}$. The **scalar multiple** $\alpha(x_n)$ is the real sequence (αx_n) defined by

$$(\alpha x)_n := \alpha x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and } \alpha \in \mathbb{R}.$$
$$\forall n \in \mathbb{N}\ \big((\alpha x)_n = \alpha x_n\big).$$

Remark (Interpretation). Scalar multiplication rescales every term by the same real number. It changes the vertical scale of the sequence but not its indexing.

Remark (Dependencies).

- Sequence

Definition (Pointwise Negation of a Sequence)

Definition 3.2.15 (Pointwise Negation of a Sequence). Let (x_n) be a real sequence. The **pointwise negation** of (x_n) is the real sequence $(-x_n)$ defined by

$$(-x)_n := -x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N}\ \big((-x)_n = -x_n\big).$$

Remark (Interpretation). Pointwise negation reflects the sequence across zero. It is the special scalar multiple corresponding to the scalar -1.

Remark (Dependencies).

- Scalar Multiple of a Sequence

Definition (Pointwise Product of Sequences)

Definition 3.2.16 (Pointwise Product of Sequences). Let (x_n) and (y_n) be real sequences. The **pointwise product** of (x_n) and (y_n) is the real sequence $(x_n y_n)$ defined by

$$(xy)_n := x_n y_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ and } (y_n) \text{ be real sequences.}$$
$$\forall n \in \mathbb{N}\ \big((xy)_n = x_n y_n\big).$$

Remark (Interpretation). The product sequence multiplies corresponding terms. The product law for limits is subtler than the sum law because one factor must be controlled while the other is approximated.

Remark (Dependencies).

- Sequence

Definition (Reciprocal Sequence)

Definition 3.2.17 (Reciprocal Sequence). Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$. The **reciprocal sequence** of (x_n) is the real sequence $(1/x_n)$ defined by

$$\left(\frac{1}{x}\right)_n := \frac{1}{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\,(x_n \neq 0).$$
$$\forall n \in \mathbb{N}\left(\left(\frac{1}{x}\right)_n = \frac{1}{x_n}\right).$$

Remark (Interpretation). The reciprocal sequence is defined only when division by each term is legal. For limit theorems, a nonzero limit guarantees eventual nonzero behavior, but the sequence-level construction itself requires nonzero terms wherever it is defined.

Remark (Dependencies).

- Sequence

Definition (Pointwise Quotient of Sequences)

Definition 3.2.18 (Pointwise Quotient of Sequences). Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$. The **pointwise quotient** of (x_n) by (y_n) is the real sequence (x_n/y_n) defined by

$$\left(\frac{x}{y}\right)_n := \frac{x_n}{y_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\,(y_n \neq 0).$$
$$\forall n \in \mathbb{N}\left(\left(\frac{x}{y}\right)_n = \frac{x_n}{y_n}\right).$$

Remark (Interpretation). The quotient sequence is the product of (x_n) with the reciprocal of (y_n). Its convergence law is therefore a product law plus a reciprocal law.

Remark (Dependencies).

- Pointwise Product of Sequences
- Reciprocal Sequence

Definition (Square Sequence)

Definition 3.2.19 (Square Sequence). Let (x_n) be a real sequence. The **square sequence** of (x_n) is the real sequence (x_n^2) defined by

$$(x^2)_n := x_n^2 \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N} \left((x^2)_n = x_n^2 \right).$$

Remark (Interpretation). The square sequence is the product of a sequence with itself. Its limit law is therefore a direct specialization of the product law.

Remark (Dependencies).

- Pointwise Product of Sequences

Definition (Absolute Value Sequence)

Definition 3.2.20 (Absolute Value Sequence). Let (x_n) be a real sequence. The **absolute value sequence** of (x_n) is the real sequence $(|x_n|)$ defined by

$$(|x|)_n := |x_n| \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence.}$$
$$\forall n \in \mathbb{N} \left((|x|)_n = |x_n| \right).$$

Remark (Interpretation). The absolute value sequence records the magnitude of each term. It removes sign information while preserving distance from zero.

Remark (Dependencies).

- Sequence

Definition (Square Root Sequence)

Definition 3.2.21 (Square Root Sequence). Let (x_n) be a real sequence such that $0 \le x_n$ for every $n \in \mathbb{N}$. The **square root sequence** of (x_n) is the real sequence $(\sqrt{x_n})$ defined by

$$(\sqrt{x})_n := \sqrt{x_n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\, (0 \le x_n).$$
$$\forall n \in \mathbb{N}\, \big((\sqrt{x})_n = \sqrt{x_n}\big).$$

Remark (Interpretation). The square root sequence is defined term by term on the nonnegative real terms. The nonnegativity hypothesis is part of the construction, not a proof artifact.

Remark (Dependencies).

- Sequence

Theorem (Limit of a Scalar Multiple)

Theorem 3.2.22 (Limit of a Scalar Multiple). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L, \alpha \in \mathbb{R}.$$
$$\big[\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \ge N\, (|x_n - L| < \varepsilon)\big]$$
$$\implies \quad \big[\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \ge N\, (|\alpha x_n - \alpha L| < \varepsilon)\big].$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \implies \mathrm{ConvergentSequence}(\alpha x_n, \alpha L).$$

Remark (Interpretation). Multiplying every term by a fixed scalar multiplies the limiting value by the same scalar. The case $\alpha = 0$ collapses the sequence to the constant zero sequence.

Remark (Dependencies).

- Convergence of a Sequence
- Scalar Multiple of a Sequence

Theorem (Limit of a Sum)

Theorem 3.2.23 (Limit of a Sum). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \ \wedge\ y_n \to M\big] \implies \big[x_n + y_n \to L + M\big].$$

Remark (Predicate reading).

$$\big(\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{ConvergentSequence}(y_n, M)\big) \implies \mathrm{ConvergentSequence}(x_n + y_n, L + M).$$

Remark (Interpretation). The sum law says that convergent sequences may be added before or after taking limits. The proof splits a target tolerance into two smaller tolerances, one for each summand.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Sum of Sequences
- Triangle Inequality

Theorem (Limit of a Negation)

Theorem 3.2.24 (Limit of a Negation). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[x_n \to L\big] \implies \big[-x_n \to -L\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(-x_n, -L).$$

Remark (Interpretation). Negation is scalar multiplication by -1, so this theorem is the scalar multiple law in its most common special case.

Remark (Dependencies).

- Limit of a Scalar Multiple
- Pointwise Negation of a Sequence

Theorem (Limit of a Difference)

Theorem 3.2.25 (Limit of a Difference). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \ \wedge \ y_n \to M\big] \implies \big[x_n - y_n \to L - M\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \implies \text{ConvergentSequence}(x_n - y_n, L - M).$$

Remark (Interpretation). The difference law follows by adding (x_n) to the negation of (y_n). It is therefore not a new analytic mechanism, but a useful named rule.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Negation
- Pointwise Difference of Sequences

Theorem (Limit of a Product)

Theorem 3.2.26 (Limit of a Product). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n y_n \to LM$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences and let } L, M \in \mathbb{R}.$$
$$\big[x_n \to L \ \wedge \ y_n \to M\big] \Longrightarrow \big[x_n y_n \to LM\big].$$

Remark (Predicate reading).

$$\big(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)\big) \Longrightarrow \text{ConvergentSequence}(x_n y_n, LM).$$

Remark (Interpretation). The product law uses both approximation and boundedness. A convergent sequence is eventually bounded, so one factor can be controlled while the other is made small.

Remark (Dependencies).

- Convergence of a Sequence
- Pointwise Product of Sequences
- Bounded Sequence

Theorem (Nonzero Limit is Eventually Nonzero)

Theorem 3.2.27 (Nonzero Limit is Eventually Nonzero). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$\big[L \neq 0 \ \wedge \ x_n \to L\big] \Longrightarrow \exists N \in \mathbb{N}\, \forall n \geq N \ (x_n \neq 0).$$

Remark (Interpretation). A sequence converging to a nonzero number eventually stays away from zero. This is the denominator-control result needed for reciprocal and quotient rules.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Reciprocal)

Theorem 3.2.28 (Limit of a Reciprocal). *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\,(x_n \neq 0), \text{ and let } L \neq 0.$$
$$[x_n \to L] \implies [\frac{1}{x_n} \to \frac{1}{L}].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}\left(\frac{1}{x_n}, \frac{1}{L}\right).$$

Remark (Interpretation). Reciprocation is continuous away from zero. The nonzero limit supplies a tail on which the denominators stay uniformly separated from zero.

Remark (Dependencies).

- Reciprocal Sequence
- Nonzero Limit is Eventually Nonzero

Theorem (Limit of a Quotient)

Theorem 3.2.29 (Limit of a Quotient). *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*
Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n), (y_n) \text{ be real sequences with } \forall n \in \mathbb{N}\,(y_n \neq 0), \text{ and let } M \neq 0.$$
$$[x_n \to L \wedge y_n \to M] \implies [\frac{x_n}{y_n} \to \frac{L}{M}].$$

Remark (Predicate reading).

$$(\text{ConvergentSequence}(x_n, L) \wedge \text{ConvergentSequence}(y_n, M)) \implies \text{ConvergentSequence}\left(\frac{x_n}{y_n}, \frac{L}{M}\right).$$

Remark (Interpretation). The quotient law is the product law applied to (x_n) and the reciprocal of (y_n). The condition $M \neq 0$ prevents the limiting denominator from collapsing.

Remark (Dependencies).

- Pointwise Quotient of Sequences
- Limit of a Product
- Limit of a Reciprocal

Theorem (Limit of a Square)

Theorem 3.2.30 (Limit of a Square). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$[x_n \to L] \implies [x_n^2 \to L^2].$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \implies \mathrm{ConvergentSequence}(x_n^2, L^2).$$

Remark (Interpretation). Squaring preserves limits because it is multiplication of a sequence by itself.

Remark (Dependencies).

- Square Sequence
- Limit of a Product

Theorem (Limit of an Absolute Value)

Theorem 3.2.31 (Limit of an Absolute Value). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L \in \mathbb{R}.$$
$$[x_n \to L] \implies [|x_n| \to |L|].$$

Remark (Predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \implies \mathrm{ConvergentSequence}(|x_n|, |L|).$$

Remark (Interpretation). Taking absolute values cannot increase distance by more than the original distance: the magnitude function is continuous on $\mathbb{R}$.

Remark (Dependencies).

- Absolute Value Sequence
- Triangle Inequality

Theorem (Positive Limit is Eventually Positive)

Theorem 3.2.32 (Positive Limit is Eventually Positive). *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence and let } L > 0.$$
$$[x_n \to L] \implies \exists N \in \mathbb{N}\, \forall n \geq N\ (0 < x_n).$$

Remark (Interpretation). Convergence to a positive limit eventually traps the sequence inside the positive half-line. This result is the sign-control counterpart of the eventual nonzero theorem.

Remark (Dependencies).

- Convergence of a Sequence

Theorem (Limit of a Square Root)

Theorem 3.2.33 (Limit of a Square Root). *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence with } \forall n \in \mathbb{N}\ (0 \leq x_n), \text{ and let } L \in \mathbb{R}.$$
$$[x_n \to L] \implies [0 \leq L \ \wedge\ \sqrt{x_n} \to \sqrt{L}].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(\sqrt{x_n}, \sqrt{L}).$$

Remark (Interpretation). The square-root function is continuous on the nonnegative real line. Near zero the proof uses the order relation; away from zero it can be rationalized by multiplying by the conjugate.

Remark (Dependencies).

- Square Root Sequence
- Convergence of a Sequence

Theorem (Polynomial Sequence Limit)

Theorem 3.2.34 (Polynomial Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p \in \mathbb{R}[t].$$
$$\big[x_n \to L\big] \implies \big[p(x_n) \to p(L)\big].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}(p(x_n), p(L)).$$

Remark (Interpretation). Polynomial limits are built from finitely many additions, scalar multiples, and products. The theorem packages those algebra laws into one reusable composition principle.

Remark (Dependencies).

- Limit of a Sum
- Limit of a Scalar Multiple
- Limit of a Product
- Limit of a Square

Theorem (Rational Sequence Limit)

Theorem 3.2.35 (Rational Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{Let } (x_n) \text{ be a real sequence, } L \in \mathbb{R}, \text{ and } p, q \in \mathbb{R}[t].$$
$$\big[q(L) \neq 0 \land \forall n \in \mathbb{N}\ (q(x_n) \neq 0) \land x_n \to L\big]$$
$$\implies \qquad \left[\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}\right].$$

Remark (Predicate reading).

$$\text{ConvergentSequence}(x_n, L) \implies \text{ConvergentSequence}\left(\frac{p(x_n)}{q(x_n)}, \frac{p(L)}{q(L)}\right).$$

Remark (Interpretation). A rational expression is a quotient of two polynomial expressions. The condition $q(L) \neq 0$ keeps the limiting denominator legitimate, and the pointwise condition $q(x_n) \neq 0$ keeps every term of the quotient sequence defined.

Remark (Dependencies).

- Polynomial Sequence Limit
- Limit of a Quotient
- Nonzero Limit is Eventually Nonzero

Definition (M-Tail of a Sequence)

Definition 3.2.36 (M-Tail of a Sequence). Let $(x_n) \subseteq \mathbb{R}$ and let $M \in \mathbb{N}$. The **M-tail** of (x_n) is the sequence

$$(x_n)_{n \geq M} := (x_M, x_{M+1}, x_{M+2}, \ldots).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall M \in \mathbb{N}\ \Big((x_n)_{n \geq M} = (x_M, x_{M+1}, x_{M+2}, \ldots)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{TailSequence}(x_n, M, y_k) := \forall k \in \mathbb{N}\ \big(y_k = x_{M+k-1}\big).$$

Remark (Interpretation). The M-tail discards the first $M - 1$ terms and retains everything from index M onward. It is the formal object underlying the phrase *all but finitely many terms*: to say that all but finitely many terms of (x_n) satisfy a property P is precisely to say that some M-tail of (x_n) satisfies P termwise. The index M is an offset into the sequence, not a bound on the terms themselves; two sequences with identical M-tails share all asymptotic properties, differing only in finitely many initial values which convergence arguments discard entirely.

Remark (Dependencies).

- Definition of sequence

Theorem (Convergence of a Tail)

Theorem 3.2.37 (Convergence of a Tail). *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n \in \mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n \to \infty} x_{m+n} = \lim_{n \to \infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N}\\ &\quad \Big(\big(x_{m+n}\big)_{n\in\mathbb{N}} \text{ converges} \iff (x_n) \text{ converges} \Big)\\ &\quad \wedge\ \Big((x_n) \text{ converges} \implies \lim_{n\to\infty} x_{m+n} = \lim_{n\to\infty} x_n \Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{TailConvergenceInvariant}(x_n, m) \ :=\ \Big(\exists L \in \mathbb{R}\ \text{ConvergentSequence}(x_n, L)\Big) \iff \Big(\exists L \in \mathbb{R}\ \text{Convergent}$$

Remark (Contrapositive quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall m \in \mathbb{N}\\ &\quad \Big(\big(x_{m+n}\big)_{n\in\mathbb{N}} \text{ diverges} \iff (x_n) \text{ diverges} \Big).\end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\,\text{TailConvergenceInvariant}(x_n, m) \ :=\ \big((x_n) \text{ diverges} \iff (x_{m+n})_{n\in\mathbb{N}} \text{ diverges}\big).$$

Remark (Interpretation). Convergence is a *tail property*: it depends only on the eventual behavior of the sequence, not on any finite initial segment. Discarding the first m terms neither creates nor destroys convergence, and when the limit exists it is preserved exactly. This theorem formalizes the informal principle that finitely many terms are irrelevant to asymptotic behavior. It is the rigorous justification for the common proof move of assuming without loss of generality that n is large enough for some auxiliary condition to hold: one simply passes to a suitable m-tail, applies the argument there, and concludes for the full sequence by this theorem. The contrapositive is equally useful: to show (x_n) diverges it suffices to show that some m-tail diverges.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence
- Uniqueness of Limits

Theorem (Convergence by Domination)

Theorem 3.2.38 (Convergence by Domination). *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n\to\infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \ \leq\ c\, a_n \qquad \forall n \geq m,$$

then $\lim_{n\to\infty} x_n = L$.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall (a_n) \subseteq (0,\infty)$$
$$\left(\left(\lim_{n\to\infty} a_n = 0\ \wedge\ \exists c > 0\ \exists m \in \mathbb{N}\ \forall n \geq m\ \left(|x_n - L| \leq c\, a_n\right)\right) \implies \lim_{n\to\infty} x_n = L\right).$$

Remark (Contrapositive quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall (a_n) \subseteq (0,\infty)$$
$$\left(\lim_{n\to\infty} x_n \neq L \implies \lim_{n\to\infty} a_n \neq 0\ \vee\ \forall c > 0\ \forall m \in \mathbb{N}\ \exists n \geq m\ \left(|x_n - L| > c\, a_n\right)\right).$$

Remark (Contrapositive predicate reading).

$$\mathrm{NotDominatedByNullError}(x_n, L, a_n) := \lim_{n\to\infty} a_n \neq 0 \vee \forall c > 0\ \forall m \in \mathbb{N}\ \exists n \geq m\left(|x_n - L| > c\, a_n\right).$$

Remark (Interpretation). The theorem is a ***domination principle***: it is not necessary to verify the ε-N condition directly for (x_n). It suffices to find a null sequence (a_n) - one converging to zero - that dominates the error $|x_n - L|$ on some tail, up to a fixed scalar c. The scalar c is irrelevant to the conclusion: since $a_n \to 0$, the product $c\,a_n \to 0$ regardless of the size of c, and any $\varepsilon > 0$ is eventually beaten by $c\,a_n$. This theorem is the rigorous engine behind the common proof pattern of bounding $|x_n - L|$ by a simpler expression and observing that the simpler expression tends to zero. The canonical instance is $a_n = 1/n$, where domination by c/n is sufficient to conclude convergence to L.

Remark (Dependencies).

- Convergence of a Sequence
- M-Tail of a Sequence

Theorem (Ratio Limit Less Than One Implies Null)

Theorem 3.2.39 (Ratio Limit Less Than One Implies Null). *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n\to\infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n\to\infty} x_n = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq (0,\infty)\ \forall L \in \mathbb{R}$$
$$\left(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L\ \wedge\ L < 1\right) \implies \lim_{n\to\infty} x_n = 0.$$

Remark (Theorem predicate reading).

$$\big(\mathrm{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1\big) \implies \mathrm{NullSequence}(x_n).$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq (0,\infty)\ \exists L \in \mathbb{R}$$
$$\Big(\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L\ \wedge\ L < 1\ \wedge\ \lim_{n\to\infty} x_n \neq 0\Big).$$

Remark (Negation predicate reading).

$$\mathrm{ConvergentSequence}\big(x_{n+1}/x_n, L\big) \wedge L < 1 \wedge \mathrm{NotConvergentTo}(x_n, 0).$$

Remark (Failure modes). A failure would be a positive sequence whose consecutive ratios settle below one in the limit but whose terms do not tend to zero. The theorem says this cannot happen: once the ratio limit is less than one, some tail has ratios bounded by a fixed number $r < 1$, forcing geometric decay.

Remark (Failure mode decomposition).

$$\underbrace{(x_n) \subseteq (0,\infty)}_{\texttt{positive terms}} \quad \underbrace{\lim_{n\to\infty} \frac{x_{n+1}}{x_n} = L < 1}_{\texttt{eventual ratio gap}} \quad \underbrace{\lim_{n\to\infty} x_n \neq 0}_{\texttt{fails to be null}}.$$

The positive-term hypothesis makes each ratio meaningful. The strict gap $L < 1$ gives a tail on which $x_{n+1} \leq r x_n$ for some $r < 1$, and repeated iteration then gives domination by a null geometric sequence.

Remark (Interpretation). This is the sequence form of the ratio test. The theorem does not merely say that the terms decrease eventually; it says they decrease at an eventually geometric rate. The strict inequality $L < 1$ is essential. When the ratio limit equals 1, the conclusion may hold, as with $1/n$, or fail, as with the constant sequence 1.

Remark (Dependencies).

- Convergence of a Sequence
- Null Sequence
- M-Tail of a Sequence
- Convergence by Domination

Monotonicity of Sequences

Monotonicity is an order condition on successive terms of a sequence. It is independent of convergence by itself, but when combined with boundedness it becomes one of the fundamental convergence mechanisms in real analysis.

Definition (Increasing Sequence)

Definition 3.2.40 (Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **increasing** if

$$x_n \le x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is increasing} \iff \forall n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) :\!= \forall n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{MonotoneIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Interpretation). An increasing sequence may stay flat. The term "increasing" is therefore used here in the non-strict sense: each term is at least as large as the one before it.

Remark (Dependencies).

- Sequence

Definition (Decreasing Sequence)

Definition 3.2.41 (Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **decreasing** if

$$x_{n+1} \le x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(x_n) :\!= \forall n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not decreasing} \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{MonotoneDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Interpretation). A decreasing sequence may stay flat. The order condition says the sequence never rises as the index advances.

Remark (Dependencies).

- Sequence

Definition (Strictly Increasing Sequence)

Definition 3.2.42 (Strictly Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly increasing** if

$$x_n < x_{n+1} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly increasing} \iff \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_n < x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly increasing} \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyIncreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_{n+1} \le x_n).$$

Remark (Interpretation). Strict increase excludes equality between consecutive terms. Every strictly increasing sequence is increasing, but the converse need not hold.

Remark (Dependencies).

- Sequence
- Increasing Sequence

Definition (Strictly Decreasing Sequence)

Definition 3.2.43 (Strictly Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **strictly decreasing** if

$$x_{n+1} < x_n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is strictly decreasing} \iff \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Definition predicate reading).

$$\text{StrictlyDecreasingSequence}(x_n) :\equiv \forall n \in \mathbb{N}\ (x_{n+1} < x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not strictly decreasing} \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{StrictlyDecreasingSequence}(x_n) \iff \exists n \in \mathbb{N}\ (x_n \le x_{n+1}).$$

Remark (Interpretation). Strict decrease excludes equality between consecutive terms. Every strictly decreasing sequence is decreasing, but the converse need not hold.

Remark (Dependencies).

- Sequence
- Decreasing Sequence

Definition (Monotone Sequence)

Definition 3.2.44 (Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **monotone** if it is increasing or decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is monotone}\\ &\iff \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \vee \big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{MonotoneSequence}(x_n) := \mathrm{MonotoneIncreasingSequence}(x_n) \vee \mathrm{MonotoneDecreasingSequence}(x_n).$$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not monotone}\\ &\iff \big[\exists n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \wedge \big[\exists m \in \mathbb{N}\ (x_m < x_{m+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{MonotoneSequence}(x_n) \iff \neg\,\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \neg\,\mathrm{MonotoneDecreasingSequence}(x_n$$

Remark (Interpretation). Monotonicity means the sequence has one global direction. It may move upward, move downward, or stay constant, but it cannot reverse direction.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence

Definition (Eventually Increasing Sequence)

Definition 3.2.45 (Eventually Increasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually increasing** if there exists $N \in \mathbb{N}$ such that

$$x_n \le x_{n+1} \quad \text{for every } n \ge N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually increasing} \iff \exists N \in \mathbb{N}\, \forall n \geq N\; (x_n \leq x_{n+1}).$$

Remark (Definition predicate reading).

$$\text{EventuallyIncreasingSequence}(x_n) :\equiv \exists N \in \mathbb{N}\, \forall n \geq N\; (x_n \leq x_{n+1}).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually increasing} \iff \forall N \in \mathbb{N}\, \exists n \geq N\; (x_{n+1} < x_n).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyIncreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\, \exists n \geq N\; (x_{n+1} < x_n).$$

Remark (Interpretation). Eventual increase allows finitely many early reversals. Only the tail of the sequence must move in a nondecreasing direction.

Remark (Dependencies).

- M-Tail of a Sequence
- Increasing Sequence

Definition (Eventually Decreasing Sequence)

Definition 3.2.46 (Eventually Decreasing Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually decreasing** if there exists $N \in \mathbb{N}$ such that

$$x_{n+1} \leq x_n \qquad \text{for every } n \geq N.$$

Remark (Standard quantified statement).

$$(x_n) \text{ is eventually decreasing} \iff \exists N \in \mathbb{N}\, \forall n \geq N\; (x_{n+1} \leq x_n).$$

Remark (Definition predicate reading).

$$\text{EventuallyDecreasingSequence}(x_n) :\equiv \exists N \in \mathbb{N}\, \forall n \geq N\; (x_{n+1} \leq x_n).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not eventually decreasing} \iff \forall N \in \mathbb{N}\, \exists n \geq N\; (x_n < x_{n+1}).$$

Remark (Negation predicate reading).

$$\neg\,\text{EventuallyDecreasingSequence}(x_n) \iff \forall N \in \mathbb{N}\, \exists n \geq N\; (x_n < x_{n+1}).$$

Remark (Interpretation). Eventual decrease allows the initial terms to behave irregularly. The order condition begins only after some finite threshold.

Remark (Dependencies).

- M-Tail of a Sequence
- Decreasing Sequence

Definition (Eventually Monotone Sequence)

Definition 3.2.47 (Eventually Monotone Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **eventually monotone** if it is eventually increasing or eventually decreasing.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is eventually monotone}\\ &\quad\iff \quad \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1})\big] \vee \big[\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n)\big].\end{aligned}$$

Remark (Definition predicate reading).

$\mathrm{EventuallyMonotoneSequence}(x_n) :\!= \mathrm{EventuallyIncreasingSequence}(x_n) \vee \mathrm{EventuallyDecreasingSequenc}$

Remark (Negated quantified statement).

$$\begin{aligned}&(x_n) \text{ is not eventually monotone}\\ &\quad\iff \quad \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_{n+1} < x_n)\big] \wedge \big[\forall N \in \mathbb{N}\ \exists n \geq N\ (x_n < x_{n+1})\big].\end{aligned}$$

Remark (Negation predicate reading).

$\neg\,\mathrm{EventuallyMonotoneSequence}(x_n) \iff \neg\,\mathrm{EventuallyIncreasingSequence}(x_n) \wedge \neg\,\mathrm{EventuallyDecreasing}$

Remark (Interpretation). Eventual monotonicity is the tail version of monotonicity. It is often the right hypothesis in applications, because finitely many early terms do not affect convergence.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail

Theorem (Monotone Convergence Theorem)

Theorem 3.2.48 (Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \sup\{x_n : n \in \mathbb{N}\} \Big), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ \Big(\lim_{n\to\infty} x_n = L \wedge L = \inf\{x_n : n \in \mathbb{N}\} \Big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \implies \mathrm{ConvergentSequence}\big(x_n, \sup\{x_n \\
&\mathrm{MonotoneDecreasingSequence}(x_n) \wedge \mathrm{BoundedBelowSequence}(x_n) \implies \mathrm{ConvergentSequence}\big(x_n, \inf\{x_n
\end{aligned}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists (x_n) \subseteq \mathbb{R}\ \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \\
&\qquad\qquad \wedge (x_n) \texttt{ does not converge} \Big), \\
&\exists (x_n) \subseteq \mathbb{R}\ \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\big] \\
&\qquad\qquad \wedge (x_n) \texttt{ does not converge} \Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\begin{aligned}
&\mathrm{MonotoneIncreasingSequence}(x_n) \wedge \mathrm{BoundedAboveSequence}(x_n) \wedge \neg\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, \\
&\mathrm{MonotoneDecreasingSequence}(x_n) \wedge \mathrm{BoundedBelowSequence}(x_n) \wedge \neg\exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n,
\end{aligned}$$

Remark (Failure modes). The theorem would fail only if a one-directional bounded sequence could avoid settling to a real limit. Completeness of $\mathbb{R}$ rules this out: an increasing bounded-above sequence is forced toward the least upper bound of its range, while a decreasing bounded-below sequence is forced toward the greatest lower bound of its range.

Remark (Failure mode decomposition).

$$\underbrace{\forall n \in \mathbb{N}\ (x_n \le x_{n+1})}_{\texttt{one direction}} \quad \underbrace{\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)}_{\texttt{ceiling}} \quad \underbrace{(x_n) \texttt{ does not converge}}_{\texttt{forbidden by completeness}}.$$

The decreasing case is dual: reverse the inequalities and replace the ceiling by a floor.

Remark (Interpretation). Boundedness prevents escape, and monotonicity prevents oscillation. Together they force convergence. The theorem is one of the standard forms in which the completeness of the real numbers becomes a sequence theorem.

Remark (Dependencies).

- Convergence of a Sequence

- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below
- Supremum
- Infimum

Theorem (Strict Monotonicity Implies Monotonicity)

Theorem 3.2.49 (Strict Monotonicity Implies Monotonicity). *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n < x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \Big), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} < x_n)\big] \implies \big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \Big).
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\mathrm{StrictlyIncreasingSequence}(x_n) \implies \mathrm{MonotoneIncreasingSequence}(x_n), \\
&\mathrm{StrictlyDecreasingSequence}(x_n) \implies \mathrm{MonotoneDecreasingSequence}(x_n).
\end{aligned}$$

Remark (Interpretation). Strict monotonicity is stronger than non-strict monotonicity because every strict inequality is also a weak inequality.

Remark (Dependencies).

- Strictly Increasing Sequence
- Strictly Decreasing Sequence
- Increasing Sequence
- Decreasing Sequence

Theorem (Bounded Monotone Sequence Equivalences)

Theorem 3.2.50 (Bounded Monotone Sequence Equivalences). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}$$
$$\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\Big),$$
$$\big[\forall n \in \mathbb{N}\ (x_{n+1} \le x_n)\big] \implies \Big(\exists L \in \mathbb{R}\ (x_n \to L) \iff \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \le x_n)\Big).$$

Remark (Theorem predicate reading).

$$\mathrm{MonotoneIncreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedAboveSeque}$$
$$\mathrm{MonotoneDecreasingSequence}(x_n) \implies \Big(\exists L \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{BoundedBelowSeque}$$

Remark (Interpretation). For monotone sequences, the only obstruction to real convergence is escape in the monotone direction. Boundedness removes that obstruction.

Remark (Dependencies).

- Monotone Convergence Theorem
- Bounded Sequence
- Convergence of a Sequence

Theorem (Eventually Monotone Convergence Theorem)

Theorem 3.2.51 (Eventually Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_n \leq x_{n+1}) \land \exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \leq M)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L), \\
&\quad \Big(\exists N \in \mathbb{N}\ \forall n \geq N\ (x_{n+1} \leq x_n) \land \exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n)\Big) \\
&\quad \implies \quad \exists L \in \mathbb{R}\ (x_n \to L).
\end{aligned}
$$

Remark (Theorem predicate reading).

$$
\begin{aligned}
&\text{EventuallyIncreasingSequence}(x_n) \land \text{BoundedAboveSequence}(x_n) \implies \exists L \in \mathbb{R}\ \ \text{ConvergentSequence}(x_n \\
&\text{EventuallyDecreasingSequence}(x_n) \land \text{BoundedBelowSequence}(x_n) \implies \exists L \in \mathbb{R}\ \ \text{ConvergentSequence}(x
\end{aligned}
$$

Remark (Interpretation). This is the tail version of the Monotone Convergence Theorem. Once the tail is monotone and bounded in the right direction, finite initial behavior is irrelevant.

Remark (Dependencies).

- Eventually Increasing Sequence
- Eventually Decreasing Sequence
- Convergence of a Tail
- Monotone Convergence Theorem

Theorem (Unbounded Monotone Divergence)

Theorem 3.2.52 (Unbounded Monotone Divergence). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R} \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \leq x_{n+1})\big] \land \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (M < x_n)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (M < x_n), \\
&\quad \Big(\big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \land \big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ (x_n < M)\big]\Big) \\
&\quad \implies \quad \forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ (x_n < M).
\end{aligned}
$$

Remark (Theorem predicate reading).

$$
\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \land \neg\,\text{BoundedAboveSequence}(x_n) \implies \text{DivergesToPositiveInfinity}(x_n) \\
&\text{MonotoneDecreasingSequence}(x_n) \land \neg\,\text{BoundedBelowSequence}(x_n) \implies \text{DivergesToNegativeInfinity}(x_n
\end{aligned}
$$

Remark (Interpretation). For a monotone sequence, unboundedness has a direction. An increasing sequence that has no upper bound eventually exceeds every real number; the decreasing case is the order-dual statement.

Remark (Dependencies).

- Divergence to Positive Infinity
- Divergence to Negative Infinity
- Increasing Sequence
- Decreasing Sequence
- Sequence Bounded Above
- Sequence Bounded Below

Theorem (Algebraic Transformations Preserve Monotonicity)

Theorem 3.2.53 (Algebraic Transformations Preserve Monotonicity). *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall (x_n) \subseteq \mathbb{R}\ \forall c, \alpha \in \mathbb{R} \\
&\quad \big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \implies \big[\forall n \in \mathbb{N}\ (x_n + c \le x_{n+1} + c)\big], \\
&\quad \big(\alpha > 0 \land \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_n \le \alpha x_{n+1})\big], \\
&\quad \big(\alpha < 0 \land \forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big) \implies \big[\forall n \in \mathbb{N}\ (\alpha x_{n+1} \le \alpha x_n)\big].
\end{aligned}$$

Remark (Theorem predicate reading).

$$\begin{aligned}
&\text{MonotoneIncreasingSequence}(x_n) \implies \text{MonotoneIncreasingSequence}(x_n + c), \\
&\text{MonotoneIncreasingSequence}(x_n) \land \alpha > 0 \implies \text{MonotoneIncreasingSequence}(\alpha x_n), \\
&\text{MonotoneIncreasingSequence}(x_n) \land \alpha < 0 \implies \text{MonotoneDecreasingSequence}(\alpha x_n).
\end{aligned}$$

Remark (Interpretation). Translation preserves order, multiplication by a positive scalar preserves order, and multiplication by a negative scalar reverses order. Negation is the special case $\alpha = -1$.

Remark (Dependencies).

- Pointwise Sum of Sequences
- Scalar Multiple of a Sequence
- Pointwise Negation of a Sequence
- Increasing Sequence
- Decreasing Sequence

Subsequences

Subsequences record what remains after passing to an infinite ordered selection of indices. They are the basic tool for extracting structure from a sequence: limits, oscillation, boundedness, compactness, and failure of convergence are all detected by suitable subsequences.

Definition (Strictly Increasing Index Map)

Definition 3.2.54 (Strictly Increasing Index Map). Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a function. The function σ is a **strictly increasing index map** if

$$k < \ell \Longrightarrow \sigma(k) < \sigma(\ell) \qquad \text{for all } k, \ell \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\sigma : \mathbb{N} \to \mathbb{N} \text{ is strictly increasing} \Longleftrightarrow \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)).$$

Remark (Definition predicate reading).

$$\text{StrictlyIncreasing}(\sigma) :\equiv \forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)).$$

Remark (Negated quantified statement).

$$\sigma \text{ is not strictly increasing} \Longleftrightarrow \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

Remark (Negation predicate reading).

$$\neg\,\text{StrictlyIncreasing}(\sigma) \Longleftrightarrow \exists k, \ell \in \mathbb{N}\ (k < \ell \wedge \sigma(\ell) \leq \sigma(k)).$$

Remark (Interpretation). The index map selects terms without repetition and without changing their order. This is the structural difference between a subsequence and an arbitrary rearrangement.

Remark (Dependencies).

- Function
- Natural Numbers
- Strict Order Successor Characterization

Definition (Subsequence)

Definition 3.2.55 (Subsequence). Let (x_n) be a real sequence. A **subsequence** of (x_n) is a sequence $(x_{\sigma(k)})_{k\in\mathbb{N}}$, where $\sigma : \mathbb{N} \to \mathbb{N}$ is a strictly increasing index map.

Remark (Standard quantified statement).

$$(y_k) \text{ is a subsequence of } (x_n)$$
$$\iff \quad \exists\sigma : \mathbb{N} \to \mathbb{N} \left(\forall k, \ell \in \mathbb{N}\, (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\, (y_k = x_{\sigma(k)})\right).$$

Remark (Definition predicate reading).

$$\text{Subsequence}(y_k, x_n) :\equiv \exists\sigma \left(\text{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\, (y_k = x_{\sigma(k)})\right).$$

Remark (Negated quantified statement).

$$(y_k) \text{ is not a subsequence of } (x_n)$$
$$\iff \quad \forall\sigma : \mathbb{N} \to \mathbb{N} \left(\exists k, \ell \in \mathbb{N}\, (k < \ell \wedge \sigma(\ell) \le \sigma(k)) \vee \exists k \in \mathbb{N}\, (y_k \neq x_{\sigma(k)})\right).$$

Remark (Negation predicate reading).

$$\neg\,\text{Subsequence}(y_k, x_n).$$

Remark (Interpretation). A subsequence keeps the original chronological order but may skip terms. The new index k counts the selected terms, while $\sigma(k)$ records where each selected term came from in the original sequence.

Remark (Dependencies).

- Sequence
- Strictly Increasing Index Map

Definition (Subsequential Limit)

Definition 3.2.56 (Subsequential Limit). Let (x_n) be a real sequence and let $L \in \mathbb{R}$. The real number L is a **subsequential limit** of (x_n) if there exists a subsequence $(x_{\sigma(k)})$ such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement).

$$L \text{ is a subsequential limit of } (x_n)$$
$$\iff \quad \exists\sigma : \mathbb{N} \to \mathbb{N} \left(\forall k, \ell \in \mathbb{N}\, (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k\to\infty} x_{\sigma(k)} = L\right).$$

Remark (Definition predicate reading).

$$\text{SubsequentialLimit}(x_n, L) :\equiv \exists\sigma \left(\text{StrictlyIncreasing}(\sigma) \wedge \text{ConvergentSequence}(x_{\sigma(k)}, L)\right).$$

Remark (Interpretation). Subsequential limits detect limiting behavior along an infinite selection of terms. A sequence may have no ordinary limit but still have one or more subsequential limits.

Remark (Dependencies).

- Subsequence
- Convergence of a Sequence

Definition (Has Convergent Subsequence)

Definition 3.2.57 (Has Convergent Subsequence). Let (x_n) be a real sequence. The sequence (x_n) **has a convergent subsequence** if there exist a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ and a real number L such that

$$\lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Standard quantified statement).

$$(x_n) \text{ has a convergent subsequence}$$
$$\iff \quad \exists\sigma : \mathbb{N} \to \mathbb{N}\, \exists L \in \mathbb{R}\, \Big(\forall k, \ell \in \mathbb{N}\, (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k\to\infty} x_{\sigma(k)} = L\Big).$$

Remark (Definition predicate reading).

$$\mathrm{HasConvergentSubsequence}(x_n) :\equiv \exists L \in \mathbb{R}\ \mathrm{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation). This property asks for at least one convergent infinite selection. It does not assert convergence of the whole sequence and does not require uniqueness of the subsequential limit.

Remark (Dependencies).

- Subsequential Limit

Theorem (Subsequence Indices Dominate the Identity)

Theorem 3.2.58 (Subsequence Indices Dominate the Identity). *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \le \sigma(k) \qquad \textit{for every } k \in \mathbb{N}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall\sigma : \mathbb{N} \to \mathbb{N}\, \Big(\big[\forall k, \ell \in \mathbb{N}\, (k < \ell \implies \sigma(k) < \sigma(\ell))\big] \implies \big[\forall k \in \mathbb{N}\, (k \le \sigma(k))\big]\Big).$$

Remark (Theorem predicate reading).

$$\mathrm{StrictlyIncreasing}(\sigma) \implies \forall k \in \mathbb{N}\, (k \le \sigma(k)).$$

Remark (Interpretation). A subsequence cannot outrun the original indexing backwards. By the time the subsequence has selected its kth term, it must have reached at least the kth original position.

Remark (Dependencies).

- Strictly Increasing Index Map
- Well-ordering of $\mathbb{N}$

Theorem (Subsequences Preserve Limits)

Theorem 3.2.59 (Subsequences Preserve Limits). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}$$
$$\Big(x_n \to L \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\Big) \implies \lim_{k\to\infty} x_{\sigma(k)} = L.$$

Remark (Theorem predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{StrictlyIncreasing}(\sigma) \implies \mathrm{ConvergentSequence}(x_{\sigma(k)}, L).$$

Remark (Interpretation). Subsequences cannot escape a limit already possessed by the whole sequence. The domination of subsequence indices by the identity transfers every tail estimate for (x_n) to a tail estimate for $(x_{\sigma(k)})$.

Remark (Dependencies).

- Subsequence
- Convergence of a Sequence
- Subsequence Indices Dominate the Identity

Theorem (Subsequential Limit of a Convergent Sequence)

Theorem 3.2.60 (Subsequential Limit of a Convergent Sequence). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(x_n \to L \wedge K \text{ is a subsequential limit of } (x_n)\Big) \implies K = L.$$

Remark (Theorem predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K) \implies K = L.$$

Remark (Interpretation). A convergent sequence has no hidden alternative limiting behavior. Every convergent subsequence inherits the same limit, and uniqueness of limits then forces equality.

Remark (Dependencies).

- Subsequences Preserve Limits
- Uniqueness of Limits

Theorem (Divergence by Two Subsequential Limits)

Theorem 3.2.61 (Divergence by Two Subsequential Limits). *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L, K \in \mathbb{R}$$
$$\Big(L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of } (x_n) \wedge L \neq K\Big)$$
$$\implies \quad (x_n) \text{ does not converge.}$$

Remark (Theorem predicate reading).

$$\mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K) \wedge L \neq K \implies \neg \exists A \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n,$$

Remark (Interpretation). Two different subsequential limits reveal persistent oscillation or splitting. If the whole sequence had a limit, every subsequence would have to inherit that same limit.

Remark (Dependencies).

- Subsequential Limit
- Subsequential Limit of a Convergent Sequence

Theorem (Boundedness Passes to Subsequences)

Theorem 3.2.62 (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}$$
$$\Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\Big)$$
$$\implies \quad \exists M > 0\ \forall k \in \mathbb{N}\ (|x_{\sigma(k)}| \leq M).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{BoundedSequence}(x_{\sigma(k)}).$$

Remark (Interpretation). A subsequence cannot create new values; it only selects from the original sequence. Any global bound for the original sequence is automatically a bound for the selection.

Remark (Dependencies).

- Bounded Sequence
- Subsequence

Theorem (Monotonicity Passes to Subsequences)

Theorem 3.2.63 (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma : \mathbb{N} \to \mathbb{N}\\ &\quad \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\big]\Big)\\ &\quad \implies \quad \forall k \in \mathbb{N}\ (x_{\sigma(k)} \le x_{\sigma(k+1)}).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{MonotoneIncreasingSequence}(x_n) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{MonotoneIncreasingSequence}(x_{\sigma(k)}).$$

Remark (Interpretation). Selecting terms in the original order preserves the order pattern already present in the sequence. This is why the index map must be strictly increasing.

Remark (Dependencies).

- Increasing Sequence
- Decreasing Sequence
- Subsequence

Theorem (Subsequence of a Subsequence)

Theorem 3.2.64 (Subsequence of a Subsequence). *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall \sigma, \tau : \mathbb{N} \to \mathbb{N}\\ &\quad \Big(\big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\big] \wedge \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \tau(k) < \tau(\ell))\big]\Big)\\ &\quad \implies \quad (x_{\sigma(\tau(k))})_{k\in\mathbb{N}} \text{ is a subsequence of } (x_n).\end{aligned}$$

Remark (Theorem predicate reading).

$$\text{StrictlyIncreasing}(\sigma) \wedge \text{StrictlyIncreasing}(\tau) \implies \text{Subsequence}(x_{\sigma(\tau(k))}, x_n).$$

Remark (Interpretation). Passing to subsequences is closed under iteration. The composed index map $\sigma \circ \tau$ is still strictly increasing.

Remark (Dependencies).

- Subsequence
- Composition of functions

Theorem (Eventual Properties Pass to Subsequences)

Theorem 3.2.65 (Eventual Properties Pass to Subsequences). *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\, \forall n \geq N\ P(n) \wedge \forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell))\Big) \\ \implies \quad \exists K \in \mathbb{N}\, \forall k \geq K\ P(\sigma(k)).$$

Remark (Theorem predicate reading).

$$\text{Eventually}(x_n, P) \wedge \text{StrictlyIncreasing}(\sigma) \implies \text{Eventually}(x_{\sigma(k)}, P).$$

Remark (Interpretation). Subsequences eventually enter every tail of the original sequence. Any property that holds on a tail of the original therefore holds on a tail of the subsequence.

Remark (Dependencies).

- Subsequence Indices Dominate the Identity
- Subsequence

Theorem (Frequent Properties Yield Subsequences)

Theorem 3.2.66 (Frequent Properties Yield Subsequences). *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall N \in \mathbb{N}\, \exists n \geq N\ P(n)\Big] \implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \forall k \in \mathbb{N}\ P(\sigma(k))\Big).$$

Remark (Theorem predicate reading).

$$\mathrm{Frequently}(x_n, P) \Longrightarrow \exists \sigma \; \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \forall k \in \mathbb{N}\; P(\sigma(k))\big).$$

Remark (Interpretation). An infinitely recurring property can be threaded into a subsequence. The construction chooses one later witness at each step.

Remark (Dependencies).

- Well-ordering of $\mathbb{N}$
- Subsequence

Theorem (Subsequential Limits Respect Bounds)

Theorem 3.2.67 (Subsequential Limits Respect Bounds). *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \le x_n \le M$ for every $n \in \mathbb{N}$, then $m \le L \le M$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\; \forall L, m, M \in \mathbb{R}$$
$$\Big(L \texttt{ is a subsequential limit of } (x_n) \wedge \forall n \in \mathbb{N}\; (m \le x_n \le M)\Big) \Longrightarrow m \le L \le M.$$

Remark (Theorem predicate reading).

$$\mathrm{SubsequentialLimit}(x_n, L) \wedge \forall n \in \mathbb{N}\; (m \le x_n \le M) \Longrightarrow m \le L \le M.$$

Remark (Interpretation). Bounds pass to subsequences, and limits preserve eventual inequalities. Hence subsequential limits remain inside any closed interval that contains all terms.

Remark (Dependencies).

- Boundedness Passes to Subsequences
- Constant Comparison for Sequence Limits

Theorem (Squeeze Passes to Subsequences)

Theorem 3.2.68 (Squeeze Passes to Subsequences). *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \le x_n \le b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)} \qquad \textit{for every } k \ge K.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big(\exists N \in \mathbb{N}\; \forall n \ge N\; (a_n \le x_n \le b_n) \wedge \forall k, \ell \in \mathbb{N}\; (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell))\Big)$$
$$\Longrightarrow \quad \exists K \in \mathbb{N}\; \forall k \ge K\; (a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)}).$$

Remark (Theorem predicate reading).

$$\mathrm{Eventually}(x_n,\ a_n \le x_n \le b_n) \wedge \mathrm{StrictlyIncreasing}(\sigma) \implies \mathrm{Eventually}(x_{\sigma(k)},\ a_{\sigma(k)} \le x_{\sigma(k)} \le b_{\sigma(k)}). \blacksquare$$

Remark (Interpretation). Squeeze hypotheses are tail hypotheses, so they survive passage to subsequences.

Remark (Dependencies).

- Eventual Properties Pass to Subsequences
- Sequence Squeeze Theorem

Theorem (Monotone Subsequence Theorem)

Theorem 3.2.69 (Monotone Subsequence Theorem). *Every real sequence has a monotone subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \big[\forall k \in \mathbb{N}\ (x_{\sigma(k)} \le x_{\sigma(k+1)}) \vee \forall k \in \mathbb{N}\ (x_{\sigma(k+1)}$$

Remark (Theorem predicate reading).

$$\exists \sigma\ \big(\mathrm{StrictlyIncreasing}(\sigma) \wedge \mathrm{MonotoneSequence}(x_{\sigma(k)})\big).$$

Remark (Interpretation). Every sequence contains an infinite ordered pattern. The proof uses the peak dichotomy: either there are infinitely many terms larger than all later terms, or there is an infinite rising chain.

Remark (Dependencies).

- Monotone Sequence
- Subsequence
- Well-ordering of $\mathbb{N}$

Theorem (Bolzano-Weierstrass Theorem for Sequences)

Theorem 3.2.70 (Bolzano-Weierstrass Theorem for Sequences). *Every bounded real sequence has a convergent subsequence.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M) \implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \big[\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \mathrm{HasConvergentSubsequence}(x_n).$$

Remark (Interpretation). Bolzano-Weierstrass is the compactness principle for bounded real sequences. Boundedness prevents escape, and the monotone subsequence theorem supplies an ordered subsequence to which the Monotone Convergence Theorem applies.

Remark (Dependencies).

- Bounded Sequence
- Monotone Subsequence Theorem
- Boundedness Passes to Subsequences
- Bounded Monotone Sequence Equivalences

Theorem (Sequential Compactness of Closed Bounded Intervals)

Theorem 3.2.71 (Sequential Compactness of Closed Bounded Intervals). *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall a, b \in \mathbb{R}\ \forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(a \leq b \wedge \forall n \in \mathbb{N}\ (a \leq x_n \leq b)\Big)\\ &\quad \implies \quad \exists \sigma : \mathbb{N} \to \mathbb{N}\ \exists L \in \mathbb{R}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \wedge \lim_{k \to \infty} x_{\sigma(k)} = L \wedge a \leq L \leq b\Big).\end{aligned}$$

Remark (Theorem predicate reading).

$$\big[\forall n \in \mathbb{N}\ (a \leq x_n \leq b)\big] \implies \exists L \in [a, b]\ \mathrm{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation). Closed bounded intervals are sequentially compact: a sequence trapped in the interval has a convergent subsequence, and closedness keeps the subsequential limit inside the interval.

Remark (Dependencies).

- Bolzano-Weierstrass Theorem for Sequences
- Subsequential Limits Respect Bounds

3.3 Divergence

Divergence Toolkit

Concept family: Failure modes for convergence of real sequences.

Atomic items in this section:

- Definition: Divergent Sequence
- Definition: Divergence to Positive Infinity
- Definition: Divergence to Negative Infinity
- Definition: Oscillatory Sequence
- Theorem: Divergence to Infinity Implies Real Divergence
- Theorem: Two Subsequential Limits Force Divergence
- Theorem: Unbounded Above Sequence Has a Positive-Infinity Subsequence
- Theorem: Unbounded Below Sequence Has a Negative-Infinity Subsequence
- Theorem: Bounded Divergence Produces Two Subsequential Limits

Divergence

Divergence is not a single behavior. A sequence can fail to converge because it escapes upward, escapes downward, oscillates between incompatible limiting patterns, or remains bounded while refusing to settle. The purpose of this section is to separate these failure modes and connect them to subsequences.

Definition (Divergent Sequence)

Definition 3.3.1 (Divergent Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **divergent** if there is no $L \in \mathbb{R}$ such that $x_n \to L$.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is divergent} \\ &\iff \quad \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DivergentSequence}(x_n) :\equiv \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Negated quantified statement).

$$(x_n) \text{ is not divergent} \iff \exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|x_n - L| < \varepsilon\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DivergentSequence}(x_n) \iff \exists L \in \mathbb{R}\ \ \mathrm{ConvergentSequence}(x_n, L).$$

Remark (Interpretation). Divergence is the formal negation of real convergence. It says that every candidate limit fails at some fixed tolerance on every tail.

Remark (Dependencies).

- Sequence
- Convergence of a Sequence

Definition (Divergence to Positive Infinity)

Definition 3.3.2 (Divergence to Positive Infinity). Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $+\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Standard quantified statement).

$$x_n \to +\infty \iff \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToPositiveInfinity}(x_n) :\equiv \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(M < x_n\big).$$

Remark (Negated quantified statement).

$$x_n \not\to +\infty \iff \exists M \in \mathbb{R}\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(x_n \leq M\big).$$

Remark (Interpretation). Divergence to $+\infty$ means every horizontal threshold is eventually left below the sequence. It is not convergence to a real number; it is escape from the real line in the positive direction.

Remark (Dependencies).

- Sequence

Definition (Divergence to Negative Infinity)

Definition 3.3.3 (Divergence to Negative Infinity). Let (x_n) be a real sequence. The sequence (x_n) **diverges to** $-\infty$ if

$$\forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Standard quantified statement).

$$x_n \to -\infty \iff \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToNegativeInfinity}(x_n) :\equiv \forall M \in \mathbb{R}\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(x_n < M\big).$$

Remark (Negated quantified statement).

$$x_n \not\to -\infty \iff \exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(M \leq x_n\big).$$

Remark (Interpretation). Divergence to $-\infty$ is escape past every lower threshold. It is the order-dual of divergence to $+\infty$.

Remark (Dependencies).

- Sequence

Definition (Oscillatory Sequence)

Definition 3.3.4 (Oscillatory Sequence). Let (x_n) be a real sequence. The sequence (x_n) is **oscillatory** if it is divergent, does not diverge to $+\infty$, and does not diverge to $-\infty$.

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is oscillatory} \\ \iff \quad &\Big[\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big)\Big] \\ &\wedge \Big[\exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(x_n \leq M\big)\Big] \\ &\wedge \Big[\exists M \in \mathbb{R}\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(M \leq x_n\big)\Big]. \end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned} \mathrm{OscillatorySequence}(x_n) \coloneqq\ &\mathrm{DivergentSequence}(x_n) \\ &\wedge\ \neg\,\mathrm{DivergesToPositiveInfinity}(x_n) \\ &\wedge\ \neg\,\mathrm{DivergesToNegativeInfinity}(x_n). \end{aligned}$$

Remark (Interpretation). Oscillation means failure to settle without one-directional escape. The sequence may remain bounded, as $(-1)^n$ does, or it may be unbounded while still repeatedly returning in incompatible directions.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Divergence to Infinity Implies Real Divergence)

Theorem 3.3.5 (Divergence to Infinity Implies Real Divergence). *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(M < x_n\big)\Big] \\ &\quad \vee \Big[\forall M \in \mathbb{R}\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(x_n < M\big)\Big] \\ &\qquad \Longrightarrow \forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - L| \geq \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{DivergesToPositiveInfinity}(x_n) \vee \mathrm{DivergesToNegativeInfinity}(x_n) \Longrightarrow \mathrm{DivergentSequence}(x_n).\blacksquare$$

Remark (Interpretation). Infinity divergence is a stronger, directional failure of real convergence. A sequence escaping past every real threshold cannot approach a finite real limit.

Remark (Dependencies).

- Divergent Sequence
- Divergence to Positive Infinity
- Divergence to Negative Infinity

Theorem (Two Subsequential Limits Force Divergence)

Theorem 3.3.6 (Two Subsequential Limits Force Divergence). *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\exists L, K \in \mathbb{R}\ \Big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential limit of} \\ &\quad \Longrightarrow \forall A \in \mathbb{R}\ \exists \varepsilon > 0\ \forall N \in \mathbb{N}\ \exists n \geq N\ \big(|x_n - A| \geq \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\begin{aligned}&\exists L, K \in \mathbb{R}\ \big(L \neq K \wedge \mathrm{SubsequentialLimit}(x_n, L) \wedge \mathrm{SubsequentialLimit}(x_n, K)\big) \\ &\quad \Longrightarrow \mathrm{DivergentSequence}(x_n).\end{aligned}$$

Remark (Interpretation). Convergence forces all subsequences to inherit the same limit. Two different subsequential limits therefore witness an irreparable conflict in the tail behavior of the original sequence.

Remark (Dependencies).

- Subsequential Limit
- Subsequences Preserve Limits
- Uniqueness of Limits
- Divergence by Two Subsequential Limits

Theorem (Unbounded Above Sequence Has a Positive-Infinity Subsequence)

Theorem 3.3.7 (Unbounded Above Sequence Has a Positive-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ \big(M < x_n\big)\Big] \\ \implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \land \forall M \in \mathbb{R}\ \exists K \in \mathbb{N}\ \forall k \geq K\ \big(M < x_{\sigma(k)}\big)\Big).$$ ∎

Remark (Definition predicate reading).

$$\neg\,\text{BoundedAboveSequence}(x_n) \implies \exists \sigma\ \big(\text{StrictlyIncreasing}(\sigma) \land \text{DivergesToPositiveInfinity}(x_{\sigma(k)})\big).$$ ∎

Remark (Interpretation). Unboundedness above may be sparse. This theorem says the high values can still be selected in increasing index order to form a subsequence that escapes to $+\infty$.

Remark (Dependencies).

- Sequence Bounded Above
- Subsequence
- Divergence to Positive Infinity

Theorem (Unbounded Below Sequence Has a Negative-Infinity Subsequence)

Theorem 3.3.8 (Unbounded Below Sequence Has a Negative-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*
Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall M \in \mathbb{R}\ \exists n \in \mathbb{N}\ \big(x_n < M\big)\Big] \\ \implies \exists \sigma : \mathbb{N} \to \mathbb{N}\ \Big(\forall k, \ell \in \mathbb{N}\ (k < \ell \implies \sigma(k) < \sigma(\ell)) \land \forall M \in \mathbb{R}\ \exists K \in \mathbb{N}\ \forall k \geq K\ \big(x_{\sigma(k)} < M\big)\Big).$$ ∎

Remark (Definition predicate reading).

$$\neg\,\text{BoundedBelowSequence}(x_n) \implies \exists \sigma\ \big(\text{StrictlyIncreasing}(\sigma) \land \text{DivergesToNegativeInfinity}(x_{\sigma(k)})\big).$$ ∎

Remark (Interpretation). Unboundedness below can be converted into a selected tail that escapes past every lower threshold. The proof is the order-reversed version of the positive-infinity subsequence construction.

Remark (Dependencies).

- Sequence Bounded Below

- Subsequence
- Divergence to Negative Infinity

Theorem (Bounded Divergence Produces Two Subsequential Limits)

Theorem 3.3.9 (Bounded Divergence Produces Two Subsequential Limits). *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\exists M > 0\, \forall n \in \mathbb{N}\, \big(|x_n| \leq M\big)\Big] \wedge \Big[\forall A \in \mathbb{R}\, \exists \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - A| \geq \varepsilon\big)\Big]$$
$$\implies \exists L, K \in \mathbb{R}\, \Big(L \neq K \wedge L \text{ is a subsequential limit of } (x_n) \wedge K \text{ is a subsequential lim}$$

Remark (Definition predicate reading).

$$\text{BoundedSequence}(x_n) \wedge \text{DivergentSequence}(x_n) \implies \exists L, K \in \mathbb{R}\, \big(L \neq K \wedge \text{SubsequentialLimit}(x_n, L) \wedge \text{Su}$$

Remark (Interpretation). Bounded divergence cannot be explained by escape to infinity. In the real line, compactness forces convergent subsequences, and divergence forces at least two incompatible subsequential limits.

Remark (Dependencies).

- Bounded Sequence
- Divergent Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences
- Subsequences Preserve Limits

3.4 Liminf and Limsup

Liminf–Limsup Toolkit

Concept family: Tail extremals and asymptotic envelopes of bounded real sequences.

Atomic items in this section:

- Definition: Tail Supremum Sequence
- Definition: Tail Infimum Sequence
- Definition: Limit Superior of a Sequence
- Definition: Limit Inferior of a Sequence
- Theorem: Tail Suprema Are Decreasing
- Theorem: Tail Infima Are Increasing
- Theorem: Liminf Is Below Limsup
- Theorem: Convergence iff Liminf Equals Limsup
- Theorem: Limsup Is the Largest Subsequential Limit
- Theorem: Liminf Is the Smallest Subsequential Limit
- Theorem: Oscillation Criterion via Liminf and Limsup
- Theorem: Limsup Comparison Under Eventual Order
- Theorem: Liminf Comparison Under Eventual Order
- Theorem: Limsup Squeeze Under Eventual Order
- Theorem: Liminf Squeeze Under Eventual Order

Liminf and Limsup

The limit superior and limit inferior measure the eventual upper and lower envelopes of a bounded sequence. Instead of asking whether all late terms settle to one number, they ask how high and how low the tails continue to reach. This makes them natural tools for detecting convergence, oscillation, and subsequential limits.

Definition (Tail Supremum Sequence)

Definition 3.4.1 (Tail Supremum Sequence). Let (x_n) be a bounded real sequence. The **tail supremum sequence** of (x_n) is the real sequence (s_n) defined by

$$s_n := \sup\{x_k : k \geq n\} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \ \big(s_n = \sup\{x_k : k \geq n\}\big).$$

Remark (Interpretation). The term s_n records the least ceiling of the n-th tail. As n grows, earlier terms are discarded, so the tail has fewer opportunities to reach large values.

Remark (Dependencies).

- Bounded Sequence
- Supremum
- M-Tail of a Sequence

Definition (Tail Infimum Sequence)

Definition 3.4.2 (Tail Infimum Sequence). Let (x_n) be a bounded real sequence. The **tail infimum sequence** of (x_n) is the real sequence (i_n) defined by

$$i_n := \inf\{x_k : k \geq n\} \quad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \ \big(i_n = \inf\{x_k : k \geq n\}\big).$$

Remark (Interpretation). The term i_n records the greatest floor of the n-th tail. As earlier terms are discarded, the tail loses opportunities to reach small values.

Remark (Dependencies).

- Bounded Sequence
- Infimum
- M-Tail of a Sequence

Definition (Limit Superior of a Sequence)

Definition 3.4.3 (Limit Superior of a Sequence). Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. The **limit superior** of (x_n) is

$$\limsup_{n\to\infty} x_n := \lim_{n\to\infty} s_n.$$

Remark (Standard quantified statement).

$$\limsup_{n\to\infty} x_n = L \iff \forall \varepsilon > 0 \ \exists N \in \mathbb{N} \ \forall n \geq N \ \big(|s_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{LimsupSequence}(x_n, L) :\equiv \text{ConvergentSequence}(s_n, L).$$

Remark (Interpretation). The limit superior is the eventual upper envelope of the sequence. It is the number to which the ceilings of the tails descend.

Remark (Dependencies).

- Tail Supremum Sequence
- Convergence of a Sequence

Definition (Limit Inferior of a Sequence)

Definition 3.4.4 (Limit Inferior of a Sequence). Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. The **limit inferior** of (x_n) is

$$\liminf_{n\to\infty} x_n := \lim_{n\to\infty} i_n.$$

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n = L \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ \big(|i_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{LiminfSequence}(x_n, L) := \text{ConvergentSequence}(i_n, L).$$

Remark (Interpretation). The limit inferior is the eventual lower envelope of the sequence. It is the number to which the floors of the tails ascend.

Remark (Dependencies).

- Tail Infimum Sequence
- Convergence of a Sequence

Theorem (Tail Suprema Are Decreasing)

Theorem 3.4.5 (Tail Suprema Are Decreasing). *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N}\ \big(s_{n+1} \leq s_n\big).$$

Remark (Definition predicate reading).

$$\text{MonotoneDecreasingSequence}(s_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot raise its supremum.

Remark (Dependencies).

- Tail Supremum Sequence
- Supremum Is Monotone Under Inclusion

Theorem (Tail Infima Are Increasing)

Theorem 3.4.6 (Tail Infima Are Increasing). *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Go to proof.

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(i_n \leq i_{n+1}\right).$$

Remark (Definition predicate reading).

$$\text{MonotoneIncreasingSequence}(i_n).$$

Remark (Interpretation). The $(n+1)$-st tail is contained in the n-th tail. Passing from a set to a smaller subset cannot lower its infimum.

Remark (Dependencies).

- Tail Infimum Sequence
- Infimum

Theorem (Liminf Is Below Limsup)

Theorem 3.4.7 (Liminf Is Below Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Remark (Interpretation). Every tail infimum lies below every corresponding tail supremum. Passing to the limits preserves this order.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Superior of a Sequence
- Constant Comparison for Sequence Limits

Theorem (Convergence iff Liminf Equals Limsup)

Theorem 3.4.8 (Convergence iff Liminf Equals Limsup). *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Go to proof.

Remark (Standard quantified statement).

$$x_n \to L \quad \Longleftrightarrow \quad \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(x_n, L) \Longleftrightarrow \big(\mathrm{LiminfSequence}(x_n, L) \wedge \mathrm{LimsupSequence}(x_n, L)\big).$$

Remark (Interpretation). Convergence occurs exactly when the eventual lower envelope and eventual upper envelope collapse to the same number.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Superior of a Sequence
- Convergence of a Sequence
- Sequence Squeeze Theorem

Theorem (Limsup Is the Largest Subsequential Limit)

Theorem 3.4.9 (Limsup Is the Largest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*

Go to proof.

Remark (Standard quantified statement).

$$S = \limsup_{n\to\infty} x_n$$
$$\Longrightarrow \quad \Big[S \text{ is a subsequential limit of } (x_n) \wedge \forall L \in \mathbb{R}\ \big(L \text{ is a subsequential limit of } (x_n$$

Remark (Definition predicate reading).

$$\mathrm{LimsupSequence}(x_n, S) \Longrightarrow \big(\mathrm{SubsequentialLimit}(x_n, S) \wedge \forall L \in \mathbb{R}\ (\mathrm{SubsequentialLimit}(x_n, L) \Longrightarrow L \leq S$$

Remark (Interpretation). The limit superior is not merely an upper bound on subsequential behavior. It is actually attained as a subsequential limit, and no subsequential limit lies above it.

Remark (Dependencies).

- Limit Superior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Liminf Is the Smallest Subsequential Limit)

Theorem 3.4.10 (Liminf Is the Smallest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Go to proof.

Remark (Standard quantified statement).

$$I = \liminf_{n\to\infty} x_n$$
$$\implies \Big[I \text{ is a subsequential limit of } (x_n) \land \forall L \in \mathbb{R}\ \big(L \text{ is a subsequential limit of } (x_n$$

Remark (Definition predicate reading).

$$\text{LiminfSequence}(x_n, I) \implies \big(\text{SubsequentialLimit}(x_n, I) \land \forall L \in \mathbb{R}\ (\text{SubsequentialLimit}(x_n, L) \implies I \leq L)$$

Remark (Interpretation). The limit inferior is attained as a subsequential limit and no subsequential limit lies below it.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Subsequential Limit
- Bolzano-Weierstrass Theorem for Sequences

Theorem (Oscillation Criterion via Liminf and Limsup)

Theorem 3.4.11 (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Go to proof.

Remark (Standard quantified statement).

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$
$$\iff \exists L, K \in \mathbb{R}\ \big(L \neq K \land L \text{ is a subsequential limit of } (x_n) \land K \text{ is a subsequential l}$$

Remark (Definition predicate reading).

$$\exists I, S \in \mathbb{R}\ \big(\text{LiminfSequence}(x_n, I) \land \text{LimsupSequence}(x_n, S) \land I < S\big)$$
$$\iff \exists L, K \in \mathbb{R}\ \big(L \neq K \land \text{SubsequentialLimit}(x_n, L) \land \text{SubsequentialLimit}(x_n, K)\big).$$

Remark (Interpretation). A strict gap between the lower and upper envelopes is exactly persistent oscillation in bounded sequences. The sequence keeps returning near at least two incompatible limiting levels.

Remark (Dependencies).

- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Two Subsequential Limits Force Divergence

Theorem (Limsup Comparison Under Eventual Order)

Theorem 3.4.12 (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists N \in \mathbb{N}\, \forall n \geq N\ (x_n \leq y_n) \\ \implies \limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Remark (Interpretation). Eventual order of sequences passes to eventual order of their upper tail envelopes. Taking the limiting ceiling preserves that comparison.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Preserves Eventual Order

Theorem (Liminf Comparison Under Eventual Order)

Theorem 3.4.13 (Liminf Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\liminf_{n\to\infty} x_n \leq \liminf_{n\to\infty} y_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\exists N \in \mathbb{N}\, \forall n \geq N\ (x_n \leq y_n) \\ \implies \liminf_{n\to\infty} x_n \leq \liminf_{n\to\infty} y_n.$$

Remark (Interpretation). Eventual order also passes to the lower tail envelopes. If one sequence is eventually below another, its eventual floor cannot sit above the other's eventual floor.

Remark (Dependencies).

- Limit Inferior of a Sequence
- Limit Preserves Eventual Order

Theorem (Limsup Squeeze Under Eventual Order)

Theorem 3.4.14 (Limsup Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \textit{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n)\\ &\quad\implies \limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.\end{aligned}$$

Remark (Interpretation). If one sequence is eventually trapped between two others, then its eventual upper envelope is trapped between their eventual upper envelopes.

Remark (Dependencies).

- Limsup Comparison Under Eventual Order

Theorem (Liminf Squeeze Under Eventual Order)

Theorem 3.4.15 (Liminf Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \textit{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\exists N \in \mathbb{N}\ \forall n \ge N\ (a_n \le x_n \le b_n)\\ &\quad\implies \liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.\end{aligned}$$

Remark (Interpretation). The lower-envelope version of the squeeze principle says that eventual two-sided order traps the eventual floors as well as the eventual ceilings.

Remark (Dependencies).

- Liminf Comparison Under Eventual Order

Remark (Illustrative cases). For the alternating sequence $x_n = (-1)^n$, the tail suprema are constantly 1 and the tail infima are constantly -1, so

$$\limsup_{n\to\infty}(-1)^n = 1, \qquad \liminf_{n\to\infty}(-1)^n = -1.$$

For any convergent sequence $x_n \to L$, both envelopes collapse to L.

3.5 Order-Theoretic Limit Constructions

Order-Theoretic Limits Toolkit

Concept family: Limits constructed from suprema and infima.

Atomic items in this section:

- Theorem: Increasing Sequence Limit as Supremum
- Theorem: Decreasing Sequence Limit as Infimum
- Theorem: Tail Suprema and Infima Converge
- Theorem: Bounded Sequence Has Limsup and Liminf

Order-Theoretic Limit Constructions

Completeness lets order data produce limits. An increasing sequence bounded above converges to the least ceiling of its range; a decreasing sequence bounded below converges to the greatest floor of its range. Applying these same ideas to tail suprema and tail infima produces limsup and liminf for every bounded sequence.

Theorem (Increasing Sequence Limit as Supremum)

Theorem 3.5.1 (Increasing Sequence Limit as Supremum). *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \Big(\big[\forall n \in \mathbb{N}\ (x_n \le x_{n+1})\big] \wedge \big[\exists M \in \mathbb{R}\ \forall n \in \mathbb{N}\ (x_n \le M)\big] \\ \implies \quad \lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}\Big).$$

Remark (Theorem predicate reading).

MonotoneIncreasingSequence$(x_n)\wedge$BoundedAboveSequence$(x_n) \implies$ ConvergentSequence$\big(x_n, \sup\{x_n :$

Remark (Interpretation). The supremum is the only possible destination: all terms lie below it, and the defining approximation property of the supremum forces the sequence eventually above every lower tolerance.

Remark (Dependencies).

- Increasing Sequence
- Sequence Bounded Above
- Supremum
- Monotone Convergence Theorem

Theorem (Decreasing Sequence Limit as Infimum)

Theorem 3.5.2 (Decreasing Sequence Limit as Infimum). *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left(\big[\forall n \in \mathbb{N}\ (x_{n+1} \leq x_n)\big] \wedge \big[\exists m \in \mathbb{R}\ \forall n \in \mathbb{N}\ (m \leq x_n)\big] \right.$$
$$\left. \implies \lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\} \right).$$

Remark (Theorem predicate reading).

MonotoneDecreasingSequence$(x_n)\wedge$BoundedBelowSequence$(x_n) \implies$ ConvergentSequence$\big(x_n, \inf\{x_n :$

Remark (Interpretation). The infimum is the decreasing analogue of the supremum target: the sequence falls toward its greatest lower barrier.

Remark (Dependencies).

- Decreasing Sequence
- Sequence Bounded Below
- Infimum
- Monotone Convergence Theorem

Theorem (Tail Suprema and Infima Converge)

Theorem 3.5.3 (Tail Suprema and Infima Converge). *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \left[\lim_{n\to\infty} \sup\{x_k : k \geq n\} = S \wedge \lim_{n\to\infty} \inf\{x_k : k \geq n\} = I \right] \right).$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \ (\mathrm{ConvergentSequence}(\sup\{x_k : k \geq n\}, S) \wedge \mathrm{ConvergentSequence}(\mathrm{i}$$

Remark (Interpretation). Tail suprema decrease because later tails contain fewer terms; tail infima increase for the same reason. Boundedness supplies the opposite bound needed for monotone convergence.

Remark (Dependencies).

- Tail Supremum Sequence
- Tail Infimum Sequence
- Tail Suprema Are Decreasing
- Tail Infima Are Increasing
- Monotone Convergence Theorem

Theorem (Bounded Sequence Has Limsup and Liminf)

Theorem 3.5.4 (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \left((x_n) \text{ is bounded} \implies \exists S, I \in \mathbb{R} \ (S = \limsup_{n\to\infty} x_n \wedge I = \liminf_{n\to\infty} x_n) \right).$$

Remark (Theorem predicate reading).

$$\mathrm{BoundedSequence}(x_n) \implies \exists S, I \in \mathbb{R} \ (\mathrm{LimsupSequence}(x_n, S) \wedge \mathrm{LiminfSequence}(x_n, I)).$$

Remark (Interpretation). For bounded sequences, limsup and liminf are not extra assumptions. They are guaranteed order-theoretic limits of the tail envelopes.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Inferior of a Sequence
- Tail Suprema and Infima Converge

3.6 Cluster Values of Sequences

Cluster Values Toolkit

Concept family: Values approached infinitely often along a sequence.

Atomic items in this section:

- Definition: Cluster Value of a Sequence
- Theorem: Cluster Values Are Subsequential Limits
- Theorem: Bounded Sequences Have Cluster Values
- Theorem: Limsup and Liminf Are Extremal Cluster Values

Cluster Values of Sequences

A convergent sequence has only one limiting value. A bounded divergent sequence may still have values that it approaches along selected subsequences. Cluster values name this repeated limiting behavior without requiring the entire sequence to converge.

Definition (Cluster Value of a Sequence)

Definition 3.6.1 (Cluster Value of a Sequence). Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The number L is a **cluster value** of (x_n) if for every $\varepsilon > 0$ and every $N \in \mathbb{N}$ there exists $n \geq N$ such that

$$|x_n - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$L \text{ is a cluster value of } (x_n) \iff \forall \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\mathrm{SequenceClusterValue}(x_n, L) := \forall \varepsilon > 0\, \forall N \in \mathbb{N}\, \exists n \geq N\, \big(|x_n - L| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$L \text{ is not a cluster value of } (x_n) \iff \exists \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(|x_n - L| \geq \varepsilon\big).$$

Remark (Interpretation). No matter how far out in the sequence one looks, some later term returns inside every neighborhood of L.

Remark (Dependencies).

- Sequence

Theorem (Cluster Values Are Subsequential Limits)

Theorem 3.6.2 (Cluster Values Are Subsequential Limits). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \Big(L \texttt{ is a cluster value of } (x_n) \iff L \texttt{ is a subsequential limit of } (x_n)\Big).$$

Remark (Theorem predicate reading).

$$\text{SequenceClusterValue}(x_n, L) \iff \text{SubsequentialLimit}(x_n, L).$$

Remark (Interpretation). The neighborhood formulation and the subsequence formulation describe the same phenomenon: infinitely many opportunities to get arbitrarily close to L.

Remark (Dependencies).

- Cluster Value of a Sequence
- Subsequential Limit
- Frequent Properties Yield Subsequences

Theorem (Bounded Sequences Have Cluster Values)

Theorem 3.6.3 (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*
Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\big[\exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \le M)\big] \implies \exists L \in \mathbb{R}\ \big(L \texttt{ is a cluster value of } (x_n)\big)\Big).$$

Remark (Theorem predicate reading).

$$\text{BoundedSequence}(x_n) \implies \exists L \in \mathbb{R}\ \text{SequenceClusterValue}(x_n, L).$$

Remark (Interpretation). Boundedness traps the sequence in a finite interval, and Bolzano-Weierstrass extracts a convergent subsequence from that trap.

Remark (Dependencies).

- Bounded Sequence

- Bolzano-Weierstrass Theorem for Sequences
- Cluster Values Are Subsequential Limits

Theorem (Limsup and Liminf Are Extremal Cluster Values)

Theorem 3.6.4 (Limsup and Liminf Are Extremal Cluster Values). *Let (x_n) be a bounded real sequence. Then $\limsup_{n\to\infty} x_n$ is the largest cluster value of (x_n), and $\liminf_{n\to\infty} x_n$ is the smallest cluster value of (x_n).*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \Big((x_n) \text{ is bounded} \implies \Big[\limsup_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \land \forall L \in \mathbb{R} \big(L \text{ is a cluster value of } (x_n) \implies L \leq \lim$$
$$\land \Big[\liminf_{n\to\infty} x_n \text{ is a cluster value of } (x_n) \land \forall L \in \mathbb{R} \big(L \text{ is a cluster value of } (x_n) \implies \liminf$$

Remark (Theorem predicate reading).

$$\begin{aligned} &\text{BoundedSequence}(x_n) \implies \\ &\quad \text{SequenceClusterValue}\big(x_n, \limsup_{n\to\infty} x_n\big) \land \forall L \in \mathbb{R} \big(\text{SequenceClusterValue}(x_n, L) \implies L \leq \limsup_{n\to\infty} x_n\big), \\ &\text{BoundedSequence}(x_n) \implies \\ &\quad \text{SequenceClusterValue}\big(x_n, \liminf_{n\to\infty} x_n\big) \land \forall L \in \mathbb{R} \big(\text{SequenceClusterValue}(x_n, L) \implies \liminf_{n\to\infty} x_n \leq L\big). \end{aligned}$$

Remark (Interpretation). Limsup and liminf describe the top and bottom edges of the sequence's subsequential behavior.

Remark (Dependencies).

- Limit Superior of a Sequence
- Limit Inferior of a Sequence
- Limsup Is the Largest Subsequential Limit
- Liminf Is the Smallest Subsequential Limit
- Cluster Values Are Subsequential Limits

Cauchy Sequences

Cauchy sequences measure internal stabilization. Instead of comparing the terms to an already named limit, the Cauchy condition compares sufficiently late terms to each other. Over the real numbers this internal criterion is equivalent to convergence, and that equivalence is one of the central sequence forms of completeness.

Definition (Cauchy Sequence)

Definition 3.6.5 (Cauchy Sequence). Let (x_n) be a real sequence. The sequence (x_n) is a **Cauchy sequence** if

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall m, n \geq N \, \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned} &(x_n) \text{ is Cauchy} \\ &\iff \quad \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall m, n \geq N \, \big(|x_m - x_n| < \varepsilon\big). \end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) :\equiv \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall m, n \geq N \, \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned} &(x_n) \text{ is not Cauchy} \\ &\iff \quad \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists m, n \geq N \, \big(|x_m - x_n| \geq \varepsilon\big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg \, \text{CauchySequence}(x_n) \iff \exists \varepsilon > 0 \, \forall N \in \mathbb{N} \, \exists m, n \geq N \, \big(|x_m - x_n| \geq \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition says that the tails of the sequence eventually have arbitrarily small internal spread. It does not name a candidate limit; it only asserts that late terms become mutually close.

Remark (Dependencies).

- Sequence

Theorem (Convergent Sequences Are Cauchy)

Theorem 3.6.6 (Convergent Sequences Are Cauchy). *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\forall (x_n) \subseteq \mathbb{R} \\ &\quad \Big(\big(\exists L \in \mathbb{R} \, \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \, (|x_n - L| < \varepsilon)\big) \\ &\qquad \implies \forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall m, n \geq N \, (|x_m - x_n| < \varepsilon) \Big). \end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R} \, \Big(\exists L \in \mathbb{R} \; \text{ConvergentSequence}(x_n, L) \implies \text{CauchySequence}(x_n) \Big).$$

Remark (Interpretation). Once every sufficiently late term is close to the same limit, any two sufficiently late terms are close to each other. This theorem is the easy half of the Cauchy criterion.

Remark (Dependencies).

- Convergence of a Sequence
- Cauchy Sequence

Theorem (Cauchy Sequences Are Bounded)

Theorem 3.6.7 (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\\ &\qquad \Longrightarrow \exists M > 0\ \forall n \in \mathbb{N}\ (|x_n| \leq M)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \big(\mathrm{CauchySequence}(x_n) \Longrightarrow \mathrm{BoundedSequence}(x_n)\big).$$

Remark (Interpretation). A Cauchy sequence has one tightly controlled tail. The finitely many terms before that tail can be absorbed into a single larger bound.

Remark (Dependencies).

- Cauchy Sequence
- Bounded Sequence

Theorem (Cauchy Sequence with a Convergent Subsequence)

Theorem 3.6.8 (Cauchy Sequence with a Convergent Subsequence). *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\\ &\quad \Big(\big[\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\big]\\ &\qquad \wedge \big[\exists \sigma : \mathbb{N} \to \mathbb{N}\ \big(\forall k, \ell \in \mathbb{N}\ (k < \ell \Longrightarrow \sigma(k) < \sigma(\ell)) \wedge \forall \varepsilon > 0\ \exists K \in \mathbb{N}\ \forall k \geq K\ (|x_{\sigma(k)} - L| < \varepsilon)\big)\big]\\ &\qquad \Longrightarrow \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall L \in \mathbb{R}\ \Big(\mathrm{CauchySequence}(x_n) \wedge \mathrm{SubsequentialLimit}(x_n, L) \implies \mathrm{ConvergentSequence}(x_n, L)\Big).$$

Remark (Interpretation). The Cauchy condition prevents the rest of the tail from wandering away from any convergent subsequence. A single subsequential limit therefore determines the limit of the whole sequence.

Remark (Dependencies).

- Cauchy Sequence
- Subsequence
- Subsequential Limit
- Convergence of a Sequence

Theorem (Cauchy Criterion for Real Sequences)

Theorem 3.6.9 (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad\Big(\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_n - L| < \varepsilon)\\ &\qquad \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon)\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\forall (x_n) \subseteq \mathbb{R}\ \Big(\exists L \in \mathbb{R}\ \mathrm{ConvergentSequence}(x_n, L) \iff \mathrm{CauchySequence}(x_n)\Big).$$

Remark (Interpretation). This theorem is the sequence form of completeness of $\mathbb{R}$. In an incomplete ambient space, a sequence may be internally stable without converging inside that space.

Remark (Dependencies).

- Convergent Sequences Are Cauchy
- Cauchy Sequences Are Bounded
- Bolzano-Weierstrass Theorem for Sequences
- Cauchy Sequence with a Convergent Subsequence

Theorem (Cauchy Criterion via Tails)

Theorem 3.6.10 (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is Cauchy}\\ &\quad\iff \quad \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall p, q \in \mathbb{N}\ \big(|x_{N+p} - x_{N+q}| < \varepsilon\big).$$

Remark (Interpretation). The Cauchy condition is a tail property. Reindexing a sufficiently late tail does not change the substance of the condition; it only changes the names of the indices.

Remark (Dependencies).

- Cauchy Sequence
- M-Tail of a Sequence

Theorem (Cauchy Tail Diameter Criterion)

Theorem 3.6.11 (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&(x_n) \text{ is Cauchy}\\ &\quad\iff \quad \forall \varepsilon > 0\ \exists N_0 \in \mathbb{N}\ \forall N \geq N_0\\ &\qquad\qquad \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \iff \forall \varepsilon > 0\ \exists N_0 \in \mathbb{N}\ \forall N \geq N_0\ \forall m, n \geq N\ \big(|x_m - x_n| < \varepsilon\big).$$

Remark (Interpretation). The tail-diameter form says that once one tail is small, every later tail is small as well. It records the monotone nature of tail control without introducing additional notation for diameters.

Remark (Dependencies).

- Cauchy Sequence

- M-Tail of a Sequence

Theorem (Cauchy Sequences Have Vanishing Successive Differences)

Theorem 3.6.12 (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \\ \Big(\forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall m, n \geq N\ (|x_m - x_n| < \varepsilon) \\ \implies \forall \varepsilon > 0\ \exists N \in \mathbb{N}\ \forall n \geq N\ (|x_{n+1} - x_n| < \varepsilon)\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \implies \mathrm{NullSequence}\big(|x_{n+1} - x_n|\big).$$

Remark (Interpretation). Cauchy control applies to every pair of late indices, so it applies in particular to consecutive late indices. The converse is false in general: small successive steps do not prevent accumulated drift.

Remark (Dependencies).

- Cauchy Sequence
- Null Sequence

Theorem (Scalar Multiples of Cauchy Sequences)

Theorem 3.6.13 (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R} \\ \Big((x_n) \texttt{ is Cauchy} \implies (\alpha x_n) \texttt{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \implies \mathrm{CauchySequence}(\alpha x_n).$$

Remark (Interpretation). Multiplying all terms by one fixed real number rescales all pairwise tail distances by the same fixed factor.

Remark (Dependencies).

- Cauchy Sequence
- Scalar Multiple of a Sequence

Theorem (Sums of Cauchy Sequences)

Theorem 3.6.14 (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \texttt{ is Cauchy} \wedge (y_n) \texttt{ is Cauchy} \Longrightarrow (x_n + y_n) \texttt{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \Longrightarrow \text{CauchySequence}(x_n + y_n).$$

Remark (Interpretation). The triangle inequality splits the tail distance of the sum into the two tail distances already controlled by the original sequences.

Remark (Dependencies).

- Cauchy Sequence
- Pointwise Sum of Sequences

Theorem (Differences of Cauchy Sequences)

Theorem 3.6.15 (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \texttt{ is Cauchy} \wedge (y_n) \texttt{ is Cauchy} \Longrightarrow (x_n - y_n) \texttt{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \wedge \text{CauchySequence}(y_n) \Longrightarrow \text{CauchySequence}(x_n - y_n).$$

Remark (Interpretation). The difference theorem is the sum theorem applied after negating one sequence. It is often the form used to compare two approximating sequences.

Remark (Dependencies).

- Sums of Cauchy Sequences

- Scalar Multiples of Cauchy Sequences
- Pointwise Difference of Sequences

Theorem (Linear Combinations of Cauchy Sequences)

Theorem 3.6.16 (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}\ \forall \alpha \in \mathbb{R}\ \forall \beta \in \mathbb{R}$$
$$\Big((x_n) \texttt{ is Cauchy} \wedge (y_n) \texttt{ is Cauchy} \implies (\alpha x_n + \beta y_n) \texttt{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(\alpha x_n + \beta y_n).$$

Remark (Interpretation). The Cauchy real sequences are closed under finite real linear combinations. This is the additive vector-space part of Cauchy algebra.

Remark (Dependencies).

- Scalar Multiples of Cauchy Sequences
- Sums of Cauchy Sequences
- Pointwise Sum of Sequences

Theorem (Products of Cauchy Sequences)

Theorem 3.6.17 (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}$$
$$\Big((x_n) \texttt{ is Cauchy} \wedge (y_n) \texttt{ is Cauchy} \implies (x_n y_n) \texttt{ is Cauchy}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \wedge \mathrm{CauchySequence}(y_n) \implies \mathrm{CauchySequence}(x_n y_n).$$

Remark (Interpretation). Products require boundedness in addition to tail closeness. The boundedness is supplied automatically because every Cauchy real sequence is bounded.

Remark (Dependencies).

- Cauchy Sequences Are Bounded
- Pointwise Product of Sequences

Theorem (Reciprocals of Cauchy Sequences)

Theorem 3.6.18 (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\\ &\quad\Big((x_n) \texttt{ is Cauchy} \land \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c)\\ &\qquad \implies (1/x_n) \texttt{ is Cauchy}\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \land \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|x_n| \geq c) \implies \mathrm{CauchySequence}(1/x_n).$$

Remark (Interpretation). The lower bound away from zero prevents division from amplifying small tail differences without control. Without it, reciprocals need not preserve Cauchy behavior.

Remark (Dependencies).

- Cauchy Sequence
- Reciprocal Sequence

Theorem (Quotients of Cauchy Sequences)

Theorem 3.6.19 (Quotients of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|y_n| \geq c$ for every $n \geq N_0$, then (x_n/y_n) is Cauchy.*
Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\forall (x_n) \subseteq \mathbb{R}\ \forall (y_n) \subseteq \mathbb{R}\\ &\quad\Big((x_n) \texttt{ is Cauchy} \land (y_n) \texttt{ is Cauchy} \land \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c)\\ &\qquad \implies (x_n/y_n) \texttt{ is Cauchy}\Big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{CauchySequence}(x_n) \land \mathrm{CauchySequence}(y_n) \land \exists c > 0\ \exists N_0 \in \mathbb{N}\ \forall n \geq N_0\ (|y_n| \geq c) \implies \mathrm{CauchySequence}($$

Remark (Interpretation). The quotient theorem is the product theorem combined with the reciprocal theorem for a denominator eventually bounded away from zero.

Remark (Dependencies).

- Products of Cauchy Sequences
- Reciprocals of Cauchy Sequences
- Pointwise Quotient of Sequences

Theorem (Absolute Values of Cauchy Sequences)

Theorem 3.6.20 (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Go to proof.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq \mathbb{R} \Big((x_n) \text{ is Cauchy} \implies (|x_n|) \text{ is Cauchy} \Big).$$

Remark (Definition predicate reading).

$$\text{CauchySequence}(x_n) \implies \text{CauchySequence}(|x_n|).$$

Remark (Interpretation). The reverse triangle inequality bounds the distance between absolute values by the original distance between terms.

Remark (Dependencies).

- Cauchy Sequence
- Absolute Value of a Sequence

3.7 Applications of Sequences

Sequence Applications Toolkit

Concept family: Numerical approximation by explicitly defined sequences.

Atomic items in this section:

- Definition: Newton Sequence for $\sqrt{2}$
- Theorem: Newton Approximation of $\sqrt{2}$
- Definition: Factorial Partial Sums
- Theorem: Factorial Partial Sums Approximate e
- Definition: Compound-Interest Sequence
- Theorem: Compound-Interest Approximation of e
- Definition: Decimal Truncation Sequence
- Theorem: Decimal Truncations Converge

Applications of Sequences

The convergence theorems for sequences become numerical tools once the sequence is computable. Newton iteration turns an equation into a convergent recursive approximation; factorial partial sums and compound-interest terms approximate the Euler number; decimal truncations approximate arbitrary real numbers by finite decimals.

Definition (Newton Sequence for $\sqrt{2}$)

Definition 3.7.1 (Newton Sequence for $\sqrt{2}$). The **Newton sequence for** $\sqrt{2}$ is the real sequence (x_n) defined by

$$x_1 := \frac{3}{2}, \qquad x_{n+1} := \frac{1}{2}\left(x_n + \frac{2}{x_n}\right) \quad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$x_1 = \frac{3}{2} \\ \land \quad \forall n \in \mathbb{N} \left(x_{n+1} = \frac{1}{2}\left(x_n + \frac{2}{x_n}\right)\right).$$

Remark (Interpretation). The recursion is Newton's method applied to the equation $x^2 - 2 = 0$. Each new term averages the current estimate with the correction $2/x_n$.

Remark (Dependencies).

- Sequence

Theorem (Newton Approximation of $\sqrt{2}$)

Theorem 3.7.2 (Newton Approximation of $\sqrt{2}$). *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \exists N \in \mathbb{N} \, \forall n \geq N \left(|x_n - \sqrt{2}| < \varepsilon\right).$$

Remark (Definition predicate reading).

$$\mathrm{ConvergentSequence}(x_n, \sqrt{2}).$$

Remark (Interpretation). The Newton sequence is decreasing after the first term and bounded below by $\sqrt{2}$, so monotone convergence supplies a limit. Passing the recursion to the limit identifies the limit as the positive solution of $L^2 = 2$.

Remark (Dependencies).

- Newton Sequence for $\sqrt{2}$
- Monotone Convergence Theorem
- Rational Sequence Limit
- Square Root Sequence

Definition (Factorial Partial Sums)

Definition 3.7.3 (Factorial Partial Sums). The **factorial partial-sum sequence** is the real sequence (s_n) defined by

$$s_n := \sum_{k=0}^{n} \frac{1}{k!} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(s_n = \sum_{k=0}^{n} \frac{1}{k!} \right).$$

Remark (Interpretation). The sequence adds the reciprocal factorial terms one at a time. The terms decrease very rapidly, so the partial sums stabilize quickly.

Remark (Dependencies).

- Sequence

Theorem (Factorial Partial Sums Approximate e)

Theorem 3.7.4 (Factorial Partial Sums Approximate e). *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N \, \big(|s_n - e| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(s_n, e).$$

Remark (Interpretation). This is the sequence form of the factorial-series definition of e. In practice the approximation is efficient because the error after n terms is controlled by a rapidly shrinking factorial tail.

Remark (Dependencies).

- Factorial Partial Sums
- The Number e
- Cauchy Criterion for Real Sequences

Definition (Compound-Interest Sequence)

Definition 3.7.5 (Compound-Interest Sequence). The **compound-interest sequence** is the real sequence (c_n) defined by

$$c_n := \left(1 + \frac{1}{n}\right)^n \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(c_n = \left(1 + \frac{1}{n}\right)^n\right).$$

Remark (Interpretation). The term c_n is the value of one unit of principal after one unit of time when interest is compounded n times at nominal rate 1.

Remark (Dependencies).

- Sequence

Theorem (Compound-Interest Approximation of e)

Theorem 3.7.6 (Compound-Interest Approximation of e). *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n\to\infty} c_n = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\, \exists N \in \mathbb{N}\, \forall n \geq N\, \big(|c_n - e| < \varepsilon\big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(c_n, e).$$

Remark (Interpretation). The compound-interest approximation gives a second natural sequence converging to the same constant as the factorial partial sums. It is slower numerically, but it explains why e appears in continuous growth.

Remark (Dependencies).

- Compound-Interest Sequence

- The Number e
- Sequence Squeeze Theorem

Definition (Decimal Truncation Sequence)

Definition 3.7.7 (Decimal Truncation Sequence). Let $\alpha \in \mathbb{R}$. The **decimal truncation sequence** of α is the real sequence (d_n) defined by

$$d_n := \frac{\lfloor 10^n \alpha \rfloor}{10^n} \qquad \text{for every } n \in \mathbb{N}.$$

Remark (Standard quantified statement).

$$\forall n \in \mathbb{N} \left(d_n = \frac{\lfloor 10^n \alpha \rfloor}{10^n} \right).$$

Remark (Interpretation). The number d_n is obtained by cutting α off after n decimal places. Truncation is one-sided: it does not round to the nearest decimal.

Remark (Dependencies).

- Sequence

Theorem (Decimal Truncations Converge)

Theorem 3.7.8 (Decimal Truncations Converge). *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \; \exists N \in \mathbb{N} \; \forall n \geq N \; \big(|d_n - \alpha| < \varepsilon \big).$$

Remark (Definition predicate reading).

$$\text{ConvergentSequence}(d_n, \alpha).$$

Remark (Interpretation). Decimal truncations converge because the truncation error is always less than 10^{-n}. This is one of the most concrete examples of convergence: finite decimal data can approximate a real number to arbitrary accuracy.

Remark (Dependencies).

- Decimal Truncation Sequence
- Sequence Squeeze Theorem

Proofs

Remark (Return). Return to Theorem

Theorem (Constant Sequence Convergence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) converges to c.*

Professional Standard Proof.
TODO: professional standard proof for constant-sequence-convergence. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-sequence-convergence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Zero Sequence is Null). *Let (x_n) be a real sequence. If $x_n = 0$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence.*

Professional Standard Proof.
TODO: professional standard proof for zero-sequence-is-null. □

Detailed Learning Proof.
TODO: detailed learning proof for zero-sequence-is-null. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Null Sequence). *Let (x_n) be a real sequence, and let $c \in \mathbb{R}$. If $x_n = c$ for every $n \in \mathbb{N}$, then (x_n) is a null sequence if and only if $c = 0$.*

Professional Standard Proof.
TODO: professional standard proof for constant-null-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-null-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Difference from the Limit is Null). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. The sequence (x_n) converges to L if and only if the sequence $(x_n - L)$ is a null sequence.*

Professional Standard Proof.
TODO: professional standard proof for difference-from-limit-is-null. $\square$

Detailed Learning Proof.
TODO: detailed learning proof for difference-from-limit-is-null. $\square$

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Ultimately Constant Sequence Convergence). *Let* (x_n) *be a real sequence. If there exist* $N_0 \in \mathbb{N}$ *and* $c \in \mathbb{R}$ *such that* $x_n = c$ *for every* $n \geq N_0$*, then* (x_n) *converges to* c*.*

Professional Standard Proof.
TODO: professional standard proof for ultimately-constant-sequence-convergence. □

Detailed Learning Proof.
TODO: detailed learning proof for ultimately-constant-sequence-convergence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Implies Ultimately Constant). *Every constant real sequence is ultimately constant.*

Professional Standard Proof.
TODO: professional standard proof for constant-implies-ultimately-constant. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-implies-ultimately-constant. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Ultimately Zero Sequence is Null). *Let (x_n) be a real sequence. If there exists $N_0 \in \mathbb{N}$ such that $x_n = 0$ for every $n \geq N_0$, then (x_n) is a null sequence.*

Professional Standard Proof.
TODO: professional standard proof for ultimately-zero-sequence-is-null. □

Detailed Learning Proof.
TODO: detailed learning proof for ultimately-zero-sequence-is-null. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Ultimately Constant Null Sequence). *Let (x_n) be a real sequence. Suppose there exist $N_0 \in \mathbb{N}$ and $c \in \mathbb{R}$ such that $x_n = c$ for every $n \geq N_0$. Then (x_n) is a null sequence if and only if $c = 0$.*

Professional Standard Proof.
TODO: professional standard proof for ultimately-constant-null-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for ultimately-constant-null-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Tail Equality Preserves Convergence). *Let (x_n) and (y_n) be real sequences. If there exists $N_0 \in \mathbb{N}$ such that $x_n = y_n$ for every $n \geq N_0$, then for every $L \in \mathbb{R}$, (x_n) converges to L if and only if (y_n) converges to L.*

Professional Standard Proof.
TODO: professional standard proof for tail-equality-preserves-convergence. □

Detailed Learning Proof.
TODO: detailed learning proof for tail-equality-preserves-convergence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventually Bounded Above Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded above if and only if there exist $N_0 \in \mathbb{N}$ and $M \in \mathbb{R}$ such that $x_n \leq M$ for every $n \geq N_0$.*

Professional Standard Proof.
TODO: professional standard proof for eventually-bounded-above-tail-formulation. □

Detailed Learning Proof.
TODO: detailed learning proof for eventually-bounded-above-tail-formulation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventually Bounded Below Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded below if and only if there exist $N_0 \in \mathbb{N}$ and $m \in \mathbb{R}$ such that $m \leq x_n$ for every $n \geq N_0$.*

Professional Standard Proof.
TODO: professional standard proof for eventually-bounded-below-tail-formulation. □

Detailed Learning Proof.
TODO: detailed learning proof for eventually-bounded-below-tail-formulation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventually Bounded Tail Formulation). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if there exist $N_0 \in \mathbb{N}$ and $M > 0$ such that $|x_n| \leq M$ for every $n \geq N_0$.*

Professional Standard Proof.
TODO: professional standard proof for eventually-bounded-tail-formulation. □

Detailed Learning Proof.
TODO: detailed learning proof for eventually-bounded-tail-formulation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Sequence If and Only If Bounded Above and Below). *Let (x_n) be a real sequence. The sequence (x_n) is bounded if and only if it is both bounded above and bounded below.*

Professional Standard Proof.
TODO: professional standard proof for bounded-sequence-iff-bounded-above-and-below. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-sequence-iff-bounded-above-and-below. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Bound Gives Upper and Lower Bounds). *Let (x_n) be a real sequence, and let $K > 0$. If $|x_n| \leq K$ for every $n \in \mathbb{N}$, then $-K \leq x_n \leq K$ for every $n \in \mathbb{N}$.*

Professional Standard Proof.
TODO: professional standard proof for absolute-bound-gives-upper-and-lower-bounds. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-bound-gives-upper-and-lower-bounds. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Upper and Lower Bounds Give an Absolute Bound). *Let (x_n) be a real sequence. If there exist $m, M \in \mathbb{R}$ such that $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then there exists $K > 0$ such that $|x_n| \leq K$ for every $n \in \mathbb{N}$.*

Professional Standard Proof.
TODO: professional standard proof for upper-and-lower-bounds-give-absolute-bound. □

Detailed Learning Proof.
TODO: detailed learning proof for upper-and-lower-bounds-give-absolute-bound. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Scalar Multiple). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If $x_n \to L$, then $\alpha x_n \to \alpha L$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-scalar-multiple. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-scalar-multiple. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Sum). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n + y_n \to L + M$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-sum. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-sum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Negation). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $-x_n \to -L$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-negation. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-negation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Difference). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. If $x_n \to L$ and $y_n \to M$, then $x_n - y_n \to L - M$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-difference. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-difference. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Product). *Let* (x_n) *and* (y_n) *be real sequences, and let* $L, M \in \mathbb{R}$. *If* $x_n \to L$ *and* $y_n \to M$, *then* $x_n y_n \to LM$.

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-product. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-product. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Nonzero Limit is Eventually Nonzero). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $x_n \neq 0$ for every $n \geq N$.*

Professional Standard Proof.
TODO: professional standard proof for nonzero-limit-eventually-nonzero. □

Detailed Learning Proof.
TODO: detailed learning proof for nonzero-limit-eventually-nonzero. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Reciprocal). *Let (x_n) be a real sequence such that $x_n \neq 0$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$ with $L \neq 0$. If $x_n \to L$, then $1/x_n \to 1/L$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-reciprocal. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-reciprocal. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Quotient). *Let (x_n) and (y_n) be real sequences such that $y_n \neq 0$ for every $n \in \mathbb{N}$, and let $L, M \in \mathbb{R}$ with $M \neq 0$. If $x_n \to L$ and $y_n \to M$, then $x_n/y_n \to L/M$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-quotient. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-quotient. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Square). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $x_n^2 \to L^2$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-square. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-square. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of an Absolute Value). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If $x_n \to L$, then $|x_n| \to |L|$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-an-absolute-value. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-an-absolute-value. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Positive Limit is Eventually Positive). *Let (x_n) be a real sequence, and let $L > 0$. If $x_n \to L$, then there exists $N \in \mathbb{N}$ such that $0 < x_n$ for every $n \geq N$.*

Professional Standard Proof.
TODO: professional standard proof for positive-limit-eventually-positive. □

Detailed Learning Proof.
TODO: detailed learning proof for positive-limit-eventually-positive. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit of a Square Root). *Let (x_n) be a real sequence such that $0 \leq x_n$ for every $n \in \mathbb{N}$, and let $L \in \mathbb{R}$. If $x_n \to L$, then $0 \leq L$ and $\sqrt{x_n} \to \sqrt{L}$.*

Professional Standard Proof.
TODO: professional standard proof for limit-of-a-square-root. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-of-a-square-root. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Polynomial Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p \in \mathbb{R}[t]$ be a polynomial. If $x_n \to L$, then $p(x_n) \to p(L)$.*

Professional Standard Proof.
TODO: professional standard proof for polynomial-sequence-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for polynomial-sequence-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Rational Sequence Limit). *Let (x_n) be a real sequence, let $L \in \mathbb{R}$, and let $p, q \in \mathbb{R}[t]$ be polynomials. Suppose $q(L) \neq 0$ and $q(x_n) \neq 0$ for every $n \in \mathbb{N}$. If $x_n \to L$, then*

$$\frac{p(x_n)}{q(x_n)} \to \frac{p(L)}{q(L)}.$$

Professional Standard Proof.
TODO: professional standard proof for rational-sequence-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for rational-sequence-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Equivalence of Convergence Formulations). *Let* $(x_n) \subseteq \mathbb{R}$ *and let* $L \in \mathbb{R}$. *The following statements are equivalent.*

(a) $x_n \to L$.

(b) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(|x_n - L| < \varepsilon\big)$.

(c) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(L - \varepsilon < x_n < L + \varepsilon\big)$.

(d) $\forall \varepsilon > 0\, \exists K \in \mathbb{N}\, \forall n \geq K\, \big(x_n \in V_\varepsilon(L)\big)$.

Professional Standard Proof.
TODO: professional standard proof for equivalence-of-convergence-formulations. □

Detailed Learning Proof.
TODO: detailed learning proof for equivalence-of-convergence-formulations. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Uniqueness of Limits). *Let $(x_n) \subseteq \mathbb{R}$. If $x_n \to L$ and $x_n \to K$, then $L = K$.*

Professional Standard Proof.
TODO: professional standard proof for uniqueness-of-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for uniqueness-of-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit Preserves Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $L, M \in \mathbb{R}$. Suppose that $x_n \to L$ and $y_n \to M$. If there exists $N_0 \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N_0$, then $L \leq M$.*

Professional Standard Proof.
TODO: professional standard proof for limit-preserves-eventual-order. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-preserves-eventual-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Strict Limit Separation Gives Eventual Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$. If $A < B$, then there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$.*

Professional Standard Proof.
TODO: professional standard proof for strict-limit-separation-gives-eventual-order. □

Detailed Learning Proof.
TODO: detailed learning proof for strict-limit-separation-gives-eventual-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventual Strict Comparison Preserves Weak Limit Order). *Let (x_n) and (y_n) be real sequences, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$ and $y_n \to B$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n < y_n$ for every $n \geq N$, then $A \leq B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n > y_n$ for every $n \geq N$, then $A \geq B$.

Professional Standard Proof.
TODO: professional standard proof for eventual-strict-comparison-preserves-weak-limit-order. □

Detailed Learning Proof.
TODO: detailed learning proof for eventual-strict-comparison-preserves-weak-limit-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Comparison for Sequence Limits). *Let (x_n) be a real sequence, and let $A, B \in \mathbb{R}$. Suppose that $x_n \to A$.*

(a) If there exists $N \in \mathbb{N}$ such that $x_n \le B$ for every $n \ge N$, then $A \le B$.

(b) If there exists $N \in \mathbb{N}$ such that $x_n < B$ for every $n \ge N$, then $A \le B$.

(c) If there exists $N \in \mathbb{N}$ such that $x_n \ge B$ for every $n \ge N$, then $A \ge B$.

(d) If there exists $N \in \mathbb{N}$ such that $x_n > B$ for every $n \ge N$, then $A \ge B$.

Professional Standard Proof.
TODO: professional standard proof for constant-comparison-for-sequence-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-comparison-for-sequence-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Squeeze Theorem). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. If there exists $N_0 \in \mathbb{N}$ such that $L \leq x_n \leq L$ for every $n \geq N_0$, then $x_n \to L$.*

Professional Standard Proof.
TODO: professional standard proof for constant-squeeze-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-squeeze-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sequence Squeeze Theorem). *Let (a_n), (x_n), and (b_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $a_n \to L$ and $b_n \to L$. If there exists $N_0 \in \mathbb{N}$ such that $a_n \leq x_n \leq b_n$ for every $n \geq N_0$, then $x_n \to L$.*

Professional Standard Proof.
TODO: professional standard proof for sequence-squeeze-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for sequence-squeeze-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute-Value Squeeze Theorem). *Let (x_n) and (u_n) be real sequences, and let $L \in \mathbb{R}$. Suppose that $u_n \to 0$. If there exists $N_0 \in \mathbb{N}$ such that $|x_n - L| \le u_n$ for every $n \ge N_0$, then $x_n \to L$.*

Professional Standard Proof.
TODO: professional standard proof for absolute-value-squeeze-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-squeeze-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Convergence of a Tail). *Let $(x_n) \subseteq \mathbb{R}$ and let $m \in \mathbb{N}$. The m-tail $(x_{m+n})_{n\in\mathbb{N}}$ converges if and only if (x_n) converges. Moreover, in this case,*

$$\lim_{n\to\infty} x_{m+n} = \lim_{n\to\infty} x_n.$$

Professional Standard Proof.
TODO: professional standard proof for convergence-of-tail. □

Detailed Learning Proof.
TODO: detailed learning proof for convergence-of-tail. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Convergence by Domination). *Let $(x_n) \subseteq \mathbb{R}$ and let $L \in \mathbb{R}$. Let (a_n) be a sequence of positive real numbers with $\lim_{n\to\infty} a_n = 0$. If there exist a constant $c > 0$ and an index $m \in \mathbb{N}$ such that*

$$|x_n - L| \;\le\; c\, a_n \qquad \forall n \ge m,$$

then $\lim_{n\to\infty} x_n = L$.

Professional Standard Proof.
TODO: professional standard proof for convergence-by-domination. □

Detailed Learning Proof.
TODO: detailed learning proof for convergence-by-domination. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Ratio Limit Less Than One Implies Null). *Let (x_n) be a sequence of positive real numbers. Suppose that the limit*

$$L := \lim_{n\to\infty} \frac{x_{n+1}}{x_n}$$

exists. If $L < 1$, then (x_n) converges and

$$\lim_{n\to\infty} x_n = 0.$$

Professional Standard Proof.
TODO: professional standard proof for ratio-limit-less-than-one-implies-null. □

Detailed Learning Proof.
TODO: detailed learning proof for ratio-limit-less-than-one-implies-null. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and bounded above, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

(b) If (x_n) is decreasing and bounded below, then (x_n) converges and

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Professional Standard Proof.
TODO: professional standard proof for monotone-convergence-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for monotone-convergence-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Strict Monotonicity Implies Monotonicity). *Let (x_n) be a real sequence.*

(a) If (x_n) is strictly increasing, then (x_n) is increasing.

(b) If (x_n) is strictly decreasing, then (x_n) is decreasing.

Professional Standard Proof.
TODO: professional standard proof for strict-monotonicity-implies-monotonicity. □

Detailed Learning Proof.
TODO: detailed learning proof for strict-monotonicity-implies-monotonicity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Monotone Sequence Equivalences). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing, then (x_n) converges if and only if (x_n) is bounded above.

(b) If (x_n) is decreasing, then (x_n) converges if and only if (x_n) is bounded below.

(c) If (x_n) is monotone, then (x_n) converges if and only if (x_n) is bounded.

Professional Standard Proof.
TODO: professional standard proof for bounded-monotone-sequence-equivalences. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-monotone-sequence-equivalences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventually Monotone Convergence Theorem). *Let (x_n) be a real sequence.*

(a) If (x_n) is eventually increasing and bounded above, then (x_n) converges.

(b) If (x_n) is eventually decreasing and bounded below, then (x_n) converges.

(c) If (x_n) is eventually monotone and bounded, then (x_n) converges.

Professional Standard Proof.
TODO: professional standard proof for eventually-monotone-convergence-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for eventually-monotone-convergence-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Unbounded Monotone Divergence). *Let (x_n) be a real sequence.*

(a) If (x_n) is increasing and is not bounded above, then $x_n \to +\infty$.

(b) If (x_n) is decreasing and is not bounded below, then $x_n \to -\infty$.

Professional Standard Proof.
TODO: professional standard proof for unbounded-monotone-divergence. □

Detailed Learning Proof.
TODO: detailed learning proof for unbounded-monotone-divergence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Algebraic Transformations Preserve Monotonicity). *Let (x_n) be a real sequence and let $c, \alpha \in \mathbb{R}$.*

(a) If (x_n) is increasing, then $(x_n + c)$ is increasing.

(b) If (x_n) is decreasing, then $(x_n + c)$ is decreasing.

(c) If (x_n) is increasing and $\alpha > 0$, then (αx_n) is increasing.

(d) If (x_n) is increasing and $\alpha < 0$, then (αx_n) is decreasing.

(e) If (x_n) is increasing, then $(-x_n)$ is decreasing.

(f) If (x_n) is decreasing, then $(-x_n)$ is increasing.

Professional Standard Proof.
TODO: professional standard proof for algebraic-transformations-preserve-monotonicity. □

Detailed Learning Proof.
TODO: detailed learning proof for algebraic-transformations-preserve-monotonicity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Subsequence Indices Dominate the Identity). *Let $\sigma : \mathbb{N} \to \mathbb{N}$ be a strictly increasing index map. Then*

$$k \leq \sigma(k) \qquad \textit{for every } k \in \mathbb{N}.$$

Professional Standard Proof.
TODO: professional standard proof for subsequence-indices-dominate-identity. □

Detailed Learning Proof.
TODO: detailed learning proof for subsequence-indices-dominate-identity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Subsequences Preserve Limits). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then every subsequence of (x_n) converges to L.*

Professional Standard Proof.
TODO: professional standard proof for subsequences-preserve-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for subsequences-preserve-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Subsequential Limit of a Convergent Sequence). *Let (x_n) be a real sequence and let $L \in \mathbb{R}$. If $x_n \to L$, then L is the only possible subsequential limit of (x_n).*

Professional Standard Proof.
TODO: professional standard proof for subsequential-limit-of-convergent-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for subsequential-limit-of-convergent-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Divergence by Two Subsequential Limits). *Let (x_n) be a real sequence. If (x_n) has two distinct subsequential limits, then (x_n) does not converge.*

Professional Standard Proof.
TODO: professional standard proof for divergence-by-two-subsequential-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for divergence-by-two-subsequential-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Boundedness Passes to Subsequences). *Every subsequence of a bounded real sequence is bounded.*

Professional Standard Proof.
TODO: professional standard proof for boundedness-passes-to-subsequences. ☐

Detailed Learning Proof.
TODO: detailed learning proof for boundedness-passes-to-subsequences. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Monotonicity Passes to Subsequences). *Every subsequence of an increasing real sequence is increasing, and every subsequence of a decreasing real sequence is decreasing.*

Professional Standard Proof.
TODO: professional standard proof for monotonicity-passes-to-subsequences. □

Detailed Learning Proof.
TODO: detailed learning proof for monotonicity-passes-to-subsequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Subsequence of a Subsequence). *Every subsequence of a subsequence of (x_n) is a subsequence of (x_n).*

Professional Standard Proof.
TODO: professional standard proof for subsequence-of-subsequence. □

Detailed Learning Proof.
TODO: detailed learning proof for subsequence-of-subsequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Eventual Properties Pass to Subsequences). *Let $P(n)$ be a property of indices. If there exists $N \in \mathbb{N}$ such that $P(n)$ holds for every $n \geq N$, then for every strictly increasing index map σ there exists $K \in \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \geq K$.*

Professional Standard Proof.
TODO: professional standard proof for eventual-properties-pass-to-subsequences. □

Detailed Learning Proof.
TODO: detailed learning proof for eventual-properties-pass-to-subsequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Frequent Properties Yield Subsequences). *Let $P(n)$ be a property of indices. If for every $N \in \mathbb{N}$ there exists $n \geq N$ such that $P(n)$ holds, then there exists a strictly increasing index map $\sigma : \mathbb{N} \to \mathbb{N}$ such that $P(\sigma(k))$ holds for every $k \in \mathbb{N}$.*

Professional Standard Proof.
TODO: professional standard proof for frequent-properties-yield-subsequences. □

Detailed Learning Proof.
TODO: detailed learning proof for frequent-properties-yield-subsequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Subsequential Limits Respect Bounds). *Let (x_n) be a real sequence and let L be a subsequential limit of (x_n). If $m \leq x_n \leq M$ for every $n \in \mathbb{N}$, then $m \leq L \leq M$.*

Professional Standard Proof.
TODO: professional standard proof for subsequential-limits-respect-bounds. □

Detailed Learning Proof.
TODO: detailed learning proof for subsequential-limits-respect-bounds. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Squeeze Passes to Subsequences). *Let (a_n), (x_n), and (b_n) be real sequences. If $a_n \leq x_n \leq b_n$ eventually, then for every subsequence $(x_{\sigma(k)})$ there exists $K \in \mathbb{N}$ such that*

$$a_{\sigma(k)} \leq x_{\sigma(k)} \leq b_{\sigma(k)} \qquad \textit{for every } k \geq K.$$

Professional Standard Proof.
TODO: professional standard proof for squeeze-passes-to-subsequences. □

Detailed Learning Proof.
TODO: detailed learning proof for squeeze-passes-to-subsequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Monotone Subsequence Theorem). *Every real sequence has a monotone subsequence.*

Professional Standard Proof.
TODO: professional standard proof for monotone-subsequence-theorem. ☐

Detailed Learning Proof.
TODO: detailed learning proof for monotone-subsequence-theorem. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bolzano-Weierstrass Theorem for Sequences). *Every bounded real sequence has a convergent subsequence.*

Professional Standard Proof.
TODO: professional standard proof for bolzano-weierstrass-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for bolzano-weierstrass-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sequential Compactness of Closed Bounded Intervals). *Let $a, b \in \mathbb{R}$ with $a \leq b$. Every sequence in $[a, b]$ has a convergent subsequence whose limit belongs to $[a, b]$.*

Professional Standard Proof.
TODO: professional standard proof for sequential-compactness-closed-bounded-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for sequential-compactness-closed-bounded-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Divergence to Infinity Implies Real Divergence). *Let (x_n) be a real sequence. If $x_n \to +\infty$ or $x_n \to -\infty$, then (x_n) is divergent.*

Professional Standard Proof.
TODO: professional standard proof for divergence-to-infinity-implies-real-divergence. $\square$

Detailed Learning Proof.
TODO: detailed learning proof for divergence-to-infinity-implies-real-divergence. $\square$

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Two Subsequential Limits Force Divergence). *Let (x_n) be a real sequence. If (x_n) has two subsequential limits $L, K \in \mathbb{R}$ with $L \neq K$, then (x_n) is divergent.*

Professional Standard Proof.
TODO: professional standard proof for two-subsequential-limits-force-divergence. □

Detailed Learning Proof.
TODO: detailed learning proof for two-subsequential-limits-force-divergence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Unbounded Above Sequence Has a Positive-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded above, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to +\infty$.*

Professional Standard Proof.
TODO: professional standard proof for unbounded-above-has-positive-infinity-subsequence. □

Detailed Learning Proof.
TODO: detailed learning proof for unbounded-above-has-positive-infinity-subsequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Unbounded Below Sequence Has a Negative-Infinity Subsequence). *Let (x_n) be a real sequence. If (x_n) is not bounded below, then there exists a subsequence $(x_{\sigma(k)})$ such that $x_{\sigma(k)} \to -\infty$.*

Professional Standard Proof.
TODO: professional standard proof for unbounded-below-has-negative-infinity-subsequence. □

Detailed Learning Proof.
TODO: detailed learning proof for unbounded-below-has-negative-infinity-subsequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Divergence Produces Two Subsequential Limits). *Let (x_n) be a bounded real sequence. If (x_n) is divergent, then there exist $L, K \in \mathbb{R}$ with $L \neq K$ such that L and K are subsequential limits of (x_n).*

Professional Standard Proof.
TODO: professional standard proof for bounded-divergence-produces-two-subsequential-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-divergence-produces-two-subsequential-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Tail Suprema Are Decreasing). *Let (x_n) be a bounded real sequence, and let (s_n) be its tail supremum sequence. Then (s_n) is decreasing.*

Professional Standard Proof.
TODO: professional standard proof for tail-suprema-are-decreasing. ☐

Detailed Learning Proof.
TODO: detailed learning proof for tail-suprema-are-decreasing. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Tail Infima Are Increasing). *Let (x_n) be a bounded real sequence, and let (i_n) be its tail infimum sequence. Then (i_n) is increasing.*

Professional Standard Proof.
TODO: professional standard proof for tail-infima-are-increasing. □

Detailed Learning Proof.
TODO: detailed learning proof for tail-infima-are-increasing. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Liminf Is Below Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Professional Standard Proof.
TODO: professional standard proof for liminf-below-limsup. □

Detailed Learning Proof.
TODO: detailed learning proof for liminf-below-limsup. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Convergence iff Liminf Equals Limsup). *Let (x_n) be a bounded real sequence, and let $L \in \mathbb{R}$. Then $x_n \to L$ if and only if*

$$\liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n = L.$$

Professional Standard Proof.
TODO: professional standard proof for convergence-iff-liminf-equals-limsup. □

Detailed Learning Proof.
TODO: detailed learning proof for convergence-iff-liminf-equals-limsup. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limsup Is the Largest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $S = \limsup_{n\to\infty} x_n$. Then S is a subsequential limit of (x_n), and every subsequential limit of (x_n) is less than or equal to S.*

Professional Standard Proof.
TODO: professional standard proof for limsup-largest-subsequential-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for limsup-largest-subsequential-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Liminf Is the Smallest Subsequential Limit). *Let (x_n) be a bounded real sequence, and let $I = \liminf_{n\to\infty} x_n$. Then I is a subsequential limit of (x_n), and every subsequential limit of (x_n) is greater than or equal to I.*

Professional Standard Proof.
TODO: professional standard proof for liminf-smallest-subsequential-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for liminf-smallest-subsequential-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Oscillation Criterion via Liminf and Limsup). *Let (x_n) be a bounded real sequence. Then*

$$\liminf_{n\to\infty} x_n < \limsup_{n\to\infty} x_n$$

if and only if (x_n) has two distinct subsequential limits.

Professional Standard Proof.
TODO: professional standard proof for oscillation-criterion-via-liminf-limsup. □

Detailed Learning Proof.
TODO: detailed learning proof for oscillation-criterion-via-liminf-limsup. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limsup Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

Professional Standard Proof.
TODO: professional standard proof for limsup-comparison-under-eventual-order. □

Detailed Learning Proof.
TODO: detailed learning proof for limsup-comparison-under-eventual-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Liminf Comparison Under Eventual Order). *Let (x_n) and (y_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that $x_n \leq y_n$ for every $n \geq N$, then*

$$\liminf_{n\to\infty} x_n \leq \liminf_{n\to\infty} y_n.$$

Professional Standard Proof.
TODO: professional standard proof for liminf-comparison-under-eventual-order. □

Detailed Learning Proof.
TODO: detailed learning proof for liminf-comparison-under-eventual-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limsup Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \text{for every } n \ge N,$$

then

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} x_n \le \limsup_{n\to\infty} b_n.$$

Professional Standard Proof.
TODO: professional standard proof for limsup-squeeze-under-eventual-order. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limsup-squeeze-under-eventual-order. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Liminf Squeeze Under Eventual Order). *Let (a_n), (x_n), and (b_n) be bounded real sequences. If there exists $N \in \mathbb{N}$ such that*

$$a_n \le x_n \le b_n \qquad \textit{for every } n \ge N,$$

then

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} x_n \le \liminf_{n\to\infty} b_n.$$

Professional Standard Proof.
TODO: professional standard proof for liminf-squeeze-under-eventual-order. □

Detailed Learning Proof.
TODO: detailed learning proof for liminf-squeeze-under-eventual-order. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Increasing Sequence Limit as Supremum). *Let (x_n) be an increasing real sequence that is bounded above. Then*

$$\lim_{n\to\infty} x_n = \sup\{x_n : n \in \mathbb{N}\}.$$

Professional Standard Proof.
TODO: professional standard proof for increasing-sequence-limit-as-supremum. □

Detailed Learning Proof.
TODO: detailed learning proof for increasing-sequence-limit-as-supremum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Decreasing Sequence Limit as Infimum). *Let (x_n) be a decreasing real sequence that is bounded below. Then*

$$\lim_{n\to\infty} x_n = \inf\{x_n : n \in \mathbb{N}\}.$$

Professional Standard Proof.
TODO: professional standard proof for decreasing-sequence-limit-as-infimum. □

Detailed Learning Proof.
TODO: detailed learning proof for decreasing-sequence-limit-as-infimum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Tail Suprema and Infima Converge). *Let (x_n) be a bounded real sequence. Define*

$$s_n = \sup\{x_k : k \geq n\}, \qquad i_n = \inf\{x_k : k \geq n\}.$$

Then (s_n) and (i_n) converge.

Professional Standard Proof.
TODO: professional standard proof for tail-suprema-and-infima-converge. □

Detailed Learning Proof.
TODO: detailed learning proof for tail-suprema-and-infima-converge. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Sequence Has Limsup and Liminf). *Every bounded real sequence has a limit superior and a limit inferior.*

Professional Standard Proof.
TODO: professional standard proof for bounded-sequence-has-limsup-and-liminf. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-sequence-has-limsup-and-liminf. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cluster Values Are Subsequential Limits). *Let (x_n) be a real sequence, and let $L \in \mathbb{R}$. Then L is a cluster value of (x_n) if and only if L is a subsequential limit of (x_n).*

Professional Standard Proof.
TODO: professional standard proof for cluster-values-are-subsequential-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for cluster-values-are-subsequential-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Sequences Have Cluster Values). *Every bounded real sequence has at least one cluster value.*

Professional Standard Proof.
TODO: professional standard proof for bounded-sequences-have-cluster-values. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-sequences-have-cluster-values. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limsup and Liminf Are Extremal Cluster Values). *Let (x_n) be a bounded real sequence. Then $\limsup_{n\to\infty} x_n$ is the largest cluster value of (x_n), and $\liminf_{n\to\infty} x_n$ is the smallest cluster value of (x_n).*

Professional Standard Proof.
TODO: professional standard proof for limsup-liminf-extremal-cluster-values. □

Detailed Learning Proof.
TODO: detailed learning proof for limsup-liminf-extremal-cluster-values. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Convergent Sequences Are Cauchy). *Let (x_n) be a real sequence. If (x_n) converges, then (x_n) is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for convergent-sequences-are-cauchy. □

Detailed Learning Proof.
TODO: detailed learning proof for convergent-sequences-are-cauchy. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Sequences Are Bounded). *Every Cauchy real sequence is bounded.*

Professional Standard Proof.
TODO: professional standard proof for cauchy-sequences-are-bounded. □

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-sequences-are-bounded. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Sequence with a Convergent Subsequence). *Let (x_n) be a Cauchy real sequence. If a subsequence of (x_n) converges to $L \in \mathbb{R}$, then (x_n) converges to L.*

Professional Standard Proof.
TODO: professional standard proof for cauchy-sequence-with-convergent-subsequence. ☐

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-sequence-with-convergent-subsequence. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Criterion for Real Sequences). *Let (x_n) be a real sequence. Then (x_n) converges if and only if (x_n) is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for cauchy-criterion-real-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-criterion-real-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Criterion via Tails). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if for every $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that the N-tail has pairwise term distances less than ε.*

Professional Standard Proof.
TODO: professional standard proof for cauchy-criterion-via-tails. □

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-criterion-via-tails. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Tail Diameter Criterion). *Let (x_n) be a real sequence. Then (x_n) is Cauchy if and only if every sufficiently late tail has arbitrarily small pairwise diameter.*

Professional Standard Proof.
TODO: professional standard proof for cauchy-tail-diameter-criterion. ☐

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-tail-diameter-criterion. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cauchy Sequences Have Vanishing Successive Differences). *If (x_n) is a Cauchy real sequence, then*

$$|x_{n+1} - x_n| \to 0.$$

Professional Standard Proof.
TODO: professional standard proof for cauchy-successive-differences-vanish. □

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-successive-differences-vanish. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Scalar Multiples of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence, and let $\alpha \in \mathbb{R}$. Then (αx_n) is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for scalar-multiple-cauchy-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for scalar-multiple-cauchy-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sums of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n + y_n)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for sum-cauchy-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for sum-cauchy-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Differences of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n - y_n)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for difference-cauchy-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for difference-cauchy-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Linear Combinations of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences, and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha x_n + \beta y_n)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for linear-combination-cauchy-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for linear-combination-cauchy-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Products of Cauchy Sequences). *Let (x_n) and (y_n) be Cauchy real sequences. Then $(x_n y_n)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for product-cauchy-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for product-cauchy-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Reciprocals of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. If there exist $c > 0$ and $N_0 \in \mathbb{N}$ such that $|x_n| \geq c$ for every $n \geq N_0$, then $(1/x_n)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for reciprocal-cauchy-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for reciprocal-cauchy-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Quotients of Cauchy Sequences). *Let* (x_n) *and* (y_n) *be Cauchy real sequences. If there exist* $c > 0$ *and* $N_0 \in \mathbb{N}$ *such that* $|y_n| \geq c$ *for every* $n \geq N_0$*, then* (x_n/y_n) *is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for quotient-cauchy-sequences. □

Detailed Learning Proof.
TODO: detailed learning proof for quotient-cauchy-sequences. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Absolute Values of Cauchy Sequences). *Let (x_n) be a Cauchy real sequence. Then $(|x_n|)$ is Cauchy.*

Professional Standard Proof.
TODO: professional standard proof for absolute-value-cauchy-sequence. □

Detailed Learning Proof.
TODO: detailed learning proof for absolute-value-cauchy-sequence. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Newton Approximation of 2). *Let (x_n) be the Newton sequence for $\sqrt{2}$. Then (x_n) converges and*

$$\lim_{n\to\infty} x_n = \sqrt{2}.$$

Professional Standard Proof.
TODO: professional standard proof for newton-approximation-sqrt-two. □

Detailed Learning Proof.
TODO: detailed learning proof for newton-approximation-sqrt-two. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Factorial Partial Sums Approximate e). *Let (s_n) be the factorial partial-sum sequence. Then (s_n) converges to the Euler number e:*

$$\lim_{n\to\infty} s_n = e.$$

Professional Standard Proof.
TODO: professional standard proof for factorial-partial-sums-approximate-e. □

Detailed Learning Proof.
TODO: detailed learning proof for factorial-partial-sums-approximate-e. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Compound-Interest Approximation of e). *Let (c_n) be the compound-interest sequence. Then*

$$\lim_{n \to \infty} c_n = e.$$

Professional Standard Proof.
TODO: professional standard proof for compound-interest-approximation-e. □

Detailed Learning Proof.
TODO: detailed learning proof for compound-interest-approximation-e. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Decimal Truncations Converge). *Let $\alpha \in \mathbb{R}$, and let (d_n) be the decimal truncation sequence of α. Then*

$$\lim_{n\to\infty} d_n = \alpha.$$

Professional Standard Proof.
TODO: professional standard proof for decimal-truncations-converge. □

Detailed Learning Proof.
TODO: detailed learning proof for decimal-truncations-converge. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Sequences, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 4

Elementary Functions

elementary-functions

Functions $\rightarrow$ **Elementary Functions** $\rightarrow$ Differentiation

4.1 Hyperbolic Functions

Toolkit — Hyperbolic Functions

- $\cosh x = (e^x + e^{-x})/2$: even part of e^x; $\cosh x \geq 1$; $\cosh 0 = 1$.
- $\sinh x = (e^x - e^{-x})/2$: odd part of e^x; bijective $\mathbb{R} \to \mathbb{R}$.
- $e^x = \cosh x + \sinh x$ for all x.
- Hyperbolic Pythagorean: $\cosh^2 x - \sinh^2 x = 1$ (note the minus sign, not plus).
- $(\sinh x)' = \cosh x$; $(\cosh x)' = \sinh x$ (no sign change, unlike trigonometric).
- $\tanh x \in (-1, 1)$ for all x; $\tanh x \to \pm 1$ as $x \to \pm\infty$.
- Inverse hyperbolic functions have explicit logarithmic formulas.

Hyperbolic Functions

Definition (Hyperbolic Functions)

Definition 4.1.1 (Hyperbolic Functions). For all $x \in \mathbb{R}$:

$$\cosh x \ := \frac{e^x + e^{-x}}{2}, \qquad \sinh x \ := \frac{e^x - e^{-x}}{2}, \qquad \tanh x \ := \frac{\sinh x}{\cosh x}.$$

Additional: $\operatorname{sech} x := 1/\cosh x$, $\operatorname{csch} x := 1/\sinh x$, $\coth x := \cosh x/\sinh x$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}: \ \cosh x = \frac{e^x + e^{-x}}{2} \geq 1, \ \sinh x = \frac{e^x - e^{-x}}{2}, \ e^x = \cosh x + \sinh x.$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicCosh}(x,y) \ :\equiv \ y = \frac{e^x + e^{-x}}{2}. \qquad \mathrm{HyperbolicSinh}(x,y) \ :\equiv \ y = \frac{e^x - e^{-x}}{2}.$$

$$\mathrm{HyperbolicTanh}(x,y) \ :\equiv \ \cosh x \neq 0 \ \wedge \ y = \frac{\sinh x}{\cosh x}.$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicPythagorean}(x) \ :\equiv \ \mathrm{HyperbolicCosh}(x,c) \wedge \mathrm{HyperbolicSinh}(x,s) \wedge c^2 - s^2 = 1.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{HyperbolicTanh}(x,y) \iff y \neq \frac{\sinh x}{\cosh x}.$$

Note $\cosh x > 0$ for all x, so HyperbolicTanh is defined on all of $\mathbb{R}$.

Remark (Contrast with trigonometric functions).

Trigonometric	**Hyperbolic**
$\cos^2 x + \sin^2 x = 1$	$\cosh^2 x - \sinh^2 x = 1$
$(\cos x)' = -\sin x$	$(\cosh x)' = \sinh x$
$(\sin x)' = \cos x$	$(\sinh x)' = \cosh x$
$e^{ix} = \cos x + i \sin x$	$e^x = \cosh x + \sinh x$
Parametrises unit circle	Parametrises unit hyperbola

The minus sign in $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$ reflects the substitution $x \mapsto ix$ in Euler's formula.

Proposition (Hyperbolic Pythagorean Identity)

Remark (Dependencies).

- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 4.1.2 (Hyperbolic Pythagorean Identity). *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \cosh^2 x - \sinh^2 x = 1.$$

Remark (Negated quantified statement).

$$\neg\,(\forall x : \cosh^2 x - \sinh^2 x = 1) \iff \exists x \in \mathbb{R} : \cosh^2 x - \sinh^2 x \neq 1.$$

The negation has no witness -- the identity holds universally.

Remark (Proof). Direct computation:

$$\cosh^2 x - \sinh^2 x = \frac{(e^x + e^{-x})^2}{4} - \frac{(e^x - e^{-x})^2}{4} = \frac{4}{4} = 1.$$

Proposition (Addition Formulas for Hyperbolic Functions)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- $\mathbb{R}$ Is a Field

Proposition 4.1.3 (Addition Formulas for Hyperbolic Functions). *For all $x, y \in \mathbb{R}$:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R}: \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y. \blacksquare$$

Remark (Definition predicate reading).

$$\mathrm{HyperbolicAddition}(x, y) \ :\equiv \ \sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y \wedge \cosh(x+y) = \cosh x \cosh y + \sin$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sinh) \iff \exists x, y \in \mathbb{R}: \ \sinh(x+y) \neq \sinh x \cosh y + \cosh x \sinh y. \blacksquare$$

The negation has no witness.

Remark (Proof strategy). Substitute the definitions and expand, or note the formulas follow from $e^{x+y} = e^x e^y$ by taking even and odd parts.

Remark (Comparison with trigonometric addition formulas). The $\sinh$ formula matches $\sin(x+y)$ exactly. The $\cosh$ formula differs from $\cos(x+y)$ only in the sign of the last term: $+\sinh x \sinh y$ versus $-\sin x \sin y$. The sign difference is a direct consequence of $\cosh^2 - \sinh^2 = 1$ versus $\cos^2 + \sin^2 = 1$.

Proposition (Limits of $\tanh$)

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Proposition 4.1.4 (Limits of $\tanh$).

$$\lim_{x \to +\infty} \tanh x = 1, \qquad \lim_{x \to -\infty} \tanh x = -1.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ \wedge\ \mathrm{LimitAtInfinity}(\tanh x,\ -\infty,\ -1).$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(\tanh x,\ +\infty,\ 1)\ :\equiv\ \forall\varepsilon>0,\ \exists M\in\mathbb{R},\ \forall x:\ x>M\Rightarrow \mathrm{OutputTolerance}(\tanh x,1,\varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to+\infty}\tanh x=1\right)\iff \exists\varepsilon_0>0,\ \forall M>0,\ \exists x>M:\ |\tanh x-1|\geq\varepsilon_0.$$

Remark (Proof strategy). Write $\tanh x = (1-e^{-2x})/(1+e^{-2x})$. As $x\to+\infty$, $e^{-2x}\to 0$, giving $\tanh x\to 1$. The limit at $-\infty$ follows by oddness: $\tanh(-x)=-\tanh(x)$.

Definition

Remark (Dependencies).

- Hyperbolic Functions
- Exponential Function
- Limit at Infinity
- Limit of a Quotient

Definition 4.1.5 (Inverse Hyperbolic Functions). Since $\sinh$ is strictly increasing and bijective $\mathbb{R}\to\mathbb{R}$, its inverse $\operatorname{arcsinh}$ is defined on all of $\mathbb{R}$. Since $\cosh$ is strictly increasing on $[0,\infty)$ with range $[1,\infty)$, its inverse $\operatorname{arccosh}$ is defined on $[1,\infty)$. Since $\tanh$ is strictly increasing with range $(-1,1)$, $\operatorname{arctanh}$ is defined on $(-1,1)$. Explicit logarithmic formulas:

$$\operatorname{arcsinh}x=\ln\left(x+\sqrt{x^2+1}\right),\quad x\in\mathbb{R}.$$
$$\operatorname{arccosh}x=\ln\left(x+\sqrt{x^2-1}\right),\quad x\geq 1.$$
$$\operatorname{arctanh}x=\tfrac{1}{2}\ln\left(\frac{1+x}{1-x}\right),\quad |x|<1.$$

Remark (Standard quantified statement).

$$\forall x\in\mathbb{R}:\ \sinh(\operatorname{arcsinh}x)=x\ \wedge\ \operatorname{arcsinh}(\sinh x)=x.$$
$$\forall x\geq 1:\ \cosh(\operatorname{arccosh}x)=x.\qquad \forall|x|<1:\ \tanh(\operatorname{arctanh}x)=x.$$

Remark (Definition predicate reading).

$$\mathrm{Arcsinh}(y,x)\ :\equiv\ x\in\mathbb{R}\ \wedge\ \sinh x=y\ :\equiv\ x=\ln\left(y+\sqrt{y^2+1}\right).$$

$$\mathrm{Arctanh}(y,x)\ :\equiv\ |y|<1\ \wedge\ \tanh x=y\ :\equiv\ x=\tfrac{1}{2}\ln\left(\frac{1+y}{1-y}\right).$$

Remark (Negated quantified statement).

$$\neg\ \mathrm{Arctanh}(y, x) \iff |y| \geq 1\ \vee\ \tanh x \neq y.$$

Remark (Derivation of arcsinh). Set $y = \sinh x = (e^x - e^{-x})/2$ and let $u = e^x > 0$. Then $y = (u - 1/u)/2$, so $u^2 - 2yu - 1 = 0$. The positive root is $u = y + \sqrt{y^2+1}$, giving $x = \ln(y + \sqrt{y^2+1})$.

Remark (Derivatives). $(\operatorname{arcsinh} x)' = 1/\sqrt{x^2+1}$; $(\operatorname{arccosh} x)' = 1/\sqrt{x^2-1}$ for $x > 1$; $(\operatorname{arctanh} x)' = 1/(1-x^2)$ for $|x| < 1$. These follow from the inverse function theorem applied to sinh, cosh, and tanh respectively.

Remark (Dependencies).

- Hyperbolic Functions
- Natural Logarithm
- Continuous Inverse Theorem
- Absolute Value on $\mathbb{R}$

4.2 The Logarithm

Toolkit — Logarithmic Functions

- $\ln : (0, \infty) \to \mathbb{R}$: inverse of e^x. Equivalently, $\ln x = \int_1^x 1/t\, dt$.
- Functional equation: $\ln(xy) = \ln x + \ln y$. Converts products to sums.
- $(\ln x)' = 1/x$; $\ln 1 = 0$; $\ln e = 1$.
- Change of base: $\log_a x = \ln x / \ln a$.
- Fundamental logarithmic limit: $\lim_{x\to 0} \ln(1+x)/x = 1$.
- Growth dominance: for all $r > 0$, $(\ln x)/x^r \to 0$ as $x \to \infty$ and $x^r \ln x \to 0$ as $x \to 0^+$.

The Logarithm

Definition (Natural Logarithm)

Definition 4.2.1 (Natural Logarithm). The **natural logarithm** $\ln : (0, \infty) \to \mathbb{R}$ is the inverse of the exponential function: for $x > 0$,

$$\ln x \ := \text{ the unique } y \in \mathbb{R} \text{ such that } e^y = x.$$

Equivalently, $\ln x$ is defined by the integral

$$\ln x \ := \int_1^x \frac{1}{t}\, dt.$$

Remark (Standard quantified statement).

$$\forall x > 0: \; e^{\ln x} = x. \qquad \forall y \in \mathbb{R}: \; \ln(e^y) = y.$$

Remark (Definition predicate reading).

$$\mathrm{NaturalLog}(x, y) \; :\equiv \; x > 0 \; \wedge \; e^y = x \; :\equiv \; x > 0 \; \wedge \; y = \int_1^x \frac{1}{t}\, dt.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\ln(1+x)}{x}, \, 0, \, 1\right) \; :\equiv \; \forall \varepsilon > 0, \; \exists \delta > 0, \; \forall x: \; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}$$

Remark (Negated quantified statement).

$$\neg \, \mathrm{NaturalLog}(x, y) \iff x \leq 0 \; \vee \; e^y \neq x.$$

Remark (Two definitions, same function). The inverse-of-exponential definition requires e^x to be established first. The integral definition is self-contained and is preferred when building analysis from scratch. Both yield the same function.

Remark (Key properties). (i) $\ln 1 = 0$; $\ln e = 1$.

(ii) **Functional equation:** $\ln(xy) = \ln x + \ln y$ for all $x, y > 0$.

(iii) $\ln(x/y) = \ln x - \ln y$; $\ln(x^r) = r \ln x$ for $r \in \mathbb{R}$.

(iv) ln is strictly increasing on $(0, \infty)$.

(v) $\ln x \to +\infty$ as $x \to \infty$; $\ln x \to -\infty$ as $x \to 0^+$.

(vi) $(\ln x)' = 1/x$ for all $x > 0$.

Remark (General logarithm). For $a > 0$, $a \neq 1$, the **logarithm base** a is $\log_a x := \ln x / \ln a$. The common logarithm is $\log_{10}$; the binary logarithm is $\log_2$.

Proposition (Fundamental Logarithmic Limit)

Remark (Dependencies).

- Exponential Function
- Inverse Bijection
- Continuous Inverse Theorem

Proposition 4.2.2 (Fundamental Logarithmic Limit).

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0:\ 0 < |x| < \delta \Rightarrow \left| \frac{\ln(1+x)}{x} - 1 \right| < \varepsilon.$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1 \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x:\ 0 < |x| < \delta \wedge \left| \frac{\ln(1+x)}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Equivalence with exponential limit). The fundamental exponential limit $\lim (e^x - 1)/x = 1$ and the fundamental logarithmic limit $\lim \ln(1+x)/x = 1$ are equivalent: each follows from the other via the substitution $x \leftrightarrow e^x - 1$. Both express the first-order Taylor approximation of the respective function at 0.

Remark (Proof strategy). From the series: $\ln(1+x) = x - x^2/2 + x^3/3 - \cdots$ (valid for $|x| < 1$). Dividing by x gives $1 - x/2 + x^2/3 - \cdots \to 1$ as $x \to 0$ by continuity of power series.

Proposition (Every Power Dominates the Logarithm)

Remark (Dependencies).

- Natural Logarithm
- Fundamental Exponential Limit
- Function Limit

Proposition 4.2.3 (Every Power Dominates the Logarithm). *For every $r > 0$:*

$$\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x \to 0^+} x^r \ln x = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0:\ \lim_{x \to \infty} x^{-r} \ln x = 0 \ \wedge\ \lim_{x \to 0^+} x^r \ln x = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^{-r} \ln x,\ +\infty,\ 0) \ :\equiv\ \forall \varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x:\ x > M \Rightarrow \mathrm{OutputTolerance}(x^{-r} \ln x, 0, \varepsilon)$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to \infty} \frac{\ln x}{x^r} = 0 \right) \iff \exists \varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M:\ x^{-r} |\ln x| \geq \varepsilon_0.$$

Remark (Proof strategy). For the first: substitute $x = e^t$ to get $t/e^{rt} \to 0$ by exponential dominance. For the second: substitute $x = 1/t$ and reduce to the first: $x^r \ln x = (-\ln t)/t^r \to 0$.

Remark (Significance). These limits establish the position of $\ln x$ in the growth hierarchy: $\ln x$ is dominated by every positive power. Combined with the exponential dominance result, the full ordering is $\ln x \ll x^r \ll e^x$ as $x \to \infty$ for any $r > 0$. The second limit says that near 0^+, the positive power x^r wins over the logarithmic blow-up $|\ln x| \to \infty$. Used in integrability arguments near 0.

Remark (Dependencies).

- Natural Logarithm
- Power Function
- Exponential Dominates Every Power
- Limit at Infinity

4.3 Power Functions and the Exponential

Toolkit — Power and Exponential Functions

- $e^x := \sum_{n=0}^{\infty} x^n/n!$: absolutely convergent for all $x \in \mathbb{R}$; functional equation $e^{x+y} = e^x e^y$; self-derivative $(e^x)' = e^x$.
- $e := \sum_{n=0}^{\infty} 1/n! = \lim_{n\to\infty}(1 + 1/n)^n$.
- Real power: $x^r := e^{r \ln x}$ for $x > 0$, $r \in \mathbb{R}$. Unifies integer, rational, and irrational exponents.
- Growth hierarchy: $\ln x \ll x^r \ll e^x$ as $x \to \infty$ for any $r > 0$.
- Fundamental exponential limit: $\lim_{x\to 0}(e^x - 1)/x = 1$.

Power Functions and the Exponential

Definition (Power Function)

Definition 4.3.1 (Power Function). Let $n \in \mathbb{N}$. The **integer power function** is $f : \mathbb{R} \to \mathbb{R}$, $f(x) := x^n$. For $r \in \mathbb{R}$ and $x > 0$, the **real power** is defined by

$$x^r := e^{r \ln x}.$$

Remark (Standard quantified statement).

$$\forall r \in \mathbb{R},\ \forall x > 0 :\ x^r = e^{r \ln x}.$$

For $r \in \mathbb{N}$ this reduces to the r-fold product $x \cdot x \cdots x$.

Remark (Definition predicate reading).

$$\mathrm{RealPower}(x, r, y) \ :\equiv\ x > 0 \ \wedge\ y = e^{r \ln x}.$$

Remark (Laws of exponents). For $x, y > 0$ and $r, s \in \mathbb{R}$:

$$x^r \cdot x^s = x^{r+s}, \quad (x^r)^s = x^{rs}, \quad (xy)^r = x^r y^r, \quad x^0 = 1, \quad x^1 = x.$$

Each follows from the corresponding property of the exponential via the definition $x^r = e^{r \ln x}$.

Remark (Interpretation). Defining $x^r := e^{r \ln x}$ unifies all three cases -- integer, rational, irrational exponents -- into a single formula and makes differentiability of x^r immediate from that of e^x and $\ln x$ via the chain rule: $(x^r)' = rx^{r-1}$.

Definition (The Number e)

Remark (Dependencies).

- Integer Powers in a Field
- Exponential Function
- Natural Logarithm
- Multiplication on $\mathbb{R}$

Definition 4.3.2 (The Number e). The **Euler number** is

$$e := \sum_{n=0}^{\infty} \frac{1}{n!} = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n \approx 2.71828\ldots$$

Both expressions define the same real number.

Remark (Standard quantified statement).

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \wedge e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n.$$

Remark (Interpretation). The series $\sum 1/n!$ converges rapidly by comparison with a geometric series. The limit $(1 + 1/n)^n$ arises from continuous compound interest and is proved equal to the series sum via the fundamental logarithmic limit. Both representations are in common use.

Remark (Dependencies).

- Sequence
- Convergent Sequence

Definition (Exponential Function)

Definition 4.3.3 (Exponential Function). The **exponential function** $\exp : \mathbb{R} \to (0, \infty)$ is defined by the absolutely convergent power series

$$e^x := \exp(x) := \sum_{n=0}^{\infty} \frac{x^n}{n!}, \qquad x \in \mathbb{R}.$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \; e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, \quad \texttt{convergent absolutely since} \quad \sum \frac{|x|^n}{n!} < \infty.$$

Remark (Definition predicate reading).

$$\mathrm{Exponential}(x, y) \;:\equiv\; y = \sum_{n=0}^{\infty} \frac{x^n}{n!}.$$

Remark (Key properties). (i) $e^0 = 1$; $e^1 = e$.

(ii) **Functional equation:** $e^{x+y} = e^x \cdot e^y$ for all $x, y \in \mathbb{R}$.

(iii) $e^x > 0$ for all x; $e^{-x} = 1/e^x$.

(iv) e^x is strictly increasing and bijective $\mathbb{R} \to (0, \infty)$.

(v) $(e^x)' = e^x$ -- the exponential is its own derivative.

Remark (General exponential). For $a > 0$, $a \neq 1$: $a^x := e^{x \ln a}$. Strictly increasing if $a > 1$, strictly decreasing if $0 < a < 1$. $(a^x)' = a^x \ln a$. Only e^x is its own derivative.

Proposition (Fundamental Exponential Limit)

Remark (Dependencies).

- The Number e
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 4.3.4 (Fundamental Exponential Limit).

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0, \; \exists \delta > 0 : \; 0 < |x| < \delta \Rightarrow \left| \frac{e^x - 1}{x} - 1 \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{e^x - 1}{x}, \, 0, \, 1\right) \;:\equiv\; \forall \varepsilon > 0, \; \exists \delta > 0, \; \forall x : \; \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\Bigg($$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{e^x - 1}{x} = 1 \right) \iff \exists \varepsilon_0 > 0, \; \forall \delta > 0, \; \exists x : \; 0 < |x| < \delta \; \wedge \; \left| \frac{e^x - 1}{x} - 1 \right| \geq \varepsilon_0.$$

Remark (Interpretation). This is the statement $(e^x)'\big|_{x=0} = 1$. The series gives $(e^x - 1)/x = 1 + x/2! + x^2/3! + \cdots \to 1$ as $x \to 0$ by continuity of power series at their center. Combined with the functional equation, this implies $(e^x)' = e^x$ everywhere.

Proposition (Compound Interest Limit)

Remark (Dependencies).

- Exponential Function
- Function Limit
- Derivative at a Point

Proposition 4.3.5 (Compound Interest Limit).

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists M > 0 :\ x > M \Rightarrow \left|\left(1+\frac{1}{x}\right)^x - e\right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{LimitAtInfinity}\big(\left(1+\tfrac{1}{x}\right)^x,\ +\infty,\ e\big)\ :\equiv\ \forall\varepsilon > 0,\ \exists M \in \mathbb{R},\ \forall x :\ x > M \Rightarrow \text{OutputTolerance}\big(\left(1+\tfrac{1}{x}\right)^x, e$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty}\left(1+\tfrac{1}{x}\right)^x = e\right) \iff \exists\varepsilon_0 > 0,\ \forall M > 0,\ \exists x > M :\ \left|\left(1+\frac{1}{x}\right)^x - e\right| \geq \varepsilon_0.$$

Remark (Proof strategy). Write $(1+1/x)^x = e^{x\ln(1+1/x)}$. Set $t = 1/x \to 0^+$ and use the fundamental logarithmic limit $\ln(1+t)/t \to 1$. Then $x\ln(1+1/x) = \ln(1+t)/t \to 1$, so $(1+1/x)^x = e^{x\ln(1+1/x)} \to e^1 = e$ by continuity of the exponential.

Proposition (Exponential Dominates Every Power)

Remark (Dependencies).

- The Number e
- Exponential Function
- Natural Logarithm
- Fundamental Logarithmic Limit
- Limit at Infinity

Proposition 4.3.6 (Exponential Dominates Every Power). *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{x^r}{e^x} = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall r > 0: \lim_{x\to\infty} x^r e^{-x} = 0.$$

Remark (Definition predicate reading).

$$\mathrm{LimitAtInfinity}(x^r e^{-x}, +\infty, 0) :\equiv \forall \varepsilon > 0, \exists M \in \mathbb{R}, \forall x: x > M \Rightarrow \mathrm{OutputTolerance}(x^r e^{-x}, 0, \varepsilon).$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to\infty} \frac{x^r}{e^x} = 0\right) \iff \exists \varepsilon_0 > 0, \forall M > 0, \exists x > M: x^r e^{-x} \geq \varepsilon_0.$$

Remark (Proof strategy). Pick integer $n > r$. The series bound $e^x \geq x^n/n!$ gives $x^r/e^x \leq n!\, x^{r-n} \to 0$ since $r - n < 0$.

Remark (Growth hierarchy). For any $r > 0$:

$$\ln x \ll x^r \ll e^x \qquad \text{as } x \to \infty.$$

The exponential eventually dominates every polynomial and every power function. This ordering is used constantly in estimates, dominated convergence arguments, and comparison tests.

Remark (Dependencies).

- Power Function
- Exponential Function
- Limit at Infinity
- Squeeze Theorem for Function Limits

4.4 Trigonometric Functions

Toolkit — Trigonometric Functions

- $\sin x$ and $\cos x$ are defined by absolutely convergent power series; properties are derived analytically, not from geometry.
- Pythagorean identity: $\sin^2 x + \cos^2 x = 1$. Proof: the derivative of the left side is 0 and its value at 0 is 1.
- Addition formulas follow from $e^{i(x+y)} = e^{ix}e^{iy}$.
- Fundamental trigonometric limit: $\lim_{x\to 0} \sin x/x = 1$.
- Second fundamental limit: $\lim_{x\to 0}(1-\cos x)/x^2 = 1/2$.
- Inverse functions arcsin, arccos, arctan are defined by restricting sin, cos, tan to monotone subdomains.

Trigonometric Functions

Definition (Sine and Cosine)

Definition 4.4.1 (Sine and Cosine via Power Series). The **sine** and **cosine** functions are defined for all $x \in \mathbb{R}$ by the absolutely convergent power series

$$\sin x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

$$\cos x := \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots$$

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R}: \ \sin x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \cos x = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!},$$

both absolutely convergent by the ratio test.

Remark (Definition predicate reading).

$$\mathrm{Sine}(x,y) :\equiv y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}. \qquad \mathrm{Cosine}(x,y) :\equiv y = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.$$

Remark (Key properties). (i) $\sin 0 = 0$; $\cos 0 = 1$.

(ii) $\sin(-x) = -\sin x$ (odd); $\cos(-x) = \cos x$ (even).

(iii) $(\sin x)' = \cos x$; $(\cos x)' = -\sin x$.

(iv) $|\sin x| \le 1$; $|\cos x| \le 1$ for all x.

(v) $\sin^2 x + \cos^2 x = 1$ (Pythagorean identity).

(vi) Periodic: $\sin(x + 2\pi) = \sin x$; $\cos(x + 2\pi) = \cos x$.

Remark (Relationship to e^{ix}). Euler's formula: $e^{ix} = \cos x + i \sin x$, where e^{ix} is defined by the complex exponential series. This gives $\cos x = \operatorname{Re}(e^{ix})$ and $\sin x = \operatorname{Im}(e^{ix})$. Euler's identity $e^{i\pi} + 1 = 0$ follows immediately.

Proposition (Pythagorean Identity)

Remark (Dependencies).

- Real Numbers
- Integer Powers in a Field
- Sequence
- Convergent Sequence

Proposition 4.4.2 (Pythagorean Identity). *For all* $x \in \mathbb{R}$*:*

$$\sin^2 x + \cos^2 x = 1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R} : \ \sin^2 x + \cos^2 x = 1.$$

Remark (Definition predicate reading).

$$\operatorname{PythagoreanIdentity}(x) \ :\equiv \ \operatorname{Sine}(x, s) \ \wedge \ \operatorname{Cosine}(x, c) \ \wedge \ s^2 + c^2 = 1.$$

Remark (Negated quantified statement).

$$\neg \left(\forall x : \sin^2 x + \cos^2 x = 1\right) \iff \exists x \in \mathbb{R} : \sin^2 x + \cos^2 x \neq 1.$$

The negation has no witness -- the identity holds universally.

Remark (Proof strategy). Define $f(x) := \sin^2 x + \cos^2 x$. Then $f'(x) = 2 \sin x \cos x + 2 \cos x(-\sin x) = 0$, so f is constant. Since $f(0) = 0 + 1 = 1$, the identity follows.

Proposition (Addition Formulas)

Remark (Dependencies).

- Sine and Cosine
- Derivative at a Point
- Zero Derivative Implies Constant

Proposition 4.4.3 (Addition Formulas). *For all* $x, y \in \mathbb{R}$*:*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x, y \in \mathbb{R} : \ \sin(x+y) = \sin x \cos y + \cos x \sin y \ \wedge \ \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Remark (Definition predicate reading).

$$\mathrm{TrigAddition}(x,y) \ :\equiv \ \sin(x+y) = \sin x \cos y + \cos x \sin y \wedge \cos(x+y) = \cos x \cos y - \sin x \sin y. \blacksquare$$

Remark (Negated quantified statement).

$$\neg\,(\text{addition formula for } \sin) \iff \exists x, y \in \mathbb{R} : \ \sin(x+y) \neq \sin x \cos y + \cos x \sin y.$$

The negation has no witness.

Remark (Proof strategy). From the Cauchy product: $e^{i(x+y)} = e^{ix}e^{iy}$. Taking real and imaginary parts yields both addition formulas simultaneously.

Remark (Derived identities).

$$\begin{aligned}
\sin(2x) &= 2\sin x \cos x \\
\cos(2x) &= \cos^2 x - \sin^2 x = 1 - 2\sin^2 x = 2\cos^2 x - 1 \\
\sin^2 x &= \tfrac{1}{2}(1 - \cos 2x) \\
\cos^2 x &= \tfrac{1}{2}(1 + \cos 2x)
\end{aligned}$$

Definition

Remark (Dependencies).

- Sine and Cosine
- Addition on $\mathbb{R}$
- Multiplication on $\mathbb{R}$

Definition 4.4.4 (Tangent, Cotangent, Secant, Cosecant).

$$\tan x := \frac{\sin x}{\cos x}, \quad \cot x := \frac{\cos x}{\sin x}, \quad \sec x := \frac{1}{\cos x}, \quad \csc x := \frac{1}{\sin x}.$$

$\tan x$ and $\sec x$ are defined where $\cos x \neq 0$, i.e., $x \neq \pi/2 + k\pi$ for $k \in \mathbb{Z}$.

Remark (Standard quantified statement).

$$\forall x \in \mathbb{R},\ \cos x \neq 0 :\ \tan x = \frac{\sin x}{\cos x}.$$

Remark (Definition predicate reading).

$$\mathrm{Tangent}(x, y)\ :\equiv\ \cos x \neq 0\ \wedge\ y = \frac{\sin x}{\cos x}.$$

Remark (Pythagorean identities for tan). Dividing $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$: $1 + \tan^2 x = \sec^2 x$. Dividing by $\sin^2 x$: $\cot^2 x + 1 = \csc^2 x$.

Definition

Remark (Dependencies).

- Sine and Cosine
- $\mathbb{R}$ Is a Field

Definition 4.4.5 (Inverse Trigonometric Functions).

- $\arcsin : [-1, 1] \to [-\pi/2, \pi/2]$ is the inverse of $\sin\big|_{[-\pi/2,\pi/2]}$.
- $\arccos : [-1, 1] \to [0, \pi]$ is the inverse of $\cos\big|_{[0,\pi]}$.
- $\arctan : \mathbb{R} \to (-\pi/2, \pi/2)$ is the inverse of $\tan\big|_{(-\pi/2,\pi/2)}$.

Remark (Standard quantified statement).

$$\forall y \in [-1, 1] :\ \sin(\arcsin y) = y. \qquad \forall x \in [-\pi/2, \pi/2] :\ \arcsin(\sin x) = x.$$

Remark (Definition predicate reading).

$$\mathrm{Arctan}(y, x)\ :\equiv\ x \in (-\pi/2, \pi/2)\ \wedge\ \tan x = y.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{\sin x}{x},\ 0,\ 1\right)\ :\equiv\ \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}\left(\frac{\sin x}{x}\right.$$

Remark (Negated quantified statement).

$$\neg\,\mathrm{Arctan}(y, x)\ \iff\ x \notin (-\pi/2, \pi/2)\ \vee\ \tan x \neq y.$$

Remark (Key limits of arctan).

$$\lim_{x\to+\infty} \arctan x = \frac{\pi}{2}, \qquad \lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}.$$

arctan is bounded, strictly increasing, and continuous on $\mathbb{R}$; the Monotone Limit Theorem gives the limits as the supremum and infimum of its range.

Theorem (Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Tangent, Cotangent, Secant, Cosecant
- Continuous Inverse Theorem
- Closed Interval

Theorem 4.4.6 (Fundamental Trigonometric Limit).

$$\lim_{x\to 0}\frac{\sin x}{x}=1.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall\varepsilon>0,\ \exists\delta>0:\ 0<|x|<\delta\Rightarrow\left|\frac{\sin x}{x}-1\right|<\varepsilon.$$

Remark (Negated quantified statement).

$$\neg\left(\lim_{x\to 0}\frac{\sin x}{x}=1\right)\iff\exists\varepsilon_0>0,\ \forall\delta>0,\ \exists x:\ 0<|x|<\delta\ \wedge\ \left|\frac{\sin x}{x}-1\right|\geq\varepsilon_0.$$

Remark (Proof strategy -- series). From the series: $\sin x/x=1-x^2/3!+x^4/5!-\cdots\to 1$ as $x\to 0$ by continuity of power series at their center.

Remark (Proof strategy -- geometric squeeze). For $0<x<\pi/2$, area comparison of two triangles and a circular sector gives $\cos x\leq\sin x/x\leq 1$. Since $\cos x\to 1$ as $x\to 0^+$, the Squeeze Theorem gives $\sin x/x\to 1$. Even symmetry extends this to $x\to 0^-$.

Remark (Significance). This is the statement $(\sin x)'\big|_{x=0}=1$, which implies $(\sin x)'=\cos x$ everywhere. It underlies all trigonometric differentiation.

Proposition (Second Fundamental Trigonometric Limit)

Remark (Dependencies).

- Sine and Cosine
- Function Limit
- Squeeze Theorem for Function Limits

Proposition 4.4.7 (Second Fundamental Trigonometric Limit).

$$\lim_{x\to 0}\frac{1-\cos x}{x^2}=\frac{1}{2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0,\ \exists \delta > 0 :\ 0 < |x| < \delta \Rightarrow \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}\left(\frac{1 - \cos x}{x^2},\ 0,\ \tfrac{1}{2} \right) :\equiv \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x :\ \mathrm{InputTolerance}(x, 0, \delta) \Rightarrow \mathrm{OutputTolerance}$$

Remark (Negated quantified statement).

$$\neg \left(\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2} \right) \iff \exists \varepsilon_0 > 0,\ \forall \delta > 0,\ \exists x :\ 0 < |x| < \delta \wedge \left| \frac{1 - \cos x}{x^2} - \frac{1}{2} \right| \geq \varepsilon_0.$$

Remark (Proof strategy). Use the half-angle identity $1 - \cos x = 2 \sin^2(x/2)$:

$$\frac{1 - \cos x}{x^2} = \frac{2 \sin^2(x/2)}{x^2} = \frac{1}{2} \left(\frac{\sin(x/2)}{x/2} \right)^2 \to \frac{1}{2} \cdot 1^2 = \frac{1}{2}.$$

Remark (Dependencies).

- Sine and Cosine
- Trigonometric Addition Formulas
- Fundamental Trigonometric Limit
- Function Limit

Proofs

Remark (Return). Return to Theorem

Proposition (Hyperbolic Pythagorean Identity). *For all $x \in \mathbb{R}$:*

$$\cosh^2 x - \sinh^2 x = 1.$$

Professional Standard Proof.
TODO: professional standard proof for hyperbolic-pythagorean. ☐

Detailed Learning Proof.
TODO: detailed learning proof for hyperbolic-pythagorean. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Addition Formulas for Hyperbolic Functions). *For all* $x, y \in \mathbb{R}$*:*

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y, \qquad \cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y.$$

Professional Standard Proof.
TODO: professional standard proof for hyperbolic-addition. ☐

Detailed Learning Proof.
TODO: detailed learning proof for hyperbolic-addition. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Limits of).

$$\lim_{x\to+\infty} \tanh x = 1, \qquad \lim_{x\to-\infty} \tanh x = -1.$$

Professional Standard Proof.
TODO: professional standard proof for tanh-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for tanh-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Fundamental Logarithmic Limit).

$$\lim_{x \to 0} \frac{\ln(1+x)}{x} = 1.$$

Professional Standard Proof.
TODO: professional standard proof for limit-logarithm-fundamental. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-logarithm-fundamental. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Every Power Dominates the Logarithm). *For every $r > 0$:*

$$\lim_{x\to\infty} \frac{\ln x}{x^r} = 0 \qquad \textit{and} \qquad \lim_{x\to 0^+} x^r \ln x = 0.$$

Professional Standard Proof.
TODO: professional standard proof for power-dominates-log. □

Detailed Learning Proof.
TODO: detailed learning proof for power-dominates-log. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Fundamental Exponential Limit).

$$\lim_{x \to 0} \frac{e^x - 1}{x} = 1.$$

Professional Standard Proof.
TODO: professional standard proof for limit-exponential-fundamental. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-exponential-fundamental. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Compound Interest Limit).

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x} = e.$$

Professional Standard Proof.
TODO: professional standard proof for limit-compound-interest. ☐

Detailed Learning Proof.
TODO: detailed learning proof for limit-compound-interest. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Exponential Dominates Every Power). *For every $r > 0$:*

$$\lim_{x \to \infty} \frac{x^r}{e^x} = 0.$$

Professional Standard Proof.
TODO: professional standard proof for exponential-dominates-power. □

Detailed Learning Proof.
TODO: detailed learning proof for exponential-dominates-power. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Pythagorean Identity). *For all* $x \in \mathbb{R}$*:*

$$\sin^2 x + \cos^2 x = 1.$$

Professional Standard Proof.
TODO: professional standard proof for pythagorean-identity. □

Detailed Learning Proof.
TODO: detailed learning proof for pythagorean-identity. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Addition Formulas). *For all $x, y \in \mathbb{R}$:*

$$\sin(x+y) = \sin x \cos y + \cos x \sin y, \qquad \cos(x+y) = \cos x \cos y - \sin x \sin y.$$

Professional Standard Proof.
TODO: professional standard proof for trig-addition-formulas. □

Detailed Learning Proof.
TODO: detailed learning proof for trig-addition-formulas. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Fundamental Trigonometric Limit).

$$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

Professional Standard Proof.
TODO: professional standard proof for fundamental-trig-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for fundamental-trig-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Second Fundamental Trigonometric Limit).

$$\lim_{x \to 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

Professional Standard Proof.
TODO: professional standard proof for limit-one-minus-cos. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-one-minus-cos. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Chapter 5

Functions

functions

Sequences $\to$ **Functions** $\to$ Continuity

5.1 Limits

Toolkit — One-Sided Limits

- RightLimit(f, c, L): approach from $x > c$ only. Input condition: $0 < x - c < \delta$.
- LeftLimit(f, c, L): approach from $x < c$ only. Input condition: $0 < c - x < \delta$.
- Sequential criteria hold for each direction separately.
- FunctionLimit$(f, c, L) \iff$ RightLimit$(f, c, L) \wedge$ LeftLimit(f, c, L).

Toolkit — Limits of Functions

- $\lim_{x \to c} f(x) = L$ requires c to be a cluster point of $\mathrm{dom}(f)$. The value $f(c)$ is irrelevant; f need not be defined at c.
- The input condition is punctured: $0 < |x - c| < \delta$ means $0 < |x - c| < \delta$. This is what separates limits from continuity, where Within(x, c, δ) permits $x = c$.
- Three equivalent forms: ε-δ, neighbourhood, sequential. The Sequential Criterion is the primary working tool.
- Bridge principle: algebra of sequence limits transfers automatically via the Sequential Criterion.
- A finite limit implies $\exists \delta > 0\ \exists M > 0\ \forall x \in A\ (0 < |x - c| < \delta \Rightarrow |f(x)| \leq M)$.
- Limits are local: behavior far from c is irrelevant.

Toolkit — Algebra of Limits

- All five rules follow from the Sequential Criterion plus algebra of sequence limits. No new ε-δ work.
- Bridge: $(x_n) \subseteq A \setminus \{c\}$, $x_n \to c \Rightarrow f(x_n) \to L$, $g(x_n) \to M \Rightarrow$ apply sequence algebra $\Rightarrow$ lift back by Sequential Criterion.
- Quotient: $M \neq 0$ forces BoundedAwayFromZeroNear(g, c), making f/g well-defined on a punctured neighbourhood.

Limits of Functions

Definition (ε-Closeness of a Function to a Value)

Definition 5.1.1 (ε-Closeness of a Function to a Value). Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, and let $\varepsilon > 0$. The function f is **ε-close to L** if $|f(x) - L| \leq \varepsilon$ for all $x \in X$.

Remark (Standard quantified statement).

$$f \ \varepsilon\text{-close to } L \iff \forall x \in X \left(|f(x) - L| \leq \varepsilon\right).$$

Remark (Interpretation). A global condition: every output lies within ε of L, regardless of where in X the input lies. This is Tao's scaffolding toward the limit definition, which localises the same output condition to a punctured neighbourhood of c.

Definition (Local δ-Closeness Near a Point)

Definition 5.1.2 (Local δ-Closeness Near a Point). Let $f : X \to \mathbb{R}$, let $L \in \mathbb{R}$, let x_0 be adherent to X, and let $\varepsilon, \delta > 0$. The function f is **ε-close to L near x_0 within δ** if

$$\forall x \in X \left(|x - x_0| < \delta \Rightarrow |f(x) - L| \leq \varepsilon\right).$$

Remark (Standard quantified statement).

$$\forall x \in X : \ |x - x_0| < \delta \ \Rightarrow \ |f(x) - L| \leq \varepsilon.$$

Remark (Interpretation). A local condition: only outputs from the δ-ball around x_0 need lie within ε of L. The input condition here is Within unpunctured. The limit definition replaces this with InputTolerance (punctured) and asserts that for every ε such a δ exists.

Remark (Dependencies).

- ε-Closeness of a Function to a Value

Definition (Limit of a Function)

Definition 5.1.3 (Limit of a Function)**.** Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. A real number L is the **limit of** f **at** c, written $\lim_{x\to c} f(x) = L$, if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Go to uniqueness proof.

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta\ \Rightarrow\ |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading). Define

$$\begin{aligned} \mathrm{OutputTol}(\varepsilon) &\Longleftrightarrow \varepsilon > 0, \\ \mathrm{InputTol}(\delta) &\Longleftrightarrow \delta > 0, \\ \mathrm{PuncturedNbhd}(x, c, \delta) &\Longleftrightarrow 0 < |x - c| < \delta, \\ \mathrm{OutputNbhd}(y, L, \varepsilon) &\Longleftrightarrow |y - L| < \varepsilon. \end{aligned}$$

Then

$$\mathrm{FunctionLimit}(f, A, c, L) \Longleftrightarrow \forall \varepsilon \Big(\mathrm{OutputTol}(\varepsilon) \Rightarrow \exists \delta \left(\mathrm{InputTol}(\delta) \wedge \forall x \in A \left[\mathrm{PuncturedNbhd}(x, c, \delta) \Rightarrow\right.\right.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\begin{aligned} \neg\, \mathrm{FunctionLimit}(f, A, c, L) &\Longleftrightarrow \exists \varepsilon_0 \Big(\mathrm{OutputTol}(\varepsilon_0) \wedge \forall \delta \left(\mathrm{InputTol}(\delta) \Rightarrow \exists x \in A \left[\mathrm{PuncturedNbhd}(x, c\right.\right. \\ &\Longleftrightarrow \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(0 < |x - c| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big). \end{aligned}$$

Remark (Failure modes).

- The proposed value L is the wrong target: the function approaches some value $M \neq L$ near c.
- The function does not settle near any single value as x approaches c through A.
- The outputs repeatedly stay at least a fixed distance from L no matter how tightly the inputs are restricted around c.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{FunctionLimit}(f, A, c, L) \Longleftrightarrow \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \Big(\ \underbrace{0 < |x - c| < \delta}_{\texttt{arbitrarily close to } c}\ \wedge\ \underbrace{|f(x) - L| \geq \varepsilon_0}_{\texttt{fixed output gap from } L}\ \Big).$$

Remark (Interpretation). The quantifier structure is asymmetric by design. The output tolerance ε is a ***demand***: the caller specifies how precise the output must be. The input tolerance δ is a ***response***: the definition asserts one always exists. The input condition is punctured (InputTolerance, not Within) because f may not be defined at c, and because continuity not limits is the concept that includes c itself. This is the deepest structural difference between the two definitions. Four critical features: (i) f need not be defined at c; (ii) c must be a cluster point of A else the condition is vacuously satisfied by every L; (iii) δ may depend on ε; (iv) the condition is about all x near c simultaneously, not just one.

Remark (Dependencies).

- Cluster Point

Definition (Sequential Definition of Limit)

Definition 5.1.4 (Sequential Definition of Limit). Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The limit $\lim_{x\to c} f(x) = L$ in the **sequential sense** holds if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, the sequence $(f(x_n))$ converges to L.

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \Rightarrow f(x_n) \to L\big).$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}_{\mathrm{seq}}(f, c, L) \ :\equiv \ \forall (x_n) \subseteq A\setminus\{c\} : \ \mathrm{ConvergentSequence}(x_n, c) \ \Rightarrow \ \mathrm{ConvergentSequence}(f($$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A \setminus \{c\} \ \big(x_n \to c \ \wedge \ f(x_n) \not\to L\big).$$

Remark (Interpretation). The sequence (x_n) must lie in $A \setminus \{c\}$: the term $x_n = c$ is excluded, mirroring the $0 <$ in the ε-δ form. The sequential form is the primary tool for both proving limits (via the bridge principle) and disproving them (exhibit one bad sequence). The ε-δ form is better for constructing explicit δ functions.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Proposition

Proposition 5.1.5 (Neighbourhood Form of the Limit). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon>0\;\exists\delta>0\;\forall x\in A\;0<|x-c|<\delta\Rightarrow|f(x)-L|<\varepsilon\Big]$$
$$\iff\Big[\forall\varepsilon>0\;\exists\delta>0\;\forall x\in V_\delta^*(c)\cap A\;f(x)\in V_\varepsilon(L)\Big].$$

Remark (Interpretation). The equivalence is purely notational: $V_\delta^*(c)\cap A$ is exactly the set of $x\in A$ satisfying $0<|x-c|<\delta$, and $f(x)\in V_\varepsilon(L)$ is exactly $\mathrm{OutputTolerance}(f(x),L,\varepsilon)$. The neighbourhood form makes the geometric content explicit: the image of every punctured δ-slice of A must fit inside the ε-ball around L. This is the form of continuity used in metric space topology.

Remark (Dependencies).

- Limit of a Function
- ε-Neighbourhood
- Deleted ε-Neighbourhood

Theorem

Theorem 5.1.6 (Uniqueness of Limits of Functions). *Let $f:A\to\mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c}f(x)=L\;\wedge\;\lim_{x\to c}f(x)=L'\;\Rightarrow\;L=L'.$$

Remark (Failure modes). Failure of uniqueness would mean that two distinct real numbers both satisfy the limit definition at the same cluster point.

Remark (Failure mode decomposition). Non-uniqueness would require two values $L\neq L'$ each satisfying the limit definition. The $\varepsilon=|L-L'|/2$ argument shows any x within $\min(\delta_1,\delta_2)$ of c satisfies both $|f(x)-L|<\varepsilon$ and $|f(x)-L'|<\varepsilon$, but then $|L-L'|\le|L-f(x)|+|f(x)-L'|<2\varepsilon=|L-L'|$, a contradiction. The cluster point condition is essential: it guarantees such an x exists.

Remark (Interpretation). The limit, when it exists, is uniquely determined by the local behavior of f near c. Uniqueness is what licenses writing $\lim_{x\to c}f(x)=L$ as an equation rather than an assertion.

Remark (Dependencies).

- Limit of a Function

Proposition

Proposition 5.1.7 (Sequential Criterion for Function Limits). *Let $f:A\to\mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c}f(x)=L$ if and only if for every sequence $(x_n)\subseteq A\setminus\{c\}$ with $x_n\to c$, one has $f(x_n)\to L$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Negated quantified statement).

$$\neg \lim_{x\to c} f(x) = L \iff \exists (x_n) \subseteq A \setminus \{c\} :\ x_n \to c\ \wedge\ f(x_n) \not\to L.$$

Remark (Interpretation). The Sequential Criterion is the engine of the algebra of limits. Once it is in hand:

$$\text{algebra of sequence limits} \xRightarrow{\text{Sequential Criterion}} \text{algebra of function limits.}$$

Every sequence result sum, product, quotient, squeeze transfers to function limits by feeding a sequence into the criterion and reading the result back. The negation is equally powerful: to disprove $\lim_{x\to c} f(x) = L$, exhibit one sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$ and $f(x_n) \not\to L$.

Remark (Dependencies).

- Limit of a Function
- Sequential Definition of Limit

Corollary

Corollary 5.1.8 (Three Equivalent Forms of the Function Limit). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\iff \Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in V_\delta^*(c) \cap A\ f(x) \in V_\varepsilon(L)\Big]$$
$$\iff \Big[\forall (x_n) \subseteq A \setminus \{c\}\ x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Interpretation). All three encodings capture the same mathematical condition. The ε-δ form is best for proving explicit bounds. The neighbourhood form is best for geometric reasoning and metric space generalisation. The sequential form is best for importing sequence results and constructing counterexamples.

Remark (Dependencies).

- Limit of a Function

- Neighbourhood Form of the Limit
- Sequential Criterion for Function Limits

Theorem

Theorem 5.1.9 (Limit Implies Local Boundedness). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \;\Rightarrow\; \exists\delta > 0\, \exists M > 0\, \forall x \in A\, \big(0 < |x - c| < \delta \Rightarrow |f(x)| \le M\big).$$

Remark (Proof sketch). Apply the definition with $\varepsilon = 1$: get $\delta > 0$ such that $|f(x) - L| < 1$ holds for all x satisfying $0 < |x - c| < \delta$. The triangle inequality then gives $|f(x)| \le |f(x) - L| + |L| < 1 + |L|$. Set $M = 1 + |L|$.

Remark (Interpretation). A finite limit does more than control the output near L: it prevents the function from growing without bound near c. Local boundedness is a coarser statement than limit existence, but it is what is needed for the product and quotient rules.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 5.1.10 (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \le f(x) \le b$ for all $x \in A$ with $x \ne c$, and $\lim_{x\to c} f(x) = L$, then $a \le L \le b$.*

Go to proof.

Remark (Standard quantified statement).

$$\big(\forall x \in A \setminus \{c\} : a \le f(x) \le b\big) \wedge \lim_{x\to c} f(x) = L \;\Rightarrow\; a \le L \le b.$$

Remark (Interpretation). If every nearby value of f is trapped in $[a, b]$, the limit cannot escape outside $[a, b]$. This is the function-limit analogue of the sequence result that limits preserve non-strict inequalities. The proof applies the Sequential Criterion and that sequence result directly.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 5.1.11 (Squeeze Theorem for Function Limits). *Let* $f, g, h : A \to \mathbb{R}$ *and let* c *be a cluster point of* A. *Suppose* $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$, *and suppose there exists* $\delta_0 > 0$ *such that*

$$\forall x \in A,\ 0 < |x-c| < \delta_0 \ \Rightarrow\ f(x) \le h(x) \le g(x).$$

Then $\lim_{x\to c} h(x) = L$.
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L \ \wedge\ f \le h \le g \text{ near } c \ \Rightarrow\ \lim_{x\to c} h(x) = L.$$

Remark (Interpretation). The proof routes entirely through the Sequential Criterion: convert to sequences, apply the sequential squeeze theorem, convert back. No new ε-δ argument is needed. The condition $f \le h \le g$ is only required near c, not globally.

Remark (Dependencies).

- Limit of a Function
- Bounded Limits

Theorem

Theorem 5.1.12 (Limits Are Local). *Let* $E \subseteq X \subseteq \mathbb{R}$, *let* x_0 *be a cluster point of* E, *let* $f : X \to \mathbb{R}$, *and let* $\delta > 0$. *Then*

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta, x_0+\delta)} f(x) = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to x_0;\, x\in E} f(x) = L \iff \lim_{x\to x_0;\, x\in E\cap(x_0-\delta,\, x_0+\delta)} f(x) = L.$$

Remark (Interpretation). The existence and value of $\mathrm{FunctionLimit}(f, x_0, L)$ depend only on the behavior of f within any fixed neighbourhood of x_0. Points outside that neighbourhood are irrelevant. This justifies focusing all limit arguments on small balls around c without any loss of generality.

Remark (Dependencies).

- Limit of a Function

Proposition

Proposition 5.1.13 (Limit of Absolute Value). *Let* $f : A \to \mathbb{R}$ *and let* c *be a cluster point of* A. *If* $\lim_{x\to c} f(x) = L$, *then* $\lim_{x\to c} |f(x)| = |L|$.
Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c} f(x) = L \ \Rightarrow\ \lim_{x\to c} |f(x)| = |L|.$$

Remark (Negated quantified statement).

$$\lim_{x\to c} |f(x)| = |L| \ \not\Rightarrow\ \lim_{x\to c} f(x) = L.$$

Remark (Failure modes). The absolute values may converge while the signs of $f(x)$ oscillate or approach the opposite sign from the proposed limiting value.

Remark (Proof strategy). The reverse triangle inequality $\big||f(x)| - |L|\big| \le |f(x) - L|$ feeds the ε-δ definition directly: the same δ that delivers $|f(x) - L| < \varepsilon$ also delivers $\big||f(x)| - |L|\big| < \varepsilon$.

Remark (Interpretation). Taking absolute values can collapse sign information, so convergence of f forces convergence of $|f|$, but convergence of $|f|$ alone does not recover the original limiting sign.

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 5.1.14 (Composition of Limits). *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x\to c_1} f(x) = c_2$ and $\lim_{y\to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x\to c_1} g(f(x)) = L$.*

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x\to c_1} f(x) = c_2 \ \wedge\ \lim_{y\to c_2} g(y) = L \ \wedge\ \big(c_2 \in B \Rightarrow g(c_2) = L\big) \ \Rightarrow\ \lim_{x\to c_1} g(f(x)) = L.$$

Remark (Failure modes). The composition conclusion can fail if the outer function is not controlled at the intermediate value reached by the inner limit.

Remark (Failure mode decomposition). The hypothesis $g(c_2) = L$ is essential. Without it the composition can fail: take $f \equiv c_2$ (constant) and $g(c_2) \neq L$. Then $g(f(x)) = g(c_2) \neq L$ for all x, so $\lim_{x\to c_1} g(f(x)) = L$ fails even though both component limits exist. The failure is that $f(x) = c_2$ is excluded from the punctured condition on g, but f hits c_2 constantly.

Remark (Interpretation). Composition of limits requires care precisely because the input condition is punctured. The extra hypothesis $g(c_2) = L$ bridges the gap: when $f(x) = c_2$, the output condition $g(f(x)) = g(c_2) = L$ is satisfied directly. This hypothesis is equivalent to requiring g to be continuous at c_2 (within B).

Remark (Dependencies).

- Limit of a Function

Theorem

Theorem 5.1.15 (Monotone Limit Theorem). *Let $a < b$ and let $f : (a, b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a, b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Go to proof.

Remark (Standard quantified statement).

f is monotone on $(a, b) \wedge \exists B > 0\ \forall x \in (a, b)\ (|f(x)| \le B) \Rightarrow \forall x_0 \in (a, b) : \lim_{x \to x_0^-} f(x) = f(x_0^-) \wedge \lim_{x \to}$

Remark (Interpretation). Boundedness guarantees the supremum and infimum used as candidate limits exist. Monotonicity forces convergence toward them from each side: for increasing f, the values $f(x)$ for $x < x_0$ form a bounded increasing net approaching their supremum. No oscillation is possible. The theorem establishes that monotone functions can only have jump discontinuities never oscillatory ones.

Remark (Dependencies).

- Monotone Function
- Bounded Function
- Left-Hand Limit
- Right-Hand Limit

Proposition

Proposition 5.1.16 (Cauchy Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x \to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \text{OutputTolerance}(\varepsilon),\ \exists\ \text{InputTolerance}(\delta),\ \forall x, y \in A :\ 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta\ \Rightarrow\ |f(x) - f$$

Go to proof.

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \left(\lim_{x \to c} f(x) = L\right) \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x, y \in A :\ 0 < |x - c| < \delta \wedge 0 < |y - c| < \delta \Rightarrow |f(x) - f($$

Remark (Interpretation). The Cauchy criterion tests limit existence without knowing L. Two outputs from any sufficiently small punctured ball must lie within ε of each other the outputs cluster even if their common limit is unknown. This is the function-limit analogue of the Cauchy completeness criterion for sequences, and relies on the completeness of $\mathbb{R}$ in exactly the same way.

Remark (Dependencies).

- Limit of a Function

Algebra of Limits

Theorem

Theorem 5.1.17 (Sum Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f+g)(x) = L + M.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x \in A\ 0 < |x-c| < \delta_f \Rightarrow |f(x) - L| < \varepsilon\Big]$$
$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x \in A\ 0 < |x-c| < \delta_g \Rightarrow |g(x) - M| < \varepsilon\Big]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f+g)(x) - (L+M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f+g, c, L+M).$$

Remark (Interpretation). Limits respect addition: sufficiently close values of f and g add to values of $f + g$ close to the sum of the two limits.

Remark (Dependencies).

- Limit of a Function
- Pointwise Sum of Functions

Theorem

Theorem 5.1.18 (Difference Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(f-g)(x) = L - M.$$

Go to proof.

Remark (Standard quantified statement).

$$\Big[\lim_{x\to c} f(x) = L\Big] \wedge \Big[\lim_{x\to c} g(x) = M\Big]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x-c| < \delta \Rightarrow |(f-g)(x) - (L-M)| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(f-g, c, L-M).$$

Remark (Interpretation). The difference rule is the sum rule applied to $f + (-g)$.

Remark (Dependencies).

- Limit of a Function
- Pointwise Difference of Functions

Theorem

Theorem 5.1.19 (Scalar Multiple Rule for Function Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then*

$$\lim_{x\to c}(\alpha f)(x) = \alpha L.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |\alpha f(x) - \alpha L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \implies \text{FunctionLimit}(\alpha f, c, \alpha L).$$

Remark (Interpretation). Multiplying a function by a fixed scalar multiplies its limiting value by that same scalar.

Remark (Dependencies).

- Limit of a Function
- Pointwise Scalar Multiple of a Function

Theorem

Theorem 5.1.20 (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then*

$$\lim_{x\to c}(fg)(x) = LM.$$

Go to proof.

Remark (Standard quantified statement).

$$\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right]$$
$$\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x)g(x) - LM| < \varepsilon.$$

Remark (Definition predicate reading).

$$\text{FunctionLimit}(f, c, L) \wedge \text{FunctionLimit}(g, c, M) \implies \text{FunctionLimit}(fg, c, LM).$$

Remark (Interpretation). The product rule combines closeness of f to L with local boundedness of g near c.

Remark (Dependencies).

- Limit of a Function
- Pointwise Product of Functions

Theorem

Theorem 5.1.21 (Quotient Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and*

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \frac{L}{M}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\left[\lim_{x\to c} f(x) = L\right] \wedge \left[\lim_{x\to c} g(x) = M\right] \wedge M \neq 0 \\ &\quad \Rightarrow \exists \delta_0 > 0\ \forall x \in A\ 0 < |x - c| < \delta_0 \Rightarrow g(x) \neq 0, \\ &\quad \text{and} \quad \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow \left|\frac{f(x)}{g(x)} - \frac{L}{M}\right| < \varepsilon. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \wedge \mathrm{FunctionLimit}(g, c, M) \wedge M \neq 0 \implies \mathrm{FunctionLimit}(f/g, c, L/M).$$ ■

Remark (Interpretation). A nonzero limiting denominator stays away from zero near the limit point, so division becomes locally legitimate and the quotient limit follows.

Remark (Dependencies).

- Limit of a Function
- Pointwise Quotient of Functions

One-Sided Limits

Definition (Right-Hand Limit)

Definition 5.1.22 (Right-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. A real number L is the **right-hand limit** of f at c if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ c < x < c + \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \; 0 < x - c < \delta \; \wedge \; |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{RightLimit}(f, c, L).$$

Remark (Failure modes). A proposed right-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the right.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \; 0 < x - c < \delta \; \wedge \; |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the right of c are tested. The value of f at c, if it is defined, is irrelevant to the right-hand limit.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Definition (Left-Hand Limit)

Definition 5.1.23 (Left-Hand Limit). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. A real number L is the **left-hand limit** of f at c if

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; c - \delta < x < c \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L) \; :\equiv \; \forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

Remark (Negated quantified statement).

$$\exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \; 0 < c - x < \delta \; \wedge \; |f(x) - L| \geq \varepsilon_0.$$

Remark (Negation predicate reading).

$$\neg \, \mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes). A proposed left-hand limit fails when some positive output tolerance is violated arbitrarily close to c from the left.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0 \, \forall \delta > 0 \, \exists x \in A \; 0 < c - x < \delta \; \wedge \; |f(x) - L| \geq \varepsilon_0.$$

Remark (Interpretation). Only inputs to the left of c are tested. This is the mirror image of the right-hand limit definition.

Remark (Dependencies).

- Limit of a Function
- Cluster Point

Theorem

Theorem 5.1.24 (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < x - c < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \iff \Big[\forall (x_n) \subseteq A \cap (c, \infty) \; x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{RightLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (c, \infty) \; x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Right-hand limits can be tested by sequences approaching c from above. Every such sequence must force the corresponding output sequence toward L.

Remark (Dependencies).

- Right-Hand Limit
- Sequential Definition of Limit

Corollary

Corollary 5.1.25 (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x \to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall \varepsilon > 0 \, \exists \delta > 0 \, \forall x \in A \; 0 < c - x < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \iff \Big[\forall (x_n) \subseteq A \cap (-\infty, c) \; x_n \to c \Rightarrow f(x_n) \to L\Big].$$

Remark (Definition predicate reading).

$$\mathrm{LeftLimit}(f, c, L) \iff \forall (x_n) \subseteq A \cap (-\infty, c) \; x_n \to c \Rightarrow f(x_n) \to L.$$

Remark (Interpretation). Left-hand limits can be tested by sequences approaching c from below. The criterion is the left-sided analogue of the right-hand sequential criterion.

Remark (Dependencies).

- Left-Hand Limit
- Sequential Definition of Limit

Theorem

Theorem 5.1.26 (Two-Sided Limit via One-Sided Limits). *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x\to c} f(x) = L$ if and only if both $\lim_{x\to c^+} f(x) = L$ and $\lim_{x\to c^-} f(x) = L$.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &\Big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ 0 < |x - c| < \delta \Rightarrow |f(x) - L| < \varepsilon\Big] \\ &\quad \iff \Big[\forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in A\ 0 < x - c < \delta_+ \Rightarrow |f(x) - L| < \varepsilon\Big] \\ &\quad\quad \wedge \Big[\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in A\ 0 < c - x < \delta_- \Rightarrow |f(x) - L| < \varepsilon\Big]. \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{FunctionLimit}(f, c, L) \iff \mathrm{RightLimit}(f, c, L) \wedge \mathrm{LeftLimit}(f, c, L).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < |x - c| < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0 \\ &\quad \iff \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < x - c < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big] \\ &\quad\quad \vee \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ 0 < c - x < \delta\ \wedge\ |f(x) - L| \geq \varepsilon_0\Big]. \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

Remark (Failure modes). A two-sided limit fails precisely when the right-hand limit fails, the left-hand limit fails, or the two one-sided behaviours do not meet at the same value.

Remark (Failure mode decomposition).

$$\neg\, \mathrm{FunctionLimit}(f, c, L) \iff \neg\, \mathrm{RightLimit}(f, c, L) \vee \neg\, \mathrm{LeftLimit}(f, c, L).$$

Remark (Interpretation). A two-sided limit is exactly the agreement of the two one-sided tests. Both sides must approach the same value.

Remark (Dependencies).

- Limit of a Function
- Right-Hand Limit
- Left-Hand Limit

5.2 Algebra of Functions

Toolkit — Algebra of Functions

- All results are set-theoretic: no limits, no topology.
- $\text{Injective}(f) \wedge \text{Injective}(g) \Rightarrow \text{Injective}(g \circ f)$.
- $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$.
- Preimage: $f^{-1}(A \cup B) = f^{-1}(A) \cup f^{-1}(B)$, $f^{-1}(A \cap B) = f^{-1}(A) \cap f^{-1}(B)$ — exact.
- Image: $f(A \cup B) = f(A) \cup f(B)$ but $f(A \cap B) \subseteq f(A) \cap f(B)$ — only $\subseteq$ in general.

Algebra of Functions

Remark (Intervals on $\mathbb{R}$). Every open interval (a, b) consists entirely of cluster points; $[a, b]$ is closed and bounded. Natural domains for functions are typically intervals or unions of intervals.

Proposition

Proposition 5.2.1 (Composition of Injections Is Injective). *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Go to proof.

Remark (Standard quantified statement).

$$\Big(\forall a_1, a_2 \in A \; (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\Big) \wedge \Big(\forall b_1, b_2 \in B \; (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\Big)$$
$$\Rightarrow \forall a_1, a_2 \in A \; (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2).$$

Remark (Definition predicate reading).

$$\text{Injective}(f) \wedge \text{Injective}(g) \implies \text{Injective}(g \circ f).$$

Remark (Negated quantified statement).

$$\exists a_1, a_2 \in A \; \big(a_1 \neq a_2 \;\wedge\; g(f(a_1)) = g(f(a_2))\big).$$

Remark (Negation predicate reading).

$$\neg \, \text{Injective}(g \circ f).$$

Remark (Failure modes). The composite fails to be injective exactly when two distinct inputs of A are sent to the same output in C after applying both functions.

Remark (Failure mode decomposition).

$$\exists a_1, a_2 \in A \; \big(a_1 \neq a_2 \;\wedge\; g(f(a_1)) = g(f(a_2))\big).$$

Remark (Interpretation). Injectivity survives composition: equality after $g \circ f$ can be pulled back first through g and then through f.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 5.2.2 (Composition of Surjections Is Surjective). $\mathrm{Surjective}(f) \wedge \mathrm{Surjective}(g) \Rightarrow \mathrm{Surjective}(g \circ f)$.

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{Surjective}(f) \wedge \mathrm{Surjective}(g) \;\Rightarrow\; \forall c \in C,\ \exists a \in A : g(f(a)) = c.$$

Remark (Definition predicate reading).

$$\mathrm{Surjective}(g \circ f, C) \;:\equiv\; \forall c \in C,\ \exists a \in A : (g \circ f)(a) = c.$$

Remark (Dependencies).

- Composition
- Function

Corollary

Corollary 5.2.3 (Composition of Bijections Is Bijective). *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\Big[\big(\forall a_1, a_2 \in A\ (f(a_1) = f(a_2) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall b \in B\ \exists a \in A\ (f(a) = b)\big)\Big]\\ &\quad\wedge \Big[\big(\forall b_1, b_2 \in B\ (g(b_1) = g(b_2) \Rightarrow b_1 = b_2)\big) \wedge \big(\forall c \in C\ \exists b \in B\ (g(b) = c)\big)\Big]\\ &\quad\Rightarrow \Big[\big(\forall a_1, a_2 \in A\ (g(f(a_1)) = g(f(a_2)) \Rightarrow a_1 = a_2)\big) \wedge \big(\forall c \in C\ \exists a \in A\ (g(f(a)) = c)\big)\Big].\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{Bijective}(f) \wedge \mathrm{Bijective}(g) \implies \mathrm{Bijective}(g \circ f).$$

Remark (Negated quantified statement).

$$\begin{gathered}\exists a_1, a_2 \in A\ \big(a_1 \neq a_2\ \wedge\ g(f(a_1)) = g(f(a_2))\big)\\ \vee\ \exists c \in C\ \forall a \in A\ \big(g(f(a)) \neq c\big).\end{gathered}$$

Remark (Negation predicate reading).

$$\neg\ \text{Bijective}(g \circ f).$$

Remark (Failure modes). A composite bijection can fail only by losing injectivity or by missing some target value in C.

Remark (Failure mode decomposition).

$$\neg\ \text{Bijective}(g \circ f) \iff \neg\ \text{Injective}(g \circ f)\ \vee\ \neg\ \text{Surjective}(g \circ f).$$

Remark (Interpretation). A bijection is an injective and surjective function. Since both of those properties are preserved by composition, bijectivity is preserved too.

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 5.2.4 (Inverse of a Bijection Is a Bijection). $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$.

Go to proof.

Remark (Standard quantified statement).

$$\text{Bijective}(f)\ \Rightarrow\ \text{InverseFunction}(f, f^{-1}),\ f^{-1} \circ f = \text{id}_A,\ f \circ f^{-1} = \text{id}_B.$$

Remark (Definition predicate reading).

$$\text{InverseFunction}(f, f^{-1})\ :\equiv\ f^{-1} \circ f = \text{id}_A\ \wedge\ f \circ f^{-1} = \text{id}_B.$$

Remark (Dependencies).

- Composition
- Function

Proposition

Proposition 5.2.5 (Preimage Distributes over Union and Intersection). *Let* $f : A \to B$, $S, T \subseteq B$. *Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}(i) \quad & x \in f^{-1}(S \cup T) \iff f(x) \in S \vee f(x) \in T \iff x \in f^{-1}(S) \cup f^{-1}(T). \\ (ii) \quad & x \in f^{-1}(S \cap T) \iff f(x) \in S \wedge f(x) \in T \iff x \in f^{-1}(S) \cap f^{-1}(T).\end{aligned}$$

Remark (Negated quantified statement).

$$x \notin f^{-1}(S) \iff f(x) \notin S \iff x \in f^{-1}(S^c).$$

Remark (Interpretation). Preimage is a complete lattice homomorphism: it preserves all set operations including complement. Contrast with image: $f(A \cap B) \subseteq f(A) \cap f(B)$ with strict containment possible when f is not injective. Preimage identities are load-bearing in topology (preimages of open sets are open) and measure theory (preimages of measurable sets are measurable).

Remark (Dependencies).

- Function

5.3 Real-Valued Functions

Toolkit — Real-Valued Functions

- Pointwise operations: sum, difference, product, quotient, scalar multiple, absolute value, max, min. All purely algebraic — no limits involved. fg is pointwise product, not composition.
- BoundedOn(f, A): $\exists B > 0,\ \forall x \in A : |f(x)| \leq B$.
- MonotoneFunction(f, A): increasing or decreasing.
- Maximum attained at x^*: $f(x^*) = \sup f(A)$, $x^* \in A$. Supremum need not be attained.

Real-Valued Functions

Definition (Pointwise Sum of Functions)

Definition 5.3.1 (Pointwise Sum of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise sum $f + g : X \to \mathbb{R}$ is the function defined by

$$(f+g)(x) = f(x) + g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f+g)(x) = f(x) + g(x)\big).$$

Remark (Interpretation). The sum is computed output by output; at each input, add the two function values at that same input.

Remark (Dependencies).

- Function

- Subset
- Addition on $\mathbb{R}$

Definition (Pointwise Difference of Functions)

Definition 5.3.2 (Pointwise Difference of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise difference $f - g : X \to \mathbb{R}$ is the function defined by

$$(f-g)(x) = f(x) - g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((f-g)(x) = f(x) - g(x)\big).$$

Remark (Interpretation). The difference subtracts the output of g from the output of f at each input separately.

Remark (Dependencies).

- Function
- Subset
- Subtraction on $\mathbb{R}$

Definition (Pointwise Product of Functions)

Definition 5.3.3 (Pointwise Product of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise product $fg : X \to \mathbb{R}$ is the function defined by

$$(fg)(x) = f(x)g(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((fg)(x) = f(x)g(x)\big).$$

Remark (Interpretation). The product is not composition: it multiplies the two real outputs at the same input.

Remark (Dependencies).

- Function
- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Scalar Multiple of a Function)

Definition 5.3.4 (Pointwise Scalar Multiple of a Function). Let $X \subseteq \mathbb{R}$, let $f : X \to \mathbb{R}$, and let $c \in \mathbb{R}$. The pointwise scalar multiple $cf : X \to \mathbb{R}$ is the function defined by

$$(cf)(x) = cf(x) \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big((cf)(x) = cf(x)\big).$$

Remark (Interpretation). Scalar multiplication stretches every output of the function by the same real scalar.

Remark (Dependencies).

- Function
- Subset
- Multiplication on $\mathbb{R}$

Definition (Pointwise Absolute Value of a Function)

Definition 5.3.5 (Pointwise Absolute Value of a Function). Let $X \subseteq \mathbb{R}$ and let $f : X \to \mathbb{R}$. The pointwise absolute value $|f| : X \to \mathbb{R}$ is the function defined by

$$|f|(x) = |f(x)| \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big(|f|(x) = |f(x)|\big).$$

Remark (Interpretation). The absolute value operation applies the ordinary real absolute value to each output.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$

Definition (Pointwise Maximum of Two Functions)

Definition 5.3.6 (Pointwise Maximum of Two Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. The pointwise maximum $\max(f, g) : X \to \mathbb{R}$ is the function defined by

$$\max(f, g)(x) = \max\{f(x), g(x)\} \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\forall x \in X \ \big(\max(f, g)(x) = \max\{f(x), g(x)\}\big).$$

Remark (Interpretation). At each input, the new function keeps the larger of the two output values.

Remark (Dependencies).

- Function

- Subset
- Maximum

Definition (Pointwise Quotient of Functions)

Definition 5.3.7 (Pointwise Quotient of Functions). Let $X \subseteq \mathbb{R}$ and let $f, g : X \to \mathbb{R}$. Suppose $g(x) \neq 0$ for every $x \in X$. The pointwise quotient $f/g : X \to \mathbb{R}$ is the function defined by

$$(f/g)(x) = \frac{f(x)}{g(x)} \qquad (x \in X).$$

Remark (Standard quantified statement).

$$\big(\forall x \in X\ (g(x) \neq 0)\big) \implies \forall x \in X\ \big((f/g)(x) = f(x)/g(x)\big).$$

Remark (Interpretation). The quotient is defined pointwise only when the denominator function never vanishes on the domain.

Remark (Dependencies).

- Function
- Subset
- Division on $\mathbb{R}$

Definition (Bounded Above Function)

Definition 5.3.8 (Bounded Above Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded above on A if there exists $M \in \mathbb{R}$ such that

$$f(x) \leq M \qquad (x \in A).$$

Remark (Standard quantified statement).

$$\exists M \in \mathbb{R}\ \forall x \in A\ \big(f(x) \leq M\big).$$

Remark (Negated quantified statement).

$$\forall M \in \mathbb{R}\ \exists x \in A\ \big(f(x) > M\big).$$

Remark (Failure modes). No proposed real ceiling works: whenever a candidate upper bound is chosen, some input has an output larger than that candidate.

Remark (Failure mode decomposition).

$$\forall M \in \mathbb{R}\ \exists x \in A\ \big(f(x) > M\big).$$

Remark (Interpretation). Bounded above means all output values lie below one common real ceiling.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Bounded Below Function)

Definition 5.3.9 (Bounded Below Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded below on A if there exists $m \in \mathbb{R}$ such that

$$m \leq f(x) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$\exists m \in \mathbb{R}\, \forall x \in A\, \big(m \leq f(x)\big).$$

Remark (Negated quantified statement).

$$\forall m \in \mathbb{R}\, \exists x \in A\, \big(f(x) < m\big).$$

Remark (Failure modes). No proposed real floor works: whenever a candidate lower bound is chosen, some input has an output smaller than that candidate.

Remark (Failure mode decomposition).

$$\forall m \in \mathbb{R}\, \exists x \in A\, \big(f(x) < m\big).$$

Remark (Interpretation). Bounded below means all output values lie above one common real floor.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Bounded Function)

Definition 5.3.10 (Bounded Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is bounded on A if there exists $B > 0$ such that

$$|f(x)| \leq B \qquad (x \in A).$$

Remark (Standard quantified statement).

$$\exists B > 0\, \forall x \in A\, \big(|f(x)| \leq B\big).$$

Remark (Definition predicate reading).

$$\mathrm{BoundedOn}(f, A) \;:=\; \exists B > 0\, \forall x \in A\, \big(|f(x)| \leq B\big).$$

Remark (Negated quantified statement).

$$\forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f, A) \iff \forall B > 0\ \exists x \in A\ \big(|f(x)| > B\big).$$

Remark (Failure modes). Every symmetric interval around 0 fails to contain all output values of f on A.

Remark (Failure mode decomposition).

$$\forall B > 0\ \exists x \in A\ \big(f(x) > B\ \lor\ f(x) < -B\big).$$

Remark (Interpretation). A bounded function has all of its values trapped inside some symmetric interval $[-B, B]$. Being bounded above alone is not enough; the function $f(x) = -x$ on $\mathbb{R}$ is bounded above by 0 but is not bounded.

Remark (Dependencies).

- Bounded Above Function
- Bounded Below Function

Definition (Supremum of a Function on a Set)

Definition 5.3.11 (Supremum of a Function on a Set). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The supremum of f on A is

$$\sup_{x\in A} f(x) = \sup f(A),$$

provided the supremum of $f(A)$ exists.

Remark (Standard quantified statement).

$$s = \sup_{x\in A} f(x) \iff \big(\forall x \in A\ (f(x) \le s)\big)\ \land\ \big(\forall u \in \mathbb{R}\ ((\forall x \in A\ (f(x) \le u)) \Rightarrow s \le u)\big).$$

Remark (Interpretation). The supremum is the least real ceiling for the output set $f(A)$; it need not be an output value of f.

Definition (Infimum of a Function on a Set)

Definition 5.3.12 (Infimum of a Function on a Set). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The infimum of f on A is

$$\inf_{x\in A} f(x) = \inf f(A),$$

provided the infimum of $f(A)$ exists.

Remark (Standard quantified statement).

$$t = \inf_{x\in A} f(x) \iff \big(\forall x \in A\ (t \le f(x))\big)\ \land\ \big(\forall \ell \in \mathbb{R}\ ((\forall x \in A\ (\ell \le f(x))) \Rightarrow \ell \le t)\big).$$

Remark (Interpretation). The infimum is the greatest real floor for the output set $f(A)$; it need not be an output value of f.

Definition (Maximum Point of a Function)

Definition 5.3.13 (Maximum Point of a Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x^* \in A$ is a maximum point of f on A if

$$f(x) \leq f(x^*) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$x^* \in A \ \wedge \ \forall x \in A \ \big(f(x) \leq f(x^*)\big).$$

Remark (Negated quantified statement).

$$x^* \notin A \ \vee \ \exists x \in A \ \big(f(x) > f(x^*)\big).$$

Remark (Failure modes). A proposed maximum point fails either by not belonging to the domain or by being beaten by another output value.

Remark (Failure mode decomposition).

$$x^* \notin A \ \vee \ \exists x \in A \ \big(f(x) > f(x^*)\big).$$

Remark (Interpretation). A maximum point is an input where the largest output value is attained.

Definition (Minimum Point of a Function)

Definition 5.3.14 (Minimum Point of a Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. A point $x_* \in A$ is a minimum point of f on A if

$$f(x_*) \leq f(x) \qquad (x \in A).$$

Remark (Standard quantified statement).

$$x_* \in A \ \wedge \ \forall x \in A \ \big(f(x_*) \leq f(x)\big).$$

Remark (Negated quantified statement).

$$x_* \notin A \ \vee \ \exists x \in A \ \big(f(x) < f(x_*)\big).$$

Remark (Failure modes). A proposed minimum point fails either by not belonging to the domain or by being undercut by another output value.

Remark (Failure mode decomposition).

$$x_* \notin A \ \vee \ \exists x \in A \ \big(f(x) < f(x_*)\big).$$

Remark (Interpretation). A minimum point is an input where the smallest output value is attained.

Definition (Increasing Function)

Definition 5.3.15 (Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is increasing on A if

$$x_1 < x_2 \implies f(x_1) \le f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \,\big(x_1 < x_2 \Rightarrow f(x_1) \le f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \,\big(x_1 < x_2 \,\wedge\, f(x_1) > f(x_2)\big).$$

Remark (Failure modes). The function fails to be increasing if some later input has a smaller output than an earlier input.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \,\big(x_1 < x_2 \,\wedge\, f(x_1) > f(x_2)\big).$$

Remark (Interpretation). As the input moves to the right in A, the output is not allowed to go down.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Increasing Function)

Definition 5.3.16 (Strictly Increasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly increasing on A if

$$x_1 < x_2 \implies f(x_1) < f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \,\big(x_1 < x_2 \Rightarrow f(x_1) < f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \,\big(x_1 < x_2 \,\wedge\, f(x_1) \ge f(x_2)\big).$$

Remark (Failure modes). Strict increase fails if some later input has an output that is no larger than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \,\big(x_1 < x_2 \,\wedge\, f(x_1) \ge f(x_2)\big).$$

Remark (Interpretation). Strict increase requires every genuine move to the right in the domain to produce a genuine increase in output.

Remark (Dependencies).

- Increasing Function

Definition (Decreasing Function)

Definition 5.3.17 (Decreasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is decreasing on A if

$$x_1 < x_2 \implies f(x_1) \geq f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \left(x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2)\right).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\right).$$

Remark (Failure modes). The function fails to be decreasing if some later input has a larger output than an earlier input.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) < f(x_2)\right).$$

Remark (Interpretation). As the input moves to the right in A, the output is not allowed to go up.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$

Definition (Strictly Decreasing Function)

Definition 5.3.18 (Strictly Decreasing Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is strictly decreasing on A if

$$x_1 < x_2 \implies f(x_1) > f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \left(x_1 < x_2 \Rightarrow f(x_1) > f(x_2)\right).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\right).$$

Remark (Failure modes). Strict decrease fails if some later input has an output that is no smaller than the earlier output.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \left(x_1 < x_2 \ \wedge \ f(x_1) \leq f(x_2)\right).$$

Remark (Interpretation). Strict decrease requires every genuine move to the right in the domain to produce a genuine decrease in output.

Remark (Dependencies).

- Decreasing Function

Definition (Monotone Function)

Definition 5.3.19 (Monotone Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is monotone on A if f is increasing on A or decreasing on A.

Remark (Standard quantified statement).

$$\big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \ \vee \ \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \geq f(x_2))\big).$$

Remark (Definition predicate reading).

$$\text{MonotoneFunction}(f, A) \ :\equiv \ \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x_1) \leq f(x_2))\big) \vee \big(\forall x_1, x_2 \in A \ (x_1 < x_2 \Rightarrow f(x$$

Remark (Negated quantified statement).

$$\big(\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2))\big) \ \wedge \ \big(\exists y_1, y_2 \in A \ (y_1 < y_2 \wedge f(y_1) < f(y_2))\big).$$

Remark (Negation predicate reading).

$$\neg \, \text{MonotoneFunction}(f, A)$$

Remark (Failure modes). A nonmonotone function has evidence of both kinds of reversal: one pair where the outputs go down as inputs increase, and another pair where the outputs go up as inputs increase.

Remark (Failure mode decomposition).

$$\begin{aligned} &\exists x_1, x_2 \in A \ (x_1 < x_2 \wedge f(x_1) > f(x_2)) \\ &\quad \wedge \ \exists y_1, y_2 \in A \ (y_1 < y_2 \wedge f(y_1) < f(y_2)). \end{aligned}$$

Remark (Interpretation). A monotone function moves consistently in one direction: nondecreasing or nonincreasing.

Remark (Dependencies).

- Increasing Function
- Decreasing Function

Definition (Constant Function)

Definition 5.3.20 (Constant Function). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is constant on A if

$$f(x_1) = f(x_2) \qquad (x_1, x_2 \in A).$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ \big(f(x_1) = f(x_2)\big).$$

Remark (Negated quantified statement).

$$\exists x_1, x_2 \in A \ \big(f(x_1) \neq f(x_2)\big).$$

Remark (Failure modes). Constancy fails as soon as two inputs in the domain have different outputs.

Remark (Failure mode decomposition).

$$\exists x_1, x_2 \in A \ \big(f(x_1) \neq f(x_2)\big).$$

Remark (Interpretation). A constant function assigns the same real value to every input in its domain.

5.4 Subsets of $\mathbb{R}$

Toolkit — Subsets of $\mathbb{R}$

- Cluster point (= limit point = accumulation point): every ε-neighbourhood contains a *distinct* point of X. Adherent point: every neighbourhood contains *some* point of X.
- Within(x, c, r): the unpunctured ball condition $|x - c| < r$. Used for adherence and continuity.
- InputTolerance(x, c, δ): the punctured ball $0 < |x - c| < \delta$. Used for cluster points and function limits.
- Closed $\iff \overline{X} = X \iff$ sequentially closed.
- Heine–Borel: X is closed and X is bounded $\iff$ every sequence in X has a subsequence converging to a limit in X.

Subsets of $\mathbb{R}$

Remark (Terminological note). Cluster point = limit point = accumulation point. Bartle and Lebl use *cluster point*; Rudin uses *limit point*; Tao uses both. Adherent point (Tao) is the weaker notion that includes isolated points.

Definition (ε-Neighbourhood)

Definition 5.4.1 (ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **ε-neighbourhood** of x is

$$V_\varepsilon(x) := (x - \varepsilon, x + \varepsilon) = \{y \in \mathbb{R} : |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon(x) \iff |y - x| < \varepsilon.$$

Remark (Interpretation). The ε-neighbourhood is the open interval of radius ε centered at x.

Remark (Dependencies).

- Open Ball on the Real Line

Definition (Deleted ε-Neighbourhood)

Definition 5.4.2 (Deleted ε-Neighbourhood)**.** Let $x \in \mathbb{R}$ and let $\varepsilon > 0$. The **deleted ε-neighbourhood** of x is

$$V_\varepsilon^*(x) := V_\varepsilon(x) \setminus \{x\} = \{y \in \mathbb{R} : 0 < |y - x| < \varepsilon\}.$$

Remark (Standard quantified statement).

$$y \in V_\varepsilon^*(x) \iff 0 < |y - x| < \varepsilon.$$

Remark (Interpretation). The deleted neighbourhood removes the base point. It is the punctured input region used in limit definitions.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Cluster Point)

Definition 5.4.3 (Cluster Point)**.** Let $X \subseteq \mathbb{R}$ and $x \in \mathbb{R}$. The point x is a **cluster point** of X if

$$\forall \varepsilon > 0, \ \exists y \in X \setminus \{x\} : \ |y - x| < \varepsilon.$$

Remark (Standard quantified statement).

$$x \text{ is a cluster point of } X \iff \forall \varepsilon > 0, \ \exists y \in X \setminus \{x\} : \ |y - x| < \varepsilon.$$

Remark (Definition predicate reading).

$$x \text{ is a cluster point of } X \ :\equiv \ \forall \varepsilon > 0, \ V_\varepsilon^*(x) \cap X \neq \emptyset.$$

Remark (Negated quantified statement).

$$\neg\, x \text{ is a cluster point of } X \iff \exists \varepsilon > 0, \ \forall\, y \in X \setminus \{x\} : \ \neg|y - x| < \varepsilon.$$

Remark (Negation predicate reading).

$$\neg x \text{ is a cluster point of } X :\equiv \exists \varepsilon > 0 : V_\varepsilon^*(x) \cap X = \emptyset.$$

Remark (Interpretation). A cluster point is approached by X via distinct witnesses arbitrarily close. A cluster point need not belong to X: 0 is a cluster point of $(0, 1)$ but $0 \notin (0, 1)$. The condition $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; without it the punctured ball condition is vacuous.

Remark (Dependencies).

- Deleted ε-Neighbourhood

Theorem

Theorem 5.4.4 (Sequential Characterization of Cluster Points). *c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.*

Go to proof.

Remark (Standard quantified statement).

$$c \text{ is a cluster point of } A \iff \exists (a_n) \subseteq A \setminus \{c\} : a_n \to c.$$

Remark (Interpretation). The forward direction: use $\varepsilon = 1/n$ to build the witness sequence. The reverse direction is immediate since the sequence witnesses the ball condition for every ε. This bridges the ε-δ definition with the sequential language.

Remark (Dependencies).

- Cluster Point

Definition (Adherent Point)

Definition 5.4.5 (Adherent Point). Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **adherent point** of X if

$$\forall \varepsilon > 0 \, \exists y \in X \, (|y - x| < \varepsilon).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \big(V_\varepsilon(x) \cap X \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\mathrm{AdherentPoint}(x, X) := \forall \varepsilon > 0 \, \exists y \in X \, (|y - x| < \varepsilon).$$

Remark (Negated quantified statement).

$$\exists \varepsilon > 0 \, \forall y \in X \, (|y - x| \geq \varepsilon).$$

Remark (Interpretation). An adherent point can be witnessed by points of X arbitrarily close to x, including possibly x itself.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Isolated Point)

Definition 5.4.6 (Isolated Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in X$. The point x is an **isolated point** of X if

$$\exists \varepsilon > 0 \left(V_\varepsilon(x) \cap X = \{x\}\right).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0\, \forall y \in X \left(|y - x| < \varepsilon \Rightarrow y = x\right).$$

Remark (Definition predicate reading).

$$\text{IsolatedPoint}(x, X) \;:=\; x \in X \wedge \exists \varepsilon > 0 \left(V_\varepsilon(x) \cap X = \{x\}\right).$$

Remark (Negated quantified statement).

$$\forall \varepsilon > 0\, \exists y \in X \setminus \{x\}\, (|y - x| < \varepsilon).$$

Remark (Interpretation). An isolated point belongs to the set but has a neighbourhood containing no other points of the set.

Remark (Dependencies).

- Adherent Point
- Cluster Point

Definition (Interior Point)

Definition 5.4.7 (Interior Point)**.** Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is an **interior point** of X if

$$\exists \varepsilon > 0 \left(V_\varepsilon(x) \subseteq X\right).$$

Remark (Standard quantified statement).

$$\exists \varepsilon > 0\, \forall y \in \mathbb{R} \left(|y - x| < \varepsilon \Rightarrow y \in X\right).$$

Remark (Definition predicate reading).

$$\text{InteriorPoint}(x, X) \;:=\; \exists \varepsilon > 0 \left(V_\varepsilon(x) \subseteq X\right).$$

Remark (Interpretation). An interior point has a full open neighbourhood contained in the set.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Boundary Point)

Definition 5.4.8 (Boundary Point). Let $X \subseteq \mathbb{R}$ and let $x \in \mathbb{R}$. The point x is a **boundary point** of X if

$$\forall \varepsilon > 0 \, \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Standard quantified statement).

$$\forall \varepsilon > 0 \, \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Definition predicate reading).

$$\mathrm{BoundaryPoint}(x, X) \;:=\; \forall \varepsilon > 0 \, \big(V_\varepsilon(x) \cap X \neq \emptyset\big) \wedge \big(V_\varepsilon(x) \cap X^c \neq \emptyset\big).$$

Remark (Interpretation). A boundary point sees both the set and its complement in every neighbourhood.

Remark (Dependencies).

- ε-Neighbourhood

Definition (Interior)

Definition 5.4.9 (Interior). Let $X \subseteq \mathbb{R}$. The **interior** of X is

$$X^\circ := \{x \in \mathbb{R} : \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X\}.$$

Remark (Standard quantified statement).

$$x \in X^\circ \iff \exists \varepsilon > 0 \; V_\varepsilon(x) \subseteq X.$$

Remark (Interpretation). The interior collects exactly the points around which the set contains a whole neighbourhood.

Remark (Dependencies).

- Interior Point

Definition (Boundary)

Definition 5.4.10 (Boundary). Let $X \subseteq \mathbb{R}$. The **boundary** of X is

$$\partial X := \{x \in \mathbb{R} : \forall \varepsilon > 0 \; (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset)\}.$$

Remark (Standard quantified statement).

$$x \in \partial X \iff \forall \varepsilon > 0 \; (V_\varepsilon(x) \cap X \neq \emptyset \wedge V_\varepsilon(x) \cap X^c \neq \emptyset).$$

Remark (Interpretation). The boundary records the points where every neighbourhood touches both inside and outside of the set.

Remark (Dependencies).

- Boundary Point

Definition (Closure)

Definition 5.4.11 (Closure).

$$\overline{X} := \{x \in \mathbb{R} : \forall \varepsilon > 0\ V_\varepsilon(x) \cap X \neq \emptyset\}.$$

Remark (Standard quantified statement).

$$x \in \overline{X} \iff x \text{ is an adherent point of } X \iff \forall \varepsilon > 0 : V_\varepsilon(x) \cap X \neq \emptyset.$$

Remark (Interpretation). The closure adds all adherent points of X, so it fills in the limiting points that can be reached by points of X.

Remark (Canonical examples). $\overline{(a,b)} = [a,b]$; $\overline{\mathbb{Q}} = \mathbb{R}$; $\overline{\mathbb{Z}} = \mathbb{Z}$; $\overline{\{1/n\}} = \{1/n\} \cup \{0\}$.

Remark (Dependencies).

- Adherent Point

Lemma

Lemma 5.4.12 (Elementary Properties of Closures). *Let $X, Y \subseteq \mathbb{R}$. Then: (i) $X \subseteq \overline{X}$; (ii) $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; (iii) $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; (iv) $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.*

Go to proof.

Remark (Standard quantified statement).

$$\overline{X \cup Y} = \overline{X} \cup \overline{Y}; \quad \overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}.$$

Remark (Failure of equality in (iii)). Take $X = (0,1)$, $Y = (1,2)$: $\overline{X \cap Y} = \emptyset$ but $\overline{X} \cap \overline{Y} = \{1\}$.

Remark (Interpretation). Closure preserves inclusions and finite unions, but intersection can lose limit points that are approached separately by the two sets.

Remark (Dependencies).

- Closure

Definition (Closed Subset of $\mathbb{R}$)

Definition 5.4.13 (Closed Subset of $\mathbb{R}$). E is closed $\iff \overline{E} = E$.

Remark (Standard quantified statement).

$$E \text{ is closed} \iff \forall x \in \mathbb{R} : x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Definition predicate reading).

$$E \text{ is closed} :\equiv \forall x : x \text{ is an adherent point of } E \Rightarrow x \in E.$$

Remark (Negated quantified statement).

$$\neg E \text{ is closed} \iff \exists x \notin E : x \text{ is an adherent point of } E.$$

Remark (Negation predicate reading).

$$\neg E \text{ is closed} :\equiv \exists x \notin E : \forall \varepsilon > 0 : V_\varepsilon(x) \cap E \neq \emptyset.$$

Remark (Interpretation). E is closed iff it contains its own boundary. Examples: $[a, b]$ closed; (a, b) not closed; $\mathbb{R}$ and $\emptyset$ both closed; $\mathbb{Q}$ not closed.

Remark (Dependencies).

- Closure

Corollary

Corollary 5.4.14 (Closed iff Sequentially Closed). *X is closed $\iff$ $\forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X$.*

Go to proof.

Remark (Standard quantified statement).

$$X \text{ is closed} \iff \forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X.$$

Remark (Interpretation). A closed set cannot be escaped by convergent sequences. Every adherent point has a witness sequence; closedness forces the limit into the set.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$
- Sequential Characterization of Cluster Points

Corollary

Corollary 5.4.15 (Every Point of an Interval Is a Cluster Point). *If I is an interval, then every $x \in I$ is a cluster point of I.*

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \forall \varepsilon > 0\ \exists y \in I \setminus \{x\}\ (|y - x| < \varepsilon).$$

Remark (Interpretation). This justifies intervals as natural domains for function limits. $\lim_{x \to c} f(x) = L$ requires $\mathrm{ClusterPoint}(c, \mathrm{dom}(f))$; on any interval this is free.

Remark (Dependencies).

- Cluster Point

Definition (Bounded Subset of $\mathbb{R}$)

Definition 5.4.16 (Bounded Subset of $\mathbb{R}$). X is bounded $\iff$ $\exists M > 0,\ \forall x \in X : |x| \leq M$.

Remark (Standard quantified statement).

$$\exists M > 0\ \forall x \in X\ (|x| \leq M).$$

Remark (Negated quantified statement).

$$\forall M > 0\ \exists x \in X\ (|x| > M).$$

Remark (Interpretation). A bounded subset of $\mathbb{R}$ fits inside some symmetric interval $[-M, M]$.

Remark (Dependencies).

- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem

Theorem 5.4.17 (Heine-Borel Theorem for $\mathbb{R}$). *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Go to proof.

Remark (Standard quantified statement).

$$X \text{ is closed} \wedge X \text{ is bounded} \iff \forall (a_n) \subseteq X,\ \exists (a_{n_k}),\ \exists L \in X : a_{n_k} \to L.$$

Remark (Definition predicate reading).

$$\text{SequentiallyCompact}(X) :\equiv X \text{ is closed} \wedge X \text{ is bounded}.$$

Remark (Negated quantified statement).

$$\neg\big(X \text{ is closed} \wedge X \text{ is bounded}\big) \iff \exists (a_n) \subseteq X : \text{no subsequence converges to a limit in}$$

Remark (Failure modes). The compactness conclusion fails when either closedness or boundedness fails.

Remark (Failure mode decomposition). Either X is bounded fails -- a sequence escapes to $\pm\infty$ -- or X is closed fails -- a convergent subsequence exists but its limit lies outside X.

Remark (Interpretation). Closed prevents limits from escaping through boundary gaps; bounded prevents escape to infinity. Both are required. Forward: bounded gives Bolzano-Weierstrass; closed keeps the limit in X. Reverse: negate either hypothesis to construct a sequence with no convergent subsequence in X.

Remark (Dependencies).

- Closed Subset of $\mathbb{R}$

- Bounded Subset of $\mathbb{R}$

Definition (True Near a Point)

Definition 5.4.18 (True Near a Point). Property $P(f(x))$ is **true near** x_0 if $\exists \delta > 0, \forall x : 0 < |x - x_0| < \delta) \Rightarrow P(f(x))$.

Remark (Standard quantified statement).

$$P(f(x)) \text{ true near } x_0 \iff \exists \delta > 0, \forall x : 0 < |x - x_0| < \delta \Rightarrow P(f(x)).$$

Remark (Interpretation). $\lim_{x \to x_0} f(x) = L$ states: for every $\varepsilon > 0$, the property $|f(x) - L| < \varepsilon$ is true near x_0.

Remark (Dependencies).

- ε-Neighbourhood

Proofs

Remark (Return). Return to Theorem

Theorem (Sum Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$, let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then $\lim_{x\to c}(f + g)(x) = L + M$.*

Professional Standard Proof.
Let $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$. By the Sequential Criterion, $f(x_n) \to L$ and $g(x_n) \to M$. By the sum rule for sequence limits, $(f + g)(x_n) = f(x_n) + g(x_n) \to L + M$. Since (x_n) was arbitrary, the Sequential Criterion gives $\lim_{x\to c}(f + g)(x) = L + M$. □

Detailed Learning Proof.
Step 1. Set up a test sequence. Let $(x_n) \subseteq A \setminus \{c\}$ be any sequence with $x_n \to c$.
Step 2. Apply the Sequential Criterion to each component. Since $\lim_{x\to c} f(x) = L$, the Sequential Criterion gives $f(x_n) \to L$. Since $\lim_{x\to c} g(x) = M$, the Sequential Criterion gives $g(x_n) \to M$.
Step 3. Apply the sum rule for sequences. The algebra of sequence limits gives

$$(f + g)(x_n) = f(x_n) + g(x_n) \to L + M.$$

Step 4. Lift back by the Sequential Criterion. Since every $(x_n) \subseteq A\setminus\{c\}$ with $x_n \to c$ satisfies $(f + g)(x_n) \to L + M$, the Sequential Criterion yields $\lim_{x\to c}(f + g)(x) = L + M$. □

Remark (Proof structure). Bridge argument: Sequential Criterion (down) → algebra of sequence limits → Sequential Criterion (up). No direct ε-δ work.

Remark (Dependencies).

- Sequential Criterion for Function Limits
- Limit of a Function

Remark (Return). Return to Theorem

Theorem (Difference Rule for Function Limits). *Let* $f, g : A \to \mathbb{R}$, *let* c *be a cluster point of* A. *If* $\lim_{x\to c} f(x) = L$ *and* $\lim_{x\to c} g(x) = M$, *then* $\lim_{x\to c}(f - g)(x) = L - M$.

Professional Standard Proof.
Let $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$. By the Sequential Criterion, $f(x_n) \to L$ and $g(x_n) \to M$. By the difference rule for sequence limits, $(f - g)(x_n) = f(x_n) - g(x_n) \to L - M$. Since (x_n) was arbitrary, the Sequential Criterion gives $\lim_{x\to c}(f - g)(x) = L - M$. □

Detailed Learning Proof.
Step 1. Set up a test sequence. Let $(x_n) \subseteq A \setminus \{c\}$ be any sequence with $x_n \to c$.
Step 2. Apply the Sequential Criterion to each component. Since $\lim_{x\to c} f(x) = L$, the Sequential Criterion gives $f(x_n) \to L$. Since $\lim_{x\to c} g(x) = M$, the Sequential Criterion gives $g(x_n) \to M$.
Step 3. Apply the difference rule for sequences. The algebra of sequence limits gives

$$(f - g)(x_n) = f(x_n) - g(x_n) \to L - M.$$

Step 4. Lift back by the Sequential Criterion. Since every such (x_n) satisfies $(f - g)(x_n) \to L - M$, the Sequential Criterion yields $\lim_{x\to c}(f - g)(x) = L - M$. □

Remark (Proof structure). Bridge argument identical to the sum rule, with the sequence difference rule in place of the sequence sum rule. The difference rule reduces to $f + (-1)g$.

Remark (Dependencies).

- Sequential Criterion for Function Limits
- Limit of a Function

Remark (Return). Return to Theorem

Theorem (Scalar Multiple Rule for Function Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $\alpha \in \mathbb{R}$. If $\lim_{x\to c} f(x) = L$, then* $\lim_{x\to c}(\alpha f)(x) = \alpha L$.

Professional Standard Proof.
Let $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$. By the Sequential Criterion, $f(x_n) \to L$. By the scalar multiple rule for sequence limits, $(\alpha f)(x_n) = \alpha f(x_n) \to \alpha L$. Since (x_n) was arbitrary, the Sequential Criterion gives $\lim_{x\to c}(\alpha f)(x) = \alpha L$. □

Detailed Learning Proof.
Step 1. Set up a test sequence. Let $(x_n) \subseteq A \setminus \{c\}$ be any sequence with $x_n \to c$.
Step 2. Apply the Sequential Criterion. Since $\lim_{x\to c} f(x) = L$, the Sequential Criterion gives $f(x_n) \to L$.
Step 3. Apply the scalar multiple rule for sequences. The algebra of sequence limits gives

$$(\alpha f)(x_n) = \alpha f(x_n) \to \alpha L.$$

Step 4. Lift back by the Sequential Criterion. Since every such (x_n) satisfies $(\alpha f)(x_n) \to \alpha L$, the Sequential Criterion yields $\lim_{x\to c}(\alpha f)(x) = \alpha L$. □

Remark (Proof structure). Bridge argument. The case $\alpha = 0$ is trivial (constant zero function); the case $\alpha \neq 0$ follows from the scalar multiple rule for sequences.

Remark (Dependencies).

- Sequential Criterion for Function Limits
- Limit of a Function

Remark (Return). Return to Theorem

Theorem (Product Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$, let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ and $\lim_{x\to c} g(x) = M$, then $\lim_{x\to c}(fg)(x) = LM$.*

Professional Standard Proof.
Let $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$. By the Sequential Criterion, $f(x_n) \to L$ and $g(x_n) \to M$. By the product rule for sequence limits, $(fg)(x_n) = f(x_n)g(x_n) \to LM$. Since (x_n) was arbitrary, the Sequential Criterion gives $\lim_{x\to c}(fg)(x) = LM$. □

Detailed Learning Proof.
Step 1. Set up a test sequence. Let $(x_n) \subseteq A \setminus \{c\}$ be any sequence with $x_n \to c$.
Step 2. Apply the Sequential Criterion to each factor. Since $\lim_{x\to c} f(x) = L$, the Sequential Criterion gives $f(x_n) \to L$. Since $\lim_{x\to c} g(x) = M$, the Sequential Criterion gives $g(x_n) \to M$.
Step 3. Apply the product rule for sequences. The algebra of sequence limits gives

$$(fg)(x_n) = f(x_n)g(x_n) \to LM.$$

The sequence product rule uses local boundedness of $(f(x_n))$ and the add-and-subtract decomposition $f(x_n)g(x_n) - LM = f(x_n)(g(x_n) - M) + M(f(x_n) - L)$.
Step 4. Lift back by the Sequential Criterion. Since every such (x_n) satisfies $(fg)(x_n) \to LM$, the Sequential Criterion yields $\lim_{x\to c}(fg)(x) = LM$. □

Remark (Proof structure). Bridge argument. The proof does no direct ε-δ work; the product rule for sequences handles the analytic content via the add-and-subtract decomposition. Local boundedness of $f(x_n)$ is provided free by convergence.

Remark (Dependencies).

- Sequential Criterion for Function Limits
- Limit of a Function

Remark (Return). Return to Theorem

Theorem (Quotient Rule for Function Limits). *Let $f, g : A \to \mathbb{R}$, let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, $\lim_{x\to c} g(x) = M$, and $M \neq 0$, then there exists $\delta_0 > 0$ such that $g(x) \neq 0$ whenever $x \in A$ and $0 < |x - c| < \delta_0$, and $\lim_{x\to c}(f/g)(x) = L/M$.*

Professional Standard Proof.
Let $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$. By the Sequential Criterion, $f(x_n) \to L$ and $g(x_n) \to M$. Since $M \neq 0$, the quotient rule for sequence limits applies: $g(x_n)$ is eventually nonzero and $f(x_n)/g(x_n) \to L/M$. Since (x_n) was arbitrary, the Sequential Criterion gives $\lim_{x\to c}(f/g)(x) = L/M$.
The existence of δ_0 follows from the Limits Bounded Away From Zero theorem applied to g: since $\lim_{x\to c} g(x) = M \neq 0$, there exists $\delta_0 > 0$ such that $|g(x)| \geq |M|/2 > 0$ for all $x \in A$ with $0 < |x - c| < \delta_0$. □

Detailed Learning Proof.
Step 1. Establish that g stays away from zero near c. Since $\lim_{x\to c} g(x) = M \neq 0$, the Limits Bounded Away From Zero theorem gives $\delta_0 > 0$ such that $|g(x)| \geq |M|/2$ for all $x \in A$ with $0 < |x - c| < \delta_0$. In particular $g(x) \neq 0$ on this punctured neighbourhood.
Step 2. Set up a test sequence. Let $(x_n) \subseteq A \setminus \{c\}$ be any sequence with $x_n \to c$.
Step 3. Apply the Sequential Criterion to each component. The Sequential Criterion gives $f(x_n) \to L$ and $g(x_n) \to M$. Since $M \neq 0$, the quotient rule for sequence limits applies:

$$\frac{f(x_n)}{g(x_n)} \to \frac{L}{M}.$$

Step 4. Lift back by the Sequential Criterion. Since every such (x_n) satisfies $(f/g)(x_n) \to L/M$, the Sequential Criterion yields $\lim_{x\to c}(f/g)(x) = L/M$. □

Remark (Proof structure). Bridge argument plus one preliminary step. The preliminary step establishes that the denominator is bounded away from zero near c, which is what licenses f/g as a well-defined function on a punctured neighbourhood. The rest is the same bridge as the other algebra-of-limits proofs.

Remark (Dependencies).

- Sequential Criterion for Function Limits
- Limit Implies Local Boundedness
- Limit of a Function

Remark (Return). Return to Theorem

Proposition (Neighbourhood Form of the Limit). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every ε-neighbourhood $V_\varepsilon(L)$ there exists a δ-neighbourhood $V_\delta(c)$ such that $x \in V_\delta^*(c) \cap A$ implies $f(x) \in V_\varepsilon(L)$.*

Professional Standard Proof.
TODO: professional standard proof for limit-neighbourhood-equiv. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-neighbourhood-equiv. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Uniqueness of Limits of Functions). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then f has at most one limit at c.*

Professional Standard Proof.
TODO: professional standard proof for limit-unique. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-unique. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Sequential Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \setminus \{c\}$ with $x_n \to c$, one has $f(x_n) \to L$.*

Professional Standard Proof.
TODO: professional standard proof for sequential-criterion-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for sequential-criterion-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Three Equivalent Forms of the Function Limit). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. The ε-δ form of $\lim_{x\to c} f(x) = L$, the neighbourhood form, and the sequential form are equivalent descriptions of the same limit condition.*

Professional Standard Proof.
TODO: professional standard proof for limit-three-equiv. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-three-equiv. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limit Implies Local Boundedness). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$ for some $L \in \mathbb{R}$, then there exist $\delta, M > 0$ such that $|f(x)| \le M$ for all $x \in A$ with $0 < |x - c| < \delta$.*

Professional Standard Proof.
TODO: professional standard proof for limit-implies-local-bounding. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-implies-local-bounding. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bounded Limits). *Let $f : A \to \mathbb{R}$, let c be a cluster point of A, and let $a, b \in \mathbb{R}$. If $a \leq f(x) \leq b$ for all $x \in A$ with $x \neq c$, and $\lim_{x \to c} f(x) = L$, then $a \leq L \leq b$.*

Professional Standard Proof.
TODO: professional standard proof for bounded-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for bounded-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Squeeze Theorem for Function Limits). *Let* $f, g, h : A \to \mathbb{R}$ *and let* c *be a cluster point of* A*. Suppose* $\lim_{x\to c} f(x) = L = \lim_{x\to c} g(x) = L$*, and suppose there exists* $\delta_0 > 0$ *such that*

$$\forall x \in A,\ 0 < |x - c| < \delta_0 \ \Rightarrow\ f(x) \le h(x) \le g(x).$$

Then $\lim_{x\to c} h(x) = L$*.*

Professional Standard Proof.
TODO: professional standard proof for squeeze-function-limits. ☐

Detailed Learning Proof.
TODO: detailed learning proof for squeeze-function-limits. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Limits Are Local). *Let $E \subseteq X \subseteq \mathbb{R}$, let x_0 be a cluster point of E, let $f : X \to \mathbb{R}$, and let $\delta > 0$. Then*

$$\lim_{x \to x_0;\, x \in E} f(x) = L \iff \lim_{x \to x_0;\, x \in E \cap (x_0 - \delta, x_0 + \delta)} f(x) = L.$$

Professional Standard Proof.
TODO: professional standard proof for limits-are-local. □

Detailed Learning Proof.
TODO: detailed learning proof for limits-are-local. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Limit of Absolute Value). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. If $\lim_{x\to c} f(x) = L$, then $\lim_{x\to c} |f(x)| = |L|$.*

Professional Standard Proof.
TODO: professional standard proof for limit-absolute-value. □

Detailed Learning Proof.
TODO: detailed learning proof for limit-absolute-value. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Composition of Limits). *Let $f : A \to B$, let $g : B \to \mathbb{R}$, let c_1 be a cluster point of A, and let $c_2 \in \mathbb{R}$ be a cluster point of B. Suppose $\lim_{x \to c_1} f(x) = c_2$ and $\lim_{y \to c_2} g(y) = L$. If $c_2 \in B$, assume also that $g(c_2) = L$. Then $\lim_{x \to c_1} g(f(x)) = L$.*

Professional Standard Proof.
TODO: professional standard proof for composition-of-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for composition-of-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Monotone Limit Theorem). *Let $a < b$ and let $f : (a,b) \to \mathbb{R}$ be monotone and bounded. Then both one-sided limits exist at every interior point. For increasing f and every $x_0 \in (a,b)$:*

$$f(x_0^-) = \sup\{\, f(x) : a < x < x_0 \,\}, \qquad f(x_0^+) = \inf\{\, f(x) : x_0 < x < b \,\}.$$

Professional Standard Proof.
TODO: professional standard proof for monotone-limit-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for monotone-limit-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Cauchy Criterion for Function Limits). *Let $f : A \to \mathbb{R}$ and let c be a cluster point of A. Then $\lim_{x\to c} f(x) = L$ exists for some $L \in \mathbb{R}$ if and only if*

$$\forall\ \mathrm{OutputTolerance}(\varepsilon),\ \exists\ \mathrm{InputTolerance}(\delta),\ \forall x, y \in A :\ 0 < |x-c| < \delta \wedge 0 < |y-c| < \delta\ \Rightarrow\ |f(x)-f($$

Professional Standard Proof.
TODO: professional standard proof for cauchy-criterion-limits. ∎

Detailed Learning Proof.
TODO: detailed learning proof for cauchy-criterion-limits. ∎

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sequential Criterion for Right-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (c, \infty)$. Then $\lim_{x \to c^+} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (c, \infty)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Professional Standard Proof.
TODO: professional standard proof for sequential-criterion-right-hand-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for sequential-criterion-right-hand-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Sequential Criterion for Left-Hand Limits). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose c is a cluster point of $A \cap (-\infty, c)$. Then $\lim_{x\to c^-} f(x) = L$ if and only if for every sequence $(x_n) \subseteq A \cap (-\infty, c)$ with $x_n \to c$, one has $f(x_n) \to L$.*

Professional Standard Proof.
TODO: professional standard proof for sequential-criterion-left-hand-limit. □

Detailed Learning Proof.
TODO: detailed learning proof for sequential-criterion-left-hand-limit. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Two-Sided Limit via One-Sided Limits). *Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. If c is a cluster point of both $A \cap (c, \infty)$ and $A \cap (-\infty, c)$, then $\lim_{x \to c} f(x) = L$ if and only if both $\lim_{x \to c^+} f(x) = L$ and $\lim_{x \to c^-} f(x) = L$.*

Professional Standard Proof.
TODO: professional standard proof for two-sided-limit-iff-matching-one-sided-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for two-sided-limit-iff-matching-one-sided-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Composition of Injections Is Injective). *Let $f : A \to B$ and let $g : B \to C$. If f is injective and g is injective, then $g \circ f : A \to C$ is injective.*

Professional Standard Proof.
TODO: professional standard proof for composition-injective. ☐

Detailed Learning Proof.
TODO: detailed learning proof for composition-injective. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Composition of Surjections Is Surjective). $\text{Surjective}(f) \wedge \text{Surjective}(g) \Rightarrow \text{Surjective}(g \circ f)$.

Professional Standard Proof.
TODO: professional standard proof for composition-surjective. □

Detailed Learning Proof.
TODO: detailed learning proof for composition-surjective. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Composition of Bijections Is Bijective). *Let $f : A \to B$ and let $g : B \to C$. If f is bijective and g is bijective, then $g \circ f : A \to C$ is bijective.*

Professional Standard Proof.
TODO: professional standard proof for composition-bijective. □

Detailed Learning Proof.
TODO: detailed learning proof for composition-bijective. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Inverse of a Bijection Is a Bijection). $\text{Bijective}(f) \Rightarrow \text{Bijective}(f^{-1})$.

Professional Standard Proof.
TODO: professional standard proof for inverse-bijection. □

Detailed Learning Proof.
TODO: detailed learning proof for inverse-bijection. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Preimage Distributes over Union and Intersection). *Let $f : A \to B$, $S, T \subseteq B$. Then:*

(i) $f^{-1}(S \cup T) = f^{-1}(S) \cup f^{-1}(T)$.

(ii) $f^{-1}(S \cap T) = f^{-1}(S) \cap f^{-1}(T)$.

(iii) $f^{-1}(S^c) = (f^{-1}(S))^c$.

Professional Standard Proof.
TODO: professional standard proof for preimage-union-intersection. □

Detailed Learning Proof.
TODO: detailed learning proof for preimage-union-intersection. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sequential Characterization of Cluster Points).
c is a cluster point of A if and only if there exists $(a_n) \subseteq A \setminus \{c\}$ with $a_n \to c$.

Professional Standard Proof.
TODO: professional standard proof for cluster-point-sequential. □

Detailed Learning Proof.
TODO: detailed learning proof for cluster-point-sequential. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Lemma (Elementary Properties of Closures). *Let* $X, Y \subseteq \mathbb{R}$. *Then: (i)* $X \subseteq \overline{X}$; *(ii)* $\overline{X \cup Y} = \overline{X} \cup \overline{Y}$; *(iii)* $\overline{X \cap Y} \subseteq \overline{X} \cap \overline{Y}$; *(iv)* $X \subseteq Y \Rightarrow \overline{X} \subseteq \overline{Y}$.

Professional Standard Proof.
TODO: professional standard proof for closure-elementary. □

Detailed Learning Proof.
TODO: detailed learning proof for closure-elementary. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Closed iff Sequentially Closed). *X is closed* $\iff$ $\forall (a_n) \subseteq X,\ a_n \to x \Rightarrow x \in X$.

Professional Standard Proof.
TODO: professional standard proof for closed-iff-seq-limits. □

Detailed Learning Proof.
TODO: detailed learning proof for closed-iff-seq-limits. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Every Point of an Interval Is a Cluster Point). *If I is an interval, then every $x \in I$ is a cluster point of I.*

Professional Standard Proof.
TODO: professional standard proof for interval-all-limit-points. □

Detailed Learning Proof.
TODO: detailed learning proof for interval-all-limit-points. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Heine-Borel Theorem for R). *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Professional Standard Proof.
TODO: professional standard proof for heine-borel. □

Detailed Learning Proof.
TODO: detailed learning proof for heine-borel. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Heine-Borel Theorem for R). *X is closed $\wedge$ X is bounded $\iff$ every $(a_n) \subseteq X$ has a subsequence converging to a limit in X.*

Professional Standard Proof.
TODO: professional standard proof for heine-borel-subsets-real-line. □

Detailed Learning Proof.
TODO: detailed learning proof for heine-borel-subsets-real-line. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Functions, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 6

Continuity

continuity

Bounds $\to$ **Continuity** $\to$ Differentiation

6.1 Point Continuity

Toolkit — Oscillation

- OscillationOn(f, E, ω): $\omega(f; E) = \sup_{x_1,x_2 \in E} |f(x_1) - f(x_2)| = \mathrm{diam}(f(E))$.
- OscillationAt(f, x_0, ω): $\omega(f; x_0) = \lim_{\delta \to 0^+} \omega(f; U_\delta(x_0) \cap E)$. The limit exists because the function $\delta \mapsto \omega(f; U_\delta(x_0) \cap E)$ is nonincreasing.
- ContinuousAt$(f, x_0) \iff \omega(f; x_0) = 0$.
- Discontinuity set: $D(f) = \{x : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x : \omega(f; x) \geq 1/n\}$.
- Each threshold set is closed; $D(f)$ is F_σ.

Toolkit — Discontinuity

- DiscontinuousAt(f, a) is the negation of ContinuousAt(f, a).
- Quantifier reversal governs failure of continuity.
- Three equivalent formulations: ε-δ, sequential, neighbourhood.
- Taxonomy: removable, jump, essential.

Toolkit — Continuity

- Three equivalent forms: ε-δ, neighbourhood, sequential. The ε-δ form is unpunctured ($x = c$ allowed), unlike function limits.
- δ may depend on both ε and c. If δ depends only on ε, the condition is uniform continuity.
- Isolated points are automatically continuous.
- Algebra: sums, differences, products, scalar multiples, and quotients (denominator nonzero) of continuous functions are continuous.
- Composition preserves continuity.

Continuity

Definition (Relative Neighborhood)

Definition 6.1.1 (Relative Neighborhood). Let $E \subseteq \mathbb{R}$ and $a \in E$. A set $U \subseteq \mathbb{R}$ is a neighborhood of a in E if and only if there exists $\delta > 0$ such that $U = E \cap (a - \delta, a + \delta)$. Equivalently, every neighborhood of a in E is a set of the form $U_E(a) := E \cap (a - \delta, a + \delta)$ for some $\delta > 0$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ is a neighborhood of } a \text{ in } E \iff \exists \delta > 0 \ \big(U = E \cap (a - \delta, a + \delta)\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{RelativeNeighborhood}(U, E, a) := \exists \delta > 0 \ \big(U = E \cap (a - \delta, a + \delta)\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } E \subseteq \mathbb{R},\ a \in E,\ \text{and } U \subseteq \mathbb{R}.\\ &U \text{ fails to be a neighborhood of } a \text{ in } E \iff \forall \delta > 0 \ \big(U \neq E \cap (a - \delta, a + \delta)\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{RelativeNeighborhood}(U, E, a) \iff \forall \delta > 0 \ \big(U \neq E \cap (a - \delta, a + \delta)\big).$$

Remark (Failure modes). Failure occurs precisely when no radius δ makes U coincide with the relative interval slice: for every $\delta > 0$ either U contains a point outside $E \cap (a - \delta, a + \delta)$ or the slice contains a point missing from U.

Remark (Failure mode decomposition).

$$\forall \delta > 0 \ \underbrace{\big(\exists x \in U : \ x \notin E \cap (a - \delta, a + \delta)\big)}_{\text{extraneous point}} \vee \underbrace{\big(\exists y \in E \cap (a - \delta, a + \delta) : \ y \notin U\big)}_{\text{missing point}}.$$

Remark (Interpretation). Relative neighborhoods capture the local behavior of E around a by intersecting E with an ordinary open interval centered at a. The construction restricts the ambient notion of openness to the geometry of E, so it is suited to relative topologies. Failure simply means that U never matches the precise slice of E at any scale, typically because U omits some nearby points of E or includes points that fall outside E near a.

Remark (Dependencies).

- Subset
- Epsilon Neighbourhood
- Order on $\mathbb{R}$

Definition (Continuity at a Point)

Definition 6.1.2 (Continuity at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is **continuous at** c if and only if

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A : \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ \ f : A \to \mathbb{R},\ \ c \in A.\\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, A, c) :\equiv \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } A \subseteq \mathbb{R},\ \ f : A \to \mathbb{R},\ \ c \in A.\\ &f \text{ is not continuous at } c \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, A, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0\big).$$

Remark (Failure modes). Continuity at c fails precisely when there exists a persistent output gap $\varepsilon_0 > 0$ such that no matter how small the chosen neighborhood of c is, a point of A within that neighborhood keeps the function values at least ε_0 away from $f(c)$. The single failure branch records this inescapable oscillation near c.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\texttt{persistent gap}}\quad \underbrace{\forall \delta > 0}_{\texttt{every neighborhood}}\quad \underbrace{\exists x \in A}_{\texttt{nearby witness}}\quad : \quad |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0.$$

Remark (Interpretation). Continuity at c means that the output $f(x)$ can be kept arbitrarily close to $f(c)$ by taking x sufficiently close to c while staying in the domain A. The negated form highlights that a discontinuity arises only when there is a fixed positive separation in output that persists no matter how tightly one zooms in on c, so nearby inputs cannot force the outputs to settle near $f(c)$. This captures the geometric intuition that the graph of a continuous function has no abrupt jumps or breaks at c.

Remark (Dependencies).

- Definition: Limit of a Function

Definition (Continuity at a Point — Neighbourhood Form)

Definition 6.1.3 (Continuity at a Point -- Neighbourhood Form). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is continuous at c if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon).$$

Remark (Standard quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A. \\ &f \text{ is continuous at } c \iff \forall \varepsilon > 0\ \exists \delta > 0 : \\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon). \end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{ContinuousAt}(f, c) :\!= \forall \varepsilon > 0\ \exists \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \subseteq (f(c)-\varepsilon, f(c)+\varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned} &\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ c \in A. \\ &\neg\big(f \text{ is continuous at } c\big) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 : \\ &\qquad f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0). \end{aligned}$$

Remark (Negation predicate reading).

$$\mathrm{DiscontinuousAt}(f, c) \iff \exists \varepsilon_0 > 0\ \forall \delta > 0 :\ f\big(A \cap (c-\delta, c+\delta)\big) \nsubseteq (f(c)-\varepsilon_0, f(c)+\varepsilon_0).$$

Remark (Failure modes). Failure occurs precisely when some fixed output tolerance cannot be matched by any neighborhood about c. This manifests by finding points of A arbitrarily close to c whose images either exceed the upper endpoint $f(c)+\varepsilon_0$ or fall below the lower endpoint $f(c)-\varepsilon_0$.

Remark (Failure mode decomposition).

$$\exists \varepsilon_0 > 0\ \forall \delta > 0\ \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \geq f(c)+\varepsilon_0\Big)}_{\text{overshoot}} \vee \underbrace{\Big(\exists x \in A \cap (c-\delta, c+\delta) : f(x) \leq f(c) -}_{\text{undershoot}}$$

Remark (Interpretation). Continuity at c demands that every output tolerance around $f(c)$ has a matching neighborhood around c whose image remains confined to the tolerance band. The negation says that one tolerance band resists every neighborhood choice, so some sequence of points from A approaches c while their images either spike above or dip below $f(c)$ by at least a fixed amount. Geometrically, arbitrarily tight vertical strips around the graph above c must be achievable by shrinking the horizontal window; failure means the graph keeps escaping that strip no matter how closely the input variable hugs c.

Definition (Continuity at a Point — Sequential Form)

Definition 6.1.4 (Continuity at a Point -- Sequential Form). Let $A \subseteq \mathbb{R}$, let $c \in A$, and let $f\colon A \to \mathbb{R}$. The function f is continuous at c if and only if for every sequence $(x_n)_{n\in\mathbb{N}}$ in A, the convergence $\lim_{n\to\infty} x_n = c$ implies $\lim_{n\to\infty} f(x_n) = f(c)$.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.$$
$$f \text{ is continuous at } c \iff \forall (x_n)_{n\in\mathbb{N}} \subseteq A \left(\lim_{n\to\infty} x_n = c \Rightarrow \lim_{n\to\infty} f(x_n) = f(c)\right).$$

Remark (Definition predicate reading).

$$\text{ContinuousAt}(f,c) :\equiv \forall x_n \left(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \Rightarrow \text{ConvergentSequence}(f(x_n$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ c \in A,\ f\colon A \to \mathbb{R}.$$
$$f \text{ is not continuous at } c \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A \left(\lim_{n\to\infty} x_n = c \wedge \lim_{n\to\infty} f(x_n) \neq f(c)\right).$$

Remark (Negation predicate reading).

$$\neg\,\text{ContinuousAt}(f,c) \iff \exists x_n \left(\text{Sequence}(x_n, A) \wedge \text{ConvergentSequence}(x_n, c) \wedge \neg\,\text{ConvergentSequence}(f$$

Remark (Failure modes). Continuity at c fails sequentially when there exists a sequence $(x_n)_{n\in\mathbb{N}}$ in A with $x_n \to c$ but $f(x_n)$ does not converge to $f(c)$. This manifests in two distinct ways: either $f(x_n)$ converges to some limit different from $f(c)$, or $f(x_n)$ fails to converge entirely (oscillating without settling or diverging to infinity).

Remark (Failure mode decomposition).

$$\neg(f \text{ continuous at } c) \iff \exists (x_n)_{n\in\mathbb{N}} \subseteq A : \underbrace{\lim_{n\to\infty} x_n = c}_{\text{valid approach}} \wedge \underbrace{\lim_{n\to\infty} f(x_n) \neq f(c)}_{\text{image sequence fails}}.$$

Remark (Interpretation). The sequential characterization asserts that information about f near c is completely encoded by its action on limits of sequences taken within the domain. Any approach to c

through admissible points must transport convergent input sequences to convergent output sequences with the expected limit; conversely, a single pathological sequence suffices to expose a discontinuity. Geometrically, this captures the idea that f cannot introduce jumps or oscillations that persist along every infinitesimal path toward c.

Remark (Dependencies).
Definition 6.1.2.

Characterization of Discontinuity

Definition (Point of Discontinuity)

Definition 6.1.5 (Point of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $a \in A$. The function f is discontinuous at a if and only if f is not continuous at a.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A.$

f is discontinuous at $a \iff \neg\Big(\forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \big(|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon\big)\Big).$

Remark (Definition predicate reading).

$$\mathrm{DiscontinuousAt}(f, a) :\!= \neg\,\mathrm{ContinuousAt}(f, a).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ a \in A.$

f is discontinuous at $a \iff \exists\varepsilon > 0\ \forall\delta > 0\ \exists x \in A\ \big(|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon\big).$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f, a) \iff \mathrm{ContinuousAt}(f, a) \iff \forall\varepsilon > 0\ \exists\delta > 0\ \forall x \in A\ \big(|x-a| < \delta \Rightarrow |f(x)-f(a)$$

Remark (Failure modes). Discontinuity arises either because the nearby values of f do not approach any single real number as $x \to a$, or because a well-defined limit exists but disagrees with the actual value $f(a)$.

Remark (Failure mode decomposition).

$$\underbrace{\neg\exists L \in \mathbb{R}\ \ \mathrm{FunctionLimit}(f, A, a, L)}_{\texttt{no limit exists}} \vee \underbrace{\big(\exists L \in \mathbb{R}\ \ \mathrm{FunctionLimit}(f, A, a, L) \wedge L \neq f(a)\big)}_{\texttt{limit-value mismatch}}.$$

Remark (Interpretation). A point of discontinuity is detected by the breakdown of the usual epsilon-delta control. Understanding removable, jump, and essential cases guides how to remedy or classify singular behavior in proofs.

Remark (Dependencies).
Definition 6.1.2.

Definition (Sequential Characterization of Discontinuity)

Definition 6.1.6 (Sequential Characterization of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a sequence $(x_n)_{n\in\mathbb{N}} \subseteq A$ such that $x_n \to c$ while $f(x_n) \not\to f(c)$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists (x_n) \subseteq A \left[x_n \to c \ \wedge \ f(x_n) \not\to f(c)\right].$$

Remark (Definition predicate reading).

$$\text{DiscontinuousAt}(f, c) := \exists (x_n) \subseteq A : \ x_n \to c \ \wedge \ \neg\big(f(x_n) \to f(c)\big).$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall (x_n) \subseteq A : \ x_n \to c \ \Rightarrow \ f(x_n) \to f(c).$$

Remark (Negation predicate reading).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \text{ContinuousAt}(f, c) \iff \forall (x_n) \subseteq A : \ x_n \to c \ \Rightarrow \ f(x_n) \to f(c).$$

Remark (Failure modes). The characterization fails -- meaning f is NOT sequentially discontinuous -- when every sequence (x_n) in A converging to c has $f(x_n) \to f(c)$. No sequential witness exists.

Remark (Failure mode decomposition).

$$\neg\,\text{DiscontinuousAt}(f, c) \iff \underbrace{\forall (x_n) \subseteq A : \ x_n \to c \ \Rightarrow \ f(x_n) \to f(c)}_{\text{every approach preserves the limit}}.$$

Remark (Interpretation). A discontinuity at c can be certified by a single sequence approaching c whose images refuse to settle at $f(c)$.

Remark (Dependencies).
Definition 6.1.4; Definition 6.1.5.

Definition (Neighbourhood Characterization of Discontinuity)

Definition 6.1.7 (Neighbourhood Characterization of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. The function f is discontinuous at c if and only if there exists a neighborhood V of $f(c)$ such that for every neighborhood U of c relative to A there exists $x \in U \cap A$ with $f(x) \notin V$.

Remark (Standard quantified statement).

$$f \text{ is discontinuous at } c \iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A\ \big(f(x) \notin V\big).$$

Remark (Definition predicate reading).

$$\text{DiscontinuousAt}(f, c) := \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A : \ f(x) \notin V.$$

Remark (Negated quantified statement).

$$f \text{ is continuous at } c \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f(x) \in V\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DiscontinuousAt}(f,c) \iff \mathrm{ContinuousAt}(f,c) \iff \forall V \text{ nbhd of } f(c)\ \exists U \text{ nbhd of } c\ \forall x \in U \cap A\ \big(f($$

Remark (Failure modes). The neighbourhood discontinuity characterization fails -- meaning f is actually continuous -- when every output neighbourhood V of $f(c)$ can be matched by some input neighbourhood U such that f maps all of $U \cap A$ into V. No perpetually-missed output neighbourhood exists.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{DiscontinuousAt}(f,c) \iff \forall V \text{ nbhd of } f(c)\ \underbrace{\exists U \text{ nbhd of } c}_{\text{some input window}} :\ \underbrace{\forall x \in U \cap A :\ f(x) \in V}_{\text{all outputs inside tolerance}}.$$

Remark (Interpretation). Discontinuity means a fixed output neighborhood V is perpetually missed no matter how tightly the input is restricted around c.

Remark (Dependencies).
Definition 6.1.1.

Lemma (Equivalent Formulations of Discontinuity at a Point)

Lemma 6.1.8 (Equivalent Formulations of Discontinuity at a Point). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. The following are equivalent:*

1. *c is a point of discontinuity in the ε-δ sense.*
2. *c is a sequential point of discontinuity.*
3. *c is a neighbourhood point of discontinuity.*

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned} &(1)\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A :\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\ \iff &(2)\ \exists (x_n) \subseteq A :\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\ \iff &(3)\ \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V. \end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned} \mathrm{DiscontinuousAt}(f,c) &\iff \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A :\ 0 < |x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon_0 \\ &\iff \exists (x_n) \subseteq A :\ x_n \to c \wedge \neg\big(f(x_n) \to f(c)\big) \\ &\iff \exists V \text{ nbhd of } f(c)\ \forall U \text{ nbhd of } c\ \exists x \in U \cap A :\ f(x) \notin V. \end{aligned}$$

Remark (Failure modes). The equivalence cannot fail for functions at limit points of A: the proof that all three characterizations are equivalent is constructive in both directions. In the degenerate case where c is not a limit point of A, the punctured-neighbourhood conditions are vacuously satisfied by every L, and the equivalence becomes trivial rather than informative.

Remark (Interpretation). The three characterizations of discontinuity are mutually convertible. Analysts select whichever viewpoint simplifies a particular argument.

Remark (Dependencies).

- Point of Discontinuity
- Sequential Characterization of Discontinuity
- Neighbourhood Characterization of Discontinuity

Definition (Types of Discontinuity)

Definition 6.1.9 (Types of Discontinuity). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $x_0 \in A$ with f discontinuous at x_0.

(i) *Removable* iff $\exists L \in \mathbb{R} : \lim_{x\to x_0} f(x) = L$ and $L \neq f(x_0)$.

(ii) *Jump* iff $\exists L_-, L_+ \in \mathbb{R} : \lim_{x\to x_0^-} f(x) = L_-$, $\lim_{x\to x_0^+} f(x) = L_+$, and $L_- \neq L_+$.

(iii) *Essential* iff it is not removable.

Remark (Standard quantified statement).

$$\text{Removable} \iff \exists L \in \mathbb{R} \ \big(\lim_{x\to x_0} f(x) = L \wedge L \neq f(x_0)\big).$$
$$\text{Jump} \iff \exists L_-, L_+ \in \mathbb{R} \ \big(\lim_{x\to x_0^-} f(x) = L_- \wedge \lim_{x\to x_0^+} f(x) = L_+ \wedge L_- \neq L_+\big).$$
$$\text{Essential} \iff \neg\exists L \in \mathbb{R} \ \big(\lim_{x\to x_0} f(x) = L \wedge L \neq f(x_0)\big).$$

Remark (Definition predicate reading).

$$\text{RemovableDiscontinuity}(f, x_0) := \exists L \in \mathbb{R} \ \big(\text{FunctionLimit}(f, A, x_0, L) \wedge L \neq f(x_0)\big).$$
$$\text{JumpDiscontinuity}(f, x_0) := \exists L_-, L_+ \in \mathbb{R} \ \big(\text{LeftLimit}(f, A, x_0, L_-) \wedge \text{RightLimit}(f, A, x_0, L_+) \wedge L$$
$$\text{EssentialDiscontinuity}(f, x_0) := \neg\, \text{RemovableDiscontinuity}(f, x_0).$$

Remark (Negated quantified statement).

$$\neg\text{Removable} \iff \forall L \in \mathbb{R} \ \big(\lim_{x\to x_0} f(x) \neq L \vee L = f(x_0)\big),$$
$$\neg\text{Jump} \iff \forall L_-, L_+ \in \mathbb{R} \ \big(\lim_{x\to x_0^-} f(x) \neq L_- \vee \lim_{x\to x_0^+} f(x) \neq L_+ \vee L_- = L_+\big),$$
$$\neg\text{Essential} \iff \text{Removable}.$$

Remark (Negation predicate reading).

$$\neg\, \text{RemovableDiscontinuity}(f, x_0) \iff \forall L \in \mathbb{R} \ \big(\neg\, \text{FunctionLimit}(f, A, x_0, L) \vee L = f(x_0)\big).$$
$$\neg\, \text{JumpDiscontinuity}(f, x_0) \iff \forall L_-, L_+ \in \mathbb{R} \ \big(\neg\, \text{LeftLimit}(f, A, x_0, L_-) \vee \neg\, \text{RightLimit}(f, A, x_0,$$
$$\neg\, \text{EssentialDiscontinuity}(f, x_0) \iff \text{RemovableDiscontinuity}(f, x_0).$$

Remark (Failure modes).

- Removable discontinuity fails when no bilateral limit exists at x_0, or when the bilateral limit exists but equals $f(x_0)$ (making the function actually continuous).
- Jump discontinuity fails when at least one of the one-sided limits does not exist, or when both one-sided limits exist but are equal (making it potentially removable or continuous).
- Essential discontinuity fails when the discontinuity is removable (the bilateral limit exists and differs from $f(x_0)$).

Remark (Failure mode decomposition).

$$\neg\,\mathrm{RemovableDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L \in \mathbb{R}\ \lim_{x\to x_0} f(x) = L}_{\texttt{no bilateral limit exists}} \lor \underbrace{\lim_{x\to x_0} f(x) = f(x_0)}_{\texttt{limit equals function value}}.$$

$$\neg\,\mathrm{JumpDiscontinuity}(f, x_0) \iff \underbrace{\neg\exists L_- \in \mathbb{R}\ \lim_{x\to x_0^-} f(x) = L_-}_{\texttt{no left limit}} \lor \underbrace{\neg\exists L_+ \in \mathbb{R}\ \lim_{x\to x_0^+} f(x) = L_+}_{\texttt{no right limit}}$$
$$\lor \underbrace{\lim_{x\to x_0^-} f(x) = \lim_{x\to x_0^+} f(x)}_{\texttt{one-sided limits agree}}.$$

Remark (Interpretation). Removable discontinuities can be healed by redefining $f(x_0)$. Jump discontinuities provide well-defined one-sided limits useful in integration theory. Essential discontinuities are pathological and require more global tools.

Remark (Dependencies).

- Point of Discontinuity
- Limit of a Function
- Left-Hand Limit
- Right-Hand Limit

Oscillation

Definition (Oscillation on a Set)

Definition 6.1.10 (Oscillation on a Set). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $E \subseteq A$. The oscillation of f on E is the extended real number

$$\omega(f; E) := \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\} = \mathrm{diam}(f(E)).$$

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].$$
$$\omega(f; E) = \omega \iff \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Definition predicate reading).

$$\mathrm{OscillationOn}(f, E, \omega) := \omega = \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R},\ E \subseteq A,\ \omega \in [0, \infty].$$
$$\omega(f; E) \neq \omega \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OscillationOn}(f, E, \omega) \iff \omega \neq \sup\{\, |f(x_1) - f(x_2)| : x_1, x_2 \in E \,\}.$$

Remark (Failure modes). The proposed value ω fails either because it is not an upper bound for pairwise differences of function values on E, or because it is an upper bound but not the least such upper bound.

Remark (Failure mode decomposition).

$$\underbrace{\exists x_1, x_2 \in E :\ |f(x_1) - f(x_2)| > \omega}_{\texttt{not an upper bound}} \ \vee\ \underbrace{\exists \eta < \omega\ \forall x_1, x_2 \in E :\ |f(x_1) - f(x_2)| \le \eta}_{\texttt{not least}}.$$

Remark (Interpretation). Oscillation on a set records the spread of the image -- the diameter of $f(E)$. Small oscillation means all values of f on E lie close together. Zero oscillation means f is constant on E.

Remark (Dependencies).

- Function
- Subset
- Supremum
- Absolute Value on $\mathbb{R}$

Definition (Oscillation at a Point)

Definition 6.1.11 (Oscillation at a Point). Let $E \subseteq \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $x_0 \in \mathbb{R}$ satisfy $U_\delta(x_0) \cap E \neq \varnothing$ for every $\delta > 0$, where $U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}$. The oscillation of f at x_0 is the extended real number

$$\omega(f; x_0) = \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big),$$

with $\omega(f; A)$ denoting the oscillation of f on the set A. Because the function $\delta \mapsto \omega\big(f;\, U_\delta(x_0) \cap E\big)$ is nonincreasing, this right-hand limit exists (possibly $+\infty$).

Remark (Standard quantified statement).

$$\text{Let } E \subseteq \mathbb{R},\ f : E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0, \infty].$$
$$\big[(\forall \delta > 0 :\ U_\delta(x_0) \cap E \neq \varnothing) \ \wedge\ \omega(f; x_0) = \omega\big]$$
$$\iff \forall \varepsilon > 0\ \exists \delta > 0\ \forall r\ (0 < r < \delta \Rightarrow |\omega(f; U_r(x_0) \cap E) - \omega| < \varepsilon).$$

Remark (Definition predicate reading).

$$\mathrm{OscillationAt}(f, x_0, \omega) :\equiv \lim_{\delta \to 0^+} \omega\big(f;\, U_\delta(x_0) \cap E\big) = \omega, \quad U_\delta(x_0) = \{x \in \mathbb{R} : |x - x_0| < \delta\}.$$

Remark (Negated quantified statement).

$$\text{Let } E \subseteq \mathbb{R},\ f: E \to \mathbb{R},\ x_0 \in \mathbb{R},\ \omega \in [0,\infty].$$
$$\neg\big[(\forall \delta > 0 : U_\delta(x_0) \cap E \neq \varnothing) \ \wedge \ \omega(f; x_0) = \omega\big]$$
$$\iff \Big[\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing\Big]$$
$$\vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap E) - \omega| \geq \varepsilon_0)\Big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{OscillationAt}(f, x_0, \omega) \iff \Big[\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing\Big] \vee \Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0)$$

Remark (Failure modes). The definition fails in two distinct ways: either the point x_0 has a punctured neighborhood missing E, so there are no arbitrarily small samples to measure, or the local oscillation values near x_0 refuse to stabilize, preventing the limit from existing.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{OscillationAt}(f, x_0, \omega) = \underbrace{\exists \delta_0 > 0 : U_{\delta_0}(x_0) \cap E = \varnothing}_{\texttt{no nearby sample points}} \vee \underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists r\ (0 < r < \delta \wedge |\omega(f; U_r(x_0) \cap}_{\texttt{oscillation limit does not exist}}$$

Remark (Interpretation). Oscillation at x_0 measures the largest possible jump in function values that survives after one zooms in arbitrarily tightly around x_0. The shrinking neighborhoods $U_r(x_0) \cap E$ supply finer and finer samples, and the oscillation on each such set records the spread of f there. If these spreads settle to a single number, that number captures the local variability; it is zero precisely when f becomes arbitrarily close to a single value, i.e., when f is continuous at x_0. Failure occurs either because x_0 has no nearby domain points or because the spreads fluctuate indefinitely, reflecting wild local behavior.

Remark (Dependencies).

Definition 6.1.10

Theorem (Continuity Characterised by Oscillation)

Theorem 6.1.12 (Continuity Characterised by Oscillation). *Let $A \subseteq \mathbb{R}$, let $f: A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f; c) = 0$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f: A \to \mathbb{R},\ c \in A.$$
$$\big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\big)$$
$$\iff$$
$$\left(\inf_{\delta > 0} \sup\big\{|f(x) - f(y)| : x, y \in A,\ |x - c| < \delta,\ |y - c| < \delta\big\} = 0\right).$$

Remark (Predicate reading).

$$\mathrm{ContinuousAt}(f, c) \iff \mathrm{OscillationAt}(f, c, 0).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ c \in A$.

$$\Big(\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A\ (|x-c| < \delta \Rightarrow |f(x)-f(c)| < \varepsilon)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x)-f(y)| : x, y \in A,\ |x-c|$$

$$\vee$$

$$\Big(\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A\ (|x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_0)\Big) \wedge \Big(\inf_{\delta>0} \sup\{|f(x)-f(y)| : x, y \in A,\ |x-c|$$

Remark (Negation predicate reading).

$$\neg\big(\mathrm{ContinuousAt}(f,c) \iff \mathrm{OscillationAt}(f,c,0)\big).$$

Remark (Failure modes). The equivalence fails precisely in two mutually exclusive ways: either f remains continuous at c while the oscillation retains a positive lower bound, or $\omega(f;c)$ vanishes even though f does not satisfy the ε-δ continuity condition at c.

Remark (Failure mode decomposition).

$$\underbrace{\mathrm{ContinuousAt}(f,c) \wedge \neg\,\mathrm{OscillationAt}(f,c,0)}_{\text{Positive oscillation despite continuity}} \vee \underbrace{\neg\,\mathrm{ContinuousAt}(f,c) \wedge \mathrm{OscillationAt}(f,c,0)}_{\text{Zero oscillation without continuity}}.$$

Remark (Interpretation). The theorem states that measuring the oscillation of f near c captures exactly the same local regularity as the classical ε-δ definition of continuity. If oscillation collapses to zero, every pair of nearby function values becomes arbitrarily close, forcing $f(x)$ to approach $f(c)$. Conversely, any discontinuity manifests as a nonzero oscillation, because nearby points can produce separated outputs. The standard failure therefore arises from detecting a residual jump or oscillatory behavior in arbitrarily small neighborhoods of c.

Remark (Dependencies).

Definition 6.1.2, Definition 6.1.11.

Proposition

Proposition 6.1.13 (Discontinuity Set via Oscillation). *Let $f : E \to \mathbb{R}$. Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f;x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f;x) \geq 1/$$

Go to proof.

Remark (Standard quantified statement).

Let $f : E \to \mathbb{R}$.

$\forall x \in E :\ x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N} :\ \omega(f;x) \geq \frac{1}{n}$.

Remark (Predicate reading).

$$\begin{aligned} x \in D(f) &\iff \mathrm{DiscontinuousAt}(f,x) \\ &\iff \exists \omega\ (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0) \\ &\iff \exists n \in \mathbb{N}\ \exists \omega\ (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega \geq \tfrac{1}{n}). \end{aligned}$$

Remark (Negated quantified statement).

Let $f : E \to \mathbb{R}$.

$$\neg\Big[\forall x \in E\ \big(x \in D(f) \iff \omega(f;x) > 0 \iff \exists n \in \mathbb{N}\ \omega(f;x) \geq \tfrac{1}{n}\big)\Big]$$
$$\iff \exists x \in E\ \Big((x \in D(f) \wedge \omega(f;x) = 0) \vee (x \notin D(f) \wedge \omega(f;x) > 0)$$
$$\vee\ (\omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \tfrac{1}{n})\Big).$$

Remark (Negation predicate reading).

$$\exists x \in E\ \Big(\exists \omega\ (\mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega = 0) \vee$$
$$\exists \omega\ (\neg\,\mathrm{DiscontinuousAt}(f,x) \wedge \mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0) \vee$$
$$\exists \omega\ (\mathrm{OscillationAt}(f,x,\omega) \wedge \omega > 0 \wedge \forall n \in \mathbb{N}\ \omega < \tfrac{1}{n})\Big).$$

Remark (Failure modes). The proposition fails precisely when some $x \in E$ is marked as discontinuous although its oscillation vanishes, or when x is declared continuous despite a positive oscillation, or when a positive oscillation never exceeds any reciprocal integer, contradicting the Archimedean step that converts $\omega(f;x) > 0$ into the countable union description.

Remark (Failure mode decomposition).

$$\exists x \in E\ \begin{cases} x \in D(f) \wedge \omega(f;x) = 0 & \text{(Type I)}, \\ x \notin D(f) \wedge \omega(f;x) > 0 & \text{(Type II)}, \\ \omega(f;x) > 0 \wedge \forall n \in \mathbb{N}\ \omega(f;x) < \frac{1}{n} & \text{(Type III)}. \end{cases}$$

Remark (Interpretation). Oscillation detects discontinuity quantitatively: every discontinuity forces $\omega(f;x)$ to stay above some positive threshold, and conversely the vanishing of oscillation characterises continuity. Because each set $\{x \in E : \omega(f;x) \geq 1/n\}$ is closed, the discontinuity set $D(f)$ is an F_σ, a countable union of closed sets. This structural description is the gateway to measure-theoretic criteria such as the Lebesgue characterisation of Riemann integrability, where the size of $D(f)$ controls integrability.

Remark (Dependencies).
Definition 6.1.5; Definition 6.1.11.

6.2 Global Theorems

Toolkit — Boundedness, Extrema, and Intermediate Values

- Implication chain on $[a,b]$: $\mathrm{ContinuousOn}(f,[a,b]) \Rightarrow \mathrm{BoundedOn}(f,[a,b]) \Rightarrow$ extrema attained $\Rightarrow$ all intermediate values attained $\Rightarrow$ $f([a,b])$ is a closed bounded interval.

- All three hypotheses are essential: closed, bounded, continuous. Drop any one and the conclusion fails.

- IVT on a general interval: continuous image of any interval is an interval (Preservation of Intervals).

- Darboux property: a continuous function cannot skip a value.

Intermediate Value Theorem

Definition (Bounded on a Set)

Definition 6.2.1 (Bounded on a Set). Let $A \subseteq \mathbb{R}$ and $f\colon A \to \mathbb{R}$. The function f is bounded on A if and only if there exists a constant $M > 0$ such that $|f(x)| \le M$ for every $x \in A$.

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.$$
$$f \text{ is bounded on } A \iff \exists M > 0\ \forall x \in A\ (|f(x)| \le M).$$

Remark (Definition predicate reading).

$$\mathrm{BoundedOn}(f,A) :\!= \exists M > 0\ \forall x \in A\ (|f(x)| \le M).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \wedge \forall x \in [a,b]\ (f(x) \ne c) \implies \Big[\big(\forall x \in [a,b]\ f(x) > c\big) \vee \big(\forall x \in [a,b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\text{Let } A \subseteq \mathbb{R} \text{ and } f\colon A \to \mathbb{R}.$$
$$\neg(f \text{ bounded on } A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BoundedOn}(f,A) \iff \forall M > 0\ \exists x \in A\ (|f(x)| > M).$$

Remark (Failure modes). Failure occurs precisely when every proposed positive bound M is exceeded by some value of $|f(x)|$ with $x \in A$.

Remark (Failure mode decomposition).

$$\underbrace{\forall M > 0}_{\text{no admissible bound}}\ \underbrace{\exists x \in A\ (|f(x)| > M)}_{\text{value exceeding } M}.$$

Remark (Interpretation). The definition packages the informal idea that the values of f stay within a fixed vertical strip over A. The witness M serves as a uniform ceiling for the magnitudes $|f(x)|$, so once such an M is known the graph of f never escapes the band $[-M, M]$. Failure means that the outputs of f grow without limit along some sequence of points inside A, so no finite strip can contain the entire image. Boundedness on a set is the foundational regularity hypothesis that underlies the boundedness and extreme value theorems for continuous functions on compact domains.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Boundedness Theorem)

Theorem 6.2.2 (Boundedness Theorem). *Let $a < b$ and let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then there exists $M > 0$ such that $|f(x)| \le M$ for every $x \in [a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b \text{ and } f : [a, b] \to \mathbb{R}.$$
$$\Big(\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)$$
$$\Longrightarrow \exists M > 0\ \forall x \in [a, b]\ (|f(x)| \le M).$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Longrightarrow \text{BoundedOn}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\text{Let } a < b \text{ and } f : [a, b] \to \mathbb{R}.$$
$$\Big(\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big)$$
$$\wedge \Big(\forall M > 0\ \exists x \in [a, b]\ (|f(x)| > M)\Big).$$

Remark (Negation predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \wedge \neg\,\text{BoundedOn}(f, [a, b]).$$

Remark (Failure modes). Each component hypothesis is indispensable: without closedness (for instance $f(x) = 1/x$ on $(0, 1]$) continuity does not prevent divergence near the missing endpoint; without bounded domain (for example $f(x) = x$ on $[0, \infty)$) linear growth forces unbounded values; without continuity (take $f(0) = 0$ and $f(x) = 1/x$ for $x \in (0, 1]$) the discontinuity at 0 again creates arbitrarily large magnitudes.

Remark (Failure mode decomposition).

$$\neg\big[\mathrm{ClosedInterval}(a,b) \wedge \mathrm{BoundedSet}([a,b]) \wedge \mathrm{ContinuousOn}(f,[a,b])\big] = \underbrace{\neg\,\mathrm{ClosedInterval}(a,b)}_{\texttt{missing endpoint}} \vee \underbrace{\neg\,\mathrm{Bou}}_{\texttt{unbo}}$$

Remark (Interpretation). Continuity ties nearby inputs to nearby outputs, and the compact interval $[a,b]$ can be covered by finitely many neighborhoods in which the oscillation of f is controlled; taking the maximal oscillation among this finite cover supplies a global bound. Failure typically arises when the domain lacks compactness: an omitted endpoint allows the function to escape control near the hole, an unbounded interval lets values diverge along the ray, and a point of discontinuity can spike the function without upsetting nearby behavior elsewhere.

Remark (Dependencies).
Definition 6.1.2; Definition 6.2.1.

Definition (Absolute Maximum and Absolute Minimum)

Definition 6.2.3 (Absolute Maximum and Absolute Minimum). Let $A \subseteq \mathbb{R}$ and let $f\colon A \to \mathbb{R}$. A point $x^* \in A$ is an absolute maximum of f on A if and only if $f(x^*) \geq f(x)$ for every $x \in A$. A point $x_* \in A$ is an absolute minimum of f on A if and only if $f(x_*) \leq f(x)$ for every $x \in A$.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in A$.
x^* is an absolute maximum of f on $A \iff \forall x \in A\ (f(x) \leq f(x^*))$.
x_* is an absolute minimum of f on $A \iff \forall x \in A\ (f(x_*) \leq f(x))$.

Remark (Definition predicate reading).

$$(x^* \in A) \wedge \mathrm{Maximum}(f(x^*), f(A)) \iff (x^* \in A) \wedge \forall x \in A\ (f(x) \leq f(x^*)),$$
$$(x_* \in A) \wedge \mathrm{Minimum}(f(x_*), f(A)) \iff (x_* \in A) \wedge \forall x \in A\ (f(x_*) \leq f(x)).$$

Remark (Negated quantified statement).

Let $A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x^*, x_* \in \mathbb{R}$.
x^* fails to be an absolute maximum of f on A
$\iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big)$.
x_* fails to be an absolute minimum of f on A
$\iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big)$.

Remark (Negation predicate reading).

$$\neg\big[(x^* \in A) \wedge \mathrm{Maximum}(f(x^*), f(A))\big] \iff (x^* \notin A) \vee \big(\exists x \in A\ (f(x) > f(x^*))\big),$$
$$\neg\big[(x_* \in A) \wedge \mathrm{Minimum}(f(x_*), f(A))\big] \iff (x_* \notin A) \vee \big(\exists x \in A\ (f(x_*) > f(x))\big).$$

Remark (Failure modes). An alleged absolute maximum can fail because the proposed point lies outside the domain A, or because some $x \in A$ produces a strictly larger function value. An alleged absolute minimum fails for the analogous reasons, either by leaving the domain or by admitting a point in A with a strictly smaller value.

Remark (Failure mode decomposition).

$$\neg\big[(x^* \in A) \wedge \text{Maximum}(f(x^*), f(A))\big] = \underbrace{(x^* \notin A)}_{\texttt{external point}} \vee \underbrace{\big(\exists x \in A\, (f(x) > f(x^*))\big)}_{\texttt{better value}},$$

$$\neg\big[(x_* \in A) \wedge \text{Minimum}(f(x_*), f(A))\big] = \underbrace{(x_* \notin A)}_{\texttt{external point}} \vee \underbrace{\big(\exists x \in A\, (f(x_*) > f(x))\big)}_{\texttt{better value}}.$$

Remark (Interpretation). An absolute maximum identifies the highest value that f attains on its domain A, together with a point of attainment, and the absolute minimum identifies the lowest attained value. These extrema describe the global ceiling and floor of the graph restricted to A, so they are the endpoints of the vertical range. Failure occurs either when the candidate point is not part of the domain or when another domain point produces a superior value, reflecting that global optimization is sensitive to both membership and comparative size. Geometrically the absolute maximum sits at the tallest point of the graph above A, while the absolute minimum sits at the lowest point.

Remark (Dependencies).

- Function
- Subset
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Extreme Value Theorem)

Theorem 6.2.4 (Extreme Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$.* *Go to proof.*

Remark (Standard quantified statement).

Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous.
$\exists x_m, x_M \in [a, b]\, \forall x \in [a, b]\, (f(x_m) \leq f(x) \leq f(x_M))$.

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \implies \exists x_m, x_M \in [a, b]\, \big(\text{Maximum}(f(x_M), f([a, b])) \wedge \text{Minimum}(f(x_m), f([a, b]))\big).$$

Remark (Failure modes). The theorem can fail if the compact interval hypothesis is lost, or if the function is not continuous on the whole interval. Without compactness an extremal value may exist only as a limiting value, and without continuity the graph may jump over its supremum or infimum.

Remark (Failure mode decomposition).

$$\underbrace{\neg(a<b)}_{\texttt{invalid interval}} \;\vee\; \underbrace{\neg\big(f:[a,b]\to\mathbb{R}\big)}_{\texttt{wrong function type}} \;\vee\; \underbrace{\exists c\in[a,b]\;\exists\varepsilon>0\;\forall\delta>0\;\exists x\in[a,b]\;\big(|x-c|<\delta\wedge|f(x)-f(c)|\geq}_{\texttt{loss of continuity}}$$

Remark (Interpretation). Continuity on a closed and bounded interval forces the image of the interval to inherit the compact features of the domain. The function cannot escape to infinity or oscillate without settling, so it must reach both its highest and lowest values somewhere in the interval. When the hypotheses fail, the typical obstruction is the loss of compactness: on an open interval the function might drift off without achieving its supremum, and without continuity it may jump over candidate extrema.

Remark (Dependencies).
Uses Definition 6.2.3 and Theorem 6.2.2.

Theorem (Location of Roots)

Theorem 6.2.5 (Location of Roots). *Let $a,b\in\mathbb{R}$ with $a<b$ and let $f:[a,b]\to\mathbb{R}$ be continuous. If either $f(a)<0<f(b)$ or $f(a)>0>f(b)$, then there exists $c\in(a,b)$ such that $f(c)=0$.* *Go to proof.*

Remark (Standard quantified statement).

Let $a,b\in\mathbb{R}$ with $a<b$ and $f:[a,b]\to\mathbb{R}$ continuous.
$$\big(f(a)<0<f(b)\vee f(a)>0>f(b)\big)\Rightarrow\exists c\in(a,b)\;\big(f(c)=0\big).$$

Remark (Failure modes). The root conclusion can fail if f is not continuous on $[a,b]$, or if the endpoint values do not have opposite signs.

Remark (Failure mode decomposition).

$$\underbrace{\exists c\in[a,b]\;\exists\varepsilon>0\;\forall\delta>0\;\exists x\in[a,b]\;\big(|x-c|<\delta\wedge|f(x)-f(c)|\geq\varepsilon\big)}_{\texttt{loss of continuity}} \;\vee\; \underbrace{\big(f(a)f(b)\geq 0\big)}_{\texttt{no sign change}}.$$

Remark (Interpretation). Continuity forces the image of the interval $[a,b]$ to contain every intermediate value between $f(a)$ and $f(b)$. A sign change therefore compels the graph to cross the horizontal axis, producing a root strictly inside the interval. Failure occurs when the endpoint values share the same sign or when f is not continuous, because then the image need not sweep across zero. Geometrically, the theorem asserts that a continuous curve connecting opposite sides of the x-axis must intersect the axis somewhere in between.

Remark (Dependencies).

Bolzano's Intermediate Value Theorem.

Theorem (Bolzano's Intermediate Value Theorem)

Theorem 6.2.6 (Bolzano's Intermediate Value Theorem). *Let $I \subseteq \mathbb{R}$ be an interval, let $f\colon I \to \mathbb{R}$ be continuous on I, and choose $a, b \in I$ with $f(a) < k < f(b)$. Then there exists $c \in I$ such that $\min\{a,b\} < c < \max\{a,b\}$ and $f(c) = k$. Go to proof.*

Remark (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\, \forall f : I \to \mathbb{R}\, \forall a, b \in I\, \forall k \in \mathbb{R} \left[\big(\forall x, y \in I\, \forall z \in \mathbb{R}\, (x < z < y \Rightarrow z \in I)\big) \wedge \big(\forall c \in I\, \forall \varepsilon > 0\, \exists \delta > 0\, \forall$$

Remark (Failure modes). The conclusion can fail only if the continuity-on-an-interval hypothesis breaks down, or if the proposed level k is not strictly between the two attained endpoint values.

Remark (Failure mode decomposition).

$$\begin{aligned} &\underbrace{\exists x, y \in I\, \exists z \in \mathbb{R}\, \big(x < z < y \wedge z \notin I\big)}_{\text{not an interval}} \\ \vee\ &\underbrace{\exists c \in I\, \exists \varepsilon > 0\, \forall \delta > 0\, \exists x \in I\, \big(|x - c| < \delta \wedge |f(x) - f(c)| \geq \varepsilon\big)}_{\text{loss of continuity}} \\ \vee\ &\underbrace{\neg\big(f(a) < k < f(b)\big)}_{\text{level not between endpoint values}}. \end{aligned}$$

Remark (Interpretation). A continuous real-valued function on an interval cannot skip any intermediate scalar between two attained values, so the graph must cross every horizontal level that lies strictly between $f(a)$ and $f(b)$. The conclusion provides an interior point c whose image equals the prescribed level k, ensuring root-finding for $f - k$. If the hypothesis fails, either f is not continuous or the level k does not lie between the endpoint values, and no such interior point need exist.

Remark (Dependencies).

- Function
- Subset
- Continuity at a Point
- Order on $\mathbb{R}$
- Maximum
- Minimum

Theorem (Image of Closed Bounded Interval Is Closed Bounded Interval)

Theorem 6.2.7 (Image of Closed Bounded Interval Is Closed Bounded Interval). *Let $f : [a,b] \to \mathbb{R}$ be continuous. Then $f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f]$. Go to proof.*

Remark (Standard quantified statement).

$$\forall a, b \in \mathbb{R}\ \forall f : [a,b] \to \mathbb{R} :$$
$$\Big(a < b \wedge \forall c \in [a,b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a,b]\ \big(|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big) \implies f([a,b]) = [\mathrm{m}$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a,b]) \implies f([a,b]) = [\min_{[a,b]} f, \max_{[a,b]} f].$$

Remark (Interpretation). The statement combines the extreme value theorem, which supplies both a minimum and a maximum on $[a,b]$, with the intermediate value theorem, which fills in every intermediate value between them. Consequently the image $f([a,b])$ is a closed interval with endpoints given by the absolute extremal values of f on $[a,b]$, showing that continuous functions map compact intervals onto compact intervals without introducing gaps.

Remark (Dependencies).
Extreme Value Theorem, Bolzano's Intermediate Value Theorem

Theorem (Preservation of Intervals)

Theorem 6.2.8 (Preservation of Intervals). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous. Then $f(I)$ is an interval.* *Go to proof.*

Remark (Standard quantified statement).

$$\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R} :$$
$$\Big[\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)$$
$$\wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)\Big] \implies \forall y_1, y_2 \in f(I)\ \forall y \in \mathbb{R}\ \big(y_1$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, I) \implies \mathrm{Interval}(f(I)).$$

Remark (Negated quantified statement).

$$\exists I \subseteq \mathbb{R}\ \exists f : I \to \mathbb{R}\ \exists y_1, y_2 \in f(I)\ \exists y \in \mathbb{R} :$$
$$\big(\forall x_1, x_2 \in I\ \forall z \in \mathbb{R}\ (x_1 < z < x_2 \Rightarrow z \in I)\big)$$
$$\wedge \big(\forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\ (|x-c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\big)$$
$$\wedge\, y_1 < y < y_2 \wedge y \notin f(I).$$

Remark (Negation predicate reading).

$$\mathrm{Interval}(I) \wedge \mathrm{ContinuousOn}(f, I) \wedge \neg\, \mathrm{Interval}(f(I)).$$

Remark (Interpretation). Continuous functions cannot tear gaps open inside their images of intervals. Once the function achieves two output values, continuity and the Intermediate Value Theorem force every intermediate value to occur as well, so the image remains an interval even when I is unbounded or half-open. Failure occurs precisely when continuity breaks, because discontinuities can skip intermediate heights and leave disconnected image sets.

Remark (Dependencies).
thm:bolzano-intermediate-value

Theorem (Darboux Property)

Theorem 6.2.9 (Darboux Property). *Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f : [a, b] \to \mathbb{R}$ be continuous, and fix $c \in \mathbb{R}$. If $f(x) \neq c$ for every $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$ or $f(x) < c$ for all $x \in [a, b]$. Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \Rightarrow \Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Negated quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \Big(\exists u \in [a, b]\ f(u) \leq c\Big) \wedge \Big(\exists v \in [a, b]\ f(v) \geq c\Big).$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \Big(\forall x \in [a, b]\ f(x) \neq c\Big) \wedge \neg\Big[\big(\forall x \in [a, b]\ f(x) > c\big) \vee \big(\forall x \in [a, b]\ f(x) < c\big)\Big].$$

Remark (Contrapositive quantified statement).

$$\text{Let } a < b,\ f : [a, b] \to \mathbb{R} \text{ be continuous, and } c \in \mathbb{R}.$$
$$\Big(\exists x \in [a, b]\ f(x) \leq c\Big) \wedge \Big(\exists y \in [a, b]\ f(y) \geq c\Big) \Rightarrow \exists z \in [a, b]\ f(z) = c.$$

Remark (Contrapositive predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{MeetsBothSides}(f, [a, b], c) \implies \exists z \in [a, b]\ (f(z) = c).$$

Remark (Interpretation). A continuous function on a compact interval cannot oscillate across a level c without actually attaining that level. Stated contrapositive fashion, if c is never taken then the entire graph lies strictly above or strictly below the horizontal line y=c. The only way this guarantee can fail is when the function produces values on both sides of c, in which case continuity forces a crossing point. Geometrically the conclusion describes the image of a connected set under a continuous map: it is an interval, so omitting c forces the image to stay in one of the open half-lines determined by c.

Remark (Dependencies).
Bolzano's Intermediate Value Theorem.

6.3 Uniform Continuity

Toolkit — Uniform Continuity

- Key distinction: $\delta = \delta(\varepsilon)$ only, not $\delta(\varepsilon, c)$. One δ serves all pairs in A.
- Input condition: $\text{Within}(x, u, \delta)$ — both x and u range freely, no puncturing.
- Implication chain: $\text{Lipschitz}(f, A) \Rightarrow \text{UniformlyContinuous}(f, A) \Rightarrow \text{ContinuousOn}(f, A)$. Both implications are strict.
- Heine–Cantor: $\text{ContinuousOn}(f, [a, b]) \Rightarrow \text{UniformlyContinuous}(f, [a, b])$.
- Failure criterion: sequences $(x_n), (u_n)$ with $x_n - u_n \to 0$ but $|f(x_n) - f(u_n)| \geq \varepsilon_0$.
- $\text{UniformlyContinuous}(f, A)$ preserves Cauchy sequences — the key lemma for continuous extensions.

Uniform Continuity

Definition (Uniform Continuity)

Definition 6.3.1 (Uniform Continuity). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The function f is uniformly continuous on A if and only if for every $\varepsilon > 0$ there exists $\delta > 0$ such that for all $x, u \in A$ the implication $|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon$ holds.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \\ \implies \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

Remark (Definition predicate reading).

$$\text{UniformlyContinuous}(f, A) := \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, u \in A\ (|x - u| < \delta \Rightarrow |f(x) - f(u)| < \varepsilon).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \\ \vee \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).$$

Remark (Negation predicate reading).

$$\neg\,\text{UniformlyContinuous}(f, A) \iff \exists \varepsilon > 0\ \forall \delta > 0\ \exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)| \geq \varepsilon).\blacksquare$$

Remark (Failure modes). The definition fails exactly when a single positive tolerance ε_0 exposes pairs of inputs arbitrarily close together whose function values remain at least ε_0 apart, so no universal δ works for that ε_0.

Remark (Failure mode decomposition).

$$\neg\,\text{UniformlyContinuous}(f, A) = \underbrace{\exists \varepsilon_0 > 0}_{\text{frozen tolerance}} \wedge \underbrace{\forall \delta > 0}_{\text{no global control}} \wedge \underbrace{\exists x, u \in A\ (|x - u| < \delta \wedge |f(x) - f(u)|}_{\text{bad witness pair}}$$

Remark (Interpretation). Uniform continuity demands that a single proximity scale δ answers any prescribed accuracy ε simultaneously across the whole domain. The negation shows what goes wrong: a fixed output tolerance resists all choices of δ because pairs of nearby inputs can always be found that stretch the function values too far apart. Geometrically, the graph of a uniformly continuous function can be traversed with a ruler of length δ horizontally and ε vertically without ever overstepping the vertical tolerance, whereas a nonuniformly continuous function exhibits spikes that defeat every fixed ruler length.

Remark (Dependencies).

- Function
- Subset
- Absolute Value on $\mathbb{R}$
- Order on $\mathbb{R}$

Theorem (Algebra of Uniform Continuity on a General Domain)

Theorem 6.3.2 (Algebra of Uniform Continuity on a General Domain). *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$

$$\Big[\big(\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\big)$$

$$\wedge \big(\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\big)\Big] \Rightarrow \Big[\forall \varepsilon > 0\ \exists \delta_1 > 0\ \forall x, y \in A\ (|x$$

$$\wedge\ \forall \varepsilon > 0\ \exists \delta_2 > 0\ \forall x, y \in A\ (|x - y| < \delta_2 \Rightarrow |(f - g)(x) - (f - g)(y)| < \varepsilon)$$

$$\wedge\ \forall \varepsilon > 0\ \exists \delta_3 > 0\ \forall x, y \in A\ (|x - y| < \delta_3 \Rightarrow |cf(x) - cf(y)| < \varepsilon)\Big],$$

$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\ \Big[\forall \varepsilon > 0\ \exists \delta_f > 0\ \forall x, y \in A\ (|x - y| < \delta_f \Rightarrow |f(x) - f(y)| < \varepsilon)\Big]$$

$$\wedge \Big[\forall \varepsilon > 0\ \exists \delta_g > 0\ \forall x, y \in A\ (|x - y| < \delta_g \Rightarrow |g(x) - g(y)| < \varepsilon)\Big]$$

$$\wedge\ \exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x, y \in A\ \big(|x - y| < \delta \wedge |f(x)g(x) - f(y)g(y)| \geq \varepsilon_0\big).$$

Remark (Predicate reading).

$$\forall A \subseteq \mathbb{R}\ \forall f, g : A \to \mathbb{R}\ \forall c \in \mathbb{R} :$$

$$\big(\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A)\big) \implies \big(\mathrm{UniformlyContinuous}(f + g, A)$$

$$\exists A \subseteq \mathbb{R}\ \exists f, g : A \to \mathbb{R} :\quad \mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \neg\, \mathrm{UniformlyCon}$$

Remark (Interpretation). Uniform continuity is stable under linear combinations on any common domain, so sums, differences, and scalar multiples inherit the same global control on oscillation from the original functions. The proof decomposes the linear expressions into epsilon-delta bounds that add and subtract without upsetting the uniform response. In contrast, products can amplify local variation when one factor lacks a bounded range, so even uniformly continuous multiplicands can produce a product that fails the global tolerance requirement. The theorem explains both the algebraic closure that analysts routinely exploit and the standard obstruction, guiding expectations when building uniformly continuous functions from simpler pieces.

Remark (Dependencies).
Definition 6.3.1.

Theorem (Algebra of Uniform Continuity on a Bounded Domain)

Theorem 6.3.3 (Algebra of Uniform Continuity on a Bounded Domain). *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Go to proof.

Remark (Standard quantified statement).

Let $A \subseteq \mathbb{R}$ be bounded and $f, g : A \to \mathbb{R}$.

$$\begin{aligned}
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big] \\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big] \\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)g(x) - f(y)g(y)| < \varepsilon), \\
&\big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon)\big] \\
&\wedge \big[\forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |g(x) - g(y)| < \varepsilon)\big] \\
&\wedge \big[\exists k > 0\ \forall x \in A\ (|g(x)| \geq k)\big] \\
&\Rightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in A\ (|x - y| < \delta \Rightarrow |f(x)/g(x) - f(y)/g(y)| < \varepsilon).
\end{aligned}$$

Remark (Predicate reading).

Assume $A \subseteq \mathbb{R}$ is bounded and $f, g : A \to \mathbb{R}$.

$$\begin{aligned}
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(fg, A), \\
&\mathrm{UniformlyContinuous}(f, A) \wedge \mathrm{UniformlyContinuous}(g, A) \wedge \exists k > 0\ \forall x \in A\ (|g(x)| \geq k) \\
&\qquad \Longrightarrow \mathrm{UniformlyContinuous}(f/g, A).
\end{aligned}$$

Remark (Failure modes). The product conclusion can fail only when one of the hypotheses used to control the product is missing: the domain is not bounded, f is not uniformly continuous on A, or g is not uniformly continuous on A. The quotient conclusion can also fail if the denominator is not bounded away from zero on A.

Remark (Failure mode decomposition).

$$\underbrace{\neg\,\mathrm{BoundedSet}(A)}_{\text{unbounded domain}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(f,A)}_{\text{first factor not uniformly continuous}} \vee \underbrace{\neg\,\mathrm{UniformlyContinuous}(g,A)}_{\text{second factor not uniformly continuous}}$$
$$\vee \underbrace{\neg\exists k>0\ \forall x\in A:\ |g(x)|\geq k}_{\text{denominator not bounded away from zero}}.$$

Remark (Interpretation). Bounded subsets of $\mathbb{R}$ prevent the product of two uniformly continuous functions from developing large oscillations, so the shared modulus of continuity for f and g can be combined to control fg. The quotient is uniformly continuous whenever g never approaches zero, because the bounded-away-from-zero hypothesis allows inversion to act as another uniformly continuous operation that preserves the common modulus after scaling. Failure of boundedness or a loss of a positive lower bound for $|g|$ are the only ways these closure properties can break, and in either case the oscillation of the algebraic combination can no longer be controlled uniformly across the entire set.

Remark (Dependencies).
Definition 6.3.1.

Proposition (Sequential Characterization of Uniform Continuity)

Proposition 6.3.4 (Sequential Characterization of Uniform Continuity). *Let $A\subseteq\mathbb{R}$ and let $f:A\to\mathbb{R}$. Then f is uniformly continuous on A if and only if for every pair of sequences (x_n) and (y_n) in A with $\lim_{n\to\infty}(x_n-y_n)=0$ we have $\lim_{n\to\infty}(f(x_n)-f(y_n))=0$.* Go to proof.

Remark (Standard quantified statement).

$$\Big[\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ \big(|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon\big)\Big] \iff \forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to 0\Rightarrow |$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f,A) \iff \forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to 0\Rightarrow|f(x_n)-f(y_n)|\to 0\big).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\exists A\subseteq\mathbb{R}\ \exists f:A\to\mathbb{R}:\\
&\Big([\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in A\ (|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon)]\\
&\qquad\wedge\exists x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to 0\wedge\neg(|f(x_n)-f(y_n)|\to 0)\big)\Big)\\
&\vee\Big([\exists\varepsilon_0>0\ \forall\delta>0\ \exists x,y\in A\ (|x-y|<\delta\wedge|f(x)-f(y)|\geq\varepsilon_0)]\wedge\forall x_n,y_n:\mathbb{N}\to A\ \big(|x_n-y_n|\to 0 \Rightarrow
\end{aligned}$$

Remark (Negation predicate reading).

$$\exists A \subseteq \mathbb{R}\ \exists f : A \to \mathbb{R} :$$
$$\Big(\text{UniformlyContinuous}(f, A) \wedge \exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)\Big)$$
$$\vee\ \Big(\neg\,\text{UniformlyContinuous}(f, A) \wedge \forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)\Big).$$

Remark (Failure modes). The equivalence can fail in exactly two ways: either f is uniformly continuous while some pair of asymptotically coincident sequences in A yields outputs that do not converge together, or every such pair of sequences does converge together while f nonetheless fails to be uniformly continuous on A.

Remark (Failure mode decomposition).

$$\underbrace{\text{UniformlyContinuous}(f, A)}_{\texttt{uniform continuity holds}} \wedge \underbrace{\exists x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \wedge \neg(|f(x_n) - f(y_n)| \to 0)\big)}_{\texttt{sequential test fails}}$$
$$\vee\ \underbrace{\neg\,\text{UniformlyContinuous}(f, A)}_{\texttt{uniform continuity fails}} \wedge \underbrace{\forall x_n, y_n : \mathbb{N} \to A\ \big(|x_n - y_n| \to 0 \Rightarrow |f(x_n) - f(y_n)| \to 0\big)}_{\texttt{sequential test holds}}.$$

Remark (Interpretation). Uniform continuity prevents the function from pulling together inputs that are asymptotically identical while letting their images drift apart. The sequential formulation packages this requirement into a tail test that ignores explicit epsilon-delta bookkeeping: whenever two points of A are eventually arbitrarily close, their function values must also be eventually arbitrarily close. Proving uniform continuity via sequences is often simpler, and conversely, producing two sequences that collide in A but whose images do not collide gives a concrete witness that uniform continuity fails. The two failure modes correspond to breaking one direction or the other of the claimed equivalence.

Remark (Dependencies).
Definition 6.3.1

Theorem (Heine–Cantor Theorem)

Theorem 6.3.5 (Heine-Cantor Theorem). *Let* $a, b \in \mathbb{R}$ *with* $a < b$ *and let* $f : [a, b] \to \mathbb{R}$ *be continuous on* $[a, b]$*. Then* f *is uniformly continuous on* $[a, b]$*.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a, b \in \mathbb{R} \text{ with } a < b \text{ and let } f : [a, b] \to \mathbb{R}.$$
$$\Big[\forall c \in [a, b]\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in [a, b]\ (|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon)\Big]$$
$$\Longrightarrow \forall \varepsilon > 0\ \exists \delta > 0\ \forall x, y \in [a, b]\ (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\text{ContinuousOn}(f, [a, b]) \Rightarrow \text{UniformlyContinuous}(f, [a, b]).$$

Remark (Negated quantified statement).

$$\text{Let } a,b\in\mathbb{R} \text{ with } a<b \text{ and let } f:[a,b]\to\mathbb{R}.$$
$$\Big[\forall c\in[a,b]\ \forall\varepsilon>0\ \exists\delta>0\ \forall x\in[a,b]\ (|x-c|<\delta\Rightarrow|f(x)-f(c)|<\varepsilon)\Big]$$
$$\wedge\Big[\exists\varepsilon_0>0\ \forall\delta>0\ \exists x,y\in[a,b]\ (|x-y|<\delta\wedge|f(x)-f(y)|\geq\varepsilon_0)\Big].$$

Remark (Negation predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b])\wedge\neg\,\mathrm{UniformlyContinuous}(f,[a,b]).$$

Remark (Interpretation). Continuity on the compact interval $[a,b]$ forces a single modulus of continuity that works simultaneously at every point, so oscillations of f cannot sharpen when inspecting smaller neighborhoods. Proofs usually argue by contradiction: assuming the negated form produces sequences of points whose distances shrink while their images stay apart, contradicting compactness-driven convergence. The theorem shows that closed bounded domains prevent the pathological behavior that obstructs uniform control on open or unbounded sets.

Remark (Dependencies).
Definition 6.1.2; Definition 6.3.1.

Proposition (Uniform Continuity Preserves Cauchy Sequences)

Proposition 6.3.6 (Uniform Continuity Preserves Cauchy Sequences).
Let $S\subseteq\mathbb{R}$, let $f:S\to\mathbb{R}$, and let (x_n) be a sequence in S. If f is uniformly continuous on S and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$. *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } S\subseteq\mathbb{R},\ f:S\to\mathbb{R},\ (x_n)_{n\in\mathbb{N}}\subseteq S.$$
$$\Big[\forall\varepsilon>0\ \exists\delta>0\ \forall x,y\in S\ (|x-y|<\delta\Rightarrow|f(x)-f(y)|<\varepsilon)\Big]$$
$$\wedge\Big[\forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall m,n\geq N\ (|x_n-x_m|<\varepsilon)\Big]$$
$$\Rightarrow\forall\varepsilon>0\ \exists N\in\mathbb{N}\ \forall m,n\geq N\ (|f(x_n)-f(x_m)|<\varepsilon).$$

Remark (Predicate reading).

$$\mathrm{UniformlyContinuous}(f,S)\wedge\mathrm{CauchySequence}(x_n)\ \Rightarrow\ \mathrm{CauchySequence}(f(x_n)).$$

Remark (Interpretation). Uniform continuity controls the change in $f(x)$ solely through the distance between inputs, so any Cauchy sequence in the domain is sent to a sequence whose tails remain uniformly close. Ordinary pointwise continuity cannot guarantee this because the necessary input tolerance could shrink along the sequence, so uniformity is exactly the additional strength needed to transport the Cauchy property. The standard failure mode occurs when f is merely continuous and the sequence approaches a boundary point where continuity loses uniform control, producing large oscillations in the image sequence. Geometrically, uniform continuity supplies a single modulus of continuity that works for the entire path traced by (x_n), ensuring that eventually all image terms lie in an arbitrarily small common interval.

Remark (Dependencies).

Definition 6.3.1; Definition 3.6.5.

Definition (Lipschitz Condition)

Definition 6.3.7 (Lipschitz Condition). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $K > 0$. The function f satisfies the Lipschitz condition on A with constant K if and only if $|f(x) - f(u)| \le K|x-u|$ for every $x, u \in A$.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \wedge (K > 0) \\ \implies \forall x, u \in A \,\big(|f(x) - f(u)| \le K|x-u|\big).$$

Remark (Definition predicate reading).

$$\text{Lipschitz}(f, A, K) :\equiv (K > 0) \wedge \forall x, u \in A \,\big(|f(x) - f(u)| \le K|x-u|\big).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0) \\ \vee \exists x, u \in A \,\big(|f(x) - f(u)| > K|x-u|\big).$$

Remark (Negation predicate reading).

$$\neg\,\text{Lipschitz}(f, A, K) \iff \neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee (K \le 0) \vee \exists x, u \in A \,\big(|f(x) - f(u)| > K|x-u|\big).$$

Remark (Failure modes). The Lipschitz condition fails when the ambient data are malformed, when the proposed constant is not positive, or when there exists a pair of points in A whose images separate faster than K times the separation of the points themselves.

Remark (Failure mode decomposition).

$$\underbrace{\neg(A \subseteq \mathbb{R})}_{\texttt{bad domain}} \vee \underbrace{\neg(f : A \to \mathbb{R})}_{\texttt{bad function type}} \vee \underbrace{K \le 0}_{\texttt{bad constant}} \vee \underbrace{\exists x, u \in A :\ |f(x) - f(u)| > K|x-u|}_{\texttt{excessive separation of outputs}}.$$

Remark (Interpretation). The Lipschitz condition requires a single constant K that simultaneously bounds the rate at which f can stretch distances across the entire set A. It encapsulates a global slope cap: every secant line through the graph has absolute slope at most K. Failure occurs once a single pair of points forces a steeper secant, signaling that no uniform control constant works on the whole domain. Geometrically, the graph of a Lipschitz function lies inside a cone of opening determined by K, so nearby points cannot be blown apart faster than K times their original spacing.

Remark (Dependencies).

- Function

Proposition (Lipschitz Implies Uniform Continuity)

Proposition 6.3.8 (Lipschitz Implies Uniform Continuity). *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}, \text{ and suppose } \exists K \geq 0\, \forall x, y \in A\, (|f(x) - f(y)| \leq K|x - y|).$$
$$\forall \varepsilon > 0\, \exists \delta > 0\, \forall x, y \in A\, (|x - y| < \delta \Rightarrow |f(x) - f(y)| < \varepsilon).$$

Remark (Predicate reading).

$$\big[\exists K \geq 0\ \text{Lipschitz}(f, A, K)\big] \Rightarrow \text{UniformlyContinuous}(f, A).$$

Remark (Contrapositive quantified statement). If f fails to be uniformly continuous on A, then f is not Lipschitz on A with any constant.

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f : A \to \mathbb{R}.\\
&\Big[\exists \varepsilon_0 > 0\, \forall \delta > 0\, \exists x, y \in A\, (|x - y| < \delta \wedge |f(x) - f(y)| \geq \varepsilon_0)\Big]\\
&\Longrightarrow \forall K \geq 0\, \exists x, y \in A\, (|f(x) - f(y)| > K|x - y|).
\end{aligned}$$

Remark (Contrapositive predicate reading).

$$\neg\, \text{UniformlyContinuous}(f, A) \implies \neg \exists K \geq 0 :\ \text{Lipschitz}(f, A, K).$$

Remark (Interpretation). A Lipschitz bound controls the variation of f by the distance between inputs with a single constant K, so the same K yields a usable $\delta = \varepsilon/K$ for every pair of points at once, proving uniform continuity. When $K = 0$ the function is constant and trivially uniform. Failure of uniform continuity would force two points arbitrarily close together whose images stay a fixed distance apart, contradicting the Lipschitz inequality.

Remark (Strict implication note). The implication Lipschitz $\Rightarrow$ uniform continuity is strict. The canonical counterexample is $f(x) = \sqrt{x}$ on $[0, 1]$, which is uniformly continuous by the Heine-Cantor theorem but satisfies $|f(x) - f(0)|/|x - 0| = 1/\sqrt{x} \to \infty$ as $x \to 0^+$, so no Lipschitz constant exists at the origin. This distinction is relevant to GPU Lipschitz-constant estimation: uniform continuity of a simulation function does not supply a computable K for the Lipschitz condition; only an explicit Lipschitz bound does.

Remark (Dependencies).
Definition 6.3.7, Definition 6.3.1.

Theorem (Algebra of Lipschitz Functions)

Theorem 6.3.9 (Algebra of Lipschitz Functions). *Let $A \subseteq \mathbb{R}$ and let $f, g : A \to \mathbb{R}$ be Lipschitz with constants $K_f, K_g \geq 0$ respectively.*

(a) For every $c \in \mathbb{R}$, the function cf is Lipschitz on A with constant $|c|K_f$.

(b) The functions $f + g$ and $f - g$ are Lipschitz on A with constant $K_f + K_g$.

(c) If $B \subseteq \mathbb{R}$, $h : B \to \mathbb{R}$ is Lipschitz with constant $K_h \geq 0$, and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$.

(d) If $|f(x)| \leq M_f$ and $|g(x)| \leq M_g$ for all $x \in A$, then fg is Lipschitz on A with constant $M_g K_f + M_f K_g$.

(e) If $|f(x)| \leq M_f$ for all $x \in A$, if $|g(x)| \geq k > 0$ for all $x \in A$, and if g is Lipschitz on A with constant K_g, then f/g is Lipschitz on A with constant $K_f/k + M_f K_g/k^2$.

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ f, g : A \to \mathbb{R},\ K_f, K_g \geq 0,\\
&\forall x, y \in A :\ |f(x) - f(y)| \leq K_f|x - y| \text{ and } |g(x) - g(y)| \leq K_g|x - y|.\\
&\text{(a) } \forall c \in \mathbb{R},\ \forall x, y \in A :\ |cf(x) - cf(y)| \leq |c|K_f|x - y|.\\
&\text{(b) } \forall x, y \in A :\ |(f \pm g)(x) - (f \pm g)(y)| \leq (K_f + K_g)|x - y|.\\
&\text{(c) If } B \subseteq \mathbb{R},\ f(A) \subseteq B,\ h : B \to \mathbb{R},\ \forall u, v \in B :\ |h(u) - h(v)| \leq K_h|u - v|,\\
&\quad\ \text{then } \forall x, y \in A :\ |h(f(x)) - h(f(y))| \leq K_h K_f|x - y|.\\
&\text{(d) If } |f(x)| \leq M_f,\ |g(x)| \leq M_g\ \forall x \in A, \text{ then } \forall x, y \in A :\\
&\quad\ |f(x)g(x) - f(y)g(y)| \leq (M_g K_f + M_f K_g)|x - y|.\\
&\text{(e) If } |f(x)| \leq M_f,\ |g(x)| \geq k > 0\ \forall x \in A, \text{ then } \forall x, y \in A :\\
&\quad\ \left|\frac{f(x)}{g(x)} - \frac{f(y)}{g(y)}\right| \leq \left(\frac{K_f}{k} + \frac{M_f K_g}{k^2}\right)|x - y|.
\end{aligned}
$$

Remark (Predicate reading).

$$
\begin{aligned}
&\mathrm{Lipschitz}(f, A, K_f) \wedge \mathrm{Lipschitz}(g, A, K_g)\\
&\implies \Big[\forall c \in \mathbb{R} : \mathrm{Lipschitz}(cf, A, |c|K_f)\Big]\\
&\qquad \wedge \mathrm{Lipschitz}(f + g, A, K_f + K_g) \wedge \mathrm{Lipschitz}(f - g, A, K_f + K_g)\\
&\qquad \wedge \Big(\mathrm{Lipschitz}(h, B, K_h) \wedge f(A) \subseteq B \Rightarrow \mathrm{Lipschitz}(h \circ f, A, K_h K_f)\Big)\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \leq M_g) \Rightarrow \mathrm{Lipschitz}(fg, A, M_g K_f + M_f K_g)\Big]\\
&\qquad \wedge \Big[(|f| \leq M_f) \wedge (|g| \geq k > 0) \Rightarrow \mathrm{Lipschitz}\Big(\frac{f}{g}, A, \frac{K_f}{k} + \frac{M_f K_g}{k^2}\Big)\Big].
\end{aligned}
$$

Remark (Interpretation). Lipschitz functions are stable under the standard algebraic operations: scaling magnifies the constant proportionally, addition and subtraction add the errors, and composition multiplies constants because distortions accumulate multiplicatively. Products and quotients require uniform control of magnitudes; bounds on the factors prevent growth in the coefficients that accompany the derivatives of the product rule heuristic, while a positive lower bound on the denominator guarantees the quotient does not amplify oscillations uncontrollably. Together, these clauses justify building rich classes of Lipschitz maps from simpler ones without losing quantitative control.

Remark (Dependencies).
Definition 6.3.7, Definition 6.2.1

Definition (Bi-Lipschitz Map)

Definition 6.3.10 (Bi-Lipschitz Map). Let $A \subseteq \mathbb{R}$ and let $f : A \to \mathbb{R}$. The map f is bi-Lipschitz on A if and only if there exist constants $0 < \alpha \leq K$ such that for every $x, u \in A$,

$$\alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|.$$

Any such pair (α, K) is called a bi-Lipschitz constant pair for f on A.

Remark (Standard quantified statement).

$$(A \subseteq \mathbb{R}) \wedge (f : A \to \mathbb{R}) \implies \exists \alpha, K \in \mathbb{R} \Big(0 < \alpha \leq K \wedge \forall x, u \in A : \ \alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|\Big).$$

Remark (Definition predicate reading).

$$\mathrm{BiLipschitz}(f, A, \alpha, K) :\equiv (0 < \alpha \leq K) \wedge \forall x, u \in A \ (\alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u|).$$

Remark (Negated quantified statement).

$$\neg(A \subseteq \mathbb{R}) \vee \neg(f : A \to \mathbb{R}) \vee \forall \alpha, K \in \mathbb{R} \Big(0 < \alpha \leq K \Rightarrow \exists x, u \in A : \ |f(x)-f(u)| < \alpha|x-u| \vee |f($$

Remark (Negation predicate reading).

$$\neg\,\mathrm{BiLipschitz}(f, A, \alpha, K) \iff \neg(0 < \alpha \leq K) \vee \exists x, u \in A \ (|f(x)-f(u)| < \alpha|x-u| \vee |f(x)-f(u)| > K$$

Remark (Failure modes). The property fails either because f collapses some pair of points faster than the allowed lower constant α, or because it stretches some pair more than the upper constant K permits. Both distortions prevent a uniform two-sided control of distances.

Remark (Failure mode decomposition).

$$\exists x, u \in A : \quad \underbrace{|f(x)-f(u)| < \alpha|x-u|}_{\text{lower-distortion failure}} \quad \text{or} \quad \underbrace{|f(x)-f(u)| > K|x-u|}_{\text{upper-distortion failure}}.$$

Remark (Interpretation). A bi-Lipschitz map distorts every distance in A by factors lying between α and K, so it is simultaneously Lipschitz and has a Lipschitz inverse defined on its image. This two-sided control guarantees injectivity and preserves geometric features such as relative spacing up to uniform scaling, while failures arise precisely from excessive collapsing or stretching of some pair of points.

Remark (Dependencies).

- Function

Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz)

Theorem 6.3.11 (Inverse of a Bi-Lipschitz Map Is Lipschitz). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and suppose there exist constants $\alpha, K > 0$ such that $\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|$ for all $x, y \in A$. Then f is injective, the inverse mapping $f^{-1} : f(A) \to A$ is well defined, and*

$$\frac{1}{K}|u-v| \le |f^{-1}(u) - f^{-1}(v)| \le \frac{1}{\alpha}|u-v| \quad \textit{for all } u, v \in f(A).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } A \subseteq \mathbb{R},\ \ f : A \to \mathbb{R},\ \ \alpha, K > 0.\\
&\Big[\forall x, y \in A\ \big(\alpha|x-y| \le |f(x)-f(y)| \le K|x-y|\big)\Big]\\
&\implies \Big[\forall x, y \in A\ (f(x) = f(y) \Rightarrow x = y)\\
&\qquad \wedge \exists g : f(A) \to A\ \big(\forall u \in f(A)\ (f(g(u)) = u)\big)\\
&\qquad \wedge \forall u, v \in f(A)\ \Big(\frac{1}{K}|u-v| \le |g(u)-g(v)| \le \frac{1}{\alpha}|u-v|\Big)\Big].
\end{aligned}$$

Remark (Predicate reading).

$$\mathrm{BiLipschitz}(f, A, \alpha, K) \Rightarrow \big(\mathrm{Injective}(f) \wedge \mathrm{BiLipschitz}(f^{-1}, f(A), 1/K, 1/\alpha)\big).$$

Remark (Interpretation). A bi-Lipschitz bound simultaneously controls the expansion and contraction of f, so no two points of A can share the same image and the inverse is automatically a well-defined function on $f(A)$. The inequalities for f^{-1} follow by applying the original bi-Lipschitz bounds to pairs of preimages, showing that reversing the mapping preserves the same type of two-sided Lipschitz control with the reciprocals of the original constants. Failure would require collapsing two distinct points or losing the two-sided bounds, either of which contradicts the defining hypothesis.

Remark (Dependencies).
Definition 6.3.7; Definition 6.3.10.

6.4 Monotone Functions

Toolkit — Monotone Functions

- Every monotone function has LeftLimit and RightLimit at every interior point.
- $\mathrm{ContinuousAt}(f, c) \iff f(c^-) = f(c) = f(c^+) \iff j_f(c) = 0$.
- All discontinuities of a monotone function are of first kind — jump only, no oscillation.
- At most countably many discontinuities: disjoint jump intervals inject into $\mathbb{Q}$.
- Strictly monotone and continuous on $I \Rightarrow f^{-1}$ exists, is strictly monotone, and is continuous.
- $\mathrm{ContinuousOn}(f, [a, b]) \wedge \mathrm{Injective}(f) \iff f$ strictly monotone.

Toolkit — Limits Superior and Inferior of Functions

- $\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup\{f(x) : 0 < |x - x_0| < \delta,\ x \in E\}$.
- Both always exist in $[-\infty, +\infty]$.
- $\limsup \geq \liminf$ always.
- $\mathrm{FunctionLimit}(f, x_0, L)$ exists $\iff \limsup = \liminf = L \in \mathbb{R}$.
- Direct analogue of sequence $\limsup/\liminf$: replace tail $\{a_n : n \geq N\}$ with punctured neighbourhood $\{f(x) : \mathrm{InputTolerance}(x, x_0, \delta)\}$.

Monotone Functions and Continuity

Theorem (One-Sided Limits of Monotone Functions)

Theorem 6.4.1 (One-Sided Limits of Monotone Functions). *Let $I \subseteq \mathbb{R}$ be an interval, let c be interior to I, and let $f : I \to \mathbb{R}$ be increasing. Define*

$$L_- := \sup\{f(x) : x \in I,\ x < c\}, \qquad L_+ := \inf\{f(x) : x \in I,\ c < x\}.$$

Then both one-sided limits exist and satisfy

$$\lim_{x \to c^-} f(x) = L_-, \qquad \lim_{x \to c^+} f(x) = L_+.$$

Go to proof.

Remark (Standard quantified statement).

$$
\begin{aligned}
&I \subseteq \mathbb{R} \land \big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \land f : I \to \mathbb{R} \\
&\land \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \land c \in I \land \exists a, b \in I\ (a < c < b) \\
&\land L_- = \sup\{f(x) : x \in I,\ x < c\} \land L_+ = \inf\{f(x) : x \in I,\ c < x\} \\
&\Longrightarrow \\
&\forall \varepsilon > 0\ \exists \delta_- > 0\ \forall x \in I\ \big((0 < c - x < \delta_- \land x < c) \Rightarrow |f(x) - L_-| < \varepsilon\big) \\
&\land \forall \varepsilon > 0\ \exists \delta_+ > 0\ \forall x \in I\ \big((0 < x - c < \delta_+ \land c < x) \Rightarrow |f(x) - L_+| < \varepsilon\big).
\end{aligned}
$$

Remark (Predicate reading).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ increasing, and c interior to I.

$$
\mathrm{LeftLimit}\Big(f, c, \sup\{f(x) : x \in I,\ x < c\}\Big) \land \mathrm{RightLimit}\Big(f, c, \inf\{f(x) : x \in I,\ x > c\}\Big).
$$

Remark (Interpretation). For an increasing function, the values just to the left of c form a monotone increasing set bounded above by any right neighborhood, so their supremum matches the left-hand limit. Symmetrically, the values just to the right form a monotone increasing tail bounded below, making the infimum coincide with the right-hand limit. The theorem shows that monotonicity alone guarantees both one-sided limits exist and equals these extremal values. Failure occurs when c is not an interior point or f is not monotone, because the comparison sets may be empty or oscillatory, preventing the suprema or infima from governing the boundary behavior.

Remark (Dependencies).

- Completeness of $\mathbb{R}$
- Monotone Function
- Supremum
- Infimum

Corollary (Continuity Criterion for Monotone Functions)

Corollary 6.4.2 (Continuity Criterion for Monotone Functions). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone increasing, and let c be an interior point of I. Then f is continuous at c if and only if its one-sided limits satisfy $f(c^-) = f(c) = f(c^+)$, where $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$.* *Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$
(f \text{ is continuous at } c) \iff \left(\lim_{x\to c^-} f(x) = f(c) = \lim_{x\to c^+} f(x)\right).
$$

Remark (Predicate reading).

$$
\mathrm{ContinuousAt}(f, c) \iff \big(\mathrm{LeftLimit}(f, c, f(c)) \land \mathrm{RightLimit}(f, c, f(c))\big).
$$

Remark (Negated quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone increasing, and c an interior point of I.

$$\neg\big(f \text{ is continuous at } c\big) \iff \left(\lim_{x \to c^-} f(x) < f(c)\right) \vee \left(f(c) < \lim_{x \to c^+} f(x)\right).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f,c) \iff \big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big) \vee \big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) <\right.$$

Remark (Failure modes). Continuity can fail only through a jump: either the graph arrives at c from the left strictly below $f(c)$, or it departs to the right strictly above $f(c)$. Both jumps reflect the inability of the one-sided limits to match the function value at the interior point.

Remark (Failure mode decomposition).

$$\neg\,\mathrm{ContinuousAt}(f,c) = \underbrace{\big(\exists L^- \left[\mathrm{LeftLimit}(f,c,L^-) \wedge L^- < f(c)\right]\big)}_{\texttt{left jump at } c} \vee \underbrace{\big(\exists L^+ \left[\mathrm{RightLimit}(f,c,L^+) \wedge f(c) <\right.}_{\texttt{right jump at } c}$$

Remark (Interpretation). A monotone graph possesses finite one-sided limits at every interior point. Continuity at c demands that these approach values exactly meet the function value, so the graph neither overshoots nor undershoots when passing through c. Any failure manifests as a jump discontinuity: the graph either leaps upward at c or remains below the right-hand approach. This criterion is indispensable when classifying discontinuities of monotone functions, since it converts continuity into a simple equality check between the function value and the adjacent plateau levels.

Remark (Dependencies).
Definition 6.1.2; Theorem 6.4.1.

Definition (Jump of a Function)

Definition 6.4.3 (Jump of a Function). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be increasing, and let c be an interior point of I. If the one-sided limits $f(c^-)$ and $f(c^+)$ exist, the jump of f at c is

$$j_f(c) := f(c^+) - f(c^-).$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R}\\
&\Big[\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big) \wedge \exists a, b \in I\ (a < c < b)\\
&\qquad \wedge \big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \wedge \exists L_-, L_+ \in \mathbb{R}\ \big(\lim_{x \to c^-} f(x) = L_- \wedge \lim_{x \to c^+} f(x) = L_+\big)\Big]\\
&\qquad \Rightarrow \big(j = j_f(c) \iff j = f(c^+) - f(c^-)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{JumpOfFunction}(f, I, c, j) := \mathrm{Interval}(I) \wedge \mathrm{MonotoneIncreasing}(f, I) \wedge \mathrm{InteriorPoint}(c, I) \wedge \exists L_-, L_+ \in \mathbb{R}$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\forall I \subseteq \mathbb{R}\ \forall f : I \to \mathbb{R}\ \forall c \in I\ \forall j \in \mathbb{R} :\\
&\neg\big(j = j_f(c)\big) \iff \neg\big(\forall x, y \in I\ \forall z \in \mathbb{R}\ (x < z < y \Rightarrow z \in I)\big)\\
&\qquad \vee \neg\big(\forall x, y \in I\ (x < y \Rightarrow f(x) \le f(y))\big) \vee \neg\big(\exists a, b \in I\ (a < c < b)\big)\\
&\qquad \vee \forall L_-, L_+ \in \mathbb{R}\ \big(\lim_{x \to c^-} f(x) \neq L_- \vee \lim_{x \to c^+} f(x) \neq L_+ \vee j \neq L_+ - L_-\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{JumpOfFunction}(f, I, c, j) \iff \neg\,\mathrm{Interval}(I) \vee \neg\,\mathrm{MonotoneIncreasing}(f, I) \vee \neg\,\mathrm{InteriorPoint}(c, I) \vee \forall L_-$$

Remark (Failure modes). A proposed jump value fails when a required one-sided limit does not exist, or when both one-sided limits exist but the proposed value is not their difference.

Remark (Failure mode decomposition).

$$\underbrace{\neg\big(f(c^-)\ \texttt{exists}\big)}_{\texttt{missing left limit}} \vee \underbrace{\neg\big(f(c^+)\ \texttt{exists}\big)}_{\texttt{missing right limit}} \vee \underbrace{j \neq f(c^+) - f(c^-)}_{\texttt{incorrect jump magnitude}}.$$

Remark (Interpretation). For a monotone function the one-sided limits exist at every interior point, and their difference measures the vertical gap in the graph at that point. A zero jump means the two limiting heights agree; a positive jump records a discontinuity of first kind.

Remark (Dependencies).
Theorem 6.4.1.

Proposition (Discontinuities of a Monotone Function Are of First Kind)

Proposition 6.4.4 (Discontinuities of a Monotone Function Are of First Kind). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x \uparrow a} f(x)$ and $\lim_{x \downarrow a} f(x)$ exist in $\mathbb{R}$.* *Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$ monotone, and $a \in I$ interior to I.

$$\big[f \text{ is discontinuous at } a\big] \implies \exists L^-, L^+ \in \mathbb{R} : \begin{cases} \lim_{x \uparrow a} f(x) = L^-, \\ \lim_{x \downarrow a} f(x) = L^+. \end{cases}$$

Remark (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \wedge \mathrm{DiscontinuousAt}(f, a) \implies \exists L^-, L^+ \in \mathbb{R}\ \big(\mathrm{LeftLimit}(f, a, L^-) \wedge \mathrm{RightLimit}(f, a$$

Remark (Interpretation). A monotone graph can jump only by settling on distinct finite one-sided limits; it cannot oscillate wildly or blow up to infinity at a point of discontinuity. Thus every discontinuity of a monotone function is a jump discontinuity of the first kind, which explains why such functions are easy to integrate and why their exceptional sets are well behaved. Geometrically, the graph approaches a finite height from the left and another finite height from the right before making the jump.

Remark (Dependencies).
Definition 6.1.5

Corollary (Jump Intervals Are Disjoint)

Corollary 6.4.5 (Jump Intervals Are Disjoint). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be nondecreasing. If $a, b \in I$ are distinct points of discontinuity of f, then the open intervals $(f(a^-), f(a^+))$ and $(f(b^-), f(b^+))$ are disjoint.* Go to proof.

Remark (Standard quantified statement).

$$\forall a, b \in I \ \Big[a \neq b \wedge \big(\exists \varepsilon_a > 0\ \forall \delta > 0\ \exists x \in I\ (|x - a| < \delta \wedge |f(x) - f(a)| \geq \varepsilon_a)\big) \wedge \big(\exists \varepsilon_b > 0\ \forall \delta > 0\ \exists y \in I\ (|y -$$

Remark (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \wedge \begin{cases} a, b \in I, \\ a \neq b, \\ \mathrm{DiscontinuousAt}(f, a), \\ \mathrm{DiscontinuousAt}(f, b) \end{cases} \implies (f(a^-), f(a^+)) \cap (f(b^-), f(b^+)) = \varnothing.$$

Remark (Interpretation). The corollary states that a nondecreasing function assigns disjoint open value intervals to distinct jump discontinuities. Because the function never reverses direction, the lower one-sided limit at the later point cannot drop below the upper one-sided limit at the earlier point, so the corresponding ranges cannot overlap. This distinction of jump intervals ensures that the cumulative vertical jumps of a monotone function stay separated, preventing the graph from revisiting the same band of values via different discontinuities.

Remark (Dependencies).
Definition 6.4.3; Proposition 6.4.4

Theorem (Discontinuities of a Monotone Function Are Countable)

Theorem 6.4.6 (Discontinuities of a Monotone Function Are Countable). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be monotone. Then the set $D_f := \{c \in I : f \text{ is discontinuous at } c\}$ is at most countable.* Go to proof.

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be monotone.
$\#\{c \in I : f$ fails to be continuous at $c\} \leq \aleph_0$.

Remark (Predicate reading).

$$\mathrm{MonotoneFunction}(f, I) \implies \#\{c \in I : \mathrm{DiscontinuousAt}(f, c)\} \leq \aleph_0.$$

Remark (Interpretation). Every discontinuity of a monotone function is a jump discontinuity with a nondegenerate vertical gap $(f(c^-), f(c^+))$. Distinct discontinuities yield pairwise disjoint gaps, and each such interval contains a rational number. Associating to every discontinuity the rational number inside its gap produces an injection of the discontinuity set into $\mathbb{Q}$, forcing the set to be countable. The only way the conclusion could fail would be the existence of uncountably many disjoint jump intervals, which is impossible because the rationals are countable.

Remark (Dependencies).
Theorem 6.4.1 and Corollary 6.4.5.

Corollary (Monotone Function Is Continuous on a Dense Set)

Corollary 6.4.7 (Monotone Function Is Continuous on a Dense Set). *Let $f : [a, b] \to \mathbb{R}$ be monotone. Then the set $\{x \in [a, b] : f$ is continuous at $x\}$ is dense in $[a, b]$.* Go to proof.

Remark (Standard quantified statement).

Let $f : [a, b] \to \mathbb{R}$ be monotone.
$\forall u, v \in [a, b]$ $(u < v \Rightarrow \exists c \in (u, v)$ such that f is continuous at $c)$.

Remark (Predicate reading).

$$\mathrm{MonotoneFunction}(f, [a, b]) \implies \mathrm{DenseSubset}\big(\{x \in [a, b] : \mathrm{ContinuousAt}(f, x)\}, [a, b]\big).$$

Remark (Negated quantified statement).

$$\exists u, v \in [a, b] \left(u < v \wedge \forall c \in (u, v)\ \exists \varepsilon_c > 0\ \forall \delta > 0\ \exists x \in [a, b]\ \big(|x-c| < \delta \wedge |f(x)-f(c)| \geq \varepsilon_c\big)\right).$$

Remark (Negation predicate reading).

$$\exists u, v \in [a, b] \left(u < v \wedge \forall c \in (u, v)\ \neg \mathrm{ContinuousAt}(f, c)\right).$$

Remark (Interpretation). A monotone function on a closed interval can only jump at most countably many times, so the points of discontinuity leave room inside every subinterval for a continuity point. Thus, although the function may fail to be continuous at isolated or countably many locations, continuity still occurs throughout any interval one inspects. This property ensures that monotone functions retain many features of continuous functions when arguments depend only on dense sets of continuity points.

Remark (Dependencies).
Discontinuities of a Monotone Function Are Countable

Proposition (Continuous Injective Is Strictly Monotone)

Proposition 6.4.8 (Continuous Injective Is Strictly Monotone). *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. Then f is injective if and only if f is strictly monotone on $[a,b]$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } f : [a,b] \to \mathbb{R} \text{ be continuous on } [a,b].$$
$$\big[\forall x,y \in [a,b]\ (f(x) = f(y) \Rightarrow x = y)\big] \iff \big[\forall x,y \in [a,b]\ (x < y \Rightarrow f(x) < f(y))\big]$$
$$\vee \big[\forall x,y \in [a,b]\ (x < y \Rightarrow f(y) < f(x))\big].$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f,[a,b]) \Longrightarrow \big(\mathrm{Injective}(f) \iff \mathrm{StrictlyMonotone}(f,[a,b])\big).$$

Remark (Interpretation). A continuous function on a closed interval cannot reverse direction without repeating a value, so injectivity enforces a single increasing or decreasing trend across the entire domain. Conversely any strictly monotone map is automatically injective. The result emphasizes that topological constraints supplied by continuity interact with order-theoretic injectivity to force global monotonicity; failure occurs precisely when the graph turns back on itself, producing repeated values. Geometrically the continuous injective graph must cross each horizontal line at most once, which is only possible when the graph rises everywhere or falls everywhere.

Remark (Dependencies).
Bolzano's Intermediate Value Theorem.

Theorem (Continuous Inverse Theorem)

Theorem 6.4.9 (Continuous Inverse Theorem). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$.* *Go to proof.*

Remark (Standard quantified statement).

Let $I \subseteq \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$.

$$\Big[(\forall x,y \in I)(x < y \Rightarrow f(x) < f(y)) \wedge \forall c \in I\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I\big(|x - c| < \delta \Rightarrow |f(x) - f(c)| < \varepsilon\big)\Big]$$
$$\Longrightarrow \Big[\exists g : f(I) \to I \text{ such that } (\forall x \in I)\, g(f(x)) = x \wedge (\forall y \in f(I))\, f(g(y)) = y$$
$$\wedge (\forall u,v \in f(I))(u < v \Rightarrow g(u) < g(v)) \wedge \forall y \in f(I)\ \forall \varepsilon > 0\ \exists \eta > 0\ \forall u \in f(I)\big(|u - y| < \eta \Rightarrow |g(u)$$

Remark (Interpretation). A continuous strictly monotone function on an interval never reverses the order of points, so it is automatically bijective onto its image. This theorem records that the inverse map retains the same order orientation and inherits continuity, reflecting the fact that no sudden jumps can occur when the original graph crosses each horizontal line exactly once. Failure would require either loss of monotonicity or a discontinuity, either of which would create a flat spot or a jump preventing the existence of a continuous inverse. Geometrically, the graph of f^{-1} is the reflection of the graph of f across the diagonal line $y = x$, so the smoothness and strict increase or decrease are visibly preserved by symmetry.

Remark (Dependencies).

- Bolzano's Intermediate Value Theorem
- Strictly Increasing Function
- Strictly Decreasing Function
- Continuity at a Point

Limits Superior and Inferior of Functions

Definition (Limits Superior and Inferior of a Function)

Definition 6.4.10 (Limits Superior and Inferior of a Function). Let $f : E \to \mathbb{R}$ and let x_0 be an accumulation point of E. For each $\delta > 0$ set

$$U(\delta) = \{\, f(x) : x \in E,\ \ 0 < |x - x_0| < \delta \,\}.$$

The extended real numbers $\limsup_{x \to x_0} f(x)$ and $\liminf_{x \to x_0} f(x)$ are defined by

$$\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup U(\delta), \qquad \liminf_{x \to x_0} f(x) = \sup_{\delta > 0} \inf U(\delta),$$

where each inner supremum or infimum is taken in $\overline{\mathbb{R}}$ and the outer infimum or supremum ranges over all $\delta > 0$.

Remark (Standard quantified statement).

Let $f : E \to \mathbb{R}$, x_0 an accumulation point of E, and $U(\delta) = \{f(x) : x \in E,\ \ 0 < |x - x_0| < \delta\}$.

$$\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup U(\delta),$$
$$\liminf_{x \to x_0} f(x) = \sup_{\delta > 0} \inf U(\delta).$$

Remark (Definition predicate reading).

$$\text{FunctionLimsup}(L, f, E, x_0) :\equiv L = \inf_{\delta > 0} \sup U(\delta),$$
$$\text{FunctionLiminf}(l, f, E, x_0) :\equiv l = \sup_{\delta > 0} \inf U(\delta),$$

where $U(\delta) = \{f(x) : x \in E,\ \ 0 < |x - x_0| < \delta\}$.

Remark (Negated quantified statement).

Let $f : E \to \mathbb{R}$, x_0 an accumulation point of E, and $U(\delta) = \{f(x) : x \in E,\ \ 0 < |x - x_0| < \delta\}$.

$$\neg\Big[\limsup_{x \to x_0} f(x) = \inf_{\delta > 0} \sup U(\delta) \ \ \wedge \ \ \liminf_{x \to x_0} f(x) = \sup_{\delta > 0} \inf U(\delta)\Big]$$
$$\iff \limsup_{x \to x_0} f(x) \neq \inf_{\delta > 0} \sup U(\delta) \vee \liminf_{x \to x_0} f(x) \neq \sup_{\delta > 0} \inf U(\delta).$$

Remark (Negation predicate reading).

$$\neg\big(\mathrm{FunctionLimsup}(L,f,E,x_0)\wedge \mathrm{FunctionLiminf}(l,f,E,x_0)\big)$$
$$\iff L\neq \inf_{\delta>0}\sup U(\delta)\vee l\neq \sup_{\delta>0}\inf U(\delta),$$

where $U(\delta)=\{f(x):x\in E,\ \ 0<|x-x_0|<\delta\}$.

Remark (Failure modes). The definition can fail only in structurally distinct ways: punctured neighborhoods might be empty despite the accumulation-point hypothesis, or the upper envelopes may blow up while a finite limsup is asserted, or the lower envelopes may dive to negative infinity while a finite liminf is asserted.

Remark (Failure mode decomposition).

$$\underbrace{\exists\delta>0:\ \ U(\delta)=\varnothing}_{\texttt{missing punctured samples}}\ \vee\ \underbrace{\Big(\inf_{\delta>0}\sup U(\delta)=+\infty\wedge L<+\infty\Big)}_{\texttt{upper envelope blow-up}}\ \vee\ \underbrace{\Big(\sup_{\delta>0}\inf U(\delta)=-\infty\wedge l>-\infty\Big)}_{\texttt{lower envelope collapse}}.$$

Remark (Interpretation). The limit superior records the smallest height that eventually dominates all nearby function values, while the limit inferior records the largest depth that eventually underestimates all nearby values. Taking suprema within punctured neighborhoods captures every oscillation that can be witnessed arbitrarily close to x_0, and the outer infimum or supremum squeezes these oscillations to the sharpest possible bounds. Failure occurs only if the neighborhood sampling breaks down or if the oscillations race off to infinite extremal heights, signaling that no finite envelope can control the function near the accumulation point.

Remark (Dependencies).

- Function
- Supremum
- Infimum
- Epsilon Neighbourhood

Proposition

Proposition 6.4.11 ($\limsup\geq\liminf$). *Let $A\subseteq\mathbb{R}$, let x_0 be a limit point of A, and let $f:A\to\mathbb{R}$. Then*

$$\limsup_{x\to x_0}f(x)\geq\liminf_{x\to x_0}f(x).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall A\subseteq\mathbb{R}\ \forall f:A\to\mathbb{R}\ \forall x_0\in\mathbb{R}:$$
$$\Big(\forall\eta>0\ \exists x\in A\setminus\{x_0\}\ (|x-x_0|<\eta)\Big)\implies\limsup_{x\to x_0}f(x)\geq\liminf_{x\to x_0}f(x).$$

Remark (Interpretation). The limsup records the largest cluster value that f attains near x_0, while the liminf records the smallest cluster value. Because every subsequential limit contributing to the liminf also contributes to the limsup, the supremum of all cluster values cannot lie below their infimum. The only way equality holds is when all cluster values coincide, i.e. when the ordinary limit exists.

Remark (Dependencies).
Definition (Limits Superior and Inferior of a Function).

Proposition

Proposition 6.4.12 (Limit Exists iff $\limsup = \liminf$). *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ (\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in A \setminus \{x_0\}\ (0 < |x - x_0| < \delta \Rightarrow |f(x) - L| < \varepsilon))\\ \iff \quad \left(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\right).\end{gathered}$$

Remark (Predicate reading).

$$\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) = L = \liminf_{x\to x_0} f(x)\big).$$

Remark (Negated quantified statement).

$$\begin{gathered}\text{Let } A \subseteq \mathbb{R},\ f\colon A \to \mathbb{R},\ x_0 \text{ a limit point of } A,\ L \in \mathbb{R}.\\ \Big[\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)\Big]\\ \iff \quad \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).\end{gathered}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{FunctionLimit}(f, A, x_0, L) \iff \big(\limsup_{x\to x_0} f(x) \neq L \vee \liminf_{x\to x_0} f(x) \neq L\big).$$

Remark (Failure modes). The criterion fails only when at least one side of the equivalence breaks: either the function fails to approach L in the epsilon-delta sense, or one of the two extremal envelopes near x_0 does not equal L.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0\ \forall \delta > 0\ \exists x \in A \setminus \{x_0\}\ \big(0 < |x - x_0| < \delta \wedge |f(x) - L| \geq \varepsilon_0\big)}_{\text{limit condition fails}} \vee \underbrace{\limsup_{x\to x_0} f(x) \neq L}_{\text{upper envelope mismatch}} \vee \underbrace{\liminf_{x\to x_0}}_{\text{lower env}}$$

Remark (Interpretation). The two-sided limit of a real-valued function at a finite limit point exists precisely when the largest subsequential accumulation value and the smallest subsequential accumulation value coincide, in which case they both equal the limit. If the limsup exceeds the liminf, as happens for $f(x) = \sin(1/x)$ near $x_0 = 0$, the function oscillates between distinct cluster heights and therefore has no finite limit. The theorem isolates the limit as the unique height compatible with every oscillatory envelope and shows that agreeing envelopes are the exact signature of convergence in this local setting.

Remark (Dependencies).
Definition 6.4.10

6.5 Limits at Infinity

Toolkit — Limits at Infinity

- $+\infty$ is adherent to X iff $\neg$ BoundedSet(X) above ($\sup X = +\infty$).
- LimitAtInfinity$(f, +\infty, L)$: replace InputTolerance(x, c, δ) with $x > M$. Same OutputTolerance$(f(x), L, \varepsilon)$.
- All standard limit laws carry over unchanged.
- Sequential characterisation: LimitAtInfinity$(f, +\infty, L) \iff f(x_n) \to L$ for every $x_n \to +\infty$ in X.

Limits at Infinity

Definition (Infinite Adherent Points)

Definition 6.5.1 (Infinite Adherent Points). Let $X \subseteq \mathbb{R}$. The point $+\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x > M$. The point $-\infty$ is adherent to X if and only if for every $M \in \mathbb{R}$ there exists $x \in X$ such that $x < M$.

Remark (Standard quantified statement).

$$+\infty \text{ is adherent to } X \iff \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),$$
$$-\infty \text{ is adherent to } X \iff \forall M \in \mathbb{R}\ \exists x \in X\ (x < M).$$

Remark (Definition predicate reading).

$$\text{PlusInfAdherent}(X) := \forall M \in \mathbb{R}\ \exists x \in X\ (x > M),$$
$$\text{MinusInfAdherent}(X) := \forall M \in \mathbb{R}\ \exists x \in X\ (x < M).$$

Remark (Negated quantified statement).

$$\neg(+\infty \text{ is adherent to } X) \iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \le M),$$
$$\neg(-\infty \text{ is adherent to } X) \iff \exists M \in \mathbb{R}\ \forall x \in X\ (x \ge M).$$

Remark (Negation predicate reading).

$$\neg\,\text{PlusInfAdherent}(X) \iff \exists M \in \mathbb{R}\; \forall x \in X\; (x \le M),$$
$$\neg\,\text{MinusInfAdherent}(X) \iff \exists M \in \mathbb{R}\; \forall x \in X\; (x \ge M).$$

Remark (Failure modes). Infinite adherence fails at $+\infty$ exactly when X is bounded above, and fails at $-\infty$ exactly when X is bounded below.

Remark (Failure mode decomposition).

$$\underbrace{\exists M \in \mathbb{R}\; \forall x \in X\; (x \le M)}_{\texttt{finite upper bound}} \qquad \underbrace{\exists M \in \mathbb{R}\; \forall x \in X\; (x \ge M)}_{\texttt{finite lower bound}}.$$

Remark (Interpretation). Both finite and infinite adherence are instances of adherence in $\overline{\mathbb{R}}$: finite adherence uses balls; infinite adherence uses rays.

Definition (Limit at Infinity)

Definition 6.5.2 (Limit at Infinity). Let $X \subseteq \mathbb{R}$ be such that for every $M \in \mathbb{R}$ there exists $x \in X$ with $x > M$, let $f : X \to \mathbb{R}$, and let $L \in \mathbb{R}$. The limit of f at $+\infty$ equals L if and only if for every $\varepsilon > 0$ there exists $M \in \mathbb{R}$ such that for every $x \in X$, $x > M$ implies $|f(x) - L| < \varepsilon$.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\;\; \forall M \in \mathbb{R}\;\; \exists x \in X\;\; (x > M),\;\; f : X \to \mathbb{R},\;\; L \in \mathbb{R}.\\ &\lim_{x \to +\infty} f(x) = L \iff \forall \varepsilon > 0\;\; \exists M \in \mathbb{R}\;\; \forall x \in X\\ &\qquad (x > M \Rightarrow |f(x) - L| < \varepsilon).\end{aligned}$$

Remark (Definition predicate reading).

$$\text{LimitAtInfinity}(f, +\infty, L) :\!= \forall \varepsilon > 0\;\; \exists M \in \mathbb{R}\;\; \forall x \in X\;\; (x > M \Rightarrow |f(x) - L| < \varepsilon).$$

Remark (Negated quantified statement).

$$\begin{aligned}&\text{Let } X \subseteq \mathbb{R},\;\; \forall M \in \mathbb{R}\;\; \exists x \in X\;\; (x > M),\;\; f : X \to \mathbb{R},\;\; L \in \mathbb{R}.\\ &\neg\Big(\lim_{x \to +\infty} f(x) = L\Big) \iff \exists \varepsilon_0 > 0\;\; \forall M \in \mathbb{R}\;\; \exists x \in X\\ &\qquad (x > M \wedge |f(x) - L| \ge \varepsilon_0).\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\text{LimitAtInfinity}(f, +\infty, L) \iff \exists \varepsilon_0 > 0\;\; \forall M \in \mathbb{R}\;\; \exists x \in X\;\; (x > M \wedge |f(x) - L| \ge \varepsilon_0).$$

Remark (Failure modes). The definition fails only when there is a fixed tolerance ε_0 that the function cannot satisfy on any tail of X, meaning every attempt to choose M leaves some point $x > M$ with $|f(x) - L| \ge \varepsilon_0$.

Remark (Failure mode decomposition).

$$\underbrace{\exists \varepsilon_0 > 0}_{\texttt{fixed tolerance}}\;\; \underbrace{\forall M \in \mathbb{R}}_{\texttt{no tail succeeds}}\;\; \underbrace{\exists x \in X\;\; (x > M \wedge |f(x) - L| \ge \varepsilon_0)}_{\texttt{far-point violation}}.$$

Remark (Interpretation). The limit at $+\infty$ demands that eventually the function values stay within any prescribed accuracy of L. Intuitively, once x is large enough, the graph of f must lie in the horizontal band of height ε around L. Failure occurs when some positive tolerance is violated infinitely often by points escaping to $+\infty$. This definition allows us to treat unbounded domains using the same epsilon control that governs finite limits, capturing the long-range behavior of f with respect to the ideal point at infinity.

Remark (Dependencies).
Definition 6.5.1.

6.6 Approximation

Toolkit — Approximation

- Hierarchy: step functions $\subset$ piecewise linear $\subset$ polynomials. All three classes are dense in $(C([a,b]), \|\cdot\|_\infty)$.
- All three results use Heine–Cantor: UniformlyContinuous$(f, [a,b])$ provides the controlling δ.
- Bernstein polynomials: $B_n(x) = \sum_{k=0}^{n} f(k/n)\binom{n}{k}x^k(1-x)^{n-k}$.
- Step function approximation feeds directly into Riemann integrability of continuous functions.

Approximation of Continuous Functions

Theorem (Step Function Approximation)

Theorem 6.6.1 (Step Function Approximation). *Let $f : [a,b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a,b] \to \mathbb{R}$ such that*

$$\sup_{x\in[a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Go to proof.

Remark (Standard quantified statement).

Let $f : [a,b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$.
$\exists s_\varepsilon : [a,b] \to \mathbb{R}$ step function $\forall x \in [a,b]$ $(|f(x) - s_\varepsilon(x)| < \varepsilon)$.

Remark (Predicate reading).

$$\text{StepApproximation}(f, s_\varepsilon, [a,b], \varepsilon) :\equiv \text{StepFunction}(s_\varepsilon, [a,b]) \wedge \forall x \in [a,b]\ (|f(x)-s_\varepsilon(x)| < \varepsilon). \blacksquare$$

Remark (Negated quantified statement).

Let $f : [a,b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$.
$\forall s : [a,b] \to \mathbb{R}$ step function $\exists x \in [a,b]$ $(|f(x) - s(x)| \geq \varepsilon)$.

Remark (Negation predicate reading).

$$\neg\exists s_\varepsilon\ \mathrm{StepApproximation}(f, s_\varepsilon, [a,b], \varepsilon) \iff \forall s : [a,b] \to \mathbb{R}\ \Big(\mathrm{StepFunction}(s, [a,b]) \Rightarrow \exists x \in [a,b]\ (|f(x)$$

Remark (Failure modes). The statement fails precisely when a continuous f and tolerance ε are fixed but every step function leaves an error of at least ε at some point in $[a,b]$.

Remark (Failure mode decomposition).

$$\underbrace{\forall s : [a,b] \to \mathbb{R} \text{ step function}}_{\text{all approximants}}\ \underbrace{\exists x \in [a,b]\ (|f(x) - s(x)| \geq \varepsilon)}_{\text{witnessed large deviation}}.$$

Remark (Interpretation). Uniform continuity on $[a,b]$ guarantees that f does not oscillate rapidly at small scales, so subdividing the interval finely enough allows a step function to track f within any prescribed uniform error. The typical construction samples f at one point of each subinterval and keeps that value constant across the piece. Failure would mean that no matter how the interval is partitioned, some subinterval forces an error at least ε, contradicting uniform continuity. The result is a foundational approximation fact used to show that continuous functions on compact intervals are Riemann integrable and to connect continuous functions with simple combinatorial models.

Remark (Dependencies).
Heine–Cantor Theorem.

Definition (Piecewise Linear Function)

Definition 6.6.2 (Piecewise Linear Function). Let $a, b \in \mathbb{R}$ with $a < b$ and let $g : [a,b] \to \mathbb{R}$. The function g is piecewise linear if and only if there exist $m \in \mathbb{N}$ with $m \geq 1$ and points $a = x_0 < x_1 < \cdots < x_m = b$ such that $g|_{[x_{k-1}, x_k]}$ is affine for every $k \in \{1, \ldots, m\}$.

Remark (Standard quantified statement).

Let $a, b \in \mathbb{R}$ with $a < b$ and $g : [a,b] \to \mathbb{R}$.

g is piecewise linear

$$\iff \exists m \in \mathbb{N}\ \Big(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a,b]\ \big[a = x_0 < x_1 < \cdots < x_m = b \\ \wedge \forall k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is affine}\big]\Big).$$

Remark (Definition predicate reading).

$$\mathrm{PiecewiseLinear}(g, [a,b]) :\!= \exists m \in \mathbb{N}\ \Big(m \geq 1 \wedge \exists x_0, \ldots, x_m \in [a,b]\ \big[a = x_0 < \cdots < x_m = b \wedge \forall k \in \{1, \ldots,$$

Remark (Negated quantified statement).

Let $a, b \in \mathbb{R}$ with $a < b$ and $g : [a,b] \to \mathbb{R}$.

g is not piecewise linear

$$\iff \forall m \in \mathbb{N}\ \Big(m \geq 1 \Rightarrow \forall x_0, \ldots, x_m \in [a,b]\ \big[a = x_0 < \cdots < x_m = b \\ \Rightarrow \exists k \in \{1, \ldots, m\}\ g|_{[x_{k-1}, x_k]} \text{ is not affine}\big]\Big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{PiecewiseLinear}(g,[a,b]) \iff \forall m \in \mathbb{N}\ \Big(m \ge 1 \Rightarrow \forall x_0,\dots,x_m \in [a,b]\ \big[a = x_0 < \cdots < x_m = b \Rightarrow \exists k \in$$

Remark (Failure modes). A proposed piecewise-linear structure can fail because no finite ordered partition of $[a,b]$ is available, or because for every such partition at least one subinterval has a restriction of g that is not affine.

Remark (Failure mode decomposition).

$$\underbrace{\forall m \in \mathbb{N}\ (m \ge 1 \Rightarrow \forall x_0,\dots,x_m \in [a,b]\ (a = x_0 < \cdots < x_m = b \Rightarrow \cdots))}_{\texttt{no valid finite affine subdivision}} \vee \underbrace{\exists k \in \{1,\dots,m\}\ g|_{[x_{k-1},x_k]}\ \texttt{is}}_{\texttt{non-affine segment}}$$

Remark (Interpretation). A piecewise linear function on $[a,b]$ has a graph made from finitely many straight line segments. The definition permits different affine rules on different subintervals, but requires a finite ordered list of nodes that covers the whole interval.

Remark (Dependencies).

- Function
- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Theorem

Theorem 6.6.3 (Piecewise Linear Approximation). *Let $f\colon [a,b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a continuous piecewise linear function $\ell_\varepsilon\colon [a,b] \to \mathbb{R}$ such that $\sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}&\text{Let } f\colon [a,b] \to \mathbb{R},\ f \text{ continuous},\ \varepsilon > 0.\\ &\exists \text{ continuous piecewise linear } \ell_\varepsilon\colon [a,b] \to \mathbb{R} : \sup_{x\in[a,b]} |f(x) - \ell_\varepsilon(x)| < \varepsilon.\end{aligned}$$

Remark (Interpretation). Every continuous function on a compact interval can be uniformly approximated by a polygonal curve matching finitely many sampled values of the original graph. The construction refines the interval into a sufficiently fine partition guaranteed by compactness, then connects the sampled values with straight segments, producing a uniformly small error over the entire domain. Failure would require a uniform oscillation exceeding ε on some subinterval, which cannot occur once the partition is fine enough. Geometrically, the theorem states that continuous graphs admit arbitrarily close polyline shadows, providing a bridge from rough step approximations to smooth polynomial approximations.

Remark (Dependencies).

- Heine–Cantor Theorem
- Piecewise Linear Function
- Continuity at a Point

Theorem (Weierstrass Approximation Theorem)

Theorem 6.6.4 (Weierstrass Approximation Theorem). *Let $f : [a, b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$ there exists a polynomial $p_\varepsilon \in \mathbb{R}[x]$ such that $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{Let } a < b \text{ in } \mathbb{R}, \text{ and let } f : [a, b] \to \mathbb{R} \text{ be continuous.}$$
$$\forall \varepsilon > 0 \ \exists p \in \mathbb{R}[x] \ (\|f - p\|_{\infty,[a,b]} < \varepsilon).$$

Remark (Predicate reading).

$$\mathrm{ContinuousOn}(f, [a, b]) \implies \forall \varepsilon > 0 \ \exists p \in \mathbb{R}[x] \ \big(\|f - p\|_{\infty,[a,b]} < \varepsilon\big).$$

Remark (Interpretation). Every continuous function on a closed interval can be uniformly approximated by polynomials, so polynomials are dense in the space of continuous functions under the supremum norm. The result holds because convolutions with Bernstein polynomials preserve continuity and converge uniformly to the original function. Failure would mean there exists a uniform tolerance that no polynomial can match, contradicting the approximation scheme built from Bernstein polynomials. Geometrically, the theorem states that the graph of a continuous function can be matched arbitrarily well by the graph of a polynomial, so polynomial graphs form a flexible mesh that can shadow any continuous curve on a compact interval.

Remark (Dependencies).
Depends on thm:bernstein-approximation.

Definition (Bernstein Polynomial)

Definition 6.6.5 (Bernstein Polynomial). Let $n \in \mathbb{N}$ and let $f : [0, 1] \to \mathbb{R}$. A function $B_n : [0, 1] \to \mathbb{R}$ is called the nth Bernstein polynomial of f if and only if, for every $x \in [0, 1]$,

$$B_n(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1 - x)^{n-k}.$$

Remark (Standard quantified statement).

$$\text{Let } n \in \mathbb{N},\ f : [0, 1] \to \mathbb{R},\ B_n : [0, 1] \to \mathbb{R}.$$
$$B_n \text{ is the } n\text{th Bernstein polynomial of } f$$
$$\iff \forall x \in [0, 1] \ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1 - x)^{n-k}\Big).$$

Remark (Definition predicate reading).

$$\mathrm{BernsteinPolynomial}(B_n, f, n) :\equiv \Big(B_n : [0,1] \to \mathbb{R}\Big) \wedge \forall x \in [0,1] \ \Big(B_n(x) = \sum_{k=0}^{n} f(k/n) \binom{n}{k} x^k (1-x)^{n-k}\Big)$$

Remark (Interpretation). The nth Bernstein polynomial samples f at the uniformly spaced nodes k/n, weights those samples by the binomial probabilities of order n, and produces for each x an average that depends polynomially on x. This construction smooths f into a polynomial that agrees with f at the endpoints and approximates f uniformly as n grows, so describing it explicitly is essential for subsequent approximation theorems. Failure arises only when the candidate polynomial disagrees with the binomial averaging recipe at some point of the unit interval, reflecting an incorrect coefficient or evaluation in the polynomial expression.

Remark (Dependencies).

- Function
- Natural Numbers
- Multiplication on $\mathbb{R}$
- Addition on $\mathbb{R}$

Theorem (Bernstein Approximation Theorem)

Theorem 6.6.6 (Bernstein Approximation Theorem). *Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n denote the nth Bernstein polynomial of f. Then for every $\varepsilon > 0$ there exists $n_\varepsilon \in \mathbb{N}$ such that $\|f - B_n\|_\infty < \varepsilon$ for all $n \geq n_\varepsilon$.* *Go to proof.*

Remark (Standard quantified statement).

Let $f : [0,1] \to \mathbb{R}$ be continuous, and for each $n \in \mathbb{N}$ let B_n be the nth Bernstein polynom
$\forall \varepsilon > 0 \ \exists n_\varepsilon \in \mathbb{N} \ \forall n \geq n_\varepsilon : \ \|f - B_n\|_\infty < \varepsilon.$

Remark (Interpretation). The theorem asserts that the Bernstein polynomials of a continuous function on $[0,1]$ converge uniformly to the original function, so the entire graph of f can be approximated to any prescribed sup-norm accuracy by explicit degree-n polynomials once n is large enough. The proof works by showing that the Bernstein operators preserve positivity and linear growth while averaging values of f, which forces the uniform error to vanish as $n \to \infty$. This matters because it gives an elementary constructive route from continuous functions to polynomials, strengthening the Weierstrass approximation theorem with a concrete approximating sequence. The approximation can fail only when f is not continuous on $[0,1]$, in which case the Bernstein polynomials may smooth out jumps and no longer capture the original function in the sup norm. Geometrically, each B_n is a convex combination of the samples $f(k/n)$, so the polygons formed by these samples become finer and eventually hug the graph of f everywhere on the interval.

Remark (Dependencies).
Definition (Continuity at a Point), Definition (Bernstein Polynomial).

6.7 Gauge

Toolkit — Gauges and Tagged Partitions

- A **gauge** $\delta : I \to (0, \infty)$ is a variable-width tolerance function allowing the mesh to depend on position.
- A tagged partition $\dot{\mathcal{P}}$ is δ-fine if each subinterval I_i lies inside the $\delta(t_i)$-neighbourhood of its tag t_i: $\mathrm{Within}(x, t_i, \delta(t_i))$ for all $x \in [x_{i-1}, x_i]$.
- **Cousin's theorem**: every gauge on $[a, b]$ admits a δ-fine tagged partition.
- Gauge proofs of Boundedness, EVT, Location of Roots, and Heine–Cantor directly construct the required constants from a δ-fine partition.

Gauges and Tagged Partitions

Definition (Partition)

Definition 6.7.1 (Partition). Let $a < b$ and set $I = [a, b]$. A finite ordered set $P = \{x_0, \dots, x_n\}$ with $n \in \mathbb{N}$ and $n \geq 1$ is a partition of I if and only if $x_0 = a$, $x_n = b$, and $x_0 < x_1 < \cdots < x_n$. In this case the closed subintervals $[x_{i-1}, x_i]$ for $i = 1, \dots, n$ subdivide I.

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I.\\
&P \text{ is a partition of } I\\
&\iff \big(x_0 = a \wedge x_n = b \wedge \forall i \in \{1, \dots, n\}\ (x_{i-1} < x_i)\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{PartitionOfInterval}(P, I) :\equiv \exists n \in \mathbb{N} \Big(n \geq 1 \wedge P = \{x_0, \dots, x_n\} \subseteq I \wedge x_0 = \min I \wedge x_n = \max I \wedge \forall i \in \{1, .$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a < b,\ I = [a, b],\ n \in \mathbb{N},\ n \geq 1, \text{ and } P = \{x_0, \dots, x_n\} \subseteq I.\\
&P \text{ is not a partition of } I\\
&\iff \big((x_0 \neq a) \vee (x_n \neq b) \vee \exists i \in \{1, \dots, n\}\ (x_{i-1} \geq x_i)\big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\, \mathrm{PartitionOfInterval}(P, I) \iff \forall n \in \mathbb{N} \Big(n \geq 1 \Rightarrow \big(P \neq \{x_0, \dots, x_n\} \vee P \not\subseteq I \vee x_0 \neq \min I \vee x_n \neq \max I \vee$$

Remark (Failure modes). The definition fails precisely in three ways: the left endpoint is omitted, the right endpoint is omitted, or the sequence of nodes ceases to be strictly increasing, leading to repeated or out-of-order points.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\texttt{left endpoint missing}} \quad \vee \quad \underbrace{x_n \neq b}_{\texttt{right endpoint missing}} \quad \vee \ \underbrace{\exists i \in \{1, \dots, n\}\ (x_{i-1} \geq x_i)}_{\texttt{nodes not strictly increasing}}.$$

Remark (Interpretation). A partition records the finite list of nodes that slice a closed interval into adjacent subintervals. The strict ordering ensures that every interior point appears exactly once in this linear traversal, so the induced subintervals $[x_{i-1}, x_i]$ tile $[a, b]$ without overlap or gaps. Computations of Riemann sums, Darboux sums, and gauge integrals use partitions as the combinatorial scaffolding that supports sampling and measurement. Failure arises when the endpoints are not captured or when the nodes regress or stagnate, which would leave gaps or redundancies in the subdivision.

Remark (Dependencies).

- Closed Interval
- Natural Numbers
- Order on $\mathbb{R}$

Definition (Tagged Partition)

Definition 6.7.2 (Tagged Partition). Let $a, b \in \mathbb{R}$ with $a < b$. A list $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ is a tagged partition of $[a, b]$ if and only if the subintervals determine a partition $P = \{[x_{i-1}, x_i]\}_{i=1}^n$ of $[a, b]$ with $a = x_0 < x_1 < \cdots < x_n = b$, and each tag satisfies $t_i \in [x_{i-1}, x_i]$ for $i = 1, \dots, n$. Equivalently, a tagged partition is exactly a partition P of $[a, b]$ together with one tag chosen inside every subinterval of P.

Remark (Standard quantified statement).

$$\begin{aligned}
&\texttt{Let } a, b \in \mathbb{R} \texttt{ with } a < b, \texttt{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\
&\dot{\mathcal{P}} \texttt{ is a tagged partition of } [a, b] \\
&\iff \big(a = x_0 < \cdots < x_n = b \wedge \forall i \in \{1, \dots, n\}\ (t_i \in [x_{i-1}, x_i])\big).
\end{aligned}$$

Remark (Definition predicate reading).

$$\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a, b]) :\equiv \Big(\mathrm{Partition}(P, [a, b]) \wedge \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n \wedge \forall i\ (t_i \in [x_{i-1}, x_i])\Big),$$

where $\mathrm{Partition}(P, [a, b])$ abbreviates $a = x_0 < \cdots < x_n = b$ for $P = \{[x_{i-1}, x_i]\}_{i=1}^n$.

Remark (Negated quantified statement).

$$\begin{aligned}
&\texttt{Let } a, b \in \mathbb{R} \texttt{ with } a < b, \texttt{ and } \dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n. \\
&\dot{\mathcal{P}} \texttt{ is not a tagged partition of } [a, b] \\
&\iff \Big(x_0 \neq a \vee x_n \neq b \vee \exists j \in \{1, \dots, n\}\ (x_{j-1} \geq x_j) \vee \exists i \in \{1, \dots, n\}\ (t_i \notin [x_{i-1}, x_i])\Big).
\end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b]) \iff (x_0 \neq a)\vee(x_n \neq b)\vee\Big(\exists j\ (x_{j-1} \geq x_j)\Big)\vee\Big(\exists i\ (t_i \notin [x_{i-1},x_i])\Big).$$

Remark (Failure modes). A list fails to be a tagged partition either because the subintervals do not form a valid partition (the mesh is disordered or has gaps) or because at least one chosen tag lies outside its designated subinterval.

Remark (Failure mode decomposition).

$$\underbrace{x_0 \neq a}_{\texttt{left endpoint defect}} \vee \underbrace{x_n \neq b}_{\texttt{right endpoint defect}} \vee \underbrace{\exists j\ (x_{j-1} \geq x_j)}_{\texttt{ordering defect}} \vee \underbrace{\exists i\ (t_i \notin [x_{i-1},x_i])}_{\texttt{tag misplaced}}.$$

Remark (Interpretation). A tagged partition augments the usual subdivision of a closed interval by selecting a representative sampling point inside each component subinterval. These tags serve as evaluation points for Riemann sums and gauge integrals. The construction encapsulates both the geometric slicing of the interval and the analytic data needed for sample values. Failure occurs when the slices no longer tile the interval in order or when a sample point is chosen outside its slice, breaking the correspondence between tags and subintervals.

Remark (Dependencies).
Definition 6.7.1.

Definition (Gauge)

Definition 6.7.3 (Gauge). Let $a,b \in \mathbb{R}$ with $a < b$ and set $I = [a,b]$. A gauge on I is a function $\delta : I \to (0,\infty)$.

Remark (Standard quantified statement).

Let $a,b \in \mathbb{R}$ with $a < b$, put $I = [a,b]$, and let $\delta : I \to \mathbb{R}$.
δ is a gauge on $I \iff \forall x \in I\ (\delta(x) > 0)$.

Remark (Definition predicate reading).

$$\mathrm{Gauge}(\delta, I) \coloneqq \big(\delta : I \to \mathbb{R}\big) \wedge \forall x \in I\ (\delta(x) > 0).$$

Remark (Negated quantified statement).

Let $a,b \in \mathbb{R}$ with $a < b$, put $I = [a,b]$, and let $\delta : I \to \mathbb{R}$.
δ fails to be a gauge on $I \iff \exists x \in I\ (\delta(x) \leq 0)$.

Remark (Negation predicate reading).

$$\neg\,\mathrm{Gauge}(\delta, I) \iff \neg\big(\delta : I \to \mathbb{R}\big) \vee \exists x \in I\ (\delta(x) \leq 0).$$

Remark (Failure modes). A purported gauge fails either because it is not a function from I to $\mathbb{R}$, or because it takes on a nonpositive value somewhere in I.

Remark (Failure mode decomposition).

$$\underbrace{\neg(\delta : I \to \mathbb{R})}_{\text{wrong function type}} \quad \vee \quad \underbrace{\exists x \in I\ (\delta(x) \leq 0)}_{\text{nonpositive gauge value}}$$

Remark (Interpretation). A gauge assigns to each point of the closed interval a positive radius, encoding how tightly a tagged subinterval must cluster around that point in fine partition arguments. The sole obstruction is positivity: if any value is zero or negative, the construction cannot deliver legitimate local scales, so the gauge structure breaks down.

Remark (Dependencies).

- Function
- Closed Interval
- Order on $\mathbb{R}$

Definition (δ-Fine Partition)

Definition 6.7.4 (δ-Fine Partition). Let $[a, b]$ be a closed interval, let $\delta : [a, b] \to (0, \infty)$ be a gauge, and let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$. The tagged partition $\dot{\mathcal{P}}$ is δ-fine if and only if for every $i \in \{1, \ldots, n\}$ and every $x \in [x_{i-1}, x_i]$ the inequality $|x - t_i| < \delta(t_i)$ holds.

Remark (Standard quantified statement).

Let $[a, b]$ be a closed interval, $\delta : [a, b] \to (0, \infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ a tagged parti
$\dot{\mathcal{P}}$ is δ-fine $\iff \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i))$.

Remark (Definition predicate reading).

$$\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) := \forall i \in \{1, \ldots, n\}\ \forall x \in [x_{i-1}, x_i]\ (|x - t_i| < \delta(t_i)).$$

Remark (Negated quantified statement).

$$\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ \big(|x - t_i| \geq \delta(t_i)\big).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta) \iff \exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]\ (|x - t_i| \geq \delta(t_i)).$$

Remark (Failure modes). A tagged partition fails to be δ-fine precisely when some subinterval contains a point whose distance from its tag reaches or exceeds the available radius $\delta(t_i)$, so the corresponding subinterval is not contained in the prescribed open neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\exists i \in \{1, \ldots, n\}\ \exists x \in [x_{i-1}, x_i]}_{\text{choose offending subinterval and point}} \quad \underbrace{(|x - t_i| \geq \delta(t_i))}_{\text{distance not controlled by } \delta} \quad .$$

Remark (Interpretation). A δ-fine tagged partition refines the interval so that each subinterval sits entirely inside the open ball centered at its tag with radius dictated by the gauge. This ensures that when summing local contributions, every evaluation point remains within a tolerance chosen for that location, a key requirement for gauge-based integral constructions. Failure occurs when a subinterval is too wide for the tolerance prescribed at its tag, so some point of the subinterval lies on or outside the allowed neighborhood.

Remark (Dependencies).
Definition 6.7.3; Definition 6.7.2

Lemma

Lemma 6.7.5 (Every Point Is Covered by a Tag). *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a,b]$ that is δ-fine on $[a,b]$, and let $x \in [a,b]$. Then there exists $i \in \{1,\dots,n\}$ such that $|x - t_i| < \delta(t_i)$. Go to proof.*

Remark (Standard quantified statement).

Let $a < b$, $\delta : [a,b] \to (0,\infty)$, and $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be δ-fine on $[a,b]$.
$\forall x \in [a,b]\ \exists i \in \{1,\dots,n\}\ (|x - t_i| < \delta(t_i))$.

Remark (Interpretation). A δ-fine tagged partition matches each subinterval to a tag that lies inside the interval and records the gauge radius allowed at that tag. The lemma asserts that every point of the underlying interval falls within the gauge ball centered at one of the tags, so the gauge control at that tag can be applied to the point. This coverage property is the mechanism that transfers local control encoded by the gauge to arbitrary points of the interval.

Remark (Dependencies).
Definition 6.7.2, Definition 6.7.4.

Theorem (Cousin's Theorem)

Theorem 6.7.6 (Cousin's Theorem). *Every gauge $\delta : [a,b] \to (0,\infty)$ on a closed interval $[a,b]$ admits a δ-fine tagged partition of $[a,b]$. Go to proof.*

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } a,b \in \mathbb{R} \text{ with } a<b \text{ and let } \delta : [a,b] \to (0,\infty).\\
&\exists m \in \mathbb{N}\ \exists x_0,\dots,x_m\ \exists t_1,\dots,t_m\\
&\qquad \big(a = x_0 < x_1 < \cdots < x_m = b \wedge \forall j \in \{1,\dots,m\}\ (t_j \in [x_{j-1}, x_j])\\
&\qquad \wedge \forall j \in \{1,\dots,m\}\ [x_{j-1}, x_j] \subseteq (t_j - \delta(t_j), t_j + \delta(t_j))\big).
\end{aligned}$$

Remark (Predicate reading).

$$\forall \delta : [a,b] \to (0,\infty) \quad \exists \dot{\mathcal{P}} \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}}, [a,b]) \wedge \mathrm{DeltaFine}(\dot{\mathcal{P}}, \delta)\Big).$$

Remark (Negated quantified statement).

$$\begin{aligned}
&\text{Let } a,b\in\mathbb{R} \text{ with } a<b \text{ and let } \delta:[a,b]\to(0,\infty).\\
&\neg\Big(\text{there exists a } \delta\text{-fine tagged partition of } [a,b]\Big)\\
&\iff \forall m\in\mathbb{N}\ \forall x_0,\dots,x_m\ \forall t_1,\dots,t_m\\
&\qquad\Big[\big(a=x_0<x_1<\cdots<x_m=b\wedge\forall j\ (t_j\in[x_{j-1},x_j])\big)\\
&\qquad\quad\Rightarrow\exists j\in\{1,\dots,m\}\ [x_{j-1},x_j]\not\subseteq(t_j-\delta(t_j),t_j+\delta(t_j))\Big].
\end{aligned}$$

Remark (Negation predicate reading).

$$\exists\delta:[a,b]\to(0,\infty)\ \forall\dot{\mathcal{P}}\ \Big(\mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b])\Rightarrow\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}},\delta)\Big).$$

Remark (Failure modes). Failure would mean that a positive gauge δ defeats every tagged partition: no matter how the interval is subdivided and tagged, at least one subinterval is not contained in the gauge neighborhood around its tag.

Remark (Failure mode decomposition).

$$\underbrace{\forall\dot{\mathcal{P}}\ \mathrm{TaggedPartition}(\dot{\mathcal{P}},[a,b])}_{\text{all tagged partitions}}\quad\Rightarrow\quad\underbrace{\neg\,\mathrm{DeltaFine}(\dot{\mathcal{P}},\delta)}_{\text{some interval escapes its gauge ball}}.$$

Remark (Interpretation). The theorem is the gauge analogue of compactness for a closed interval. Although the allowed radius may vary from point to point, finitely many tagged subintervals can still be chosen so that each lies inside the neighborhood dictated by its own tag. The standard bisection proof uses the Nested Interval Property to rule out a gauge that defeats all finite tagged partitions.

Remark (Dependencies).
Definition 6.7.3, Definition 6.7.2, Definition 6.7.4.

Remark (Gauge proofs of the four classical theorems). Each proof follows the same pattern: define a gauge δ from the continuity hypothesis, invoke Cousin's theorem for a δ-fine partition, apply the Covering Lemma to reduce to finitely many tags, derive the global conclusion.

Boundedness. At each t, continuity gives $\delta(t)$ with $\mathrm{Within}(x,t,\delta(t))\Rightarrow|f(x)-f(t)|\le 1$. A δ-fine partition with tags $t_1,\dots,t_n$ gives $K=\max_i|f(t_i)|$; the Covering Lemma yields $|f(x)|\le 1+K$ for all x.

EVT. Let $M=\sup f(I)$; suppose $f(x)<M$ everywhere. At each t, take $\delta(t)$ with $\mathrm{Within}(x,t,\delta(t))\Rightarrow f(x)<\frac{1}{2}(M+f(t))$. A δ-fine partition produces $\widetilde{M}=\frac{1}{2}(M+\max_i f(t_i))<M$ bounding f everywhere -- contradiction.

Location of Roots. Assume $f(t)\neq 0$ for all t. At each t, take $\delta(t)$ preserving $\mathrm{sgn}(f(t))$. A δ-fine partition forces consistent sign from $f(a)<0$ to $f(b)>0$ -- contradiction.

Heine-Cantor. Given $\varepsilon>0$, at each t let $\delta(t)$ satisfy $\mathrm{Within}(x,t,2\delta(t))\Rightarrow|f(x)-f(t)|\le\varepsilon/2$. A δ-fine partition yields $\delta_\varepsilon=\min_i\delta(t_i)>0$. For $|x-u|<\delta_\varepsilon$: find t_i covering x; then $|u-t_i|\le|u-x|+|x-t_i|<2\delta(t_i)$; so $|f(x)-f(u)|\le\varepsilon$.

Tagged Partitions and Mesh

Definition (Mesh of a Partition)

Definition 6.7.7 (Mesh of a Partition). Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. A real number m is the *mesh* (or *norm*) of P if and only if

$$m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

The mesh of P is denoted by $\|P\|$ and is the unique m satisfying this equivalence.

Remark (Standard quantified statement).

Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. m is the mesh of $P \iff m = \max_{1 \le i \le n} (x_i - x_{i-1})$.

Remark (Definition predicate reading).

$$\operatorname{NormOfPartition}(P, m) :\!= m = \max_{1 \le i \le n} (x_i - x_{i-1}).$$

Remark (Negated quantified statement).

Let $a, b \in \mathbb{R}$ with $a < b$, and let $P = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. $\neg(m$ is the mesh of $P) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \vee \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big)$.

Remark (Negation predicate reading).

$$\neg \operatorname{NormOfPartition}(P, m) \iff \big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big) \vee \big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big).$$

Remark (Failure modes). Failure occurs either when m underestimates the largest subinterval of P or when m overshoots that largest subinterval.

Remark (Failure mode decomposition).

$$\neg \operatorname{NormOfPartition}(P, m) = \underbrace{\big(m < \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\texttt{Underestimate}} \vee \underbrace{\big(m > \max_{1 \le i \le n} (x_i - x_{i-1})\big)}_{\texttt{Overestimate}}.$$

Remark (Interpretation). The mesh of a finite partition records the length of the widest subinterval and thus measures the resolution of the partition. Refining a partition strictly decreases or preserves this value, so a small mesh signals uniformly short subintervals, a critical input for Riemann and gauge integral estimates. Failure arises when a proposed number does not coincide with the actual maximum subinterval length, either because some subinterval is longer than the proposal or because all subintervals are shorter.

Remark (Dependencies).
Definition 6.7.1.

Definition (Refinement)

Definition 6.7.8 (Refinement). Let $[a, b] \subseteq \mathbb{R}$ and let P and Q be partitions of $[a, b]$. Then Q is a refinement of P if and only if every point of P is also a point of Q; equivalently, $P \subseteq Q$.

Remark (Standard quantified statement).

$$\forall x\ (x \in P \Rightarrow x \in Q).$$

Remark (Definition predicate reading).

$$\text{Refinement}(Q, P) := P \subseteq Q.$$

Remark (Negated quantified statement).

$$\exists x\ (x \in P \wedge x \notin Q).$$

Remark (Negation predicate reading).

$$\neg\,\text{Refinement}(Q, P) \iff \exists x \in P\ (x \notin Q).$$

Remark (Failure modes). Failure occurs precisely when some point listed in P fails to appear in Q, meaning Q omits at least one of the partition points it was supposed to inherit from P.

Remark (Failure mode decomposition).

$$\underbrace{\exists x \in P\ (x \notin Q)}_{\text{point of } P \text{ omitted by } Q}.$$

Remark (Interpretation). Refinement formalizes the idea of subdividing an interval more finely without losing any previously chosen partition points. The condition $P \subseteq Q$ guarantees that every subinterval defined by P is still represented when one passes to Q, allowing comparisons of sums or integrals computed on different meshes. Failure means that Q discards at least one point of P, so information tied to that breakpoint would be lost when transitioning between partitions.

Remark (Dependencies).
Definition 6.7.1.

Definition (Common Refinement)

Definition 6.7.9 (Common Refinement). Let $a < b$ and let P and Q be partitions of $[a, b]$. Their common refinement is the partition R obtained by taking the union of their node sets, so $R = P \cup Q$ as sets of points in $[a, b]$.

Remark (Standard quantified statement).

Let $a < b$ in $\mathbb{R}$, and let P, Q be partitions of $[a, b]$.
$\forall R$ partition of $[a, b]$: R is the common refinement of P and $Q \iff R = P \cup Q$.

Remark (Definition predicate reading).

$$\mathrm{CommonRefinement}(R, P, Q) :\!= (R = P \cup Q).$$

Remark (Negated quantified statement).

$$\begin{aligned} &a < b \wedge P, Q \subseteq [a, b] \wedge R \subseteq [a, b] \\ &\implies \neg\big(\forall x \in \mathbb{R}\ (x \in R \iff x \in P \cup Q)\big) \\ &\iff \big(\exists x \in R\ (x \notin P \cup Q)\big) \vee \big(\exists x \in P \cup Q\ (x \notin R)\big). \end{aligned}$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{CommonRefinement}(R, P, Q) \iff R \neq P \cup Q.$$

Remark (Failure modes). The claim that a partition R is the common refinement of P and Q fails exactly when R either contains a node not present in $P \cup Q$ or omits a node that $P \cup Q$ contributes. These are the only two structural ways in which R can differ from $P \cup Q$.

Remark (Failure mode decomposition).

$$R \neq P \cup Q \iff \underbrace{\exists x \in R \setminus (P \cup Q)}_{\texttt{extra node}} \vee \underbrace{\exists x \in (P \cup Q) \setminus R}_{\texttt{missing node}}.$$

Remark (Interpretation). Forming the common refinement of two partitions simply collects every partition point that either partition already uses, producing the unique partition whose nodes are exactly those existing ones. No new nodes are created beyond those already in P or Q, and none are discarded; the construction merely merges the lists so that later comparisons, such as evaluating tagged sums on both partitions simultaneously, can be performed without further adjustment.

Remark (Dependencies).
Definition 6.7.1

Proofs

Remark (Return). Return to Theorem

Lemma (Equivalent Formulations of Discontinuity at a Point). *Let $E \subseteq \mathbb{R}$, $f : E \to \mathbb{R}$, $x_0 \in E$. The ε-δ, sequential, and neighbourhood formulations of discontinuity at x_0 are mutually equivalent.*

Professional Standard Proof.
Let $E \subseteq \mathbb{R}$, $f : E \to \mathbb{R}$, and $x_0 \in E$. The three formulations are:

(i) ε-δ discontinuity: $\exists \varepsilon_0 > 0\, \forall \delta > 0\, \exists x \in E : |x - x_0| < \delta \wedge |f(x) - f(x_0)| \geq \varepsilon_0$

(ii) Sequential discontinuity: $\exists (x_n) \subseteq E : x_n \to x_0 \wedge f(x_n) \not\to f(x_0)$

(iii) Neighbourhood discontinuity: $\exists V$ neighbourhood of $f(x_0)$ $\forall U$ neighbourhood of x_0 $\exists x \in U \cap E : f(x) \notin V$

For (i)⇒(ii): Given $\varepsilon_0 > 0$ from (i), construct x_n by choosing $\delta = 1/n$ and selecting $x_n \in E$ with $|x_n - x_0| < 1/n$ and $|f(x_n) - f(x_0)| \geq \varepsilon_0$. Then $x_n \to x_0$ but $f(x_n) \not\to f(x_0)$ since $|f(x_n) - f(x_0)| \geq \varepsilon_0$.
For (ii)⇒(i): Given sequence $(x_n) \subseteq E$ with $x_n \to x_0$ and $f(x_n) \not\to f(x_0)$, there exists $\varepsilon_0 > 0$ and subsequence with $|f(x_{n_k}) - f(x_0)| \geq \varepsilon_0$. For any $\delta > 0$, choose k large enough so $|x_{n_k} - x_0| < \delta$. Then x_{n_k} witnesses the ε-δ discontinuity condition.
For (i)⇔(iii): Set $V = (f(x_0) - \varepsilon_0, f(x_0) + \varepsilon_0)$ and $U = (x_0 - \delta, x_0 + \delta)$. The condition $|f(x) - f(x_0)| \geq \varepsilon_0$ is equivalent to $f(x) \notin V$, establishing the equivalence. □

Detailed Learning Proof.
Step 1. Establish the three equivalent formulations of discontinuity at x_0. The ε-δ formulation states there exists a fixed $\varepsilon_0 > 0$ such that for every $\delta > 0$, some point $x \in E$ within distance δ of x_0 has $|f(x) - f(x_0)| \geq \varepsilon_0$. The sequential formulation requires a sequence in E converging to x_0 whose image under f fails to converge to $f(x_0)$. The neighbourhood formulation uses open sets: there exists a neighbourhood V of $f(x_0)$ such that every neighbourhood U of x_0 contains a point $x \in E$ with $f(x) \notin V$.
Step 2. Prove (i)⇒(ii) by constructing a bad sequence. Assume ε-δ discontinuity with witness $\varepsilon_0 > 0$. For each $n \in \mathbb{N}$, apply the discontinuity condition with $\delta = 1/n$ to obtain $x_n \in E$ such that $|x_n - x_0| < 1/n$ and $|f(x_n) - f(x_0)| \geq \varepsilon_0$. The first inequality gives $x_n \to x_0$, while the second prevents $f(x_n) \to f(x_0)$ since the sequence $(f(x_n))$ cannot approach $f(x_0)$ when each term remains at least distance ε_0 away.
Step 3. Prove (ii)⇒(i) using the negation of convergence. Assume sequential discontinuity with sequence $(x_n) \subseteq E$ where $x_n \to x_0$ but $f(x_n) \not\to f(x_0)$. The non-convergence of $f(x_n)$ means there exists $\varepsilon_0 > 0$ and infinitely many indices with $|f(x_n) - f(x_0)| \geq \varepsilon_0$. Given any $\delta > 0$, the convergence $x_n \to x_0$ ensures some x_n satisfies $|x_n - x_0| < \delta$. Choose such an x_n that also satisfies $|f(x_n) - f(x_0)| \geq \varepsilon_0$. This x_n witnesses the ε-δ discontinuity condition.
Step 4. Establish (i)⇔(iii) by translating between metric and neighbourhood language. The ε-δ condition uses the neighbourhood $V =$

$(f(x_0)-\varepsilon_0, f(x_0)+\varepsilon_0)$ of $f(x_0)$ and arbitrary neighbourhoods $U=(x_0-\delta, x_0+\delta)$ of x_0. The condition $|f(x)-f(x_0)| \geq \varepsilon_0$ is precisely equivalent to $f(x) \notin V$, and $|x-x_0| < \delta$ means $x \in U$. Since every neighbourhood of x_0 contains some interval $(x_0 - \delta, x_0 + \delta)$ and every neighbourhood of $f(x_0)$ contains some interval $(f(x_0)-\varepsilon, f(x_0)+\varepsilon)$, the two formulations are equivalent. □

Remark (Proof structure). This equivalence proof uses the standard technique of cyclic implications between three conditions. The connection between ε-δ and sequential formulations relies on the fundamental construction of bad sequences using $\delta = 1/n$ and the characterization of non-convergence through eventual bounds. The neighbourhood formulation is simply a topological restatement of the metric condition using open intervals, demonstrating how continuity concepts translate between metric and topological settings.

Remark (Dependencies).

- Definition of Point of Discontinuity
- Definition of Sequential Discontinuity at a Point
- Definition of Neighbourhood Discontinuity at a Point
- Definition of Convergent Sequence

Remark (Return). Return to Theorem

Theorem (Continuity Characterised by Oscillation). *Let $A \subseteq \mathbb{R}$, let $f\colon A \to \mathbb{R}$, and let $c \in A$. Then f is continuous at c if and only if $\omega(f;c) = 0$.*

Professional Standard Proof.
TODO: professional standard proof for continuity-iff-zero-oscillation. □

Detailed Learning Proof.
TODO: detailed learning proof for continuity-iff-zero-oscillation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Discontinuity Set via Oscillation). *Let* $f : E \to \mathbb{R}$. *Then*

$$D(f) := \{x \in E : f \text{ is discontinuous at } x\} = \{x \in E : \omega(f; x) > 0\} = \bigcup_{n=1}^{\infty} \{x \in E : \omega(f; x) \geq 1/n\}$$

Professional Standard Proof.
TODO: professional standard proof for discontinuity-set-via-oscillation. ☐

Detailed Learning Proof.
TODO: detailed learning proof for discontinuity-set-via-oscillation. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Boundedness Theorem). *Every function continuous on a closed bounded interval $[a, b]$ is bounded on $[a, b]$.*

Professional Standard Proof.
Let $I = [a, b]$ be a closed bounded interval and let $f : I \to \mathbb{R}$ be continuous on I. Suppose for contradiction that f is not bounded on I. For each $n \in \mathbb{N}$, choose $x_n \in I$ such that $|f(x_n)| > n$. The sequence $(x_n) \subseteq [a, b]$ is bounded, so by the Bolzano-Weierstrass Theorem, it has a convergent subsequence $x_{n_k} \to x^*$ for some $x^* \in \mathbb{R}$. Since I is closed and each $x_{n_k} \in I$, we have $x^* \in I$. By continuity of f at x^*, we have $f(x_{n_k}) \to f(x^*)$, so $(f(x_{n_k}))$ is bounded. But $|f(x_{n_k})| > n_k \geq k$ for all $k \in \mathbb{N}$, so $(f(x_{n_k}))$ is unbounded. This contradiction establishes that f is bounded on I. □

Detailed Learning Proof.
Step 1. Assume for contradiction that f is unbounded on $I = [a, b]$. This means that for each positive integer n, the set $\{x \in I : |f(x)| > n\}$ is nonempty. Therefore, for each $n \in \mathbb{N}$, there exists $x_n \in I$ such that $|f(x_n)| > n$. This constructs a sequence (x_n) in I with the property that the function values $f(x_n)$ grow without bound.
Step 2. Apply the Bolzano-Weierstrass Theorem to extract a convergent subsequence. Since $(x_n) \subseteq [a, b]$ and $[a, b]$ is bounded, the sequence (x_n) is bounded. By the Bolzano-Weierstrass Theorem, every bounded sequence in $\mathbb{R}$ has a convergent subsequence. Therefore, there exists a subsequence $x_{n_k} \to x^*$ for some $x^* \in \mathbb{R}$.
Step 3. Show that the limit point belongs to the interval. Since $I = [a, b]$ is closed and each term x_{n_k} belongs to I, the limit x^* must also belong to I. This uses the fundamental property that closed sets contain the limits of all their convergent sequences.
Step 4. Apply continuity to obtain convergence in the image. Since f is continuous at $x^* \in I$ and $x_{n_k} \to x^*$, the definition of continuity implies that $f(x_{n_k}) \to f(x^*)$. Therefore, the sequence $(f(x_{n_k}))$ converges to the finite value $f(x^*)$.
Step 5. Derive the contradiction. Every convergent sequence is bounded, so $(f(x_{n_k}))$ is bounded. However, from our construction in Step 1, we have $|f(x_{n_k})| > n_k$ for all k. Since n_k is a subsequence of natural numbers, $n_k \geq k$, which means $|f(x_{n_k})| > k$ for all $k \in \mathbb{N}$. This shows that $(f(x_{n_k}))$ is unbounded, contradicting the fact that it is bounded. Therefore, our assumption that f is unbounded must be false, and f is bounded on I. □

Remark (Proof structure). The proof proceeds by contradiction, assuming f is unbounded and deriving a contradiction through a three-stage argument: image to domain to image. First, unboundedness in the image allows construction of a sequence (x_n) with $|f(x_n)| > n$. Second, the Bolzano-Weierstrass Theorem operates on this sequence in the domain $[a, b]$ to extract a convergent subsequence whose limit remains in the closed interval. Third, continuity pulls the convergent subsequence back to the image, where convergence implies boundedness, contradicting the constructed unboundedness. The compactness of $[a, b]$ (closed and bounded) is essential: boundedness enables Bolzano-Weierstrass, and closedness ensures the limit stays in the domain.

Remark (Dependencies).

- Definition of bounded on a set
- Definition of continuity on a set
- Bolzano–Weierstrass Theorem
- Convergent sequences are bounded
- Closed sets contain limits of convergent sequences

Remark (Return). Return to Theorem

Theorem (Extreme Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$, and let $f : [a, b] \to \mathbb{R}$ be continuous. There exist points $x_m, x_M \in [a, b]$ such that $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$. Equivalently, f attains both an absolute maximum and an absolute minimum on $[a, b]$.*

Professional Standard Proof.
TODO: professional standard proof for extreme-value-theorem. ☐

Detailed Learning Proof.
TODO: detailed learning proof for extreme-value-theorem. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Location of Roots Theorem). *Let $f : [a,b] \to \mathbb{R}$ be continuous on $[a,b]$. If $f(a) < 0 < f(b)$ or $f(a) > 0 > f(b)$, then there exists $c \in (a,b)$ such that $f(c) = 0$.*

Professional Standard Proof.
Without loss of generality assume $f(a) < 0 < f(b)$. Define $S := \{x \in [a,b] \mid f(x) < 0\}$. Since $f(a) < 0$, the set S is nonempty and bounded above by b. By the Completeness Axiom, $s := \sup S$ exists and satisfies $s \in [a,b]$.
If $f(s) > 0$, then Sign Preservation provides $\varepsilon > 0$ such that $f(x) > 0$ for all $x \in (s-\varepsilon, s+\varepsilon) \cap [a,b]$. In particular, $f(x) > 0$ on $(s-\varepsilon, s]$, so $s-\varepsilon$ is an upper bound for S, contradicting $s = \sup S$.
If $f(s) < 0$, then Sign Preservation provides $\varepsilon > 0$ such that $f(x) < 0$ for all $x \in (s-\varepsilon, s+\varepsilon) \cap [a,b]$. Since $s < b$ (otherwise $f(b) < 0$ contradicts $f(b) > 0$), the point $s+\varepsilon/2$ lies in this interval for sufficiently small ε. Thus $s+\varepsilon/2 \in S$ and $s+\varepsilon/2 > s$, contradicting s being an upper bound for S.
Therefore $f(s) = 0$. Since $f(a) \neq 0$ and $f(b) \neq 0$, the point $c := s$ lies in (a,b). ☐

Detailed Learning Proof.
Step 1. Without loss of generality, assume $f(a) < 0 < f(b)$. The case $f(a) > 0 > f(b)$ follows by considering $-f$. Define $S := \{x \in [a,b] \mid f(x) < 0\}$. Since $f(a) < 0$, the element a belongs to S, so S is nonempty. Since $S \subseteq [a,b]$, the set S is bounded above by b.
Step 2. By the Completeness Axiom, the supremum $s := \sup S$ exists and satisfies $s \in [a,b]$. The goal is to establish that $f(s) = 0$ by ruling out the alternatives $f(s) > 0$ and $f(s) < 0$.
Step 3. Suppose $f(s) > 0$. Since f is continuous at s, the Sign Preservation theorem provides $\varepsilon > 0$ such that $f(x) > 0$ for all $x \in (s-\varepsilon, s+\varepsilon) \cap [a,b]$. In particular, $f(x) > 0$ for all $x \in (s-\varepsilon, s]$. This means no point of the interval $(s-\varepsilon, s]$ belongs to S, so $s-\varepsilon$ is an upper bound for S. Since $s-\varepsilon < s = \sup S$, this contradicts the definition of supremum as the least upper bound.
Step 4. Suppose $f(s) < 0$. Since f is continuous at s, the Sign Preservation theorem provides $\varepsilon > 0$ such that $f(x) < 0$ for all $x \in (s-\varepsilon, s+\varepsilon) \cap [a,b]$. If $s = b$, then $f(b) < 0$, contradicting the hypothesis $f(b) > 0$. Therefore $s < b$. For sufficiently small ε, the point $s+\varepsilon/2$ satisfies $s < s+\varepsilon/2 < b$, so $s+\varepsilon/2 \in [a,b]$ and lies in the neighborhood where $f(x) < 0$. Thus $s+\varepsilon/2 \in S$ and $s+\varepsilon/2 > s$, contradicting s being an upper bound for S.
Step 5. Since neither $f(s) > 0$ nor $f(s) < 0$ is possible, it follows that $f(s) = 0$. Finally, $s \neq a$ because $f(a) < 0 \neq 0 = f(s)$, and $s \neq b$ because $f(b) > 0 \neq 0 = f(s)$. Therefore $c := s \in (a,b)$ and $f(c) = 0$. ☐

Remark (Proof structure). This is a proof by contradiction using the supremum construction. The strategy defines S as the negative region of f and considers $s = \sup S$. Continuity at s combined with Sign Preservation rules out both $f(s) > 0$ (which would push the supremum strictly leftward) and $f(s) < 0$ (which would push the supremum strictly rightward), leaving only $f(s) = 0$.

Remark (Dependencies).

- Definition of Continuity
- Definition of Supremum
- Completeness Axiom
- Sign Preservation Theorem

Remark (Return). Return to Theorem

Theorem (Bolzano's Intermediate Value Theorem). *Let I be an interval, let $f : I \to \mathbb{R}$ be continuous on I, let $a, b \in I$, and let $k \in \mathbb{R}$. If $f(a) < k < f(b)$ or $f(a) > k > f(b)$, then there exists c between a and b such that $f(c) = k$.*

Professional Standard Proof.
Without loss of generality assume $f(a) < k < f(b)$; the case $f(a) > k > f(b)$ follows by applying the same argument to $-f$, which is continuous whenever f is. Define $g : I \to \mathbb{R}$ by $g(x) := f(x) - k$. Since f is continuous on I and k is constant, the algebra of continuous functions gives that g is continuous on I. Observe $g(a) = f(a) - k < 0$ and $g(b) = f(b) - k > 0$. Since g is continuous on I and $g(a) < 0 < g(b)$, the Location of Roots Theorem produces a point c strictly between a and b such that $g(c) = 0$. Therefore $f(c) - k = 0$, which gives $f(c) = k$. □

Detailed Learning Proof.
Step 1. Reduce to the case $f(a) < k < f(b)$. If instead $f(a) > k > f(b)$, apply the argument to the function $-f$. Since f is continuous on I, the function $-f$ is also continuous on I. Moreover, $(-f)(a) = -f(a) < -k$ and $(-f)(b) = -f(b) > -k$, so $(-f)(a) < -k < (-f)(b)$. If the theorem holds for this case, then there exists c between a and b such that $(-f)(c) = -k$, which means $f(c) = k$.
Step 2. Transform the problem into a root-finding problem. Define $g : I \to \mathbb{R}$ by $g(x) = f(x) - k$. The function g is continuous on I because it is the difference of the continuous function f and the constant function k, and the algebra of continuous functions guarantees that such differences are continuous.
Step 3. Verify the sign conditions for the Location of Roots Theorem. By construction, $g(a) = f(a) - k < 0$ since $f(a) < k$. Similarly, $g(b) = f(b) - k > 0$ since $f(b) > k$. Therefore $g(a) < 0 < g(b)$.
Step 4. Apply the Location of Roots Theorem. Since g is continuous on the interval I, $a, b \in I$, and $g(a) < 0 < g(b)$, the Location of Roots Theorem guarantees the existence of a point c strictly between a and b such that $g(c) = 0$.
Step 5. Translate back to the original function. The condition $g(c) = 0$ means $f(c) - k = 0$, which gives $f(c) = k$. This completes the proof. □

Remark (Proof structure). This is a direct proof that reduces the intermediate value problem to a root-finding problem. The key insight is to translate the function f vertically by $-k$ to create a new function $g = f - k$ that crosses zero precisely when f achieves the value k. The continuity of g follows from the algebra of continuous functions, and the Location of Roots Theorem provides the desired point. The symmetry between the two cases ($f(a) < k < f(b)$ and $f(a) > k > f(b)$) is handled cleanly by observing that $-f$ inherits continuity from f.

Remark (Dependencies).

- Definition of Continuous Function

- Algebra of Continuous Functions
- Location of Roots Theorem

Remark (Return). Return to Theorem

Theorem (Image of a Closed Bounded Interval Is a Closed Bounded Interval). *Let $f : [a, b] \to \mathbb{R}$ be continuous on the closed bounded interval $[a, b]$. Then $f([a, b]) = [m, M]$, where $m = \inf f([a, b])$ and $M = \sup f([a, b])$.*

Professional Standard Proof.
Since $[a, b]$ is closed and bounded and f is continuous, the Extreme Value Theorem guarantees the existence of points $x_m, x_M \in [a, b]$ such that $f(x_m) = m := \min f([a, b])$ and $f(x_M) = M := \max f([a, b])$. Every value of f lies in $[m, M]$, so $f([a, b]) \subseteq [m, M]$.
For the reverse inclusion, let $k \in [m, M]$. If $k = m$, then $f(x_m) = k$, so $k \in f([a, b])$. If $k = M$, then $f(x_M) = k$, so $k \in f([a, b])$. If $m < k < M$, then $f(x_m) = m < k < M = f(x_M)$, and the Intermediate Value Theorem applied to f on the subinterval $[\min(x_m, x_M), \max(x_m, x_M)] \subseteq [a, b]$ produces c between x_m and x_M with $f(c) = k$. Therefore $[m, M] \subseteq f([a, b])$, and hence $f([a, b]) = [m, M]$. □

Detailed Learning Proof.
Step 1. Apply the Extreme Value Theorem to establish endpoints.
Since f is continuous on the closed bounded interval $[a, b]$, the Extreme Value Theorem guarantees that f attains its minimum and maximum values. Therefore there exist points $x_m, x_M \in [a, b]$ such that $f(x_m) = m := \min f([a, b])$ and $f(x_M) = M := \max f([a, b])$. By definition of minimum and maximum, every value $f(x)$ for $x \in [a, b]$ satisfies $m \leq f(x) \leq M$, which establishes $f([a, b]) \subseteq [m, M]$.
Step 2. Prove the reverse inclusion by exhaustive case analysis.
To show $[m, M] \subseteq f([a, b])$, consider any $k \in [m, M]$ and establish that $k \in f([a, b])$. There are three exhaustive cases: $k = m$, $k = M$, or $m < k < M$.
Step 3. Handle the boundary cases using EVT witnesses.
If $k = m$, then $k = f(x_m)$ by the choice of x_m, so $k \in f([a, b])$. If $k = M$, then $k = f(x_M)$ by the choice of x_M, so $k \in f([a, b])$. The Extreme Value Theorem witnesses directly provide these boundary values.
Step 4. Handle the strict interior case using the Intermediate Value Theorem.
If $m < k < M$, then $f(x_m) = m < k < M = f(x_M)$. The interval $[\min(x_m, x_M), \max(x_m, x_M)]$ is a subinterval of $[a, b]$ on which f remains continuous. Since f takes values m and M at the endpoints of this subinterval and k lies strictly between these values, the Intermediate Value Theorem guarantees the existence of some c in this subinterval with $f(c) = k$. Therefore $k \in f([a, b])$.
Step 5. Conclude the equality of sets.
Since every $k \in [m, M]$ belongs to $f([a, b])$ by the case analysis in Steps 3-4, the inclusion $[m, M] \subseteq f([a, b])$ holds. Combined with the reverse inclusion from Step 1, this establishes $f([a, b]) = [m, M]$. □

Remark (Proof structure). This proof employs a direct approach using two set containments. The inclusion $f([a, b]) \subseteq [m, M]$ follows immediately from the definition of minimum and maximum values guaranteed by the Extreme Value Theorem. The reverse inclusion $[m, M] \subseteq f([a, b])$ requires exhaustive case analysis: the boundary cases $k = m$ and $k = M$ are handled by the EVT witnesses directly, while the strict interior case $m < k < M$ invokes the Intermediate Value Theorem on an appropriate subinterval. This partition cleanly separates the roles of EVT and IVT.

Remark (Dependencies).

- Extreme Value Theorem
- Bolzano's Intermediate Value Theorem
- Definition of Continuous Function
- Definition of Closed Bounded Interval

Remark (Return). Return to Theorem

Theorem (Preservation of Intervals). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be continuous on I. Then $f(I)$ is an interval.*

Professional Standard Proof.
By the Characterization of Intervals, it suffices to show that $f(I)$ is closed under betweenness: for any $y_1, y_2 \in f(I)$ with $y_1 < y_2$, every k satisfying $y_1 < k < y_2$ belongs to $f(I)$.
Let $y_1, y_2 \in f(I)$ with $y_1 < y_2$. By definition of the image, there exist $x_1, x_2 \in I$ with $f(x_1) = y_1$ and $f(x_2) = y_2$. Since I is an interval, the subinterval $J := [\min(x_1, x_2), \max(x_1, x_2)] \subseteq I$. The function f is continuous on J as the restriction of a continuous function. Since $f(x_1) = y_1 < k < y_2 = f(x_2)$, the Intermediate Value Theorem applied to f on J produces $c \in J$ with $f(c) = k$. Since $J \subseteq I$, we have $c \in I$, so $k = f(c) \in f(I)$. Therefore $f(I)$ is closed under betweenness and is an interval. □

Detailed Learning Proof.
Step 1. Establish the strategy using interval characterization. By the Characterization of Intervals, to prove $f(I)$ is an interval, it suffices to show that $f(I)$ is closed under betweenness. This means for any $y_1, y_2 \in f(I)$ with $y_1 < y_2$, every k satisfying $y_1 < k < y_2$ must belong to $f(I)$.
Step 2. Set up the betweenness verification. Let $y_1, y_2 \in f(I)$ with $y_1 < y_2$, and let k satisfy $y_1 < k < y_2$. Since $y_1, y_2 \in f(I)$, by definition of the image there exist $x_1, x_2 \in I$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$.
Step 3. Construct the connecting subinterval. Since I is an interval, it contains all points between any two of its elements. Therefore the closed interval $J := [\min(x_1, x_2), \max(x_1, x_2)] \subseteq I$. This construction handles both cases: whether $x_1 < x_2$ or $x_2 < x_1$.
Step 4. Apply the Intermediate Value Theorem. The function f is continuous on J as the restriction of a continuous function to a subset of its domain. Since $f(x_1) = y_1 < k < y_2 = f(x_2)$, the values $f(x_1)$ and $f(x_2)$ bracket the target value k. By the Intermediate Value Theorem applied to the continuous function f on the interval J, there exists $c \in J$ such that $f(c) = k$.
Step 5. Conclude that the intermediate value lies in the image. Since $J \subseteq I$ and $c \in J$, we have $c \in I$. Therefore $k = f(c) \in f(I)$. Since $y_1, y_2 \in f(I)$ and k were arbitrary, $f(I)$ is closed under betweenness and is therefore an interval. □

Remark (Proof structure). This is a direct proof that verifies the interval characterization. The strategy involves pulling back two points in $f(I)$ to their preimages in I, using the interval structure of I to construct a connecting subinterval, then applying the Intermediate Value Theorem to fill in every intermediate value. The argument works for any type of interval because it relies only on the Intermediate Value Theorem, not the Extreme Value Theorem.

Remark (Dependencies).

- Definition of Interval
- Characterization of Intervals
- Bolzano's Intermediate Value Theorem
- Definition of Continuous Function

Remark (Return). Return to Theorem

Theorem (Darboux Property). *Let $f : [a, b] \to \mathbb{R}$ be continuous on the closed interval $[a, b]$. For any $c \in \mathbb{R}$, if $f(x) \neq c$ for all $x \in [a, b]$, then either $f(x) > c$ for all $x \in [a, b]$, or $f(x) < c$ for all $x \in [a, b]$.*

Professional Standard Proof.
Let f be continuous on $[a, b]$ and define $S = \{f(x) \mid x \in [a, b]\}$. By the Extreme Value Theorem, f attains its maximum and minimum on $[a, b]$, so $S = [\inf S, \sup S]$ by the Intermediate Value Theorem. Let $c \in \mathbb{R}$ with $f(x) \neq c$ for all $x \in [a, b]$. Then $c \notin S$. Since $S = [\inf S, \sup S]$ is an interval, either $c < \inf S$ or $c > \sup S$. If $c < \inf S$, then $f(x) \geq \inf S > c$ for all $x \in [a, b]$. If $c > \sup S$, then $f(x) \leq \sup S < c$ for all $x \in [a, b]$. □

Detailed Learning Proof.
Step 1. Define the range of f and establish its structure. Let $S = \{f(x) \mid x \in [a, b]\}$ be the range of f on $[a, b]$. Since f is continuous on the closed and bounded interval $[a, b]$, the Extreme Value Theorem guarantees that f attains its maximum and minimum values on $[a, b]$. Therefore, $\inf S$ and $\sup S$ exist and are finite.
Step 2. Show that S forms a closed interval. By the Intermediate Value Theorem, since f is continuous on $[a, b]$, the range S contains every value between $\inf S$ and $\sup S$. Therefore, $S = [\inf S, \sup S]$, which is a closed interval.
Step 3. Analyze the position of c relative to S. Given that $f(x) \neq c$ for all $x \in [a, b]$, it follows that $c \notin S$. Since $S = [\inf S, \sup S]$ is an interval, the value c cannot lie between $\inf S$ and $\sup S$. Therefore, either $c < \inf S$ or $c > \sup S$.
Step 4. Conclude the dichotomy. If $c < \inf S$, then for all $x \in [a, b]$, $f(x) \geq \inf S > c$, so $f(x) > c$. If $c > \sup S$, then for all $x \in [a, b]$, $f(x) \leq \sup S < c$, so $f(x) < c$. Thus, either $f(x) > c$ for all $x \in [a, b]$, or $f(x) < c$ for all $x \in [a, b]$. □

Remark (Proof structure). This is a direct proof utilizing the topological properties of continuous functions. The key insight is that the range of a continuous function on a closed interval forms a closed interval itself, which means it has no gaps. When a value c is excluded from this range, it must lie entirely outside the interval, creating the strict dichotomy between being above or below all function values.

Remark (Dependencies).

- Extreme Value Theorem
- Bolzano's Intermediate Value Theorem
- Definition of Continuous Function

Remark (Return). Return to Theorem

Theorem (Algebra of Uniform Continuity on a General Domain). *(a) For every set $A \subseteq \mathbb{R}$, every pair of functions $f, g : A \to \mathbb{R}$ that are uniformly continuous on A, and every scalar $c \in \mathbb{R}$, the functions $f + g$, $f - g$, and cf are uniformly continuous on A.*

(b) There exist a set $A \subseteq \mathbb{R}$ and uniformly continuous functions $f, g : A \to \mathbb{R}$ such that the product fg is not uniformly continuous on A.

Professional Standard Proof.
TODO: professional standard proof for algebra-of-uniform-continuity-general. □

Detailed Learning Proof.
TODO: detailed learning proof for algebra-of-uniform-continuity-general. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Algebra of Uniform Continuity on a Bounded Domain). *Let $A \subseteq \mathbb{R}$ be bounded, and let $f, g : A \to \mathbb{R}$ be uniformly continuous on A.*

(a) The product $x \mapsto f(x)g(x)$ is uniformly continuous on A.

(b) If there exists $k > 0$ such that $|g(x)| \geq k$ for all $x \in A$, then the quotient $x \mapsto f(x)/g(x)$ is uniformly continuous on A.

Professional Standard Proof.
TODO: professional standard proof for algebra-of-uniform-continuity-bounded. □

Detailed Learning Proof.
TODO: detailed learning proof for algebra-of-uniform-continuity-bounded. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Sequential Characterization of Uniform Continuity). *Let $X \subseteq \mathbb{R}$ and $f : X \to \mathbb{R}$. Then f is uniformly continuous on X if and only if for every pair of sequences (x_n), (y_n) in X with $|x_n - y_n| \to 0$, we have $|f(x_n) - f(y_n)| \to 0$.*

Professional Standard Proof.
($\Rightarrow$) Assume f is uniformly continuous on X. Let (x_n) and (y_n) be sequences in X with $|x_n - y_n| \to 0$. Fix $\varepsilon > 0$. By uniform continuity, there exists $\delta > 0$ such that for all $x, y \in X$, $|x - y| < \delta$ implies $|f(x) - f(y)| < \varepsilon$. Since $|x_n - y_n| \to 0$, there exists $N \in \mathbb{N}$ such that $|x_n - y_n| < \delta$ for all $n > N$. Applying the uniform continuity condition gives $|f(x_n) - f(y_n)| < \varepsilon$ for all $n > N$. Since $\varepsilon > 0$ was arbitrary, $|f(x_n) - f(y_n)| \to 0$.
($\Leftarrow$) Suppose f is not uniformly continuous on X. By the Non-Uniform Continuity Criteria, there exist $\varepsilon_0 > 0$ and sequences (x_n), (y_n) in X such that $|x_n - y_n| \to 0$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. But then $|f(x_n) - f(y_n)| \not\to 0$, contradicting the hypothesis. Therefore f is uniformly continuous on X. $\square$

Detailed Learning Proof.
Step 1. Establish the forward implication by ε-δ convergence control. Assume f is uniformly continuous on X, and let (x_n), (y_n) be sequences in X with $|x_n - y_n| \to 0$. To show $|f(x_n) - f(y_n)| \to 0$, fix arbitrary $\varepsilon > 0$. The uniform continuity of f provides $\delta > 0$ such that whenever $x, y \in X$ satisfy $|x - y| < \delta$, then $|f(x) - f(y)| < \varepsilon$. This uniform bound applies to all pairs in X, not just those near a specific point.
Step 2. Apply convergence hypothesis to obtain eventual proximity of sequence terms. Since $|x_n - y_n| \to 0$, the convergence definition provides $N \in \mathbb{N}$ such that $|x_n - y_n| < \delta$ for all $n > N$. This bridges the gap between the convergence assumption and the uniform continuity condition by ensuring that for sufficiently large indices, the sequence terms (x_n, y_n) satisfy the distance requirement for the uniform bound.
Step 3. Conclude convergence of function value differences. For all $n > N$, the uniform continuity condition applies to give $|f(x_n) - f(y_n)| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, this establishes $|f(x_n) - f(y_n)| \to 0$, completing the forward direction.
Step 4. Establish the reverse implication by contrapositive using the Non-Uniform Continuity Criteria. Assume the sequential condition holds: for every pair of sequences (x_n), (y_n) in X with $|x_n - y_n| \to 0$, we have $|f(x_n) - f(y_n)| \to 0$. Suppose toward contradiction that f is not uniformly continuous on X. The Non-Uniform Continuity Criteria characterizes this failure: there exist $\varepsilon_0 > 0$ and sequences (x_n), (y_n) in X such that $|x_n - y_n| \to 0$ but $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$.
Step 5. Derive contradiction and conclude uniform continuity. The sequences (x_n), (y_n) from the Non-Uniform Continuity Criteria satisfy $|x_n - y_n| \to 0$, so by the hypothesis, $|f(x_n) - f(y_n)| \to 0$. But this contradicts $|f(x_n) - f(y_n)| \geq \varepsilon_0 > 0$ for all n. Therefore f must be uniformly continuous on X. $\square$

Remark (Proof structure). Bidirectional equivalence proved by direct implication and contrapositive. The forward direction employs a standard ε-δ argument where uniform continuity provides the distance tolerance δ, convergence $|x_n - y_n| \to 0$ ensures eventual satisfaction of this tolerance, and the uniform bound controls the function value differences. The reverse direction uses contrapositive reasoning: failure of uniform continuity is characterized sequentially by the Non-Uniform Continuity Criteria, and this characterization directly contradicts the sequential hypothesis.

Remark (Dependencies).

- Definition of uniform continuity
- Non-Uniform Continuity Criteria

Remark (Return). Return to Theorem

Theorem (Heine-Cantor Theorem). *Every function continuous on a closed bounded interval $[a, b]$ is uniformly continuous on $[a, b]$.*

Professional Standard Proof.
The argument proceeds by contradiction. Suppose $f: [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ but not uniformly continuous. Then there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exist points $x, y \in [a, b]$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon_0$. Applying this to $\delta = 1/n$ yields sequences (x_n) and (y_n) in $[a, b]$ satisfying $|x_n - y_n| < 1/n$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all $n \in \mathbb{N}$. The Bolzano-Weierstrass theorem produces a subsequence (x_{n_k}) converging to some $c \in [a, b]$. The triangle inequality gives $|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < 1/n_k + |x_{n_k} - c|$, so $y_{n_k} \to c$. Since f is continuous at c, both $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$, hence $|f(x_{n_k}) - f(y_{n_k})| \to 0$. This contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0 > 0$ for all k. □

Detailed Learning Proof.
Step 1. Assume for contradiction that $f: [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ but not uniformly continuous. By negating the definition of uniform continuity, there exists $\varepsilon_0 > 0$ such that for every $\delta > 0$ there exist points $x, y \in [a, b]$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon_0$. This gives us a way to construct problematic pairs of points.
Step 2. Construct two sequences by applying the negated uniform continuity to $\delta = 1/n$ for each $n \in \mathbb{N}$. This produces sequences (x_n) and (y_n) in $[a, b]$ satisfying $|x_n - y_n| < 1/n$ and $|f(x_n) - f(y_n)| \geq \varepsilon_0$ for all n. The sequences have inputs that get arbitrarily close while their outputs remain separated by at least ε_0.
Step 3. Apply the Bolzano-Weierstrass theorem to extract convergence. Since (x_n) is bounded in $[a, b]$, it has a convergent subsequence (x_{n_k}) with limit $c \in \mathbb{R}$. Since $[a, b]$ is closed, the limit point c belongs to $[a, b]$.
Step 4. Show that (y_{n_k}) converges to the same limit c. By the triangle inequality, $|y_{n_k} - c| \leq |y_{n_k} - x_{n_k}| + |x_{n_k} - c| < 1/n_k + |x_{n_k} - c|$. As $k \to \infty$, both $1/n_k \to 0$ and $|x_{n_k} - c| \to 0$, so $y_{n_k} \to c$.
Step 5. Apply continuity of f at c to derive the contradiction. Since f is continuous at c and both subsequences converge to c, the sequential characterization of continuity gives $f(x_{n_k}) \to f(c)$ and $f(y_{n_k}) \to f(c)$. Therefore $|f(x_{n_k}) - f(y_{n_k})| \to 0$. But this contradicts $|f(x_{n_k}) - f(y_{n_k})| \geq \varepsilon_0 > 0$ for all k, since limits preserve non-strict inequalities. The assumption fails, proving uniform continuity. □

Remark (Proof structure). The proof uses contradiction by negating uniform continuity to construct two sequences whose inputs converge to the same point while their outputs remain separated by a fixed positive distance. The Bolzano-Weierstrass theorem provides convergent subsequences, and continuity forces the output gap to collapse to zero, yielding the required contradiction.

Remark (Dependencies).

- Definition of Uniform Continuity
- Definition of Continuity
- Bolzano–Weierstrass Theorem
- Sequential Characterization of Continuity
- Limits Preserve Non-Strict Inequalities

Remark (Return). Return to Proposition

Proposition (Uniform Continuity Preserves Cauchy Sequences). *If $f\colon S \to \mathbb{R}$ is uniformly continuous and (x_n) is a Cauchy sequence in S, then $(f(x_n))$ is a Cauchy sequence in $\mathbb{R}$.*

Professional Standard Proof.
Let $\varepsilon > 0$ be given. By uniform continuity of f on S, there exists $\delta > 0$ such that for all $x, y \in S$ with $|x - y| < \delta$, one has $|f(x) - f(y)| < \varepsilon$. Since (x_n) is Cauchy, there exists $N \in \mathbb{N}$ such that for all $m, n \geq N$, one has $|x_m - x_n| < \delta$. For such m, n, uniform continuity yields $|f(x_m) - f(x_n)| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, $(f(x_n))$ is Cauchy. □

Detailed Learning Proof.
Step 1. Apply uniform continuity to convert the target tolerance into a domain distance constraint. Given $\varepsilon > 0$, uniform continuity of f on S provides $\delta > 0$ such that whenever $x, y \in S$ satisfy $|x - y| < \delta$, the image points satisfy $|f(x) - f(y)| < \varepsilon$. This establishes the connection between domain proximity and image proximity at the required tolerance level.
Step 2. Apply the Cauchy property to achieve the domain distance constraint. Since (x_n) is Cauchy in S, the distance δ from Step 1 determines an index $N \in \mathbb{N}$ such that for all indices $m, n \geq N$, the sequence terms satisfy $|x_m - x_n| < \delta$. This achieves the domain proximity required by uniform continuity.
Step 3. Combine the constraints to establish the Cauchy property for the image sequence. For indices $m, n \geq N$, the domain points $x_m, x_n \in S$ satisfy $|x_m - x_n| < \delta$, so uniform continuity from Step 1 yields $|f(x_m) - f(x_n)| < \varepsilon$. Since $\varepsilon > 0$ was arbitrary, the sequence $(f(x_n))$ satisfies the definition of Cauchy. □

Remark (Proof structure). Direct proof using the definition of Cauchy sequences. The strategy chains uniform continuity with the Cauchy property through an intermediate tolerance δ: uniform continuity converts the target tolerance ε into a domain constraint δ, while the Cauchy hypothesis provides eventual satisfaction of this domain constraint. The δ serves as a bridge connecting the two conditions rather than an independent parameter.

Remark (Dependencies).

- Definition of Uniform Continuity
- Definition of Cauchy Sequence

Remark (Return). Return to Theorem

Proposition (Lipschitz Implies Uniform Continuity). *Let $A \subseteq \mathbb{R}$ and $f : A \to \mathbb{R}$. If there exists $K \geq 0$ such that $|f(x) - f(y)| \leq K|x - y|$ for all $x, y \in A$, then f is uniformly continuous on A.*

Professional Standard Proof.
TODO: professional standard proof for lipschitz-implies-uc. ☐

Detailed Learning Proof.
TODO: detailed learning proof for lipschitz-implies-uc. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Algebra of Lipschitz Functions). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$ be Lipschitz on A with constants K_f and K_g respectively, and let $c \in \mathbb{R}$. Then:*

(a) cf is Lipschitz with constant $|c|K_f$

(b) $f + g$ and $f - g$ are Lipschitz with constant $K_f + K_g$

(c) If $h : B \to \mathbb{R}$ is Lipschitz on B with constant K_h and $f(A) \subseteq B$, then $h \circ f$ is Lipschitz on A with constant $K_h K_f$

(d) If $|f| \leq M_f$ and $|g| \leq M_g$ on A, then fg is Lipschitz with constant $M_g K_f + M_f K_g$

(e) If $|f| \leq M_f$ and $|g(x)| \geq k > 0$ for all $x \in A$, then f/g is Lipschitz with constant $K_f/k + M_f K_g/k^2$

Professional Standard Proof.
(a) For $x, u \in A$: $|cf(x) - cf(u)| = |c|\,|f(x) - f(u)| \leq |c|K_f|x - u|$.
(b) Triangle inequality gives $|(f + g)(x) - (f + g)(u)| \leq |f(x) - f(u)| + |g(x) - g(u)| \leq (K_f + K_g)|x - u|$. Same bound for $f - g$.
(c) Chain inequalities: $|h(f(x)) - h(f(u))| \leq K_h|f(x) - f(u)| \leq K_h K_f|x - u|$.
(d) Add-subtract identity: $(fg)(x) - (fg)(u) = f(x)(g(x) - g(u)) + g(u)(f(x) - f(u))$. Then $|(fg)(x) - (fg)(u)| \leq |f(x)|\,|g(x) - g(u)| + |g(u)|\,|f(x) - f(u)| \leq M_f K_g|x - u| + M_g K_f|x - u|$.
(e) First show $1/g$ is Lipschitz with constant K_g/k^2: $|1/g(x) - 1/g(u)| = |g(x) - g(u)|/|g(x)g(u)| \leq K_g|x - u|/k^2$. Then apply (d) to $f \cdot (1/g)$ with $M_{1/g} = 1/k$. □

Detailed Learning Proof.
Step 1. Establish the scalar multiple property. For any $x, u \in A$, the definition of scalar multiplication gives

$$|cf(x) - cf(u)| = |c|\,|f(x) - f(u)|.$$

Since f is Lipschitz with constant K_f, one has $|f(x) - f(u)| \leq K_f|x - u|$. Substituting yields $|cf(x) - cf(u)| \leq |c|K_f|x - u|$, establishing that cf is Lipschitz with constant $|c|K_f$.
Step 2. Prove the sum and difference properties. The Lipschitz conditions on f and g give $|f(x) - f(u)| \leq K_f|x - u|$ and $|g(x) - g(u)| \leq K_g|x - u|$ for all $x, u \in A$. The triangle inequality applied to the sum yields

$$|(f + g)(x) - (f + g)(u)| \leq |f(x) - f(u)| + |g(x) - g(u)| \leq (K_f + K_g)|x - u|.$$

For the difference, since $|-(g(x) - g(u))| = |g(x) - g(u)|$, the same bound $(K_f + K_g)|x - u|$ applies.
Step 3. Establish the composition property. For the composite function $h \circ f$, one applies the Lipschitz condition for h first, then for f:

$$|h(f(x)) - h(f(u))| \leq K_h|f(x) - f(u)| \leq K_h K_f|x - u|.$$

This establishes that $h \circ f$ is Lipschitz with constant $K_h K_f$.

Step 4. Prove the product property for bounded functions. The key insight is the add-subtract identity: $(fg)(x) - (fg)(u) = f(x)(g(x) - g(u)) + g(u)(f(x) - f(u))$. Taking absolute values and applying the triangle inequality:

$$|(fg)(x) - (fg)(u)| \leq |f(x)|\,|g(x) - g(u)| + |g(u)|\,|f(x) - f(u)|.$$

The boundedness conditions $|f| \leq M_f$ and $|g| \leq M_g$ together with the Lipschitz conditions yield

$$|(fg)(x) - (fg)(u)| \leq M_f K_g |x - u| + M_g K_f |x - u| = (M_g K_f + M_f K_g)|x - u|.$$

Step 5. Establish the quotient property. First, establish that $1/g$ is Lipschitz. For the difference quotient:

$$\left| \frac{1}{g(x)} - \frac{1}{g(u)} \right| = \frac{|g(x) - g(u)|}{|g(x)g(u)|} \leq \frac{K_g |x - u|}{k^2},$$

using $|g(x)|, |g(u)| \geq k$. Thus $1/g$ is Lipschitz with constant K_g/k^2. Since $f/g = f \cdot (1/g)$ and $|1/g| \leq 1/k$, applying the product rule from Step 4 gives the Lipschitz constant $K_f/k + M_f K_g/k^2$. $\square$

Remark (Proof structure). The proof employs direct verification of the Lipschitz condition for each operation. Parts (a) and (b) use elementary properties of absolute values and the triangle inequality. Part (c) chains two Lipschitz inequalities in succession. Part (d) relies on an algebraic identity to separate the product into manageable terms that can be bounded using the given conditions. Part (e) reduces to part (d) by first establishing that the reciprocal of a bounded-away-from-zero Lipschitz function is itself Lipschitz.

Remark (Dependencies).

- Definition of Lipschitz Condition
- Definition of Bounded on a Set

Remark (Return). Return to Theorem

Theorem (Inverse of a Bi-Lipschitz Map Is Lipschitz). *Let $A \subseteq \mathbb{R}$ and let $f : A \to f(A)$ be bi-Lipschitz with bounds α and K, meaning*

$$\alpha|x-u| \leq |f(x)-f(u)| \leq K|x-u| \quad \textit{for all } x,u \in A.$$

Then (a) f is injective, so $f^{-1} : f(A) \to A$ exists; (b) f^{-1} is Lipschitz with constant $1/\alpha$; (c) f^{-1} is bi-Lipschitz with bounds $1/K$ and $1/\alpha$.

Professional Standard Proof.
Part (a): If $x \neq u$ then $|x-u| > 0$, so the lower bound gives $|f(x)-f(u)| \geq \alpha|x-u| > 0$, hence $f(x) \neq f(u)$. Thus f is injective and f^{-1} exists.
Part (b): Let $y, v \in f(A)$ with $y = f(x)$ and $v = f(u)$ for unique $x, u \in A$. The lower bound gives $\alpha|f^{-1}(y) - f^{-1}(v)| = \alpha|x-u| \leq |f(x)-f(u)| = |y-v|$. Dividing by $\alpha > 0$ yields $|f^{-1}(y) - f^{-1}(v)| \leq \frac{1}{\alpha}|y-v|$.
Part (c): The upper bound gives $|y-v| = |f(x)-f(u)| \leq K|x-u| = K|f^{-1}(y) - f^{-1}(v)|$. Dividing by $K > 0$ yields $\frac{1}{K}|y-v| \leq |f^{-1}(y)-f^{-1}(v)|$. Combined with part (b), this establishes the bi-Lipschitz bounds. □

Detailed Learning Proof.
Step 1. Establish injectivity of f. Suppose $x, u \in A$ with $x \neq u$. Then $|x-u| > 0$. The lower bi-Lipschitz bound gives $|f(x)-f(u)| \geq \alpha|x-u| > 0$ since $\alpha > 0$. Therefore $f(x) \neq f(u)$, proving f is injective. Since $f : A \to f(A)$ is surjective by definition, the inverse function $f^{-1} : f(A) \to A$ exists and is well-defined.
Step 2. Prove f^{-1} is Lipschitz with constant $1/\alpha$. Let $y, v \in f(A)$ be arbitrary. Since f is surjective onto $f(A)$, there exist unique $x, u \in A$ such that $y = f(x)$ and $v = f(u)$. This means $f^{-1}(y) = x$ and $f^{-1}(v) = u$. Applying the lower bi-Lipschitz bound to f: $\alpha|x-u| \leq |f(x)-f(u)|$. Substituting our expressions: $\alpha|f^{-1}(y) - f^{-1}(v)| \leq |y-v|$. Since $\alpha > 0$, dividing both sides by α gives $|f^{-1}(y) - f^{-1}(v)| \leq \frac{1}{\alpha}|y-v|$. This establishes that f^{-1} is Lipschitz with constant $1/\alpha$.
Step 3. Establish the lower bi-Lipschitz bound for f^{-1}. Using the same $y, v \in f(A)$ and $x = f^{-1}(y)$, $u = f^{-1}(v)$, apply the upper bi-Lipschitz bound to f: $|f(x)-f(u)| \leq K|x-u|$. Substituting: $|y-v| \leq K|f^{-1}(y) - f^{-1}(v)|$. Since $K > 0$, dividing both sides by K gives $\frac{1}{K}|y-v| \leq |f^{-1}(y) - f^{-1}(v)|$. Combined with the upper bound from Step 2, this shows f^{-1} is bi-Lipschitz with bounds $1/K$ and $1/\alpha$. □

Remark (Proof structure). The proof uses a direct approach based on variable substitution. The key insight is that the bi-Lipschitz bounds on f translate into bi-Lipschitz bounds on f^{-1} by exchanging the roles of domain and codomain through the substitution $y = f(x)$, $v = f(u)$. The lower bound on f becomes the upper bound on f^{-1}, and vice versa, with the constants reciprocated.

Remark (Dependencies).

- Definition of bi-Lipschitz map
- Definition of Lipschitz Condition

Remark (Return). Return to Theorem

Proposition (). *Let $A \subseteq \mathbb{R}$, let x_0 be a limit point of A, and let $f : A \to \mathbb{R}$. Then*

$$\limsup_{x \to x_0} f(x) \geq \liminf_{x \to x_0} f(x).$$

Professional Standard Proof.
TODO: professional standard proof for limsup-geq-liminf-function. □

Detailed Learning Proof.
TODO: detailed learning proof for limsup-geq-liminf-function. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Proposition (Limit Exists iff =). *Let $A \subseteq \mathbb{R}$, let $f \colon A \to \mathbb{R}$, let x_0 be a limit point of A with $x_0 \in \mathbb{R}$, and let $L \in \mathbb{R}$. Then $\lim_{x\to x_0} f(x)$ exists and equals L if and only if $\limsup_{x\to x_0} f(x) = \liminf_{x\to x_0} f(x) = L$.*

Professional Standard Proof.
TODO: professional standard proof for limsup-liminf-limit-criterion. □

Detailed Learning Proof.
TODO: detailed learning proof for limsup-liminf-limit-criterion. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (One-Sided Limits of Monotone Functions). *Let $f : I \to \mathbb{R}$ be monotone increasing on the interval I, and let $c \in I$ be an interior point. Then both one-sided limits of f at c exist and are given by*

$$\lim_{x\to c^-} f(x) = \sup\{f(x) : x < c,\, x \in I\} \qquad \text{and} \qquad \lim_{x\to c^+} f(x) = \inf\{f(x) : x > c,\, x \in I\}.$$

Moreover, $f(c^-) \le f(c) \le f(c^+)$.

Professional Standard Proof.
Define $S := \{f(x) : x \in I, x < c\}$ and $J := \{f(x) : x \in I, x > c\}$. Since c is interior, there exist $a, b \in I$ with $a < c < b$. Monotonicity implies $S \subseteq (-\infty, f(b)]$ and $J \subseteq [f(a), +\infty)$, so $s := \sup S$ and $j := \inf J$ exist. For the left limit: given $\varepsilon > 0$, since $s - \varepsilon$ is not an upper bound of S, there exists $x_0 < c$ with $f(x_0) > s - \varepsilon$. Set $\delta := c - x_0 > 0$. For $x \in (c - \delta, c) \cap I$, monotonicity gives $s - \varepsilon < f(x_0) \le f(x) \le s$, hence $|f(x) - s| < \varepsilon$. For the right limit: given $\varepsilon > 0$, there exists $x_1 > c$ with $f(x_1) < j + \varepsilon$. Set $\delta := x_1 - c > 0$. For $x \in (c, c+\delta) \cap I$, monotonicity gives $j \le f(x) \le f(x_1) < j + \varepsilon$, hence $|f(x) - j| < \varepsilon$. The ordering follows since $f(c)$ is an upper bound of S and a lower bound of J. □

Detailed Learning Proof.
Step 1. Define the sets $S := \{f(x) : x \in I, x < c\}$ and $J := \{f(x) : x \in I, x > c\}$. Since c is an interior point of I, there exist points $a, b \in I$ with $a < c < b$. The monotonicity of f ensures that every element of S satisfies $f(x) \le f(b)$ and every element of J satisfies $f(x) \ge f(a)$. Therefore both sets are bounded, and the supremum $s := \sup S$ and infimum $j := \inf J$ exist in $\mathbb{R}$.
Step 2. Establish the left-hand limit $\lim_{x\to c^-} f(x) = s$. Let $\varepsilon > 0$ be given. Since s is the supremum of S, the value $s - \varepsilon$ fails to be an upper bound of S. Therefore there exists some $x_0 \in I$ with $x_0 < c$ and $f(x_0) > s - \varepsilon$. Define $\delta := c - x_0 > 0$. For any $x \in I$ satisfying $c - \delta < x < c$, equivalently $x_0 < x < c$, the monotonicity of f gives $f(x_0) \le f(x)$, so $f(x) > s - \varepsilon$. Since $f(x) \in S$, the definition of supremum ensures $f(x) \le s$. Combined, this gives $|f(x) - s| < \varepsilon$.
Step 3. Establish the right-hand limit $\lim_{x\to c^+} f(x) = j$. Let $\varepsilon > 0$ be given. Since j is the infimum of J, the value $j + \varepsilon$ fails to be a lower bound of J. Therefore there exists some $x_1 \in I$ with $x_1 > c$ and $f(x_1) < j + \varepsilon$. Define $\delta := x_1 - c > 0$. For any $x \in I$ satisfying $c < x < c + \delta$, equivalently $c < x < x_1$, the monotonicity of f gives $f(x) \le f(x_1)$, so $f(x) < j + \varepsilon$. Since $f(x) \in J$, the definition of infimum ensures $f(x) \ge j$. Combined, this gives $|f(x) - j| < \varepsilon$.
Step 4. Verify the ordering $f(c^-) \le f(c) \le f(c^+)$. For every $x \in I$ with $x < c$, monotonicity gives $f(x) \le f(c)$, making $f(c)$ an upper bound of the set S. Since $s = \sup S$ is the least upper bound, this forces $s \le f(c)$. Similarly, for every $x \in I$ with $x > c$, monotonicity gives $f(x) \ge f(c)$, making $f(c)$ a lower bound of the set J. Since $j = \inf J$ is the greatest lower bound, this forces $f(c) \le j$. □

Remark (Proof structure). The argument is a direct ε-δ proof driven by the supremum and infimum characterizations of the one-sided limits. For the left-hand limit, the supremum s defines the target value; the defining property that $s - \varepsilon$ is not an upper bound supplies the witness point x_0; setting $\delta = c - x_0$ ensures that every x in the punctured left neighborhood lies above x_0, whereupon monotonicity squeezes $f(x)$ into the interval $(s - \varepsilon, s]$. The right-hand argument mirrors this construction using the infimum. The final ordering relationship follows immediately from applying monotonicity to characterize $f(c)$ as both an upper bound for the left set and a lower bound for the right set.

Remark (Dependencies).

- Completeness of $\mathbb{R}$
- Definition of supremum
- Definition of infimum
- Definition of monotone function
- Definition of one-sided limit

Remark (Return). Return to Corollary

Corollary (Continuity Criterion for Monotone Functions). *Let $f : I \to \mathbb{R}$ be monotone increasing and let $c \in I$ be an interior point. Then f is continuous at c if and only if $f(c^-) = f(c^+)$.*

Professional Standard Proof.
Let $f : I \to \mathbb{R}$ be monotone increasing and let $c \in I$ be an interior point.
($\Rightarrow$) Suppose f is continuous at c. Then $\lim_{x\to c} f(x)$ exists and equals $f(c)$. By the criterion that a two-sided limit exists if and only if both one-sided limits exist and are equal, both $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ exist and equal $f(c)$. In particular $f(c^-) = f(c^+)$.
($\Leftarrow$) Suppose $f(c^-) = f(c^+)$. By the One-Sided Limits of Monotone Functions theorem, these one-sided limits equal $\sup\{f(x) : x < c\}$ and $\inf\{f(x) : x > c\}$ respectively, and the same theorem gives the ordering $f(c^-) \le f(c) \le f(c^+)$. Since $f(c^-) = f(c^+)$, this ordering forces $f(c^-) = f(c) = f(c^+)$. Hence both one-sided limits exist and equal $f(c)$, so the two-sided limit $\lim_{x\to c} f(x) = f(c)$ exists, and f is continuous at c. $\square$

Detailed Learning Proof.
Step 1. Establish the forward direction: continuity implies matching one-sided limits. If f is continuous at c, then the two-sided limit $\lim_{x\to c} f(x)$ exists and equals $f(c)$. By the characterization of two-sided limits in terms of one-sided limits, this means both $\lim_{x\to c^-} f(x)$ and $\lim_{x\to c^+} f(x)$ exist and both equal $f(c)$. Therefore $f(c^-) = f(c) = f(c^+)$, so in particular $f(c^-) = f(c^+)$.
Step 2. Establish the reverse direction: matching one-sided limits imply continuity. Assume $f(c^-) = f(c^+)$. By the One-Sided Limits of Monotone Functions theorem, both one-sided limits exist and satisfy the ordering $f(c^-) \le f(c) \le f(c^+)$. Since $f(c^-) = f(c^+)$ by hypothesis, this ordering forces all three values to be equal: $f(c^-) = f(c) = f(c^+)$.
Step 3. Conclude continuity from matching one-sided limits. Since both one-sided limits exist and equal $f(c)$, the criterion for two-sided limits guarantees that $\lim_{x\to c} f(x)$ exists and equals $f(c)$. This is precisely the definition of continuity at c. $\square$

Remark (Proof structure). A direct biconditional proof. The forward direction unpacks the definition of continuity via the equivalence of two-sided and matching one-sided limits. The reverse direction applies the ordering $f(c^-) \le f(c) \le f(c^+)$ from the One-Sided Limits of Monotone Functions theorem: the hypothesis $f(c^-) = f(c^+)$ squeezes $f(c)$ to the common value, after which the two-sided limit criterion closes the argument.

Remark (Dependencies).

- Definition of Continuity at a Point
- One-Sided Limits of Monotone Functions
- Two-Sided Limit Iff Matching One-Sided Limits

Remark (Return). Return to Theorem

Proposition (Discontinuities of a Monotone Function Are of First Kind). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$ be monotone, and let $a \in I$ be such that there exist points of I strictly less and strictly greater than a. If f is discontinuous at a, then both one-sided limits $\lim_{x\uparrow a} f(x)$ and $\lim_{x\downarrow a} f(x)$ exist in $\mathbb{R}$.*

Professional Standard Proof.
TODO: professional standard proof for monotone-discontinuities-first-kind. □

Detailed Learning Proof.
TODO: detailed learning proof for monotone-discontinuities-first-kind. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Corollary

Corollary (Jump Intervals of a Monotone Function Are Disjoint). *Let $f : I \to \mathbb{R}$ be a monotone function on an interval I, and let $D \subseteq I$ be the set of all discontinuities of f. For each $c \in D$, the jump interval at c is $J_c(f) := (f(c^-), f(c^+))$. If $c, d \in D$ with $c \neq d$, then $J_c(f) \cap J_d(f) = \emptyset$.*

Professional Standard Proof.
Without loss of generality assume f is monotone increasing and $c < d$. For any $x \in (c, d)$, monotonicity yields $f(c^+) \leq f(x) \leq f(d^-)$, hence $f(c^+) \leq f(d^-)$. Since every element of $J_c(f) = (f(c^-), f(c^+))$ is strictly less than $f(c^+)$ and every element of $J_d(f) = (f(d^-), f(d^+))$ is strictly greater than $f(d^-)$, the intervals are disjoint. $\square$

Detailed Learning Proof.
Step 1. Without loss of generality, assume f is monotone increasing. The case where f is decreasing follows by applying the result to $-f$. Let $c, d \in D$ be distinct discontinuities, and without loss of generality assume $c < d$.
Step 2. The jump intervals are $J_c(f) = (f(c^-), f(c^+))$ and $J_d(f) = (f(d^-), f(d^+))$. To show they are disjoint, establish the key inequality $f(c^+) \leq f(d^-)$.
Step 3. For any $x \in (c, d)$, monotonicity of f gives $f(c^+) \leq f(x)$ and $f(x) \leq f(d^-)$. This follows because $f(c^+) = \inf\{f(y) : y > c\}$ and $f(d^-) = \sup\{f(y) : y < d\}$, and x satisfies both $x > c$ and $x < d$.
Step 4. Taking any $x \in (c, d)$ in the chain of inequalities from Step 3 yields $f(c^+) \leq f(x) \leq f(d^-)$, hence $f(c^+) \leq f(d^-)$.
Step 5. By definition, every element of $J_c(f)$ is strictly less than $f(c^+)$, and every element of $J_d(f)$ is strictly greater than $f(d^-)$. Since $f(c^+) \leq f(d^-)$, no element can belong to both intervals, so $J_c(f) \cap J_d(f) = \emptyset$. $\square$

Remark (Proof structure). The argument reduces to establishing one monotonicity inequality: $f(c^+) \leq f(d^-)$ whenever $c < d$. This follows by taking the infimum over values just right of c and the supremum over values just left of d, both of which are bounded by any $f(x)$ with $c < x < d$. Once this inequality is established, the two open intervals are separated and hence disjoint.

Remark (Dependencies).

- One-Sided Limits of Monotone Functions
- Discontinuities of a Monotone Function Are of the First Kind

Remark (Return). Return to Theorem

Theorem (Discontinuities of a Monotone Function Are Countable). *Let $f : I \to \mathbb{R}$ be a monotone function on an interval I. Then the set of discontinuities of f is at most countable.*

Professional Standard Proof.
Let $D = \{c \in I : f \text{ is discontinuous at } c\}$. Without loss of generality, assume f is monotone increasing. For each $c \in D$, the jump interval $J_c(f) = (f(c^-), f(c^+))$ is non-empty by the first kind nature of monotone discontinuities. By density of $\mathbb{Q}$ in $\mathbb{R}$, choose $r(c) \in J_c(f) \cap \mathbb{Q}$. Since jump intervals of distinct discontinuities are disjoint, the function $r : D \to \mathbb{Q}$ is injective. Therefore D injects into the countable set $\mathbb{Q}$, hence D is at most countable. □

Detailed Learning Proof.
Step 1. Define the discontinuity set and reduce to the increasing case. Let $D = \{c \in I : f \text{ is discontinuous at } c\}$. Without loss of generality, assume f is monotone increasing; if f is decreasing, apply the result to $-f$ and observe that f is discontinuous at c if and only if $-f$ is discontinuous at c.
Step 2. Construct jump intervals and select rational witnesses. For each discontinuity $c \in D$, the jump interval $J_c(f) = (f(c^-), f(c^+))$ is a non-empty open interval since monotone functions have only first kind discontinuities. By density of $\mathbb{Q}$ in $\mathbb{R}$, there exists a rational number $r(c) \in J_c(f)$ with $f(c^-) < r(c) < f(c^+)$.
Step 3. Establish injectivity of the rational assignment. The function $r : D \to \mathbb{Q}$ defined by $c \mapsto r(c)$ is injective. To see this, let $c, d \in D$ with $c \neq d$. Then the jump intervals $J_c(f)$ and $J_d(f)$ are disjoint. Since $r(c) \in J_c(f)$ and $r(d) \in J_d(f)$, it follows that $r(c) \neq r(d)$.
Step 4. Conclude countability from the injection. The injection $r : D \hookrightarrow \mathbb{Q}$ into the countable set $\mathbb{Q}$ implies that D is at most countable. □

Remark (Proof structure). This is a counting argument via injection into a countable set. The strategy exploits the geometric structure of monotone discontinuities: each discontinuity creates a non-empty "gap" (jump interval), and these gaps are disjoint. Density of rationals allows selection of distinct rational witnesses from each gap, creating an injection into $\mathbb{Q}$.

Remark (Dependencies).

- Discontinuities of a Monotone Function Are of the First Kind
- Jump Intervals of a Monotone Function Are Disjoint
- Density of $\mathbb{Q}$ in $\mathbb{R}$
- Injection into a Countable Set Is Countable

Remark (Return). Return to Corollary

Corollary (Monotone Function Is Continuous on a Dense Set). *Let $f : I \to \mathbb{R}$ be a monotone function on an interval I. Then the set of points of continuity of f is dense in I.*

Professional Standard Proof.
Let $D = \{x \in I : f \text{ is not continuous at } x\}$ be the set of discontinuities. By the theorem that discontinuities of a monotone function are countable, D is at most countable. The set of points of continuity is $I \setminus D$. To show $I \setminus D$ is dense in I, let $(a, b) \subseteq I$ be any open subinterval with $a < b$. Since (a, b) is uncountable while D is at most countable, $(a, b) \not\subseteq D$. Therefore there exists $x \in (a, b)$ with $x \notin D$, so $(a, b) \cap (I \setminus D) \neq \emptyset$. Since every open subinterval of I meets $I \setminus D$, the set $I \setminus D$ is dense in I. □

Detailed Learning Proof.
Step 1. Define the set of discontinuities $D = \{x \in I : f \text{ is not continuous at } x\}$. By the theorem on discontinuities of monotone functions, D is at most countable. This gives us precise control over the size of the set where f fails to be continuous.
Step 2. The set of continuity points is $I \setminus D$. To prove this is dense in I, consider any open subinterval $(a, b) \subseteq I$ with $a < b$. The interval (a, b) has uncountably many points, while D has at most countably many points.
Step 3. Since an uncountable set cannot be contained in a countable set, $(a, b) \not\subseteq D$. Therefore there exists at least one point $x \in (a, b)$ such that $x \notin D$, which means x is a point of continuity and $(a, b) \cap (I \setminus D) \neq \emptyset$.
Step 4. Since every open subinterval of I contains a point from $I \setminus D$, the set $I \setminus D$ is dense in I by definition. □

Remark (Proof structure). This is a direct proof using a cardinality argument. The strategy exploits the fact that discontinuities form a countable set while open intervals are uncountable, creating an inevitable overlap between any open interval and the complement of the discontinuity set. The key insight is translating the countability constraint into a density conclusion through the intermediate step of examining open subintervals.

Remark (Dependencies).

- Discontinuities of a Monotone Function Are Countable
- Intervals of $\mathbb{R}$ Are Uncountable
- Definition of Dense Subset

Remark (Return). Return to Theorem

Proposition (Continuous Injective Is Strictly Monotone). *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$. Then f is injective if and only if f is strictly monotone on $[a, b]$.*

Professional Standard Proof.
TODO: professional standard proof for continuous-injective-iff-strictly-monotone. □

Detailed Learning Proof.
TODO: detailed learning proof for continuous-injective-iff-strictly-monotone. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Continuous Inverse Theorem). *Let $I \subseteq \mathbb{R}$ be an interval and let $f : I \to \mathbb{R}$ be strictly monotone and continuous on I. Then the inverse function $f^{-1} : f(I) \to I$ exists, is strictly monotone, and is continuous on $f(I)$.*

Professional Standard Proof.
TODO: professional standard proof for continuous-inverse-theorem. □

Detailed Learning Proof.
TODO: detailed learning proof for continuous-inverse-theorem. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Step Function Approximation). *Let $f : [a, b] \to \mathbb{R}$ be continuous and let $\varepsilon > 0$. Then there exists a step function $s_\varepsilon : [a, b] \to \mathbb{R}$ such that*

$$\sup_{x \in [a,b]} |f(x) - s_\varepsilon(x)| < \varepsilon.$$

Professional Standard Proof.
TODO: professional standard proof for step-function-approximation. □

Detailed Learning Proof.
TODO: detailed learning proof for step-function-approximation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Piecewise Linear Approximation). *TODO: restate the theorem.*

Professional Standard Proof.
TODO: professional standard proof for piecewise-linear-approximation. □

Detailed Learning Proof.
TODO: detailed learning proof for piecewise-linear-approximation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Weierstrass Approximation Theorem). *Let* $f : [a,b] \to \mathbb{R}$ *be continuous. For every* $\varepsilon > 0$ *there exists a polynomial* $p_\varepsilon \in \mathbb{R}[x]$ *such that* $\|f - p_\varepsilon\|_{\infty,[a,b]} < \varepsilon$*.*

Professional Standard Proof.
TODO: professional standard proof for weierstrass-approximation. □

Detailed Learning Proof.
TODO: detailed learning proof for weierstrass-approximation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Bernstein Approximation Theorem). *TODO: restate the theorem.*

Professional Standard Proof.
TODO: professional standard proof for bernstein-approximation. □

Detailed Learning Proof.
TODO: detailed learning proof for bernstein-approximation. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Lemma (Every Point Is Covered by a Tag). *Let $\dot{\mathcal{P}} = \{([x_{i-1}, x_i], t_i)\}_{i=1}^n$ be a tagged partition of $[a, b]$ that is δ-fine on $[a, b]$, and let $x \in [a, b]$. Then there exists $i \in \{1, \ldots, n\}$ such that $|x - t_i| < \delta(t_i)$.*

Professional Standard Proof.
TODO: professional standard proof for every-point-covered-by-tag. ☐

Detailed Learning Proof.
TODO: detailed learning proof for every-point-covered-by-tag. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Cousin's Theorem). *Every gauge* $\delta : [a,b] \to (0,\infty)$ *on a closed interval* $[a,b]$ *admits a* δ*-fine tagged partition of* $[a,b]$.

Professional Standard Proof.
TODO: professional standard proof for cousins-theorem. ☐

Detailed Learning Proof.
TODO: detailed learning proof for cousins-theorem. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Continuity, using only concepts at or before this chapter in the registry.

`*Remark* (Dependency ceiling). The capstone may use only results routed at or before this chapter.`

Chapter 7

Differentiation

differentiation

Continuity $\to$ **Differentiation** $\to$ Riemann Integration

7.1 Derivative Definition

Derivative Definition — toolkit

Concept family: The derivative at a point, and its three equivalent formulations.

Atomic items in this section:

- Definition: Derivative at a Point (epsilon–delta form)
- Definition: Topological Definition of Derivative at a Point
- Definition: Sequential Definition of Derivative at a Point
- Theorem: Equivalence of Definitions of the Derivative at a Point
- Proposition: Derivative: h-Form Equivalence
- Theorem: Differentiable Implies Continuous
- Theorem: Uniqueness of the Derivative
- Proposition: Derivative of the Identity

Definition (Derivative at a Point)

Definition 7.1.1 (Derivative at a Point)**.** Let $f : D \to \mathbb{R}$ be a real-valued function and $c \in \operatorname{int}(D)$. We say f is *differentiable at c with derivative* $L = f'(c)$ if and only if:

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Standard quantified statement)*.*

$$\forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in D \Big(0 < |x - c| < \delta \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}(f,c,L) \iff \mathrm{FunctionLimit}\left(\lambda x.\,\frac{f(x)-f(c)}{x-c},\ D,\ c,\ L\right).$$

The predicate $\mathrm{DerivativeAt}(f,c,L)$ asserts that L is the value of the limit of the difference quotient of f at c.

Remark (Negated quantified statement).

$$\forall L\in\mathbb{R}\ \exists\varepsilon>0\ \forall\delta>0\ \exists x\in D\left(0<|x-c|<\delta\ \wedge\ \left|\frac{f(x)-f(c)}{x-c}-L\right|\geq\varepsilon\right).$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DerivativeAt}(f,c,L) \iff \neg\,\mathrm{FunctionLimit}\left(\lambda x.\,\frac{f(x)-f(c)}{x-c},\ D,\ c,\ L\right).$$

Remark (Failure modes).

- The difference quotient has no finite limit at c (e.g. $|f(x)-f(c)|/|x-c|\to\infty$).
- The difference quotient oscillates without settling near any single value.
- The proposed value L is the wrong candidate.

Remark (Interpretation). The derivative is a limit of slopes of secant lines. The ε-δ form demands that the difference quotient can be made arbitrarily close to L by taking x sufficiently close to c. The input condition $0<|x-c|$ is punctured: $x=c$ is excluded because $(f(c)-f(c))/(c-c)$ is $0/0$.

Remark (Dependencies).

- Limit of a Function

Definition (Topological Definition of Derivative at a Point)

Definition 7.1.2 (Topological Definition of Derivative at a Point). Let $f:D\to\mathbb{R}$ and $c\in\mathrm{int}(D)$. f is differentiable at c with derivative L if for every neighbourhood V of L, there exists a neighbourhood U of c such that for all $x\in(U\setminus\{c\})\cap D$:

$$\frac{f(x)-f(c)}{x-c}\in V.$$

Remark (Standard quantified statement).

$$\forall V\ \text{neighbourhood of}\ L\ \exists U\ \text{neighbourhood of}\ c\ \forall x\in(U\setminus\{c\})\cap D:\ \frac{f(x)-f(c)}{x-c}\in V.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{top}}(f,c,L) \iff \forall V\ni L\ \exists U\ni c\ \forall x\in(U\setminus\{c\})\cap D:\ \frac{f(x)-f(c)}{x-c}\in V.$$

Remark (Negated quantified statement).

$$\exists V \text{ neighbourhood of } L\, \forall U \text{ neighbourhood of } c\, \exists x \in (U\setminus\{c\})\cap D: \frac{f(x)-f(c)}{x-c} \notin V.$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DerivativeAt}_{\mathrm{top}}(f,c,L) \iff \exists V \ni L\, \forall U \ni c\, \exists x \in (U\setminus\{c\})\cap D: \frac{f(x)-f(c)}{x-c} \notin V.$$

Remark (Failure modes). The topological derivative fails when a fixed output neighbourhood of L cannot be entered by the difference quotient no matter how tightly the input is restricted around c. Geometric reading: no punctured neighbourhood around c maps all secant slopes into any prescribed window around L.

Remark (Interpretation). The topological form replaces ε-balls with abstract neighbourhoods, making the definition coordinate-free and suited for generalisation to metric spaces and manifolds.

Remark (Dependencies).

- Function
- Relative Neighborhood
- Subtraction on $\mathbb{R}$
- Division on $\mathbb{R}$

Definition (Sequential Definition of Derivative at a Point)

Definition 7.1.3 (Sequential Definition of Derivative at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. We say that f is differentiable at c with derivative L if

$$\forall (x_n) \subseteq A\setminus\{c\}\ \left(x_n \to c \Rightarrow \tfrac{f(x_n)-f(c)}{x_n-c} \to L\right).$$

Remark (Standard quantified statement).

$$\forall (x_n) \subseteq A\setminus\{c\}:\ x_n \to c\ \Rightarrow\ \frac{f(x_n)-f(c)}{x_n-c} \to L.$$

Remark (Definition predicate reading).

$$\mathrm{DerivativeAt}_{\mathrm{seq}}(f,c,L) \iff \forall (x_n) \subseteq A\setminus\{c\}:\ x_n \to c\ \Rightarrow\ \frac{f(x_n)-f(c)}{x_n-c} \to L.$$

Remark (Negated quantified statement).

$$\exists (x_n) \subseteq A\setminus\{c\}:\ x_n \to c\ \wedge\ \frac{f(x_n)-f(c)}{x_n-c} \not\to L.$$

Remark (Negation predicate reading).

$$\neg\,\mathrm{DerivativeAt}_{\mathrm{seq}}(f,c,L) \iff \exists (x_n) \subseteq A\setminus\{c\}:\ x_n \to c\ \wedge\ \frac{f(x_n)-f(c)}{x_n-c} \not\to L.$$

Remark (Failure modes). Failure occurs when a single approach sequence $(x_n) \to c$ exists along which the difference quotients either: (i) diverge to infinity (vertical tangent); (ii) oscillate without settling; (iii) converge but to a value other than L (wrong slope candidate). A single such sequence certifies non-differentiability.

Remark (Failure mode decomposition).

$$\underbrace{x_n \to c}_{\text{approach condition}} \quad \wedge \quad \underbrace{\frac{f(x_n) - f(c)}{x_n - c} \not\to L}_{\text{quotient failure}}.$$

The two canonical witness types are: two subsequences with different limiting difference quotients (e.g., $f(x) = |x|$ at $c = 0$), or quotients growing without bound.

Remark (Interpretation). The sequential form is the primary tool for disproving differentiability: exhibit a single sequence $x_n \to c$ along which the difference quotient fails to converge to any single value.

Remark (Dependencies).

- Sequence
- Convergent Sequence
- Derivative at a Point
- Limit of a Function

Theorem (Equivalence of Definitions of the Derivative at a Point)

Theorem 7.1.4 (Equivalence of Definitions of the Derivative at a Point). *Let $f : A \to \mathbb{R}$ and $c \in A$ be a limit point of A. The following are equivalent:*

1. *f is differentiable at c in the ε-δ sense with derivative L.*
2. *f is differentiable at c in the topological sense with derivative L.*
3. *f is differentiable at c in the sequential sense with derivative L.*

Go to proof.

Remark (Standard quantified statement).

$$\text{(i)} \iff \text{(ii)} \iff \text{(iii)}.$$

Remark (Predicate reading).

$$\text{DerivativeAt}(f, c, L) \iff \text{DerivativeAt}_{\text{top}}(f, c, L) \iff \text{DerivativeAt}_{\text{seq}}(f, c, L).$$

All three predicates name the same underlying condition on the difference quotient of f at c.

Remark (Negated quantified statement).

$$\begin{aligned} \neg(\text{i}) \iff \neg(\text{ii}) \iff \neg(\text{iii}) \\ \iff \exists (x_n) \subseteq A \setminus \{c\} : x_n \to c \wedge \frac{f(x_n) - f(c)}{x_n - c} \not\to L. \end{aligned}$$

Remark (Failure modes). The equivalence cannot produce contradictory verdicts: if c is a limit point of A, all three forms agree. Degenerate case: if c is not a limit point of A, the punctured-ball and sequence conditions are vacuously satisfied by every candidate L simultaneously, making the equivalence trivially true for all L rather than characterizing a unique derivative.

Remark (Interpretation). The three forms are notational restatements of the same underlying limit condition on the difference quotient. Analysts choose whichever form is most convenient for the proof at hand.

Remark (Dependencies).

- Derivative at a Point
- Topological Definition
- Sequential Definition

Proposition (Derivative: h-Form Equivalence)

Proposition 7.1.5 (Derivative: h-Form Equivalence). *Let $f : A \to \mathbb{R}$ and $c \in \operatorname{int}(A)$. Then f is differentiable at c with derivative L if and only if*

$$\lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Go to proof.

Remark (Standard quantified statement).

$$\operatorname{DerivativeAt}(f, c, L) \iff \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} = L.$$

Remark (Negated quantified statement).

$$\neg \operatorname{DerivativeAt}(f, c, L) \iff \lim_{h \to 0} \frac{f(c+h) - f(c)}{h} \neq L.$$

Remark (Failure modes). The h-form fails in two branches: (i) The incremental ratio $(f(c+h) - f(c))/h$ has no finite limit as $h \to 0$ (corner, cusp, or vertical tangent at c). (ii) The ratio converges to a finite value different from L (proposed L is the wrong slope).

Remark (Interpretation). The h-form substitutes $x = c + h$ in the difference quotient. It is the standard notation in calculus and makes the incremental structure of the derivative explicit.

Remark (Dependencies).

- Derivative at a Point

Theorem (Differentiable Implies Continuous)

Theorem 7.1.6 (Differentiable Implies Continuous). *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, then f is continuous at c.*

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivableAt}(f, c) \;\Rightarrow\; \mathrm{ContinuousAt}(f, c).$$

Remark (Predicate reading).

$$\mathrm{DifferentiableAt}(f, c) \implies \mathrm{ContinuousAt}(f, c).$$

Remark (Failure modes). The implication can only fail in the reverse direction: continuity at c does not force differentiability at c.

Remark (Failure mode decomposition). The converse fails: $f(x) = |x|$ is continuous at 0 but not differentiable there.

Remark (Contrapositive quantified statement).

$$\neg\texttt{ContinuousAt}(f, c) \;\Rightarrow\; \neg\texttt{DifferentiableAt}(f, c).$$

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{ContinuousAt}(f, c) \implies \neg\,\mathrm{DifferentiableAt}(f, c).$$

Remark (Interpretation). Differentiability is strictly stronger than continuity: the existence of a well-defined tangent slope forces the function values to coalesce at $f(c)$ as $x \to c$.

Remark (Dependencies).

- Derivative at a Point
- Continuity at a Point

Theorem (Uniqueness of the Derivative)

Theorem 7.1.7 (Uniqueness of the Derivative). *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \,\wedge\, \mathrm{DerivativeAt}(f, c, L') \;\Rightarrow\; L = L'.$$

Remark (Predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \,\wedge\, \mathrm{DerivativeAt}(f, c, L') \implies L = L'.$$

Remark (Failure modes). The uniqueness proof assumes $L \neq L'$ and takes $\varepsilon = |L - L'|/2 > 0$. The definition supplies δ making the difference quotient within ε of both L and L' simultaneously, forcing $|L - L'| < 2\varepsilon = |L - L'|$, a contradiction. The cluster-point hypothesis on c is essential: without it, the punctured-ball condition is vacuous and the limit is trivially satisfied by every value.

Remark (Contrapositive quantified statement).

$$L \neq L' \;\Rightarrow\; \neg\big(\mathrm{DerivativeAt}(f, c, L) \;\wedge\; \mathrm{DerivativeAt}(f, c, L')\big).$$

Two distinct candidates cannot both be the derivative of f at c.

Remark (Contrapositive predicate reading).

$$L \neq L' \implies \neg\,\mathrm{DerivativeAt}(f, c, L) \;\vee\; \neg\,\mathrm{DerivativeAt}(f, c, L').$$

Remark (Interpretation). Uniqueness licenses writing $f'(c)$ as a definite value rather than a set of candidates.

Remark (Dependencies).

- Derivative at a Point

Proposition (Derivative of the Identity)

Proposition 7.1.8 (Derivative of the Identity). *Let $f(x) = x$ for all $x \in \mathbb{R}$. Then $f'(c) = 1$ for every $c \in \mathbb{R}$.*

Go to proof.

Remark (Standard quantified statement).

$$\forall c \in \mathbb{R} : \frac{d}{dx}[x]\Big|_{x=c} = 1.$$

Remark (Interpretation). The identity function has slope 1 everywhere. It is the base case for computing derivatives of polynomials via the algebra of derivatives.

Remark (Dependencies).

- Derivative at a Point
- Function

7.2 Secant Tangent

Secant and Tangent — toolkit

Concept family: The geometric objects underlying the derivative: secant lines and tangent lines.

Atomic items in this section:

- Definition: Secant Line
- Definition: Tangent Line

Definition (Secant Line)

Definition 7.2.1 (Secant Line). Let $f : A \to \mathbb{R}$, and let $x_1, x_2 \in A$ with $x_1 \neq x_2$. The *secant line* determined by the points $(x_1, f(x_1))$ and $(x_2, f(x_2))$ is the unique line passing through these two points. Its slope is

$$m_{\text{sec}}(x_1, x_2) := \frac{f(x_2) - f(x_1)}{x_2 - x_1}.$$

Remark (Standard quantified statement).

$$\forall x_1, x_2 \in A \ (x_1 \neq x_2 \Rightarrow \text{SecantLine}(f, x_1, x_2) \text{ has slope } \tfrac{f(x_2)-f(x_1)}{x_2-x_1}).$$

Remark (Interpretation). With increments $\Delta x := x_2 - x_1$ and $\Delta y := f(x_2) - f(x_1)$, the secant slope is $\Delta y/\Delta x$ -- the average rate of change of f over the interval $[x_1, x_2]$. This is an exact algebraic identity; no limit is involved. As $x_2 \to x_1$, if the secant slope converges, its limit is the derivative $f'(x_1)$ -- the instantaneous rate of change. Geometrically, the secant line rotates about the fixed point $(x_1, f(x_1))$ as x_2 approaches x_1, and the limit position is the tangent line.

Definition (Tangent Line)

Definition 7.2.2 (Tangent Line). Let $f : A \to \mathbb{R}$, let $c \in A$, and suppose that f is differentiable at c. The *tangent line* to the graph of f at the point $(c, f(c))$ is the line through $(c, f(c))$ with slope $f'(c)$. Its equation is

$$y = f(c) + f'(c)(x - c).$$

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \ \Rightarrow \ \text{LinearApproximationAt}(f, c) = f(c) + f'(c)(x - c).$$

Remark (Interpretation). The tangent line is the limit of the family of secant lines through $(c, f(c))$ and $(x, f(x))$ as $x \to c$. More precisely: the slope $m_{\text{sec}}(c, x) = (f(x) - f(c))/(x - c)$ converges to $f'(c)$, so the tangent line is the line whose slope is that limit. It is also the unique linear function $\ell(x) = f(c) + f'(c)(x - c)$ satisfying $\ell(c) = f(c)$ and $\ell'(c) = f'(c)$ -- the best first-order approximation to f near c. The error $f(x) - \ell(x) = o(x - c)$ as $x \to c$, meaning it vanishes faster than the displacement.

Remark (Dependencies).

- Definition: Secant Line
- Definition: Derivative at a Point

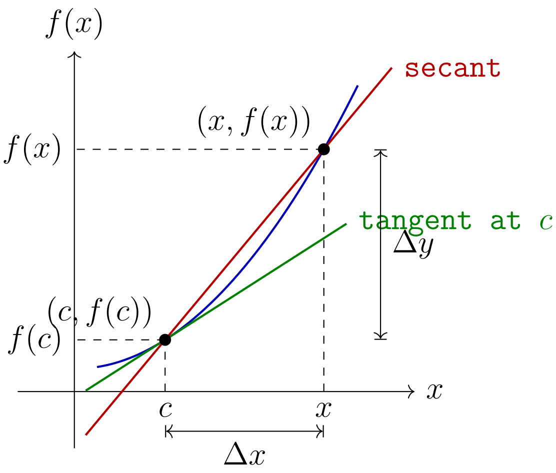


Figure 7.1: The secant line through $(c, f(c))$ and $(x, f(x))$ has slope $\Delta y/\Delta x$. As $x \to c$, the secant line rotates toward the tangent line at $(c, f(c))$, whose slope is $f'(c)$.

7.3 One Sided Derivatives

One-Sided Derivatives — toolkit

Concept family: Left-hand and right-hand derivatives, and their relation to full differentiability.

Atomic items in this section:

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Definition (Left-Hand Derivative)

Definition 7.3.1 (Left-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a left-hand limit point of I. We say that f has a *left-hand derivative* at c if the limit

$$\lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_-(c)$:

$$f'_-(c) := \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\ \forall \varepsilon > 0\ \exists \delta > 0\ \forall x \in I \Big(-(\delta) < x - c < 0 \implies \left| \frac{f(x) - f(c)}{x - c} - L \right| < \varepsilon \Big).$$

In this case $f'_-(c) = L$.

Remark (Definition predicate reading).

$$\text{LeftDerivativeAt}(f, c, L) \iff \text{LeftLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, c,\, L\right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\, \exists \varepsilon > 0\, \forall \delta > 0\, \exists x \in I\left(-\delta < x - c < 0 \;\wedge\; \left|\frac{f(x) - f(c)}{x - c} - L\right| \geq \varepsilon\right).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\, \neg\, \text{LeftDerivativeAt}(f, c, L).$$

No finite L is the left-hand limit of the difference quotient.

Remark (Interpretation). The left-hand derivative $f'_-(c)$ is the limit of the difference quotient as x approaches c *strictly from the left*. It measures the instantaneous rate of change of f at c as seen from the left side. The right-hand derivative $f'_+(c)$ is the symmetric object from the right. At interior points of an interval, both one-sided derivatives are defined, and the full derivative $f'(c)$ exists if and only if they agree. At endpoints, only the inward one-sided derivative is accessible, and it serves as the boundary derivative. The canonical example: for $f(x) = |x|$ at $c = 0$, $f'_-(0) = -1$ and $f'_+(0) = +1$; since these disagree, $f'(0)$ does not exist.

Remark (Dependencies).

- Definition: Derivative at a Point

Definition (Right-Hand Derivative)

Definition 7.3.2 (Right-Hand Derivative). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$ be such that c is a right-hand limit point of I. We say that f has a *right-hand derivative* at c if the limit

$$\lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}$$

exists. In that case, the value of this limit is denoted $f'_+(c)$:

$$f'_+(c) := \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c}.$$

Remark (Standard quantified statement).

$$\exists L \in \mathbb{R}\, \forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in I\left(0 < x - c < \delta \implies \left|\frac{f(x) - f(c)}{x - c} - L\right| < \varepsilon\right).$$

In this case $f'_+(c) = L$.

Remark (Definition predicate reading).

$$\text{RightDerivativeAt}(f, c, L) \iff \text{RightLimit}\left(\lambda x.\, \frac{f(x) - f(c)}{x - c},\, c,\, L\right).$$

Remark (Negated quantified statement).

$$\forall L \in \mathbb{R}\ \exists \varepsilon > 0\ \forall \delta > 0\ \exists x \in I \Big(0 < x - c < \delta\ \wedge\ \left| \frac{f(x) - f(c)}{x - c} - L \right| \geq \varepsilon \Big).$$

Remark (Negation predicate reading).

$$\forall L \in \mathbb{R}\ \neg\, \mathrm{RightDerivativeAt}(f, c, L).$$

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Left-Hand Derivative

Theorem

Theorem 7.3.3 (Differentiability and One-Sided Derivatives). *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Then f is differentiable at $c \in I$ if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal. In that case,*

$$f'(c) = f'_-(c) = f'_+(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \iff \Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L) \Big) \text{ for some } L \in \mathbb{R}$$

Remark (Negated quantified statement).

$$\neg\, \mathrm{DifferentiableAt}(f, c) \iff \neg \Big(\mathrm{LeftDerivativeAt}(f, c, L) \wedge \mathrm{RightDerivativeAt}(f, c, L) \Big) \text{ for all } L \in \mathbb{R}$$

That is: either a one-sided derivative fails to exist, or both exist but are unequal.

Remark (Interpretation). This theorem decomposes differentiability at an interior point into two one-sided conditions. It is the standard tool for proving non-differentiability: to show f is not differentiable at c, compute $f'_-(c)$ and $f'_+(c)$ and observe they disagree. For $f(x) = |x|$ at $c = 0$: $f'_-(0) = -1 \neq +1 = f'_+(0)$, so f is not differentiable at 0. The theorem also provides the correct definition of differentiability on a closed interval $[a, b]$: differentiable on the open interior (a, b) in the full sense, plus $f'_+(a)$ and $f'_-(b)$ both exist at the endpoints.

Remark (Dependencies).

- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Definition: Derivative at a Point

7.4 Algebra Of Derivatives

Algebra of Derivatives — toolkit

Concept family: The pointwise algebra rules for differentiation at a point, and their global (interval) forms.

Pointwise rules (at a point c):

- Theorem: Constant Multiple Rule
- Theorem: Sum Rule
- Theorem: Product Rule
- Theorem: Quotient Rule
- Corollary: Finite Sum Rule
- Corollary: Extended Product Rule
- Corollary: Power Rule for Integer Powers of a Function
- Theorem: Finite Linear Combination Rule

Global rules (on an interval I):

- Theorem: Constant Multiple Rule on an Interval
- Theorem: Sum Rule on an Interval
- Theorem: Product Rule on an Interval
- Theorem: Quotient Rule on an Interval
- Corollary: Finite Sum Rule on an Interval
- Corollary: Extended Product Rule on an Interval
- Corollary: Power Rule for Integer Powers of a Function on an Interval
- Theorem: Finite Linear Combination Rule on an Interval

Theorem (Constant Multiple Rule)

Theorem 7.4.1 (Constant Multiple Rule). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Go to proof. *Go to proof.*

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \;\Rightarrow\; \mathrm{DerivativeAt}(\alpha f,\, c,\, \alpha f'(c)).$$

Remark (Interpretation). Multiplying a function by a constant scales its derivative by the same constant. The proof is one line: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits passes it through the limit. Differentiation is therefore homogeneous of degree one with respect to scalar multiplication.

Remark (Dependencies).

- Definition: Derivative at a Point
- Constant Multiple Rule for Limits

Theorem (Sum Rule)

Theorem 7.4.2 (Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then $f + g$ is differentiable at c, and*

$$(f + g)'(c) = f'(c) + g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f, c) \wedge \mathrm{DifferentiableAt}(g, c) \;\Rightarrow\; \mathrm{DerivativeAt}(f + g,\, c,\, f'(c) + g'(c)).$$

Remark (Interpretation). The derivative of a sum is the sum of the derivatives. The proof is immediate: the difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients, and the Sum Rule for Limits distributes the limit across the sum. Together with the Constant Multiple Rule, this establishes that differentiation is a linear operation: $(\alpha f + \beta g)' = \alpha f' + \beta g'$.

Remark (Dependencies).

- Definition: Derivative at a Point
- Sum Rule for Limits

Theorem (Product Rule)

Theorem 7.4.3 (Product Rule). *Let $A \subseteq \mathbb{R}$, let $f, g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If f and g are differentiable at c, then fg is differentiable at c, and*

$$(fg)'(c) = f'(c)g(c) + f(c)g'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f,c) \wedge \mathrm{DifferentiableAt}(g,c) \;\Rightarrow\; \mathrm{DerivativeAt}(fg,\ c,\ f'(c)g(c)+f(c)g'(c)). \blacksquare$$

Remark (Interpretation). The derivative of a product is not simply the product of the derivatives. The correct formula adds two terms: the rate of change of f weighted by the current value of g, plus the rate of change of g weighted by the current value of f. The proof inserts the auxiliary term $f(x)g(c)$ to decouple the increments of f and g, isolating the individual difference quotients. Continuity of f at c (which follows from differentiability) is used to evaluate $\lim_{x\to c} f(x) = f(c)$ in one of the factors.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits
- Product Rule for Limits

Theorem (Quotient Rule)

Theorem 7.4.4 (Quotient Rule). *Let $A \subseteq \mathbb{R}$, let $f,g : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Suppose that f and g are differentiable at c and that $g(c) \neq 0$. Then f/g is differentiable at c, and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DifferentiableAt}(f,c) \wedge \mathrm{DifferentiableAt}(g,c) \wedge g(c) \neq 0 \;\Rightarrow\; \mathrm{DerivativeAt}\left(\frac{f}{g},\ c,\ \frac{f'(c)g(c)-f(c)g'(c)}{(g(c))^2}\right). \blacksquare$$

Remark (Interpretation). The quotient rule is the product rule applied to $f \cdot g^{-1}$, but stated directly. The hypothesis $g(c) \neq 0$ ensures the quotient is defined near c continuity of g (from its differentiability) and the quotient rule for limits guarantee $g(x) \neq 0$ in a deleted neighbourhood of c, so the difference quotient of f/g is well-defined throughout. The add-and-subtract technique with $f(c)g(c)$ decouples the increments of f and g in the numerator, exactly as in the product rule proof.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Differentiable Implies Continuous
- Sum Rule for Limits

- Product Rule for Limits
- Quotient Rule for Limits

Corollary

Corollary 7.4.5 (Finite Sum Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at c, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \in \{1, \ldots, n\}\ \text{DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)).$$

Remark (Dependencies).

- Theorem: Sum Rule
- Theorem: Constant Multiple Rule

Corollary

Corollary 7.4.6 (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^{n} f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \text{DifferentiableAt}(f_i, c) \Rightarrow \text{DerivativeAt}\Big(\textstyle\prod_i f_i,\ c,\ \sum_k f_k'(c) \prod_{i \neq k} f_i(c)\Big).$$

Remark (Dependencies).

- Theorem: Product Rule

Corollary

Corollary 7.4.7 (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f,c) \ \wedge \ n \in \mathbb{N} \ \Rightarrow \ \text{DerivativeAt}(f^n,\ c,\ n(f(c))^{n-1}f'(c)).$$

Remark (Dependencies).

- Corollary: Extended Product Rule

Theorem

Theorem 7.4.8 (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \dots, f_n : A \to \mathbb{R}$, let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \text{DifferentiableAt}(f_i, c) \ \Rightarrow \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\ c,\ \sum_i \alpha_i f_i'(c)).$$

Remark (Interpretation). The Finite Linear Combination Rule is the full statement that differentiation is a *linear operator*: it commutes with scalar multiplication and addition simultaneously, for any finite collection of differentiable functions. This is not a separate theorem from the Constant Multiple and Sum Rules it is their inductive closure. Stated in operator language: if $\mathcal{D}(I)$ denotes the vector space of functions differentiable at c, then $D_c : \mathcal{D}(I) \to \mathbb{R}$ defined by $D_c(f) = f'(c)$ is a linear functional. The Finite Linear Combination Rule says $D_c(\sum_i \alpha_i f_i) = \sum_i \alpha_i D_c(f_i)$.

Remark (Dependencies).

- Theorem: Constant Multiple Rule
- Theorem: Sum Rule

Theorem

Theorem 7.4.9 (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \text{DifferentiableAt}(f, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\alpha f,\ x,\ \alpha f'(x)).$$

Remark (Dependencies).

- Theorem: Constant Multiple Rule

Theorem

Theorem 7.4.10 (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f+g$ is differentiable on I, and*

$$(f+g)'(x) = f'(x) + g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \big(\mathrm{DifferentiableAt}(f,x) \wedge \mathrm{DifferentiableAt}(g,x)\big) \Rightarrow \forall x \in I\ \mathrm{DerivativeAt}(f+g,\ x,\ f'(x)+g'(x)).$$

Remark (Dependencies).

- Theorem: Sum Rule

Theorem

Theorem 7.4.11 (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \big(\mathrm{DifferentiableAt}(f,x) \wedge \mathrm{DifferentiableAt}(g,x)\big) \Rightarrow \forall x \in I\ \mathrm{DerivativeAt}(fg,\ x,\ f'(x)g(x)+f(x)$$

Remark (Dependencies).

- Theorem: Product Rule

Theorem

Theorem 7.4.12 (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I\ \big(\mathrm{DifferentiableAt}(f,x) \wedge \mathrm{DifferentiableAt}(g,x) \wedge g(x) \neq 0\big) \Rightarrow \forall x \in I\ \mathrm{DerivativeAt}\Big(\tfrac{f}{g},\ x,\ \tfrac{f'(x)g(x}{(}$$

Remark (Dependencies).

- Theorem: Quotient Rule

Corollary

Corollary 7.4.13 (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \mathrm{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \mathrm{DerivativeAt}\big(\textstyle\sum_i \alpha_i f_i,\ x,\ \sum_i \alpha_i f_i'(x)\big).$$

Remark (Dependencies).

- Corollary: Finite Sum Rule

Corollary

Corollary 7.4.14 (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \dots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^{n} f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i\ \forall x \in I\ \ \mathrm{DifferentiableAt}(f_i, x)\ \Rightarrow\ \forall x \in I\ \ \mathrm{DerivativeAt}\Big(\textstyle\prod_i f_i,\ x,\ \sum_k f_k'(x) \prod_{i \neq k} f_i(x)\Big). \blacksquare$$

Remark (Dependencies).

- Corollary: Extended Product Rule

Corollary

Corollary 7.4.15 (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1} f'(x) \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in I \ \text{DifferentiableAt}(f, x) \wedge n \in \mathbb{N} \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(f^n,\, x,\, n(f(x))^{n-1} f'(x)).$$

Remark (Dependencies).

- Corollary: Power Rule for Integer Powers of a Function

Theorem

Theorem 7.4.16 (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i \right)' (x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall i \ \forall x \in I \ \text{DifferentiableAt}(f_i, x) \ \Rightarrow \ \forall x \in I \ \text{DerivativeAt}(\textstyle\sum_i \alpha_i f_i,\, x,\, \sum_i \alpha_i f_i'(x)).$$

Remark (Interpretation). The pointwise statements above -- each proved at a single point c -- extend to global statements when the hypotheses hold at every point of I. If f is differentiable on I, then $f' : I \to \mathbb{R}$ is itself a function and differentiation acts as a linear operator on the vector space $\mathcal{D}(I)$ of functions differentiable on I:

$$(\alpha f + \beta g)' = \alpha f' + \beta g', \qquad (fg)' = f'g + fg', \qquad \left(\frac{f}{g} \right)' = \frac{f'g - fg'}{g^2}.$$

These equalities hold as functions on I. The Finite Linear Combination Rule in its global form is precisely the statement that $D : \mathcal{D}(I) \to \mathcal{F}(I)$ defined by $D(f) = f'$ is a linear map between function spaces, where $\mathcal{F}(I)$ denotes functions on I. The interval variants are not separate theorems -- they are the pointwise results collected into function notation by applying the pointwise result at each $x \in I$.

Remark (Dependencies).

- Theorem: Finite Linear Combination Rule

Remark (Inverse function -- forward note). When f is strictly monotone and differentiable on I with $f'(x) \neq 0$ for all $x \in I$, the inverse $g : J \to I$ is differentiable on $J = f(I)$ and

$$g' = \frac{1}{f' \circ g},$$

as functions on J. This is the global form of the Inverse Function Theorem for derivatives and is treated in the inverse function section.

7.5 Chain Rule

Carathéodory and Chain Rule — toolkit

Concept family: The Carathéodory factorization of the derivative, its characterization theorem, and the Chain Rule.

Atomic items in this section:

- Definition: Carathéodory Factorization
- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Chain Rule

Definition (Carathéodory Factorization)

Definition 7.5.1 (Carathéodory Factorization). Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. We say that f admits a *Carathéodory factorization* at c if there exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, f is differentiable at c and $f'(c) = \phi(c)$.

Remark (Standard quantified statement).

$$\exists \phi : I \to \mathbb{R}\ \big(\mathrm{ContinuousAt}(\phi, c)\ \wedge\ \forall x \in I\ [f(x) = f(c) + (x - c)\phi(x)]\big).$$

Remark (Definition predicate reading). The existence of a Carathéodory factorization at c is equivalent to $\mathrm{DifferentiableAt}(f, c)$ by the characterization theorem below. The factorizing function is explicit:

$$\phi(x) = \begin{cases} \dfrac{f(x) - f(c)}{x - c} & x \neq c, \\ f'(c) & x = c. \end{cases}$$

Differentiability of f at c is precisely the statement that this piecewise-defined ϕ is continuous at c.

Remark (Interpretation). The Carathéodory factorization replaces the difference quotient -- a function with a removable discontinuity at c -- with a function ϕ that is genuinely continuous at c. The factorization $f(x) - f(c) = (x - c)\phi(x)$ holds everywhere on I, including at $x = c$ where both sides equal zero. The derivative is recovered as $\phi(c)$ without ever dividing by zero. This reframing converts analytical limit conditions into topological continuity conditions, which compose and factor more cleanly. It is the preferred tool for proving the product rule, quotient rule, and chain rule, because continuity of a product or composition is easier to establish than the existence of a limit of a quotient.

Carathéodory vs. Lipschitz. Both control the increment $f(x) - f(c)$ by the factor $(x - c)$, but differently. The Carathéodory factorization requires the slope function $\phi(x)$ to extend *continuously* to c; it controls the *behaviour* of the slope, not just its size. The Lipschitz condition $|f(x) - f(c)| \leq L|x - c|$ only bounds the size of the increment. Carathéodory implies Lipschitz: if ϕ is continuous at c it is bounded near c, so $|f(x) - f(c)| = |\phi(x)||x - c| \leq K|x - c|$ for some K near c. The converse fails: $f(x) = |x|$ is Lipschitz on $\mathbb{R}$ with constant 1 but the slope function takes values ± 1 on either side of 0 with no continuous extension.

Remark (Dependencies).

- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 7.5.2 (Carathéodory Characterization of Differentiability). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$. *Go to proof.* *Go to proof.*

Remark (Standard quantified statement).

$$\text{DifferentiableAt}(f, c) \iff \exists \phi : I \to \mathbb{R} \,\big(\text{ContinuousAt}(\phi, c) \wedge \forall x \in I \,[f(x) = f(c) + (x - c)\phi(x)]\big).$$

In this case $\phi(c) = f'(c)$.

Remark (Interpretation). This theorem establishes that differentiability and the existence of a Carathéodory factorization are exactly the same condition. The proof is a two-direction construction: in the forward direction, define ϕ as the difference quotient extended by $f'(c)$ at c and verify continuity from the definition of the derivative; in the backward direction, recover the difference quotient from ϕ and read off its limit as $\phi(c)$. The theorem's power is that it converts every future differentiability argument into a continuity argument: to show $g \circ f$ is differentiable, find a continuous Carathéodory function for it; composing the Carathéodory functions of f and g yields one automatically.

Remark (Dependencies).

- Definition: Carathéodory Factorization
- Definition: Derivative at a Point
- Definition: Continuous at a Point

Theorem

Theorem 7.5.3 (Chain Rule). *Let $I, J \subseteq \mathbb{R}$ be intervals, let $f : I \to \mathbb{R}$ and $g : J \to \mathbb{R}$, and suppose that $f(I) \subseteq J$. Let $c \in I$. If f is differentiable at c and g is differentiable at $f(c)$, then the composition $g \circ f$ is differentiable at c, and*

$$(g \circ f)'(c) = g'(f(c))\, f'(c).$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{DifferentiableAt}(f, c) \ \wedge \ \text{DifferentiableAt}(g, f(c)) \\ &\quad \Rightarrow \ \text{DerivativeAt}(g \circ f,\, c,\, g'(f(c)) \cdot f'(c)).\end{aligned}$$

Remark (Interpretation). The chain rule computes the derivative of a composition by multiplying the derivatives at the respective points: the outer function g is differentiated at the intermediate value $f(c)$, and the inner function f is differentiated at c. The Carathéodory proof is the cleanest: let ϕ_f and ϕ_g be the Carathéodory functions of f at c and g at $f(c)$ respectively. Then

$$g(f(x)) - g(f(c)) = \phi_g(f(x)) \cdot (f(x) - f(c)) = \phi_g(f(x)) \cdot \phi_f(x) \cdot (x - c).$$

The function $x \mapsto \phi_g(f(x)) \cdot \phi_f(x)$ is continuous at c -- the product of two continuous functions -- and its value at c is $g'(f(c)) \cdot f'(c)$. By the characterization theorem, $g \circ f$ is differentiable with the stated derivative. The naive proof via difference quotients fails because $f(x) = f(c)$ for x near c is possible, causing division by zero; Carathéodory avoids this entirely.

Remark (Dependencies).

- Theorem: Carathéodory Characterization of Differentiability
- Theorem: Differentiable Implies Continuous
- Theorem: Composition of Limits
- Theorem: Product Rule for Function Limits

7.6 Derivative Geometry

Derivative Geometry — toolkit

Concept family: Relative extrema, the Interior Extremum Theorem, and the necessary condition for extrema.

Atomic items in this section:

- Definition: Relative Minimum
- Definition: Relative Maximum
- Definition: Relative Extremum
- Theorem: Interior Extremum Theorem
- Theorem: Necessary Condition for an Interior Extremum
- Corollary: Necessary Condition for an Extremum

Understanding Derivatives — toolkit

Concept family: Monotonicity, local behavior, and qualitative consequences of the derivative.

Atomic items in this section:

- Definition: Increasing at a Point
- Definition: Decreasing at a Point
- Theorem: Zero Derivative Implies Constant
- Corollary: Equal Derivatives Imply Difference Is Constant
- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Lemma: Positive Derivative Implies Locally Increasing
- Lemma: Negative Derivative Implies Locally Decreasing
- Theorem: First Derivative Test: Relative Maximum
- Theorem: First Derivative Test: Relative Minimum
- Theorem: Darboux's Theorem

Cross-references (canonical definitions in the functions chapter):

- Definition: Strictly Increasing Function
- Definition: Strictly Decreasing Function

Definition (Relative Minimum)

Definition 7.6.1 (Relative Minimum). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative minimum* at c if there exists $\delta > 0$ such that

$$f(c) \le f(x) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(c) < f(x) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative minimum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0 \, \forall x \in A \, \big(|x-c| < \delta \implies f(c) \le f(x)\big).$$

Remark (Definition predicate reading).

$$\text{LocalMinimum}(f, c) \iff \exists \delta > 0 \, \forall x \in A \, \big(\text{Within}(x, c, \delta) \implies f(c) \le f(x)\big).$$

Remark (Interpretation). f has a relative minimum at c if $f(c)$ is the smallest value of f in some open neighbourhood of c within the domain. The word *relative* (or *local*) signals that we are comparing $f(c)$ only to nearby values, not to all values of f globally. The strict variant excludes the degenerate case where nearby points share the minimum value. The symmetric notion -- relative maximum -- reverses the inequality. Together they constitute a relative extremum. These are purely neighbourhood conditions; no derivative appears in the definitions.

Remark (Dependencies).

- Function
- Subset
- Open Interval
- Order on $\mathbb{R}$

Definition (Relative Maximum)

Definition 7.6.2 (Relative Maximum). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative maximum* at c if there exists $\delta > 0$ such that

$$f(x) \le f(c) \qquad \text{for all } x \in A \cap (c-\delta, c+\delta).$$

If, in addition,

$$f(x) < f(c) \qquad \text{for all } x \in \big(A \cap (c-\delta, c+\delta)\big) \setminus \{c\},$$

then f has a *strict relative maximum* at c.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A\ \big(|x - c| < \delta \implies f(x) \leq f(c)\big).$$

Remark (Definition predicate reading).

$$\mathrm{LocalMaximum}(f, c) \iff \exists \delta > 0\ \forall x \in A\ \big(\mathrm{Within}(x, c, \delta) \implies f(x) \leq f(c)\big).$$

Remark (Dependencies).

- Definition: Relative Minimum

Definition (Relative Extremum)

Definition 7.6.3 (Relative Extremum). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f has a *relative extremum* at c if f has either a relative minimum at c or a relative maximum at c.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Definition predicate reading).

$$\mathrm{LocalExtremum}(f, c) \iff \mathrm{LocalMinimum}(f, c) \lor \mathrm{LocalMaximum}(f, c).$$

Remark (Dependencies).

- Definition: Relative Minimum
- Definition: Relative Maximum

Theorem

Theorem 7.6.4 (Interior Extremum Theorem). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and f is differentiable at c, then*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \land \mathrm{DifferentiableAt}(f, c) \Rightarrow \mathrm{DerivativeAt}(f, c, 0).$$

Remark (Contrapositive quantified statement).

$$\mathrm{DerivativeAt}(f, c, L) \land L \neq 0 \Rightarrow \neg \mathrm{LocalExtremum}(f, c).$$

If $f'(c)$ exists and is nonzero, then c is not a relative extremum.

Remark (Contrapositive predicate reading).

$$\mathrm{DerivativeAt}(f, c, L) \land L \neq 0 \Rightarrow \neg \mathrm{LocalMinimum}(f, c) \land \neg \mathrm{LocalMaximum}(f, c).$$

Remark (Interpretation). The theorem gives a necessary condition for interior extrema under differentiability: the derivative must vanish. The geometric picture is immediate -- at a local max or min, the tangent line must be horizontal, since the function is neither consistently increasing nor consistently decreasing. The proof is a sign argument: at a relative maximum, the difference quotient is non-positive for $x > c$ and non-negative for $x < c$; since the derivative is a single limit, it is squeezed to zero. The converse fails: $f'(c) = 0$ does not imply an extremum ($f(x) = x^3$ at $c = 0$). Critical points -- where $f'(c) = 0$ or $f'(c)$ does not exist -- are candidates for extrema, not guarantees.

Remark (Dependencies).

- Definition: Relative Extremum
- Definition: Derivative at a Point
- Definition: Left-Hand Derivative
- Definition: Right-Hand Derivative
- Theorem: Differentiability and One-Sided Derivatives

Theorem

Theorem 7.6.5 (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\mathrm{LocalExtremum}(f, c) \Rightarrow \neg\,\mathrm{DifferentiableAt}(f, c) \lor \mathrm{DerivativeAt}(f, c, 0).$$

Remark (Interpretation). This is a weakening of the Interior Extremum Theorem that drops the differentiability hypothesis. Without assuming $f'(c)$ exists, the conclusion is a disjunction: either the derivative fails to exist at c, or it exists and equals zero. This is the correct form for identifying *critical points* -- points that are candidates for extrema. The set of critical points is the union of points where f' is zero and points where f' does not exist.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem

Corollary

Corollary 7.6.6 (Necessary Condition for an Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then*

$$f'(c) = 0 \quad \text{or} \quad f'(c) \text{ does not exist.}$$

Go to proof. Go to proof.

Remark (Standard quantified statement).

$$\text{LocalExtremum}(f, c) \Rightarrow \text{DerivativeAt}(f, c, 0) \vee \neg\,\text{DifferentiableAt}(f, c).$$

Remark (Dependencies).

- Theorem: Necessary Condition for an Interior Extremum

Definition (Increasing at a Point)

Definition 7.6.7 (Increasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *increasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \le f(c) \le f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A \cap (c - \delta, c)\ \forall y \in A \cap (c, c + \delta)\ \big(f(x) \le f(c) \le f(y)\big).$$

Remark (Definition predicate reading). f is increasing at c if c lies between the values of f on either side in some punctured neighbourhood: f is smaller to the left and larger to the right (weakly). This is a strictly local notion -- it says nothing about f outside the δ-neighbourhood of c. A function can be increasing at c without being increasing (globally or on any interval). The non-strict inequalities permit the function to be flat on one or both sides of c, distinguishing this from the notion of a strict local increase.

Remark (Dependencies).

- Definition: Strictly Increasing Function (functions chapter)

Definition (Decreasing at a Point)

Definition 7.6.8 (Decreasing at a Point). Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$. We say that f is *decreasing at c* if there exists $\delta > 0$ such that for all

$$x \in A \cap (c - \delta, c) \quad \text{and} \quad y \in A \cap (c, c + \delta),$$

we have $f(x) \ge f(c) \ge f(y)$.

Remark (Standard quantified statement).

$$\exists \delta > 0\ \forall x \in A \cap (c-\delta, c)\ \forall y \in A \cap (c, c+\delta)\ \big(f(x) \geq f(c) \geq f(y)\big).$$

Remark (Dependencies).

- Definition: Increasing at a Point

Theorem (Zero Derivative Implies Constant)

Theorem 7.6.9 (Zero Derivative Implies Constant). *Let $I := [a,b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a,b). If*

$$f'(x) = 0 \qquad \textit{for all } x \in (a,b),$$

then f is constant on I. Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a,b)\ \mathrm{DerivativeAt}(f,x,0)\ \Rightarrow\ \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = C).$$

Remark (Contrapositive quantified statement).

$$\neg\big(\exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = C)\big)\ \Rightarrow\ \exists x \in (a,b)\ \neg\,\mathrm{DerivativeAt}(f,x,0).$$

If f is not constant on I, then f' is not identically zero on (a,b).

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f,I)\ \Rightarrow\ \exists x \in (a,b)\ (f'(x) \neq 0).$$

Remark (Interpretation). The MVT is the engine here: if $f'(x) = 0$ everywhere on (a,b), then for any two points $x_1 < x_2$ in I, the MVT supplies a c with $f(x_2) - f(x_1) = f'(c)(x_2 - x_1) = 0$, so $f(x_1) = f(x_2)$. Since this holds for all pairs, f is constant. The theorem is the foundation of antiderivative uniqueness: two antiderivatives of the same function differ by a constant. The contrapositive is used in ODE uniqueness arguments -- if a solution is not identically the zero solution, its derivative is not identically zero somewhere.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Corollary

Corollary 7.6.10 (Equal Derivatives Imply Difference Is Constant). *Let $I := [a,b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a,b). If*

$$f'(x) = g'(x) \qquad \textit{for all } x \in (a,b),$$

then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$. Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a,b)\ (f'(x) = g'(x)) \ \Rightarrow\ \exists C \in \mathbb{R}\ \forall x \in I\ (f(x) = g(x) + C).$$

Remark (Contrapositive quantified statement).

$$\forall C \in \mathbb{R}\ \exists x \in I\ (f(x) \neq g(x) + C) \ \Rightarrow\ \exists x \in (a,b)\ (f'(x) \neq g'(x)).$$

If $f - g$ is not a constant function, then f' and g' disagree somewhere.

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{ConstantOn}(f - g,\ I) \ \Rightarrow\ \exists x \in (a,b)\ (f'(x) \neq g'(x)).$$

Remark (Interpretation). This is the corollary that makes antiderivatives well-defined up to a constant: any two antiderivatives of the same function on an interval differ by a constant. The proof is one line -- apply the previous theorem to $h = f - g$, whose derivative is $(f-g)' = f' - g' = 0$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Theorem (Nondecreasing Iff Nonnegative Derivative)

Theorem 7.6.11 (Nondecreasing Iff Nonnegative Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if*

$$f'(x) \geq 0 \qquad \textit{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{NondecreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \geq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\mathrm{NondecreasingOn}(f, I).$$

If the derivative is negative at some point, the function is not nondecreasing.

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) < 0) \ \Rightarrow\ \neg\,\mathrm{NondecreasingOn}(f, I).$$

Remark (Interpretation). The derivative is the instantaneous rate of change; requiring it to be nonnegative everywhere exactly characterises functions that never decrease. The forward direction (f nondecreasing $\Rightarrow f' \geq 0$) follows from the definition: the difference quotient is nonnegative when f is nondecreasing, so the limit is nonnegative. The

reverse direction ($f' \geq 0 \Rightarrow$ nondecreasing) is a consequence of the MVT: for $x < y$, the MVT gives $f(y) - f(x) = f'(c)(y - x) \geq 0$.
Note carefully: $f' \geq 0$ characterises *nondecreasing* (weak), not *strictly increasing* (strict). Strict monotonicity requires $f' > 0$ at all but isolated points. The function $f(x) = x^3$ satisfies $f'(0) = 0$ but is strictly increasing; $f' \geq 0$ with equality only at isolated points is sufficient for strict increase, but $f' > 0$ everywhere is not necessary.

Remark (Dependencies).

- Theorem: Mean Value Theorem
- Definition: Derivative at a Point

Theorem

Theorem 7.6.12 (Nonincreasing Iff Nonpositive Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if*

$$f'(x) \leq 0 \qquad \text{for all } x \in I.$$

Go to proof.

Remark (Standard quantified statement).

$$\text{NonincreasingOn}(f, I) \iff \forall x \in I\ (f'(x) \leq 0).$$

Remark (Contrapositive quantified statement).

$$\exists x \in I\ (f'(x) > 0) \Rightarrow \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Contrapositive predicate reading).

$$\exists x \in I\ (f'(x) > 0) \Rightarrow \neg\,\text{NonincreasingOn}(f, I).$$

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Lemma

Lemma 7.6.13 (Positive Derivative Implies Locally Increasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that*

$$f(x) < f(c) \quad \text{for all } x \in I \text{ with } c - \delta < x < c,$$

and

$$f(x) > f(c) \quad \text{for all } x \in I \text{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f,c,L) \land L > 0 \;\Rightarrow\; \exists \delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) < f(c)\big) \land \big(c < x < c{+}\delta \Rightarrow f(x$$

Remark (Contrapositive quantified statement).

$$\forall \delta > 0\, \exists x \in I\, \big((c{-}\delta < x < c \land f(x) \geq f(c)) \lor (c < x < c{+}\delta \land f(x) \leq f(c))\big) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{LocallyIncreasingAt}(f,c) \;\Rightarrow\; f'(c) \leq 0.$$

Remark (Interpretation). If the instantaneous rate of change is strictly positive at c, the function is genuinely climbing through c: it is strictly below $f(c)$ immediately to the left and strictly above immediately to the right. The proof is a direct ε-δ argument: take $\varepsilon = f'(c)/2 > 0$; the definition of the derivative supplies δ such that the difference quotient is within ε of $f'(c)$, forcing it to be positive, which then forces the correct sign on $f(x) - f(c)$. This lemma is the key ingredient in the proof of the First Derivative Test.

Remark (Dependencies).

- Definition: Derivative at a Point

Lemma

Lemma 7.6.14 (Negative Derivative Implies Locally Decreasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose that f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that*

$$f(x) > f(c) \quad \textit{for all } x \in I \textit{ with } c - \delta < x < c,$$

and

$$f(x) < f(c) \quad \textit{for all } x \in I \textit{ with } c < x < c + \delta.$$

Go to proof.

Remark (Standard quantified statement).

$$\mathrm{DerivativeAt}(f,c,L) \land L < 0 \;\Rightarrow\; \exists \delta > 0\, \forall x \in I\, \big(c{-}\delta < x < c \Rightarrow f(x) > f(c)\big) \land \big(c < x < c{+}\delta \Rightarrow f(x$$

Remark (Contrapositive quantified statement).

$$\neg\,\mathrm{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{LocallyDecreasingAt}(f,c) \;\Rightarrow\; f'(c) \geq 0.$$

Remark (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

Theorem (First Derivative Test: Relative Maximum)

Theorem 7.6.15 (First Derivative Test: Relative Maximum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \geq 0 \quad \textit{for all } x \in (c - \delta, c), \qquad f'(x) \leq 0 \quad \textit{for all } x \in (c, c + \delta),$$

then f has a relative maximum at c. Go to proof.

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \ \wedge\ \forall x \in (c - \delta, c)\ (f'(x) \geq 0) \ \wedge\ \forall x \in (c, c + \delta)\ (f'(x) \leq 0)\right] \\ \Rightarrow \text{LocalMaximum}(f, c).$$

Remark (Interpretation). If the function is nondecreasing immediately to the left of c and nonincreasing immediately to the right, then c is a local maximum -- the function climbs to c and then falls away. The proof integrates: by the nondecreasing characterisation applied on $(c - \delta, c)$, $f(x) \leq f(c)$ for $x < c$; by the nonincreasing characterisation on $(c, c + \delta)$, $f(x) \leq f(c)$ for $x > c$. The sign condition on f' replaces the Interior Extremum Theorem's converse (which did not hold in general) -- here we are asserting a sufficient condition, not proving a necessary one. The test is applicable even when $f'(c) = 0$ or $f'(c)$ does not exist.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Theorem

Theorem 7.6.16 (First Derivative Test: Relative Minimum). *Let $I := [a, b] \subseteq \mathbb{R}$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a, b)$. Suppose f is differentiable on (a, c) and (c, b). If there exists $\delta > 0$ such that $(c - \delta, c + \delta) \subseteq I$ and*

$$f'(x) \leq 0 \quad \textit{for all } x \in (c - \delta, c), \qquad f'(x) \geq 0 \quad \textit{for all } x \in (c, c + \delta),$$

then f has a relative minimum at c. Go to proof.

Remark (Standard quantified statement).

$$\exists \delta > 0 \left[(c - \delta, c + \delta) \subseteq I \ \wedge\ \forall x \in (c - \delta, c)\ (f'(x) \leq 0) \ \wedge\ \forall x \in (c, c + \delta)\ (f'(x) \geq 0)\right] \\ \Rightarrow \text{LocalMinimum}(f, c).$$

Remark (Dependencies).

- Theorem: First Derivative Test: Relative Maximum

- Definition: Relative Minimum

Theorem (Darboux's Theorem)

Theorem 7.6.17 (Darboux's Theorem). *Let $I := [a,b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a,b)$ such that*

$$f'(c) = k.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall k \left(\min(f'(a), f'(b)) < k < \max(f'(a), f'(b)) \Rightarrow \exists c \in (a,b)\ (f'(c) = k)\right).$$

Remark (Contrapositive quantified statement).

$$\forall c \in (a,b)\ (f'(c) \neq k) \Rightarrow k \leq \min(f'(a), f'(b)) \vee k \geq \max(f'(a), f'(b)).$$

If f' skips the value k, then k is not strictly between $f'(a)$ and $f'(b)$.

Remark (Contrapositive predicate reading).

$$\neg\,\mathrm{IVP}(f', [a,b]) \Rightarrow f \text{ is not differentiable on } [a,b].$$

A function whose derivative fails the intermediate value property cannot be a derivative.

Remark (Interpretation). The Intermediate Value Theorem (IVT) says continuous functions have the intermediate value property (IVP). Darboux's theorem says derivatives also have the IVP -- even without being continuous. This is a striking rigidity result: the class of functions that can appear as derivatives is severely constrained. A derivative cannot have a jump discontinuity (it may be discontinuous, but only via oscillation). The proof does not use continuity of f'; it uses differentiability of f to construct an auxiliary function $g(x) = f(x) - kx$ whose derivative changes sign, then applies the Extreme Value Theorem and Interior Extremum Theorem to force an interior extremum where $g'(c) = f'(c) - k = 0$.

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

7.7 Mean Value Theorem

Mean Value Theorem — toolkit

Concept family: Rolle's Theorem and the Mean Value Theorem — the chapter's load-bearing existence results.

Atomic items in this section:

- Theorem: Rolle's Theorem
- Theorem: Mean Value Theorem

Theorem (Rolle's Theorem)

Theorem 7.7.1 (Rolle's Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a, b)$ such that*

$$f'(c) = 0.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b) \ \text{DifferentiableAt}(f, x) \ \wedge \ f(a) = f(b)\\ &\quad \Rightarrow \ \exists c \in (a, b) \ \text{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \text{DifferentiableOn}(f, (a, b)) \ \wedge \ f(a) = f(b)\\ &\quad \Longrightarrow \ \exists c \in (a, b) \ : \ \text{DerivativeAt}(f, c, 0).\end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \operatorname{sgn}(x - 1/2)$ on $[0, 1]$. Then $f(0) = f(1) = -1$, but f has a jump discontinuity at $x = 1/2$. The derivative does not exist anywhere, so there is no horizontal tangent.
- **Differentiability fails:** Let $f(x) = |x - 1/2|$ on $[0, 1]$. Then $f(0) = f(1) = 1/2$, and f is continuous, but f has a corner at $x = 1/2$ where the derivative is undefined. For all $c \neq 1/2$, we have $f'(c) = \pm 1 \neq 0$.
- **Equal endpoints fail:** Let $f(x) = x$ on $[0, 1]$. Then f is continuous and differentiable with $f'(x) = 1 > 0$ everywhere, but $f(0) = 0 \neq 1 = f(1)$.

Remark (Contrapositive quantified statement).

$\big(\forall c \in (a, b) \ : \ f'(c) \neq 0\big) \ \Rightarrow \ f$ not continuous on $[a, b] \vee f$ not differentiable on $(a, b) \vee f(a)$

Remark (Contrapositive predicate reading).

$$\begin{aligned}&\big(\forall c \in (a, b) \ : \ \neg\,\text{DerivativeAt}(f, c, 0)\big)\\ &\quad \Longrightarrow \ \neg\,\text{ContinuousOn}(f, [a, b]) \ \vee \ \neg\,\text{DifferentiableOn}(f, (a, b)) \ \vee \ f(a) \neq f(b).\end{aligned}$$

Remark (Interpretation). If a differentiable function returns to the same value at both endpoints of an interval, the derivative must vanish somewhere in between. Geometrically: if the curve starts and ends at the same height, at some interior point the tangent line is horizontal. The proof combines two ingredients -- the Extreme Value Theorem guarantees that a continuous function on a closed bounded interval attains its maximum and minimum, and the Interior Extremum Theorem guarantees that at an interior extremum the derivative is zero. If both extrema occur at the endpoints then f is constant and $f' \equiv 0$ throughout. Rolle's Theorem is the special case of the MVT where the secant slope is zero.

Remark (Dependencies).

- Theorem: Interior Extremum Theorem
- Theorem: Extreme Value Theorem
- Definition: Derivative at a Point

Theorem (Mean Value Theorem)

Theorem 7.7.2 (Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that*

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Go to proof.

Remark (Standard quantified statement).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \forall x \in (a, b)\ \text{DifferentiableAt}(f, x) \\ &\Rightarrow \ \exists c \in (a, b)\ \text{DerivativeAt}\Big(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Predicate reading).

$$\begin{aligned}&\text{ContinuousOn}(f, [a, b]) \ \wedge \ \text{DifferentiableOn}(f, (a, b)) \\ &\Longrightarrow \ \exists c \in (a, b)\ :\ \text{DerivativeAt}\Big(f,\ c,\ \tfrac{f(b)-f(a)}{b-a}\Big).\end{aligned}$$

Remark (Failure modes). Each hypothesis is essential:

- **Continuity fails:** Let $f(x) = \begin{cases} 0 & \text{if } x \in [0, 1/2) \\ 1 & \text{if } x \in [1/2, 1] \end{cases}$. Then f has a jump discontinuity at $x = 1/2$, and the secant slope is $\frac{f(1)-f(0)}{1-0} = 1$. But $f'(x) = 0$ wherever the derivative exists, so no interior point achieves the average rate of change.
- **Differentiability fails:** Let $f(x) = |x|$ on $[-1, 1]$. Then f is continuous and the secant slope is $\frac{f(1)-f(-1)}{1-(-1)} = \frac{1-1}{2} = 0$. But $f'(x) = \pm 1$ everywhere f is differentiable (i.e., for $x \neq 0$), so no interior point has derivative equal to the secant slope.

The MVT is an existence result. It guarantees some c exists but gives no formula for it.

Remark (Contrapositive quantified statement).

$$\left(\forall c \in (a,b) \;:\; f'(c) \neq \frac{f(b)-f(a)}{b-a}\right) \Rightarrow f \text{ not continuous on } [a,b] \vee f \text{ not differentiable on } ($$

Remark (Contrapositive predicate reading).

$$\Big(\forall c \in (a,b) \;:\; \neg\,\mathrm{DerivativeAt}\Big(f,\, c,\, \tfrac{f(b)-f(a)}{b-a}\Big)\Big)$$
$$\Longrightarrow \neg\,\mathrm{ContinuousOn}(f,[a,b]) \;\vee\; \neg\,\mathrm{DifferentiableOn}(f,(a,b)).$$

Remark (Interpretation). The MVT asserts that the instantaneous rate of change equals the average rate of change at some interior point. Geometrically: the tangent line at some point c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. The proof reduces to Rolle's Theorem by subtracting the secant: define $g(x) = f(x) - \frac{f(b)-f(a)}{b-a}(x-a)$, which satisfies $g(a) = g(b) = f(a)$, then apply Rolle's. The theorem is an existence result, not a construction: it guarantees some c exists but gives no formula for it. Its immediate corollaries -- the constant function corollary, the monotonicity corollary, and the Lipschitz bound corollary -- each flow from the MVT by a one-line argument and are the theorem's primary applications.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

7.8 Taylor Expansion

Taylor Expansion Toolkit

Concept family: Taylor polynomials, Taylor remainders, and Taylor's theorem.

Atomic items in this section:

- Definition: Taylor Polynomial at a Point
- Definition: Taylor Remainder
- Definition: Maclaurin Polynomial
- Theorem: Taylor's Theorem with Lagrange Remainder
- Corollary: Taylor Expansion with Peano Remainder

Taylor Expansion

Taylor expansion approximates a differentiable function near a point by a polynomial whose coefficients are forced by the derivatives at that point. The polynomial records the local data, while the remainder records the error left after truncating the expansion.

Definition (Taylor Polynomial at a Point)

Definition 7.8.1 (Taylor Polynomial at a Point). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives $f^{(k)}(a)$ for $0 \le k \le n$. The **Taylor polynomial of order** n for f at a is

$$T_{n,a}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k.$$

Remark (Standard quantified statement).

$$\begin{aligned}
&\text{Let } I \subseteq \mathbb{R},\ a \in I,\ n \in \mathbb{N},\ f : I \to \mathbb{R}.\\
&\text{If } f^{(k)}(a) \text{ exists for every } 0 \le k \le n, \text{ then}\\
&\forall x \in I \left(T_{n,a}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k \right).
\end{aligned}$$

Remark (Interpretation). The Taylor polynomial is the unique polynomial of degree at most n whose first n derivatives at a agree with those of f. Its coefficients are not chosen by fitting sample points; they are dictated by derivative data at one point.

Remark (Dependencies).

- Derivative at a Point
- Power Rule for Integer Powers of a Function
- Finite Sum Rule

Definition (Taylor Remainder)

Definition 7.8.2 (Taylor Remainder). Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have a Taylor polynomial $T_{n,a}f$. The **Taylor remainder of order** n at a is the function $R_{n,a}f : I \to \mathbb{R}$ defined by

$$R_{n,a}f(x) := f(x) - T_{n,a}f(x).$$

Remark (Standard quantified statement).

$$\forall x \in I \left(R_{n,a}f(x) = f(x) - T_{n,a}f(x) \right).$$

Remark (Interpretation). The remainder is the exact error made by replacing f with its Taylor polynomial. Taylor's theorem is valuable because it gives usable formulas or estimates for this error.

Remark (Dependencies).

- Taylor Polynomial at a Point

Definition (Maclaurin Polynomial)

Definition 7.8.3 (Maclaurin Polynomial). Let $n \in \mathbb{N}$, and let f be defined on an interval containing 0 with derivatives $f^{(k)}(0)$ for $0 \leq k \leq n$. The **Maclaurin polynomial of order** n for f is

$$T_{n,0}f(x) := \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k.$$

Remark (Standard quantified statement).

$$\forall x \in I \left(T_{n,0}f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(0)}{k!} x^k \right).$$

Remark (Interpretation). Maclaurin polynomials are Taylor polynomials centered at the origin. They are not a separate approximation theory; they are the centered-at-zero case of the same construction.

Remark (Dependencies).

- Taylor Polynomial at a Point

Theorem (Taylor's Theorem with Lagrange Remainder)

Theorem 7.8.4 (Taylor's Theorem with Lagrange Remainder). *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Go to proof.

Remark (Standard quantified statement).

$$\forall x \in (a, b)\ \exists c \in (a, x) \left(f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \right).$$

Remark (Interpretation). Taylor's theorem says that the error after truncating at degree n has the same shape as the next Taylor term, but with the next derivative evaluated at some intermediate point rather than at the center. The point c is generally not known explicitly; the theorem is most often used to estimate the size of the remainder.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder

- Mean Value Theorem
- Finite Sum Rule
- Power Rule for Integer Powers of a Function

Corollary (Taylor Expansion with Peano Remainder)

Corollary 7.8.5 (Taylor Expansion with Peano Remainder). *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \text{as } x \to a.$$

Go to proof.

Remark (Standard quantified statement).

$$\lim_{x \to a} \frac{f(x) - \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k}{(x-a)^n} = 0.$$

Remark (Interpretation). The Peano form is a local asymptotic statement. It says that the Taylor polynomial captures all terms through order n, and the remaining error is smaller than $(x-a)^n$ near the center.

Remark (Dependencies).

- Taylor Polynomial at a Point
- Taylor Remainder
- Limit of a Function
- Derivative at a Point

Proofs

Remark (Return). Return to Theorem

Theorem (Equivalence of Definitions of the Derivative at a Point). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. Let $L \in \mathbb{R}$. The following are equivalent:*

(i) ***(Epsilon-Delta)*** $\forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in A\, \left(0 < |x - c| < \delta \Rightarrow \left|\frac{f(x)-f(c)}{x-c} - L\right| < \varepsilon\right)$.

(ii) ***(Neighbourhood)*** *For every $\varepsilon > 0$, there exists a neighbourhood V of c such that $x \in (A \cap V) \setminus \{c\} \Rightarrow \left|\frac{f(x)-f(c)}{x-c} - L\right| < \varepsilon$.*

(iii) ***(Sequential)*** $\forall (x_n) \subseteq A \setminus \{c\}\, \left(x_n \to c \Rightarrow \frac{f(x_n)-f(c)}{x_n-c} \to L\right)$.

Professional Standard Proof.
Let $g : A \setminus \{c\} \to \mathbb{R}$ denote the difference quotient function $g(x) = \frac{f(x)-f(c)}{x-c}$. The three conditions are precisely the epsilon-delta, neighbourhood, and sequential characterisations of $\lim_{x\to c} g(x) = L$, applied to g at the limit point c of $A\setminus\{c\}$. Their equivalence is therefore exactly the equivalence of the three standard definitions of a function limit, which is a theorem in the limits chapter. Since the equivalence of those three definitions is already established, the equivalence of (i)-(iii) follows immediately by applying that result to g. □

Detailed Learning Proof.
Step 1. Reduce to a function limit. Define the difference quotient function $g : A \setminus \{c\} \to \mathbb{R}$ by

$$g(x) := \frac{f(x) - f(c)}{x - c}.$$

This function is well-defined on $A \setminus \{c\}$, and c is a limit point of $A \setminus \{c\}$ since c is a limit point of A. All three conditions (i)-(iii) assert $\lim_{x\to c} g(x) = L$ in three different equivalent formulations.
Step 2. Identify (i) as the epsilon-delta definition of $\lim_{x\to c} g(x) = L$. Condition (i) states:

$$\forall \varepsilon > 0\, \exists \delta > 0\, \forall x \in A\, \left(0 < |x - c| < \delta \Rightarrow |g(x) - L| < \varepsilon\right).$$

This is verbatim the epsilon-delta definition of $\mathrm{FunctionLimit}(g, A \setminus \{c\}, c, L)$.
Step 3. Identify (ii) as the neighbourhood definition of $\lim_{x\to c} g(x) = L$. A neighbourhood V of c with $|c' - c| < r$ for some $r > 0$ corresponds to the punctured ball $B_r(c) \cap A = A \cap (c - r, c + r)$. The condition $x \in (A \cap V) \setminus \{c\}$ with $|g(x) - L| < \varepsilon$ is the neighbourhood formulation of the function limit. The epsilon-delta and neighbourhood formulations of function limits are equivalent by the standard theorem on function limits.
Step 4. Identify (iii) as the sequential criterion for $\lim_{x\to c} g(x) = L$. Condition (iii) asserts that for every sequence $(x_n) \subseteq A \setminus \{c\}$ converging to c, the sequence $g(x_n)$ converges to L. This is exactly the sequential criterion for function limits. The sequential criterion is equivalent to the epsilon-delta definition by the standard theorem on function limits.

Step 5. Conclude the three-way equivalence. Since (i), (ii), and (iii) are respectively the epsilon-delta, neighbourhood, and sequential formulations of the same statement $\lim_{x \to c} g(x) = L$, and these three formulations are mutually equivalent for any function limit at a limit point, all three are equivalent. □

Remark (Proof structure). This is a reduction proof. The key insight is that all three conditions are formulations of the same underlying object: the limit of the difference quotient function $g(x) = (f(x) - f(c))/(x - c)$ at c. Once this identification is made, the three-way equivalence follows immediately from the already-established equivalence of the three standard function limit definitions. The proof requires no ε-δ calculation; it is purely structural.

Remark (Dependencies).

- Definition: Derivative at a Point (epsilon–delta)
- Definition: Topological Definition of Derivative at a Point
- Definition: Sequential Definition of Derivative at a Point
- Equivalence of Function Limit Definitions

Remark (Return). Return to Proposition

Proposition (Derivative: h-Form Equivalence). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

1. *f is differentiable at c.*
2. *The limit $\lim\limits_{h\to 0} \dfrac{f(c+h)-f(c)}{h}$ exists.*

In this case, $f'(c) = \lim\limits_{h\to 0} \dfrac{f(c+h)-f(c)}{h}$.

Professional Standard Proof.
The substitution $h = x - c$ is a bijection from $\{x \in I : x \neq c\}$ to $\{h \in \mathbb{R} : h \neq 0, c + h \in I\}$. Under this substitution, $x \to c$ if and only if $h \to 0$, and $\frac{f(x)-f(c)}{x-c} = \frac{f(c+h)-f(c)}{h}$ identically. Therefore the two limits exist simultaneously and are equal when either exists. □

Detailed Learning Proof.
Step 1. Identify the substitution relating the two forms. Set $h = x - c$, equivalently $x = c + h$. For $x \in I$ with $x \neq c$, this gives $h \neq 0$ and $c + h \in I$. The substitution is bijective: every admissible x corresponds to a unique admissible h and vice versa.
Step 2. Verify the difference quotients are identical under the substitution. For any such x and h:

$$\frac{f(x)-f(c)}{x-c} = \frac{f(c+h)-f(c)}{h}.$$

This is a pure algebraic identity requiring no limits.
Step 3. Verify the limit variable transforms correctly. As $x \to c$ through $I \setminus \{c\}$, $h = x - c \to 0$; conversely as $h \to 0$ with $c + h \in I$, $x = c + h \to c$. The correspondence between the limit processes is exact.
Step 4. Conclude the equivalence. Since the same function of the same variable is approached through the same deleted neighbourhood, the two limits exist simultaneously and agree:

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = L \iff \lim_{h\to 0} \frac{f(c+h)-f(c)}{h} = L.$$

Hence f is differentiable at c (in the x-form sense) if and only if the h-form limit exists, and both equal $f'(c)$ when they do. □

Remark (Proof structure). This is a direct equivalence proof by change of variables. The substitution $h = x - c$ is a translation that converts one parameterisation of the punctured neighbourhood of c into another. Because the substitution is bijective and the difference quotients are identical under it, the two limit definitions are literally the same limit under different notation. No ε-δ argument is needed beyond the observation that the substitution preserves the limit process.

Remark (Dependencies).

- Definition: Derivative at a Point

Remark (Return). Return to Theorem

Theorem (Differentiability Implies Continuity). *Let f be defined on an open interval I containing c. If f is differentiable at c, then f is continuous at c.*

Professional Standard Proof.
Assume f is differentiable at c. For any $x \in I$ with $x \neq c$, the identity $f(x) - f(c) = \frac{f(x)-f(c)}{x-c} \cdot (x - c)$ holds. Taking limits as $x \to c$ and applying the Product Rule for Limits yields $\lim_{x\to c}[f(x) - f(c)] = \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \cdot \lim_{x\to c}(x - c) = f'(c) \cdot 0 = 0$. Therefore $\lim_{x\to c} f(x) = f(c)$, establishing continuity at c. □

Detailed Learning Proof.
Step 1. Factor the vertical displacement using secant identity. For any $x \in I$ with $x \neq c$, the vertical displacement admits the factorization $f(x) - f(c) = \left(\frac{f(x)-f(c)}{x-c}\right) \cdot (x - c)$. This identity holds on all of $I \setminus \{c\}$ by arithmetic; no limit has been taken.
Step 2. Distribute the limit over the product. Taking the limit as $x \to c$ and applying the Product Rule for Limits: $\lim_{x\to c}[f(x) - f(c)] = \left(\lim_{x\to c} \frac{f(x)-f(c)}{x-c}\right) \cdot (\lim_{x\to c}(x - c))$. The Product Rule applies because both limits on the right exist -- the first by the differentiability hypothesis, the second trivially.
Step 3. Evaluate each factor. By definition of differentiability at a point, the first factor equals $f'(c)$, a finite real number. The second factor satisfies $\lim_{x\to c}(x - c) = 0$ by the Identity Limit. Therefore: $\lim_{x\to c}[f(x) - f(c)] = f'(c) \cdot 0 = 0$.
Step 4. Conclude continuity. Since $\lim_{x\to c}[f(x) - f(c)] = 0$, it follows that $\lim_{x\to c} f(x) = f(c)$. By definition of continuity at a point, f is continuous at c. □

Remark (Proof structure). The proof employs a direct approach, hinging on the algebraic factorization of $f(x) - f(c)$ into the product of the difference quotient and the displacement. This converts the continuity problem into a limit-of-product problem, where the derivative hypothesis controls one factor and the displacement annihilates the product.

Remark (Dependencies).

- Derivative at a Point
- Product Rule for Limits
- Identity Limit
- Definition of Continuity at a Point

Remark (Return). Return to Theorem

Theorem (Uniqueness of the Derivative). *Let $f : A \to \mathbb{R}$ and $c \in A$. If f is differentiable at c, its derivative is unique.*

Professional Standard Proof.
TODO: professional standard proof for uniqueness-of-the-derivative. □

Detailed Learning Proof.
TODO: detailed learning proof for uniqueness-of-the-derivative. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Proposition

Proposition (Derivative of the Identity). *Let $I \subseteq \mathbb{R}$ be an open interval, and let $\mathrm{id}_I : I \to \mathbb{R}$ be the identity function defined by $\mathrm{id}_I(x) := x$. Then id_I is differentiable at every $c \in I$, and $\mathrm{id}_I'(c) = 1$.*

Professional Standard Proof.
For any $c \in I$ and $x \in I$ with $x \neq c$, the difference quotient of id_I is

$$\frac{\mathrm{id}_I(x) - \mathrm{id}_I(c)}{x - c} = \frac{x - c}{x - c} = 1.$$

Since the difference quotient is identically 1 on $I \backslash \{c\}$, its limit as $x \to c$ is 1. Therefore $\mathrm{id}_I'(c) = 1$. □

Detailed Learning Proof.
Step 1. Write out the difference quotient. Fix $c \in I$. For any $x \in I$ with $x \neq c$, substitute $\mathrm{id}_I(x) = x$ and $\mathrm{id}_I(c) = c$:

$$\frac{\mathrm{id}_I(x) - \mathrm{id}_I(c)}{x - c} = \frac{x - c}{x - c}.$$

Step 2. Simplify the difference quotient. Since $x \neq c$, the denominator $x - c \neq 0$, so cancellation is valid:

$$\frac{x - c}{x - c} = 1 \qquad \text{for all } x \in I \setminus \{c\}.$$

The difference quotient is the constant function 1 on the entire punctured domain $I \setminus \{c\}$.
Step 3. Evaluate the limit. A constant function has that constant as its limit at every point. Therefore

$$\mathrm{id}_I'(c) = \lim_{x \to c} \frac{\mathrm{id}_I(x) - \mathrm{id}_I(c)}{x - c} = \lim_{x \to c} 1 = 1.$$

Step 4. Conclude. Since $c \in I$ was arbitrary, id_I is differentiable at every point of I with derivative 1 throughout. □

Remark (Proof structure). This is a direct computation from the definition of the derivative. The difference quotient of the identity function simplifies to 1 identically, making the limit trivial. No limit law beyond the constant function limit is required.

Remark (Dependencies).

- Definition: Derivative at a Point

Remark (Return). Return to Theorem

Theorem (Uniqueness of the Derivative). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f is differentiable at c, then $f'(c)$ is unique.*

Professional Standard Proof.
Suppose both L_1 and L_2 satisfy the definition of the derivative at c. Then both are values of the limit $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$. By the Uniqueness of Limits of Functions, this limit has at most one value. Therefore $L_1 = L_2$. □

Detailed Learning Proof.
Step 1. Suppose both L_1 and L_2 satisfy the definition of the derivative at c. By definition of the derivative, this means both L_1 and L_2 equal $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$. The difference quotient $\frac{f(x)-f(c)}{x-c}$ is a function of x defined for all $x \in I$ with $x \neq c$.
Step 2. Since I is an open interval and $c \in I$, the point c is a cluster point of the domain $I \setminus \{c\}$. The Uniqueness of Limits of Functions theorem states that if a function has a limit at a cluster point, then that limit is unique.
Step 3. Therefore, the limit $\lim_{x\to c} \frac{f(x)-f(c)}{x-c}$ can have at most one value. Since both L_1 and L_2 equal this limit, it follows that $L_1 = L_2$. □

Remark (Proof structure). This is a direct proof by reduction to a previously established result. The strategy is to recognize that the derivative is defined as a limit of the difference quotient, then apply the general theorem that limits are unique. The proof works because it reduces a specific question about derivatives to the general principle of uniqueness of limits.

Remark (Dependencies).

- Derivative at a Point
- Uniqueness of Limits of Functions

Remark (Return). Return to Theorem

Theorem (Differentiability and One-Sided Derivatives). *Let $I \subseteq \mathbb{R}$ be an open interval and let $f : I \to \mathbb{R}$. Fix $c \in I$. Then f is differentiable at c if and only if both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and are equal, in which case $f'(c) = f'_-(c) = f'_+(c)$.*

Professional Standard Proof.
($\Rightarrow$) If f is differentiable at c, then by definition the two-sided limit $\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = f'(c)$ exists. By the characterization of two-sided limits in terms of one-sided limits, both $f'_-(c)$ and $f'_+(c)$ exist and equal $f'(c)$.
($\Leftarrow$) If both $f'_-(c)$ and $f'_+(c)$ exist and equal some $L \in \mathbb{R}$, then both one-sided limits of the difference quotient exist and equal L. By the characterization of two-sided limits, the two-sided limit $\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = L$ exists, so f is differentiable at c with $f'(c) = L$. $\square$

Detailed Learning Proof.
Step 1. Establish the forward implication: differentiability implies equal one-sided derivatives.
Assume f is differentiable at c. By the definition of differentiability, the two-sided limit

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = f'(c)$$

exists. The fundamental theorem relating two-sided and one-sided limits states that a two-sided limit exists if and only if both corresponding one-sided limits exist and are equal. Applying this characterization to the difference quotient, both

$$f'_-(c) = \lim_{x\to c^-} \frac{f(x)-f(c)}{x-c} \quad \text{and} \quad f'_+(c) = \lim_{x\to c^+} \frac{f(x)-f(c)}{x-c}$$

exist and satisfy $f'_-(c) = f'_+(c) = f'(c)$.
Step 2. Establish the reverse implication: equal one-sided derivatives imply differentiability.
Assume both one-sided derivatives $f'_-(c)$ and $f'_+(c)$ exist and equal some common value $L \in \mathbb{R}$. This means

$$\lim_{x\to c^-} \frac{f(x)-f(c)}{x-c} = L \quad \text{and} \quad \lim_{x\to c^+} \frac{f(x)-f(c)}{x-c} = L.$$

Since both one-sided limits of the difference quotient exist and have the same value L, the characterization of two-sided limits guarantees that the two-sided limit exists and equals L:

$$\lim_{x\to c} \frac{f(x)-f(c)}{x-c} = L.$$

By the definition of differentiability, this establishes that f is differentiable at c with derivative $f'(c) = L = f'_-(c) = f'_+(c)$. $\square$

Remark (Proof structure). This is a pure equivalence proof that reduces entirely to the characterization of two-sided limits in terms of one-sided limits. The derivative is defined as a two-sided limit of the difference quotient, so the equivalence between differentiability and equal one-sided derivatives follows immediately from applying the one-sided limit criterion to this defining limit. No calculation or construction is required the result is a direct translation of the limit-theoretic characterization.

Remark (Dependencies).

- Definition of Differentiability at a Point
- Two-Sided Limit and One-Sided Limits

Remark (Return). Return to Theorem

Theorem (Constant Multiple Rule). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $\alpha \in \mathbb{R}$. If f is differentiable at c, then αf is differentiable at c, and*

$$(\alpha f)'(c) = \alpha f'(c).$$

Professional Standard Proof.
TODO: professional standard proof for constant-multiple-rule. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-multiple-rule. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sum Rule). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, and let $f, g : I \to \mathbb{R}$ be differentiable at c. Then $f + g$ is differentiable at c and $(f+g)'(c) = f'(c) + g'(c)$.*

Professional Standard Proof.
Let $h := f + g$. For $x \in I$ with $x \neq c$, the difference quotient expands as

$$\frac{h(x) - h(c)}{x - c} = \frac{(f(x) + g(x)) - (f(c) + g(c))}{x - c} = \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c}.$$

By the Sum Rule for Limits,

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} + \lim_{x \to c} \frac{g(x) - g(c)}{x - c} = f'(c) + g'(c),$$

where the individual limits exist and equal $f'(c)$ and $g'(c)$ by differentiability of f and g at c. □

Detailed Learning Proof.
Step 1. Form and split the difference quotient of $h := f + g$. For $x \in I$ with $x \neq c$, substituting $h(x) = f(x) + g(x)$ and grouping terms gives

$$\frac{h(x) - h(c)}{x - c} = \frac{(f(x) + g(x)) - (f(c) + g(c))}{x - c} = \frac{f(x) - f(c)}{x - c} + \frac{g(x) - g(c)}{x - c}.$$

This step relies purely on arithmetic of real numbers and the linearity of addition.
Step 2. Pass to the limit using the Sum Rule for Limits. Since both f and g are differentiable at c, the limits $\lim_{x \to c} \frac{f(x) - f(c)}{x - c}$ and $\lim_{x \to c} \frac{g(x) - g(c)}{x - c}$ exist and equal $f'(c)$ and $g'(c)$ respectively. By the Sum Rule for Limits,

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \lim_{x \to c} \frac{f(x) - f(c)}{x - c} + \lim_{x \to c} \frac{g(x) - g(c)}{x - c} = f'(c) + g'(c).$$

Therefore h is differentiable at c and $h'(c) = f'(c) + g'(c)$. □

Remark (Proof structure). This is a direct proof using algebraic manipulation and limit properties. The difference quotient of $f + g$ splits linearly into the sum of the individual difference quotients by arithmetic. The Sum Rule for Limits then distributes the limit across the sum, and differentiability of f and g supplies the two derivative values.

Remark (Dependencies).

- Definition of Differentiability at a Point
- Sum Rule for Limits

Remark (Return). Return to Theorem

Theorem (Product Rule). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, and let $f, g : I \to \mathbb{R}$ be differentiable at c. Then fg is differentiable at c and $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$.*

Professional Standard Proof.
Set $h := fg$. For $x \in I$ with $x \neq c$, the difference quotient is

$$\frac{h(x) - h(c)}{x - c} = \frac{f(x)g(x) - f(c)g(c)}{x - c}.$$

Add and subtract $f(x)g(c)$ to obtain

$$\frac{f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)}{x - c} = f(x) \cdot \frac{g(x) - g(c)}{x - c} + g(c) \cdot \frac{f(x) - f(c)}{x - c}.$$

As $x \to c$, the sum rule and product rule for limits give

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \left(\lim_{x \to c} f(x)\right) g'(c) + g(c)f'(c).$$

Since differentiability implies continuity, $\lim_{x \to c} f(x) = f(c)$, yielding $(fg)'(c) = f(c)g'(c) + g(c)f'(c)$. □

Detailed Learning Proof.
Step 1. Form the difference quotient of the product function $h = fg$. For $x \in I$ with $x \neq c$, the difference quotient becomes

$$\frac{h(x) - h(c)}{x - c} = \frac{f(x)g(x) - f(c)g(c)}{x - c}.$$

This expression involves the product of two functions in both the numerator terms, making direct application of limit laws difficult.
Step 2. Insert the auxiliary term $f(x)g(c)$ by adding and subtracting it in the numerator:

$$\frac{f(x)g(x) - f(x)g(c) + f(x)g(c) - f(c)g(c)}{x - c}.$$

Factor the first pair as $f(x)[g(x) - g(c)]$ and the second pair as $g(c)[f(x) - f(c)]$ to obtain

$$f(x) \cdot \frac{g(x) - g(c)}{x - c} + g(c) \cdot \frac{f(x) - f(c)}{x - c}.$$

This decomposition isolates the difference quotients of f and g separately, making them accessible to the definition of derivative.
Step 3. Apply limit laws as $x \to c$. The sum rule for limits distributes the limit across the sum, and the product rule for limits handles the first term:

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \left(\lim_{x \to c} f(x)\right) \left(\lim_{x \to c} \frac{g(x) - g(c)}{x - c}\right) + g(c) \left(\lim_{x \to c} \frac{f(x) - f(c)}{x - c}\right).$$

Since f is differentiable at c, it is continuous at c, so $\lim_{x \to c} f(x) = f(c)$. The difference quotient limits equal $g'(c)$ and $f'(c)$ by the definition of differentiability. Therefore $h'(c) = f(c)g'(c) + g(c)f'(c)$. □

Remark (Proof structure). The proof uses a direct approach via the definition of derivative. The key insight is the add-and-subtract technique that inserts $f(x)g(c)$ to decouple the increments of f and g. This splits the difference quotient of the product into a sum of two terms, each containing exactly one difference quotient. The continuity of f at c, which follows from its differentiability, provides the crucial factor $\lim_{x\to c} f(x) = f(c)$ needed to evaluate the limit.

Remark (Dependencies).

- Definition of Differentiability at a Point
- Differentiability Implies Continuity
- Sum Rule for Limits
- Product Rule for Limits

Remark (Return). Return to Theorem

Theorem (Quotient Rule). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, and let $f, g : I \to \mathbb{R}$ be differentiable at c with $g(c) \neq 0$. Then f/g is differentiable at c and*

$$\left(\frac{f}{g}\right)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

Professional Standard Proof.
Set $h := f/g$. Since g is differentiable at c, it is continuous at c, so $\lim_{x\to c} g(x) = g(c) \neq 0$. By the quotient rule for limits, $g(x) \neq 0$ for all x in some deleted neighbourhood of c, so h is defined near c. For $x \neq c$ with $g(x) \neq 0$:

$$\frac{h(x) - h(c)}{x - c} = \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x - c)} = \frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)(x - c)}.$$

Factoring yields

$$\frac{1}{g(x)g(c)}\left[g(c) \cdot \frac{f(x) - f(c)}{x - c} - f(c) \cdot \frac{g(x) - g(c)}{x - c}\right].$$

Taking the limit as $x \to c$ and applying limit laws:

$$h'(c) = \frac{1}{g(c)^2}[g(c) \cdot f'(c) - f(c) \cdot g'(c)] = \frac{f'(c)g(c) - f(c)g'(c)}{(g(c))^2}.$$

$\square$

Detailed Learning Proof.
Step 1. Form the difference quotient of $h = f/g$ and find a common denominator. Since g is differentiable at c, it is continuous at c, so $\lim_{x\to c} g(x) = g(c) \neq 0$. By the quotient rule for limits, $g(x) \neq 0$ for all x in some deleted neighbourhood of c, ensuring h is defined near c. For $x \neq c$ with $g(x) \neq 0$:

$$\frac{h(x) - h(c)}{x - c} = \frac{\frac{f(x)}{g(x)} - \frac{f(c)}{g(c)}}{x - c} = \frac{f(x)g(c) - f(c)g(x)}{g(x)g(c)(x - c)}.$$

Step 2. Insert the auxiliary term $f(c)g(c)$ to decouple the increments of f and g. Add and subtract $f(c)g(c)$ in the numerator:

$$\frac{f(x)g(c) - f(c)g(c) + f(c)g(c) - f(c)g(x)}{g(x)g(c)(x - c)}.$$

Factor each pair and separate over the denominator:

$$= \frac{1}{g(x)g(c)}\left[g(c) \cdot \frac{f(x) - f(c)}{x - c} - f(c) \cdot \frac{g(x) - g(c)}{x - c}\right].$$

This isolates the difference quotients of f and g, which approach $f'(c)$ and $g'(c)$ respectively.

Step 3. Pass to the limit using limit laws. By the sum, product, and quotient rules for limits:

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \frac{1}{g(c) \cdot g(c)} \left[g(c) \cdot f'(c) - f(c) \cdot g'(c) \right],$$

where $\lim_{x \to c} g(x) = g(c)$ by continuity of g, and the difference quotient limits equal $f'(c)$ and $g'(c)$ by definition of differentiability. Therefore:

$$h'(c) = \frac{g(c)f'(c) - f(c)g'(c)}{(g(c))^2}.$$

□

Remark (Proof structure). The proof follows a direct approach mirroring the product rule strategy. After combining fractions in the difference quotient of f/g, the add-and-subtract technique with the auxiliary term $f(c)g(c)$ decouples the increments of f and g in the numerator, isolating the individual difference quotients. Continuity of g at c provides both the limit $\lim g(x) = g(c)$ in the denominator and the guarantee that $g(x) \neq 0$ near c so the quotient is well-defined throughout.

Remark (Dependencies).

- Definition of Differentiability at a Point
- Differentiability Implies Continuity
- Sum Rule for Limits
- Product Rule for Limits
- Quotient Rule for Limits

Remark (Return). Return to Corollary

Corollary (Finite Sum Rule). *If $f_1, f_2, \ldots, f_n$ are functions on an interval I to $\mathbb{R}$ that are differentiable at $c \in I$, then:*

$$\left(\sum_{i=1}^{n} f_i\right)'(c) = \sum_{i=1}^{n} f_i'(c)$$

Professional Standard Proof.
The proof proceeds by mathematical induction on n. For the base case $n = 1$, the statement $(f_1)'(c) = f_1'(c)$ holds trivially. Assume the statement holds for some $k \in \mathbb{N}$, so $\left(\sum_{i=1}^{k} f_i\right)'(c) = \sum_{i=1}^{k} f_i'(c)$. For $n = k+1$, let $g = \sum_{i=1}^{k} f_i$. Then $\left(\sum_{i=1}^{k+1} f_i\right)'(c) = (g + f_{k+1})'(c)$. By the sum rule for derivatives, $(g+f_{k+1})'(c) = g'(c)+f_{k+1}'(c)$. Applying the inductive hypothesis yields $g'(c) + f_{k+1}'(c) = \sum_{i=1}^{k} f_i'(c) + f_{k+1}'(c) = \sum_{i=1}^{k+1} f_i'(c)$. By mathematical induction, the statement holds for all $n \in \mathbb{N}$. □

Detailed Learning Proof.
Step 1. Establish the base case for induction. For $n = 1$, the statement becomes $(f_1)'(c) = f_1'(c)$, which is an identity and therefore holds trivially.
Step 2. Formulate the inductive hypothesis. Assume the statement holds for some $k \in \mathbb{N}$, meaning that for any collection of k functions differentiable at c, their sum satisfies $\left(\sum_{i=1}^{k} f_i\right)'(c) = \sum_{i=1}^{k} f_i'(c)$.
Step 3. Prove the inductive step for $n = k+1$. Define $g = \sum_{i=1}^{k} f_i$. Then the sum of $k+1$ functions can be written as $\sum_{i=1}^{k+1} f_i = g + f_{k+1}$, allowing us to reduce to the two-function case.
Step 4. Apply the sum rule for derivatives. Since g and f_{k+1} are both differentiable at c (the former by the inductive hypothesis and the latter by assumption), the sum rule gives $(g + f_{k+1})'(c) = g'(c) + f_{k+1}'(c)$.
Step 5. Substitute the inductive hypothesis. By the inductive hypothesis, $g'(c) = \left(\sum_{i=1}^{k} f_i\right)'(c) = \sum_{i=1}^{k} f_i'(c)$. Therefore, $g'(c) + f_{k+1}'(c) = \sum_{i=1}^{k} f_i'(c) + f_{k+1}'(c) = \sum_{i=1}^{k+1} f_i'(c)$.
Step 6. Conclude by mathematical induction. Having established both the base case and the inductive step, the principle of mathematical induction guarantees that the statement holds for all $n \in \mathbb{N}$. □

Remark (Proof structure). This proof employs mathematical induction to extend the sum rule from two functions to an arbitrary finite number of functions. The strategy reduces the general case to repeated application of the two-function sum rule, leveraging the inductive structure to build from simple cases to the general result.

Remark (Dependencies).

- Derivative at a Point
- Sum Rule for Derivatives
- Induction Principle for a Peano System

Remark (Return). Return to Theorem

Corollary (Extended Product Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then $\prod_{i=1}^n f_i$ is differentiable at c, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{k=1}^{n} f_k'(c) \prod_{\substack{i=1 \\ i\neq k}}^{n} f_i(c).$$

Professional Standard Proof.
TODO: professional standard proof for extended-product-rule. □

Detailed Learning Proof.
TODO: detailed learning proof for extended-product-rule. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Power Rule for Integer Powers of a Function). *Let $A \subseteq \mathbb{R}$, let $f : A \to \mathbb{R}$, let $c \in A$ be a limit point of A, and let $n \in \mathbb{N}$. If f is differentiable at c, then f^n is differentiable at c, and*

$$(f^n)'(c) = n(f(c))^{n-1} f'(c).$$

Professional Standard Proof.
TODO: professional standard proof for power-rule-special-case. □

Detailed Learning Proof.
TODO: detailed learning proof for power-rule-special-case. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Finite Linear Combination Rule). *Let $A \subseteq \mathbb{R}$, let $f_1, \ldots, f_n : A \to \mathbb{R}$, let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$, and let $c \in A$ be a limit point of A. If each f_i is differentiable at c, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Professional Standard Proof.
TODO: professional standard proof for finite-linear-combination-rule. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-linear-combination-rule. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Multiple Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $\alpha \in \mathbb{R}$. If f is differentiable on I, then αf is differentiable on I, and*

$$(\alpha f)'(x) = \alpha f'(x) \qquad \textit{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for constant-multiple-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for constant-multiple-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then $f + g$ is differentiable on I, and*

$$(f + g)'(x) = f'(x) + g'(x) \qquad \textit{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for sum-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for sum-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. If f and g are differentiable on I, then fg is differentiable on I, and*

$$(fg)'(x) = f'(x)g(x) + f(x)g'(x) \qquad \textit{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for product-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for product-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Quotient Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f, g : I \to \mathbb{R}$. Suppose that f and g are differentiable on I and that $g(x) \neq 0$ for all $x \in I$. Then f/g is differentiable on I, and*

$$\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2} \qquad \text{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for quotient-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for quotient-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Finite Sum Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable on I, and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for finite-sum-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-sum-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Extended Product Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f_1, \dots, f_n : I \to \mathbb{R}$. If each f_i is differentiable on I, then $\prod_{i=1}^n f_i$ is differentiable on I, and*

$$\left(\prod_{i=1}^{n} f_i\right)'(x) = \sum_{k=1}^{n} f_k'(x) \prod_{\substack{i=1 \\ i \neq k}}^{n} f_i(x) \qquad \text{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for extended-product-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for extended-product-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Power Rule for Integer Powers of a Function on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $n \in \mathbb{N}$. If f is differentiable on I, then f^n is differentiable on I, and*

$$(f^n)'(x) = n(f(x))^{n-1}f'(x) \qquad \textit{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for power-rule-special-case-interval. ☐

Detailed Learning Proof.
TODO: detailed learning proof for power-rule-special-case-interval. ☐

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Finite Linear Combination Rule on an Interval). *Let $I \subseteq \mathbb{R}$ be an interval, let $f_1, \dots, f_n : I \to \mathbb{R}$, and let $\alpha_1, \dots, \alpha_n \in \mathbb{R}$. If each f_i is differentiable on I, then*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(x) = \sum_{i=1}^{n} \alpha_i f_i'(x) \qquad \text{for all } x \in I.$$

Professional Standard Proof.
TODO: professional standard proof for finite-linear-combination-rule-interval. □

Detailed Learning Proof.
TODO: detailed learning proof for finite-linear-combination-rule-interval. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Constant Multiple Rule). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, let $f : I \to \mathbb{R}$ be differentiable at c, and let $\alpha \in \mathbb{R}$. Then αf is differentiable at c and $(\alpha f)'(c) = \alpha f'(c)$.*

Professional Standard Proof.
Set $h := \alpha f$. For $x \in I$ with $x \neq c$, the difference quotient becomes

$$\frac{h(x) - h(c)}{x - c} = \frac{\alpha f(x) - \alpha f(c)}{x - c} = \alpha \cdot \frac{f(x) - f(c)}{x - c}.$$

By the Constant Multiple Rule for Limits,

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \alpha \cdot \lim_{x \to c} \frac{f(x) - f(c)}{x - c} = \alpha f'(c),$$

where the final equality follows from the differentiability of f at c. Therefore $h'(c) = \alpha f'(c)$. □

Detailed Learning Proof.
Step 1. Form the difference quotient of $h := \alpha f$. For $x \in I$ with $x \neq c$, substituting the definition of h gives

$$\frac{h(x) - h(c)}{x - c} = \frac{\alpha f(x) - \alpha f(c)}{x - c} = \alpha \cdot \frac{f(x) - f(c)}{x - c}.$$

The scalar α factors out by basic arithmetic properties of real numbers.
Step 2. Apply the limit properties to establish differentiability. By the Constant Multiple Rule for Limits, the scalar α can be moved outside the limit:

$$\lim_{x \to c} \frac{h(x) - h(c)}{x - c} = \alpha \cdot \lim_{x \to c} \frac{f(x) - f(c)}{x - c}.$$

Since f is differentiable at c, the limit $\lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ exists and equals $f'(c)$ by definition of differentiability at a point. Therefore $h'(c) = \alpha f'(c)$. □

Remark (Proof structure). This is a direct proof using the definition of differentiability. The strategy exploits the linearity of limits: the scalar α factors out of the difference quotient algebraically, then the Constant Multiple Rule for Limits allows passage of the scalar through the limit operation. The differentiability of the original function f supplies the required limit value.

Remark (Dependencies).

- Definition of Differentiability at a Point
- Constant Multiple Rule for Limits

Remark (Return). Return to Corollary

Corollary (Finite Product Rule). *If $f_1, f_2, \ldots, f_n$ are functions on an interval I to $\mathbb{R}$ that are differentiable at $c \in I$, then:*

$$\left(\prod_{i=1}^{n} f_i\right)'(c) = \sum_{i=1}^{n} \left(f_i'(c) \prod_{j \neq i} f_j(c)\right)$$

Professional Standard Proof.
By induction on n. Base case $n = 1$ is trivial. Assume the formula holds for $n = k$. For $n = k+1$, let $g = \prod_{i=1}^{k} f_i$. Then $\prod_{i=1}^{k+1} f_i = g \cdot f_{k+1}$. By the product rule, $(g \cdot f_{k+1})'(c) = g'(c) f_{k+1}(c) + g(c) f_{k+1}'(c)$. The inductive hypothesis gives $g'(c) = \sum_{i=1}^{k} \left(f_i'(c) \prod_{j \neq i, j \leq k} f_j(c)\right)$. Substituting and distributing $f_{k+1}(c)$ into the sum yields $\sum_{i=1}^{k} \left(f_i'(c) \prod_{j \neq i, j \leq k+1} f_j(c)\right) + f_{k+1}'(c) \prod_{j=1}^{k} f_j(c)$, which equals $\sum_{i=1}^{k+1} \left(f_i'(c) \prod_{j \neq i} f_j(c)\right)$. $\square$

Detailed Learning Proof.
Step 1. Establish the base case for $n = 1$. The formula states $(f_1)'(c) = f_1'(c)$, which is trivially true since the product contains only one function and the sum contains only one term.
Step 2. State the inductive hypothesis. Assume the formula holds for $n = k$:

$$\left(\prod_{i=1}^{k} f_i\right)'(c) = \sum_{i=1}^{k} \left(f_i'(c) \prod_{j \neq i, j=1}^{k} f_j(c)\right)$$

Step 3. Set up the inductive step for $n = k+1$. Define $g(x) = \prod_{i=1}^{k} f_i(x)$, so the product of $k+1$ functions becomes $g(x) \cdot f_{k+1}(x)$. This reduces the problem to applying the two-function product rule.
Step 4. Apply the product rule for two functions. The derivative is:

$$(g \cdot f_{k+1})'(c) = g'(c) f_{k+1}(c) + g(c) f_{k+1}'(c)$$

Step 5. Substitute the inductive hypothesis for $g'(c)$. This gives:

$$\left[\sum_{i=1}^{k} \left(f_i'(c) \prod_{j \neq i, j=1}^{k} f_j(c)\right)\right] f_{k+1}(c) + \left[\prod_{i=1}^{k} f_i(c)\right] f_{k+1}'(c)$$

Step 6. Distribute $f_{k+1}(c)$ into the sum. Each term $f_i'(c) \prod_{j \neq i, j=1}^{k} f_j(c)$ becomes $f_i'(c) \prod_{j \neq i, j=1}^{k+1} f_j(c)$ when multiplied by $f_{k+1}(c)$:

$$\sum_{i=1}^{k} \left(f_i'(c) \prod_{j \neq i, j=1}^{k+1} f_j(c)\right) + f_{k+1}'(c) \prod_{j=1}^{k} f_j(c)$$

Step 7. Recognize that the second term fits the pattern for $i = k+1$. The term $f_{k+1}'(c) \prod_{j=1}^{k} f_j(c)$ equals $f_{k+1}'(c) \prod_{j \neq k+1, j=1}^{k+1} f_j(c)$, completing the sum from $i = 1$ to $i = k+1$. $\square$

Remark (Proof structure). This is a proof by mathematical induction. The strategy works by reducing the $(k+1)$-function product to a two-function product using the substitution $g = \prod_{i=1}^{k} f_i$, then applying the standard product rule and the inductive hypothesis. The key insight is that each term in the final sum represents the derivative of exactly one factor while keeping all other factors constant.

Remark (Dependencies).

- Product Rule
- Definition of Differentiability

Remark (Return). Return to Theorem

Corollary (Power Rule (Special Case)). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, let $f : I \to \mathbb{R}$ be differentiable at c, and let $n \in \mathbb{N}$. Then f^n is differentiable at c and $(f^n)'(c) = n(f(c))^{n-1} f'(c)$.*

Professional Standard Proof.
Set $f_1 = f_2 = \cdots = f_n = f$ in the Extended Product Rule to obtain $(f^n)'(c) = \sum_{k=1}^n f_k'(c) \prod_{\substack{i=1 \\ i\neq k}}^n f_i(c)$. Since each $f_k = f$, each derivative $f_k'(c) = f'(c)$ and each remaining product equals $(f(c))^{n-1}$. Therefore each term in the sum equals $f'(c)(f(c))^{n-1}$. Summing n identical terms yields $(f^n)'(c) = n \cdot f'(c) \cdot (f(c))^{n-1} = n(f(c))^{n-1} f'(c)$. □

Detailed Learning Proof.
Step 1. Apply the Extended Product Rule with all factors equal to f.
Set $f_1 = f_2 = \cdots = f_n = f$. By the Extended Product Rule,

$$(f^n)'(c) = (f_1 f_2 \cdots f_n)'(c) = \sum_{k=1}^n f_k'(c) \prod_{\substack{i=1 \\ i\neq k}}^n f_i(c).$$

This expresses the derivative of the n-fold product as a sum where each term differentiates exactly one factor and evaluates the remaining factors at c.
Step 2. Simplify each term in the sum.
Since $f_k = f$ for every k, each factor $f_k'(c) = f'(c)$. The remaining product contains $n - 1$ factors each equal to $f(c)$:

$$\prod_{\substack{i=1 \\ i\neq k}}^n f_i(c) = (f(c))^{n-1}.$$

Each term in the sum is therefore $f'(c) \cdot (f(c))^{n-1}$, independently of the choice of k. The index k determines which factor is differentiated, but since all factors are the same function, this choice does not affect the result.
Step 3. Evaluate the sum.
Since the summand is constant in k, summing over $k = 1, \ldots, n$ gives a factor of n:

$$(f^n)'(c) = \sum_{k=1}^n f'(c)(f(c))^{n-1} = n \cdot f'(c) \cdot (f(c))^{n-1} = n(f(c))^{n-1} f'(c).$$

This completes the derivation of the power rule formula. □

Remark (Proof structure). The argument is a direct specialization of the Extended Product Rule. Setting all n factors equal to f causes every term in the product rule sum to collapse to the same value $f'(c)(f(c))^{n-1}$. Counting the n identical terms yields the power rule formula. No induction is needed beyond what is already encoded in the Extended Product Rule.

Remark (Dependencies).

- Extended Product Rule

Remark (Return). Return to Theorem

Theorem (Finite Linear Combination Rule). *Let $I \subseteq \mathbb{R}$ be an open interval, let $c \in I$, let $n \in \mathbb{N}$, let $f_1, \ldots, f_n : I \to \mathbb{R}$ be differentiable at c, and let $\alpha_1, \ldots, \alpha_n \in \mathbb{R}$. Then $\sum_{i=1}^{n} \alpha_i f_i$ is differentiable at c and*

$$\left(\sum_{i=1}^{n} \alpha_i f_i\right)'(c) = \sum_{i=1}^{n} \alpha_i f_i'(c).$$

Professional Standard Proof.
The proof proceeds by induction on n. For $n = 1$, the result follows from the Constant Multiple Rule. For the inductive step, assume the statement holds for $n = k$ and consider $n = k + 1$. Write $\sum_{i=1}^{k+1} \alpha_i f_i = \left(\sum_{i=1}^{k} \alpha_i f_i\right) + \alpha_{k+1} f_{k+1}$. By the inductive hypothesis, $\sum_{i=1}^{k} \alpha_i f_i$ is differentiable at c with derivative $\sum_{i=1}^{k} \alpha_i f_i'(c)$. By the Constant Multiple Rule, $\alpha_{k+1} f_{k+1}$ is differentiable at c with derivative $\alpha_{k+1} f_{k+1}'(c)$. The Sum Rule then gives the desired result. □

Detailed Learning Proof.
Step 1. Base case $n = 1$. Define $h = \alpha_1 f_1$. By the Constant Multiple Rule, h is differentiable at c and $h'(c) = \alpha_1 f_1'(c)$. This equals $\sum_{i=1}^{1} \alpha_i f_i'(c)$, establishing the base case.
Step 2. Inductive hypothesis. Assume the statement holds for some $n = k \geq 1$: for any k functions differentiable at c and any scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{R}$,

$$\left(\sum_{i=1}^{k} \alpha_i f_i\right)'(c) = \sum_{i=1}^{k} \alpha_i f_i'(c).$$

Step 3. Inductive step for $n = k + 1$. Consider $h = \sum_{i=1}^{k+1} \alpha_i f_i$. Write this as $h = \left(\sum_{i=1}^{k} \alpha_i f_i\right) + \alpha_{k+1} f_{k+1}$. Let $g = \sum_{i=1}^{k} \alpha_i f_i$. By the inductive hypothesis, g is differentiable at c with $g'(c) = \sum_{i=1}^{k} \alpha_i f_i'(c)$. By the Constant Multiple Rule, $\alpha_{k+1} f_{k+1}$ is differentiable at c with derivative $\alpha_{k+1} f_{k+1}'(c)$.
Step 4. Application of the Sum Rule. Since $h = g + \alpha_{k+1} f_{k+1}$ where both g and $\alpha_{k+1} f_{k+1}$ are differentiable at c, the Sum Rule gives

$$h'(c) = g'(c) + \alpha_{k+1} f_{k+1}'(c) = \sum_{i=1}^{k} \alpha_i f_i'(c) + \alpha_{k+1} f_{k+1}'(c) = \sum_{i=1}^{k+1} \alpha_i f_i'(c).$$

Step 5. Conclusion by induction. The statement holds for all $n \in \mathbb{N}$. □

Remark (Proof structure). The proof uses induction on n, establishing linearity of differentiation over finite linear combinations. The base case invokes the Constant Multiple Rule, while the inductive step decomposes the $(k + 1)$-term sum by peeling off the last term, applying the inductive hypothesis to the first k terms, the Constant Multiple Rule to the final term, and the Sum Rule to combine them. This structure reveals that finite linearity builds systematically from the two fundamental linearity properties: scalar multiplication and addition.

Remark (Dependencies).

- Constant Multiple Rule
- Sum Rule

Remark (Return). Return to Theorem

Theorem (Differentiation Algebra on Sets). *Let $I \subseteq \mathbb{R}$ be an open interval and let $f, g : I \to \mathbb{R}$ be differentiable on I. Let $\alpha \in \mathbb{R}$.*

1. ***Constant Multiple Rule:*** *$(\alpha f)' = \alpha f'$ on I.*
2. ***Sum Rule:*** *$(f + g)' = f' + g'$ on I.*
3. ***Product Rule:*** *$(fg)' = f'g + fg'$ on I.*
4. ***Quotient Rule:*** *If $g(x) \neq 0$ for all $x \in I$, then $(f/g)' = (f'g - fg')/g^2$ on I.*

Professional Standard Proof.
Each result follows by pointwise application of the corresponding differentiation rule at arbitrary points $c \in I$.
Constant Multiple Rule: Let $h := \alpha f$ and fix $c \in I$. Since f is differentiable at c, the pointwise Constant Multiple Rule gives $h'(c) = \alpha f'(c)$. Since $c \in I$ was arbitrary, $h' = \alpha f'$ on I.
Sum Rule: Let $h := f + g$ and fix $c \in I$. Since f and g are differentiable at c, the pointwise Sum Rule gives $h'(c) = f'(c) + g'(c)$. Since $c \in I$ was arbitrary, $h' = f' + g'$ on I.
Product Rule: Let $h := fg$ and fix $c \in I$. Since f and g are differentiable at c, the pointwise Product Rule gives $h'(c) = f'(c)g(c) + f(c)g'(c)$. Since $c \in I$ was arbitrary, $h' = f'g + fg'$ on I.
Quotient Rule: Let $h := f/g$ and fix $c \in I$. Since $g(c) \neq 0$ and f, g are differentiable at c, the pointwise Quotient Rule gives $h'(c) = (f'(c)g(c) - f(c)g'(c))/(g(c))^2$. Since $c \in I$ was arbitrary and $g(x) \neq 0$ on all of I, $h' = (f'g - fg')/g^2$ on I. □

Detailed Learning Proof.
Step 1. Establish the universal strategy. Each global differentiation rule is obtained by applying the corresponding pointwise rule at an arbitrary point $c \in I$ and concluding the function identity holds on the entire interval I.
Step 2. Prove the Constant Multiple Rule. Let $h := \alpha f$ and fix arbitrary $c \in I$. Since f is differentiable at c, the pointwise Constant Multiple Rule ensures h is differentiable at c with $h'(c) = \alpha f'(c)$. Since $c \in I$ was arbitrary, this equality holds at every point of I, giving $h' = \alpha f'$ on I.
Step 3. Prove the Sum Rule. Let $h := f+g$ and fix arbitrary $c \in I$. Since both f and g are differentiable at c, the pointwise Sum Rule ensures h is differentiable at c with $h'(c) = f'(c) + g'(c)$. Since $c \in I$ was arbitrary, this equality holds at every point of I, giving $h' = f' + g'$ on I.
Step 4. Prove the Product Rule. Let $h := fg$ and fix arbitrary $c \in I$. Since both f and g are differentiable at c, the pointwise Product Rule ensures h is differentiable at c with $h'(c) = f'(c)g(c) + f(c)g'(c)$. Since $c \in I$ was arbitrary, this equality holds at every point of I, giving $h' = f'g + fg'$ on I.
Step 5. Prove the Quotient Rule. Let $h := f/g$ and fix arbitrary $c \in I$. The global hypothesis ensures $g(c) \neq 0$, so the pointwise Quotient Rule

applies: h is differentiable at c with $h'(c) = (f'(c)g(c) - f(c)g'(c))/(g(c))^2$. Since $c \in I$ was arbitrary and $g(x) \neq 0$ throughout I, this equality holds at every point of I, giving $h' = (f'g - fg')/g^2$ on I. □

Remark (Proof structure). Universal instantiation strategy. Each result lifts the corresponding pointwise differentiation rule to a global statement on the interval I by applying the pointwise rule at an arbitrary point $c \in I$ and leveraging the arbitrariness to conclude the function identity holds throughout I. The Quotient Rule requires the additional global hypothesis that g is nonzero on all of I to ensure the pointwise condition is satisfied at every application.

Remark (Dependencies).

- Constant Multiple Rule (Pointwise)
- Sum Rule (Pointwise)
- Product Rule (Pointwise)
- Quotient Rule (Pointwise)

Remark (Return). Return to Theorem

Theorem (Carathéodory Characterization of Differentiability). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, and let $c \in I$. The following are equivalent:*

(i) f is differentiable at c.

(ii) There exists a function $\phi : I \to \mathbb{R}$, continuous at c, such that

$$f(x) = f(c) + (x - c)\,\phi(x) \qquad \text{for all } x \in I.$$

In this case, $\phi(c) = f'(c)$.

Professional Standard Proof.
TODO: professional standard proof for caratheodory-characterization-of-differentiability. □

Detailed Learning Proof.
TODO: detailed learning proof for caratheodory-characterization-of-differentiability. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Chain Rule). *Let $J, I \subseteq \mathbb{R}$ be open intervals, let $f : J \to \mathbb{R}$ and $g : I \to \mathbb{R}$ be functions with $f(J) \subseteq I$, and let $c \in J$. If f is differentiable at c and g is differentiable at $f(c)$, then $g \circ f$ is differentiable at c and $(g \circ f)'(c) = g'(f(c)) \cdot f'(c)$.*

Professional Standard Proof.
Let $d = f(c)$. By Carathéodory's theorem, there exist continuous functions $\varphi : J \to \mathbb{R}$ and $\psi : I \to \mathbb{R}$ such that $f(x) - f(c) = \varphi(x)(x - c)$ with $\varphi(c) = f'(c)$ and $g(y) - g(d) = \psi(y)(y - d)$ with $\psi(d) = g'(d)$. Substituting $y = f(x)$ gives $g(f(x)) - g(f(c)) = \psi(f(x))[f(x) - f(c)] = \psi(f(x)) \cdot \varphi(x) \cdot (x - c)$. Define $\Phi(x) := \psi(f(x)) \cdot \varphi(x)$. Since f is differentiable at c, it is continuous at c, so $\psi \circ f$ is continuous at c by continuity of composition. Since φ is continuous at c, their product Φ is continuous at c. By Carathéodory's theorem, $g \circ f$ is differentiable at c with $(g \circ f)'(c) = \Phi(c) = \psi(f(c)) \cdot \varphi(c) = g'(f(c)) \cdot f'(c)$. □

Detailed Learning Proof.
Step 1. Apply Carathéodory's theorem to linearize f at c. Since f is differentiable at c, there exists $\varphi : J \to \mathbb{R}$, continuous at c, such that $f(x) - f(c) = \varphi(x)(x - c)$ for all $x \in J$ and $\varphi(c) = f'(c)$. This factorization replaces the potentially problematic difference quotient with a continuous function.
Step 2. Apply Carathéodory's theorem to linearize g at $d = f(c)$. Since g is differentiable at d, there exists $\psi : I \to \mathbb{R}$, continuous at d, such that $g(y) - g(d) = \psi(y)(y - d)$ for all $y \in I$ and $\psi(d) = g'(d)$. This provides a factorization of increments in g around the point $f(c)$.
Step 3. Compose the two factorizations to linearize $g \circ f$ at c. Substitute $y = f(x)$ into the identity for g: $g(f(x)) - g(f(c)) = \psi(f(x))[f(x) - f(c)]$. Substitute the identity for $f(x) - f(c)$ from Step 1: $g(f(x)) - g(f(c)) = \psi(f(x)) \cdot \varphi(x) \cdot (x - c)$.
Step 4. Identify the Carathéodory function for $g \circ f$. Define $\Phi : J \to \mathbb{R}$ by $\Phi(x) := \psi(f(x)) \cdot \varphi(x)$. The identity from Step 3 becomes $(g \circ f)(x) - (g \circ f)(c) = \Phi(x)(x - c)$ for all $x \in J$. By Carathéodory's theorem, $g \circ f$ is differentiable at c if and only if Φ is continuous at c.
Step 5. Verify that Φ is continuous at c. Since f is differentiable at c, it is continuous at c. Since ψ is continuous at $d = f(c)$ and f is continuous at c, the composition $\psi \circ f$ is continuous at c. Since φ is continuous at c from Step 1, the product $\Phi = (\psi \circ f) \cdot \varphi$ is continuous at c.
Step 6. Conclude differentiability and compute the derivative. By Carathéodory's theorem, $g \circ f$ is differentiable at c and $(g \circ f)'(c) = \Phi(c)$. Evaluating: $(g \circ f)'(c) = \Phi(c) = \psi(f(c)) \cdot \varphi(c) = g'(f(c)) \cdot f'(c)$. □

Remark (Proof structure). The proof uses a constructive approach via Carathéodory's theorem. Both f and g are linearized using Carathéodory factorizations, yielding continuous functions φ and ψ. The key insight is that composing these factorizations produces a single Carathéodory factorization for $g \circ f$ with Carathéodory function $\Phi = (\psi \circ f) \cdot \varphi$. The proof succeeds because this product function is continuous at c, which follows from combining the continuity properties of f, ψ, and φ using standard continuity theorems. The derivative formula emerges naturally by evaluating Φ at c.

Remark (Dependencies).

- Carathéodory's Theorem
- Differentiability Implies Continuity
- Composition of Limits
- Product Rule for Function Limits

Remark (Return). Return to Theorem

Theorem (Interior Extremum Theorem). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c and is differentiable at c, then $f'(c) = 0$.*

Professional Standard Proof.
Consider the case where f has a relative maximum at c. By definition, there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and $f(x) - f(c) \leq 0$ for all $x \in (c-\delta, c+\delta)$.
For $x \in (c, c+\delta)$, the quotient $\frac{f(x)-f(c)}{x-c} \leq 0$ since the numerator is nonpositive and the denominator is positive. By order preservation for limits, $f'_+(c) \leq 0$.
For $x \in (c-\delta, c)$, the quotient $\frac{f(x)-f(c)}{x-c} \geq 0$ since the numerator is nonpositive and the denominator is negative. By order preservation for limits, $f'_-(c) \geq 0$.
Since f is differentiable at c, the one-sided derivatives exist and agree: $f'(c) = f'_+(c) = f'_-(c)$. Therefore $0 \geq f'(c) \geq 0$, which forces $f'(c) = 0$.
The case of a relative minimum follows by applying the argument to $-f$ at c. □

Detailed Learning Proof.
Step 1. Establish the constraint from the relative maximum. Since f has a relative maximum at c, by definition there exists $\delta > 0$ such that $(c-\delta, c+\delta) \subseteq I$ and $f(x) - f(c) \leq 0$ for all $x \in (c-\delta, c+\delta)$. This inequality means $f(c)$ is the largest value in the interval, so every other value satisfies $f(x) \leq f(c)$.
Step 2. Analyze the difference quotient for $x > c$. For any $x \in (c, c+\delta)$, the increment $f(x) - f(c) \leq 0$ by Step 1, and the displacement $x - c > 0$. Therefore the difference quotient satisfies $\frac{f(x)-f(c)}{x-c} \leq 0$. This inequality holds for all x in the right half-neighborhood of c.
Step 3. Apply order preservation to obtain $f'_+(c) \leq 0$. Since the difference quotient is nonpositive throughout the interval $(c, c+\delta)$, and the right-hand derivative is the limit of this quotient as x approaches c from the right, order preservation for limits gives $f'_+(c) = \lim_{x\to c^+} \frac{f(x)-f(c)}{x-c} \leq 0$.
Step 4. Analyze the difference quotient for $x < c$. For any $x \in (c-\delta, c)$, the increment $f(x) - f(c) \leq 0$ by Step 1, and the displacement $x - c < 0$. When a nonpositive number is divided by a negative number, the result is nonnegative, so $\frac{f(x)-f(c)}{x-c} \geq 0$. This inequality holds for all x in the left half-neighborhood of c.
Step 5. Apply order preservation to obtain $f'_-(c) \geq 0$. Since the difference quotient is nonnegative throughout the interval $(c-\delta, c)$, order preservation for limits gives $f'_-(c) = \lim_{x\to c^-} \frac{f(x)-f(c)}{x-c} \geq 0$.
Step 6. Use differentiability to conclude $f'(c) = 0$. Since f is differentiable at c, the one-sided derivatives exist and are equal: $f'(c) = f'_+(c) = f'_-(c)$. From Steps 3 and 5, this common value satisfies both $f'(c) \leq 0$ and $f'(c) \geq 0$. The only real number satisfying both inequalities is 0, so $f'(c) = 0$.

Step 7. Handle the minimum case. If f has a relative minimum at c, then $-f$ has a relative maximum at c. Applying the above argument to $-f$ gives $(-f)'(c) = 0$, which means $-f'(c) = 0$, so $f'(c) = 0$. □

Remark (Proof structure). The proof uses a direct approach that reads the sign of the difference quotient separately on each side of the critical point. The relative maximum condition provides a uniform constraint $f(x) - f(c) \leq 0$ throughout a neighborhood of c. However, the sign of the difference quotient depends on the sign of the displacement $x - c$, which changes across c. By analyzing the left and right sides separately, the proof obtains complementary inequalities on the one-sided derivatives that together force the derivative to be zero. No contradiction argument is neededthe conclusion emerges directly from the sandwich $0 \geq f'(c) \geq 0$.

Remark (Dependencies).

- Definition of Relative Maximum
- Order Preservation for Limits
- Differentiability and One-Sided Derivatives

Remark (Return). Return to Theorem

Theorem (Necessary Condition for an Interior Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or*

$$f'(c) = 0.$$

Professional Standard Proof.
TODO: professional standard proof for necessary-condition-extremum. □

Detailed Learning Proof.
TODO: detailed learning proof for necessary-condition-extremum. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Theorem (Zero Derivative Implies Constant). *Let $I := [a,b] \subseteq \mathbb{R}$ be a closed interval, and let $f : I \to \mathbb{R}$ be continuous on I and differentiable on (a,b). If $f'(x) = 0$ for all $x \in (a,b)$, then f is constant on I.*

Professional Standard Proof.
Fix any $x \in I$. By the Mean Value Theorem applied to f on $[a,x]$, there exists $c \in (a,x)$ such that $f(x) - f(a) = f'(c)(x-a) = 0 \cdot (x-a) = 0$. Therefore $f(x) = f(a)$ for all $x \in I$. □

Detailed Learning Proof.
Step 1. Fix an arbitrary point in I. Let $x \in I$ be arbitrary. The goal is to show $f(x) = f(a)$.
Step 2. Apply the Mean Value Theorem on $[a,x]$. Since f is continuous on $[a,b] \supseteq [a,x]$ and differentiable on $(a,b) \supseteq (a,x)$, the MVT applies to f on $[a,x]$: there exists $c \in (a,x)$ such that

$$f(x) - f(a) = f'(c)(x-a).$$

Step 3. Evaluate using the hypothesis. Since $c \in (a,x) \subseteq (a,b)$, the hypothesis gives $f'(c) = 0$. Therefore

$$f(x) - f(a) = 0 \cdot (x-a) = 0,$$

so $f(x) = f(a)$.
Step 4. Conclude. Since $x \in I$ was arbitrary, $f(x) = f(a)$ for all $x \in I$. Setting $C = f(a)$, the function f is identically C on I. □

Remark (Proof structure). A direct application of the MVT. The hypothesis $f' \equiv 0$ annihilates the right-hand side of the MVT identity, collapsing $f(x) - f(a) = 0$ for every x. The only non-trivial input is the MVT itself; everything else is arithmetic.

Remark (Dependencies).

- Theorem: Mean Value Theorem

Remark (Return). Return to Corollary

Corollary (Equal Derivatives Imply Difference Is Constant). *Let $I := [a,b] \subseteq \mathbb{R}$, and let $f, g : I \to \mathbb{R}$ be continuous on I and differentiable on (a,b). If $f'(x) = g'(x)$ for all $x \in (a,b)$, then there exists $C \in \mathbb{R}$ such that $f(x) = g(x) + C$ for all $x \in I$.*

Professional Standard Proof.
Define $h := f - g$. Then h is continuous on I and differentiable on (a,b), with $h'(x) = f'(x) - g'(x) = 0$ for all $x \in (a,b)$. By the Zero Derivative Implies Constant theorem, h is constant on I: there exists $C \in \mathbb{R}$ with $h(x) = C$ for all $x \in I$. Therefore $f(x) = g(x) + C$. □

Detailed Learning Proof.
Step 1. Define the auxiliary function. Set $h := f - g$. Since f and g are both continuous on I and differentiable on (a,b), so is h.
Step 2. Compute the derivative of h. For every $x \in (a,b)$, the Sum Rule gives $h'(x) = f'(x) - g'(x) = 0$ by hypothesis.
Step 3. Apply the Zero Derivative Implies Constant theorem. Since h is continuous on I, differentiable on (a,b), and $h' \equiv 0$ on (a,b), there exists $C \in \mathbb{R}$ such that $h(x) = C$ for all $x \in I$.
Step 4. Recover the conclusion. From $h(x) = C$, we have $f(x) - g(x) = C$, so $f(x) = g(x) + C$ for all $x \in I$. □

Remark (Proof structure). A one-line reduction to the preceding theorem via the auxiliary function $h = f - g$. The hypothesis $f' = g'$ becomes $h' = 0$, and the conclusion $f = g + C$ becomes $h = C$.

Remark (Dependencies).

- Theorem: Zero Derivative Implies Constant

Remark (Return). Return to Theorem

Theorem (Nondecreasing Iff Nonnegative Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nondecreasing on I if and only if $f'(x) \geq 0$ for all $x \in I$.*

Professional Standard Proof.
($\Rightarrow$) Suppose f is nondecreasing. For any $c \in I$ and $x \in I$ with $x > c$, the hypothesis gives $f(x) \geq f(c)$, so the difference quotient $\frac{f(x)-f(c)}{x-c} \geq 0$. By order preservation for limits, $f'(c) = \lim_{x\to c} \frac{f(x)-f(c)}{x-c} \geq 0$.
($\Leftarrow$) Suppose $f'(x) \geq 0$ for all $x \in I$. For any $a, b \in I$ with $a < b$, the MVT supplies $c \in (a, b)$ with $f(b) - f(a) = f'(c)(b - a) \geq 0$. Therefore $f(a) \leq f(b)$, so f is nondecreasing. $\square$

Detailed Learning Proof.
Step 1. Prove the forward direction: nondecreasing implies $f' \geq 0$. Fix $c \in I$. For any $x \in I$ with $x > c$, the nondecreasing hypothesis gives $f(x) \geq f(c)$, so the numerator $f(x) - f(c) \geq 0$ and the denominator $x - c > 0$, making the difference quotient $\frac{f(x)-f(c)}{x-c} \geq 0$. For $x < c$, both numerator and denominator are ≤ 0 and < 0 respectively, so the quotient is again ≥ 0. By order preservation for limits, $f'(c) \geq 0$.
Step 2. Prove the reverse direction: $f' \geq 0$ implies nondecreasing. Let $a, b \in I$ with $a < b$. The function f is continuous on $[a, b]$ (since it is differentiable) and differentiable on (a, b). By the Mean Value Theorem, there exists $c \in (a, b)$ such that

$$f(b) - f(a) = f'(c)(b - a).$$

Since $f'(c) \geq 0$ and $b - a > 0$, the right-hand side is ≥ 0, so $f(b) \geq f(a)$.
Step 3. Conclude. Since $a < b$ in I was arbitrary and $f(a) \leq f(b)$ holds for all such pairs, f is nondecreasing on I. $\square$

Remark (Proof structure). A biconditional proof with two direct arguments. The forward direction uses order preservation for limits applied to the non-negative difference quotient. The reverse direction is a one-application of the MVT: the MVT converts the derivative condition into a sign condition on $f(b) - f(a)$.

Remark (Dependencies).

- Definition: Derivative at a Point
- Theorem: Mean Value Theorem
- Order Preservation for Limits

Remark (Return). Return to Theorem

Theorem (Nonincreasing Iff Nonpositive Derivative). *Let $I \subseteq \mathbb{R}$ be an interval, and let $f : I \to \mathbb{R}$ be differentiable on I. Then f is nonincreasing on I if and only if $f'(x) \leq 0$ for all $x \in I$.*

Professional Standard Proof.
Apply the Nondecreasing theorem to $g := -f$. Then $g'(x) = -f'(x)$, and g is nondecreasing iff f is nonincreasing. The theorem gives: g is nondecreasing iff $g'(x) = -f'(x) \geq 0$ for all x, which is equivalent to $f'(x) \leq 0$ for all x. □

Detailed Learning Proof.
Step 1. Reduce to the nondecreasing case via negation. Define $g := -f$. Then g is differentiable on I with $g'(x) = -f'(x)$ for all $x \in I$. Observe that f is nonincreasing if and only if $g = -f$ is nondecreasing: for $a < b$, $f(a) \geq f(b) \iff -f(a) \leq -f(b) \iff g(a) \leq g(b)$.
Step 2. Apply the Nondecreasing Iff Nonnegative Derivative theorem to g. The theorem gives: g is nondecreasing iff $g'(x) \geq 0$ for all $x \in I$.
Step 3. Translate back to f. Substituting $g'(x) = -f'(x)$: $g'(x) \geq 0$ iff $-f'(x) \geq 0$ iff $f'(x) \leq 0$. Combining the equivalences: f is nonincreasing iff $f'(x) \leq 0$ for all $x \in I$. □

Remark (Proof structure). A one-line reduction to the symmetric theorem via $g = -f$.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative

Remark (Return). Return to Lemma

Lemma (Positive Derivative Implies Locally Increasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose f is differentiable at c. If $f'(c) > 0$, then there exists $\delta > 0$ such that $f(x) < f(c)$ for all $x \in I$ with $c-\delta < x < c$, and $f(x) > f(c)$ for all $x \in I$ with $c < x < c+\delta$.*

Professional Standard Proof.
Set $\varepsilon = f'(c)/2 > 0$. By differentiability of f at c, there exists $\delta > 0$ such that for all $x \in I$ with $0 < |x-c| < \delta$,

$$\left|\frac{f(x)-f(c)}{x-c} - f'(c)\right| < \varepsilon = \frac{f'(c)}{2}.$$

This gives $\frac{f(x)-f(c)}{x-c} > f'(c)-\varepsilon = \frac{f'(c)}{2} > 0$ for all such x. For $x \in (c, c+\delta)$, the denominator $x-c > 0$, so $f(x)-f(c) > 0$. For $x \in (c-\delta, c)$, the denominator $x-c < 0$, so $f(x)-f(c) < 0$. □

Detailed Learning Proof.
Step 1. Choose ε to control the sign of the difference quotient. Since $f'(c) > 0$, set $\varepsilon := f'(c)/2$. Then $\varepsilon > 0$ and $f'(c)-\varepsilon = f'(c)/2 > 0$.
Step 2. Extract δ from the definition of differentiability. Since f is differentiable at c with $f'(c) > 0$, there exists $\delta > 0$ such that for all $x \in I$ with $0 < |x-c| < \delta$:

$$\left|\frac{f(x)-f(c)}{x-c} - f'(c)\right| < \varepsilon.$$

This rearranges to:

$$f'(c)-\varepsilon < \frac{f(x)-f(c)}{x-c} < f'(c)+\varepsilon.$$

In particular, $\dfrac{f(x)-f(c)}{x-c} > f'(c)-\varepsilon = \dfrac{f'(c)}{2} > 0$ for all $x \in I$ with $0 < |x-c| < \delta$.
Step 3. Deduce the right-side inequality. For $x \in I$ with $c < x < c+\delta$, the difference quotient is positive and the denominator $x-c > 0$, so the numerator $f(x)-f(c) > 0$, giving $f(x) > f(c)$.
Step 4. Deduce the left-side inequality. For $x \in I$ with $c-\delta < x < c$, the difference quotient is positive and the denominator $x-c < 0$, so the numerator $f(x)-f(c) < 0$, giving $f(x) < f(c)$. □

Remark (Proof structure). A direct ε-δ argument. The key insight is to take $\varepsilon = f'(c)/2$, which forces the difference quotient to be bounded away from zero (positive) throughout the punctured neighbourhood. The sign of the denominator $x-c$ then determines the sign of the numerator $f(x)-f(c)$ on each side of c.

Remark (Dependencies).

- Definition: Derivative at a Point

Remark (Return). Return to Lemma

Lemma (Negative Derivative Implies Locally Decreasing). *Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \to \mathbb{R}$, let $c \in I$, and suppose f is differentiable at c. If $f'(c) < 0$, then there exists $\delta > 0$ such that $f(x) > f(c)$ for all $x \in I$ with $c - \delta < x < c$, and $f(x) < f(c)$ for all $x \in I$ with $c < x < c + \delta$.*

Professional Standard Proof.
Apply the Positive Derivative lemma to $-f$ at c. Since $(-f)'(c) = -f'(c) > 0$, there exists $\delta > 0$ such that $-f(x) < -f(c)$ for $x \in (c, c + \delta)$ and $-f(x) > -f(c)$ for $x \in (c - \delta, c)$. Multiplying by -1 reverses the inequalities and gives the result. □

Detailed Learning Proof.
Step 1. Reduce to the positive derivative case. Define $h := -f$. Then h is differentiable at c and $h'(c) = -f'(c) > 0$ since $f'(c) < 0$.
Step 2. Apply the Positive Derivative Implies Locally Increasing lemma to h. Since $h'(c) > 0$, there exists $\delta > 0$ such that:

- $h(x) < h(c)$ for all $x \in I$ with $c - \delta < x < c$, and
- $h(x) > h(c)$ for all $x \in I$ with $c < x < c + \delta$.

Step 3. Translate back to f. Substituting $h = -f$ and $h(c) = -f(c)$: $-f(x) < -f(c)$ gives $f(x) > f(c)$ for $x \in (c - \delta, c)$, and $-f(x) > -f(c)$ gives $f(x) < f(c)$ for $x \in (c, c + \delta)$. □

Remark (Proof structure). A one-line reduction to the symmetric lemma via $h = -f$. No new ε-δ argument is needed.

Remark (Dependencies).

- Lemma: Positive Derivative Implies Locally Increasing

Remark (Return). Return to Theorem

Theorem (First Derivative Test: Relative Maximum). *Let $I := [a,b]$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$ with f differentiable on (a,c) and (c,b). If there exists $\delta > 0$ with $(c-\delta, c+\delta) \subseteq I$ and $f'(x) \geq 0$ for $x \in (c-\delta, c)$ and $f'(x) \leq 0$ for $x \in (c, c+\delta)$, then f has a relative maximum at c.*

Professional Standard Proof.
By the Nondecreasing theorem applied on $[c-\delta, c]$, $f'(x) \geq 0$ on $(c-\delta, c)$ implies f is nondecreasing on $[c-\delta, c]$, so $f(x) \leq f(c)$ for all $x \in [c-\delta, c]$. By the Nonincreasing theorem applied on $[c, c+\delta]$, $f'(x) \leq 0$ on $(c, c+\delta)$ implies f is nonincreasing on $[c, c+\delta]$, so $f(x) \leq f(c)$ for all $x \in [c, c+\delta]$. Therefore $f(x) \leq f(c)$ for all $x \in (c-\delta, c+\delta)$, establishing a relative maximum at c. □

Detailed Learning Proof.
Step 1. Handle the left side of c. The function f is continuous on $[c-\delta, c]$ and differentiable on $(c-\delta, c)$ with $f'(x) \geq 0$. By the Nondecreasing Iff Nonnegative Derivative theorem, f is nondecreasing on $[c-\delta, c]$. Therefore $f(x) \leq f(c)$ for all $x \in [c-\delta, c]$.
Step 2. Handle the right side of c. The function f is continuous on $[c, c+\delta]$ and differentiable on $(c, c+\delta)$ with $f'(x) \leq 0$. By the Nonincreasing Iff Nonpositive Derivative theorem, f is nonincreasing on $[c, c+\delta]$. Therefore $f(x) \leq f(c)$ for all $x \in [c, c+\delta]$.
Step 3. Combine the two sides. For any $x \in (c-\delta, c+\delta)$, either $x \leq c$ (covered by Step 1) or $x \geq c$ (covered by Step 2), giving $f(x) \leq f(c)$ throughout the punctured neighbourhood.
Step 4. Conclude relative maximum. By definition of relative maximum, the existence of $\delta > 0$ with $f(x) \leq f(c)$ for all $x \in I \cap (c-\delta, c+\delta)$ means f has a relative maximum at c. □

Remark (Proof structure). A direct application of the monotonicity theorems on each side of c separately. The sign condition on f' is converted to a monotonicity condition, which is then used to bound $f(x)$ against $f(c)$ from above on both sides.

Remark (Dependencies).

- Theorem: Nondecreasing Iff Nonnegative Derivative
- Theorem: Nonincreasing Iff Nonpositive Derivative
- Definition: Relative Maximum

Remark (Return). Return to Theorem

Theorem (First Derivative Test: Relative Minimum). *Let $I := [a,b]$, let $f : I \to \mathbb{R}$ be continuous on I, and let $c \in (a,b)$ with f differentiable on (a,c) and (c,b). If there exists $\delta > 0$ with $(c-\delta, c+\delta) \subseteq I$ and $f'(x) \leq 0$ for $x \in (c-\delta, c)$ and $f'(x) \geq 0$ for $x \in (c, c+\delta)$, then f has a relative minimum at c.*

Professional Standard Proof.
Apply the First Derivative Test for Relative Maximum to $-f$ at c. The sign conditions for $(-f)' = -f'$ are reversed: $-f'(x) \geq 0$ on $(c-\delta, c)$ and $-f'(x) \leq 0$ on $(c, c+\delta)$. Therefore $-f$ has a relative maximum at c, meaning $-f(x) \leq -f(c)$, i.e. $f(x) \geq f(c)$, for all $x \in (c-\delta, c+\delta)$. Hence f has a relative minimum at c. □

Detailed Learning Proof.
Step 1. Define $h := -f$ and verify the sign conditions flip. Since $h'(x) = -f'(x)$: on $(c-\delta, c)$, $f'(x) \leq 0$ implies $h'(x) = -f'(x) \geq 0$; on $(c, c+\delta)$, $f'(x) \geq 0$ implies $h'(x) = -f'(x) \leq 0$.
Step 2. Apply the First Derivative Test for Relative Maximum to h.
The sign conditions are satisfied, so h has a relative maximum at c:
$h(x) \leq h(c)$ for all $x \in (c-\delta, c+\delta)$.
Step 3. Translate back to f. From $h(x) \leq h(c)$: $-f(x) \leq -f(c)$, so $f(x) \geq f(c)$ for all $x \in (c-\delta, c+\delta)$. By definition, f has a relative minimum at c. □

Remark (Proof structure). A one-line reduction to the First Derivative Test for Relative Maximum via $h = -f$.

Remark (Dependencies).

- Theorem: First Derivative Test: Relative Maximum
- Definition: Relative Minimum

Remark (Return). Return to Theorem

Theorem (Darboux's Theorem). *Let $I := [a,b] \subseteq \mathbb{R}$ and let $f : I \to \mathbb{R}$ be differentiable on I. If k is a real number strictly between $f'(a)$ and $f'(b)$, then there exists $c \in (a,b)$ such that $f'(c) = k$.*

Professional Standard Proof.
Without loss of generality assume $f'(a) < k < f'(b)$. Define $g(x) := f(x) - kx$. Then g is continuous on $[a,b]$, differentiable on $[a,b]$, and $g'(x) = f'(x) - k$. In particular $g'(a) = f'(a) - k < 0$ and $g'(b) = f'(b) - k > 0$. By the Extreme Value Theorem, g attains its minimum on $[a,b]$ at some point c. Since $g'(a) < 0$, the Negative Derivative Locally Decreasing lemma gives points $x > a$ near a with $g(x) < g(a)$, so $c \neq a$. Since $g'(b) > 0$, the Positive Derivative Locally Increasing lemma gives points $x < b$ near b with $g(x) < g(b)$, so $c \neq b$. Therefore $c \in (a,b)$, and the Interior Extremum Theorem gives $g'(c) = 0$, i.e. $f'(c) = k$. □

Detailed Learning Proof.
Step 1. Reduce to the case $f'(a) < k < f'(b)$. If $f'(a) > k > f'(b)$, apply the argument to $-f$ (whose derivative satisfies $(-f)'(a) = -f'(a) < -k < -f'(b) = (-f)'(b)$) and note that a zero of $(-f)' + k$ is a zero of $f' - k$. Henceforth assume $f'(a) < k < f'(b)$.
Step 2. Construct the auxiliary function. Define $g : [a,b] \to \mathbb{R}$ by $g(x) := f(x) - kx$. Since f is differentiable on $[a,b]$ and kx is differentiable everywhere, g is differentiable on $[a,b]$ with $g'(x) = f'(x) - k$. The target condition $f'(c) = k$ becomes $g'(c) = 0$.
Step 3. Identify the sign conditions on g' at the endpoints.

$$g'(a) = f'(a) - k < 0 \qquad \text{and} \qquad g'(b) = f'(b) - k > 0.$$

Step 4. Locate the global minimum of g in the interior. Since g is continuous on the closed bounded interval $[a,b]$, the Extreme Value Theorem guarantees that g attains its minimum at some point $c \in [a,b]$.
Step 5. Exclude the left endpoint. Since $g'(a) < 0$, the Negative Derivative Implies Locally Decreasing lemma applies: there exists $\delta_1 > 0$ such that $g(x) < g(a)$ for all $x \in (a, a + \delta_1)$. In particular, g takes values strictly less than $g(a)$ near a, so the minimum of g cannot be at a. Therefore $c \neq a$.
Step 6. Exclude the right endpoint. Since $g'(b) > 0$, the Positive Derivative Implies Locally Increasing lemma applies: there exists $\delta_2 > 0$ such that $g(x) < g(b)$ for all $x \in (b - \delta_2, b)$. In particular, g takes values strictly less than $g(b)$ near b, so the minimum cannot be at b. Therefore $c \neq b$.
Step 7. Conclude. Since $c \in [a,b]$ with $c \neq a$ and $c \neq b$, we have $c \in (a,b)$. Since c is an interior minimum and f is differentiable on (a,b), the Interior Extremum Theorem gives $g'(c) = 0$, hence $f'(c) = k$. □

Remark (Proof structure). The proof constructs an auxiliary function $g = f - kx$ that shifts the target value k to a zero of g'. The strategy then has two phases. First, the EVT guarantees g attains its minimum on

$[a, b]$. Second, the sign conditions $g'(a) < 0$ and $g'(b) > 0$ -- derived from $f'(a) < k < f'(b)$ -- are used via the local behavior lemmas to show the minimum lies strictly in the interior: near a the function is decreasing (so a is not a min), and near b it is increasing (so b is not a min). The Interior Extremum Theorem then converts the interior minimum into the equation $g'(c) = 0$. The proof does not require continuity of f'; differentiability of f alone is sufficient.

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Lemma: Positive Derivative Implies Locally Increasing
- Lemma: Negative Derivative Implies Locally Decreasing
- Definition: Derivative at a Point

Remark (Return). Return to Theorem

Corollary (Necessary Condition for an Extremum). *Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. If f has a relative extremum at c, then either $f'(c)$ does not exist, or $f'(c) = 0$.*

Professional Standard Proof.
Let $I \subseteq \mathbb{R}$ be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$. Assume f has a relative extremum at c. Consider two cases. If $f'(c)$ does not exist, the conclusion holds immediately. If $f'(c)$ exists, then by the Interior Extremum Theorem, $f'(c) = 0$. In either case the disjunction holds. □

Detailed Learning Proof.
Step 1. Establish the case structure. Since $f'(c)$ either exists or does not exist, the proof considers two exhaustive cases: $f'(c)$ exists, or $f'(c)$ does not exist.
Step 2. Handle the case where $f'(c)$ does not exist. If $f'(c)$ does not exist, then the first disjunct of the conclusion holds immediately, making the entire disjunction true.
Step 3. Handle the case where $f'(c)$ exists. If $f'(c)$ exists and f has a relative extremum at the interior point c, then by the Interior Extremum Theorem, $f'(c) = 0$. This makes the second disjunct true, so the entire disjunction holds.
Step 4. Conclude. Since the disjunction holds in both exhaustive cases, the theorem is proved. □

Remark (Proof structure). This is a two-case exhaustion proof based on whether the derivative exists. The first case is trivial by the definition of disjunction. The second case applies the Interior Extremum Theorem directly. The proof structure is purely logical with no calculation required.

Remark (Dependencies).

- Interior Extremum Theorem

Remark (Return). Return to Theorem

Theorem (Rolle's Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a,b] \to \mathbb{R}$ is continuous on $[a,b]$, differentiable on (a,b), and satisfies $f(a) = f(b)$. Then there exists $c \in (a,b)$ such that $f'(c) = 0$.*

Professional Standard Proof.
By the Extreme Value Theorem, f attains a maximum M and a minimum m on $[a,b]$. If $M = m$ then f is constant on $[a,b]$, so $f' \equiv 0$ on (a,b) and any $c \in (a,b)$ serves. Otherwise $M > m$, so at least one of M, m differs from $f(a) = f(b)$ and is therefore attained at some interior point $c \in (a,b)$. Since c is a relative extremum and f is differentiable at c, the Interior Extremum Theorem gives $f'(c) = 0$. □

Detailed Learning Proof.
Step 1. Apply the Extreme Value Theorem. Since f is continuous on the closed bounded interval $[a,b]$, the Extreme Value Theorem guarantees that f attains both its maximum value $M = \max_{[a,b]} f$ and its minimum value $m = \min_{[a,b]} f$ on $[a,b]$.
Step 2. Handle the degenerate case. If $M = m$, then f is constant on $[a,b]$ -- every value of f equals $M = m$. A constant function has derivative zero everywhere it is differentiable. Therefore $f'(c) = 0$ for every $c \in (a,b)$, and the theorem holds.
Step 3. Handle the non-degenerate case: locate an interior extremum. Assume $M > m$. Since $f(a) = f(b)$, at most one of $\{M, m\}$ can equal the common endpoint value $f(a)$. Therefore at least one of M, m is not equal to $f(a) = f(b)$. Call it E (either the maximum or minimum). Any point at which f attains the value E cannot be an endpoint (since $E \neq f(a) = f(b)$), so it must lie in the open interval (a,b). Let $c \in (a,b)$ be a point where $f(c) = E$.
Step 4. Identify c as a relative extremum. If $E = M$ then $f(c) = M \geq f(x)$ for all $x \in [a,b]$, so in particular $f(c) \geq f(x)$ for all x in any neighbourhood of c within $[a,b]$. Thus f has a relative maximum at c. If $E = m$ then $f(c) = m \leq f(x)$ for all $x \in [a,b]$, so f has a relative minimum at c. In either case c is a relative extremum of f.
Step 5. Apply the Interior Extremum Theorem. Since $c \in (a,b)$, c is an interior point of $[a,b]$. Since f is differentiable on (a,b) and $c \in (a,b)$, f is differentiable at c. By the Interior Extremum Theorem, differentiability at a relative extremum implies $f'(c) = 0$. □

Remark (Proof structure). The proof is a case split on whether f is constant, followed by an existence argument for an interior extremum. The constant case is immediate. The non-constant case uses a pigeonhole observation: with $f(a) = f(b)$, both the maximum and minimum cannot simultaneously equal the endpoint value, so at least one extreme value is attained strictly inside the interval. The Interior Extremum Theorem then converts the extremum into a zero of the derivative. The three ingredients are the Extreme Value Theorem (existence of extreme values), a boundary exclusion argument (the extreme value is not at the `endpoints), and the Interior Extremum Theorem (differentiability at an interior extremum forces zero derivative).`

Remark (Dependencies).

- Theorem: Extreme Value Theorem
- Theorem: Interior Extremum Theorem
- Definition: Derivative at a Point

Remark (Return). Return to Theorem

Theorem (Mean Value Theorem). *Let $a, b \in \mathbb{R}$ with $a < b$. Suppose that $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists $c \in (a, b)$ such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$.*

Professional Standard Proof.
Define the auxiliary function

$$g(x) := f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Then g is continuous on $[a, b]$, differentiable on (a, b), and satisfies $g(a) = 0 = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ with $g'(c) = 0$. Since $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$, this gives $f'(c) = \frac{f(b)-f(a)}{b-a}$. □

Detailed Learning Proof.
Step 1. Construct the auxiliary function. The secant line through $(a, f(a))$ and $(b, f(b))$ has slope $\frac{f(b)-f(a)}{b-a}$ and passes through $(a, f(a))$, giving the line $\ell(x) = f(a) + \frac{f(b)-f(a)}{b-a}(x - a)$. Define

$$g(x) := f(x) - \ell(x) = f(x) - f(a) - \frac{f(b) - f(a)}{b - a}(x - a).$$

Geometrically, $g(x)$ is the signed vertical distance from the graph of f to the secant line at x.
Step 2. Verify the hypotheses of Rolle's Theorem for g. Continuity: g is the difference of f (continuous on $[a, b]$) and the linear function ℓ (continuous everywhere), so g is continuous on $[a, b]$. Differentiability: $g'(x) = f'(x) - \frac{f(b)-f(a)}{b-a}$ exists on (a, b) since f is differentiable there. Equal endpoint values: $g(a) = f(a) - f(a) - 0 = 0$ and $g(b) = f(b) - f(a) - (f(b) - f(a)) = 0$, so $g(a) = g(b) = 0$.
Step 3. Apply Rolle's Theorem. Since g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b)$, Rolle's Theorem guarantees the existence of $c \in (a, b)$ with $g'(c) = 0$.
Step 4. Recover the conclusion for f. From Step 2, $g'(c) = f'(c) - \frac{f(b)-f(a)}{b-a}$. Setting this to zero gives

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

□

Remark (Proof structure). The proof reduces the MVT to Rolle's Theorem by a linear subtraction trick. Subtracting the secant line from f produces an auxiliary function g that vanishes at both endpoints, making Rolle's Theorem applicable. The derivative of g at the Rolle point c then directly yields the MVT conclusion. The auxiliary function $g(x) = f(x) - \ell(x)$ measures the deviation of f from its secant; the proof asserts that this deviation has a horizontal tangent somewhere in the interior.

Remark (Dependencies).

- Theorem: Rolle's Theorem
- Definition: Derivative at a Point

Remark (Return). Return to Theorem

Theorem (Taylor's Theorem with Lagrange Remainder). *Let $a, b \in \mathbb{R}$ with $a < b$, let $n \in \mathbb{N}$, and let $f : [a, b] \to \mathbb{R}$. Suppose that $f, f', \ldots, f^{(n)}$ are continuous on $[a, b]$ and that $f^{(n+1)}$ exists on (a, b). Then, for every $x \in (a, b)$, there exists c strictly between a and x such that*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Professional Standard Proof.
TODO: professional standard proof for taylor-theorem-lagrange-remainder. □

Detailed Learning Proof.
TODO: detailed learning proof for taylor-theorem-lagrange-remainder. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Remark (Return). Return to Theorem

Corollary (Taylor Expansion with Peano Remainder). *Let $I \subseteq \mathbb{R}$ be an interval, let $a \in I$, let $n \in \mathbb{N}$, and let $f : I \to \mathbb{R}$ have derivatives up to order n in a neighbourhood of a. If $f^{(n)}$ is continuous at a, then*

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + o\big((x-a)^n\big) \qquad \textit{as } x \to a.$$

Professional Standard Proof.
TODO: professional standard proof for taylor-expansion-peano-remainder. □

Detailed Learning Proof.
TODO: detailed learning proof for taylor-expansion-peano-remainder. □

Remark (Proof structure).

Remark (Dependencies).

- TODO: list mathematical dependencies.

Capstone

Capstone Problem

Problem. TODO: state one synthesizing problem for Differentiation, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Chapter 8

Riemann Integration

riemann-integration

Differentiation → **Riemann Integration** → Function Sequences

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Riemann Integration, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 9

Measure Theory

measure-theory

Functional Analysis → **Measure Theory** → Probability Theory

9.1 Measure Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Measure Theory, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 10

Probability Theory

probability-theory

Measure Theory → **Probability Theory** → Quantum Calculus

10.1 Probability Theory Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Probability Theory, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 11

Functional Analysis

functional-analysis

Complex Analysis → **Functional Analysis** → Measure Theory

11.1 Functional Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Functional Analysis, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Chapter 12

Complex Analysis

complex-analysis

Real Analysis $\rightarrow$ **Complex Analysis** $\rightarrow$ Functional Analysis

12.1 Complex Analysis Notes

Remark (Status). This chapter is planned. Active mathematical content has not yet been added.

Proofs

Capstone Problem

Problem. TODO: state one synthesizing problem for Complex Analysis, using only concepts at or before this chapter in the registry.

Remark (Dependency ceiling). The capstone may use only results routed at or before this chapter.

Capstone

Bibliography